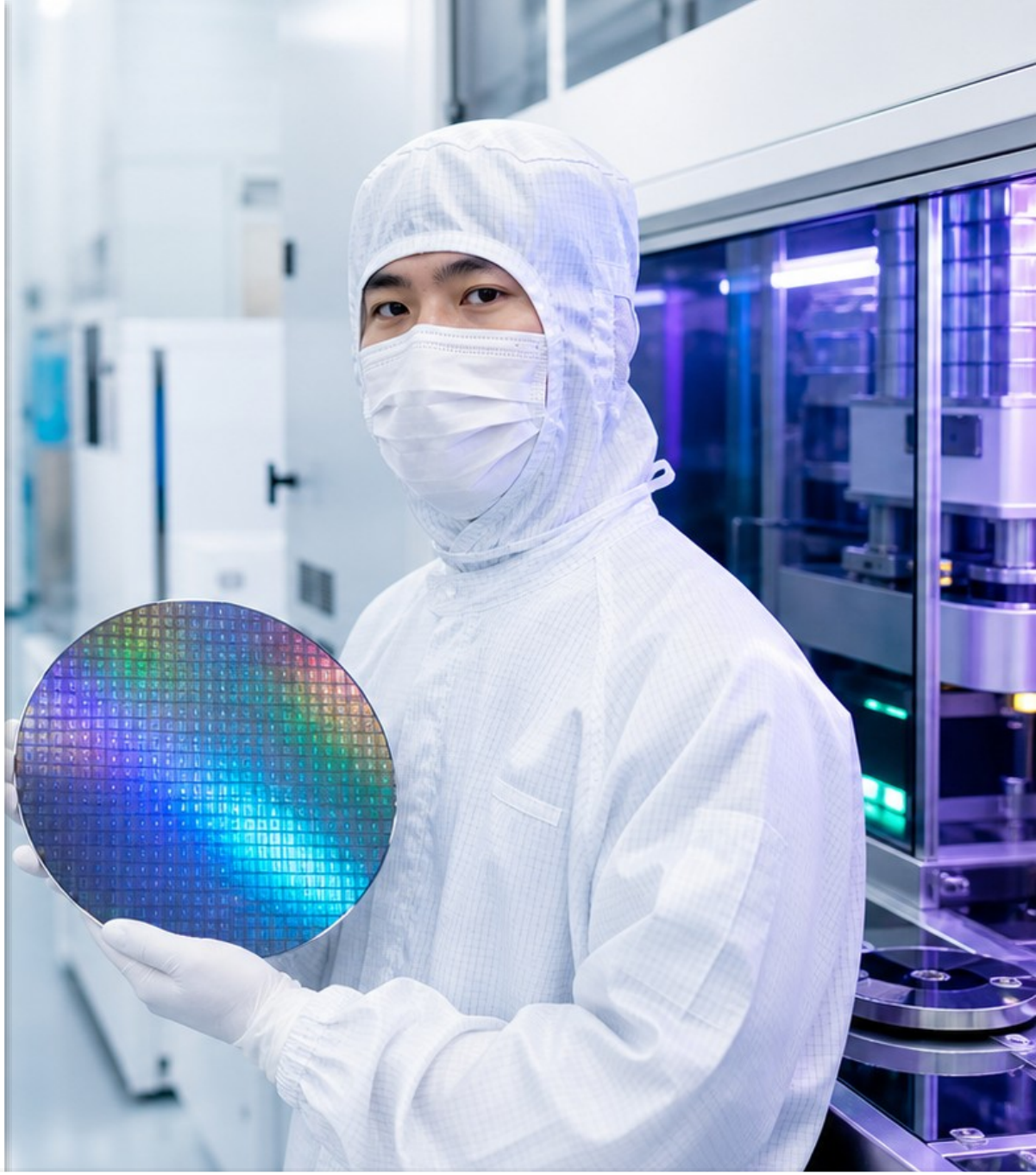
NON-WSQ COURSE

Semiconductor Device and Process Integration Masterclass

Connect device behaviour to the process decisions that create it.

C605
Version 1.0 · 1 Oct 2026
Trainer: _____

Tertiary Infotech Academy Pte Ltd
UEN 201200696W



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About the Trainer

General Trainer

Complete before delivery

NAME	_____
TITLE / DESIGNATION	_____
QUALIFICATIONS	_____
AREAS OF EXPERTISE	_____
TRAINING & INDUSTRY EXPERIENCE	_____
CONTACT	_____

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ADMIN · TRAINER

About the Trainer

Dr Alfred Ang

Founder · Senior Educator · PhD in Physics

NAME	Dr Alfred Ang
TITLE / DESIGNATION	Founder and Senior Educator
QUALIFICATIONS	PhD in Physics · ACTA/ACLP equivalent adult education experience
AREAS OF EXPERTISE	Device physics · semiconductor processing · applied AI
TRAINING & INDUSTRY EXPERIENCE	Decades of STEM teaching and technical training
CONTACT	enquiry@tertiaryinfotech.com

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Let's Know Each Other

1

Your role

What part of semiconductor design, manufacturing, test or quality do you work with?

2

Your experience

Which process module or device topic is most familiar to you?

3

Your challenge

What yield, performance or reliability problem would you like to understand better?

4

Your goal

What decision should you be able to make by the end of Day 2?

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Ground Rules

1 Be present

Set mobile phones to silent and step out quietly for urgent calls.

2 Participate

Ask questions, test ideas and contribute process or device examples.

3 Respect evidence

Challenge the reasoning, not the person; distinguish data from assumptions.

4 Protect information

Use only approved, de-identified workplace examples during discussion.

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ADMIN · TWO LEARNING UNITS

Course Outline

LU1

Fundamentals of Device Physics and Device Fabrication

Semiconductor physics · process modules · manufacturing flows · layout and design rules

Topic 1

LU2

Process Integration Issues

Yield · device performance · reliability · packaging · follow-up actions

Topic 2

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ADMIN · 2 DAYS · 15 HOURS

Lesson Plan

1

Day 1 AM

Admin, materials, carrier physics, junctions and devices

2

Day 1 PM

Process modules, CMOS flow, design rules and Activities 1-3

3

Day 2 AM

Integration interactions, yield, performance and Activities 4-6

4

Day 2 PM

Reliability, packaging, Activities 7-8 and recap

Each day runs 9:30 AM – 5:30 PM · 7.5 instructional hours

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ADMIN · LEARNER MATERIALS

Access the Course Activities

OPTION A

Clone the repository

git clone https://github.com/tertiarycourses/C605-Semiconductor-Device-and-Process-Integration-Masterclass.git

OPTION B

Download the ZIP

Open GitHub · choose Code · Download ZIP · extract the activities folder

https://github.com/tertiarycourses/C605-Semiconductor-Device-and-Process-Integration-Masterclass

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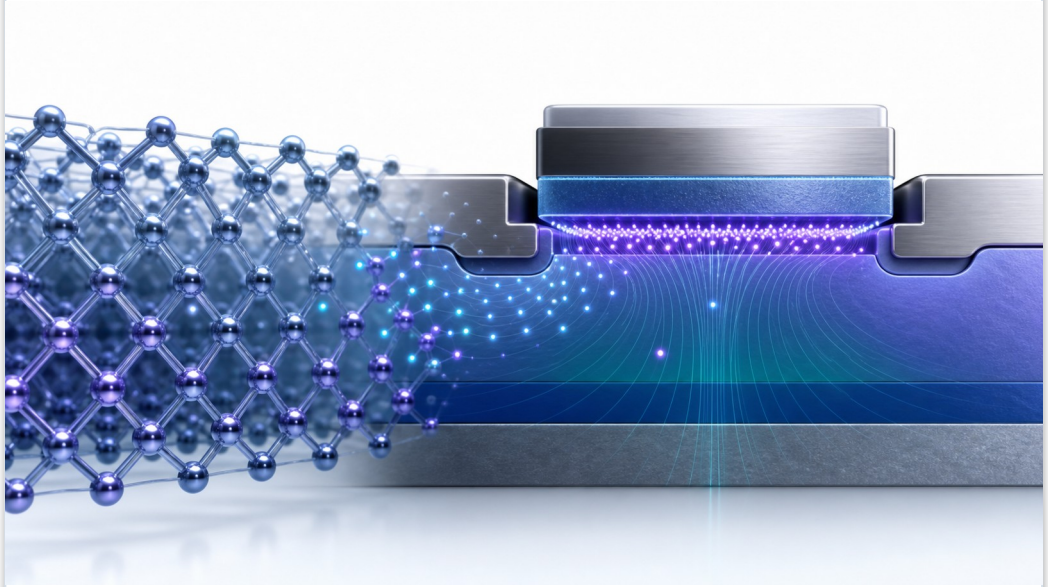
TOPIC 01

01

Fundamentals of Device Physics and Device Fabrication

Concepts · mechanisms · process evidence · integration decisions



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TOPIC 01

Topics

1 Introduction to semiconductor device physics

Introduction to semiconductor device physics
Introduction to process modules
Basic manufacturing process flows
Layout and design rules

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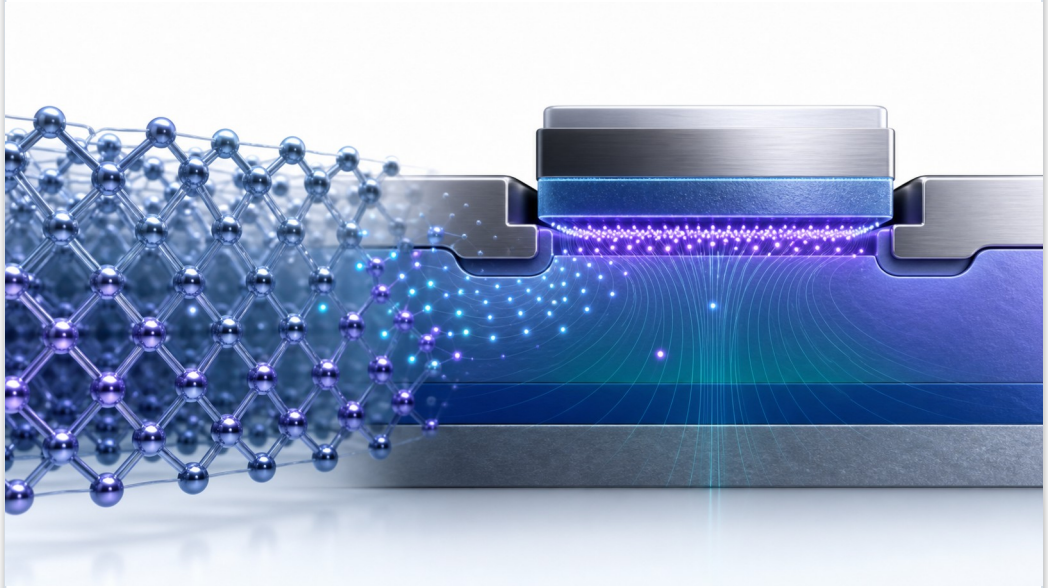
TOPIC 01

Semiconductor Materials and Carrier Physics

Concepts · mechanisms · process evidence · integration decisions



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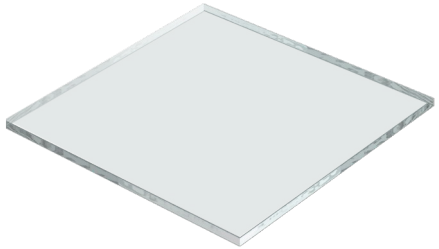
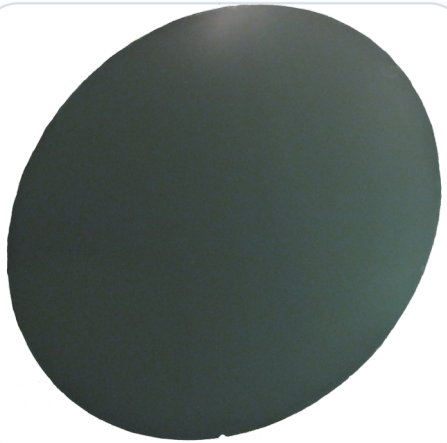
TOPIC 01

What are Semiconductors?





1 Evidence 1

Semiconductors are materials whose electrical properties lie between Conductors and Insulators
Eg. Silicon and Germanium

2 Conductors

Conductors

3 Insulators

Insulators

4 Semiconductors

Semiconductors

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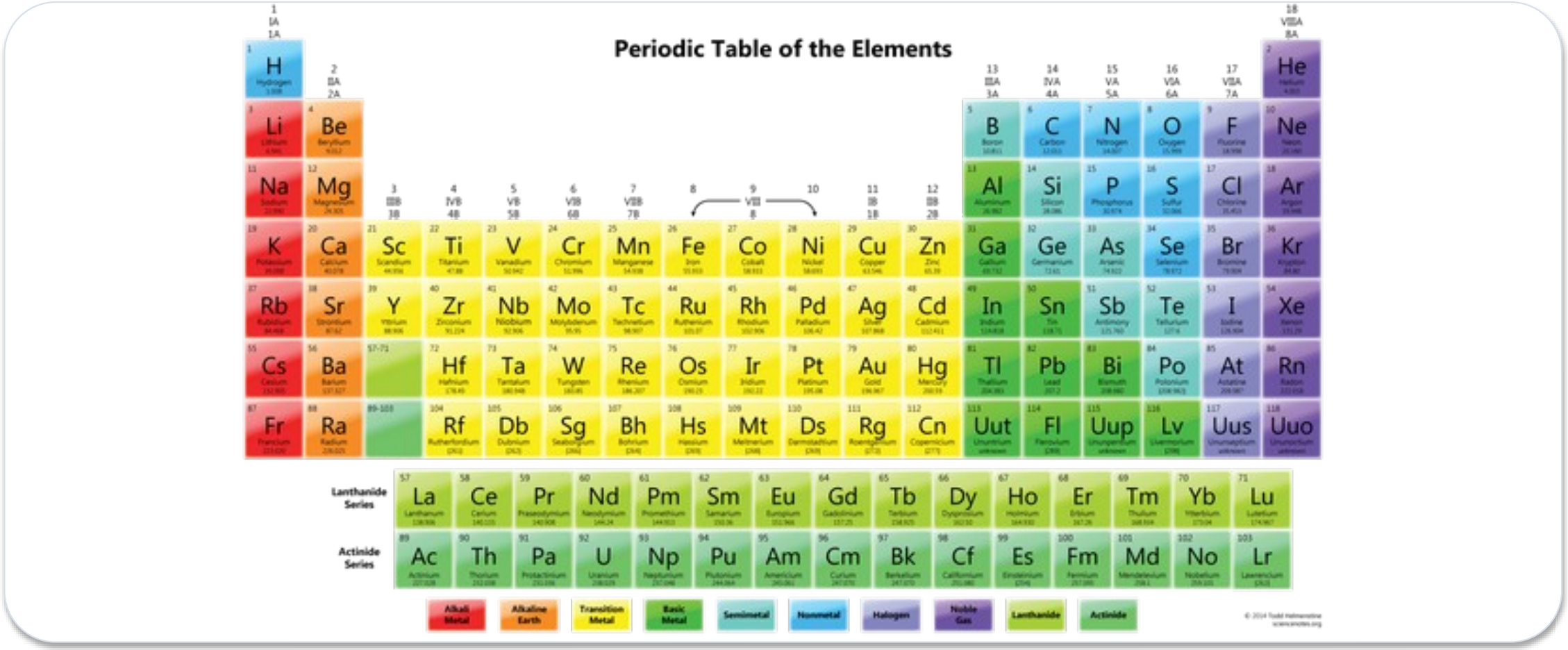
TOPIC 01

Semiconductors are elements with 4 valence electrons



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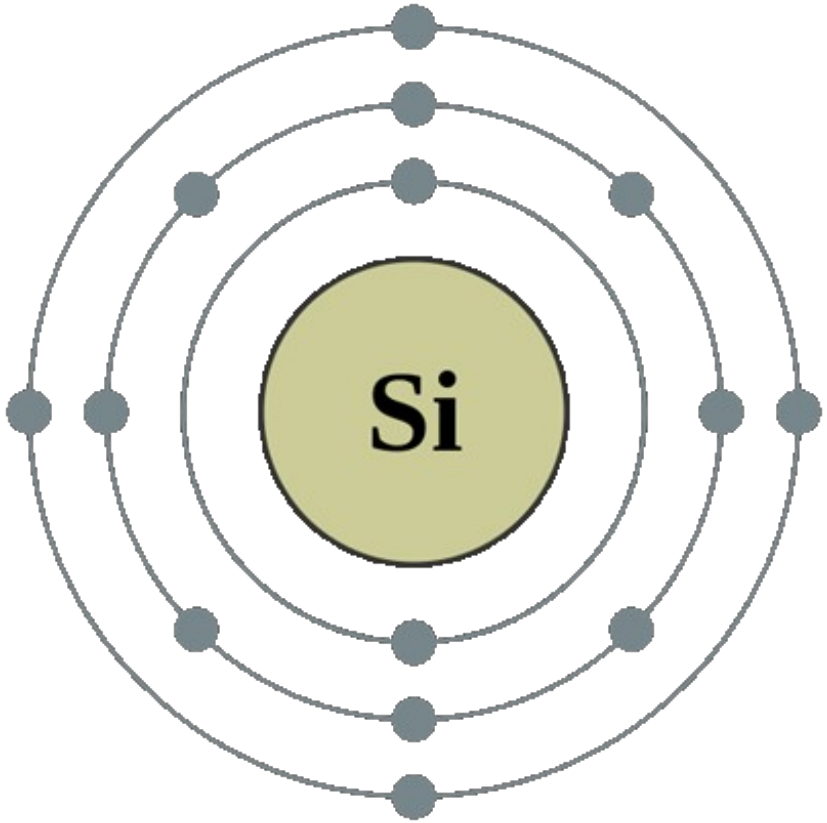
TOPIC 01

Silicon



1

Silicon is the most widely used semiconductor

Silicon is the most widely used semiconductor
It is cheap and readily available
Silicon dioxide is also a very stable insulator

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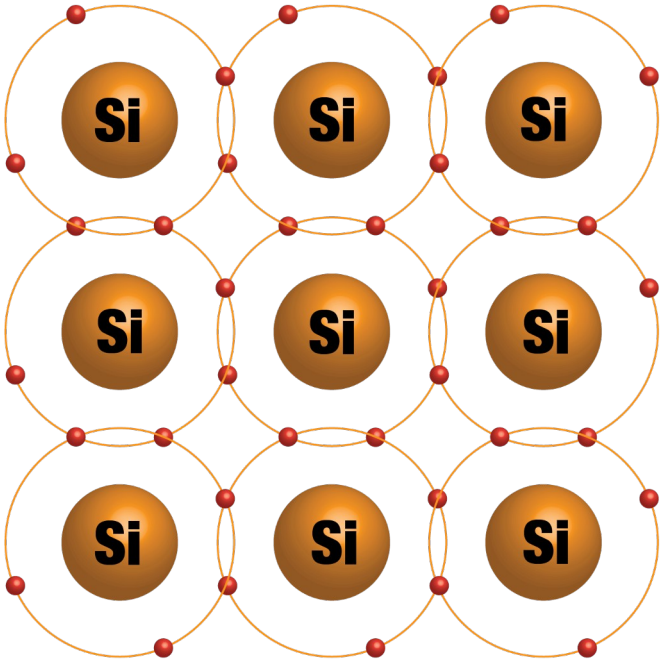
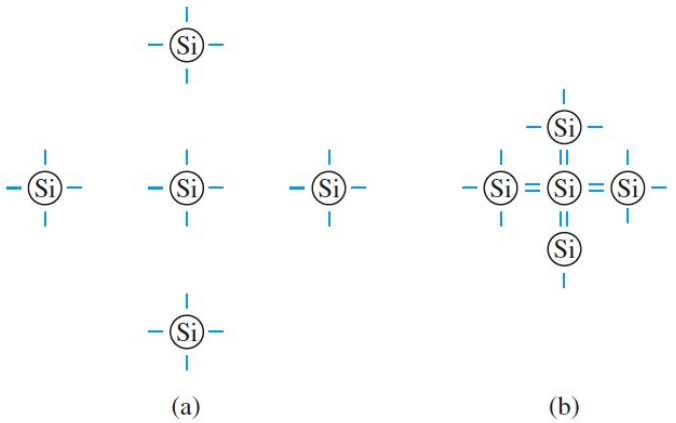
TOPIC 01

Silicon Crystal





1

Evidence 1

Silicon atoms in a crystal are bonded by covalent bonds

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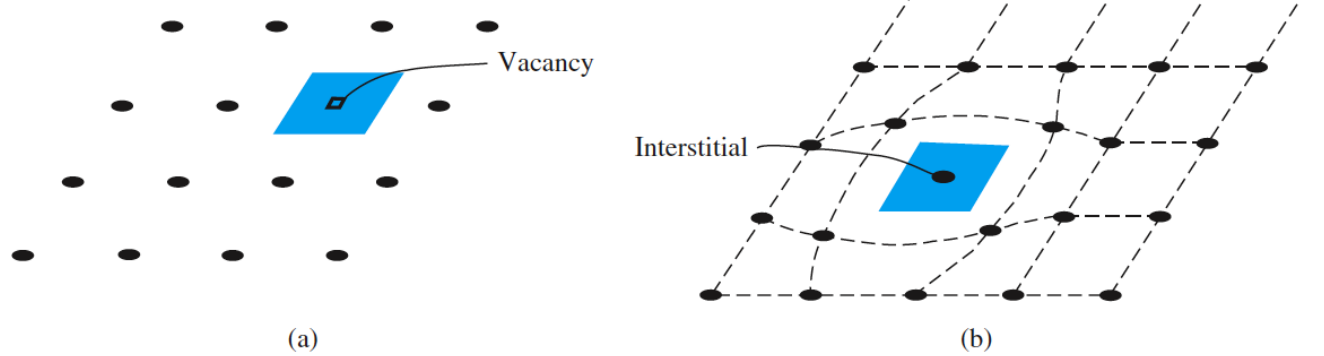
TOPIC 01

Defects



1

Evidence 1

Two-dimensional representation of a single-crystal lattice showing (a) a vacancy defect and (b) an interstitial defect.

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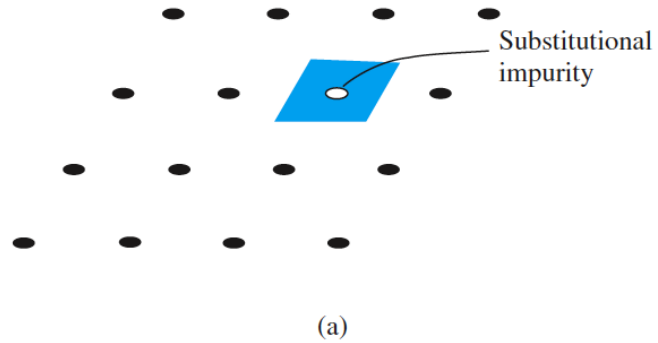
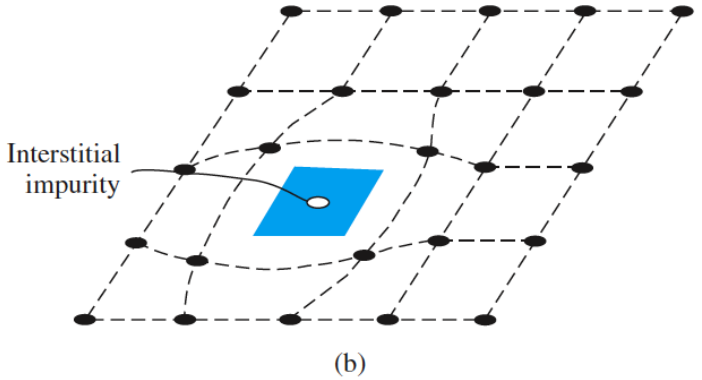
TOPIC 01

Impurities





1

Evidence 1

Two-dimensional representation of a single-crystal lattice showing (a) a substitutional impurity and (b) an intersititial impurity.

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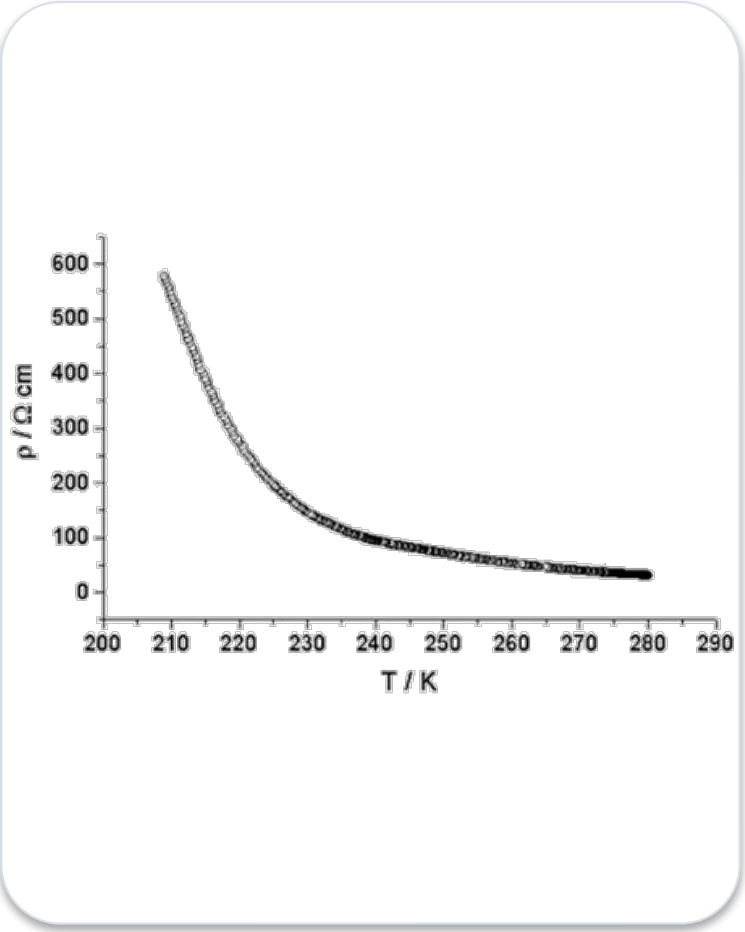
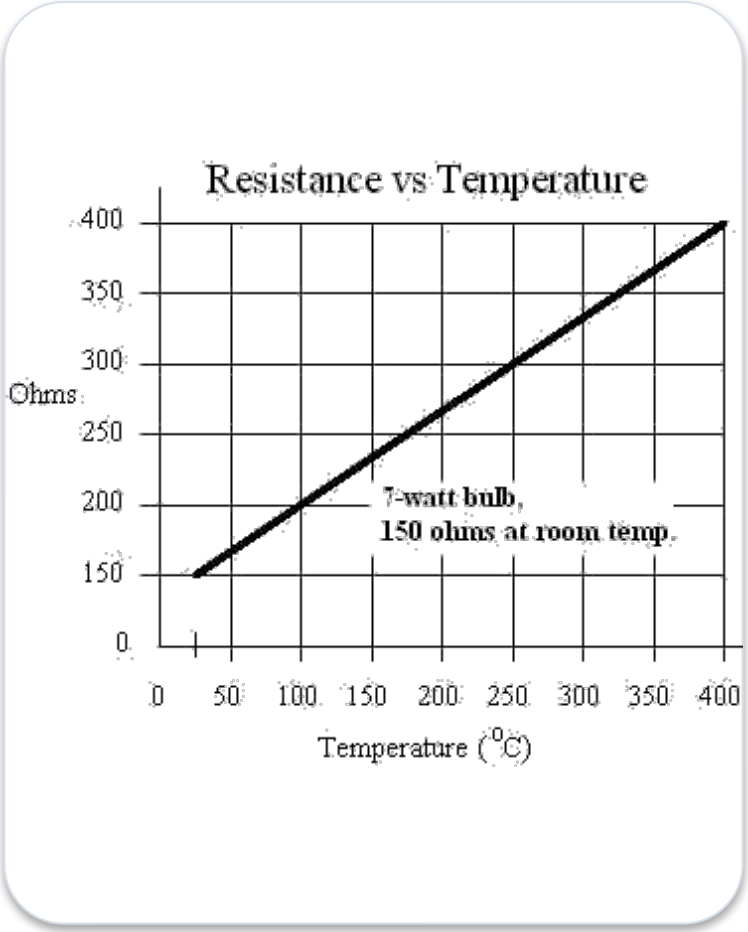
TOPIC 01

Semiconductor Resistivity





1 **Evidence 1**

A semiconductor crystal with no impurities is known as an intrinsic semiconductor. Unlike conductor, the conductivity of an intrinsic semiconductor increases with temperature.

2 **Conductor**

Conductor

3 **Semiconductor**

Semiconductor

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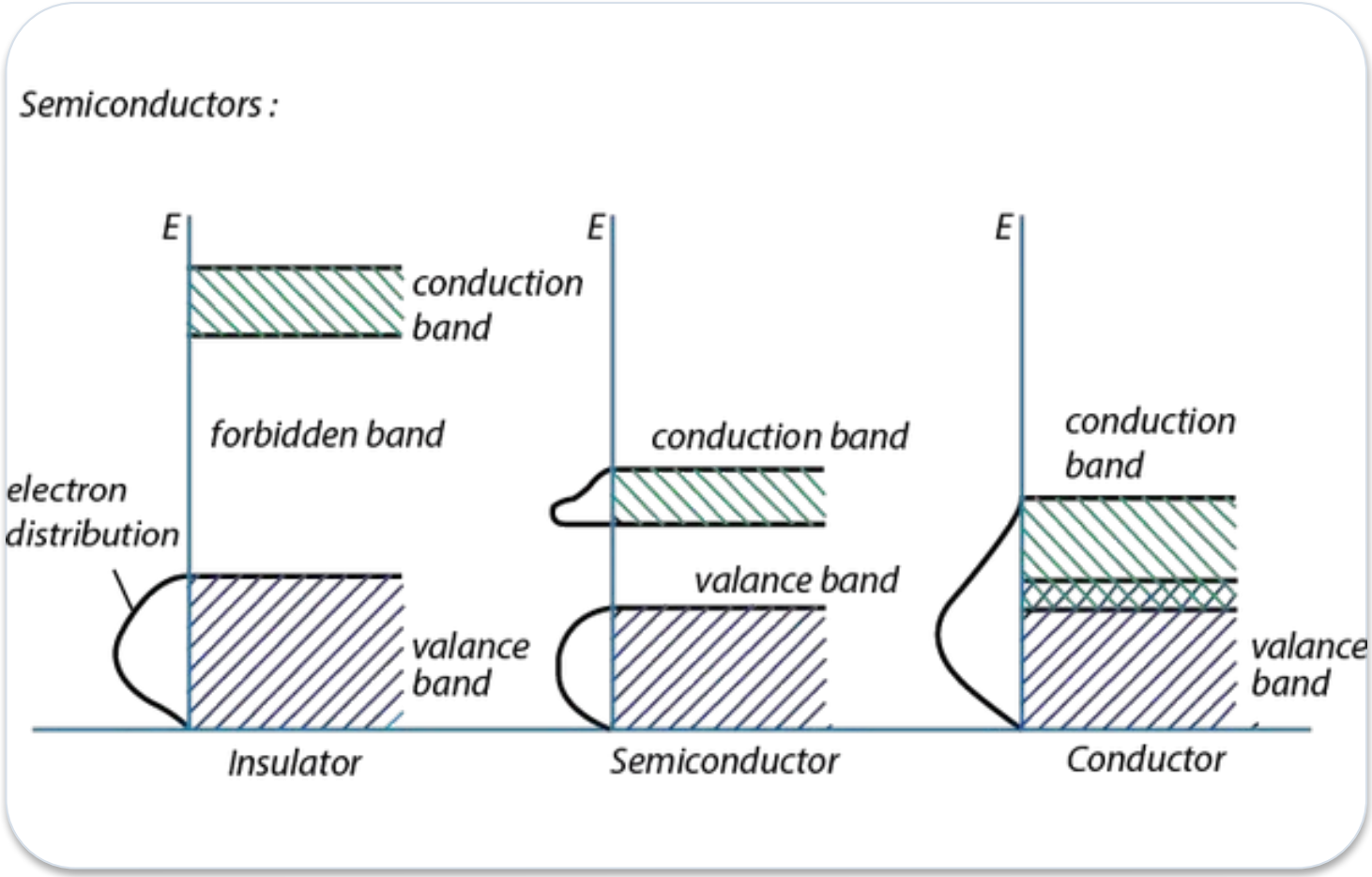
TOPIC 01

Energy Band Model



1

Evidence 1

To explain the semiconductor resistivity, a new theory known as the band theory is introduced in 1930's.

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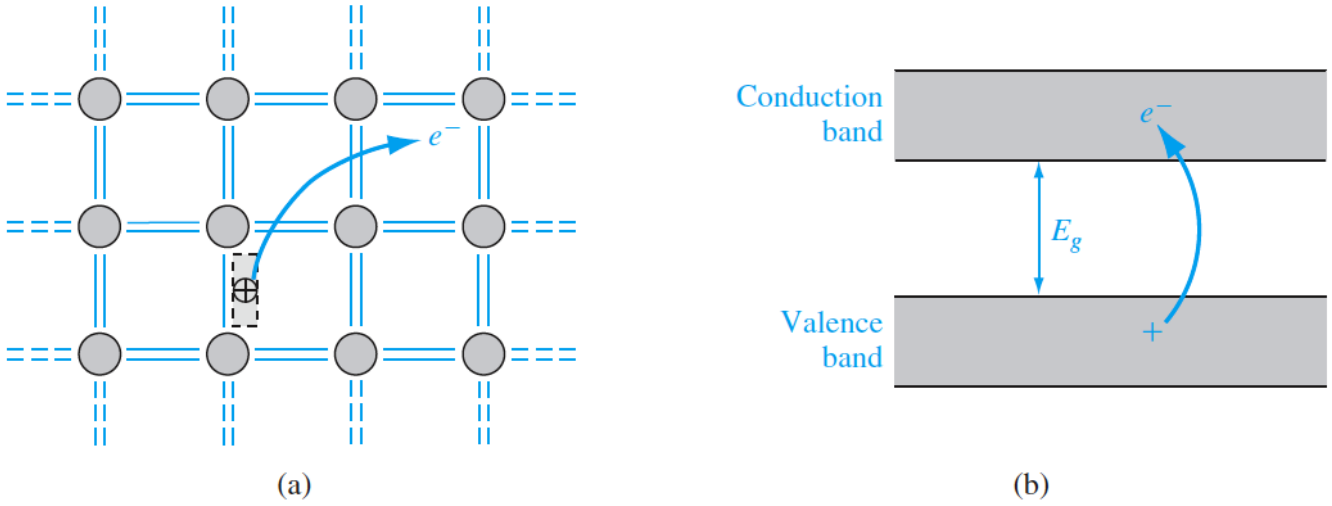
TOPIC 01

Semiconductor Band Theory



1

Evidence 1

Semiconductor energy gaps is typically smaller than about 1 electron volt, therefore it is possible for enough electrons to be excited thermally into the conduction band, so that an applied electric field can produce a modest current.
This explains why the resistivity reduces with increasing temperature

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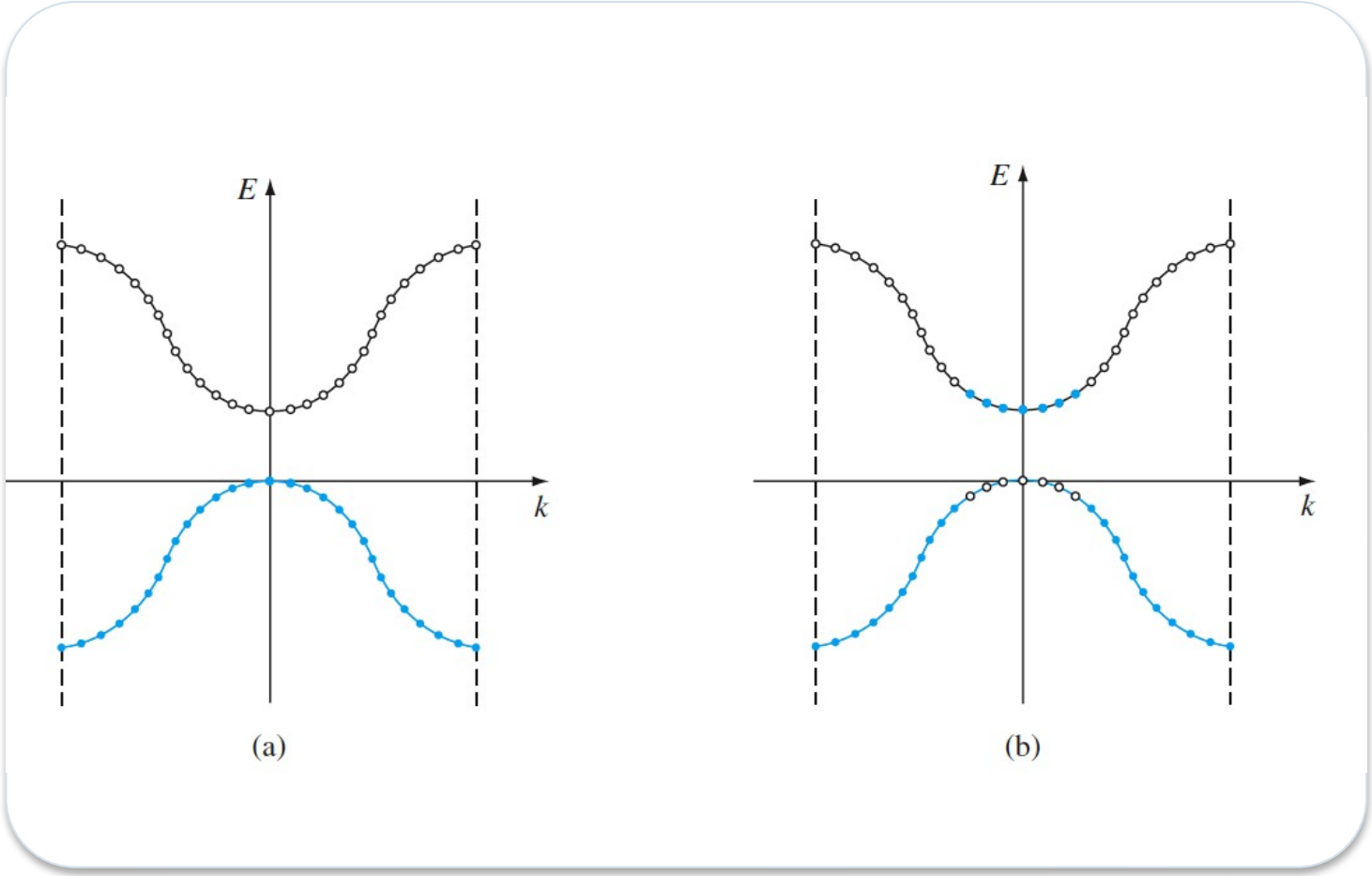
Semiconductor Band Theory



1

Evidence 1

The E versus k diagram of the conduction and valence bands of a semiconductor at (a) T=0 K and (b) T >0 K.

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TOPIC 01

Holes

Positively charged "empty state" created when a valence electron moves to the conduction band
At T > 0 K, valence electrons gain thermal energy and may hop into empty states
Movement of electrons into empty states creates the appearance of positive charge movement
This moving positive charge is called a hole, acting as a second charge carrier
Holes can be modeled as classical particles using Newtonian mechanics

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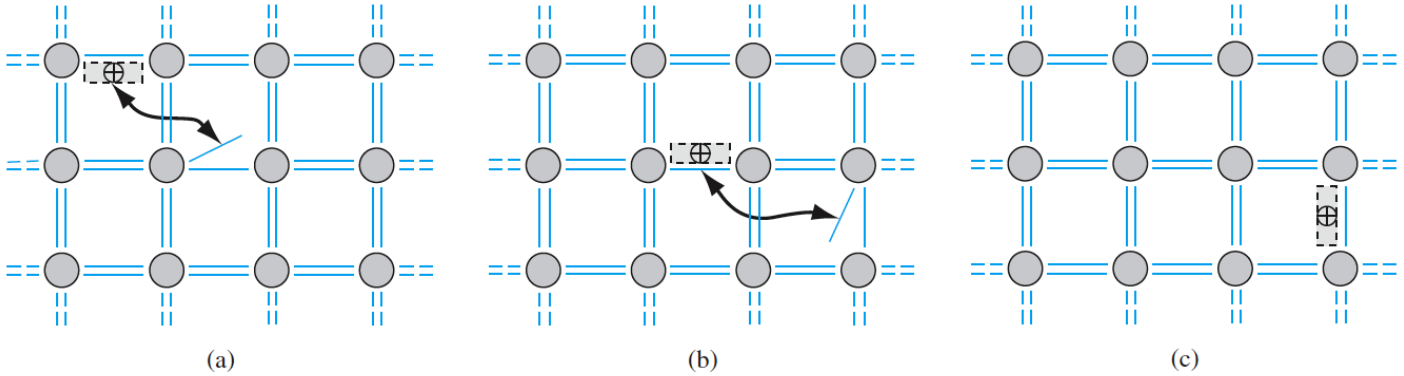
Holes



1

Evidence 1

Visualization of the movement of a hole in a semiconductor.

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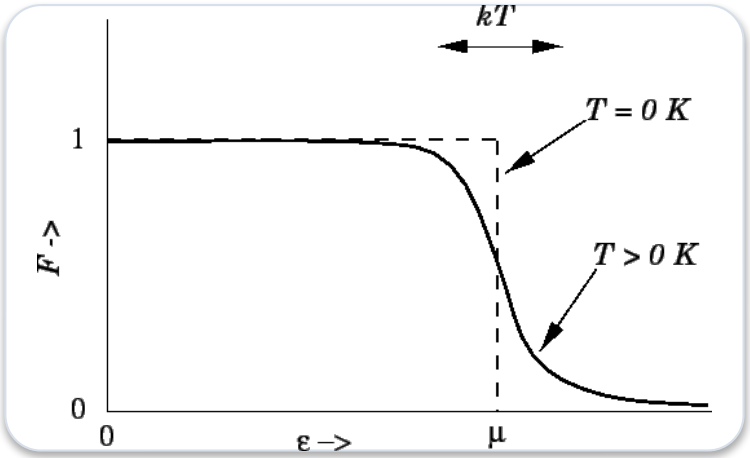
TOPIC 01

Fermi-Dirac Distribution

$$f(E) = \frac{1}{1 + \exp\left[(E - E_F)/kT\right]}$$



1 **EF is called Fermi level**

EF is called Fermi level

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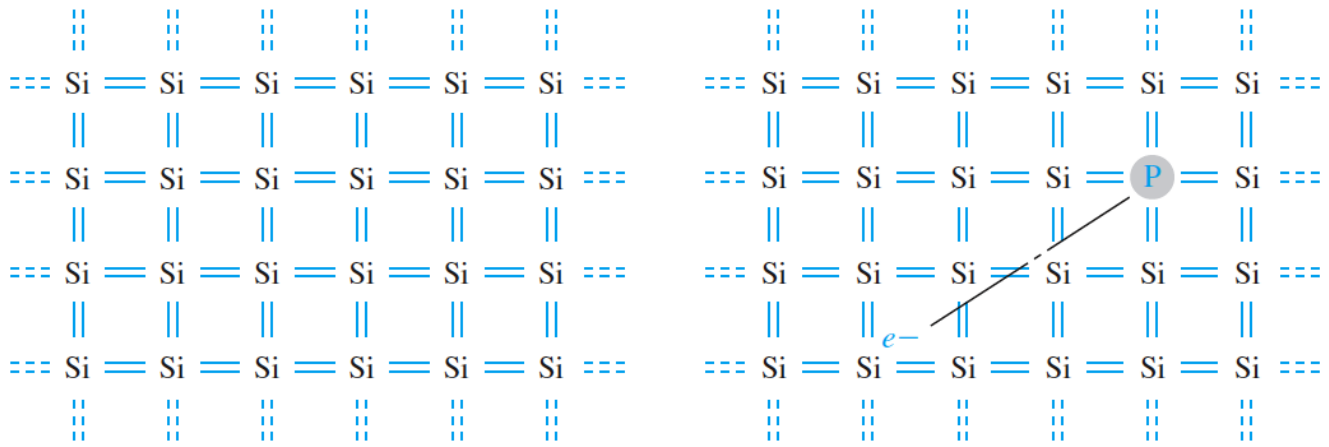
Doping



1

Evidence 1

By doping the intrinsic material, the material can be altered to have more electrons or holes.
There are two types of doped material
n-type (doped with P => electrons)
p-type (doped with B => holes)

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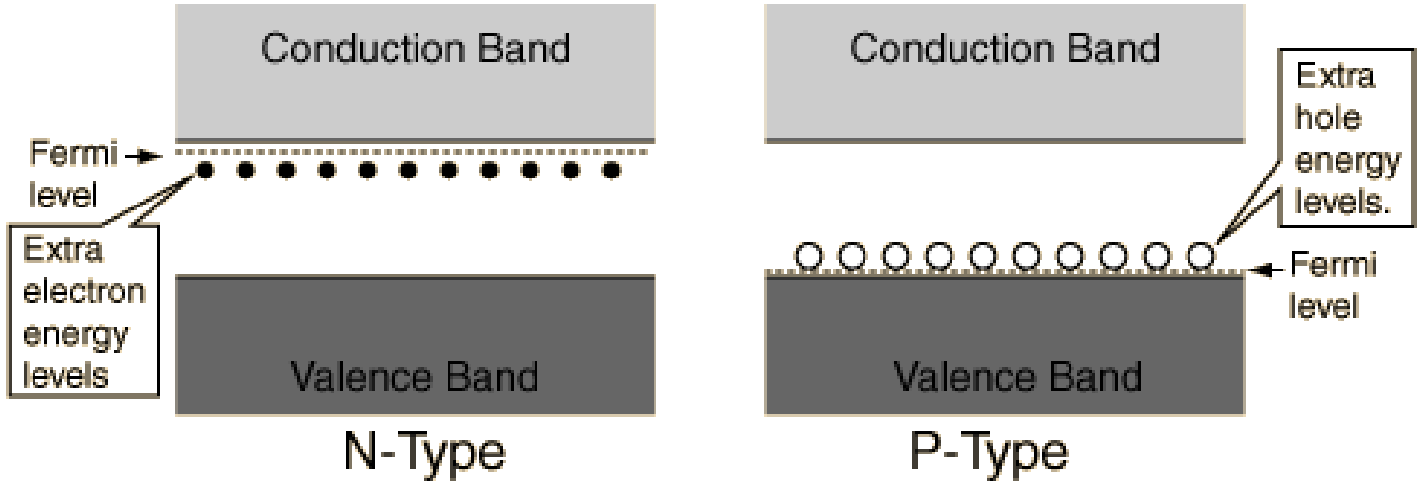
TOPIC 01

Doping on Energy Bands



1

Evidence 1

For N-type semiconductor, the fermi level is near to the conduction band
For P-type semiconductor, the fermi level is near to the valence band

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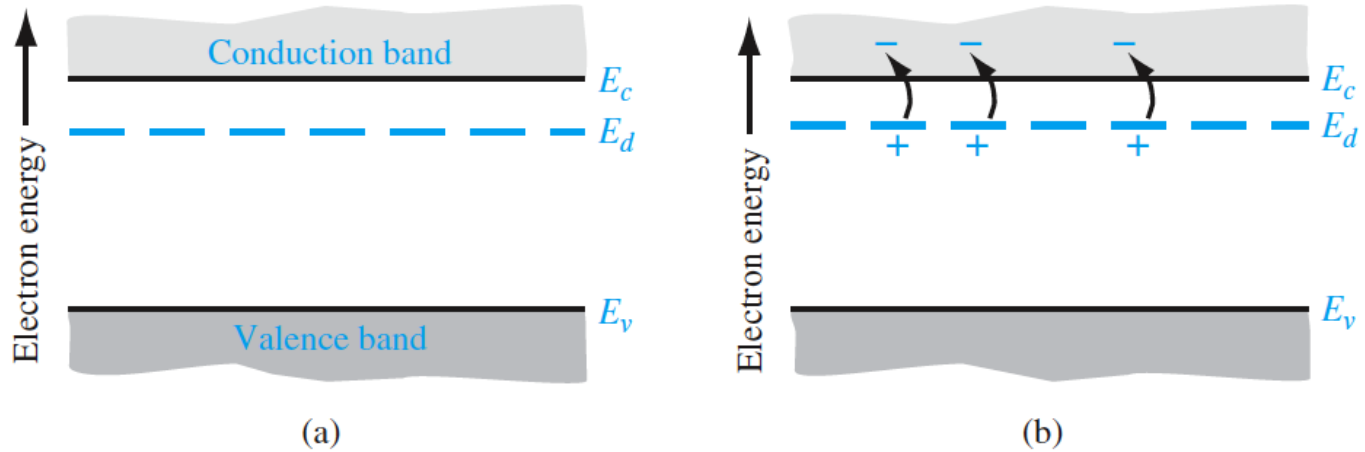
Doping on Energy Bands



1

Evidence 1

The energy-band diagram showing (a) the discrete donor energy state and (b) the effect of a donor state being ionized.

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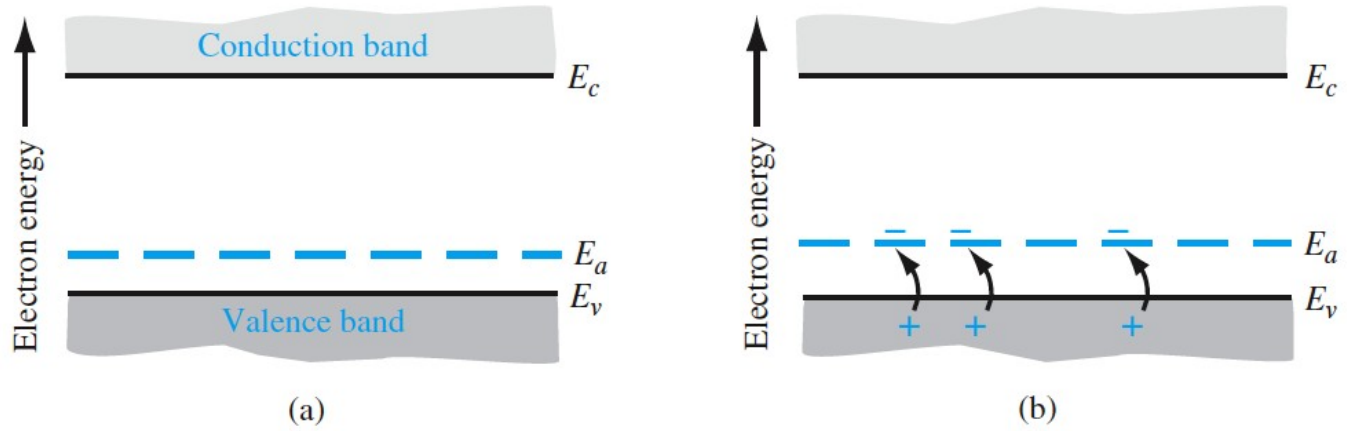
Doping on Energy Bands



1 **Evidence 1**

The energy-band diagram showing (a) the discrete acceptor energy state and (b) the effect of an acceptor state being ionized.

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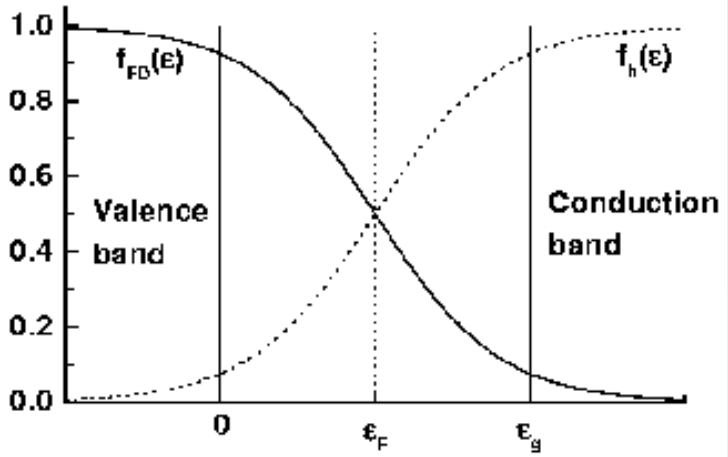
TOPIC 01

Electron Concentration

$$n_0 = \int_{E_c}^{\infty} f(E)N(E)dE$$



1

Evidence 1

The Fermi distribution function can be used to calculate the concentrations of electrons and holes in a semiconductor if the densities of available states in the valence and conduction bands are known.

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TOPIC 01

Why Does Doping Matter?

1 **N-Type**

N-Type

2 **P-Type**

P-Type

3 **Diode**

Diode
Doping allows one to build complex devices like diodes and transistors.

4 **Transistor**

Transistor

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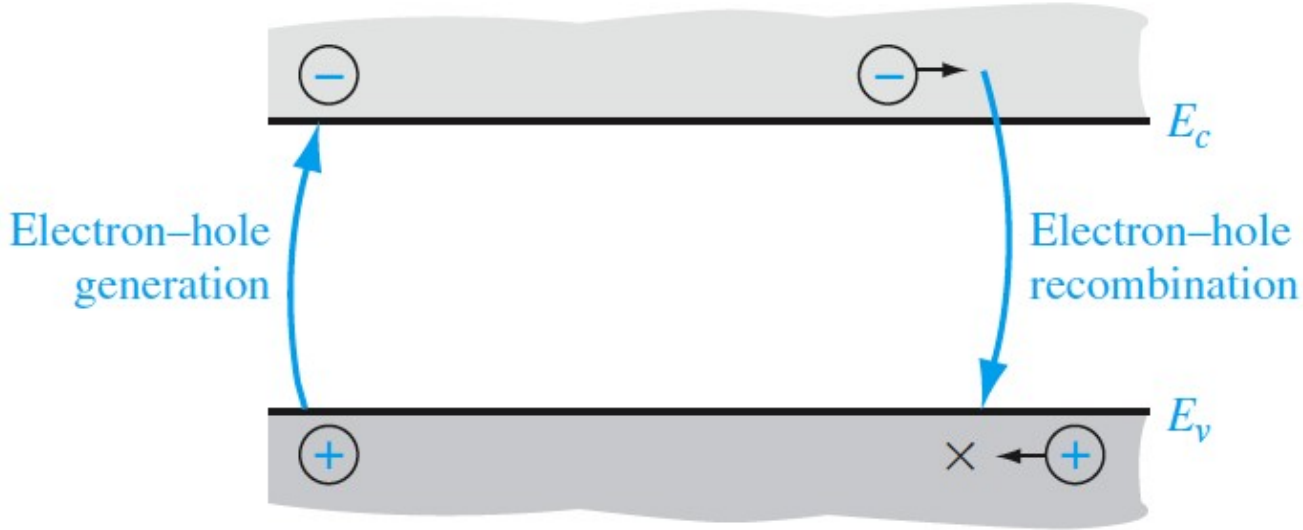
TOPIC 01

Electron–hole generation and recombination



1

Evidence 1

Thermal equilibrium maintains constant electron and hole concentrations
Electrons are thermally excited from the valence to conduction band
Recombination occurs when conduction electrons fall into holes in the valence band
In equilibrium, generation and recombination rates of electrons and holes are equal

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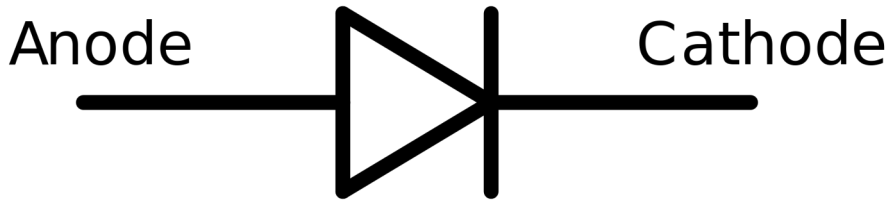
TOPIC 01

PN Junctions and Diode Behaviour

Concepts · mechanisms · process evidence · integration decisions



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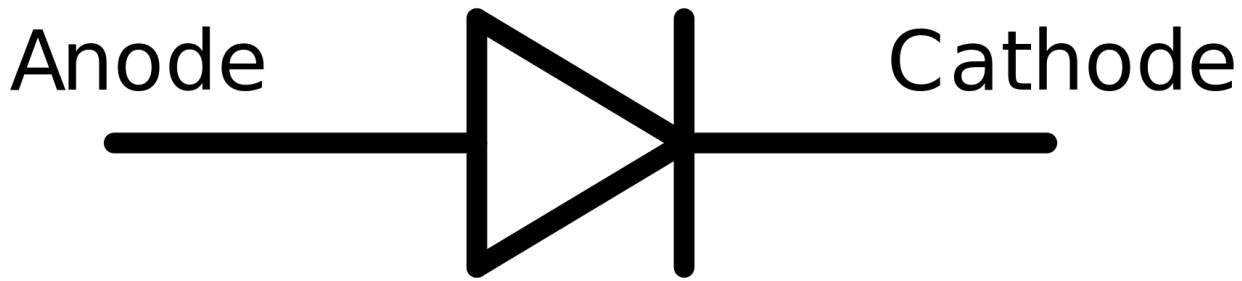
TOPIC 01

What is a Diode?



1 **The diode is formed by a PN junction**

The diode is formed by a PN junction

2 **N-Type**

N-Type

3 **P-Type**

P-Type

4 **+**

+

-

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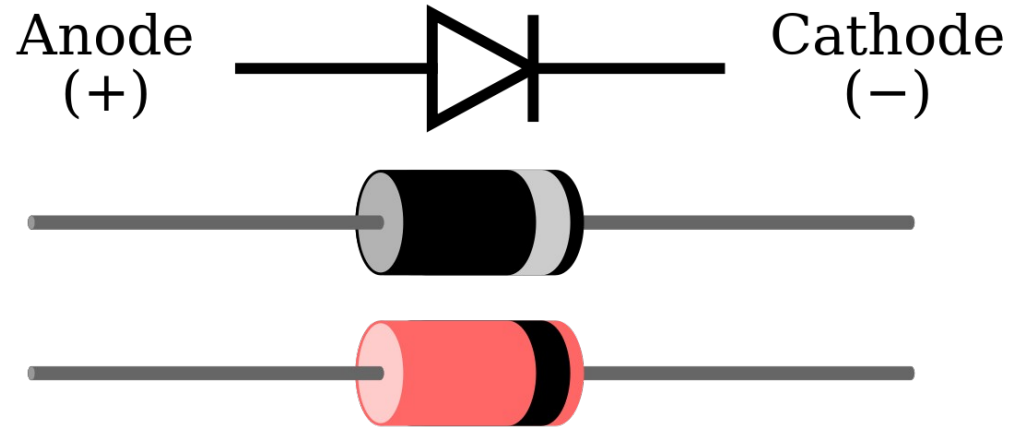
Diode



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TOPIC 01

BASIC STRUCTURE OF THE pn JUNCTION

Step junction has uniform doping with an abrupt change at the junction
Large initial density gradient of electrons and holes at the junction
Majority electrons from the n region diffuse into the p region, and holes from p diffuse into n
Diffusion creates net positive charge in n region (donor atoms) and net negative charge in p region (acceptor atoms)
Resulting positive and negative charges induce an electric field near the junction

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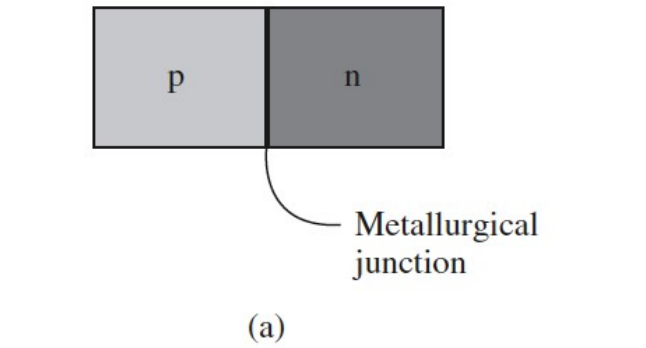
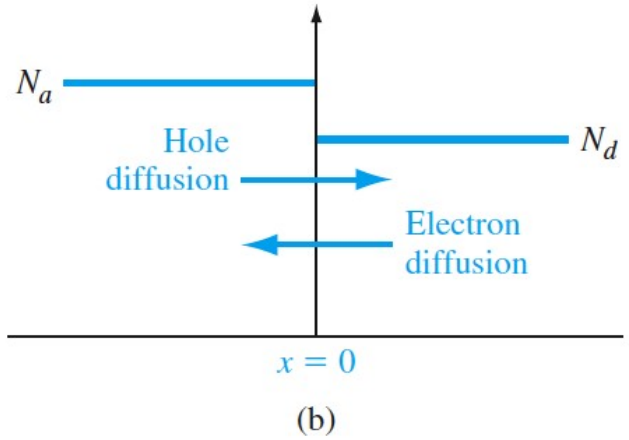
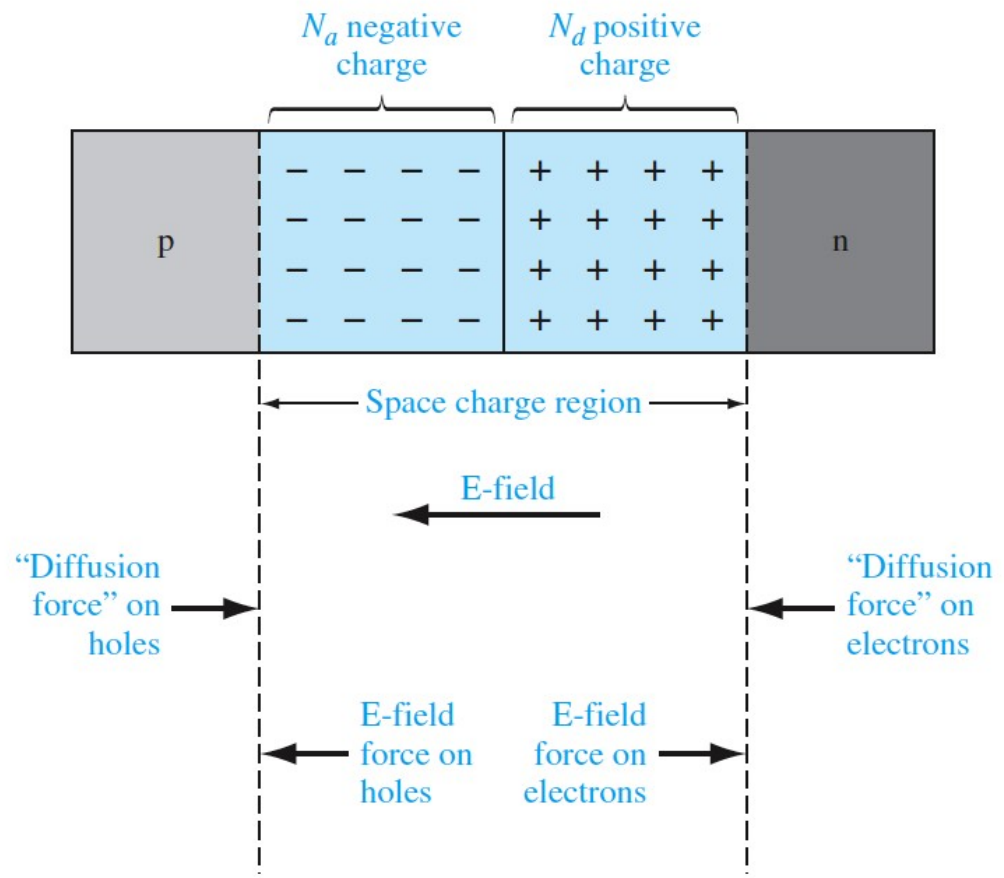
BASIC STRUCTURE OF THE pn JUNCTION







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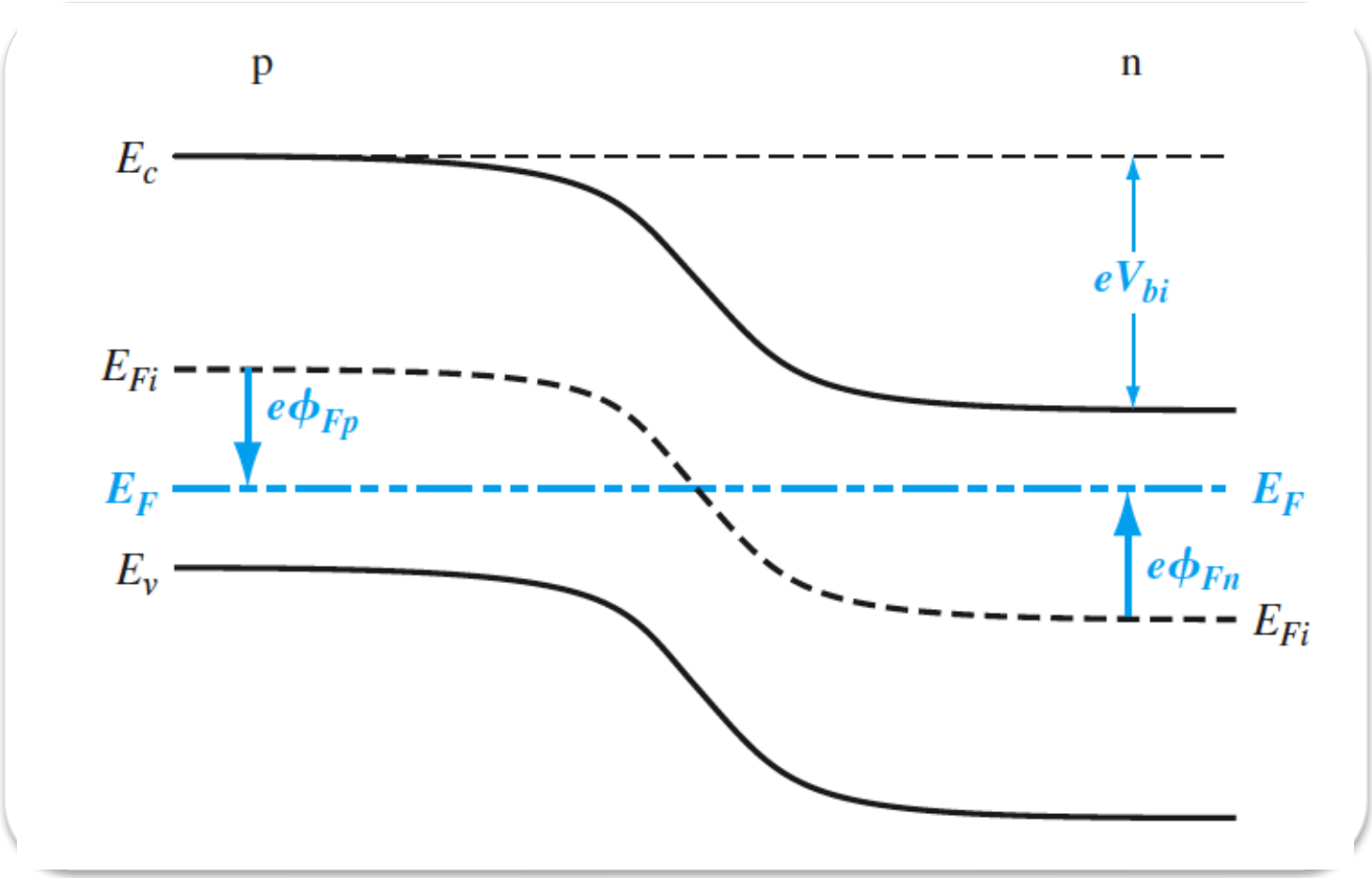
ZERO APPLIED BIAS



Evidence 1

In thermal equilibrium, no voltage is applied across the pn junction
Fermi energy level remains constant throughout the system
Conduction and valence band energies bend across the space charge region
Band bending occurs due to changes in the relative position of the conduction and valence bands between p and n regions

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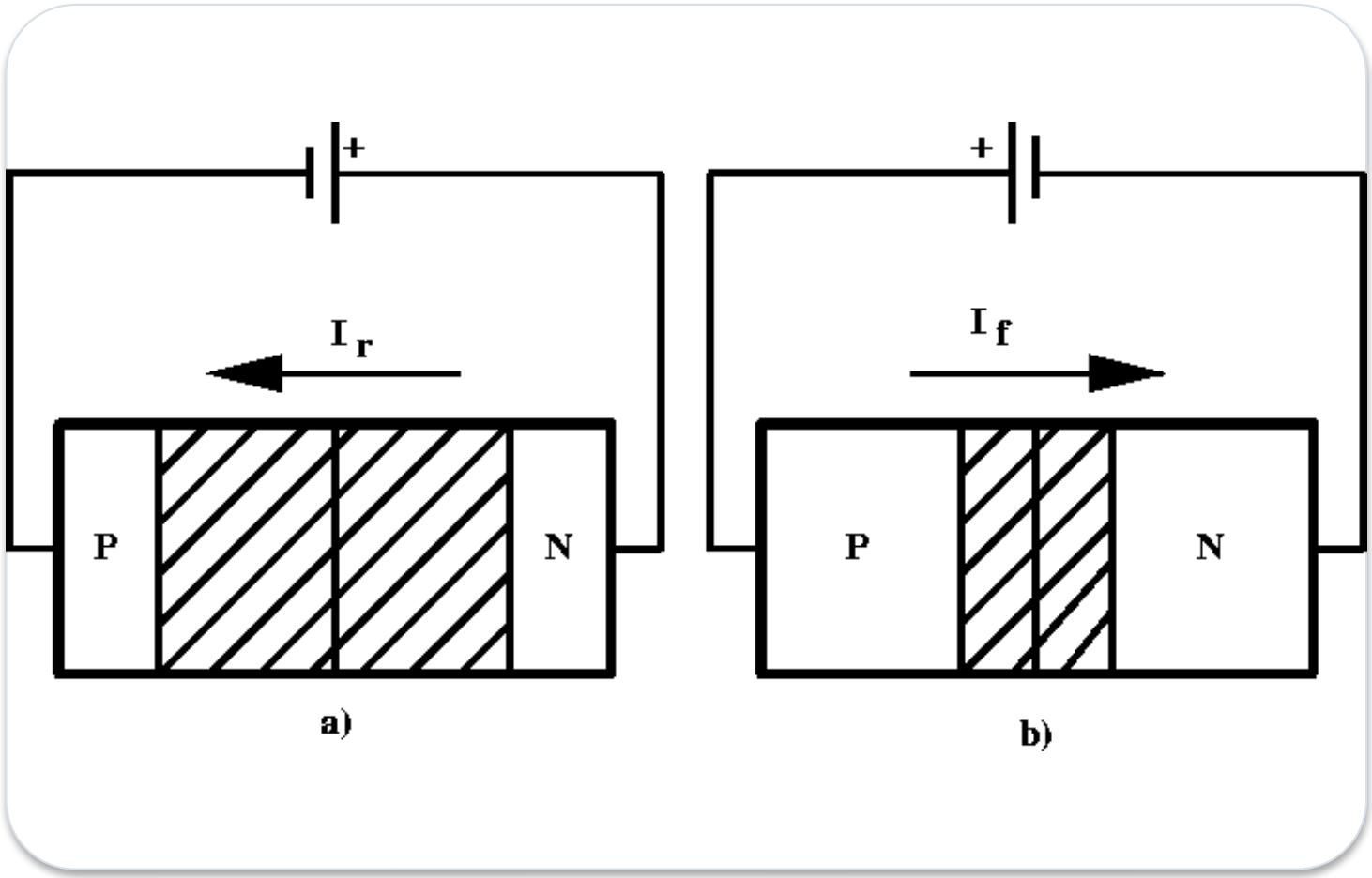
Forward Bias and Reverse Bias



1 **Forward Bias: Connect positive to Anode - ON**

Forward Bias: Connect positive to Anode - ON
Reverse Bias: Connect positive to Cathode - OFF

2 **Reverse Bias**

Reverse Bias

3 **Forward Bias**

Forward Bias

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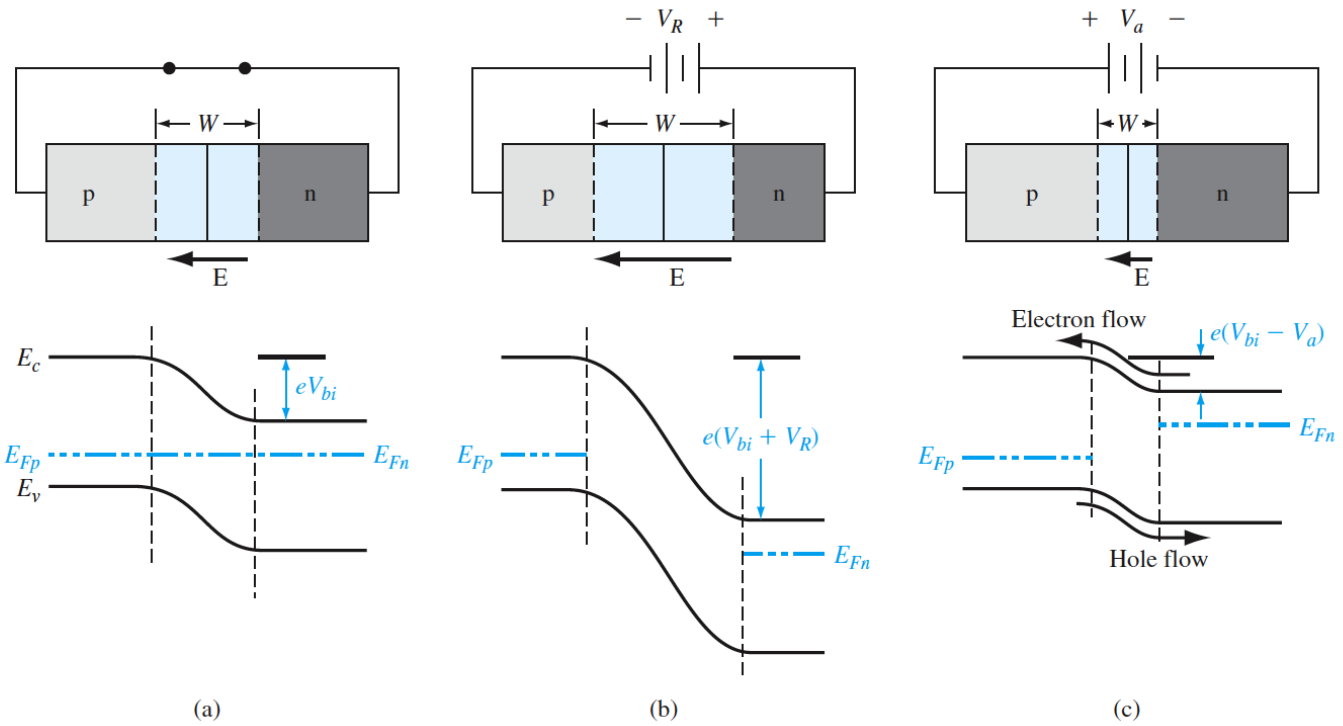
Charge Flow in a pn Junction



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Evidence 1

A pn junction and its associated energy-band diagram for (a) zero bias, (b) reverse bias, and (c) forward bias.

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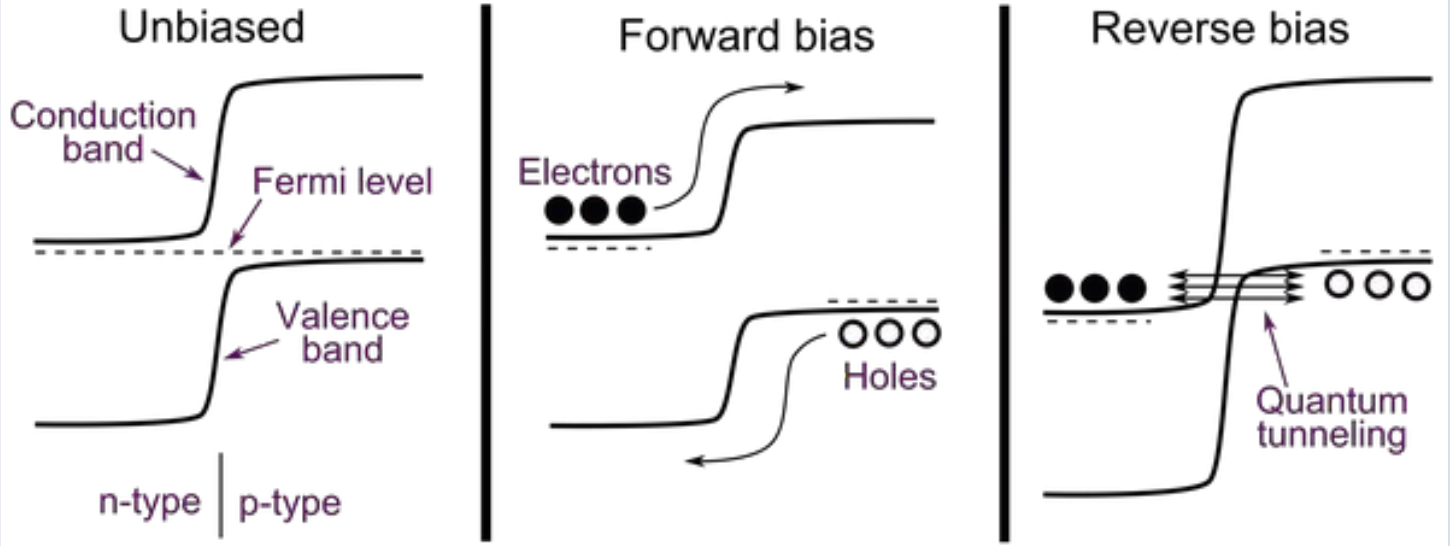
TOPIC 01

How Does Diode Work?



1

Evidence 1

Fermi level is like the water level of a lake. If two different parts of the lake have different water level, water will flow until the water level of the lake is uniform. It's the same with electrons and the Fermi level.

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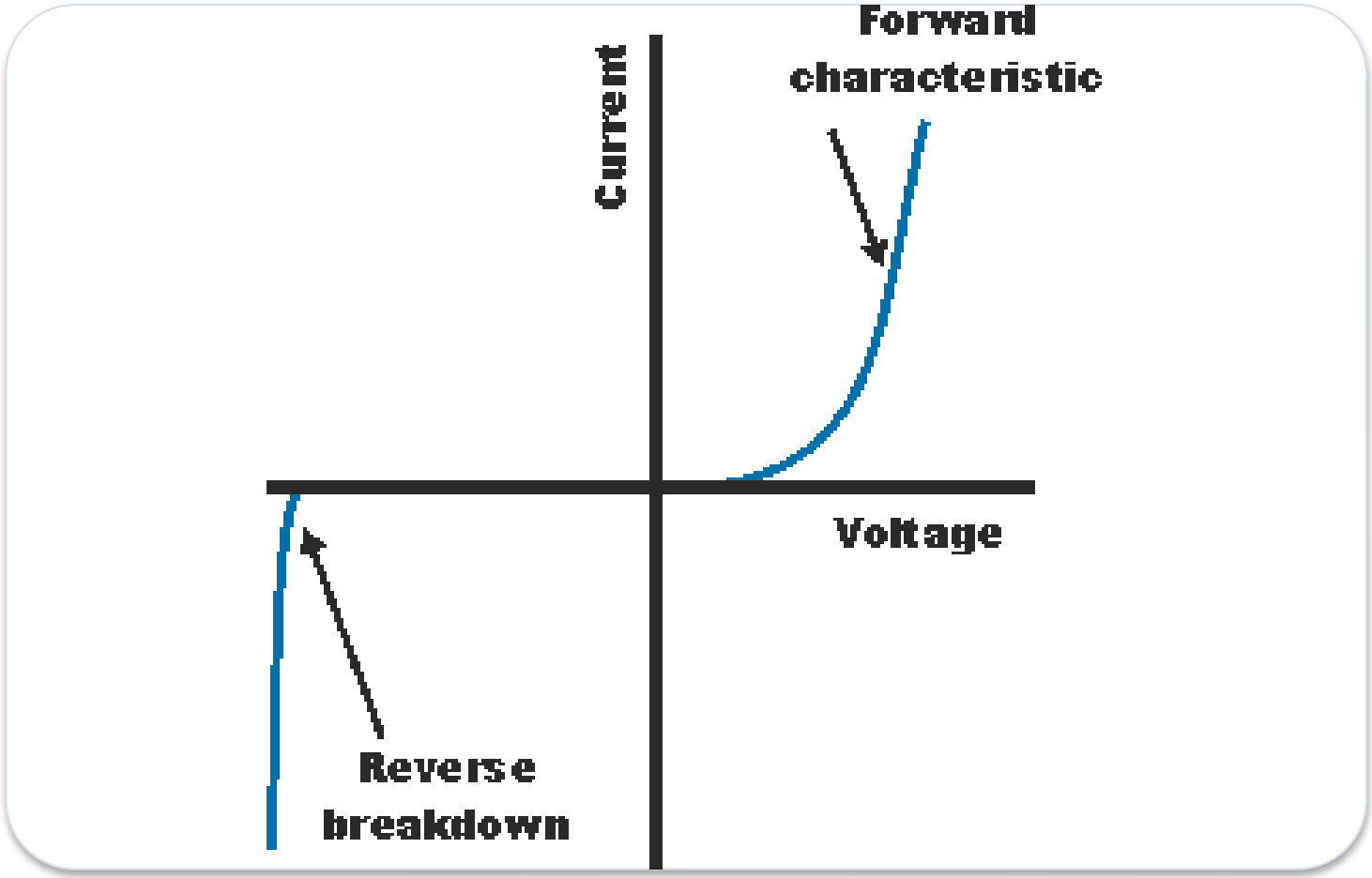
IV Characteristics of Diode



1 **VF ~ 0.7V**

VF ~ 0.7V

2 **VBR ~ -100V**

VBR ~ -100V

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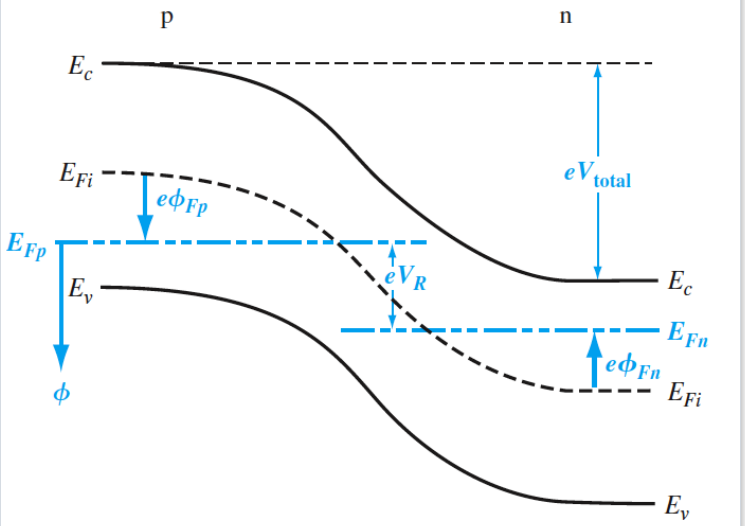
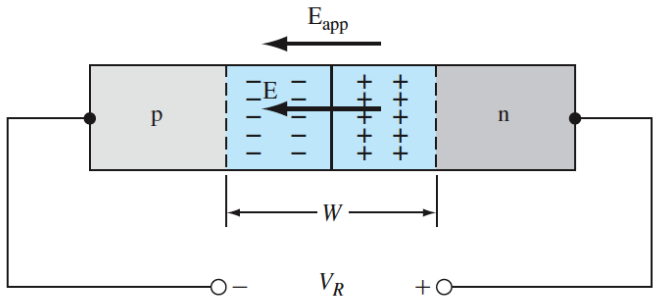
REVERSE APPLIED BIAS





1

Evidence 1

Applying a voltage across the pn junction disrupts thermal equilibrium
Fermi energy level is no longer constant throughout the system
Positive voltage applied to the n region lowers the Fermi level on the n side below that of the p side
The difference in Fermi levels equals the applied voltage (in energy units

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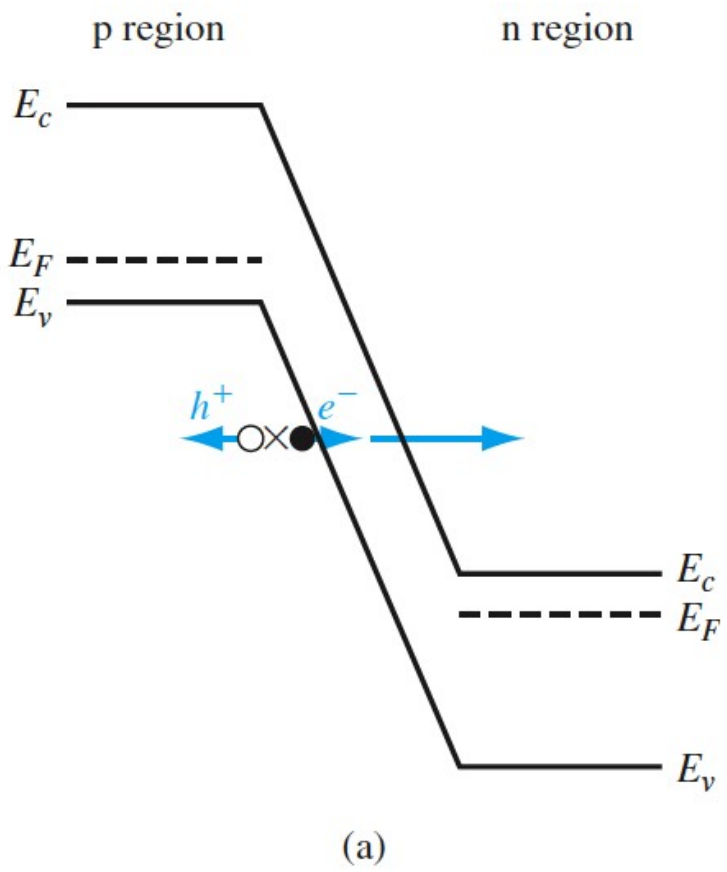
TOPIC 01

JUNCTION BREAKDOWN



1

Evidence 1

Reverse-biased breakdown in a pn junction occurs via the Zener effect and avalanche effect
Zener breakdown happens in highly doped junctions through electron tunneling
In reverse bias, conduction and valence bands are close enough for electrons to tunnel from the p-side valence band to the n-side conduction band

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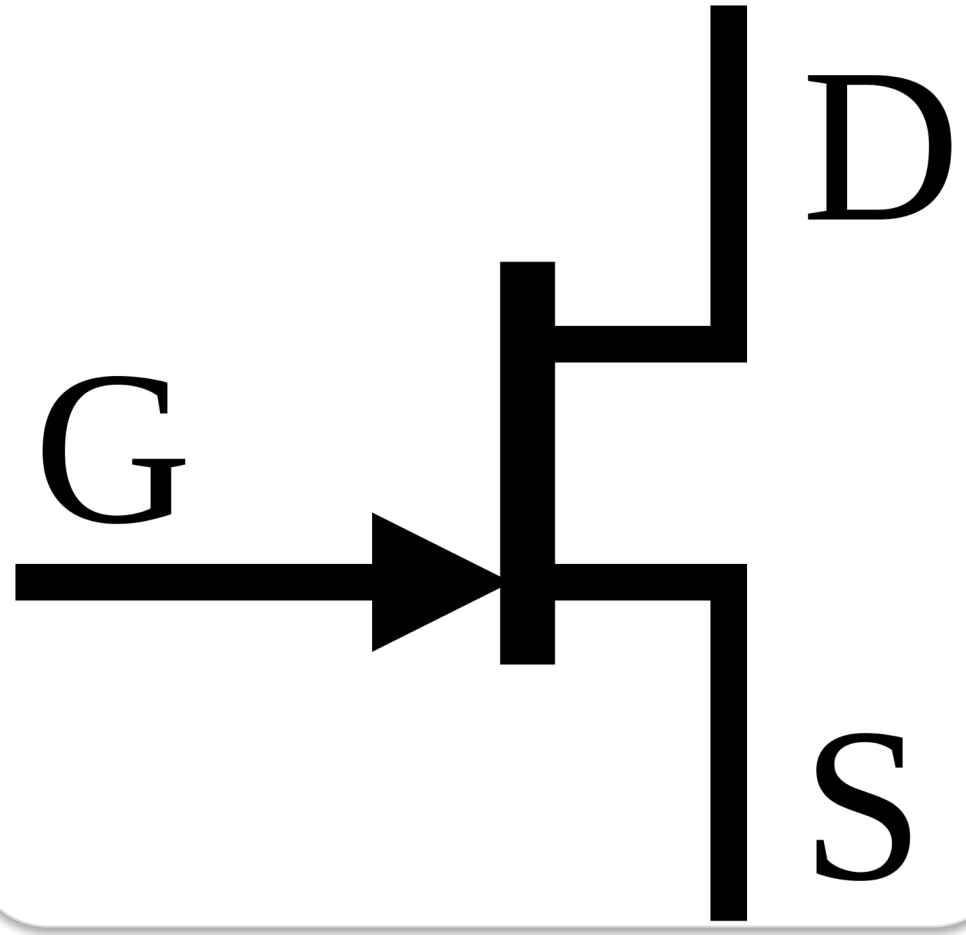
TOPIC 01

Field Effect Transistor

Concepts · mechanisms · process evidence · integration decisions



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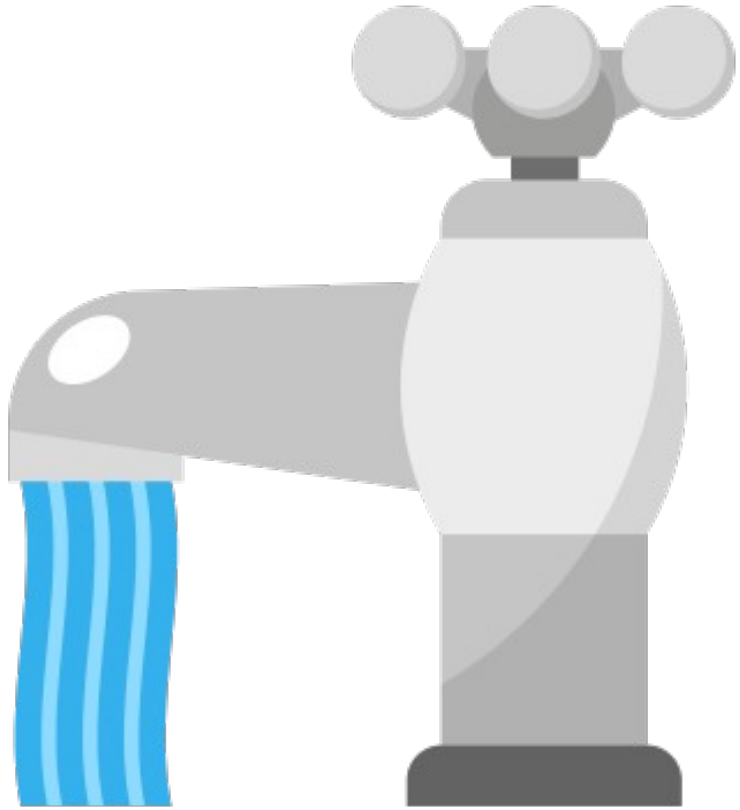
TOPIC 01

What is a Transistor



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Transistor is like a water faucet. There are 3 modes

Transistor is like a water faucet. There are 3 modes

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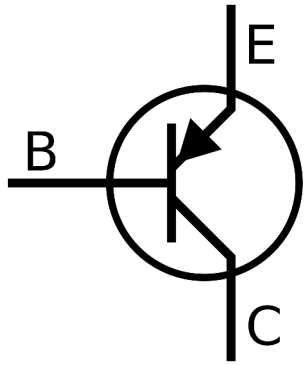
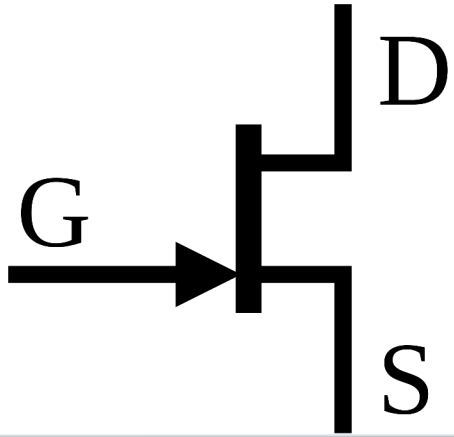
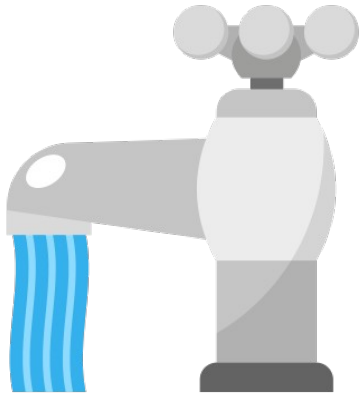
TOPIC 01

BJT and FET







1 **Evidence 1**

The base of BJT and the gate of FET acts like the valve.
They control the current flow

2 **Current**

Current

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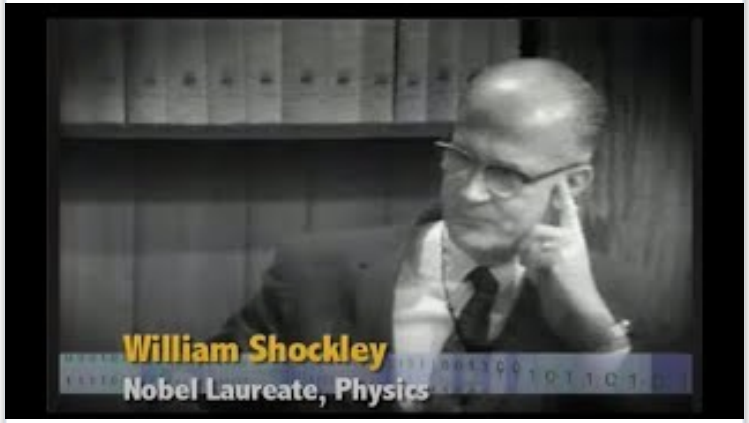
TOPIC 01

Who invented Solid State Transistor?



1

Evidence 1

Solid State Transistor was invented by John Bardeen, Walter Brattain and William Shockley at AT&T's Bell Labs in 1948
They were awarded the Nobel prize in Physics on 1956

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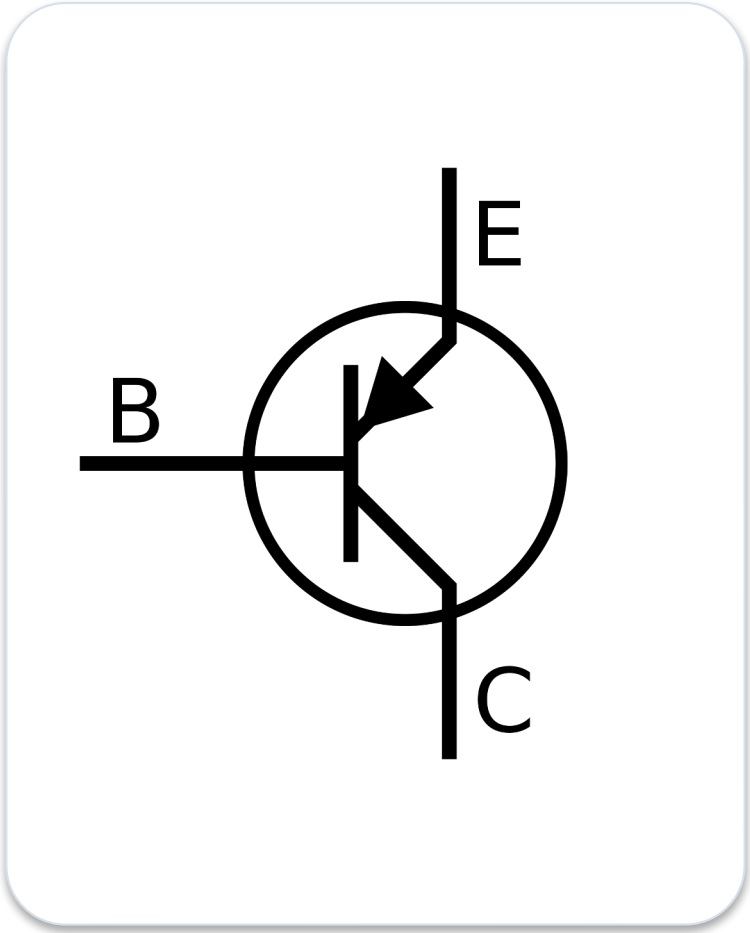
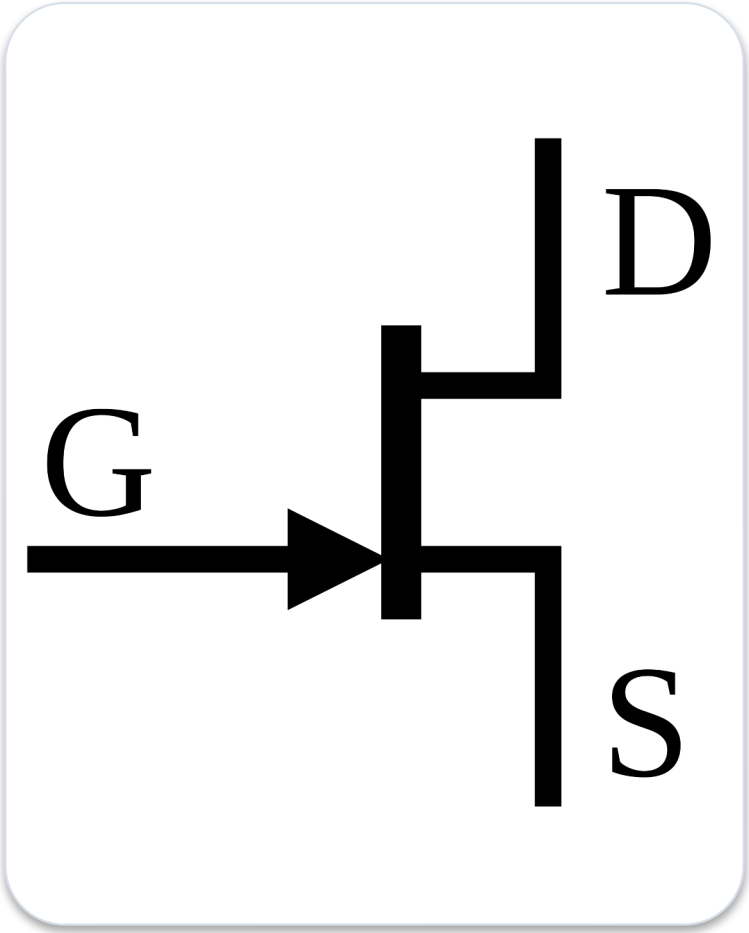
TOPIC 01

BJT and FET





1 **There are two types of transistors**

There are two types of transistors
Bipolar Junction Transistor (BJT)
Field Effect Transistor (FET)

2 **BJT**

BJT
(Collector)
(Source)

3 **FET**

FET
(Base)
(Gate)

4 **(Emitter)**

(Emitter)
(Drain)

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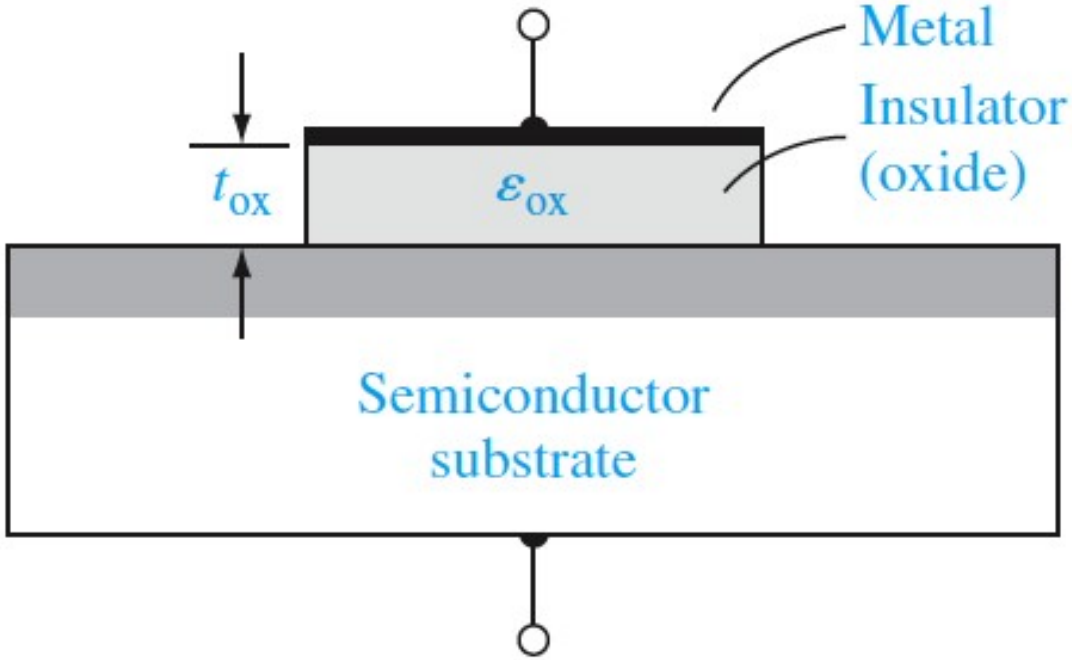
TOPIC 01

MOS Capacitor



1

Evidence 1

MOS structure can be understood using a parallel-plate capacitor model
Negative voltage on the top plate induces a negative charge, with positive charge on the bottom plate
An electric field is created between the plates
Capacitance per unit area: $C = \epsilon/d$
Charge per unit area: $Q = C \times V$

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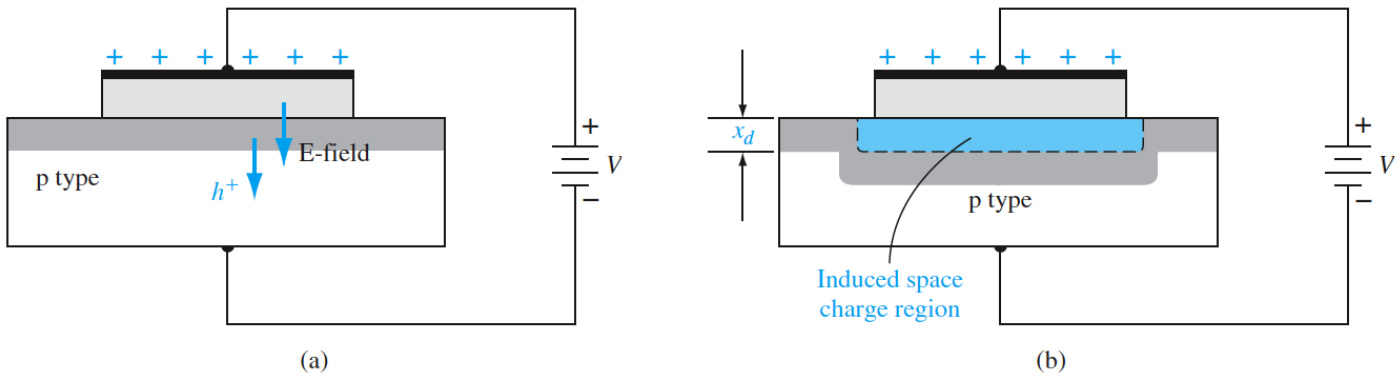
TOPIC 01

MOS Capacitor



1

Evidence 1

The MOS capacitor with a moderate positive gate bias, showing (a) the electric field and charge flow and (b) the induced space charge region.

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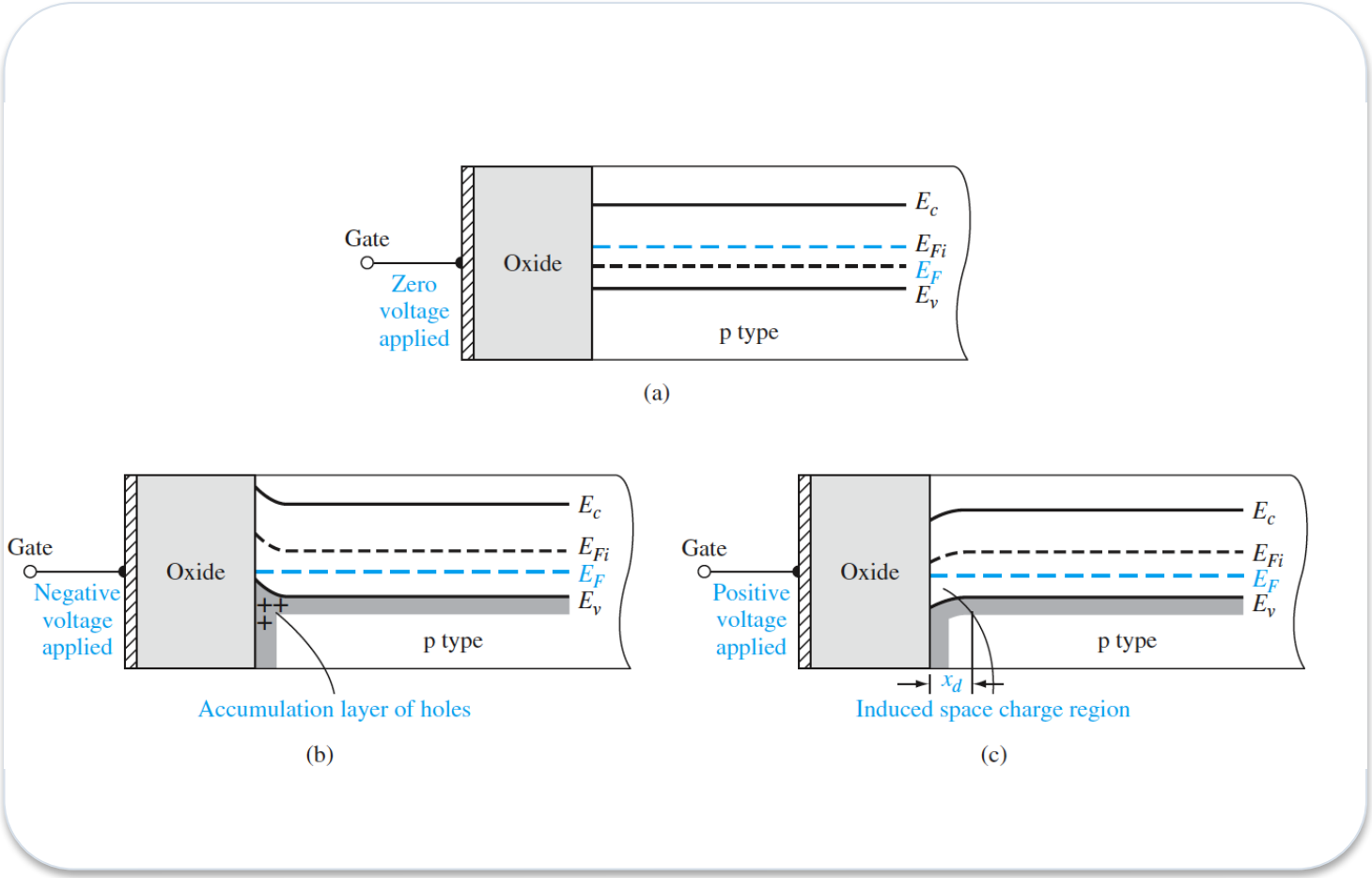
TOPIC 01

Applied Voltage on MOS Capacitor



1

Evidence 1

The energy-band diagram of a MOS capacitor with a p-type substrate for (a) a zero applied gate bias showing the ideal case, (b) a negative gate bias, and (c) a moderate positive gate bias.

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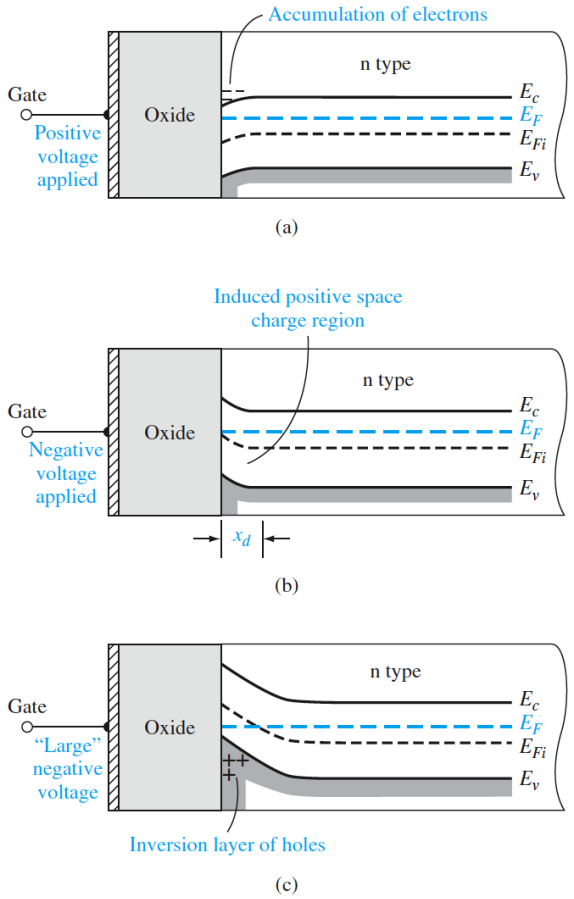
Applied Voltage on MOS Capacitor



1

Evidence 1

A larger positive voltage on the MOS capacitor gate increases the electric field and charge
Increased negative charge leads to more band bending and a larger space charge region
Surface inversion occurs: p-type surface becomes n-type due to band bending near the oxide interface
Positive gate voltage creates an electron inversion layer at the oxide–semiconductor interface
For an n-type substrate, a positive gate voltage induces an electron accumulation layer
A negative gate voltage on an n-type substrate induces a positive space charge region

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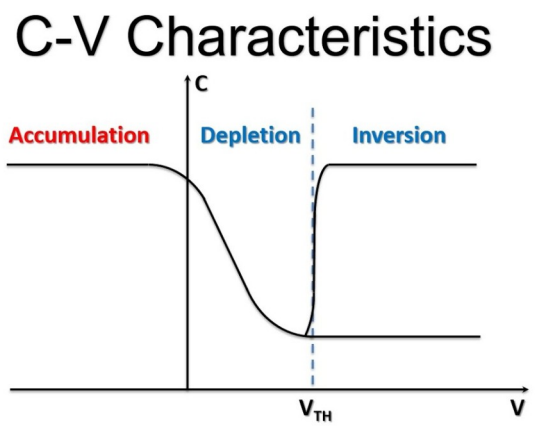
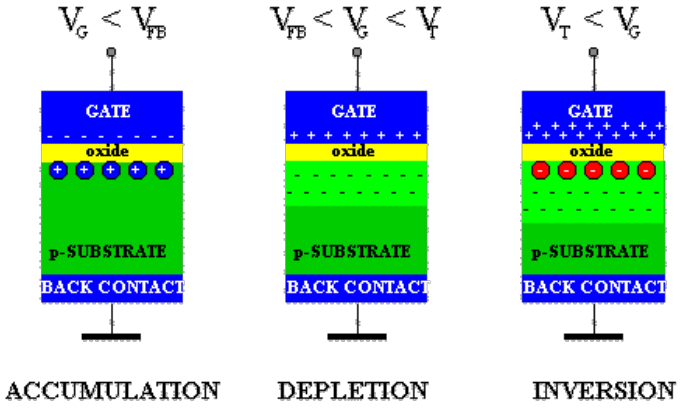
TOPIC 01

Applied Voltage on MOS Capacitor





1

Vg > Vth -> Inversion

Vg > Vth -> Inversion

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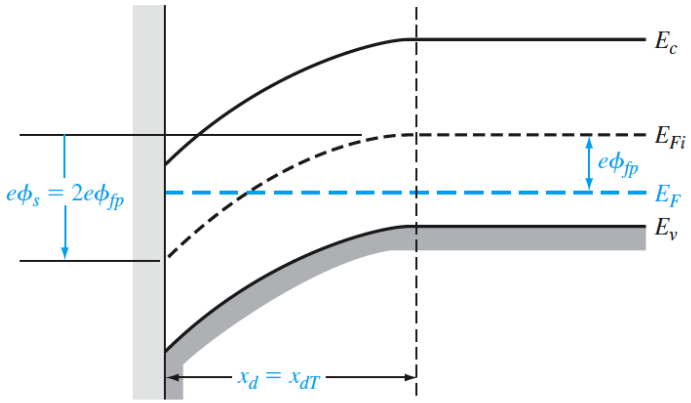
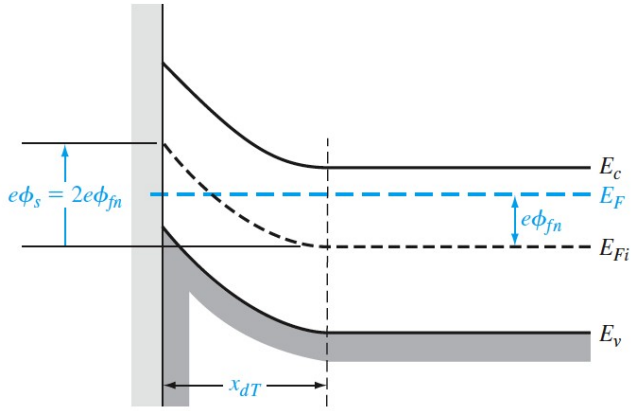
TOPIC 01

Inversion in MOSFET





1 Evidence 1

The surface Fermi level is symmetrically positioned relative to the intrinsic level, like in the bulk semiconductor
Electron concentration at the surface equals hole concentration in the bulk
This condition is called the threshold inversion point, with the applied gate voltage known as the threshold voltage
Beyond this threshold, the conduction band bends closer to the Fermi level, but only slightly changes with further gate voltage increase
Electron concentration at the surface increases exponentially with surface potential, while the space charge region width stabilizes near its maximum

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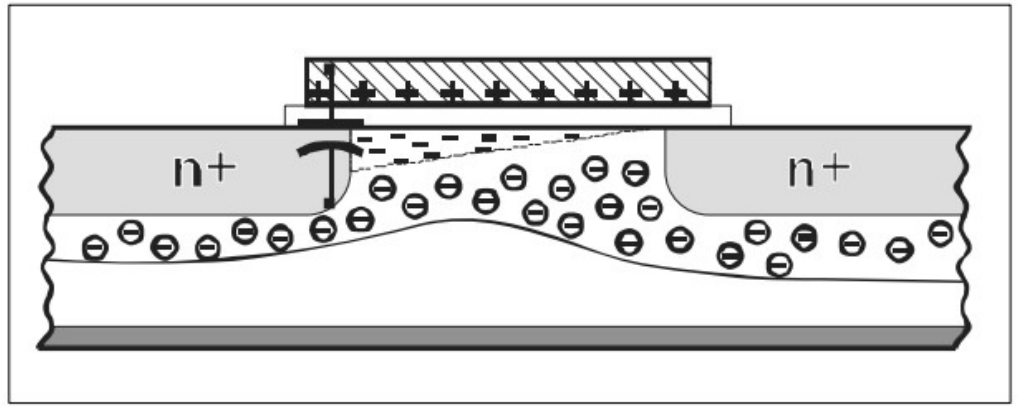
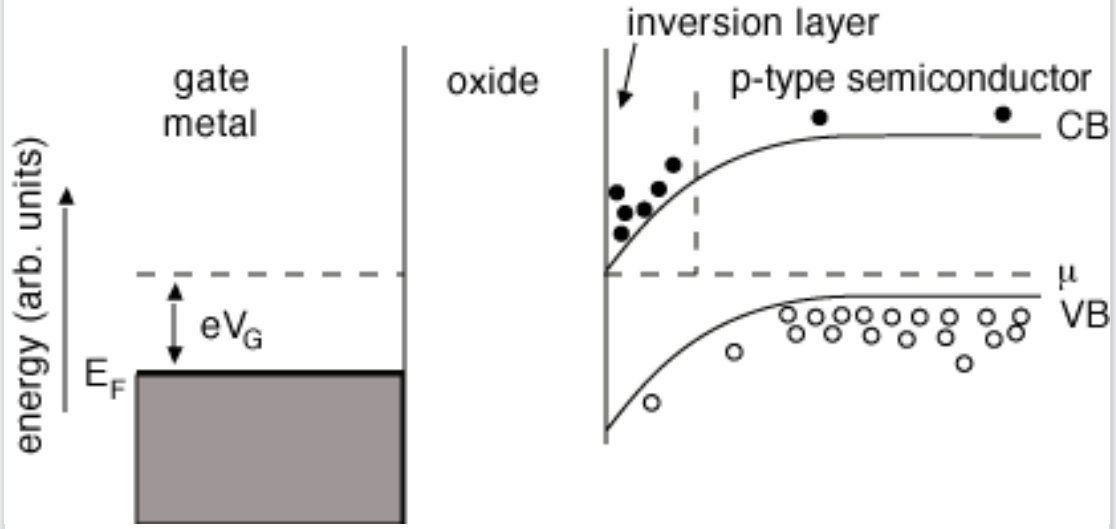
Inversion in MOSFET

MOSCAP in INVERSION

- As V_{GS} is increased further, the inversion layer forms and the channel to bulk capacitance is “shielded”. Most of the MOSCAP capacitance is from C_{GS}.





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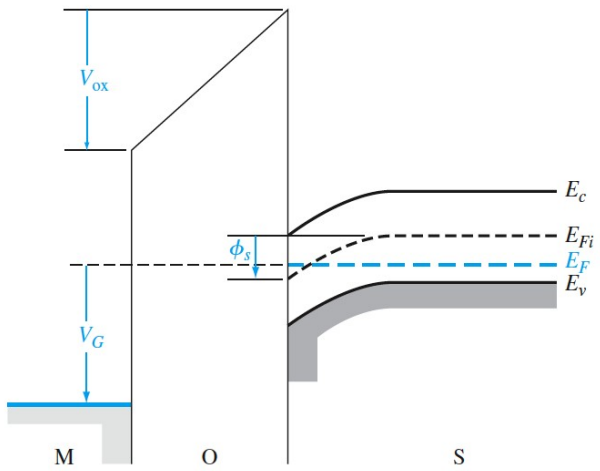
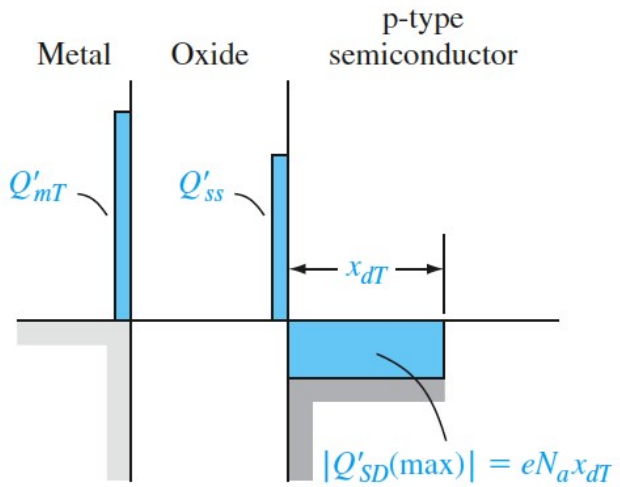
TOPIC 01

Threshold Voltage





1

Evidence 1

Threshold voltage is the gate voltage needed to reach the threshold inversion point

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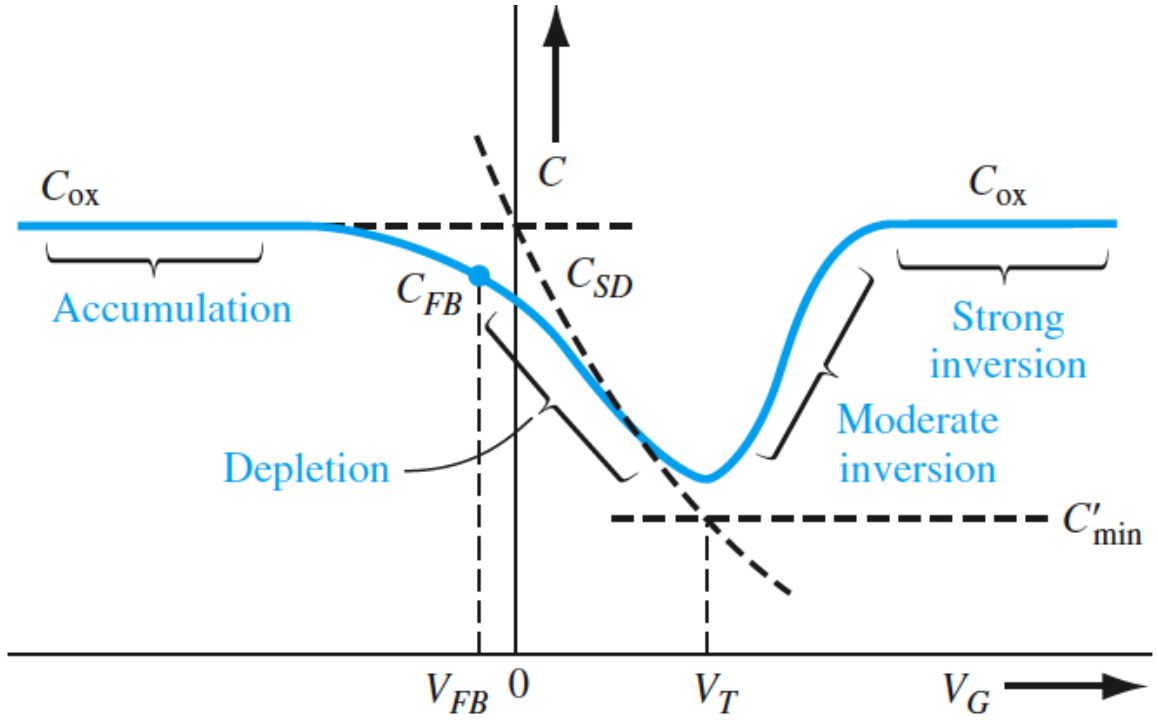
TOPIC 01

CV Characteristics



1

Evidence 1

Ideal low-frequency capacitance versus gate voltage of a MOS capacitor with a p-type substrate.

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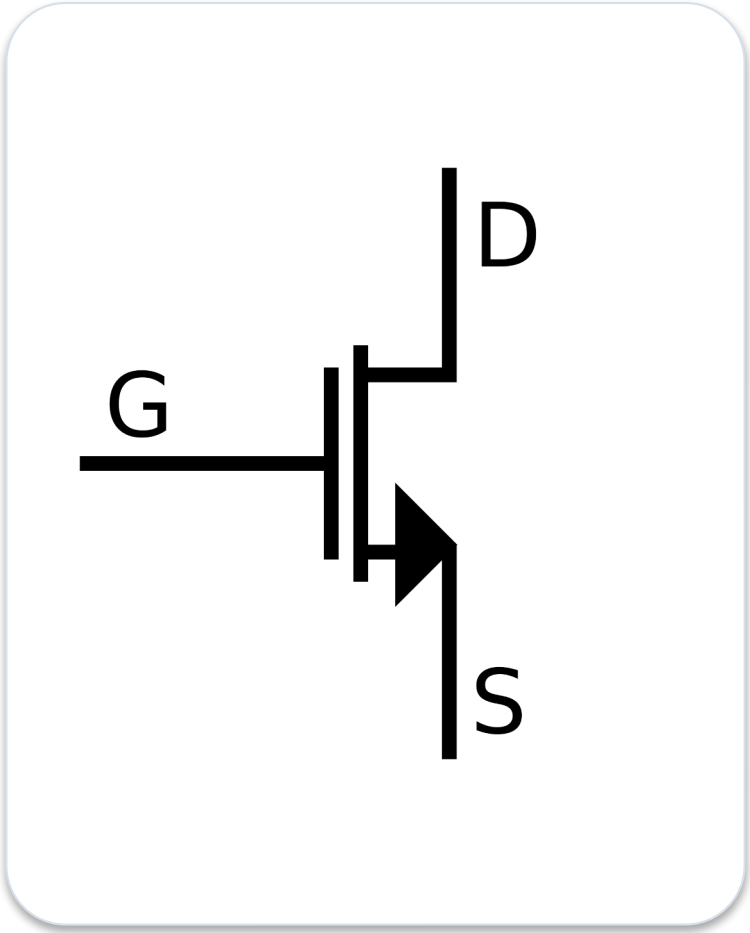
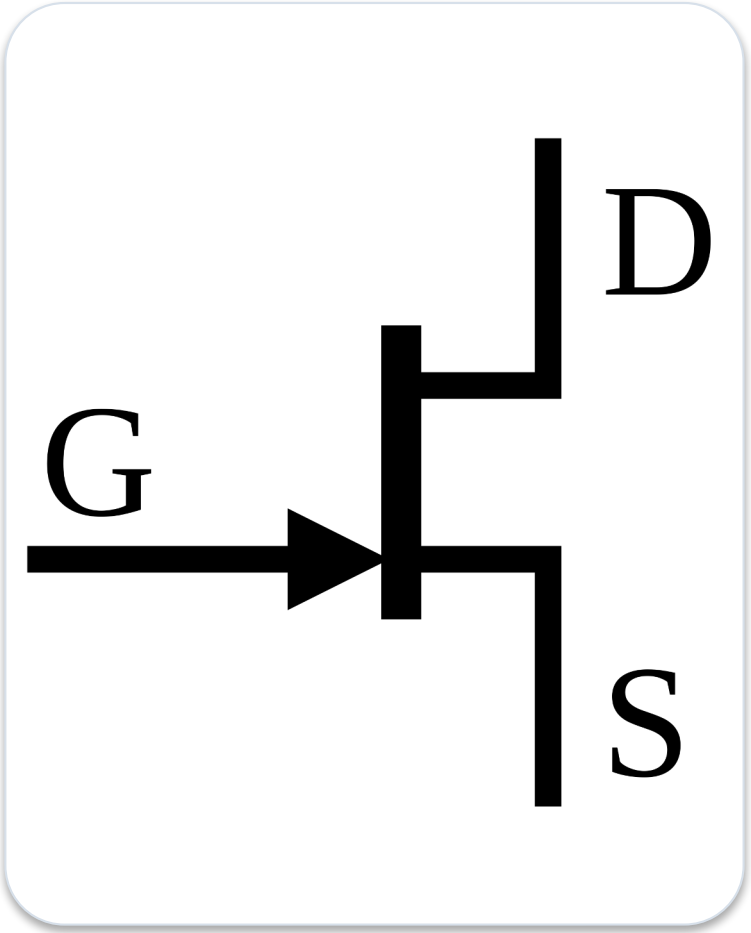
TOPIC 01

What is Field Effect Transistor





1

Evidence 1

Unlike BJT use current to control, Field Effect Transistor (FET) use electric field to control the current.
There are 2 types of FETs
JFET
MOSFET

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TOPIC 01

Who invented FET?

1

Evidence 1

Metal-oxide-semiconductor field-effect transistor (MOSFET) are invented by Dawon Kahng and Martin M. (John) Atalla at Bell Labs in 1959

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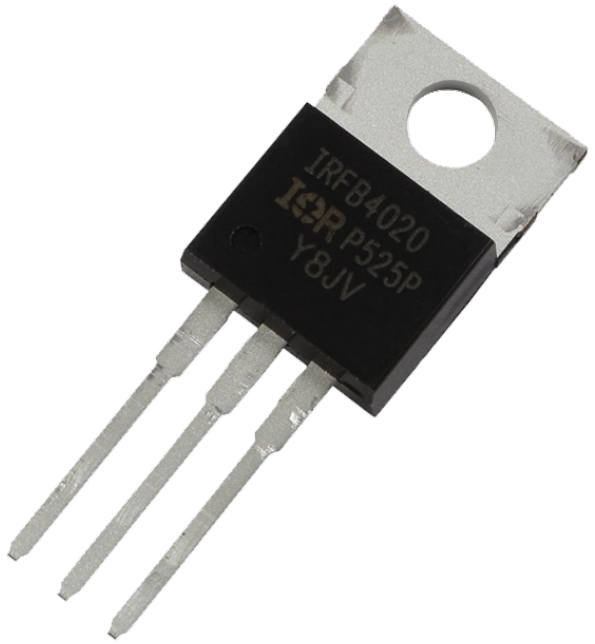
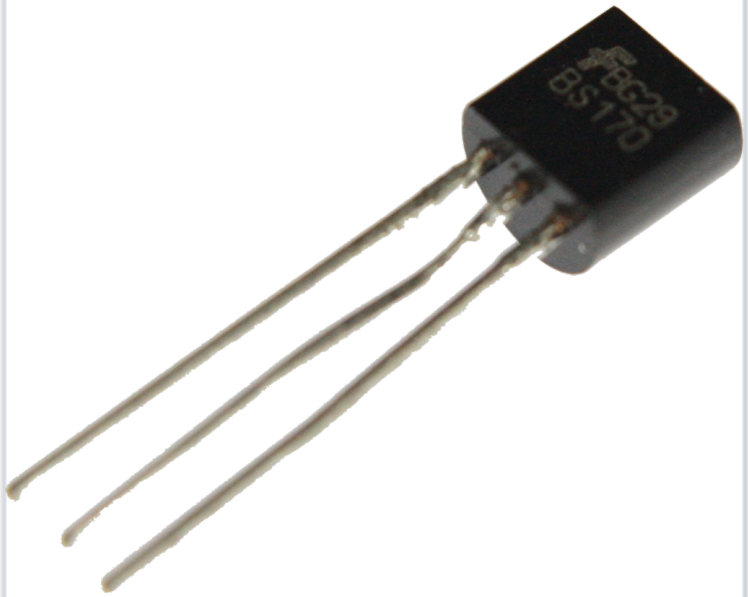
TOPIC 01

MOSFET Components





1 **Source**

Source

2 **Drain**

Drain

3 **Gate**

Gate

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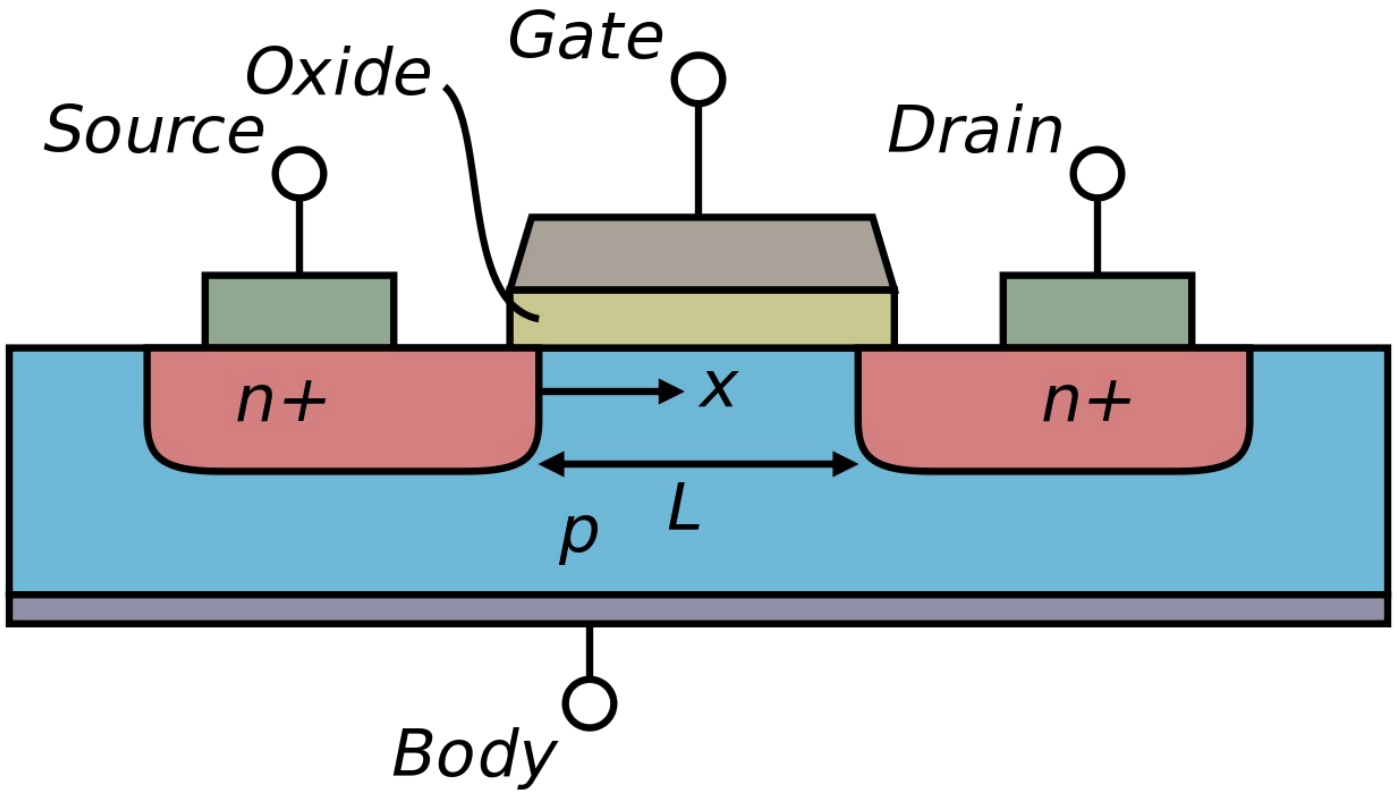
TOPIC 01

MOSFET Cross Section Structure



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A typical n-channel MOSFET structure is shown below

A typical n-channel MOSFET structure is shown below

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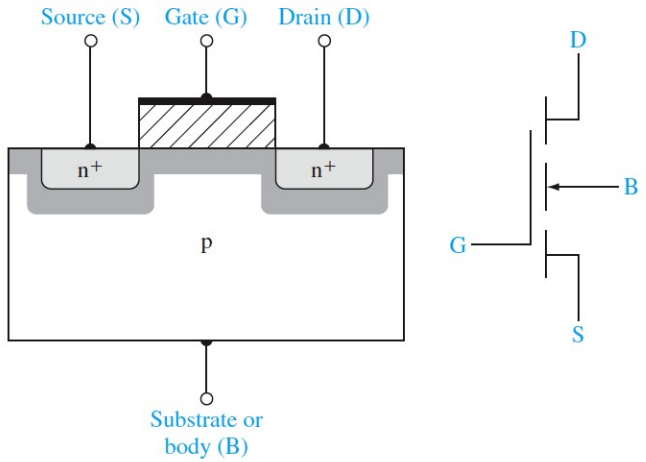
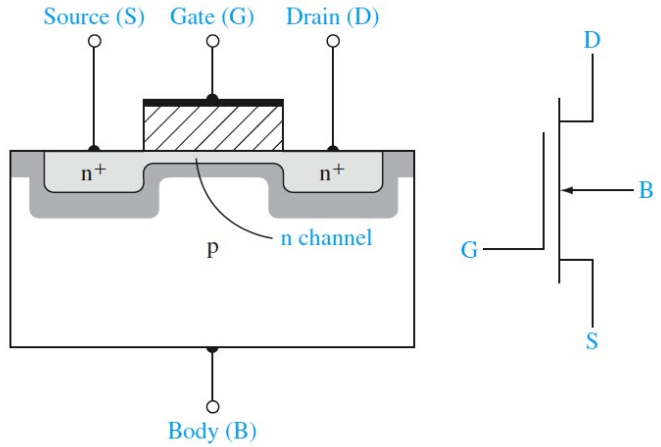
TOPIC 01

Types of MOSFET





1 **There are four basic MOSFET device types.**

There are four basic MOSFET device types.

2 **n-channel enhancement mode MOSFET.**

n-channel enhancement mode MOSFET.

3 **n-channel depletion mode MOSFET.**

n-channel depletion mode MOSFET.

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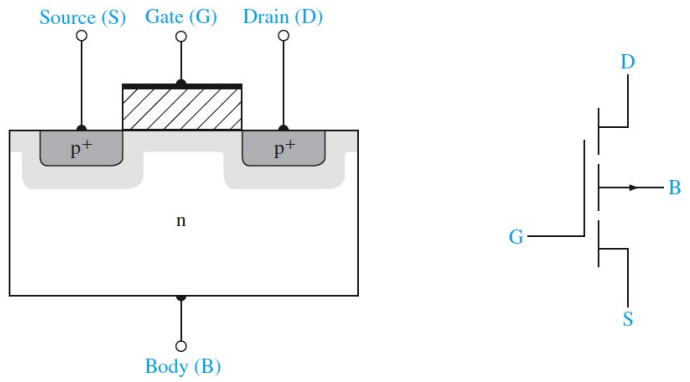
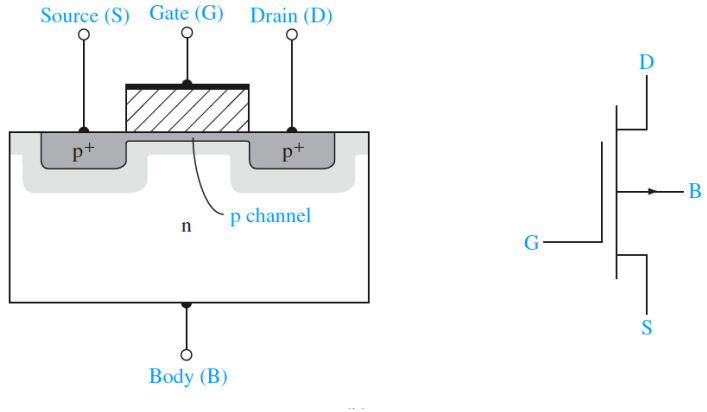
TOPIC 01

Types of MOSFET





1 **There are four basic MOSFET device types.**

There are four basic MOSFET device types.

2 **p-channel enhancement mode MOSFET.**

p-channel enhancement mode MOSFET.

3 **p-channel depletion mode MOSFET.**

p-channel depletion mode MOSFET.

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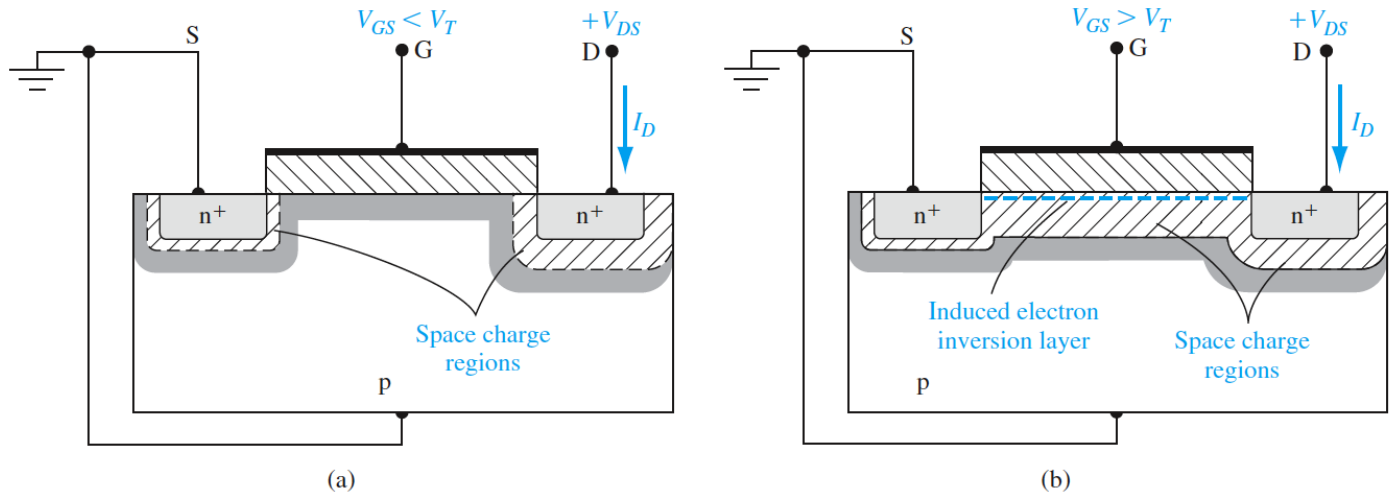
Enhancement Mode MOSFET



1 **When VGS > Vth, the channel is turned on**

When VGS > Vth, the channel is turned on
When VGS < Vth, the channel is turned off

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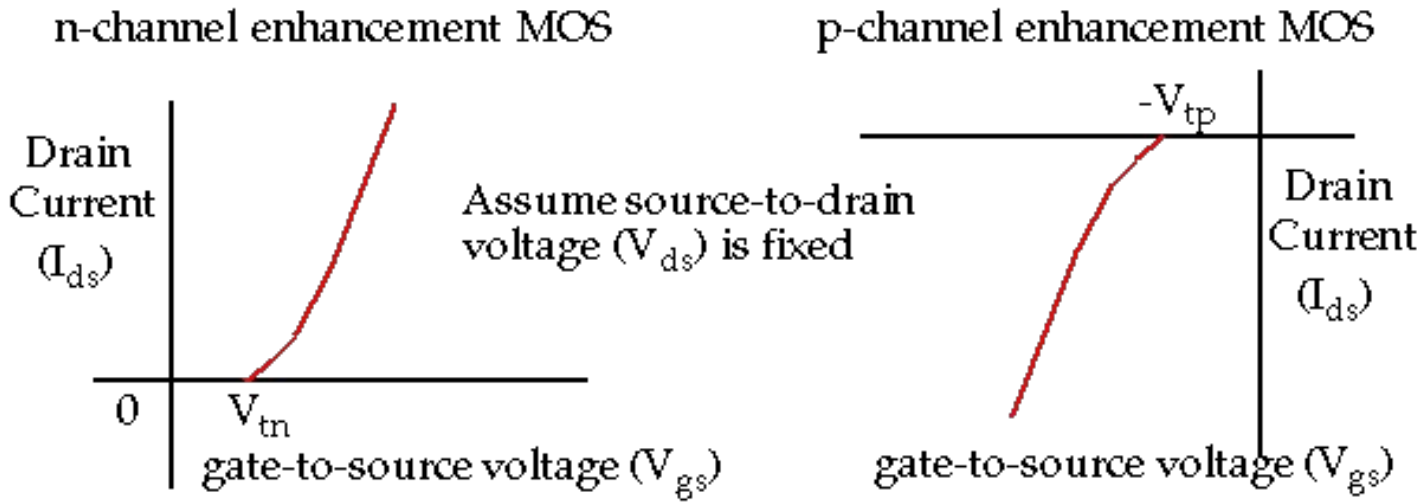
Enhancement Mode MOSFET



1

When VGS > Vth, the channel is turned on

When VGS > Vth, the channel is turned on
When VGS < Vth, the channel is turned off

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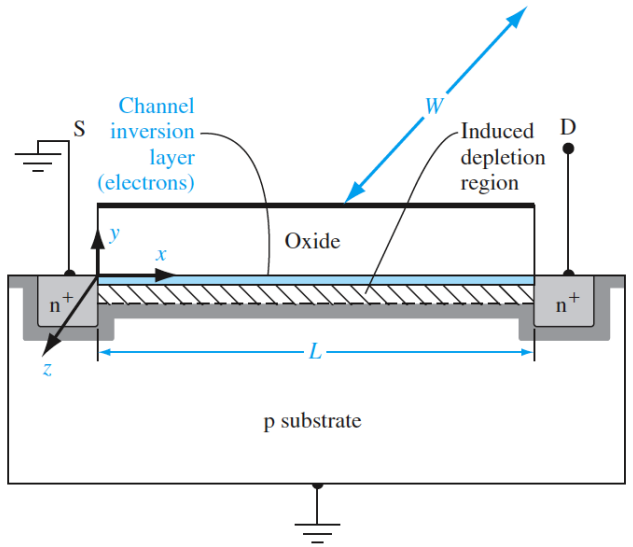
TOPIC 01

Drain Current Equation

$$I_D = 0 \begin{cases} V_{GS} > V_T \end{cases}$$

$$I_{D,lin} = \frac{\mu_p \cdot C_{ox}}{2} \cdot \frac{W}{L} \cdot [2 \cdot (V_{GS} - V_T)V_{DS} - V_{DS}^2] \begin{cases} V_{GS} \leqslant V_T \\ V_{DS} > V_{GS} - V_T \end{cases}$$

$$I_{D,sat} = \frac{\mu_p \cdot C_{ox}}{2} \cdot \frac{W}{L} \cdot (V_{GS} - V_T)^2 \begin{cases} V_{GS} \leqslant V_T \\ V_{DS} \leqslant V_{GS} - V_T \end{cases}$$



1 **There are 3 operating region of MOSFET**

There are 3 operating region of MOSFET
Off Region (ID = 0)
Linear Region (ID < IDsat)
Saturation Region (ID = IDsat)

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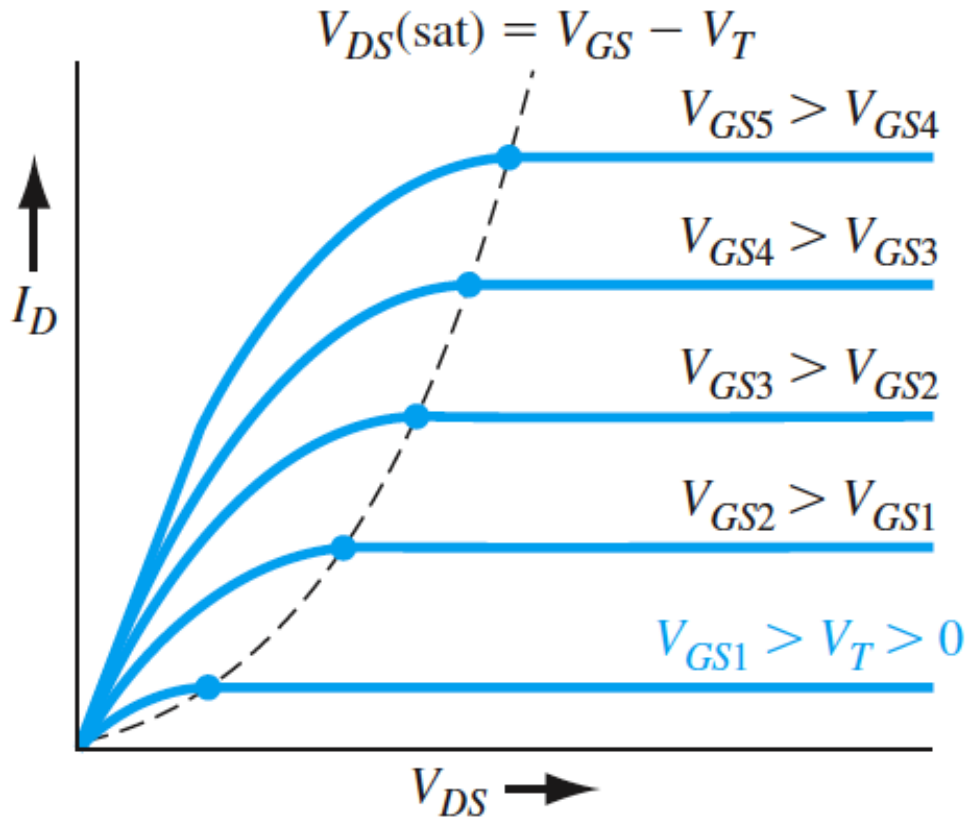
IV Characteristics of n-channel MOSFET



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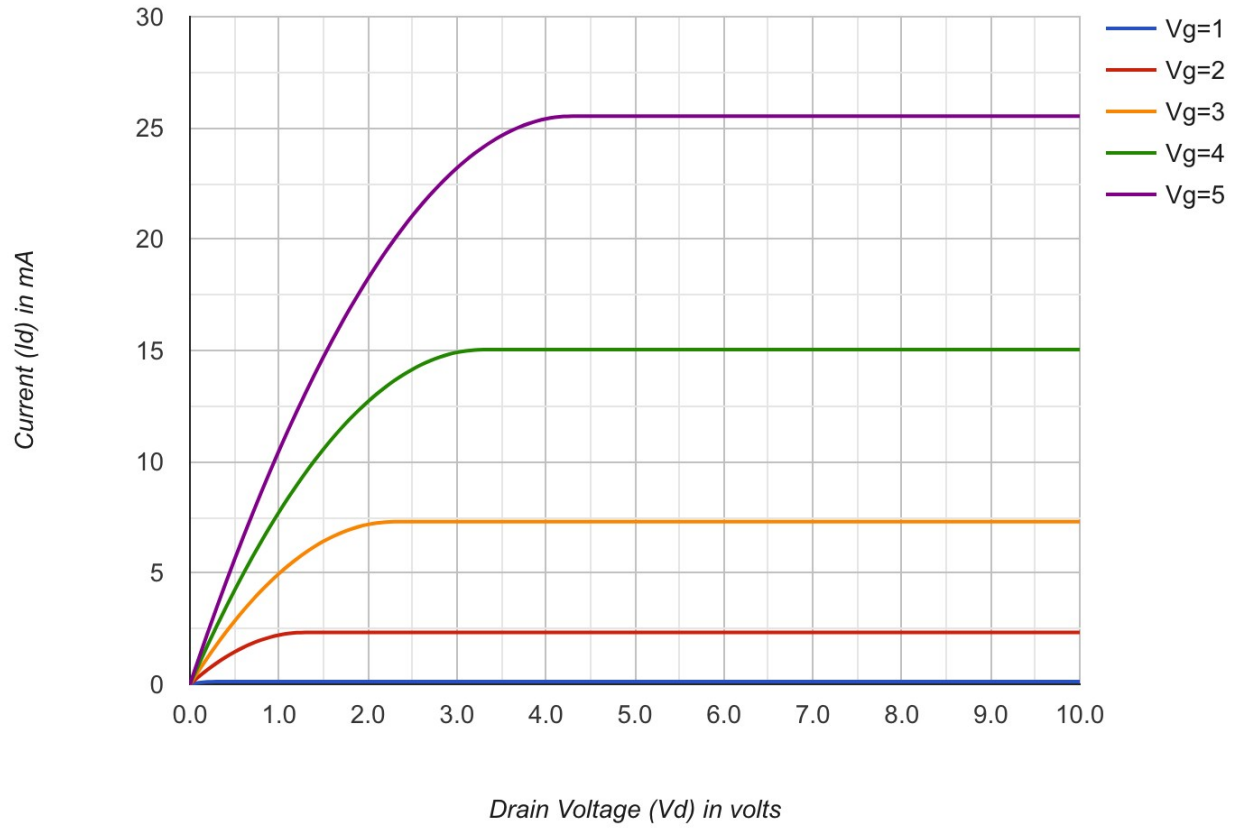
Activity: MOSFET



1

Try the MOSFET calculator below

Try the MOSFET calculator below
https://cleanroom.byu.edu/mosfet_calc

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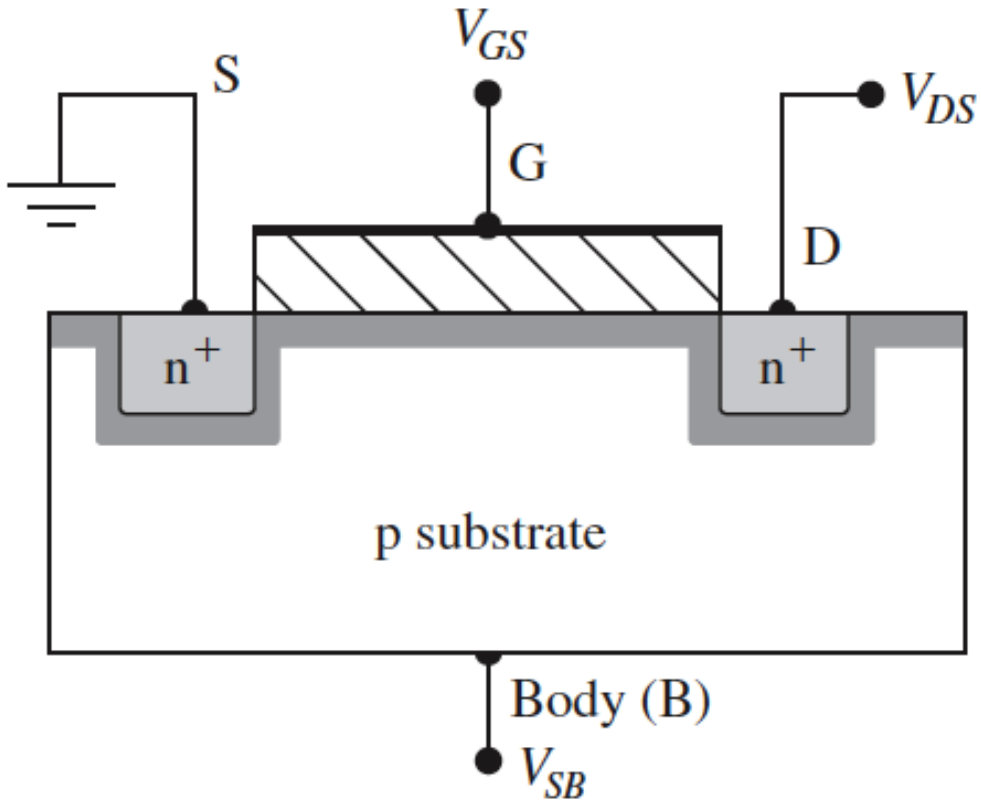
Substrate Bias Effects



1

Evidence 1

When a reverse-biased VSB>0 is applied, the space charge region width under the oxide increases from the original xdT

More charge accumulates in the space charge region due to the reverse bias

To maintain charge neutrality, the positive charge on the gate must increase to compensate for the increased negative space charge

As a result, the threshold voltage of the n-channel MOSFET increases when VSB>0

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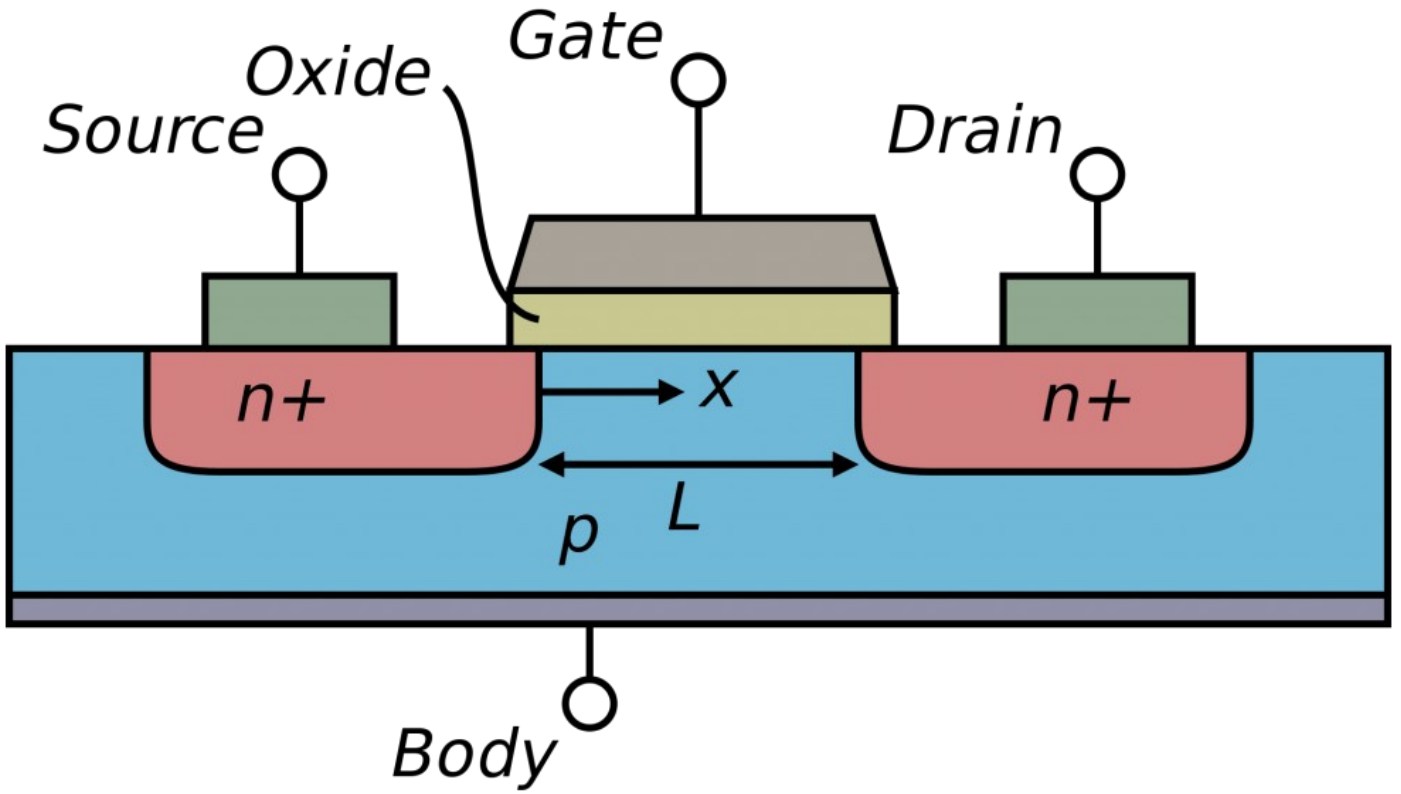
TOPIC 01

MOSFET Design



1

Evidence 1

To design a MOSFET, you have to consider the following 2 key parameters
Gate oxide thickness Tox
Channel Length Lg

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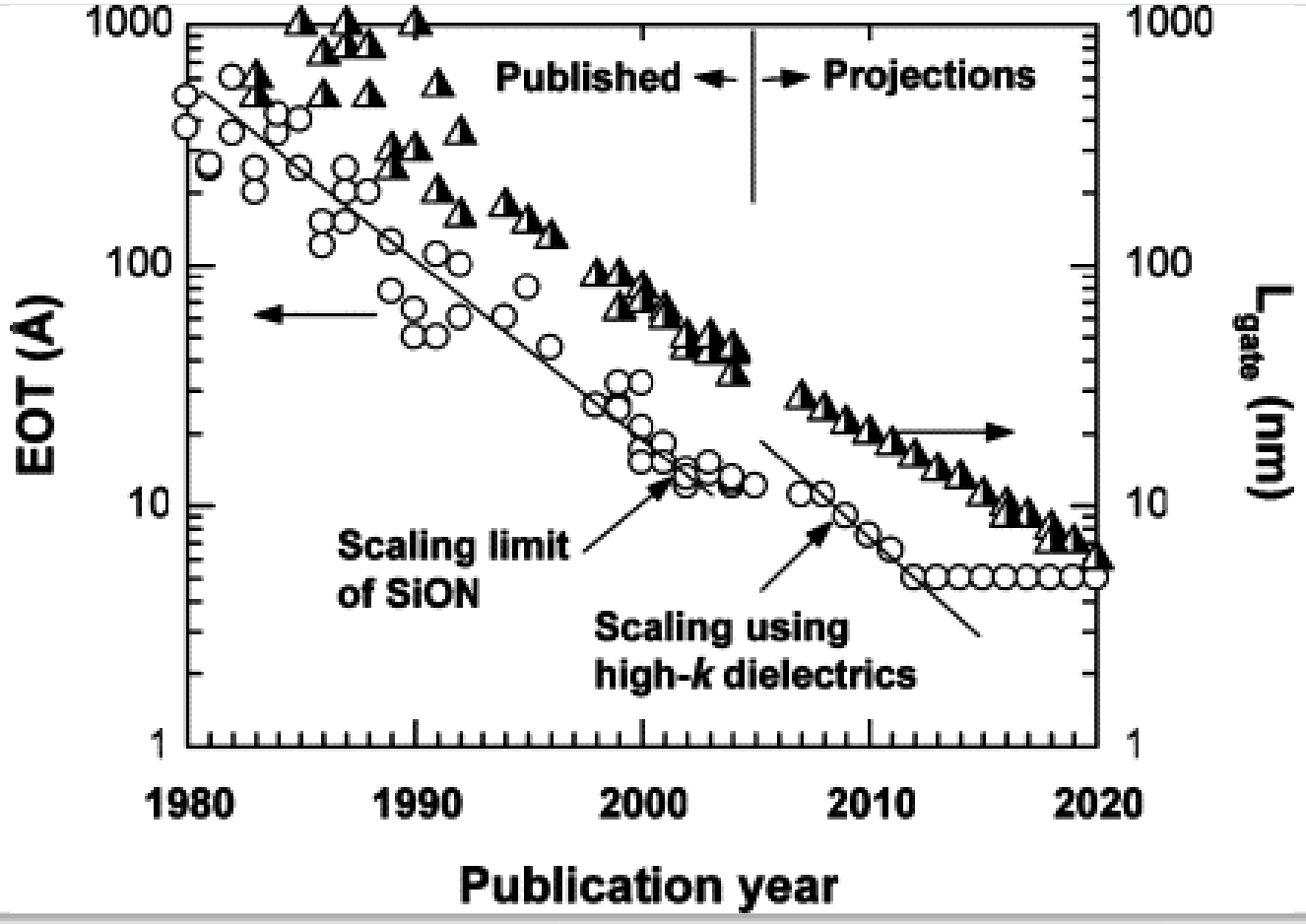
Tox and Lg Scaling



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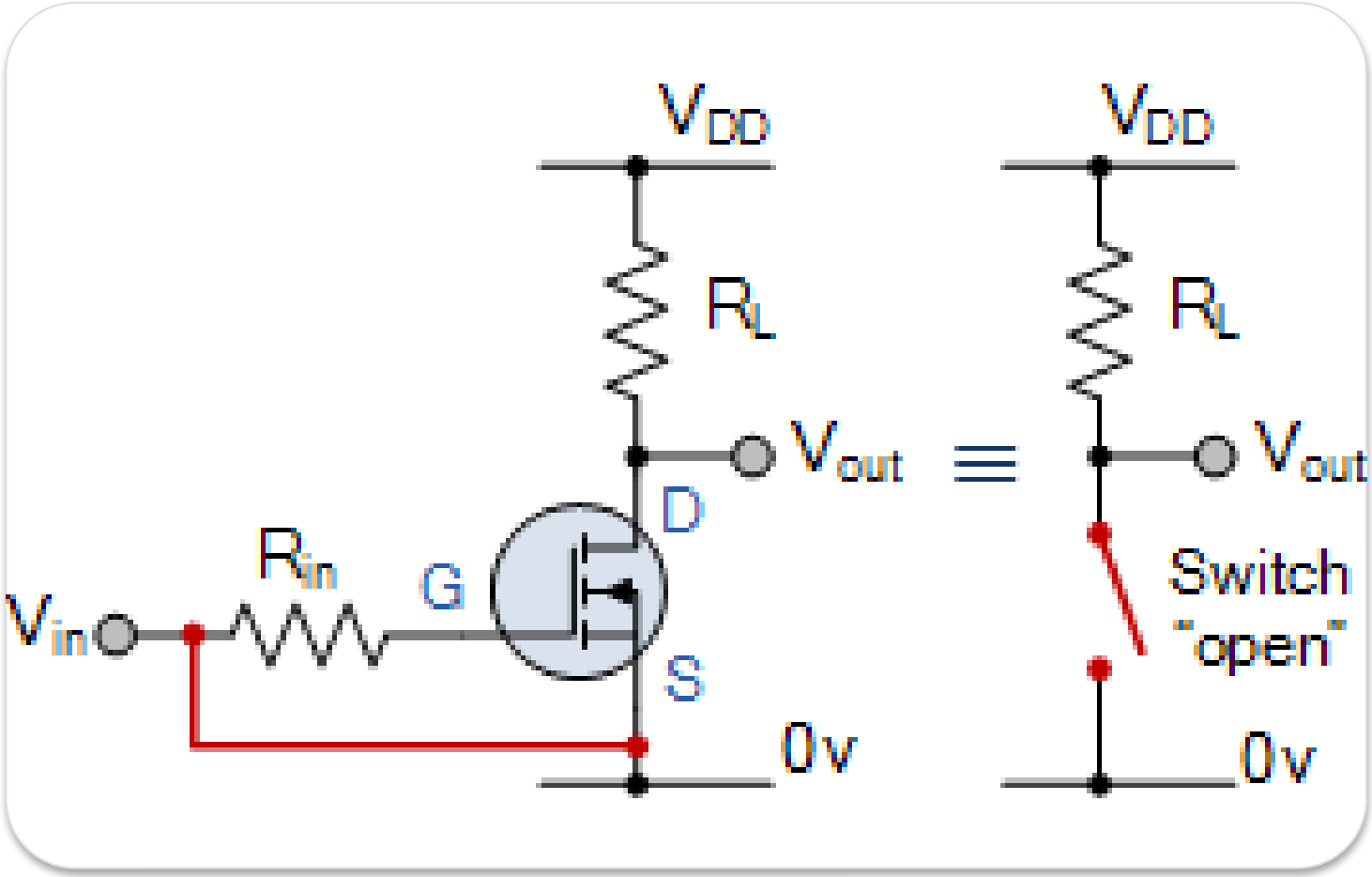
TOPIC 01

Use MOSFET as a Switch



1

Evidence 1

MOSFET is used widely as a switch by simply control the gate voltage

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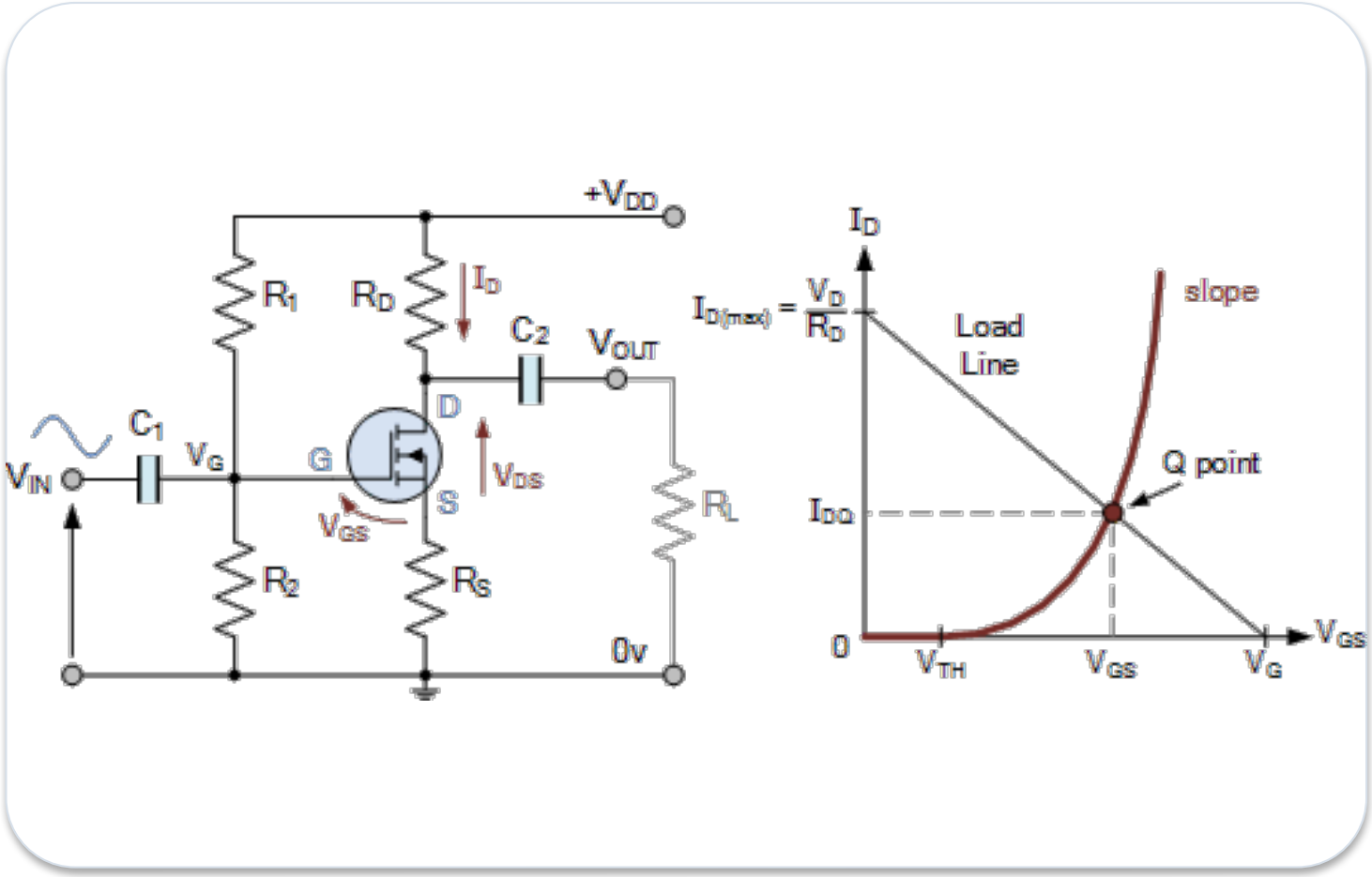
TOPIC 01

Use MOSFET as Amplifier



1

Evidence 1

Similar to BJT, MOSFET can also be used as amplifier operating in the linear mode

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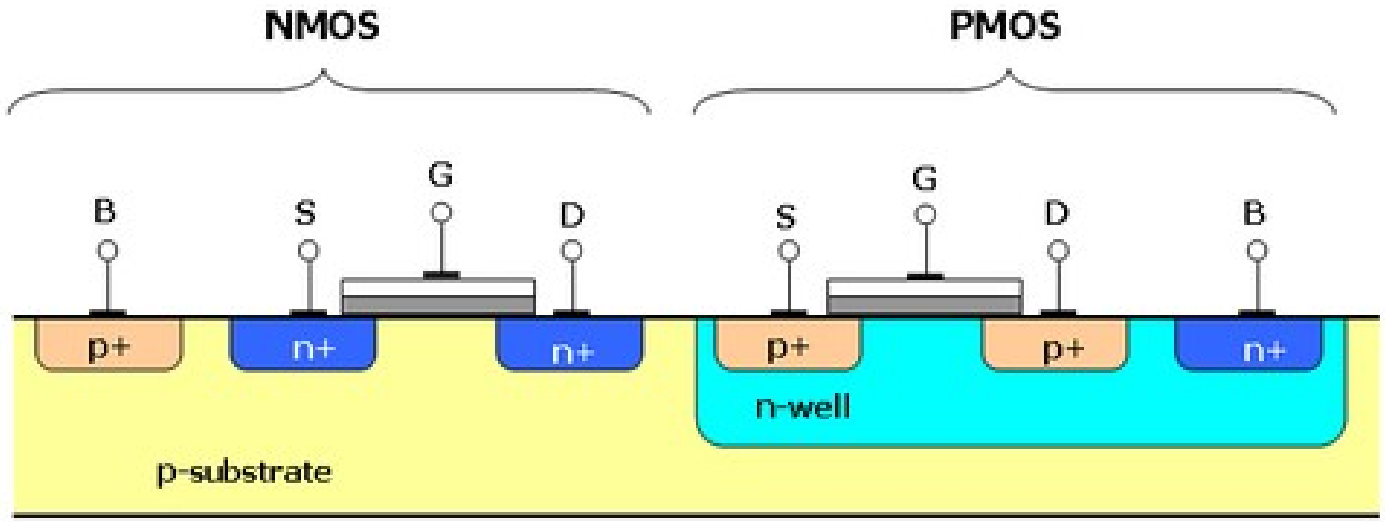
TOPIC 01

CMOS



1

Evidence 1

Complementary MOS (CMOS) consists of both NMOSFET and PMOSFET.

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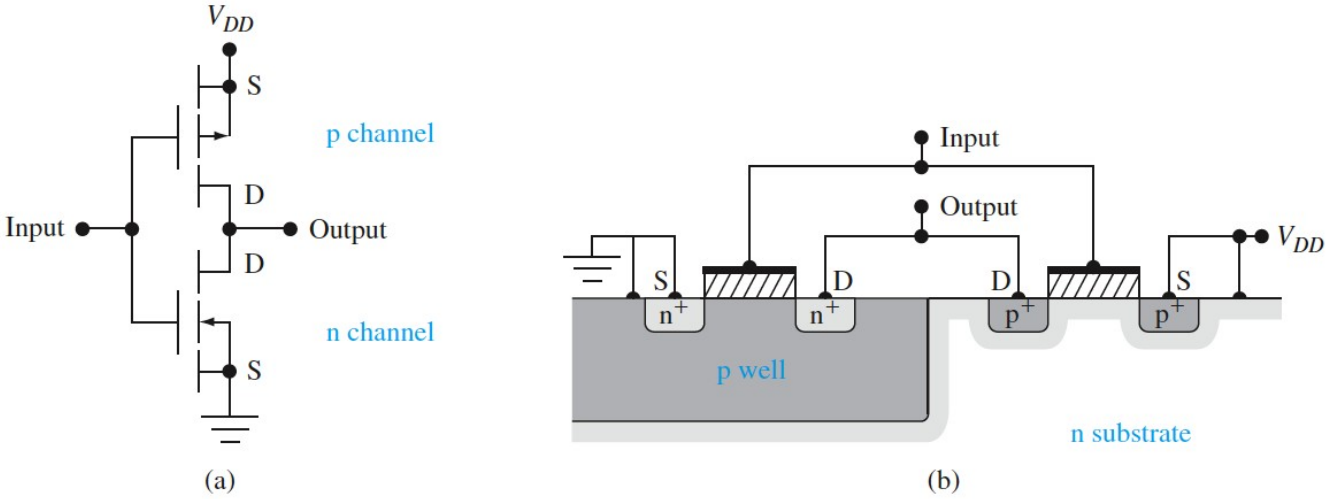
Use CMOS as Inverter



1

CMOS is used widely in inverter.

CMOS is used widely in inverter.
input = 1 -> output = 0
input = 0 -> output= 1

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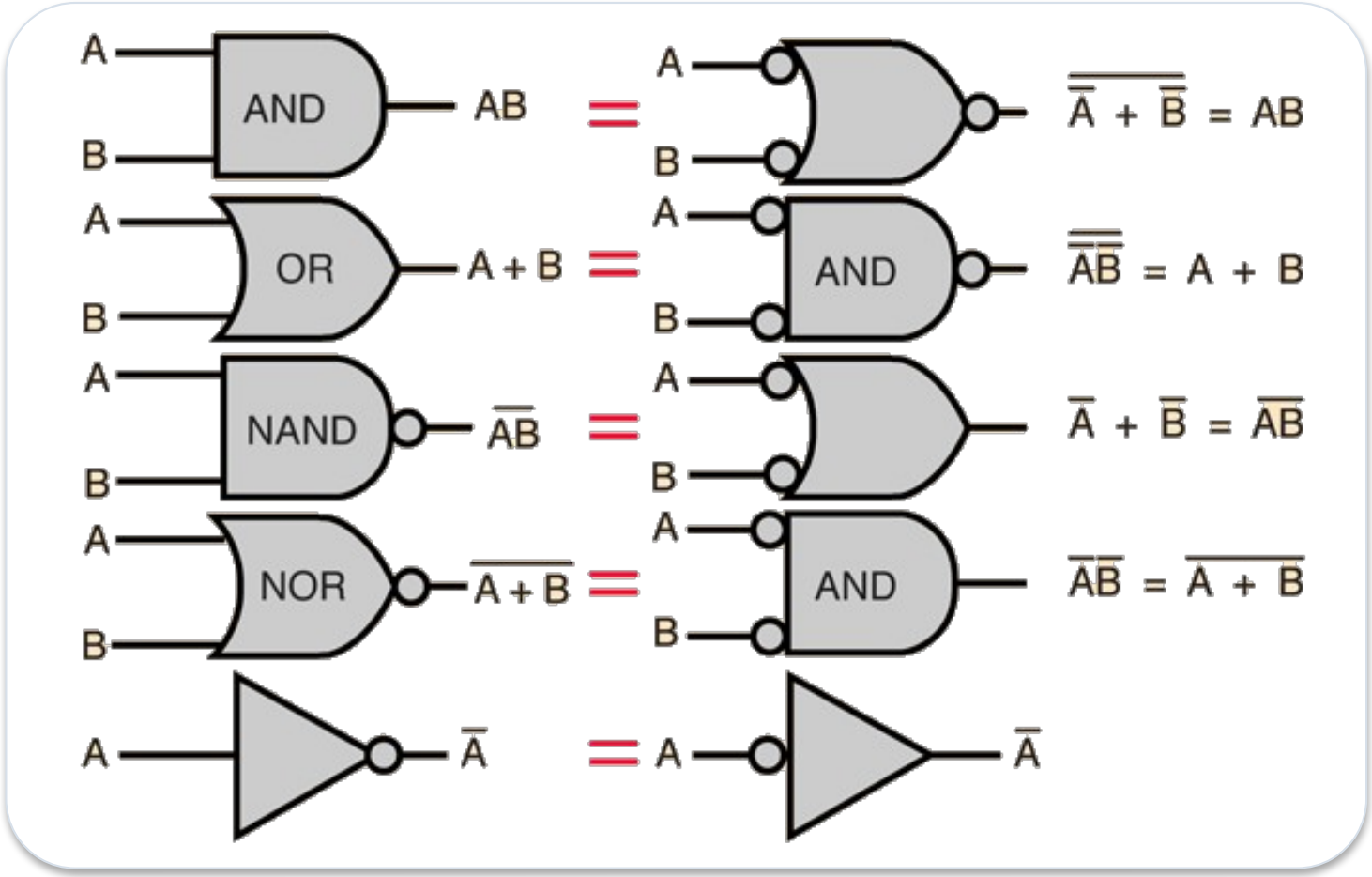
TOPIC 01

Logic Gates



1

Logic gates can be formed by CMOS inverters

Logic gates can be formed by CMOS inverters
All the calculator or computers are using logic gates

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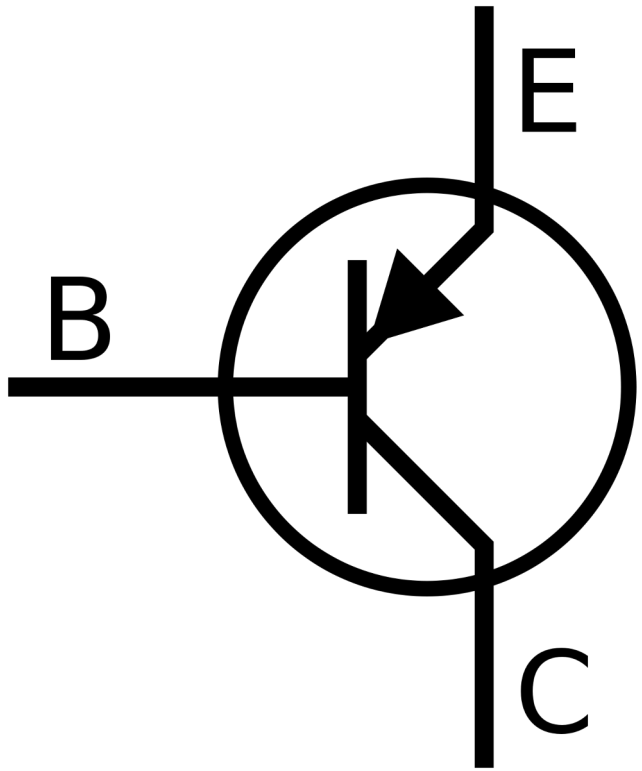
TOPIC 01

Bipolar Junction Transistors

Concepts · mechanisms · process evidence · integration decisions



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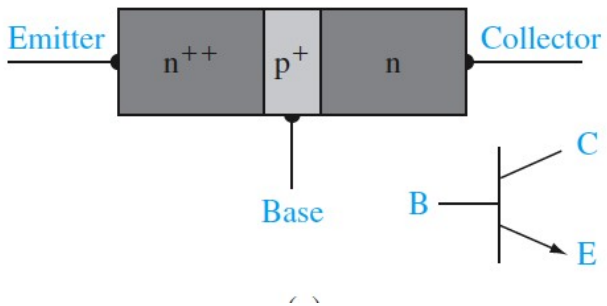
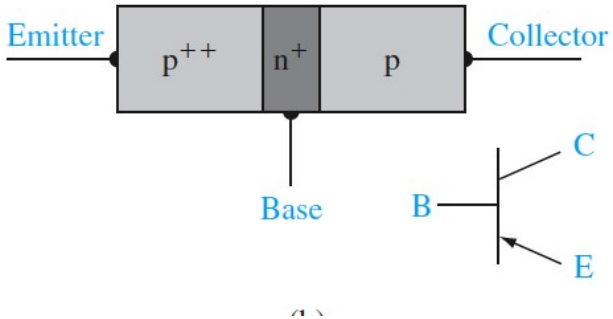
TOPIC 01

Bipolar transistor





1

Evidence 1

Bipolar transistors have three doped regions: emitter, base, and collector
Two pn junctions are present in both npn and pnp transistors
Base width is small relative to the minority carrier diffusion length
Emitter has the highest doping concentration, and the collector has the lowest
Narrow base and doping levels are key to transistor operation
Concepts from pn junction theory apply to bipolar transistors

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TOPIC 01

NPN and PNP BJT

1

There are two types of BJT transistors

There are two types of BJT transistors
NPN BJT
PNP BJT

2

N-Type

N-Type

3

P-Type

P-Type

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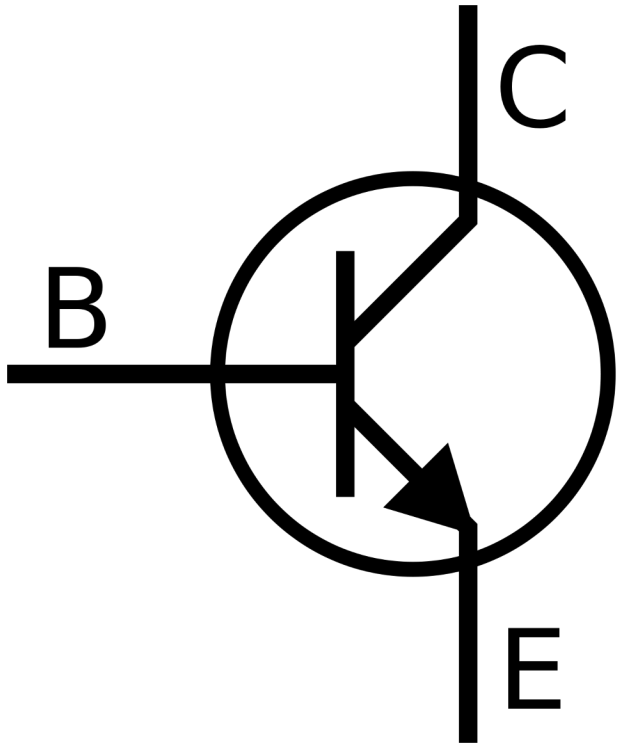
TOPIC 01

NPN BJT Symbol



1 **Evidence 1**

There are 3 terminals to the NPN transistor - Base, Emitter and Collector.
Current flow from collector to emitter.

2 **N-Type**

N-Type
Emitter

3 **P-Type**

P-Type
Base

4 **Collector**

Collector

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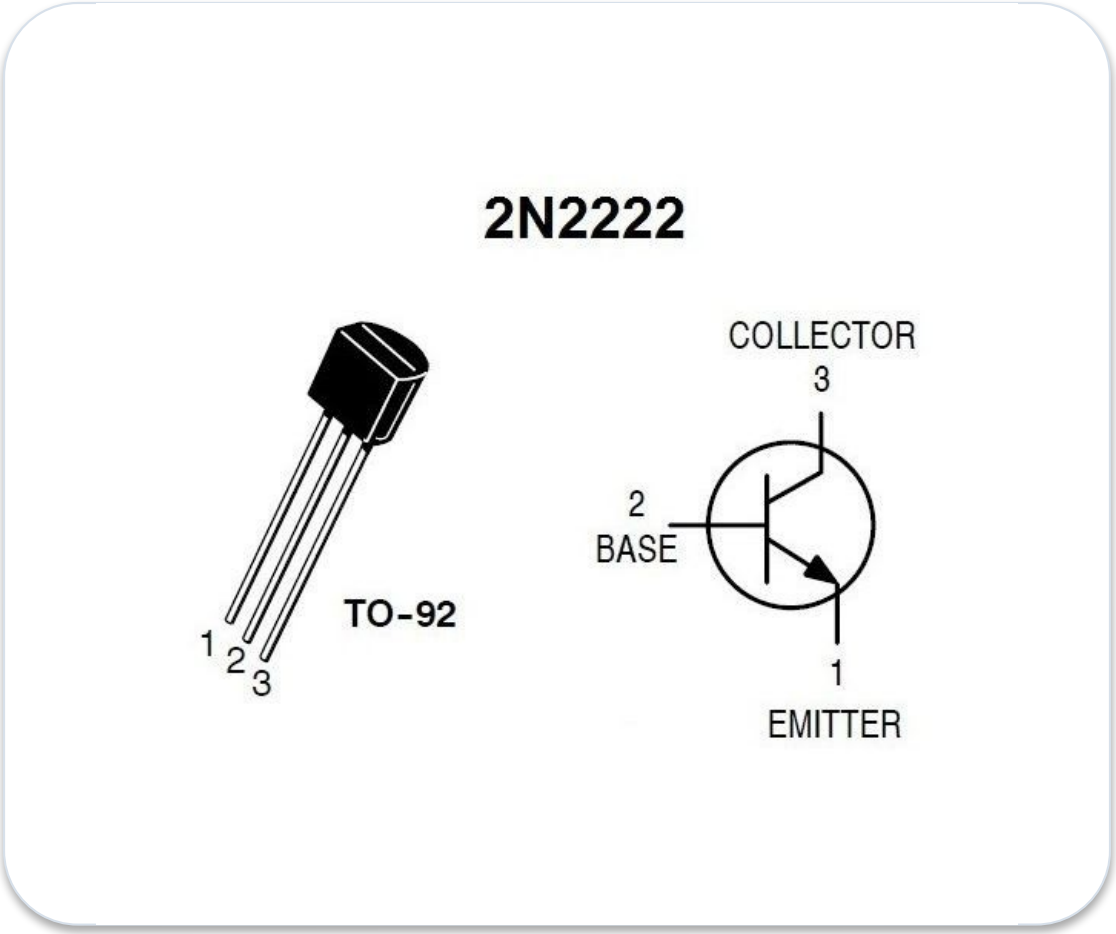
TOPIC 01

NPN Transistor Component



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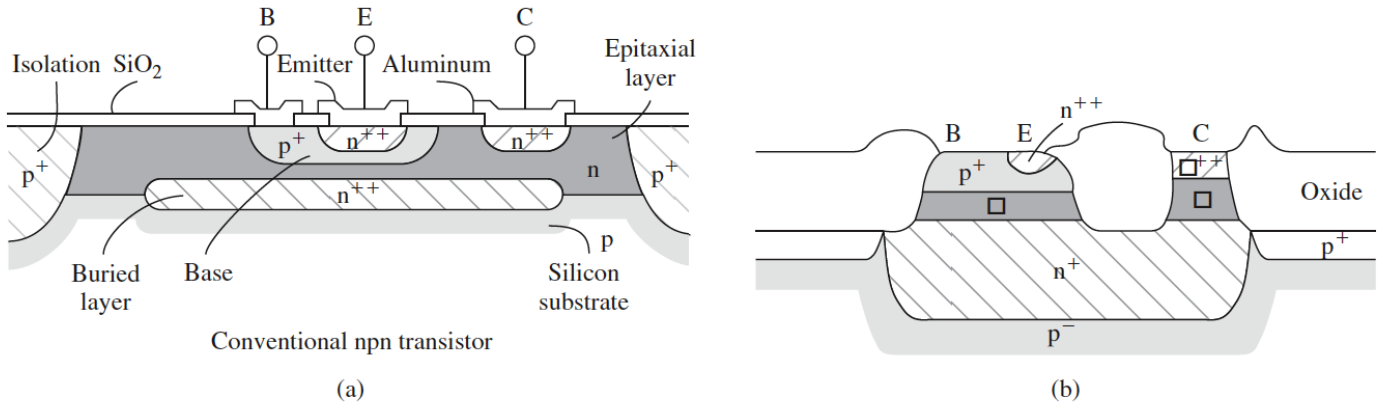
TOPIC 01

Cross Section of Bipolar transistor



1

Evidence 1

Cross section of (a) a conventional integrated circuit npn bipolar transistor and (b) an oxide-isolated npn bipolar transistor.

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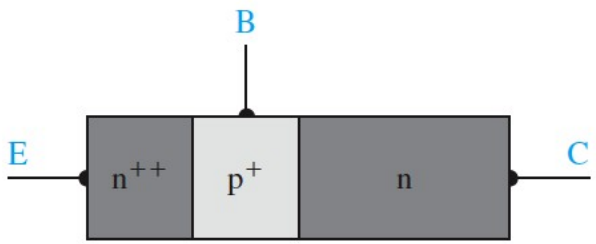
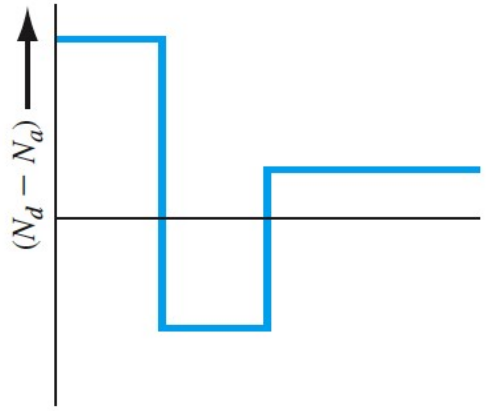
TOPIC 01

Basic Principle of Operation



(a)



(b)

1

Evidence 1

Idealized doping profile of a uniformly doped npn bipolar transistor.

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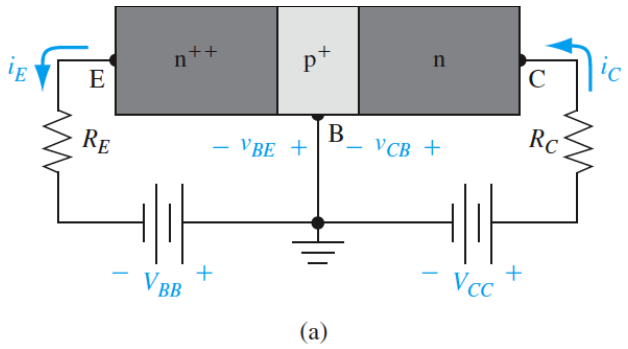
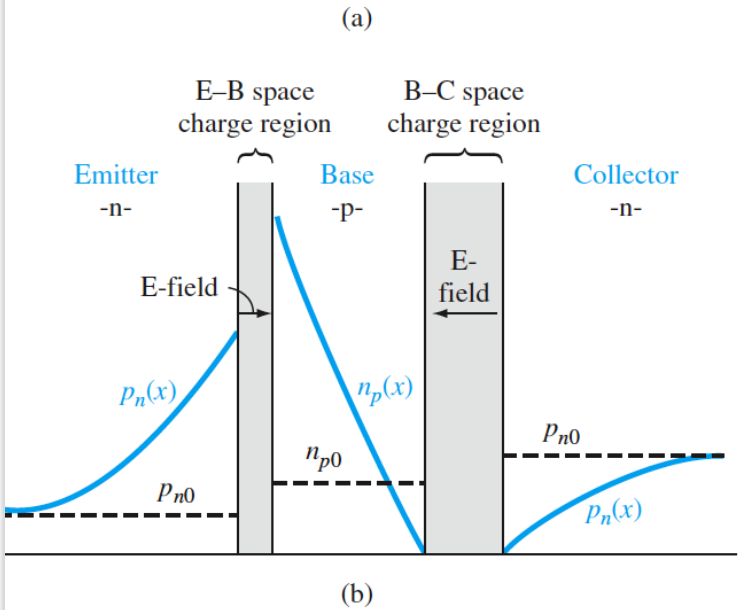
Basic Principle of Operation



(a)



1

Evidence 1

(a) Biasing of an npn bipolar transistor in the forward-active mode, (b) minority carrier distribution in an npn bipolar transistor operating in the forward-active mode

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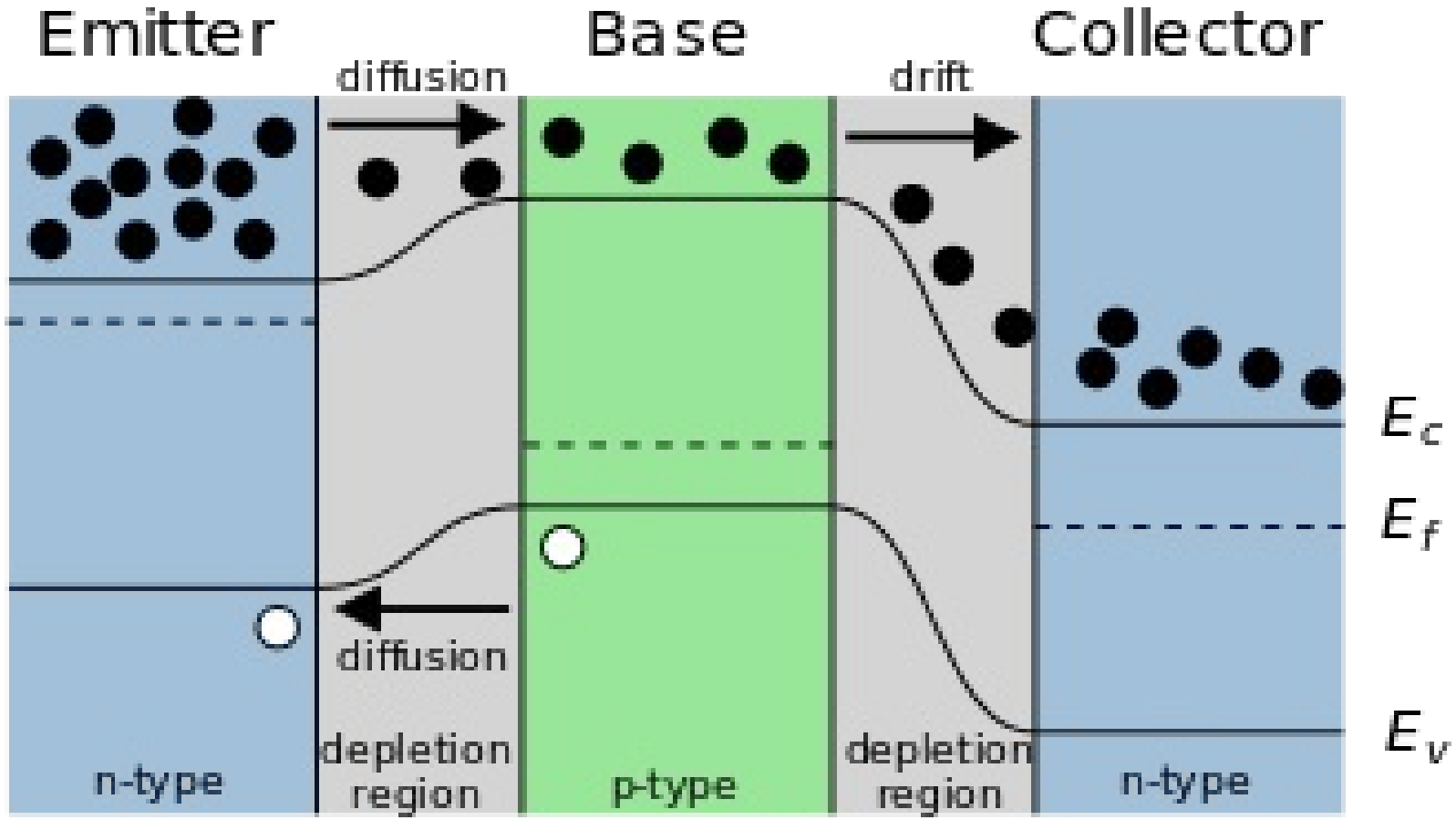
Basic Operation of BJT



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TOPIC 01

Basic Operation of BJT

In the forward-active mode, the base-emitter (B–E) junction is forward biased, allowing electrons (for an NPN transistor) or holes (for a PNP transistor) to flow from the emitter into the base. This results in the injection of a large number of majority carriers (electrons in NPN, holes in PNP) from the emitter across the B–E junction into the base.
Once these carriers enter the base, they become minority carriers (electrons in a P-type base, or holes in an N-type base) and create an excess concentration of minority carriers. The base is designed to be thin and lightly doped, allowing most of these carriers to diffuse across the base toward the base-collector (B–C) junction.
The B–C junction is reverse biased, which creates a strong electric field that rapidly sweeps the minority carriers (electrons or holes) from the base into the collector.

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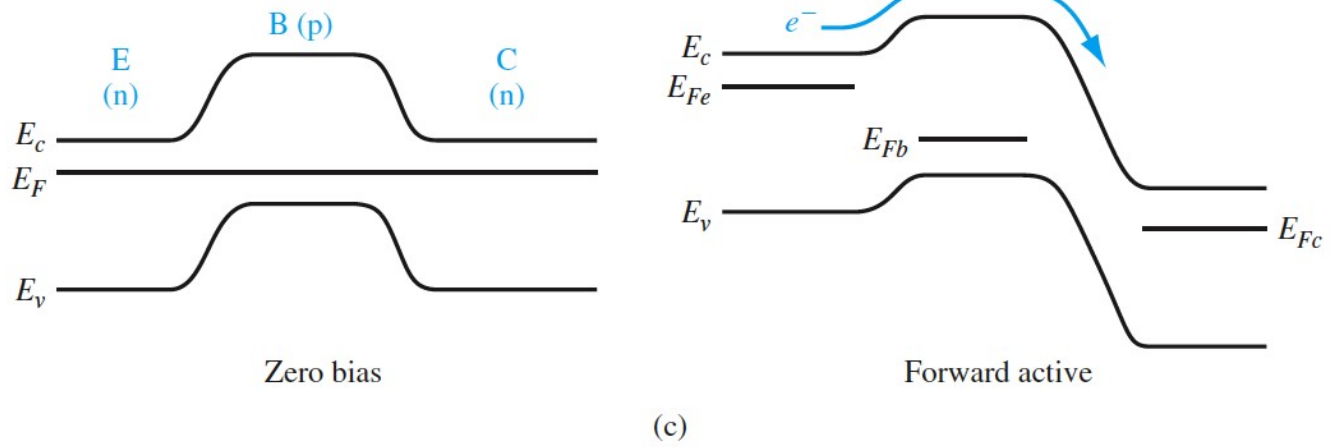
TOPIC 01

Basic Principle of BJT



(c)

1 **Evidence 1**

Energy-band diagram of the npn bipolar transistor under zero bias and under a forward-active mode bias.

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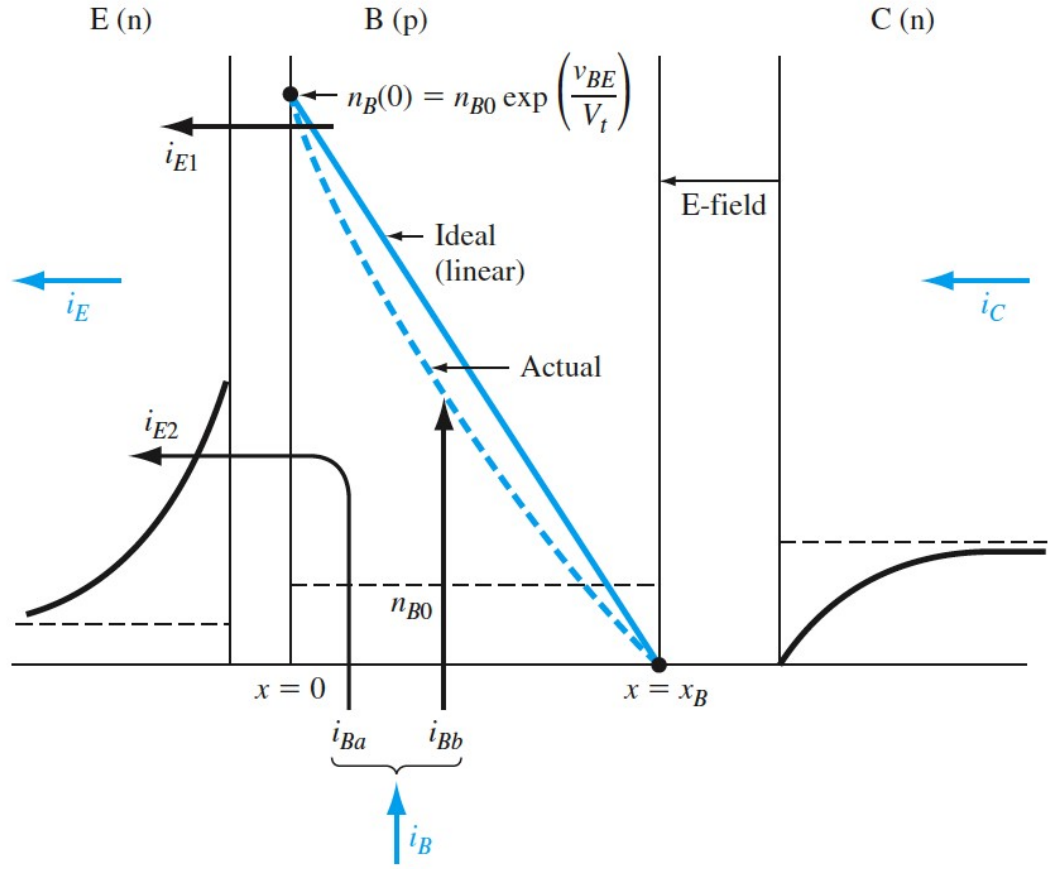
TOPIC 01

Minority carrier distributions for BJT



Evidence 1

Minority carrier distributions and basic currents in a forward-biased npn bipolar transistor.

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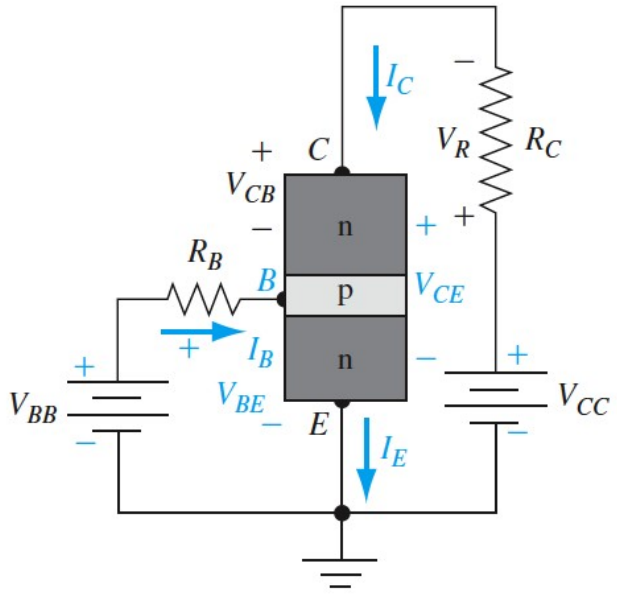
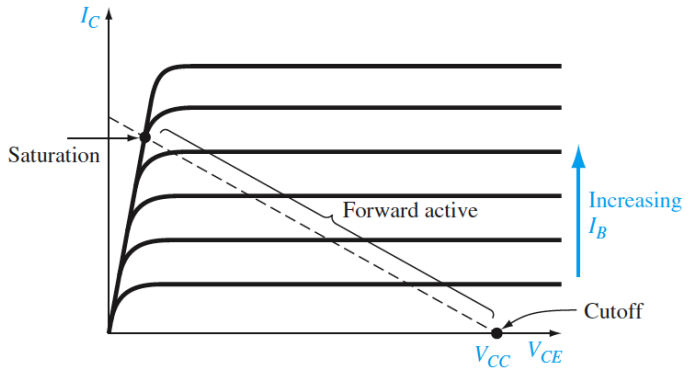
TOPIC 01

IV Characteristics of BJT





1 **Evidence 1**

An npn bipolar transistor in a common-emitter circuit configuration. Bipolar transistor common-emitter current–voltage characteristics with load line superimposed

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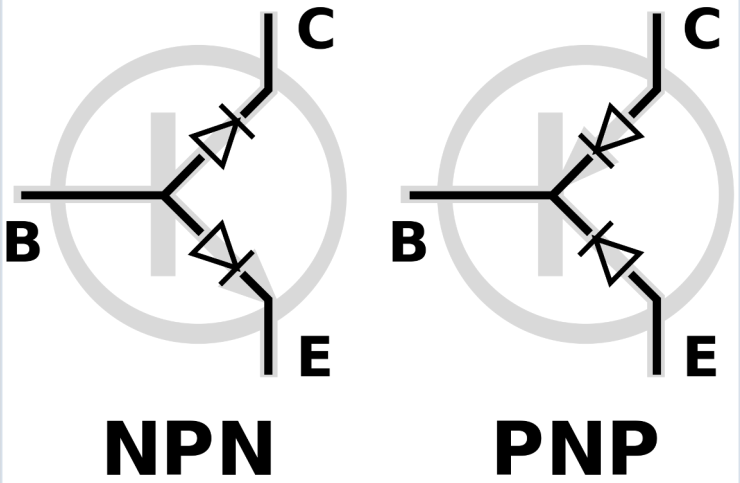
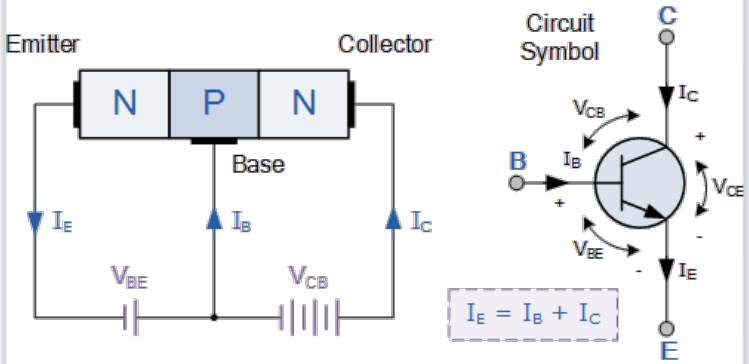
TOPIC 01

Active Mode (Linear Region)





1

Evidence 1

When VBE ~0.7V, BE junction is short circuited, the transistor is in ACTIVE mode

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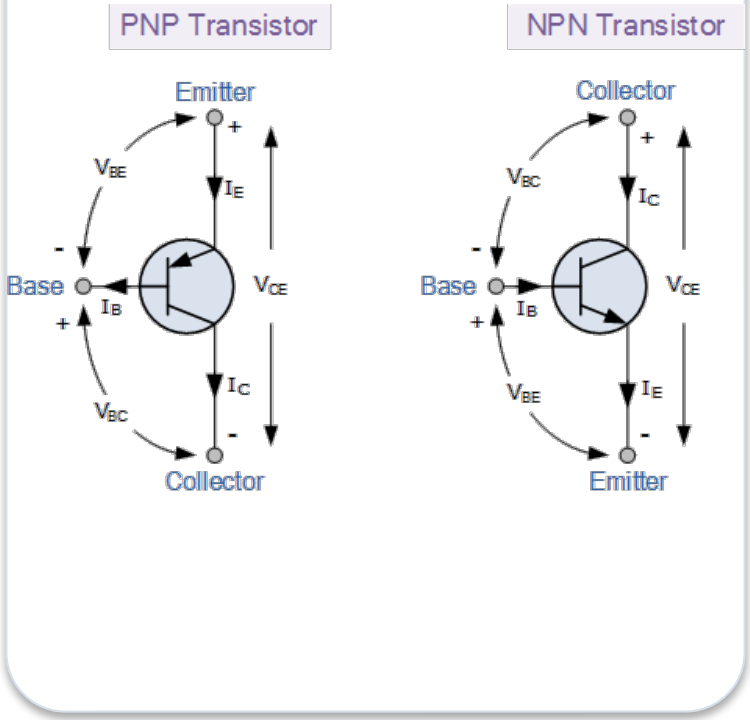
TOPIC 01

Active Mode (Linear Region)



$$I_C = \beta I_B$$

$$\beta \approx 100$$

$$I_E = I_C + I_B$$

$$V_{BE} \approx 0.7V$$

1. **1mA**
   1mA
2. **100mA**
   100mA
3. **99mA**
   99mA

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TOPIC 01

Saturation Mode

1

Key idea 1

There is a maximum collector current Ic can flow. Once Ic reach the maximum, the transistor is in the saturation mode or short circuit mode.
Current flows freely from collector to emitter like a resistor.
Ideally VCE = 0V. In reality, VCE(Sat) ~ 0.2V
VB > VC>VE

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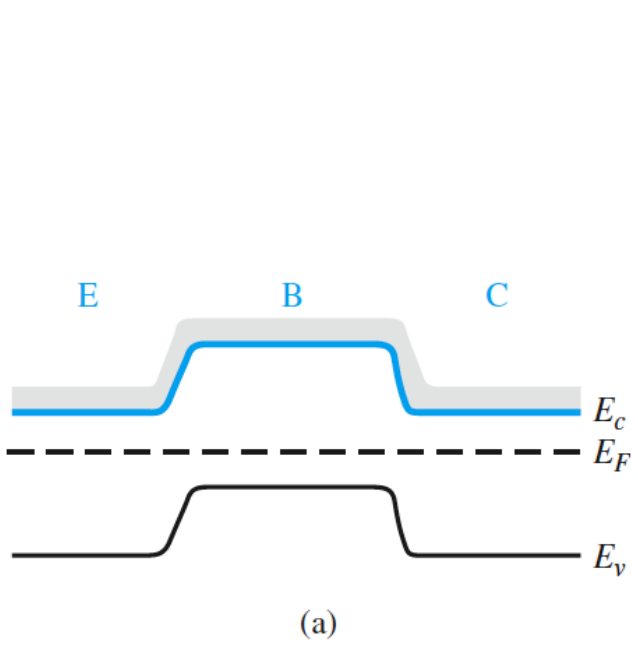
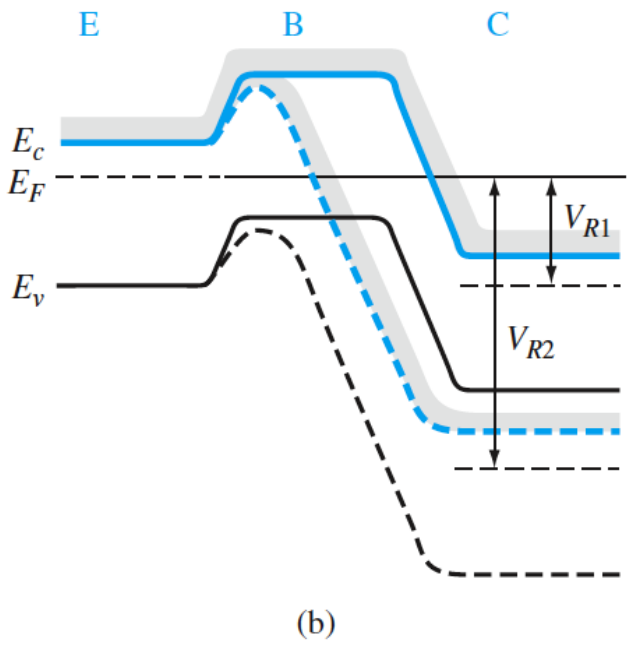
TOPIC 01

Breakdown Voltage





1

Evidence 1

Energy-band diagram of an npn bipolar transistor (a) in thermal equilibrium, and (b) with a reverse-biased B–C voltage before punch-through, VR1, and after punch-through, VR2.

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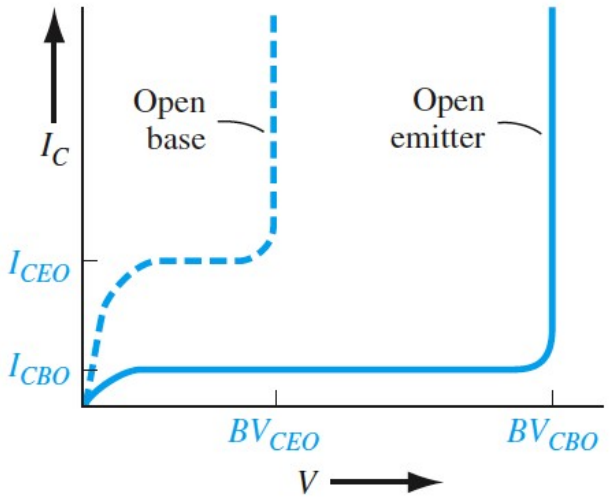
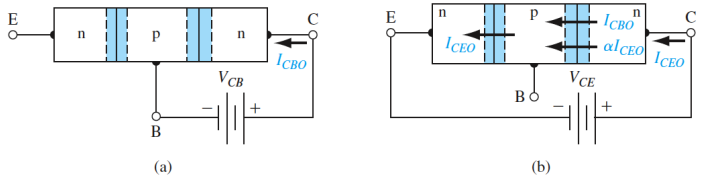
TOPIC 01

Breakdown Voltage





1

Evidence 1

Relative breakdown voltages and saturation currents of the open-base and open-emitter configurations.

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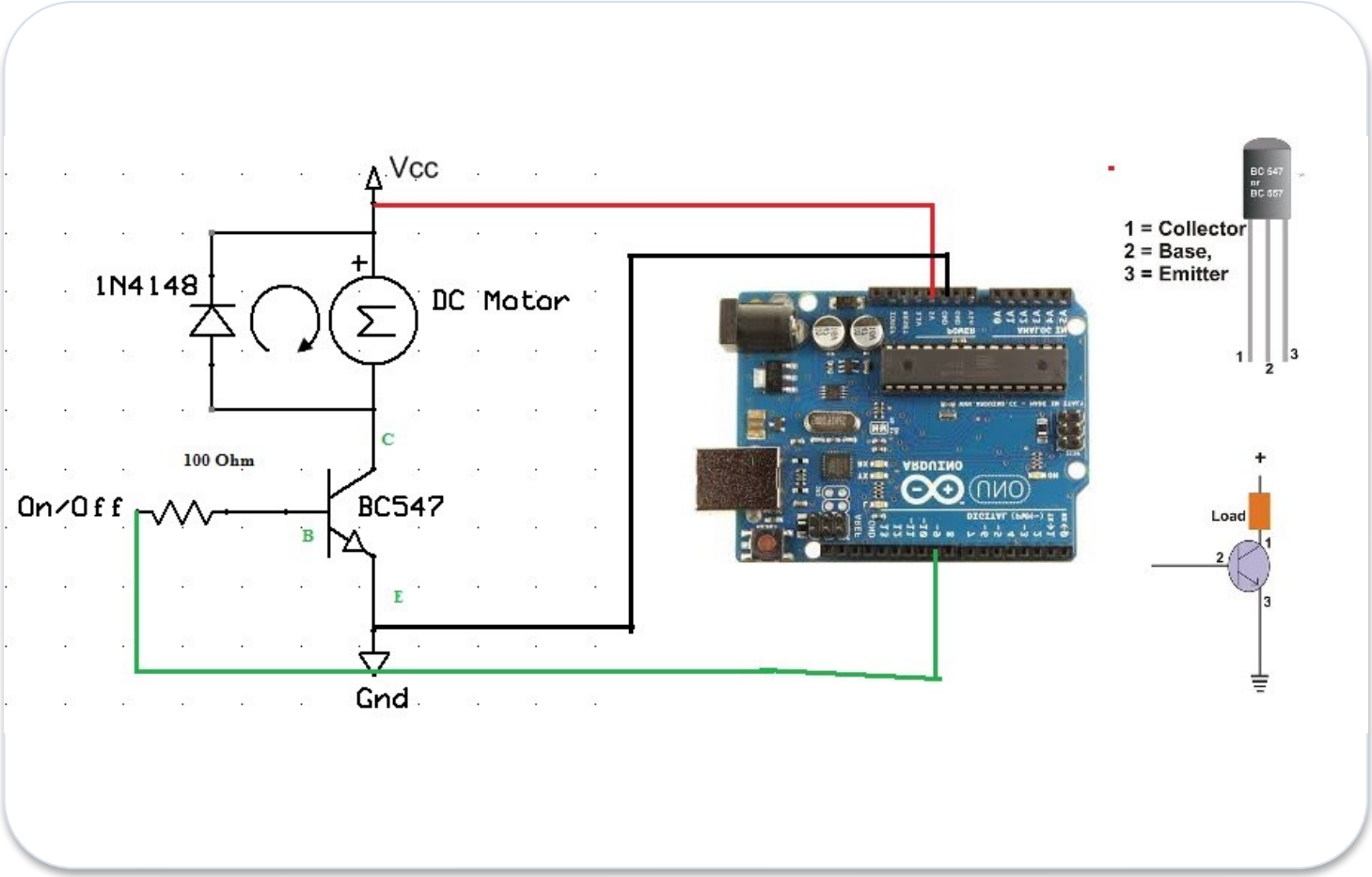
TOPIC 01

Use BJT as a Switch



1 **BJT is used widely as a switch**

BJT is used widely as a switch
ON - Saturation Mode
OFF - Cutoff Mode
Eg: Use BJT to drive a higher voltage needed for a motor to turn

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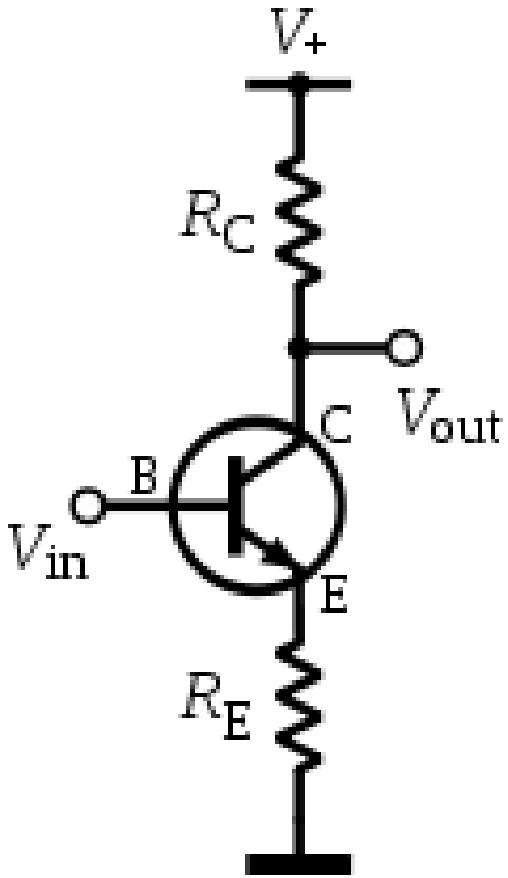
TOPIC 01

Use BJT as Amplifier



1

Evidence 1

You can use BJT as a amplifier when operating the BJT in the active mode
There are 3 configuration
Common Emitter
Common Collector
Common Base

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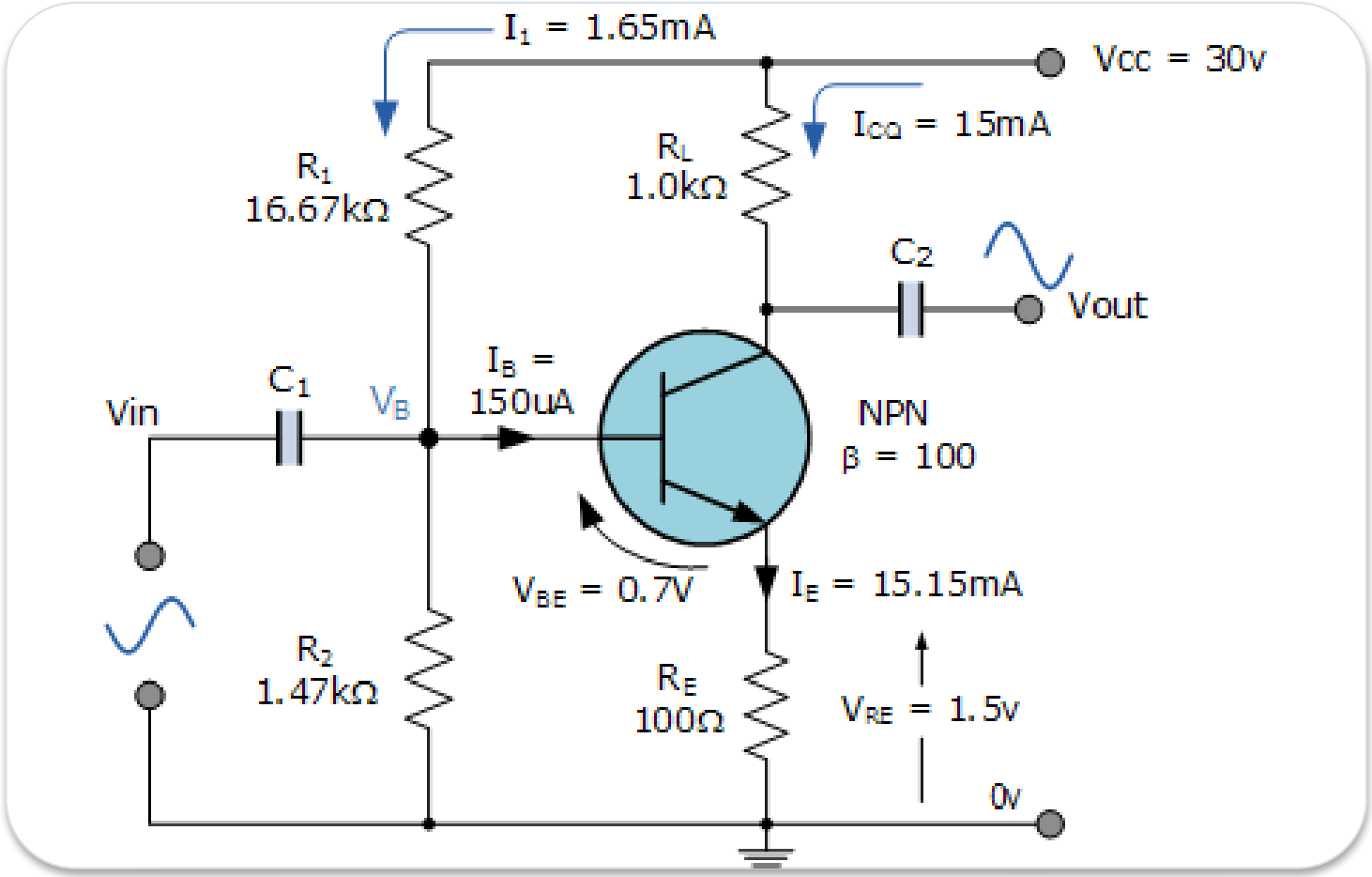
TOPIC 01

Common Emitter NPN Amplifier



1

Evidence 1

A typical common emitter NPN amplifier is shown below

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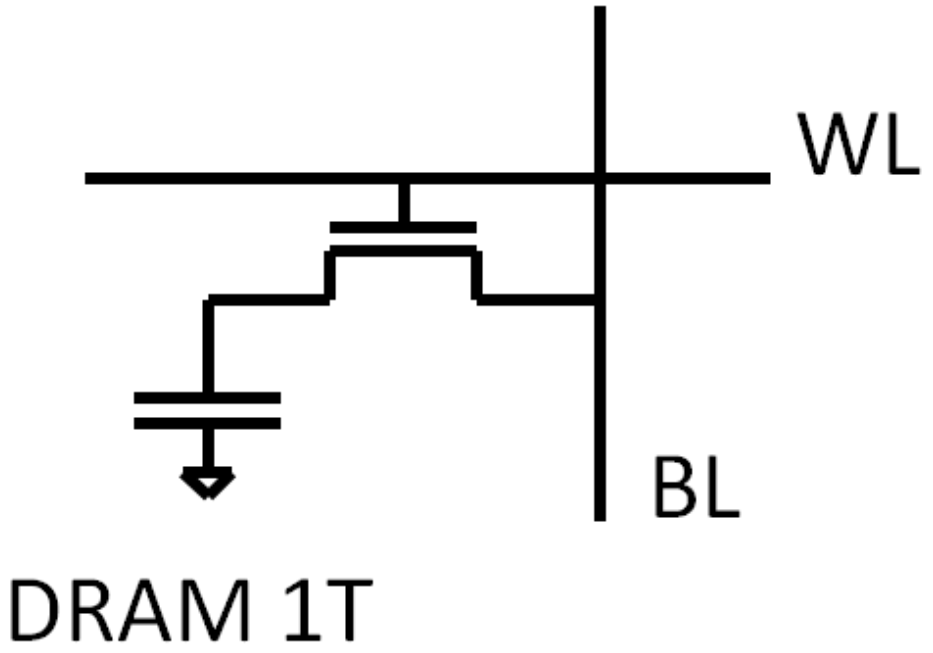
TOPIC 01

Memory Devices and Integrated Circuits

Concepts · mechanisms · process evidence · integration decisions



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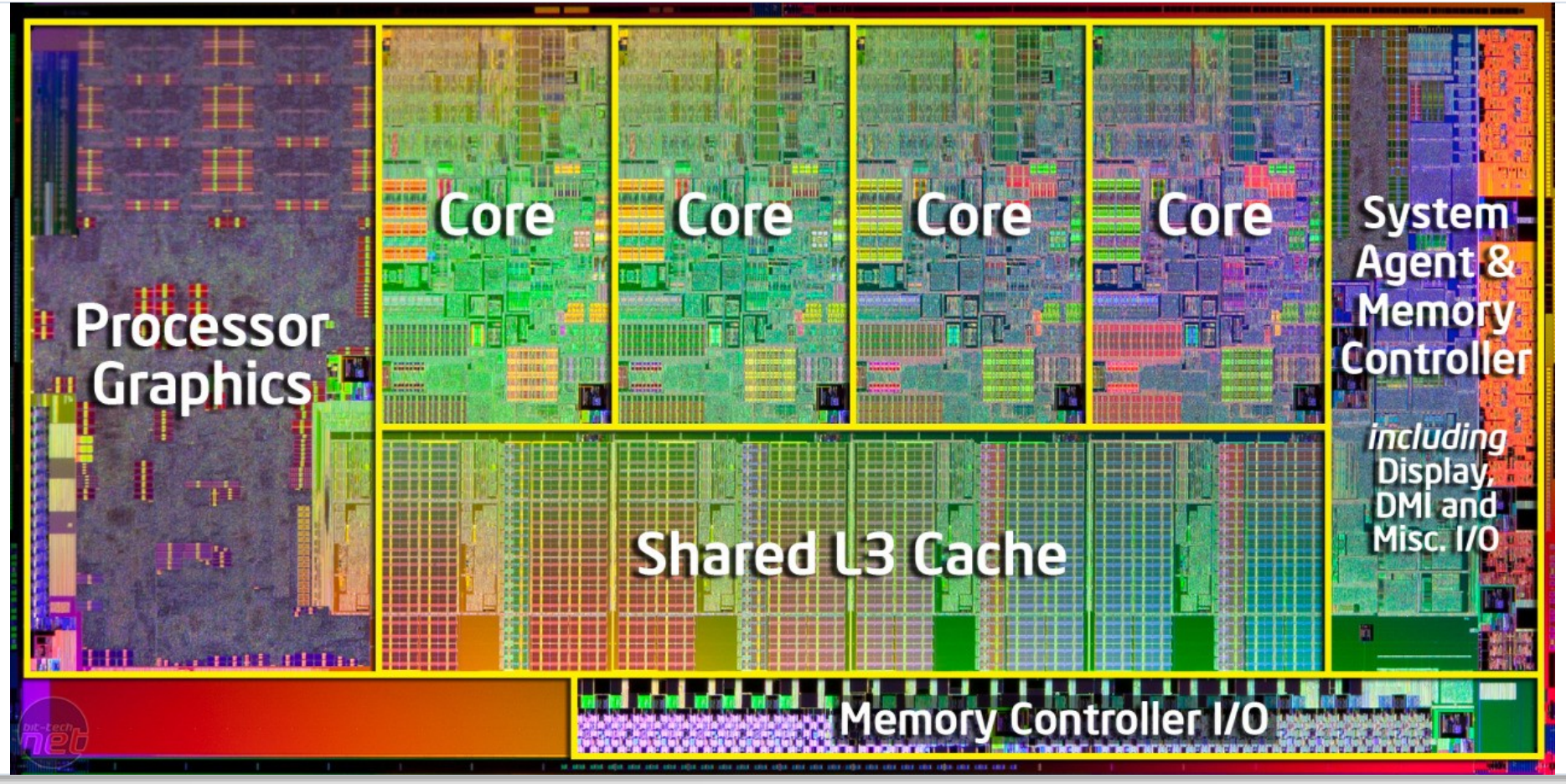
Memory in CPU



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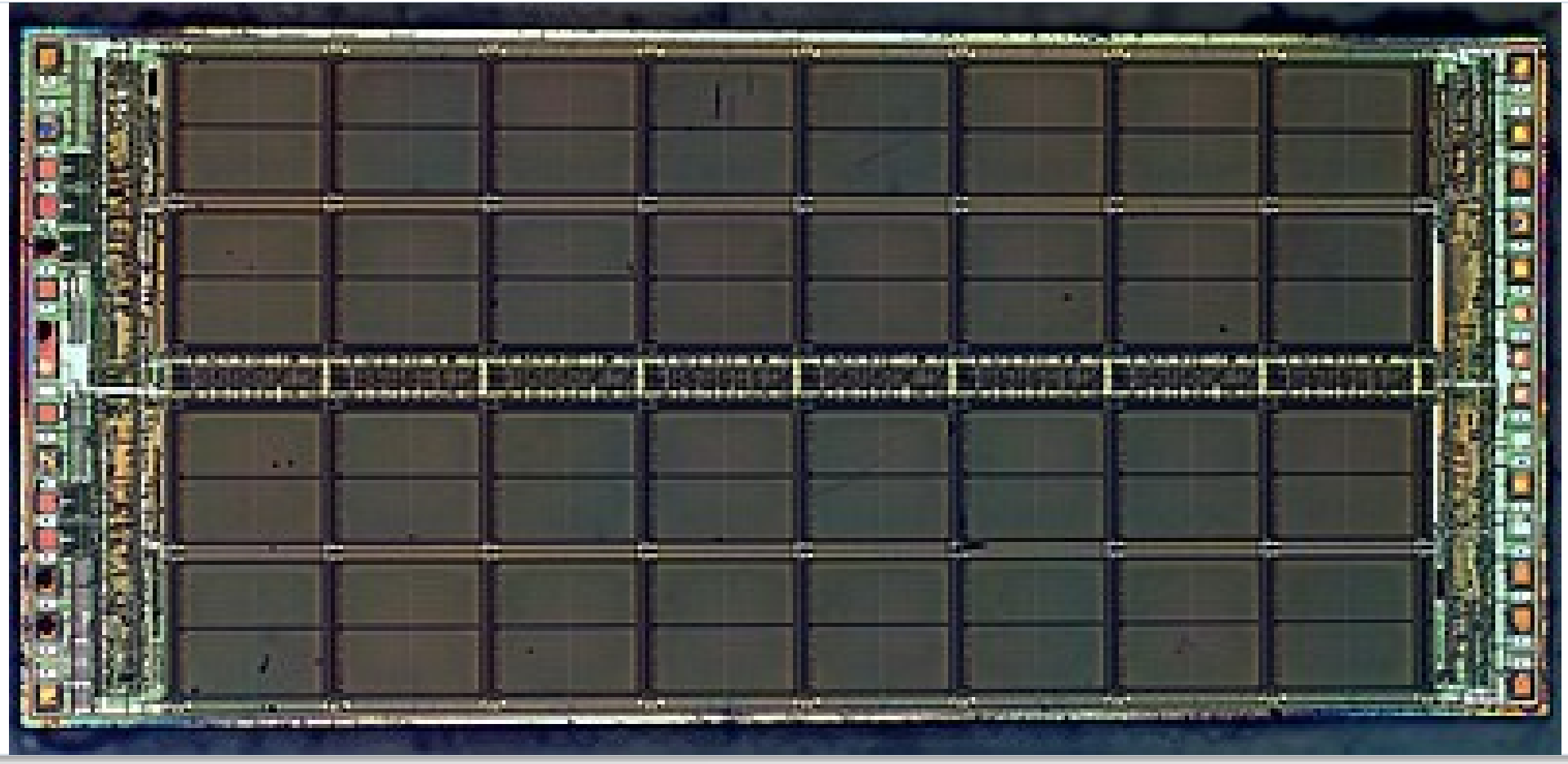
TOPIC 01

Dynamic RAM (DRAM)



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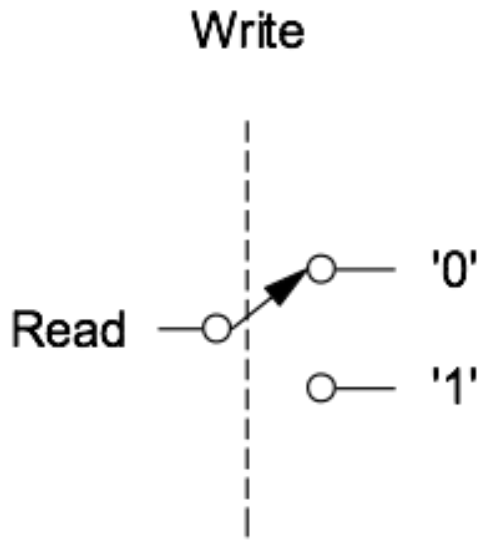
TOPIC 01

Basic Concepts of Memory



1

Write '0' and '1' to an element

Write '0' and '1' to an element
Read '0' and '1' from an element

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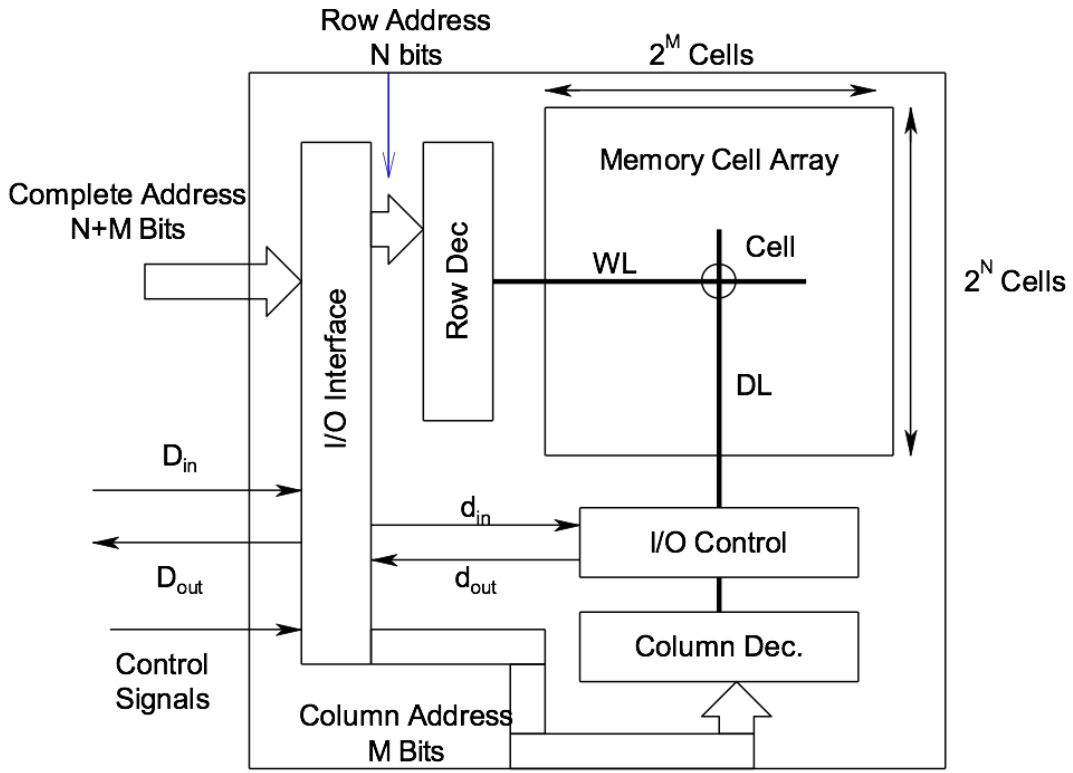
TOPIC 01

Memory Chip



1

Evidence 1

A typical memory chip consist of memory cell array, row decoder, and column decoder, and sense amplifiers

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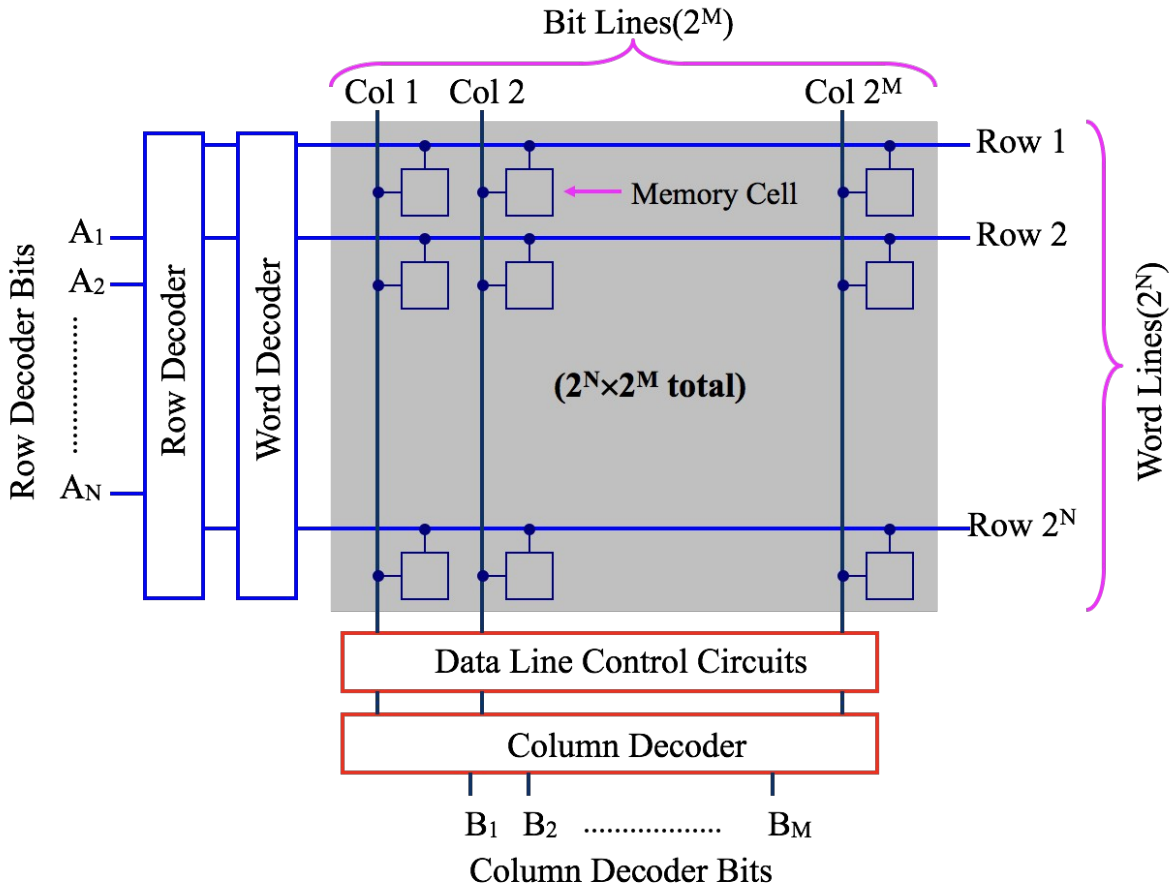
Memory Cell Array



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TOPIC 01

Type of Memory Devices

1 **Memory Devices**

Memory Devices
ROM
EEPROM

2 **Read Only Memory (ROM)**

Read Only Memory (ROM)

3 **Random Access Memory (RAM)**

Random Access Memory (RAM)

4 **DRAM**

DRAM
SRAM
NOR Flash
NAND Flash

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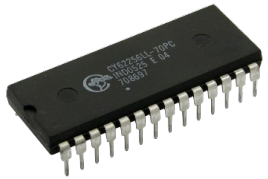
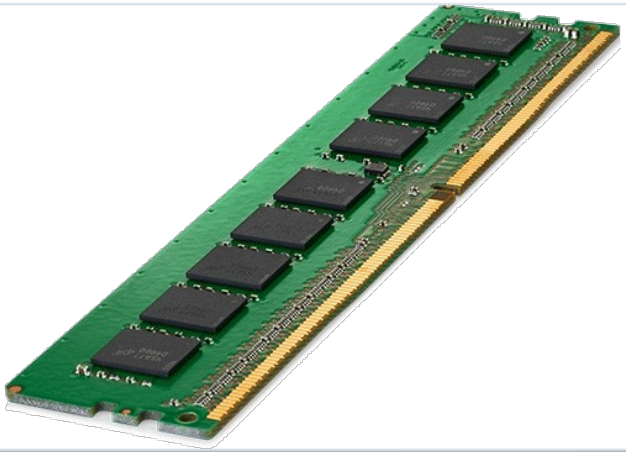
TOPIC 01

Memory Components





1. **SRAM**
   SRAM
2. **Flash**
   Flash
3. **EEPROM**
   EEPROM
4. **DRAM**
   DRAM

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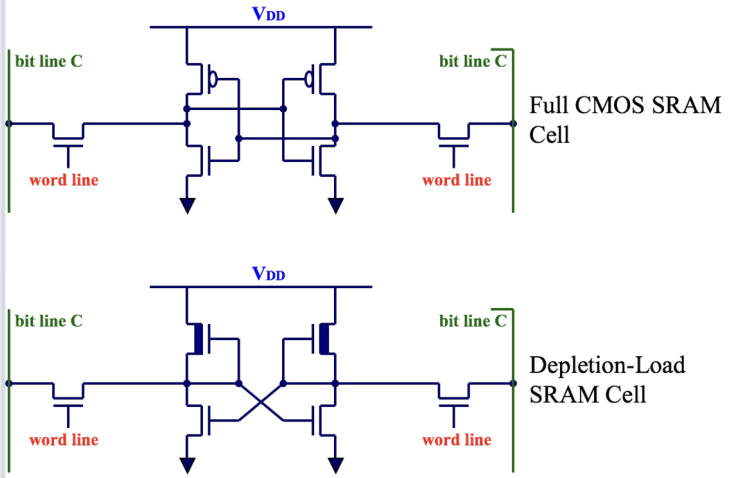
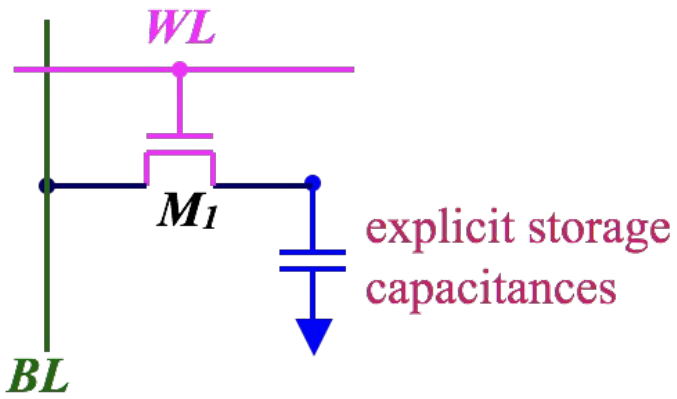
TOPIC 01

DRAM vs SRAM





1 **DRAM**

DRAM

2 **SRAM**

SRAM

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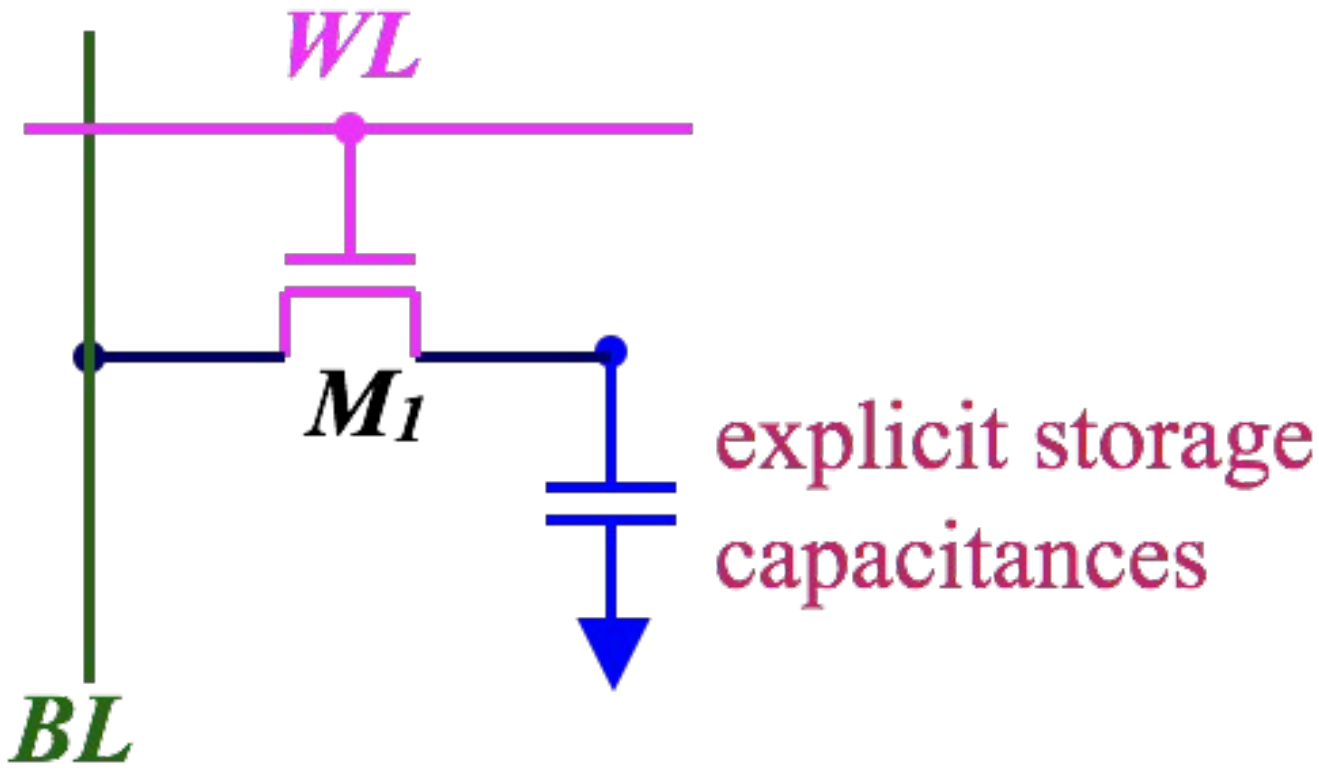
TOPIC 01

How DRAM Work?



1

When writing, the WL will turn on the M1.

When writing, the WL will turn on the M1.
The BL will write a "0" or "1" to the capacitor
When reading, the sense-amplifier determines the level of charge in the capacitor

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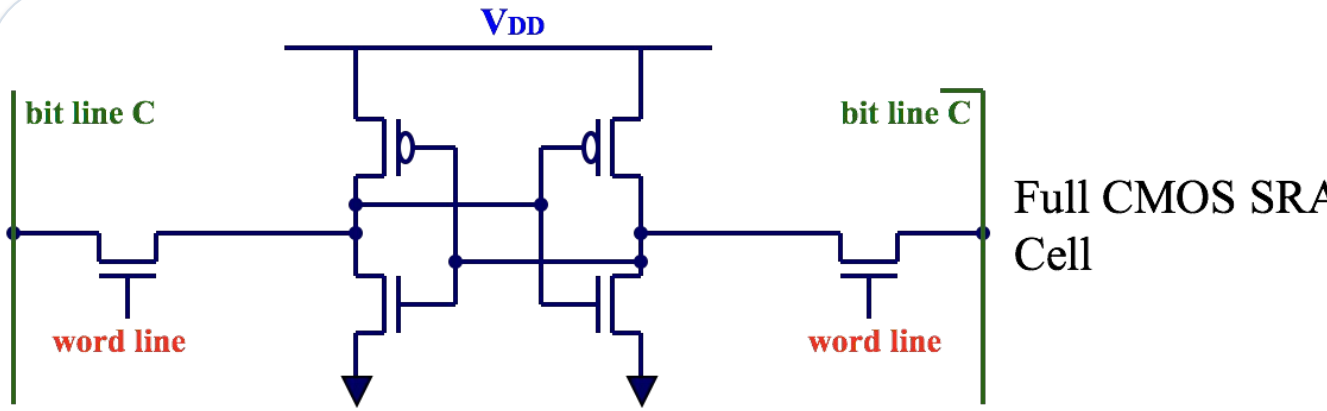
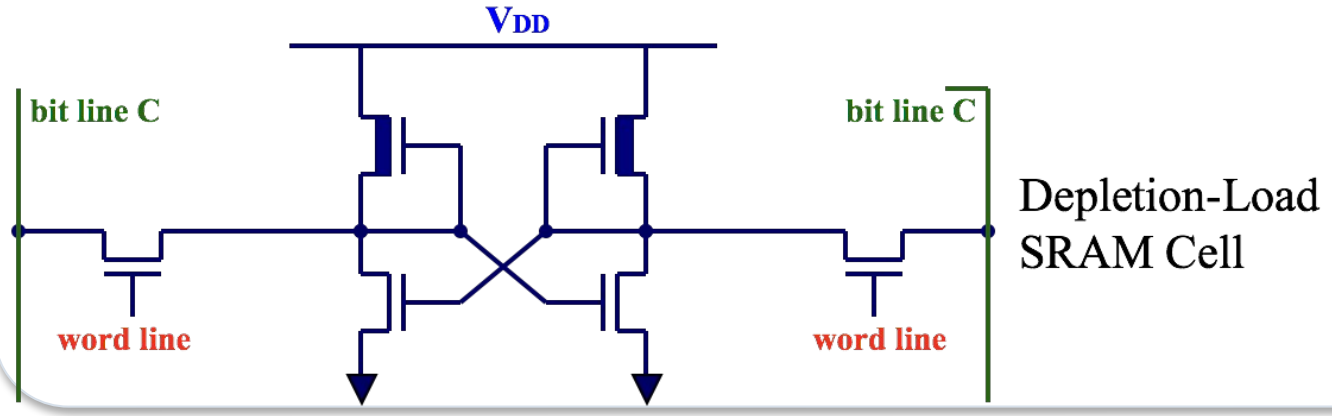
How SRAM Work?



Full CMOS SRAM Cell



Depletion-Load SRAM Cell

1

Evidence 1

When the cell is selected, the value to be written is stored in the cross-coupled flip-flops.
Most SRAM memories select an entire row of cells at a time, and read out the contents of all the cells in the row along the column lines.

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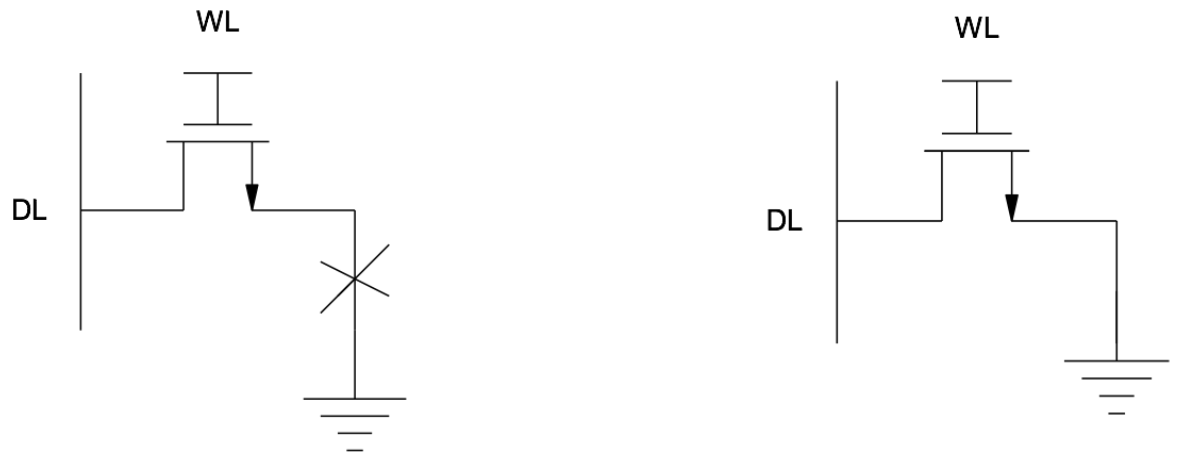
TOPIC 01

Fuse ROM vs EEPROM



1 **Fuse ROM**

Fuse ROM

2 **EEPROM**

EEPROM

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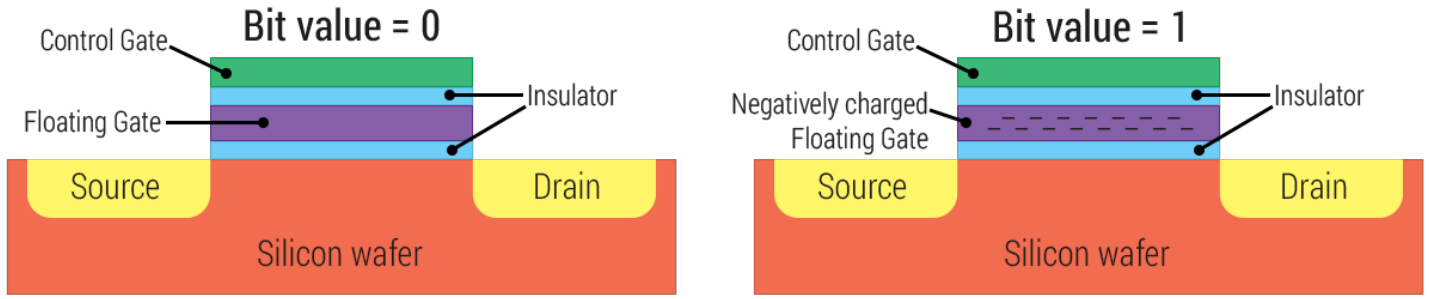
TOPIC 01

Floating Gate Transistor



1

Evidence 1

A floating gate transistor can be used for storing charges
EEPROM and Flash use floating gate transistors

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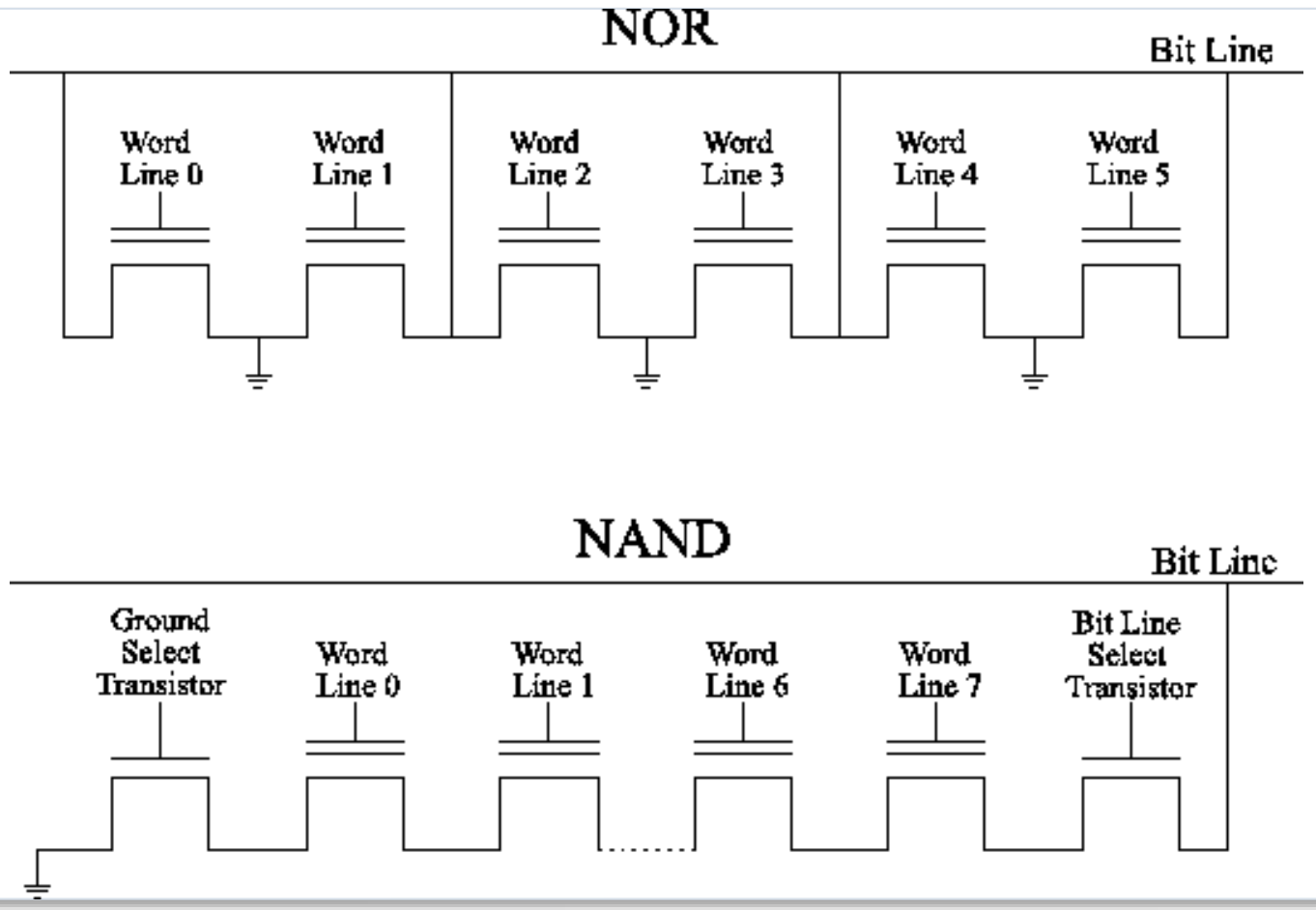
NOR vs NAND Flash Architecture



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TOPIC 01

Introduction to process modules

Concepts · mechanisms · process evidence · integration decisions

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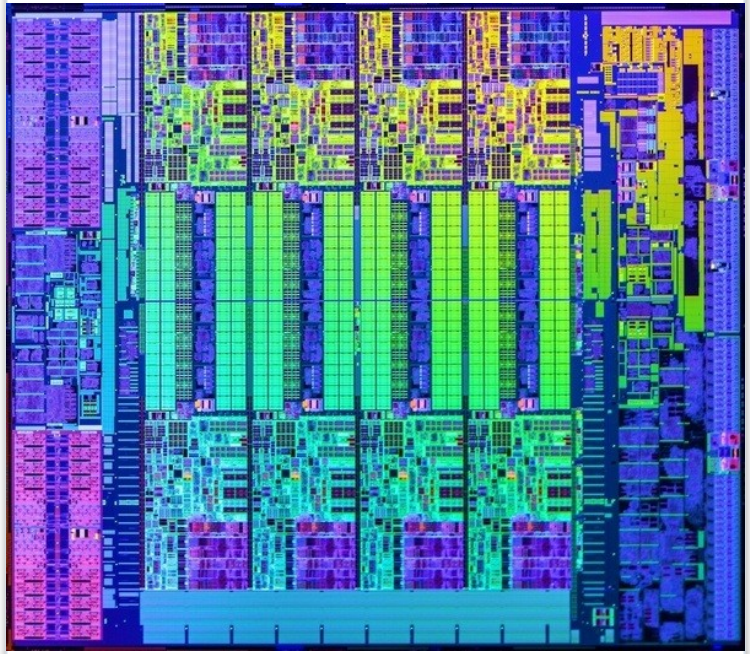
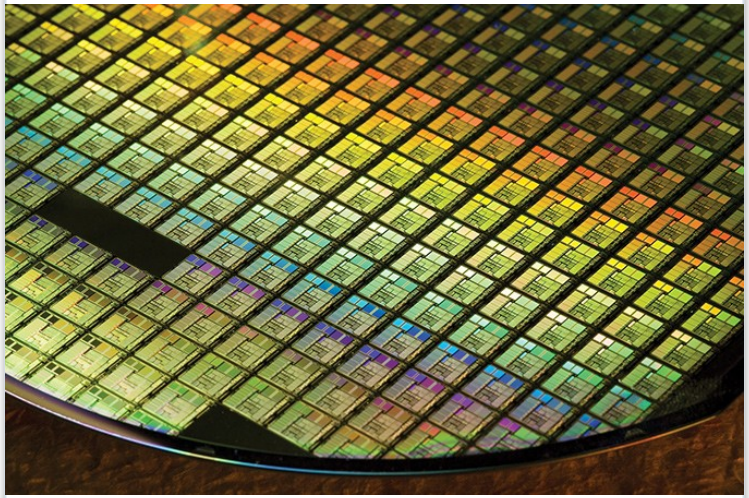
TOPIC 01

Integrated Circuit





1

Evidence 1

Integrated circuits (ICs) are semiconductor devices that made up of all the active and passive components in a silicon chip.

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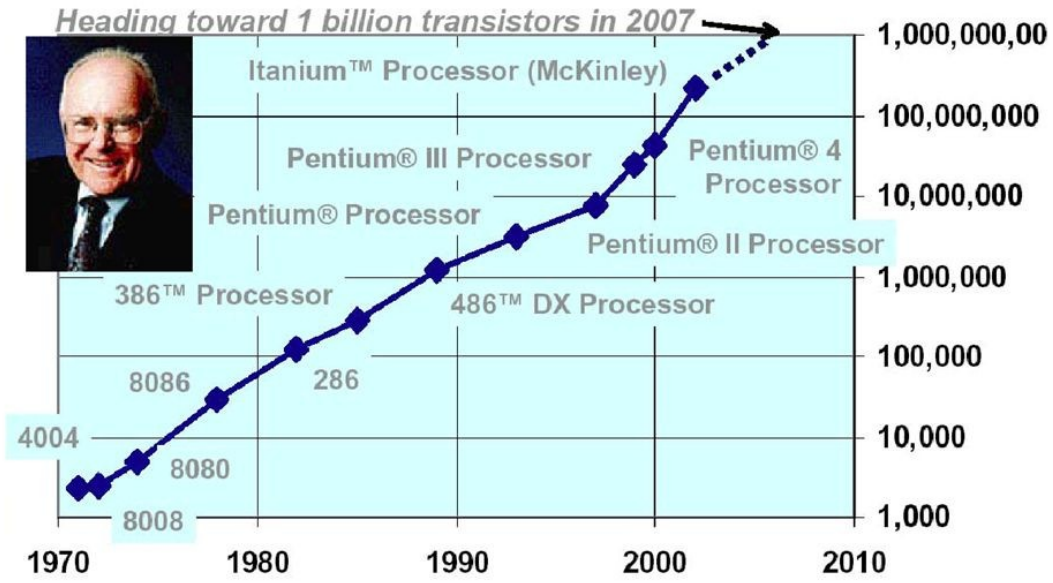
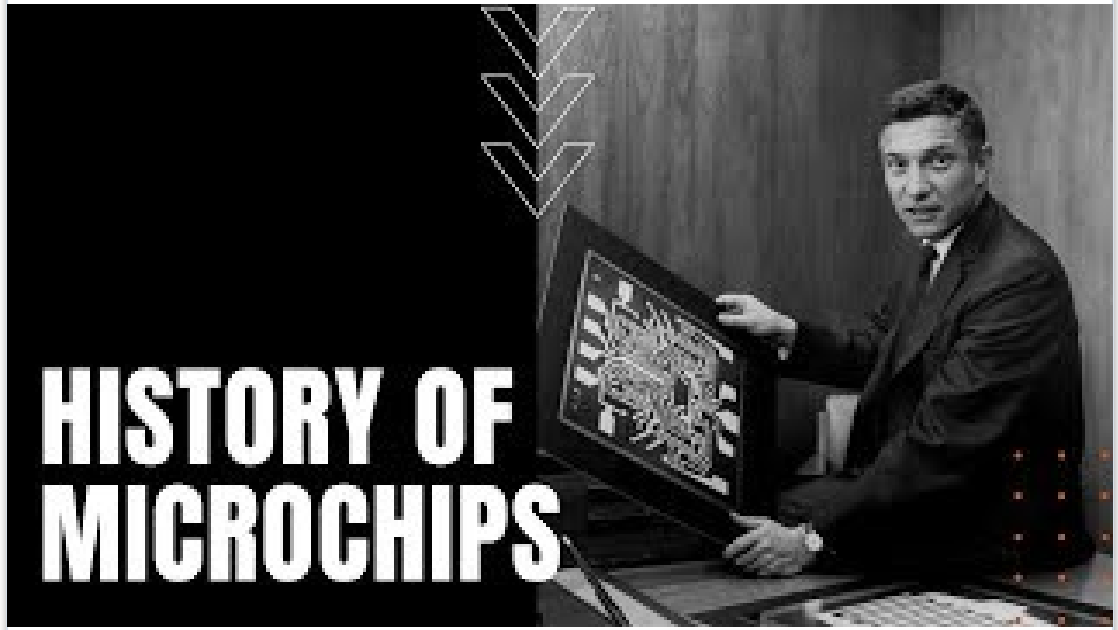
TOPIC 01

Moore’s Law





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TOPIC 01

Wafer Fabrication

Chips are built in huge factories called fabs consisting of clean rooms

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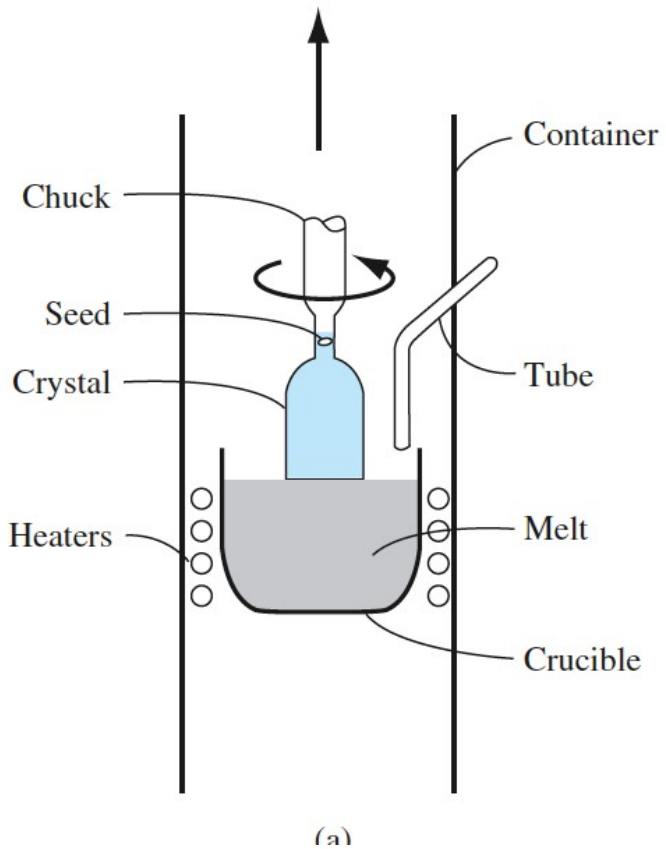
TOPIC 01

Growing Silicon Wafers



1

Evidence 1

Czochralski method used for growing single-crystal materials
Seed crystal contacts liquid phase and is slowly pulled from the melt
Solidification occurs at the solid–liquid interface
Crystal is rotated during pulling for uniform temperature
Impurity atoms (e.g., boron, phosphorus) can be added for intentional doping
Produces a silicon ingot (boule)

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TOPIC 01

Growing Silicon Wafers

1

Key idea 1

Czochralski method used for growing single-crystal materials
Seed crystal contacts liquid phase and is slowly pulled from the melt
Solidification occurs at the solid–liquid interface
Crystal is rotated during pulling for uniform temperature
Impurity atoms (e.g., boron, phosphorus) can be added for intentional doping
Produces a silicon ingot (boule)

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TOPIC 01

Thermal Oxidation

Silicon ICs succeed due to the excellent native oxide (SiO2) formed on silicon
SiO2 used as gate and field insulator in MOSFETs
Metal interconnects placed on top of field oxide
Most semiconductors lack high-quality native oxides for device fabrication
Silicon oxidizes at room temperature, forming a thin oxide (25 Å thick)
Oxidation accelerated at high temperatures for oxygen diffusion to the silicon surface
44% of the final oxide thickness comes from consumed silicon

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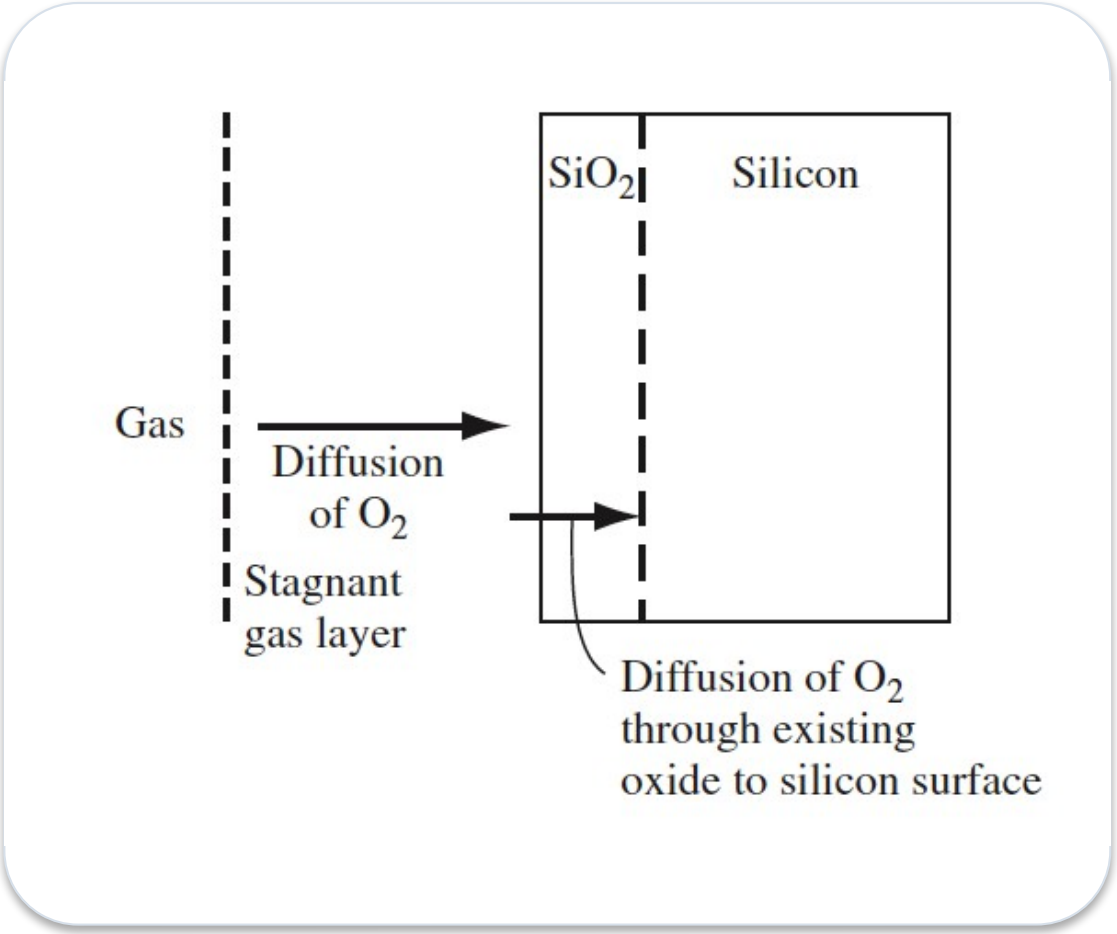
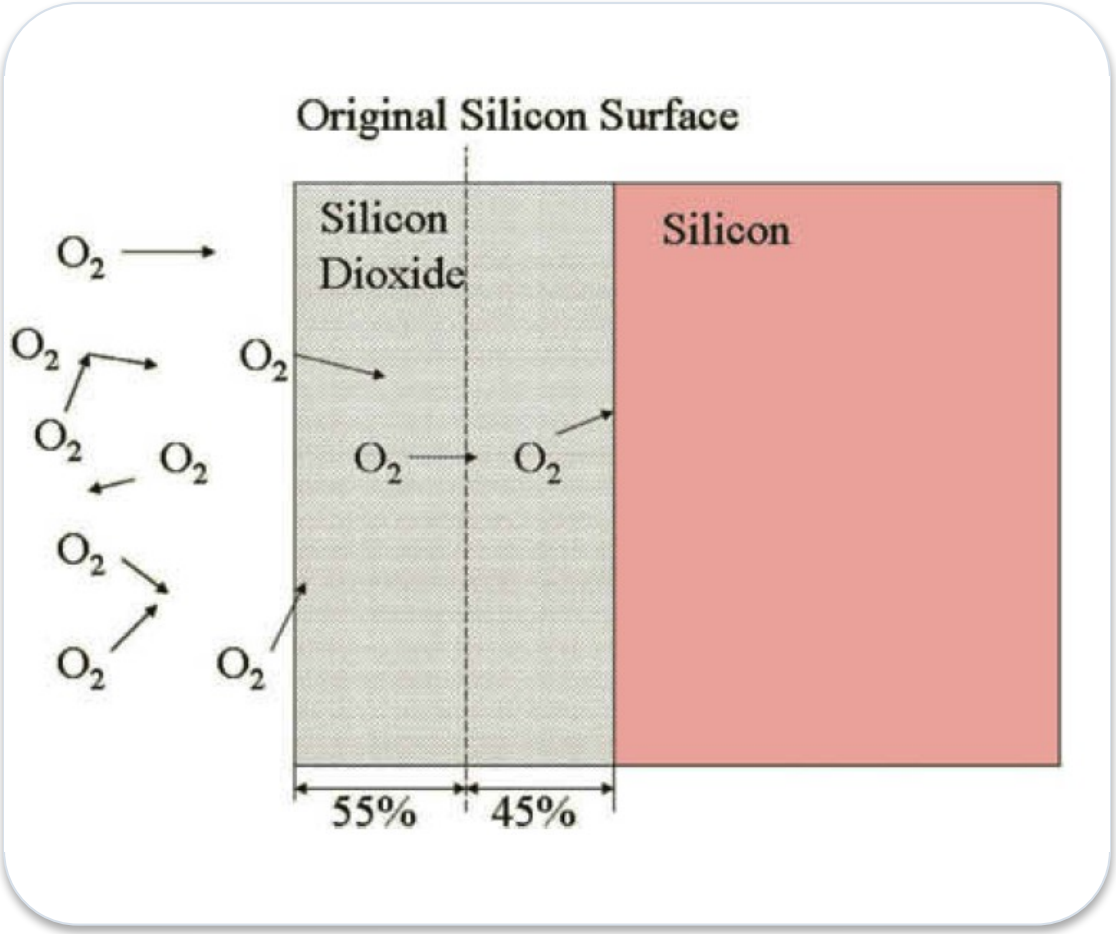
Thermal Oxidation Process





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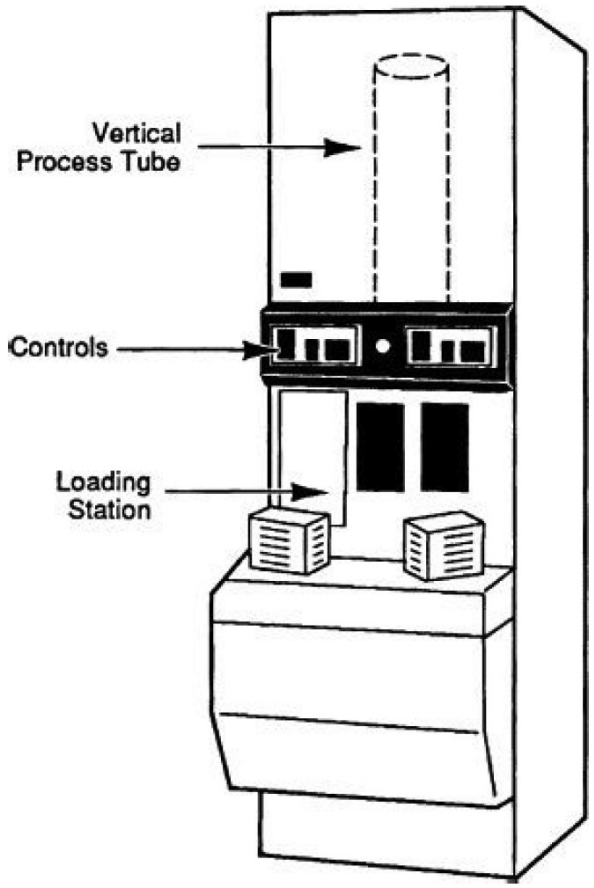
TOPIC 01

Vertical Tube Furnaces



1

Evidence 1

Horizontal tube furnaces are preferred for larger-diameter wafers but present challenges, such as maintaining laminar gas flow to ensure uniform reactions.
To address these issues, vertical tube furnaces (VTFs) have become the preferred choice for high-production processes with large wafers. In VTFs, the tube is positioned vertically, with loading from the top or bottom, while using similar materials and heating systems as horizontal furnaces.

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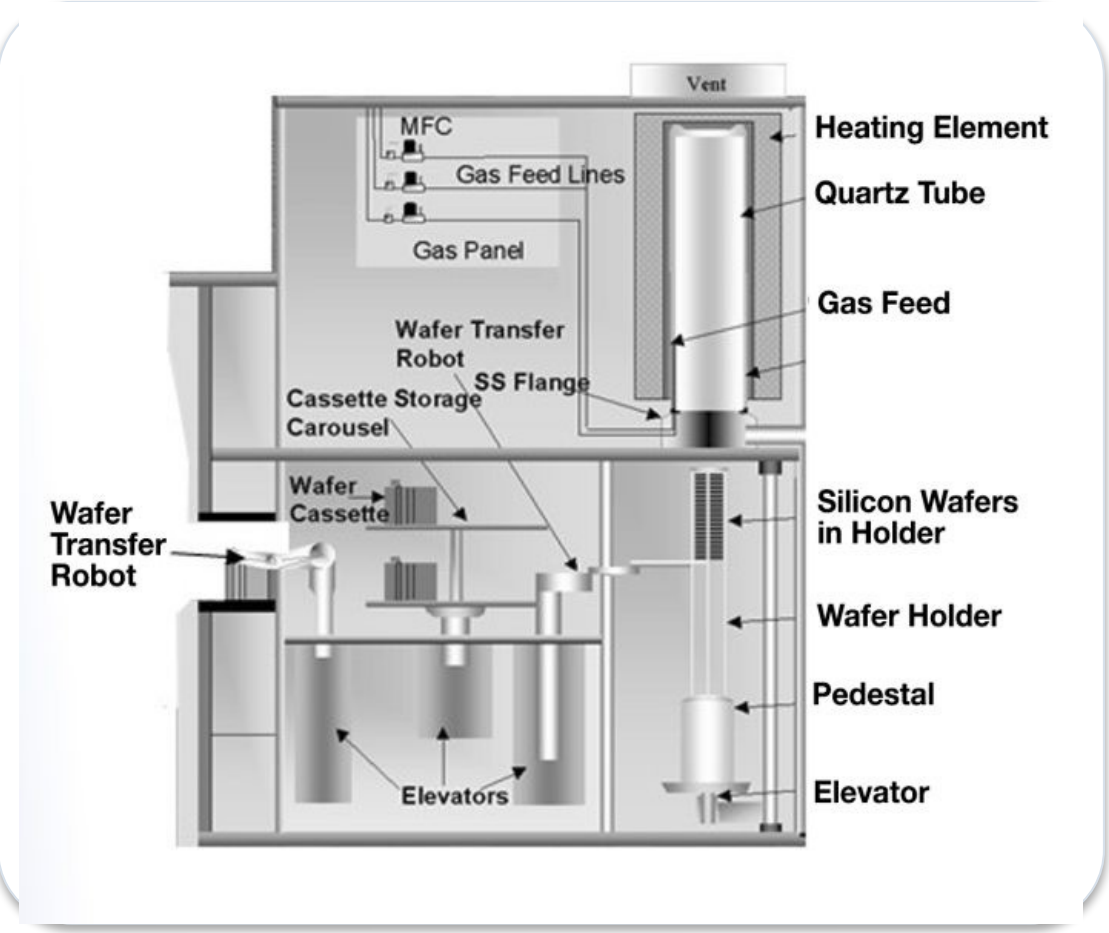
Vertical Tube Furnaces



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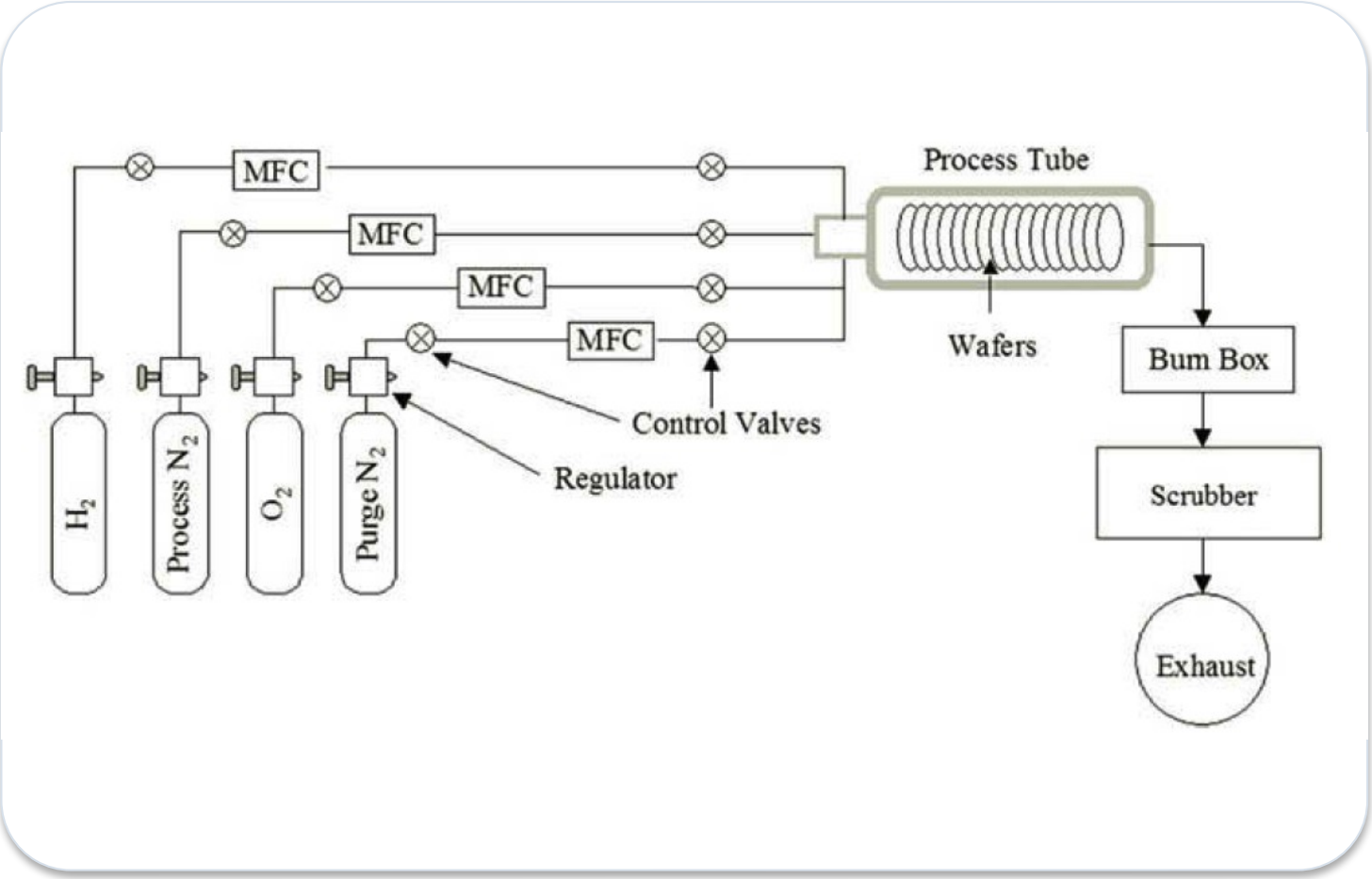
TOPIC 01

Dry Oxidation



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Evidence 1

Dry oxidation has a lower growth rate than wet oxidation, yet it has better oxide film quality. Therefore, thin oxides such as screen, pad, and especially gate oxides, use a dry oxidation process.

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TOPIC 01

Wafer Warping

1

Evidence 1

Larger-diameter wafers are prone to warping due to different expansion rates of silicon and silicon dioxide, especially at high temperatures above 1150°C. Two methods help minimize warping:
Ramping: Wafers are inserted at a lower temperature, then gradually heated to the process level and cooled before removal.
Slow loading: Wafers are loaded at 1 inch per minute to reduce warping.
For large wafers, both methods are used, though slow loading extends process time. Vertical tube furnaces mitigate this issue.

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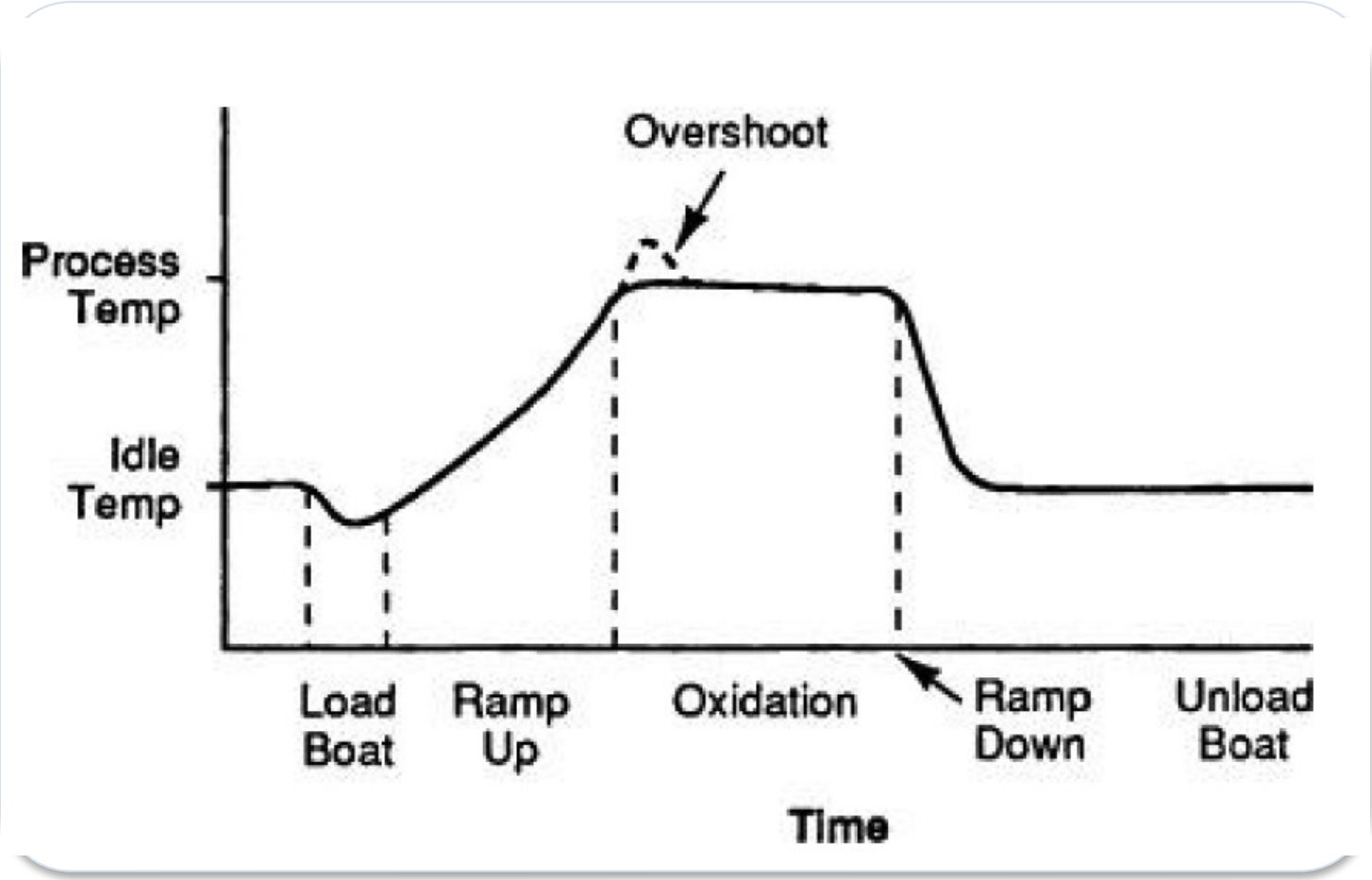
TOPIC 01

Temperature Control System



1

Evidence 1

The temperature control system uses thermocouples connected to proportional band controllers to regulate the power fed to heating coils.
The system adjusts power based on the deviation from the set-point temperature, providing more power when further from the target and less as it approaches.
This allows for rapid temperature recovery without overshoot (when the temperature exceeds the desired level due to excessive power).
Controllers are fine-tuned to maintain the desired temperature in the processing section of the tube.

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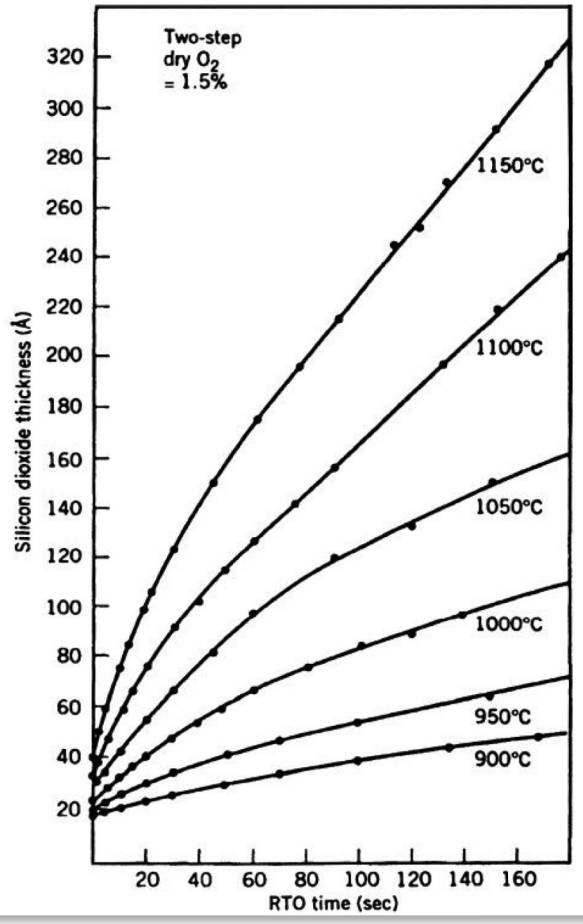
TOPIC 01

Rapid Thermal Process (RTP)



1

Evidence 1

Ion implantation requires annealing to repair crystal damage, traditionally done in tube furnaces, but this can lead to unwanted dopant spreading. Rapid Thermal Process (RTP) technology was developed to address this.

RTP uses radiation heating, typically from tungsten halogen lamps, to heat the wafer surface quickly to 800–1050°C at rates of 50–100°C per second. Unlike tube furnaces, RTP achieves high temperatures in seconds and cools rapidly.

The short heating time limits wafer-wide heating, allowing crystal damage to be repaired without causing dopant diffusion.

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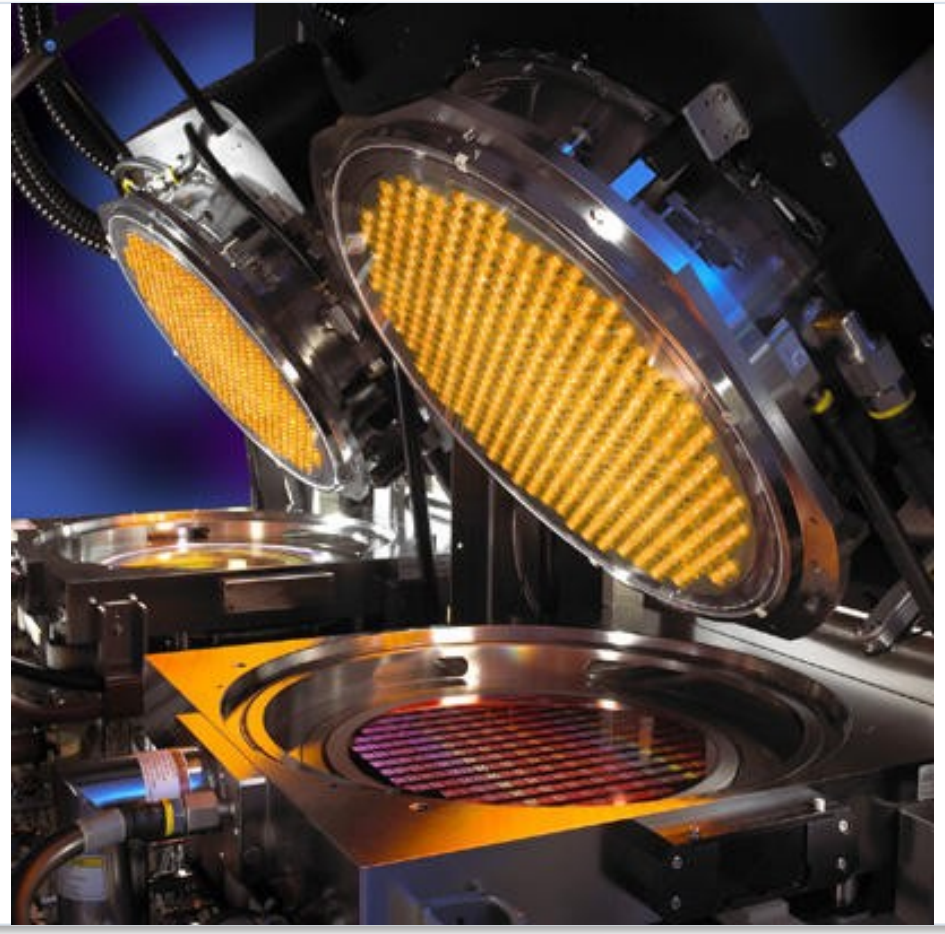
TOPIC 01

Rapid Thermal Process (RTP)



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TOPIC 01

Preoxidation Wafer Cleaning

Surface cleaning before oxidation involves mechanical scrubbing, RCA wet-cleaning, and an HF etch (HF-last process) to remove contaminants and native oxides.
Steam is used for oxides >1200 Å, while pure oxygen is used for thinner oxides (<1200 Å) for better control and density.
Thin MOS gate oxides are grown at 900°C, sometimes in a wet oxygen environment at reduced pressure to reduce process time.
Chlorine improves cleanliness and performance when incorporated into the oxidation process.
Dry nitrogen is used to terminate the oxidation by removing the oxidant.

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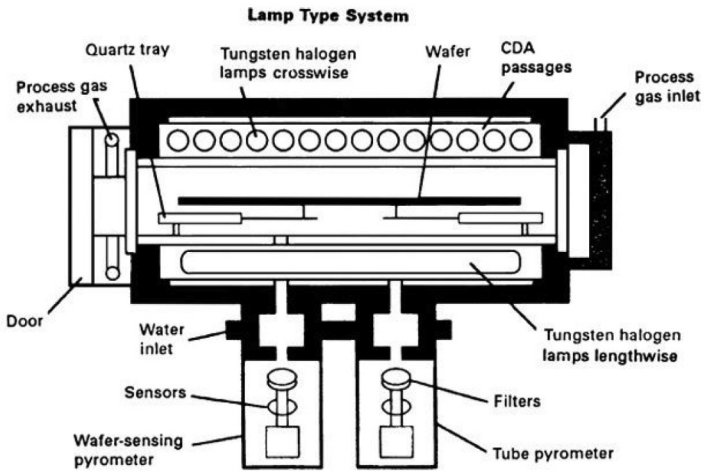
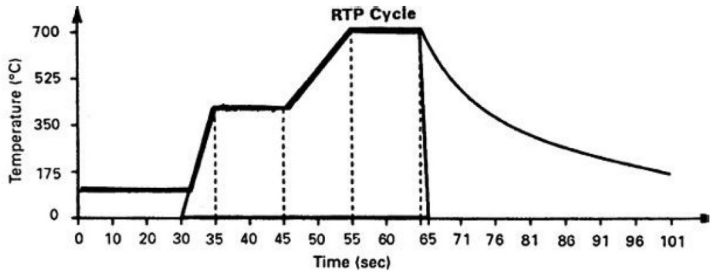
TOPIC 01

Rapid Thermal Process (RTP)





1

Evidence 1

RTP (Rapid Thermal Processing) reduces the thermal budget in wafer processing by minimizing the time wafers are exposed to high temperatures, which limits dopant diffusion and crystal dislocations.

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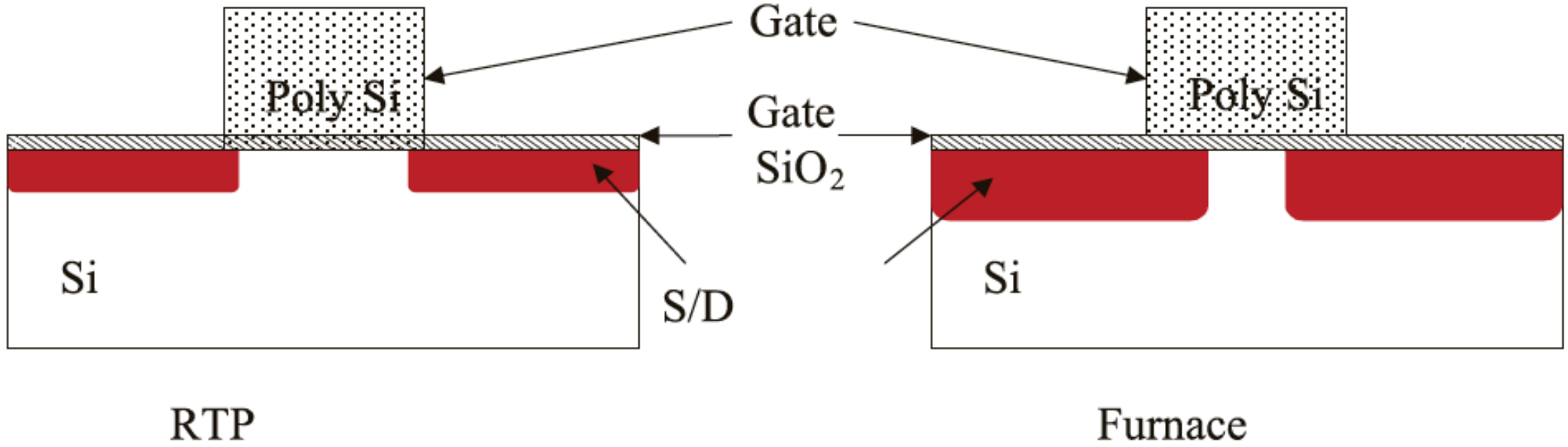
Application of RTP in SDE



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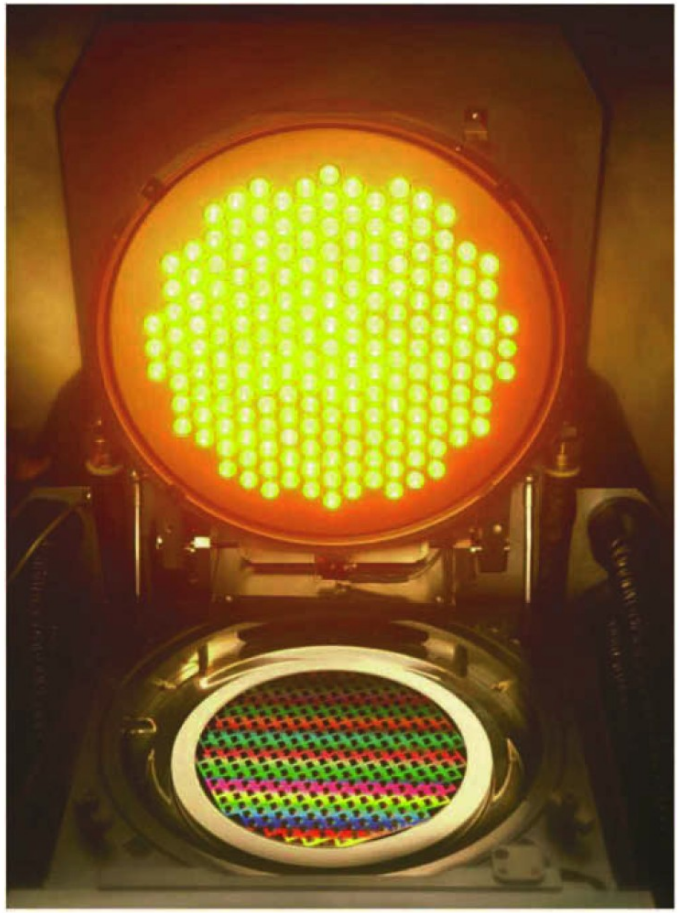
TOPIC 01

Advantages of RTP



1

Evidence 1

Reduced dopant spreading: Less time at high temperatures enables denser designs.
Fewer dislocations: Fewer heating/cooling cycles minimize crystal damage.
Single-wafer processing: Enhances uniformity, especially for larger wafers.
RTP for thin oxides: Ideal for MOS gates with shrinking feature sizes and thinner layers.
RTO systems: Offer precise control for ultra-thin oxides (as thin as 10 Å), unlike conventional furnaces, which struggle with slower temperature changes and oxygen control.

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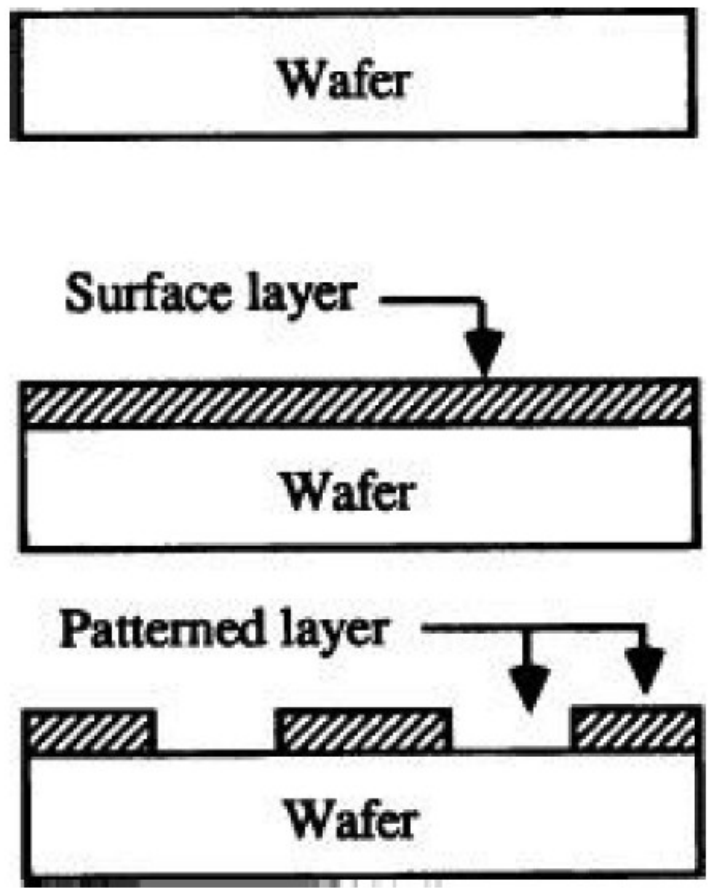
TOPIC 01

Photolithography

Photolithography: Creating circuits using photomasks and UV light.
Photomask: Contains opaque regions blocking UV.
Photoresist: Polymer on semiconductor surface, changes with UV exposure.
Developer: Removes unwanted photoresist, creating patterns.



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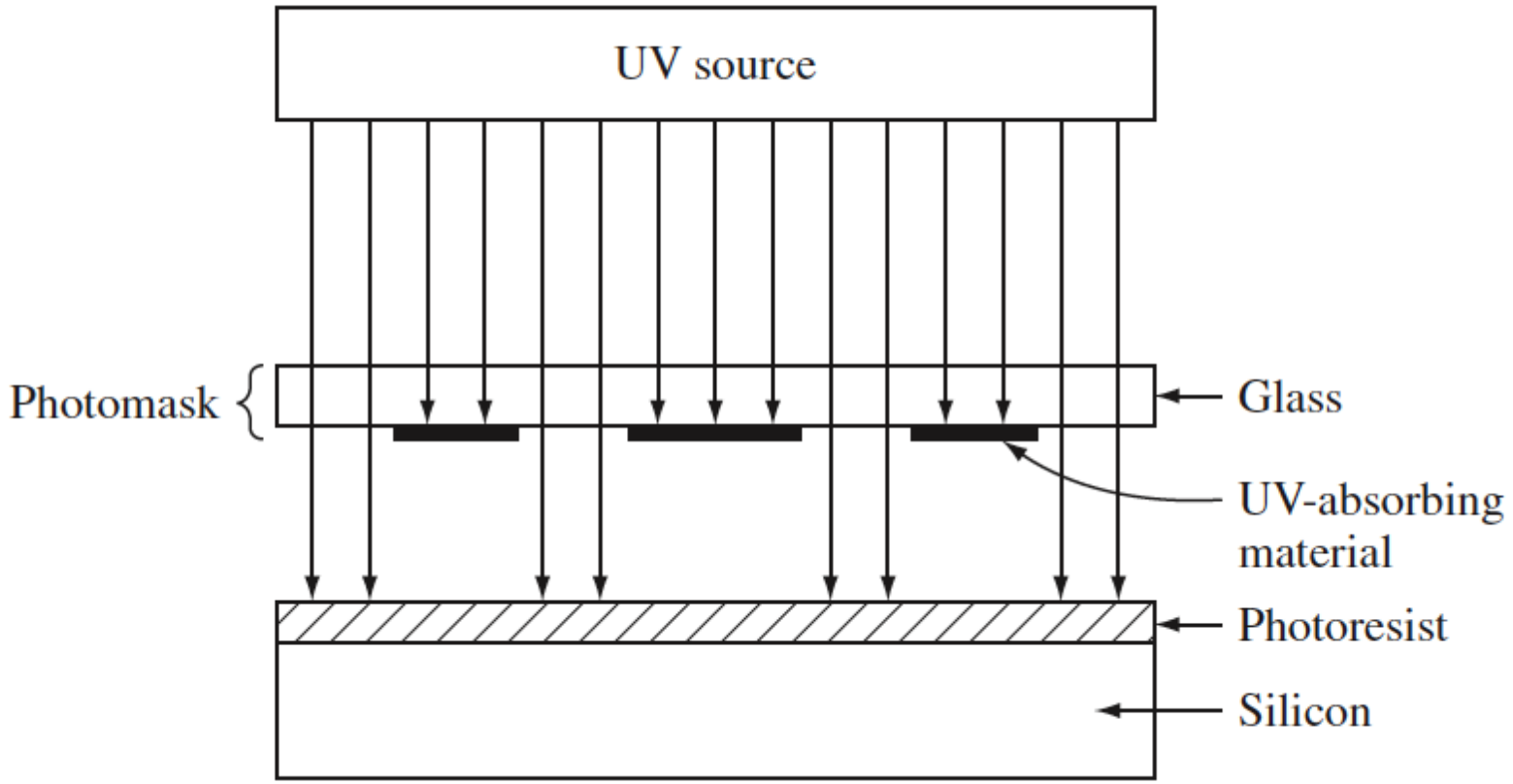
Photolithography



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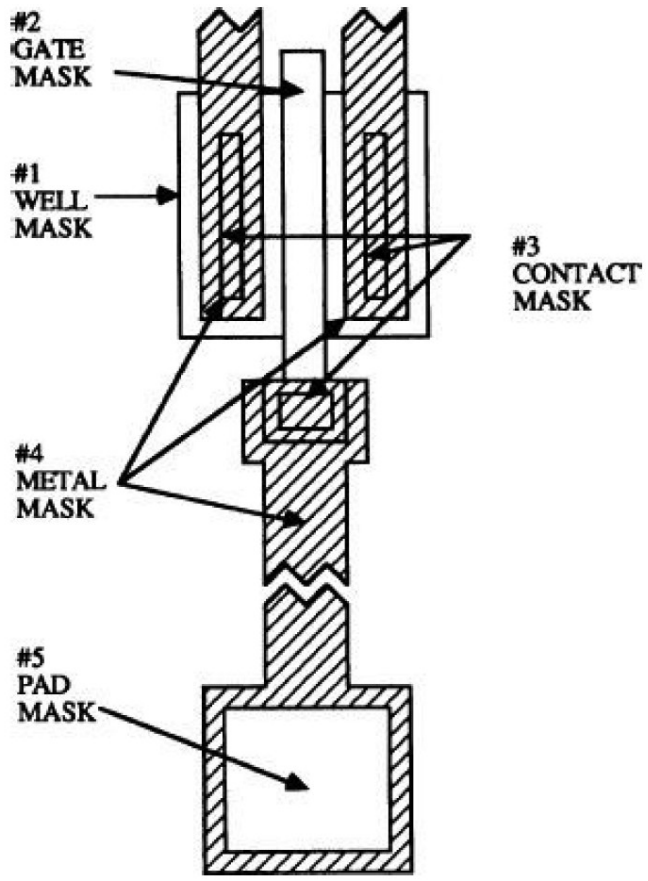
TOPIC 01

Photolithography



1

Evidence 1

Photolithography Outcomes: Holes or islands created on wafer surface.
Alignment (Registration): Precise placement of patterns, crucial for circuit function.
Multiple Steps: IC fabrication can require 40+ masking steps.
Defect Control: Precise dimensions and minimal defects are essential.
Masking as Main Defect Source: Variations in each masking step contribute to defects.

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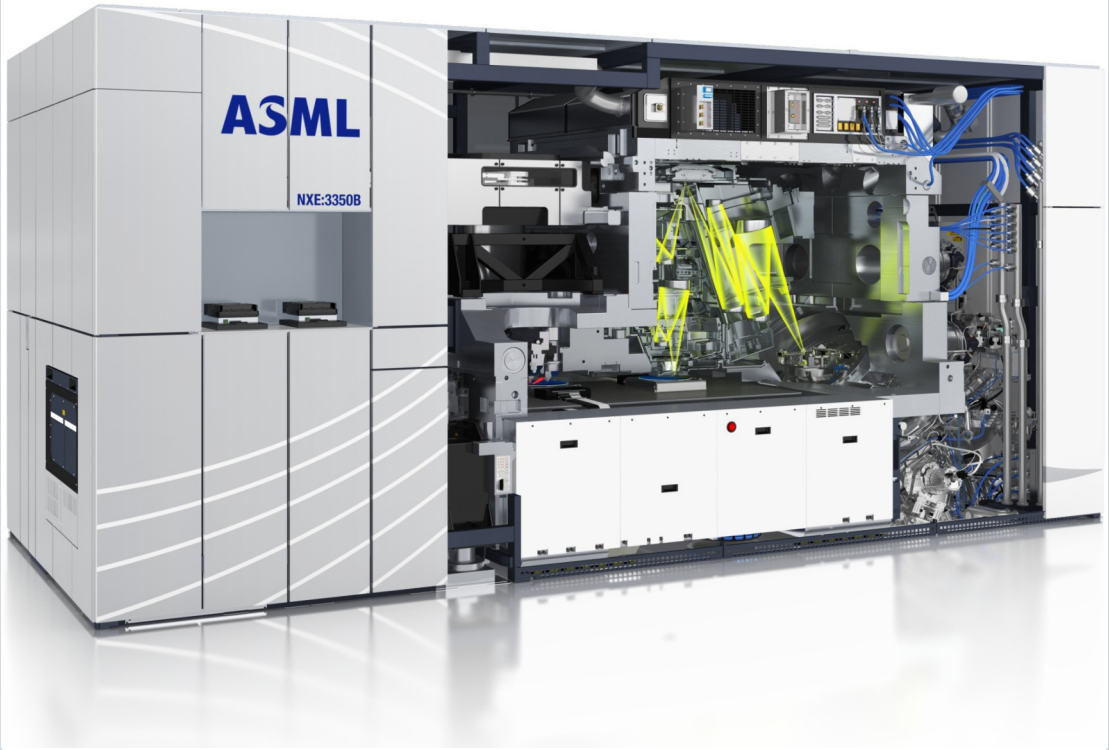
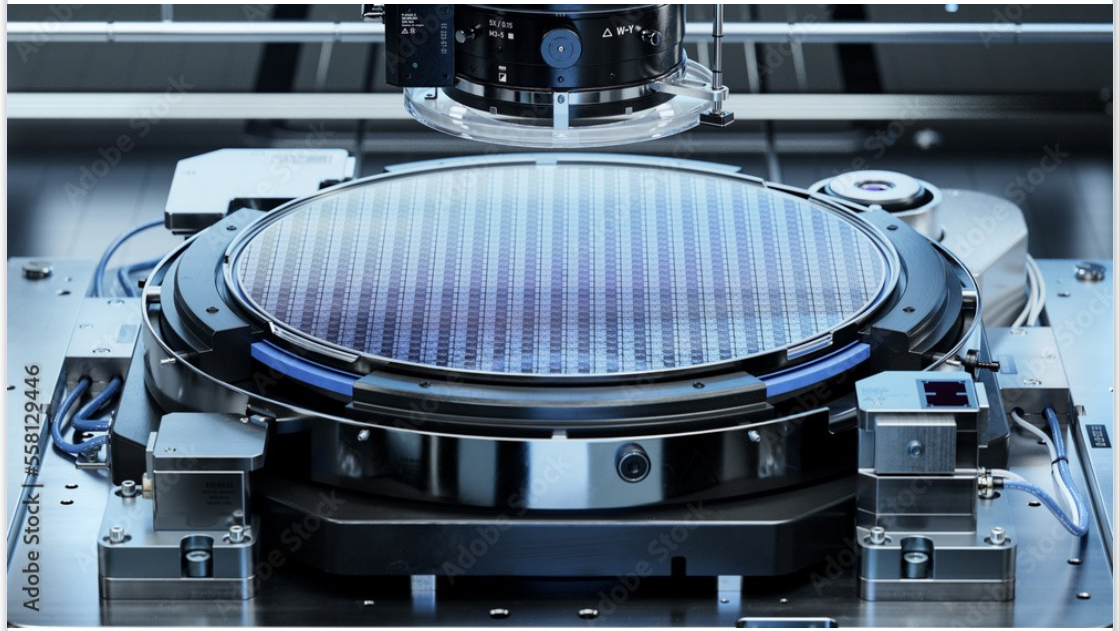
TOPIC 01

Photolithography





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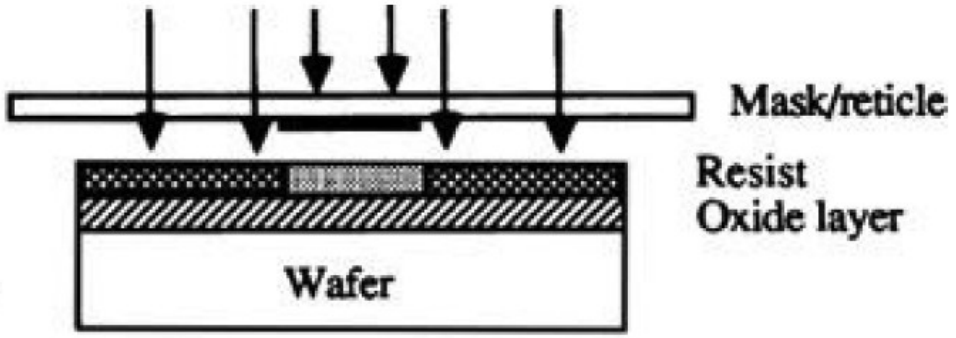
Pattern Transfer Steps

PROCESS STEP	PURPOSE
Alignment and Exposure	Precise alignment of mask/reticle to wafer and exposure of photoresist. Negative resist is polymerized.
Development	Removal of unpolymerized resist.



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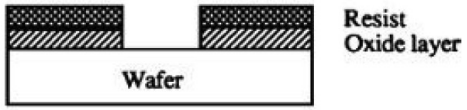
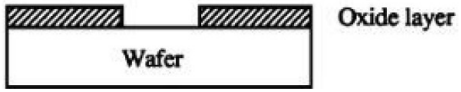
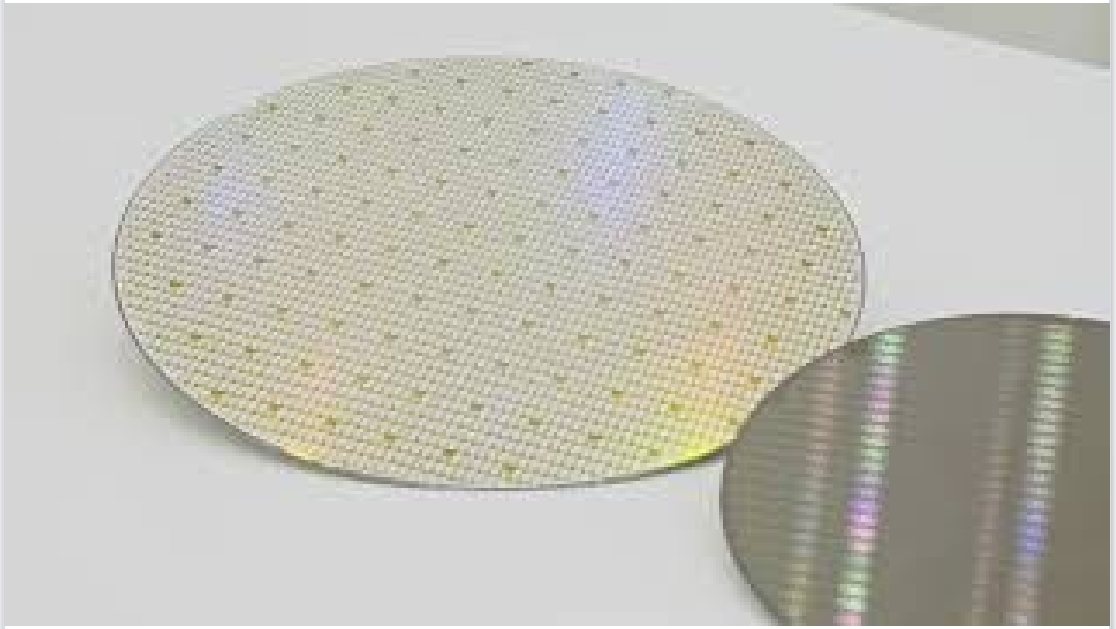
Pattern Transfer Steps

PROCESS STEP	PURPOSE	
Etch	Top layer of wafer is removed through opening in resist layer.	
Photoresist removal (strip)	Remove photoresist layer from wafer.	



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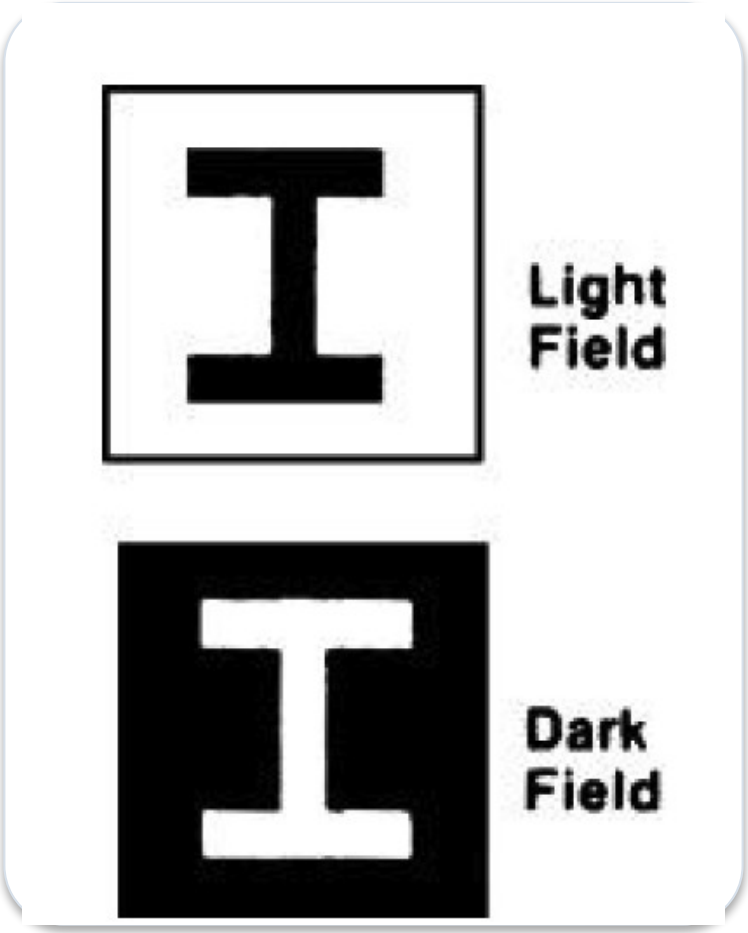
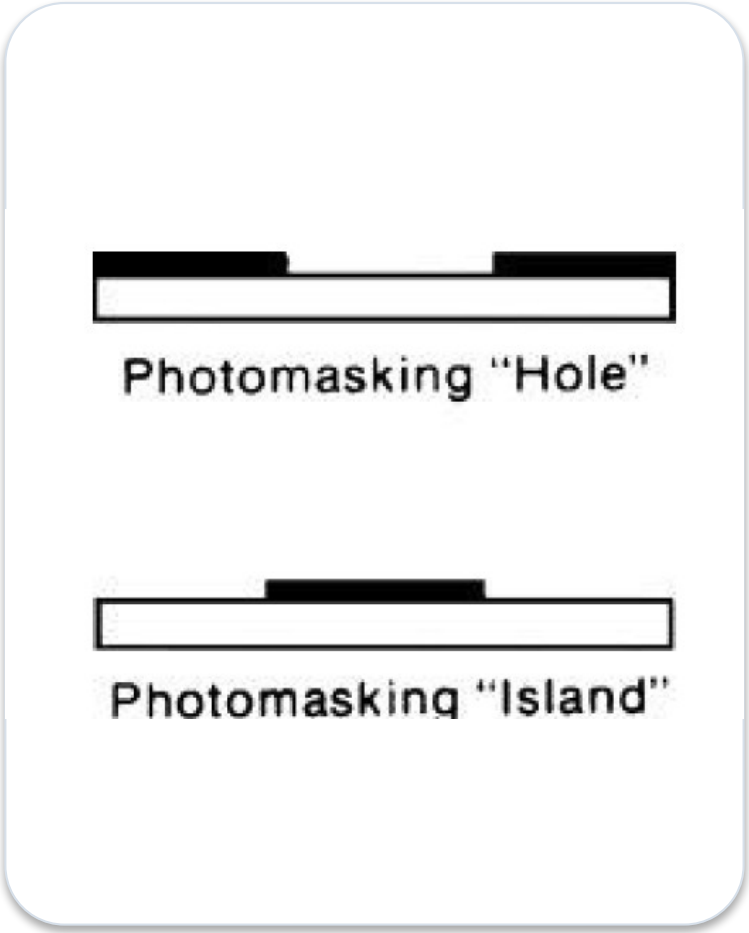
TOPIC 01

Mask Types





1

Evidence 1

Mask Types: Clear-field masks have opaque patterns, creating holes. Dark-field masks have transparent patterns, creating islands. Pattern & Outcome: The mask's pattern determines whether material is removed (hole) or left behind (island) on the wafer.

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TOPIC 01

Photoresist

Component	Function
Polymer	Polymer structure changes from soluble to polymerized (or vice versa) when exposed by the exposure source in an aligner.
Solvent	Thins resist to allow application of thin layers by spinning.
Sensitizers	Controls and/or modifies chemical reaction of resist during exposure.
Additives	Various added chemical to achieve process results, such as dyes.

1

Evidence 1

Photoresists are manufactured for both general and specific applications. They are tuned to respond to specific wavelengths of light and different exposing sources.
They are given specific thermal flow characteristics and formulated to adhere to specific surfaces.

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TOPIC 01

Photoresist Types

1

Key idea 1

Photoresist Types: Negative-acting resists become less soluble after exposure, positive-acting resists become more soluble.
Photosolubilization: Light changes the chemical structure of positive resists to make them more soluble in developer.
Positive Resist Outcome: Using a light-field mask with a positive resist creates an island of material on the wafer.

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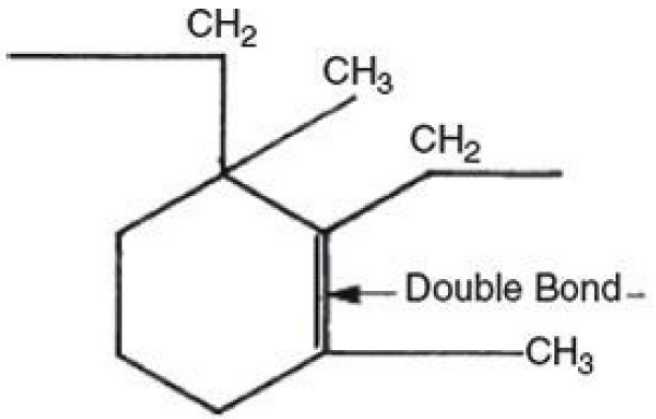
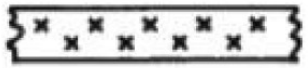
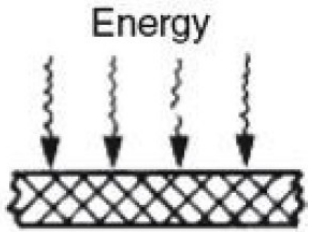
TOPIC 01

Negative Photoresist Chemistry





Unpolymerized



Polymerized

1

Evidence 1

Exposure causes polymerization, making the resist more resistant to etching.
Yellow Light Protection: Used in negative resist processing to prevent accidental exposure.
.

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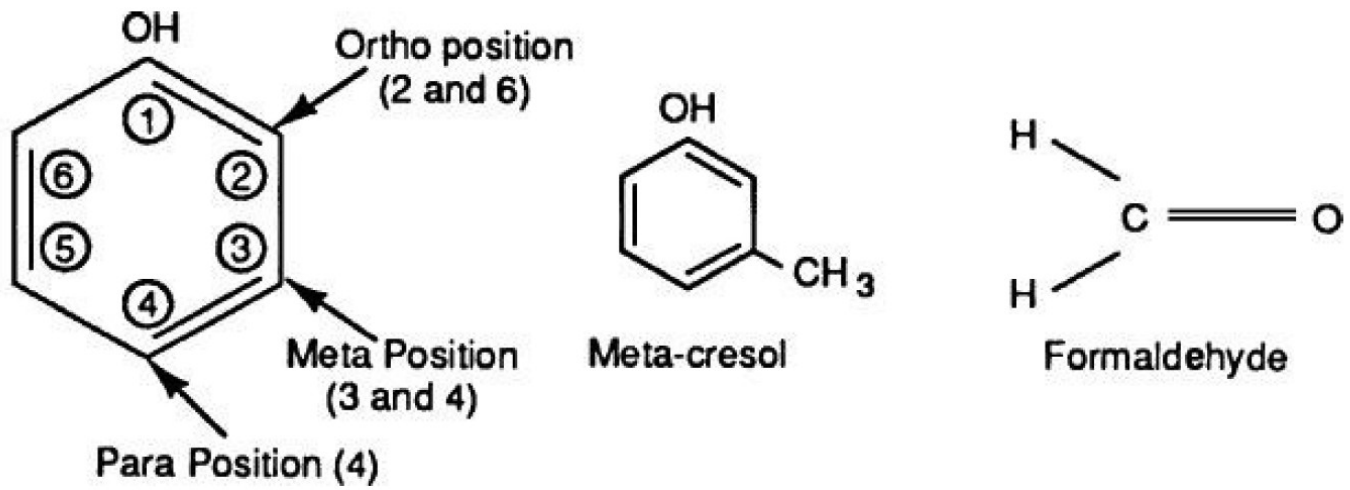
TOPIC 01

Positive Photoresist Chemistry



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Evidence 1

Phenol-formaldehyde novolak resin is the primary polymer used in positive photoresists.
Exposure to light makes the polymer more soluble, allowing developer to remove the exposed areas
.

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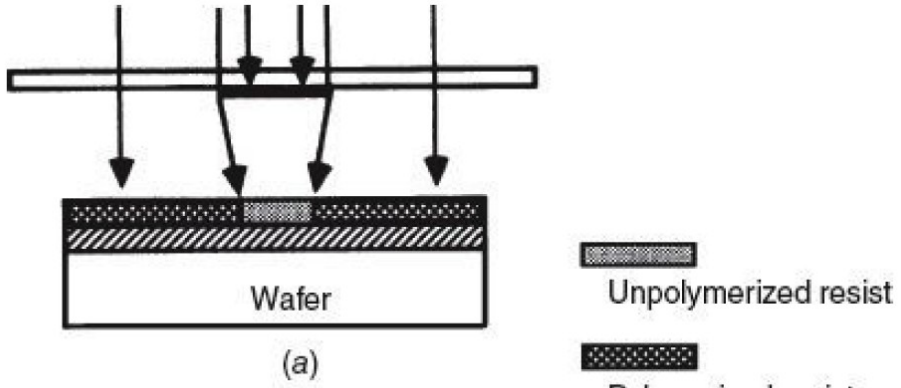
TOPIC 01

Comparison of Positive and Negative Resists



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Evidence 1

Positive Resist Rise: Became dominant in the 1980s due to superior resolution for VLSI circuits.
Mask Polarity Shift: Switching to positive resist required changing from light-field to dark-field masks.
Dimensional Differences: Positive resist and dark-field masks result in wider image dimensions due to diffraction.
Process Transition: Shifting from negative to positive resist requires a complete process redesign.

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Comparison of negative and positive resists

Parameter	Negative	Positive
Aspect Ratio (Resolution)		Higher
Adhesion	Better	
Exposure Speed	Faster	
Pinhole Count		Lower
Step Coverage		Better
Cost		Higher
Developers	Organic Solvents	Aqueous
Strippers		
Oxide Steps	Acid	Acid
Metal Steps	Chlorinated Solvent Compounds	Simple Solvents

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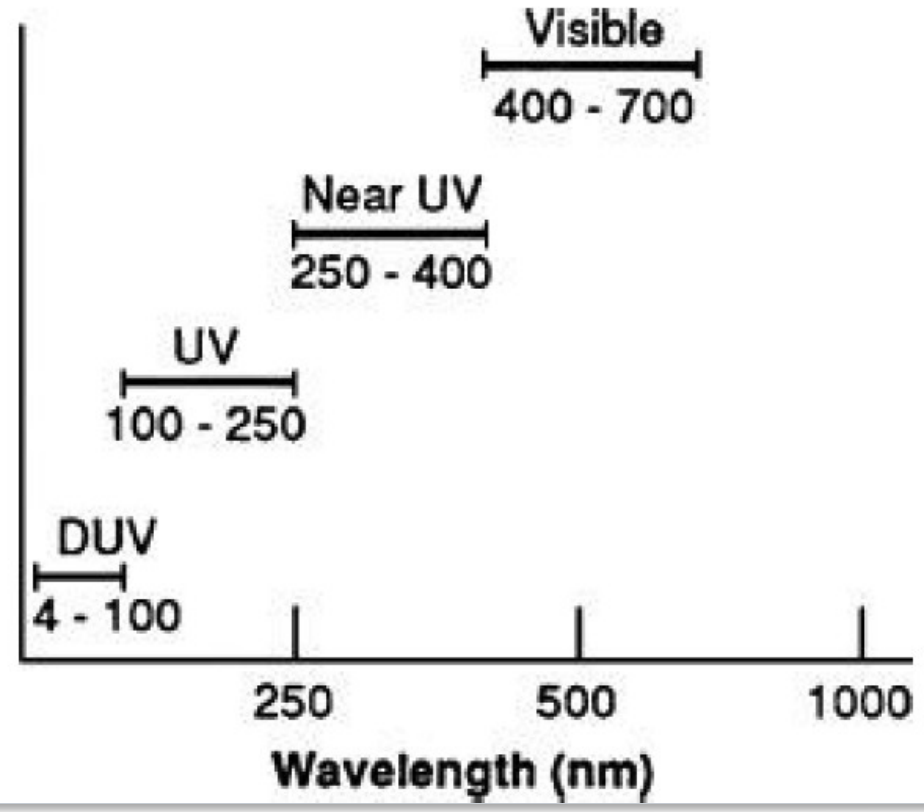
TOPIC 01

Photoresist Sensitivity



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Evidence 1

Determines how quickly the resist reacts to exposure.
Most resists are sensitive to UV and DUV light.

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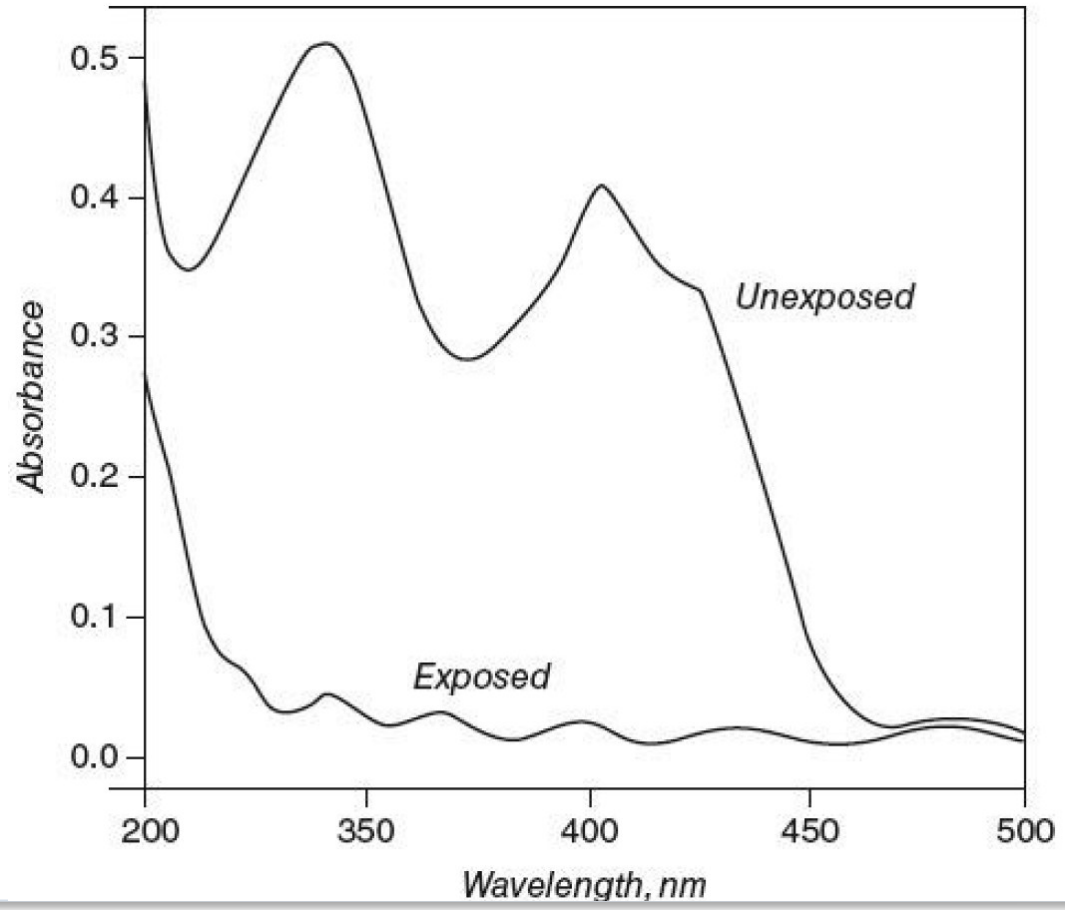
TOPIC 01

Photoresist Sensitivity Measurement



1

Evidence 1

Measured in mJ/cm², indicating the energy required to initiate the resist reaction.
The specific wavelengths to which a resist is sensitive.
Regions in the spectral response with higher energy levels correspond to specific wavelengths.

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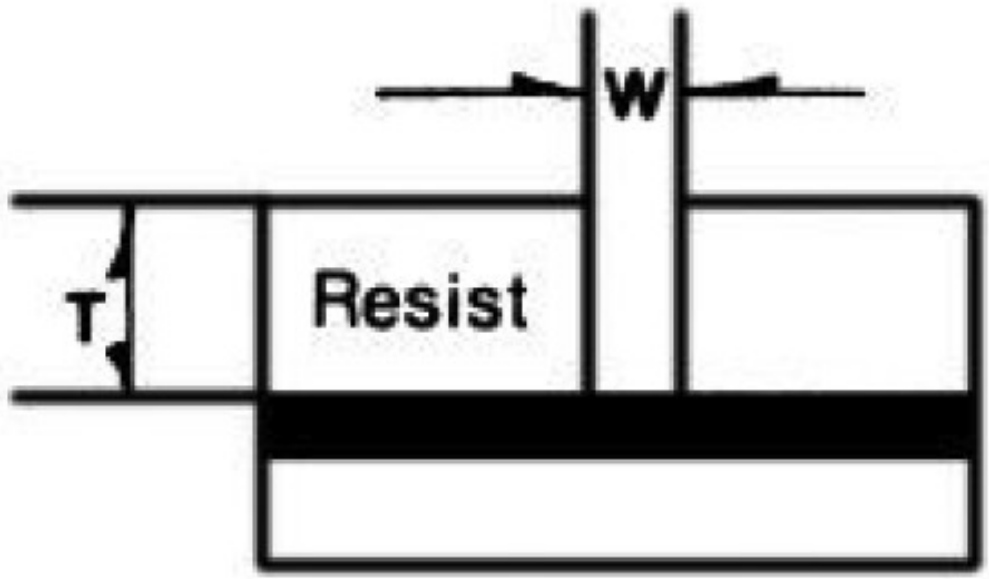
TOPIC 01

Resolution Capability



$$\text{Aspect Ratio} = \frac{W}{T}$$

1

Evidence 1

Resolution Capability: The smallest feature a resist can produce.
Critical Dimension (CD): The target size for a device or circuit feature.
Aspect Ratio: Ratio of resist thickness to image opening, a measure of resist capability.
Pattern Density & Shape: Influence resist design; specialized resists exist for high-density patterns and small contact holes.

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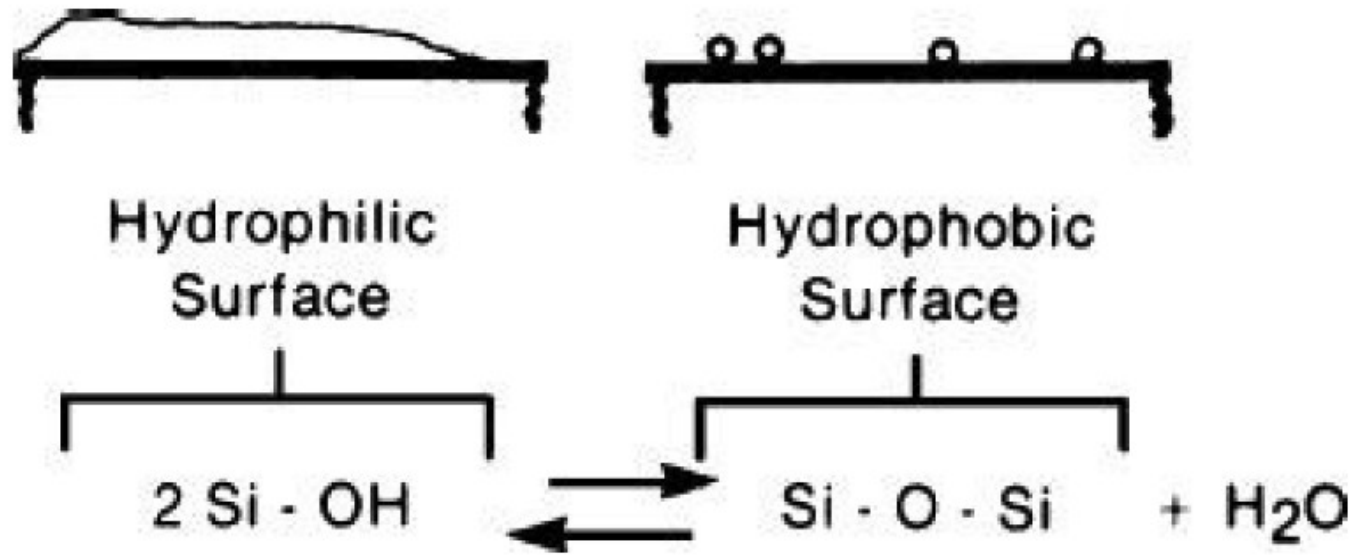
TOPIC 01

Dehydration Baking



1

Evidence 1

Hydrophobic Surface: Dry surface, promotes good photoresist adhesion.
Hydrophilic Surface: Wet surface, reduces resist adhesion.
Dehydration Bake: Heating to remove surface moisture, improving adhesion.
Low-Temperature Bake: Most common due to ease of implementation and compatibility with spin process.

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TOPIC 01

Soft Bake

Soft Bake: Heating to evaporate solvents in the resist, improving image formation and adhesion.
Soft Bake Goals: Correct image definition and resist adhesion during etching.
Soft Bake Parameters: Time and temperature are critical, determined by the resist manufacturer.
Heat Transfer Methods: Conduction (hot plates), convection (ovens), and radiation (RTP systems).

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Photoresist removal

Parameter	Negative	Positive
Aspect Ratio (Resolution)		Higher
Adhesion	Better	
Exposure Speed	Faster	
Pinhole Count		Lower
Step Coverage		Better
Cost		Higher
Developers	Organic Solvents	Aqueous
Strippers		
Oxide Steps	Acid	Acid
Metal Steps	Chlorinated Solvent Compounds	Simple Solvents

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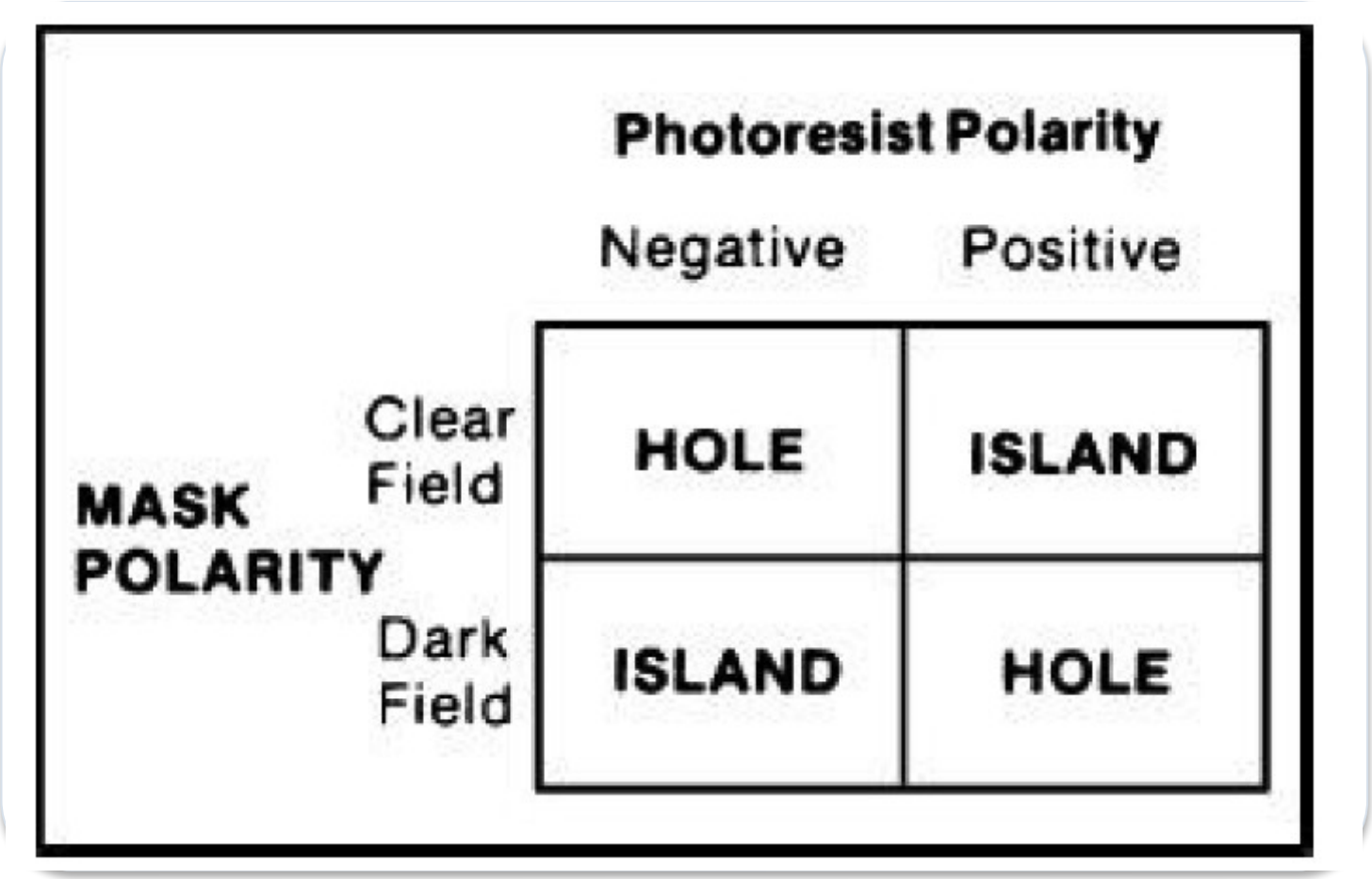
TOPIC 01

Mask & Resist Combinations



1

Evidence 1

Mask & Resist Combinations: Different combinations of clear/dark-field masks and positive/negative resists produce varying outcomes on the wafer.
Choice Factors: The choice depends on dimensional control and defect protection required for proper circuit function

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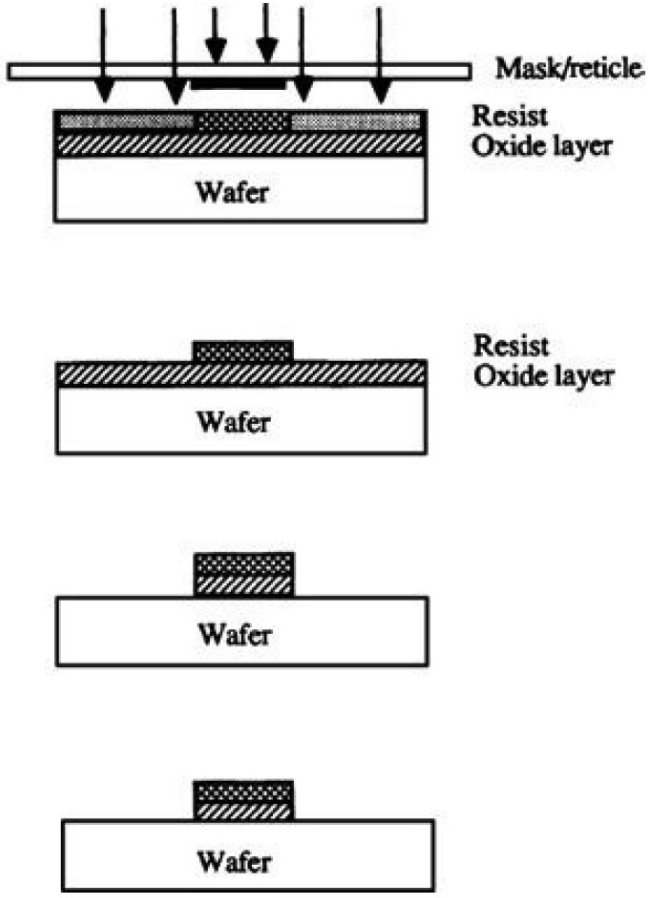
TOPIC 01

Image transfer from a light-field mask with a positive photoresist to create an island.

PROCESS STEP	PURPOSE
Alignment and Exposure	Precise alignment of mask/reticle to wafer and exposure of photoresist Positive resist is photo-solubilized
Development	Removal of unpolymerized resist
Etch	Removal of surface layer
Photo resist removal	Removal of photo-resist



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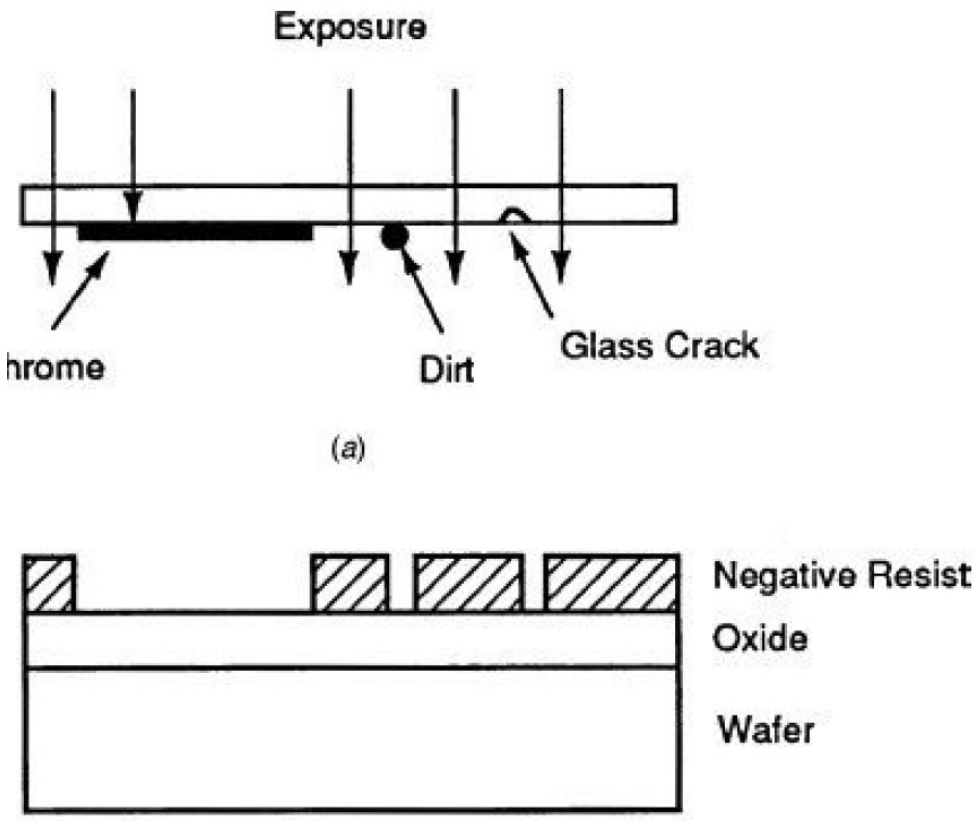
Clear-Field Mask Issue



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Evidence 1

Prone to glass damage and dirt particles creating unwanted holes.

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TOPIC 01

Pinhole Issue

Pinholes: Small voids in the resist layer that allow etching to reach the wafer surface.
Causes: Particulate contamination, spin process, and resist structure.
Thickness Tradeoff: Thinner resists have more pinholes, thicker resists are harder to resolve small features.
Positive Resist Advantage: Higher aspect ratio allows for thicker films with fewer pinholes.

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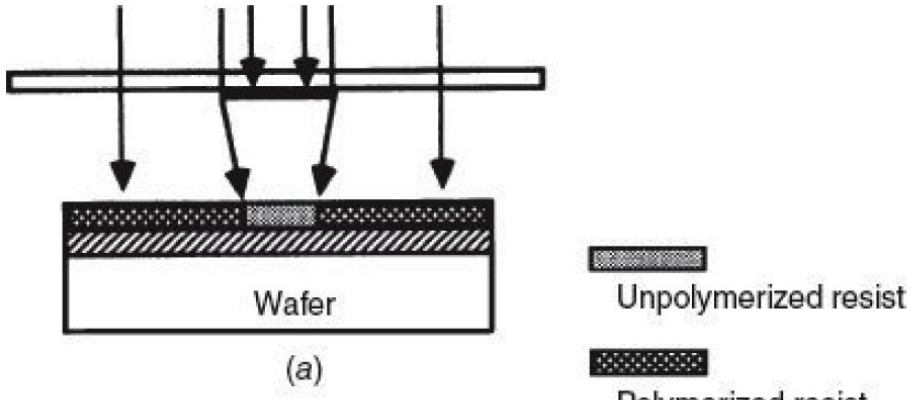
Pinhole Protection



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Evidence 1

Dark-field masks with positive resist offer better pinhole protection due to the chrome layer covering most of the surface.
Positive resist is preferred due to better pinhole control.

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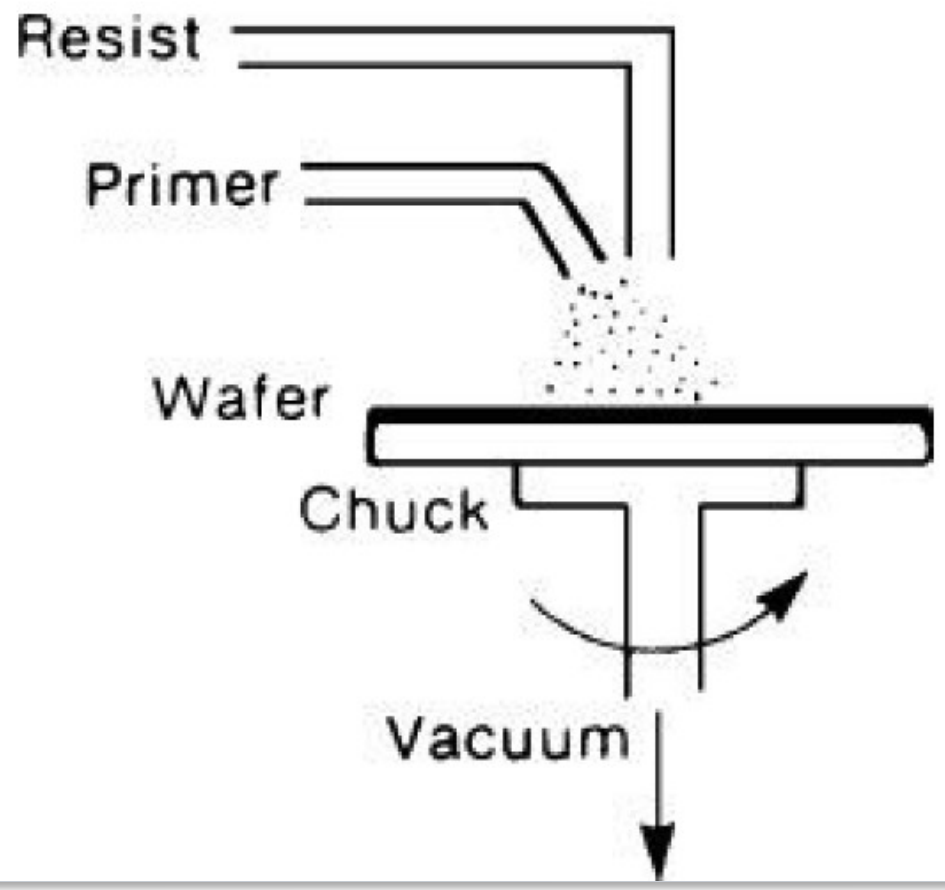
TOPIC 01

Spin Priming



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Evidence 1

Primer Application: Liquid primers, like HMDS, are applied to wafers during the spin process.
Spin Priming Advantages: Takes place in-line with resist spinning, streamlining the process

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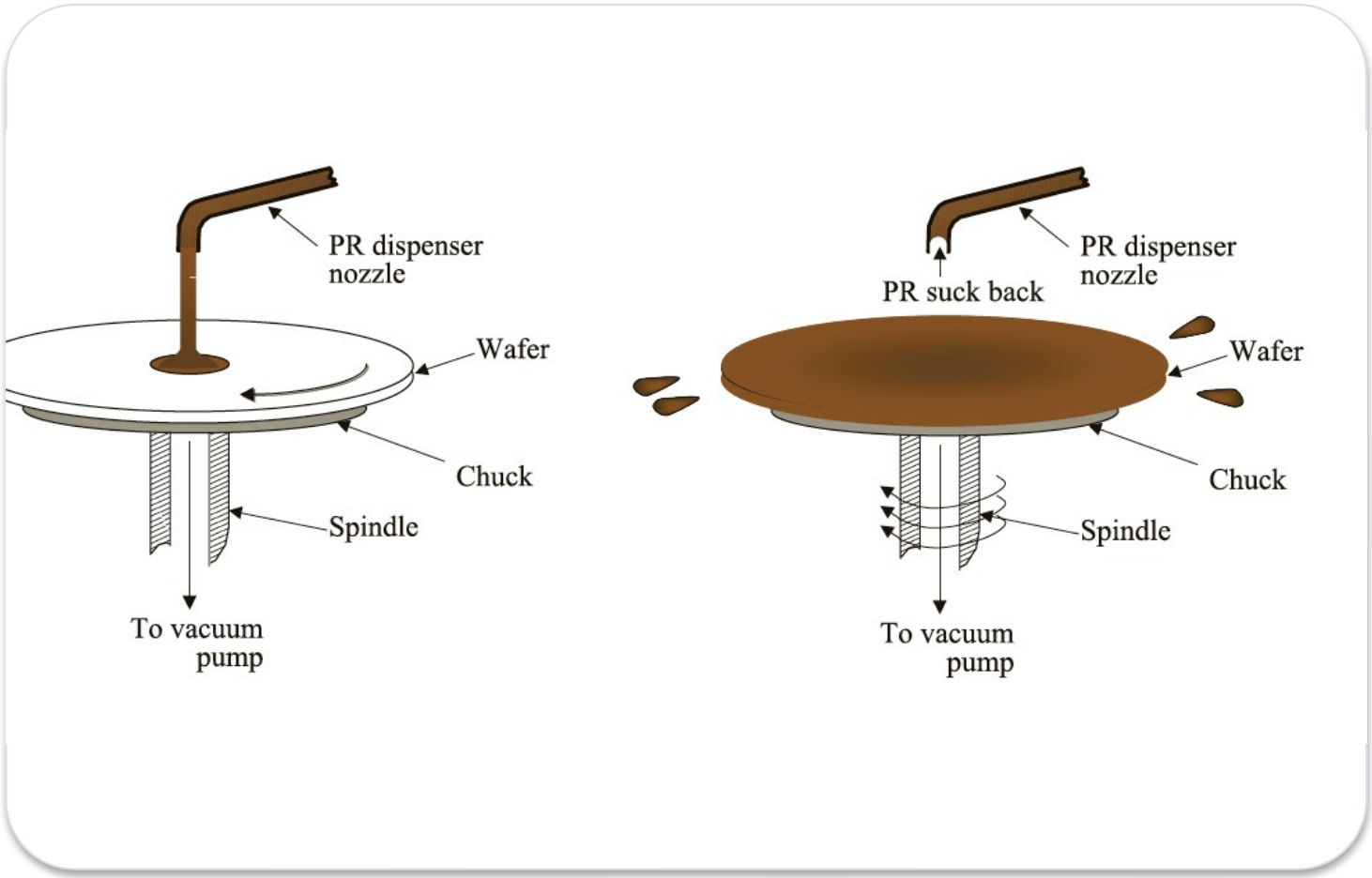
TOPIC 01

Spin Process



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Evidence 1

Resist Application: Resist is deposited onto the wafer in a puddle, allowing it to spread evenly. Resist Volume Control: Too little resist results in incomplete coverage, too much causes buildup or backside contamination.

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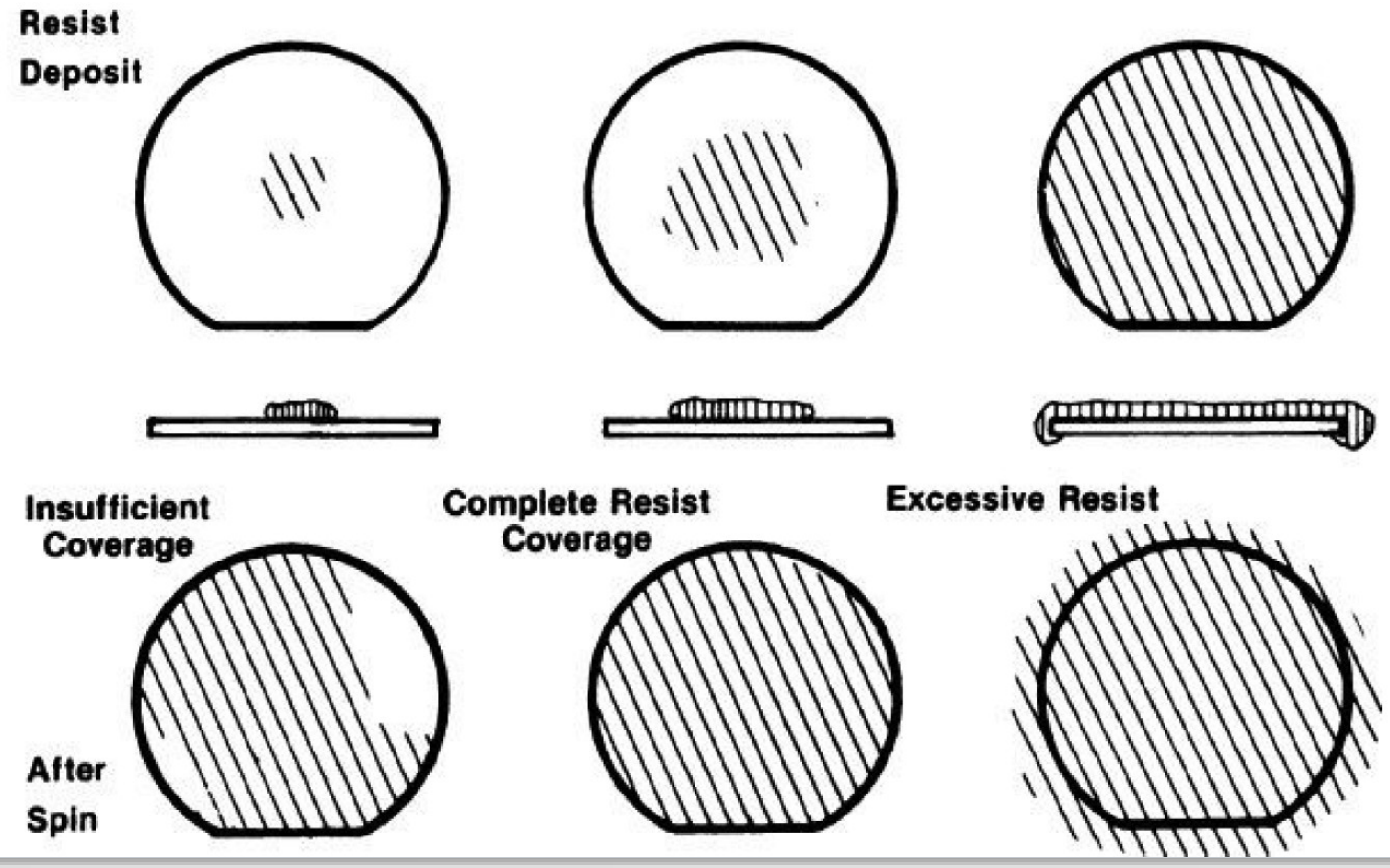
Resist Coverage Issue



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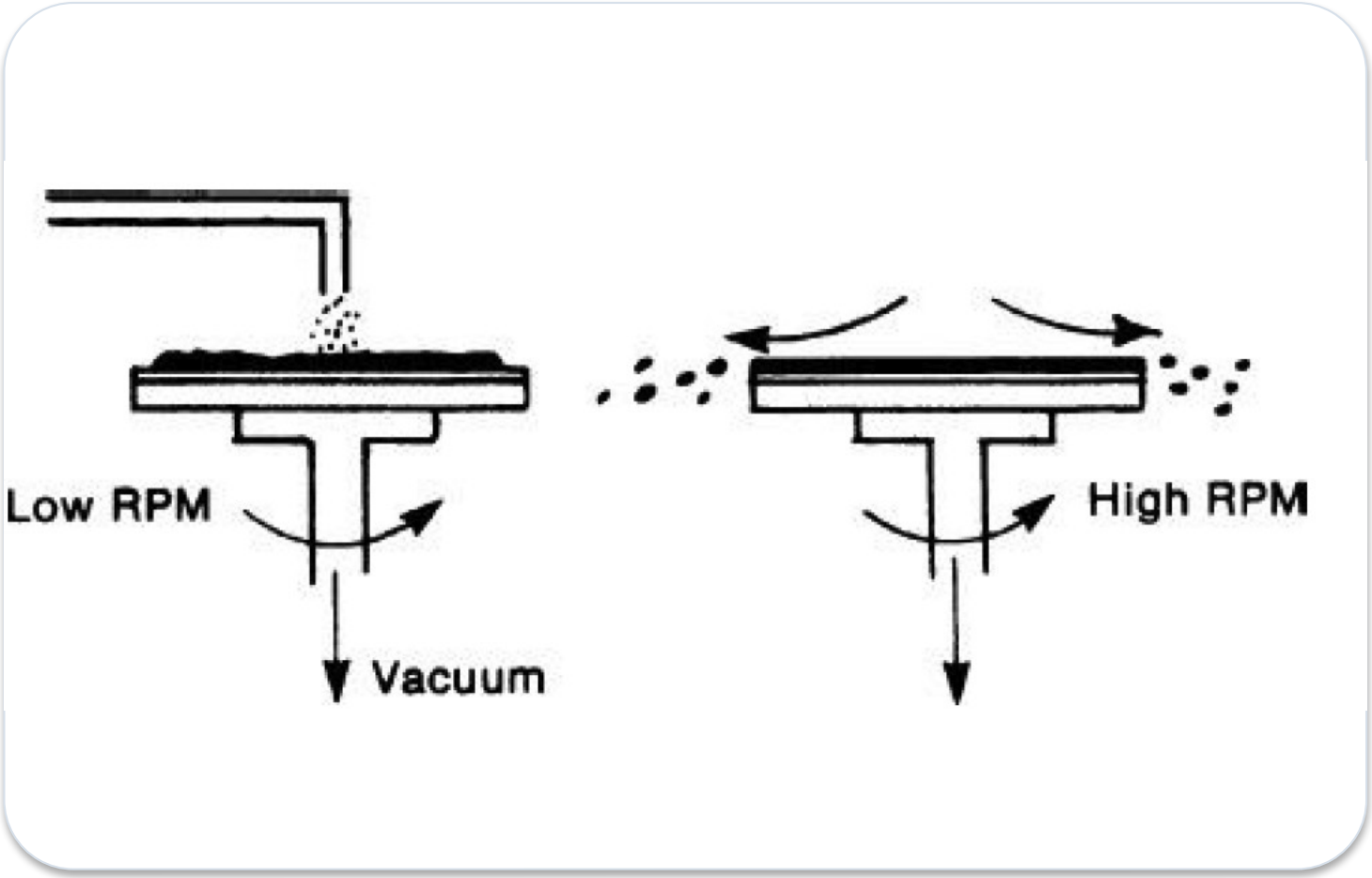
TOPIC 01

Dynamic Dispense



1

Evidence 1

Dynamic Spin Dispensing: Technique for applying resist to larger wafers, using low-speed rotation to aid initial spreading.
Benefits: Uses less resist and achieves a more uniform layer
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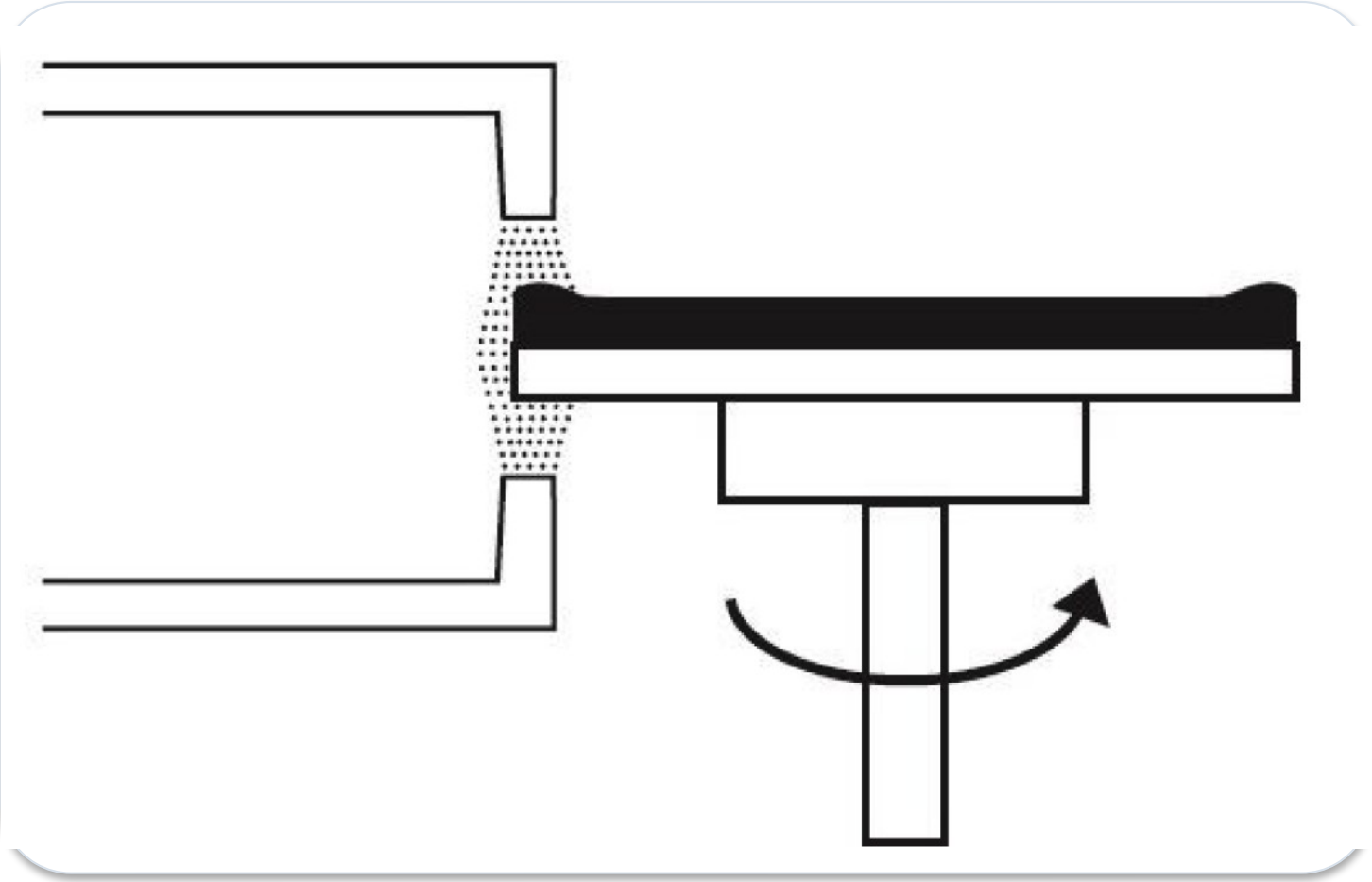
TOPIC 01

Edge Bead Removal



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Evidence 1

Edge Bead: A buildup of resist around the wafer edge caused by high-speed spinning.
Removal: Solvent spray is used to remove the edge bead.
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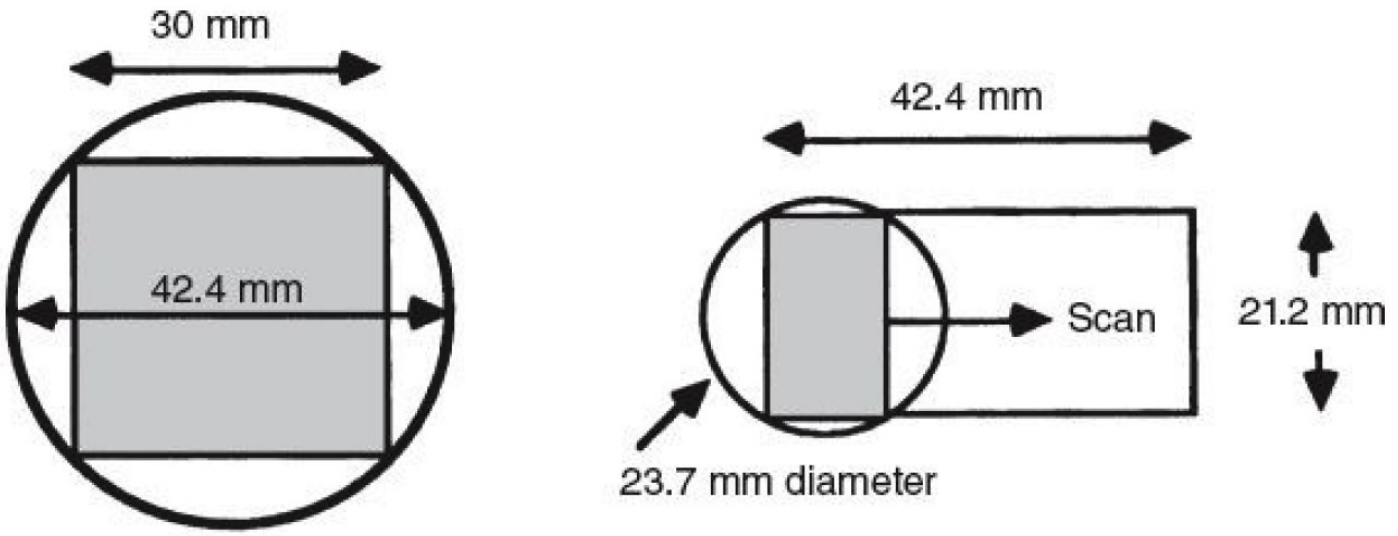
TOPIC 01

Step and Scan Aligners (Steppers)



1

Evidence 1

Larger die sizes would normally require larger lens systems with larger fields of vision. Increasing the field of vision shortens the time of alignment and exposure. However, larger lenses become expensive. An alternative is a stepper with a smaller lens and the capability to scan the smaller field over the required area

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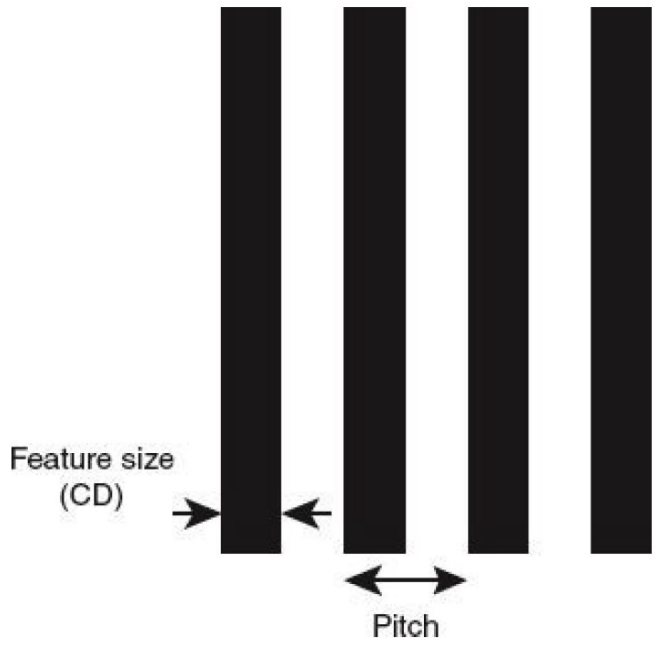
TOPIC 01

Next Generation Lithography

$$\sigma = k\frac{\lambda}{NA}$$



1

Evidence 1

There is a basic relationship between the components of an exposure system. The relationship is:
where σ = the minimum feature size
k = constant (sometimes called the Rayleigh constant)
λ = the wavelength of the exposing light
NA = is the numerical aperture of the lens
The term k (or k1 in some formulas) relates

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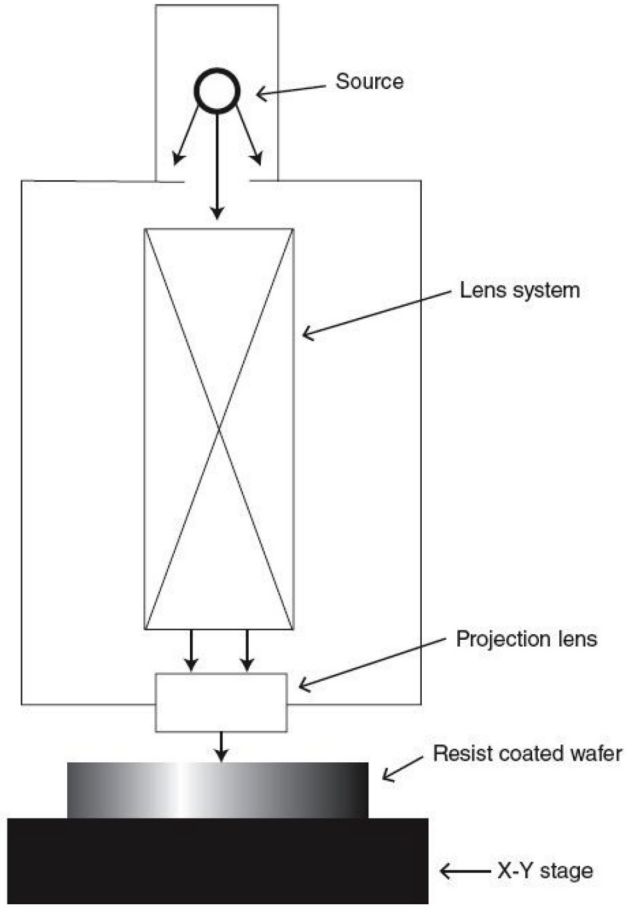
TOPIC 01

Next Generation Lithography



1

Evidence 1

'k' Factor: Represents the lens' ability to distinguish between adjacent images.
Diffraction Effects: Cause blurring of images, even with perfect lenses.
Smaller Image Size: Achieved by decreasing wavelength or increasing numerical aperture (NA).

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TOPIC 01

Next Generation Lithography

Technology Generation	Wavelength (mm)	Exposure Source
DUV G line	436	Mercury (Hg)
DUV H line	405	"
DUV I line	365	"
Excimer Laser	248	KrF
"	193	ArF
"	157	F_2
Extreme UV (EU)	135	Tin Vapor

1

Evidence 1

Rayleigh Formula: Shows that smaller wavelengths enable the patterning of smaller features.
Industry Progression: The industry has transitioned to shorter wavelengths to meet the demand for smaller feature sizes.

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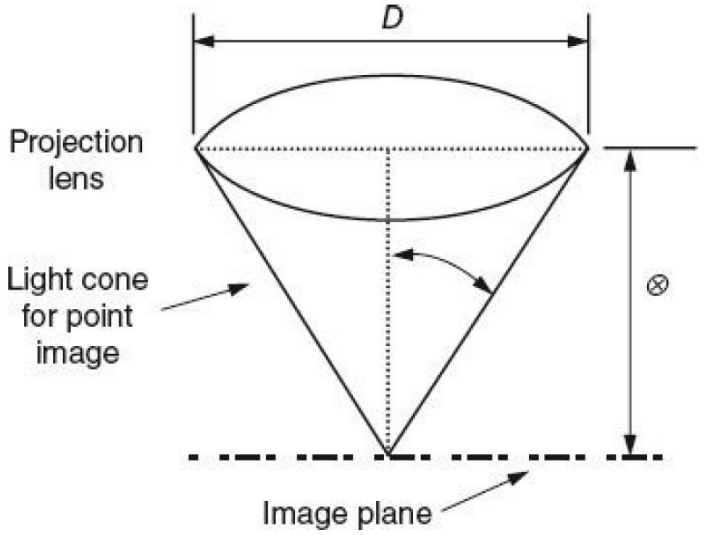
TOPIC 01

Numerical Aperture and Resolution Enhancement



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Evidence 1

Projection Systems: Used for exposing small features, with the mask/reticle and wafer separated.
Resolution & Dimensional Control: Challenges in projection optics to minimize loss of resolution and dimensional control.
Numerical Aperture (NA): Lens' ability to gather light, a key factor in resolution.
Depth of Focus (DoF): Trade-off parameter limiting NA increases, as higher NA results in shallower DoF.

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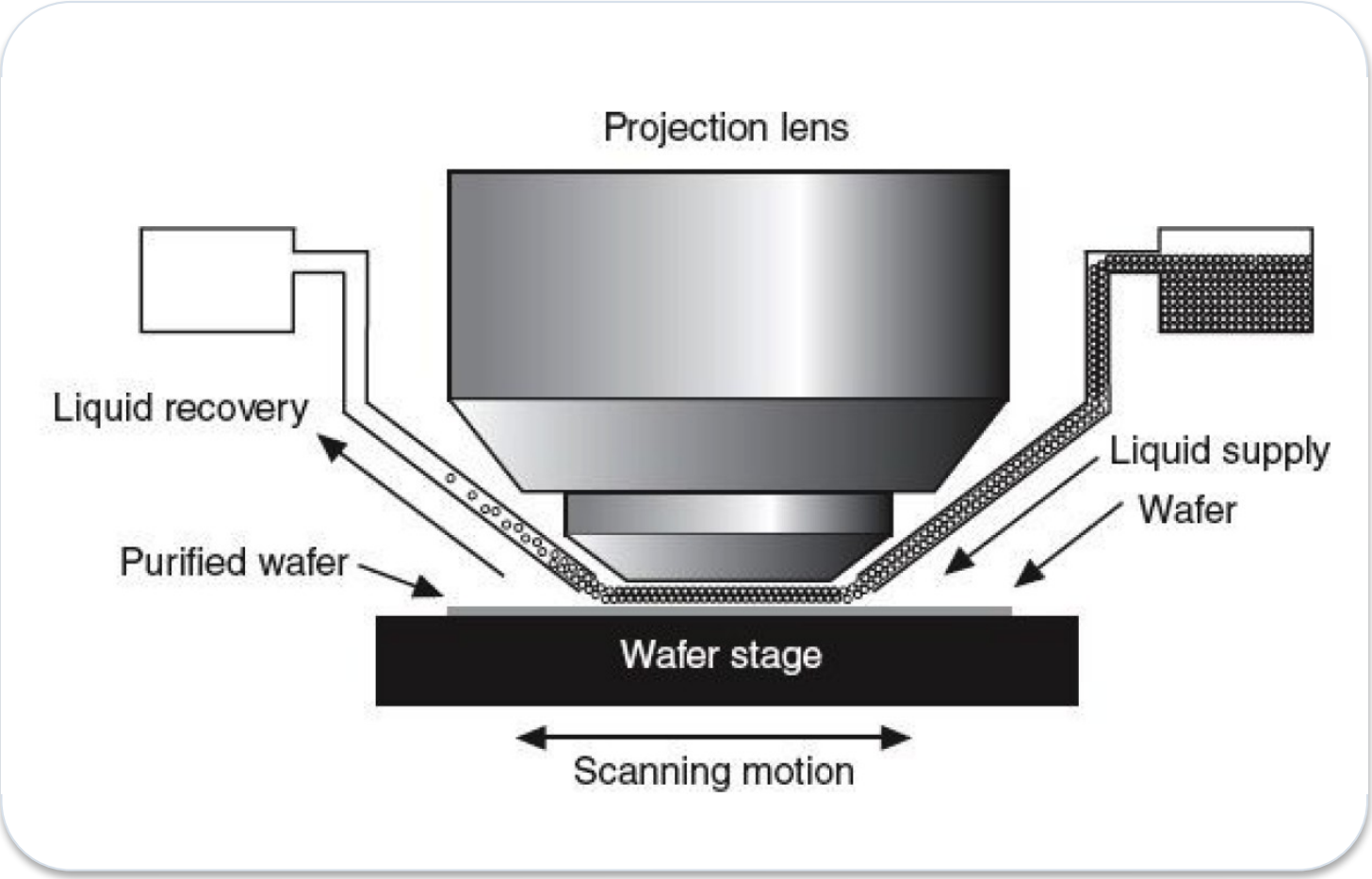
TOPIC 01

Immersion Exposure System



1

Evidence 1

Light Refraction & Diffraction: Impact image fidelity, with diffraction causing bending of light around edges.
Immersion Lithography: Using purified water between the lens and wafer increases NA and allows for smaller printed features.
Benefits: Reduces image size, enabling smaller feature sizes.

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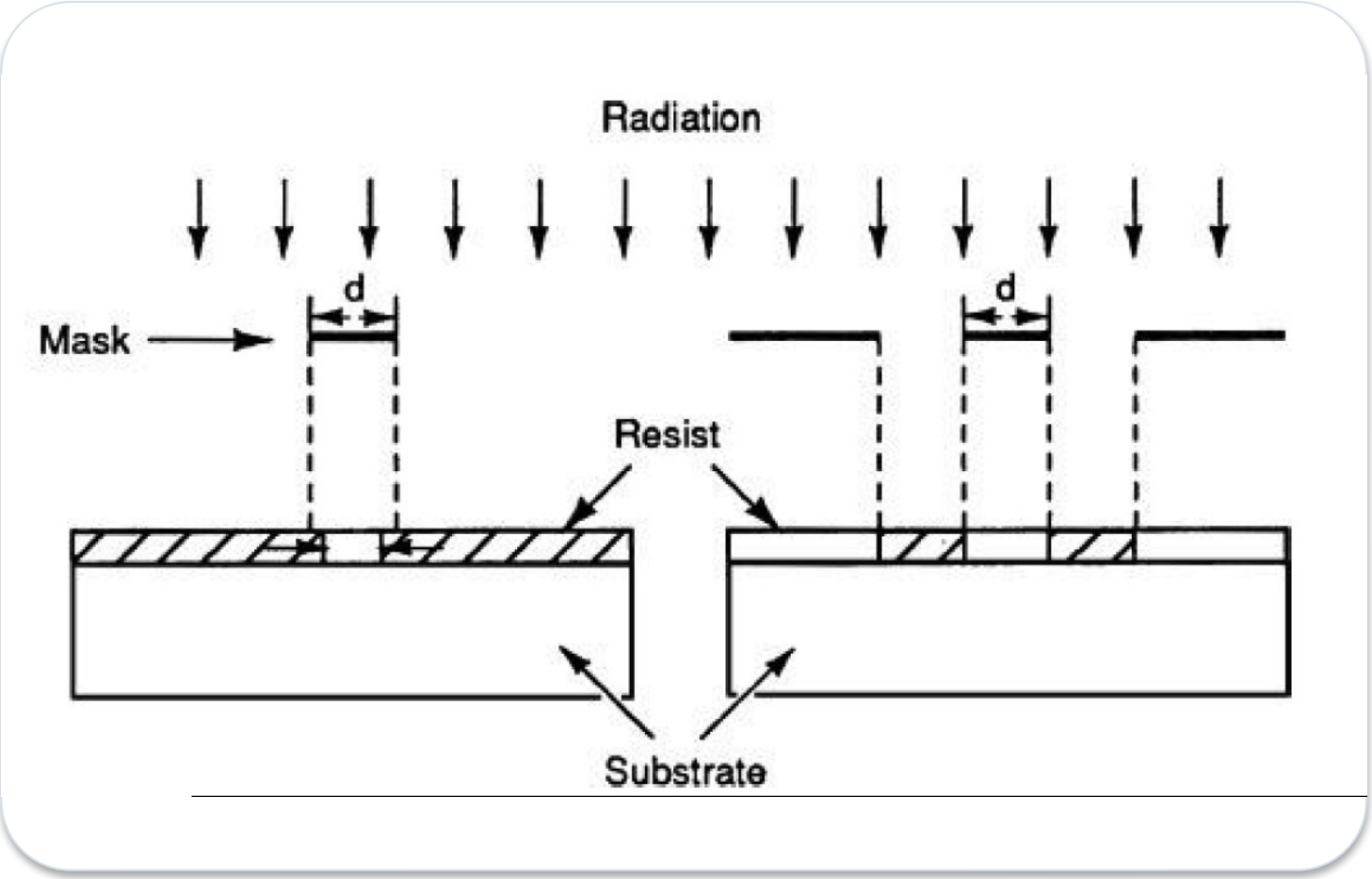
TOPIC 01

Contrast Effects



1

Evidence 1

Good resolution becomes difficult in regions of the mask where an opaque line is surrounded by a large clear area. The large amount of radiation coming around the opaque line tends to shrink the dimension of the line in the resist layer as the exposing rays diffract around the edge of the pattern. This problem is called a proximity effect.

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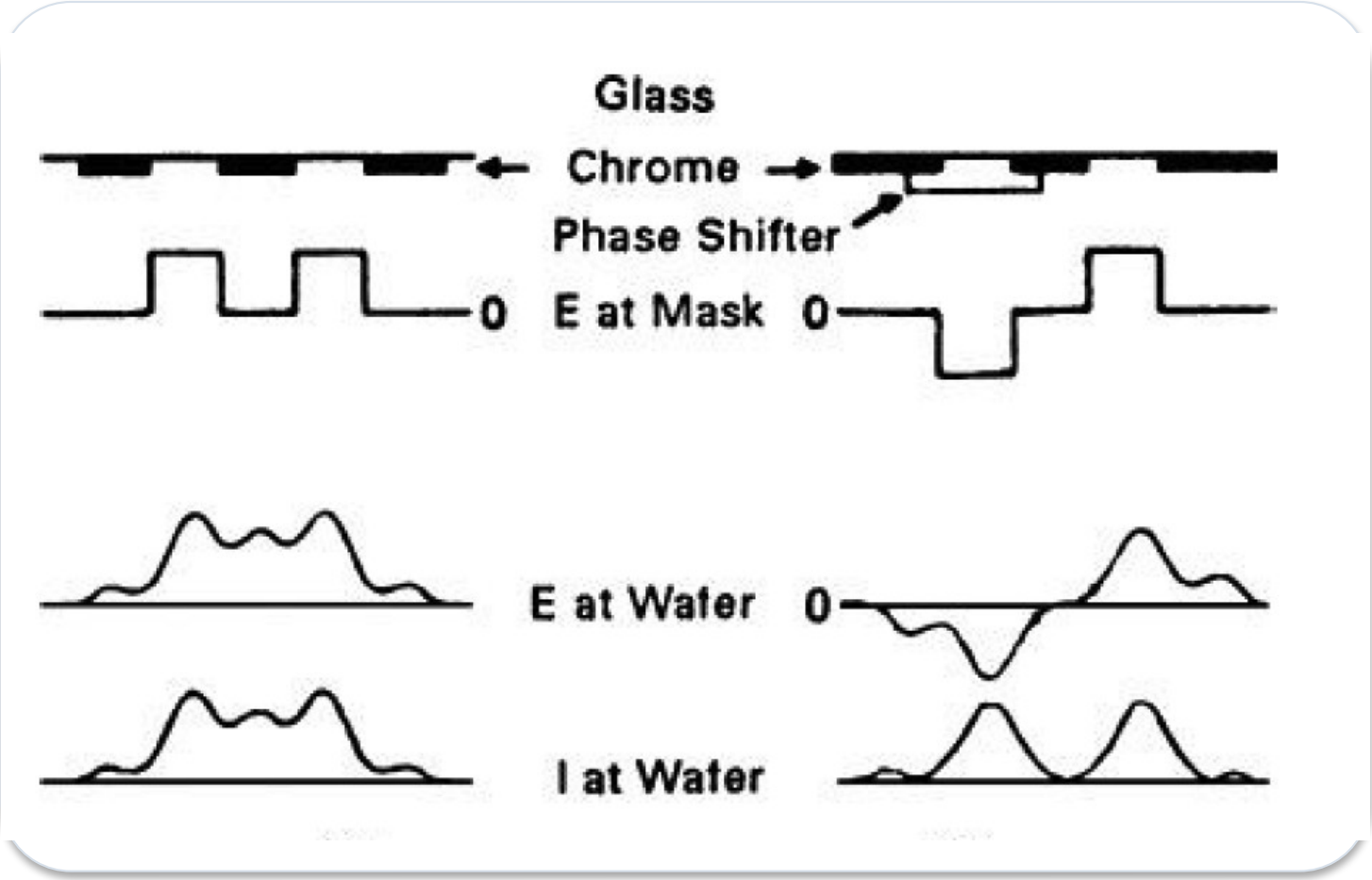
TOPIC 01

Phase-Shift Masks



1

Evidence 1

Phase-Shift Masks: Used to improve image fidelity by shifting the phase of exposure rays, preventing diffraction patterns from blending.
Alternating Phase-Shift Mask (AAPSM): Uses a silicon dioxide layer to shift the phase of alternate patterns, enhancing resolution.

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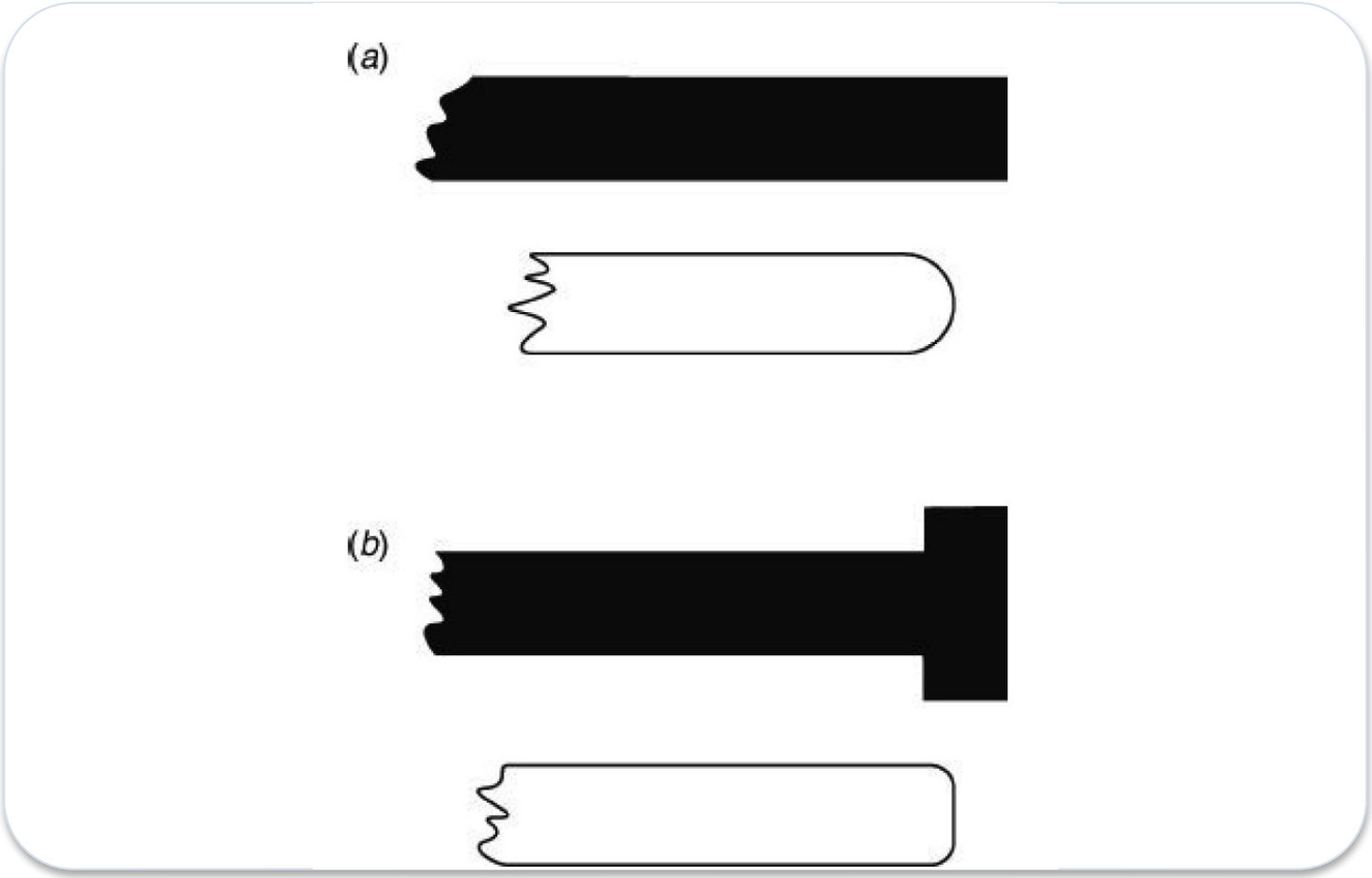
TOPIC 01

Optical Proximity Correction



1

Evidence 1

Optical Proximity Correction (OPC): Uses distorted patterns on the mask to compensate for diffraction effects, improving image fidelity.
OPC Applications: Addressing distorted patterns in dense areas, rounded corners, and contact holes.
Double Masking: Combines a phase mask with a trim mask to achieve the desired pattern.

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TOPIC 01

Mask Making

Chrome layers block energy at wavelengths of 365 nm, 248 nm, and 193 nm
Smaller dimensions require new exposure sources (EUV, X-ray, electrons, ions) and materials
Mask/reticle making varies by exposure method (e.g., pattern generator, laser, e-beam)
Pattern generator uses motor-driven shutters and light source to create patterns on a resist-covered chrome mask
Reticle pattern transferred via step-and-repeat to make a master plate
Master plate used in contact printer to create multiple working mask plates
After exposure, masks go through development, inspection, etch, strip, and final inspection
Inspections are crucial to avoid defects that could result in large-scale wafer scrap

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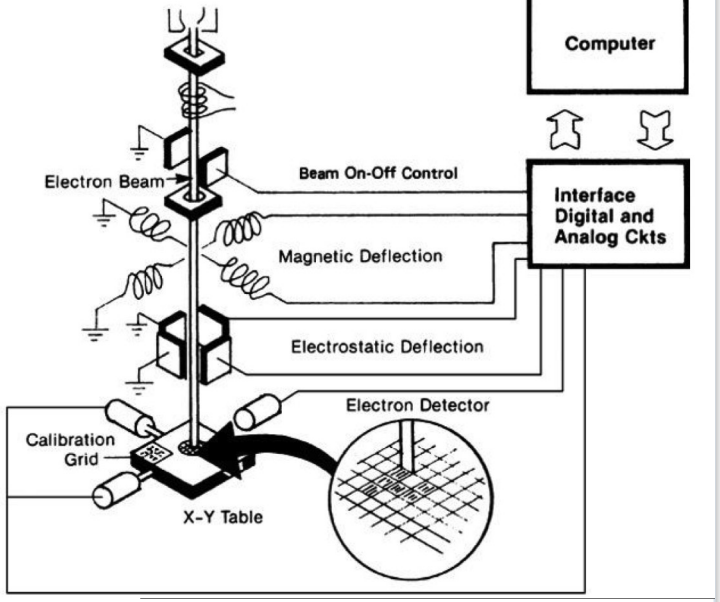
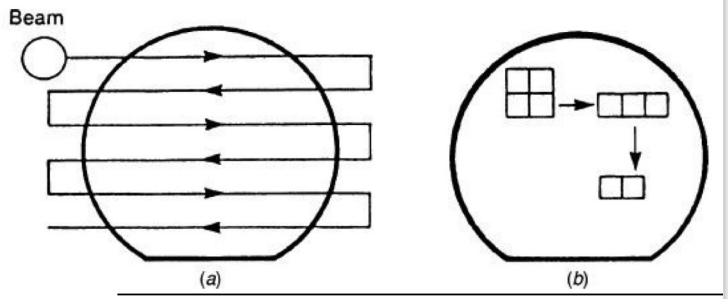
TOPIC 01

Electron Beam or Direct Writing





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Evidence 1

A mature technology used for creating high-quality masks/reticles.
Electron source and a "blanker" for turning the beam on and off.

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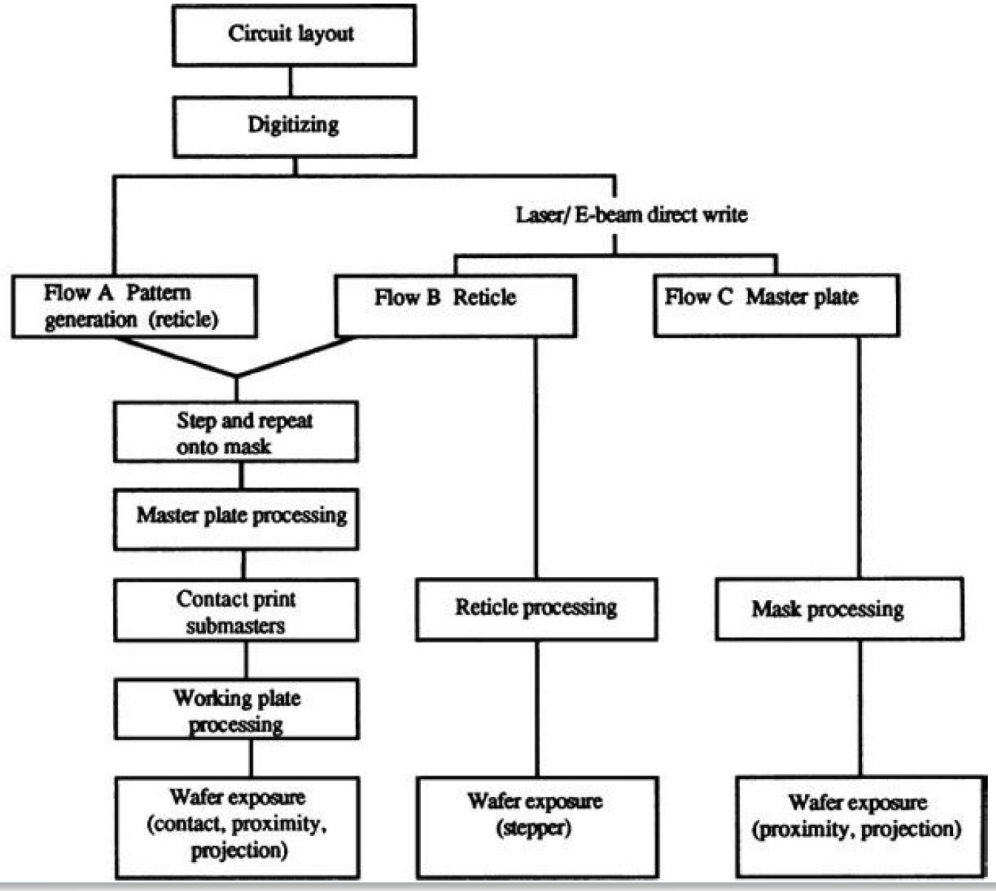
TOPIC 01

Mask-or reticle-making processing flows.



1

Evidence 1

High-Quality Masks/Reticles: Required for advanced products with small features and tight alignment tolerances.
Laser/E-Beam Direct Write: Methods used to create masks/reticles, with laser systems using a 364 nm wavelength.
Defect-Free Requirements: VLSI/ULSI circuits require virtually defect-free masks/reticles with tight dimensional control.
Repair Technologies: Laser zapping and FIBs are used to repair defects on masks/reticles.

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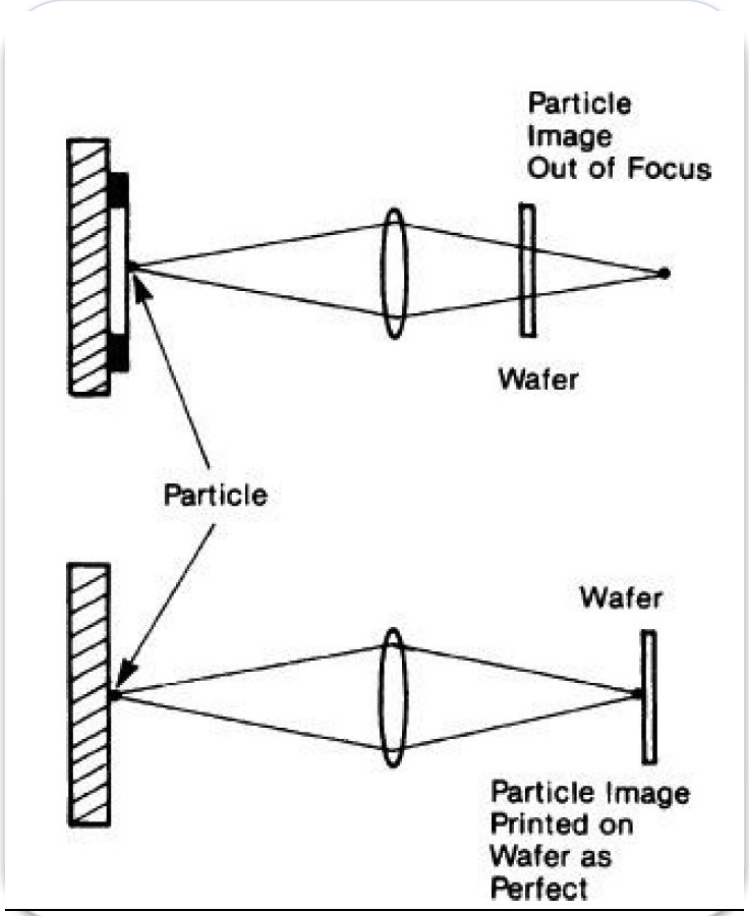
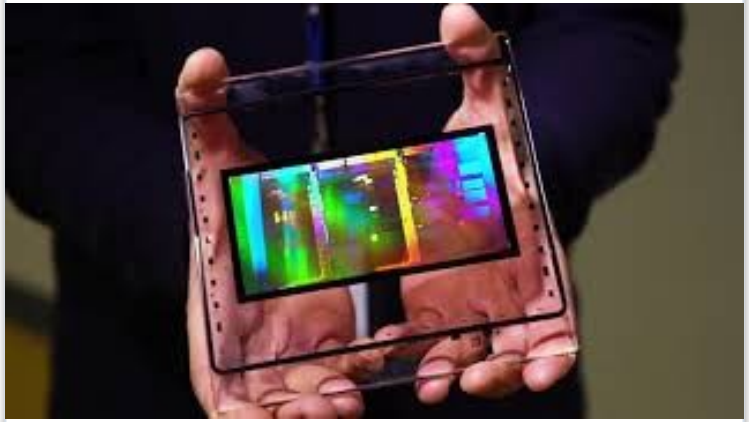
TOPIC 01

Pellicles





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Evidence 1

Projection exposure systems (aligners and steppers) increased mask/reticle lifetime
Higher-quality masks/reticles needed due to extended use
Yield loss from dirt/scratches during handling and cleaning
Cleaning introduces contamination, scratches, and breakage ("Catch-22")
Solution: pellicle—a thin, optically neutral polymer on a frame, fitted over the mask
Pellicle protects the mask by holding dirt/dust out of the focal plane, making particles transparent to exposing rays

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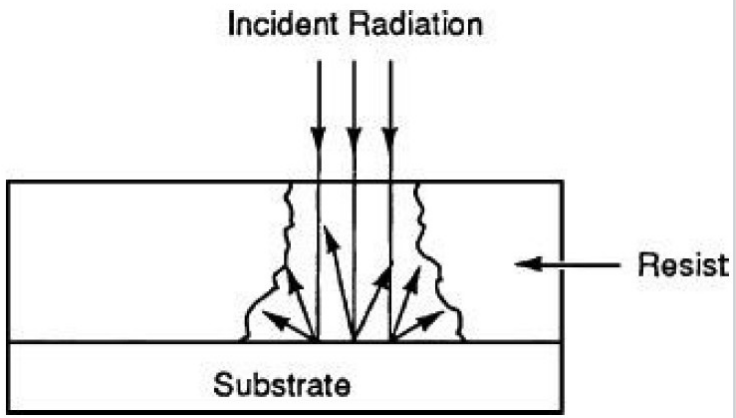
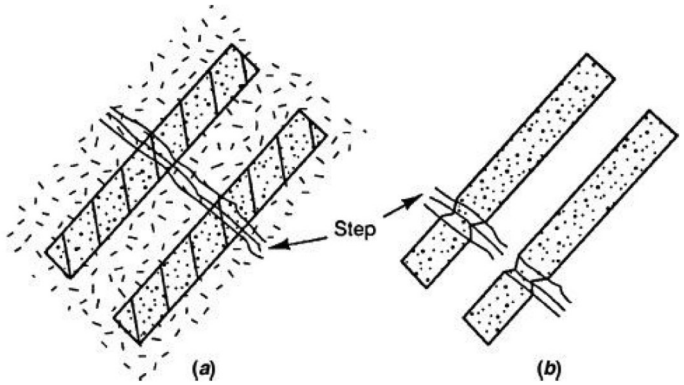
TOPIC 01

Subsurface Reflectivity





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Evidence 1

Ideal exposure: radiation at 90° to wafer surface for clear image in resist
In reality, some waves travel at angles, exposing unwanted resist areas
Reflectivity varies by surface material and smoothness; metals like aluminum have high reflectivity
Goal: smooth, consistent surfaces to control reflections
Reflection issues worsen with varied topography (wafer steps)
Step sidewalls reflect radiation at angles, reducing image resolution
Light interference at steps causes "notching" in the pattern

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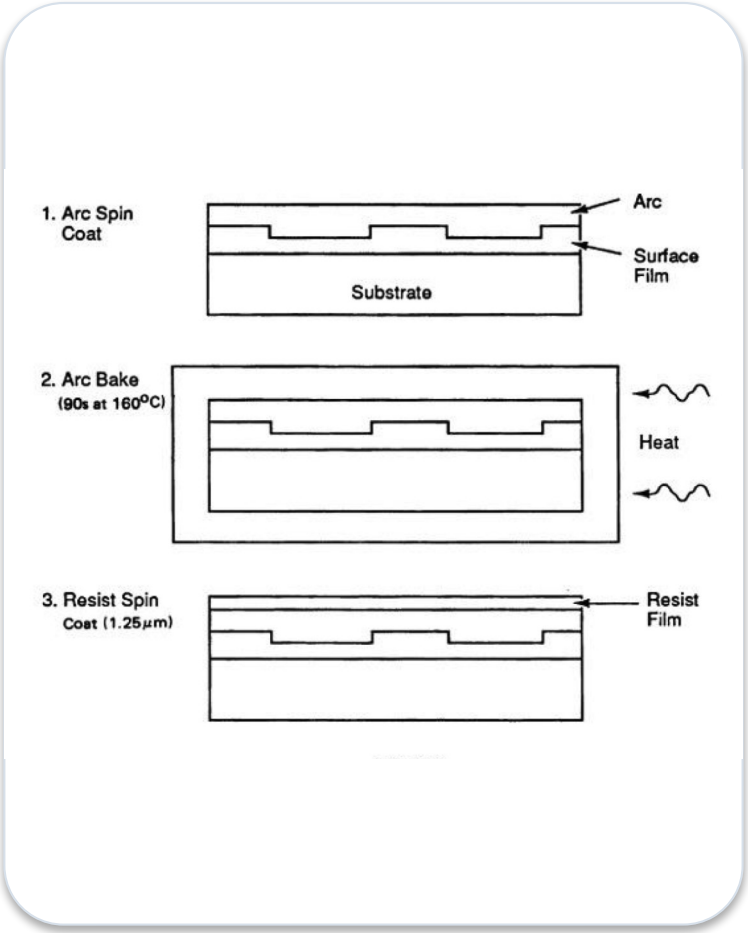
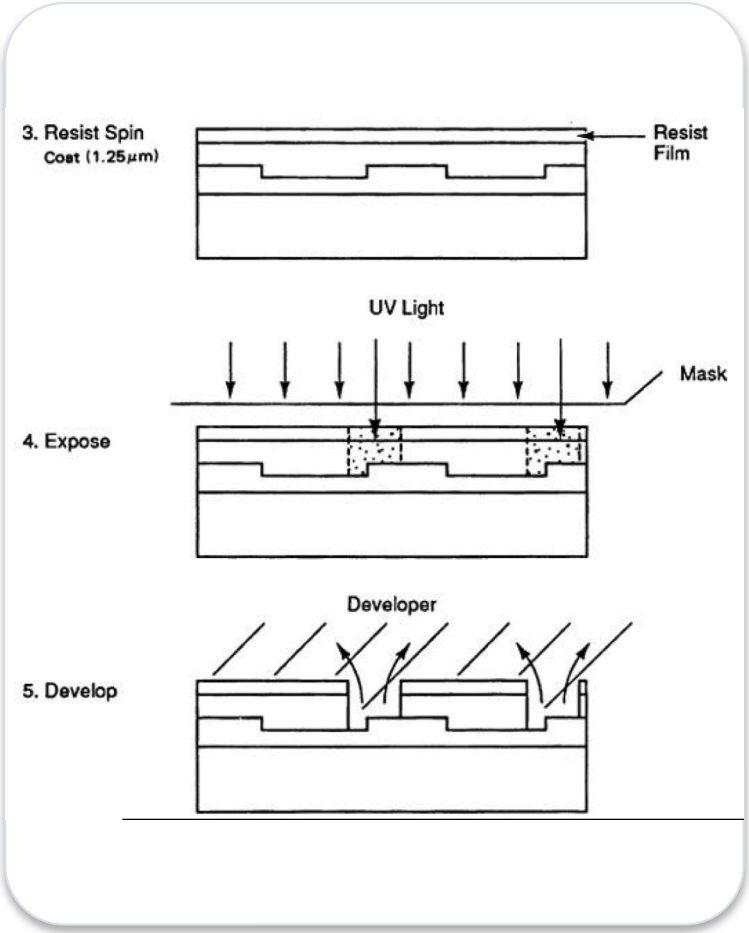
TOPIC 01

Antireflective Coatings





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Evidence 1

Antireflective coatings (ARCs) applied before resist help pattern small images
ARC planarizes the surface, improving resist uniformity
Reduces light scattering, enhancing image definition
Minimizes standing wave effects and improves image contrast
Increases exposure latitude for better patterning quality

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TOPIC 01

Chemical-mechanical polishing (CMP)

Planarization techniques smooth surfaces but don't achieve full global flatness
Small-dimension images still face light scattering and metal-step coverage issues
Chemical-mechanical polishing (CMP) is used for global planarization of wafer surfaces
CMP advantages:
Global planarization across the wafer
Removes all materials
Works on multimaterial surfaces
Enables high-definition damascene process and copper metallization
No hazardous gases
Low-cost process
Challenges differ between wafer polishing (removing microns of silicon) and metallization (removing sub-micron layers)

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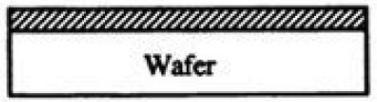
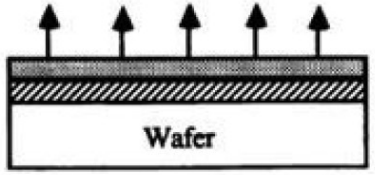
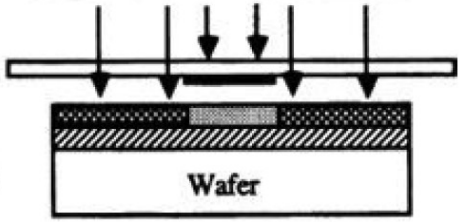
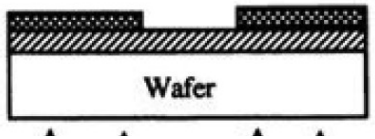
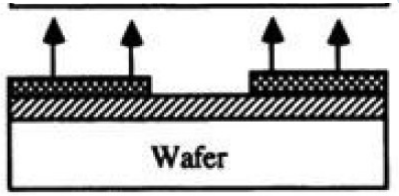
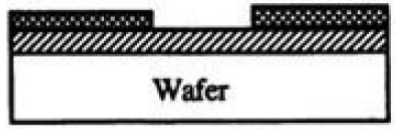
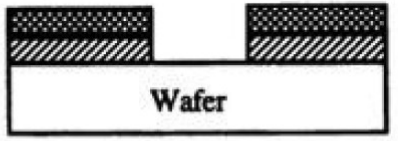
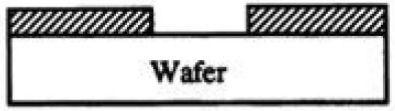
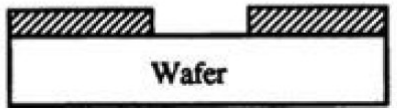
Typical Litho Process Steps

1. Surface preparation	Clean and dry wafer surface	
2. Photoresist apply	Spin coat a thin layer of photoresist on surface	
3. Soft bake	Partial evaporation of photoresist solvents by heating	
4. Alignment and Exposure	Precise alignment of mask/reticle to wafer and exposure of photoresist. Negative resist is polymerized	
5. Development	Removal of unpolymerized resist	
6. Hard bake	Additional evaporation of solvents	
7. Develop inspect	Inspect surface for alignment and defects	
8. Etch	Top layer of wafer is removed through opening in resist layer	
9. Photoresist removal (strip)	Remove photoresist layer from wafer	
10. Final inspection	Surface inspection for etch irregularities and other problems	

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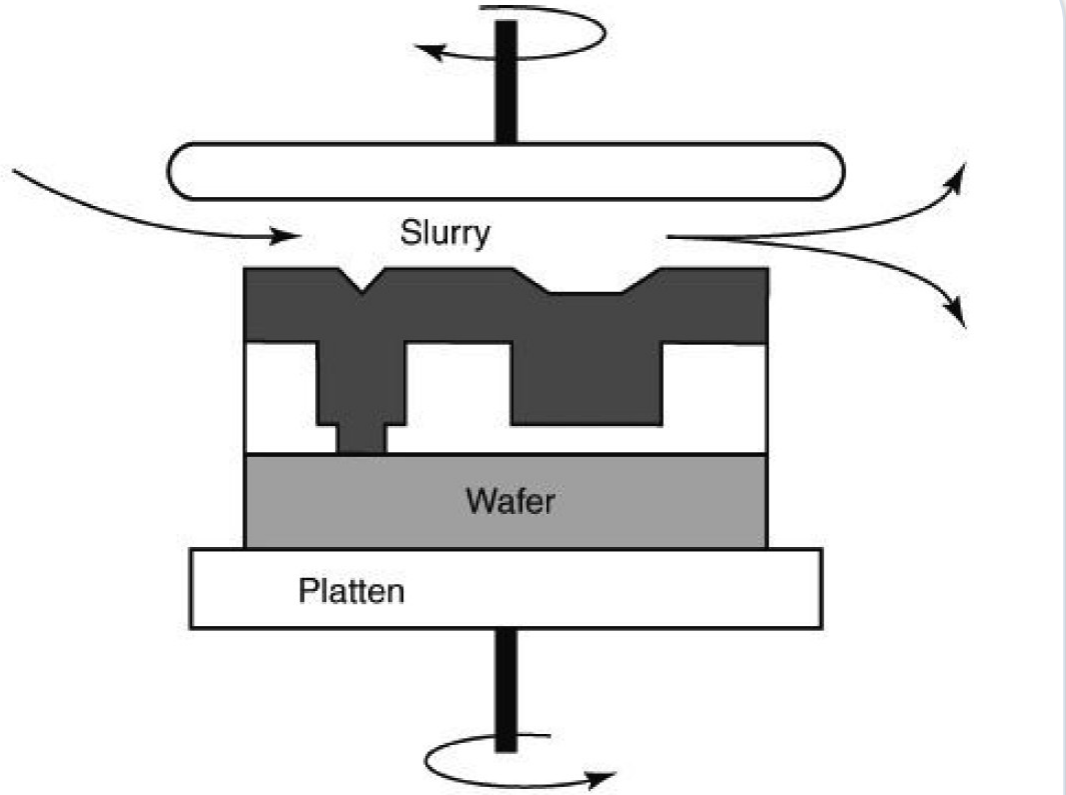
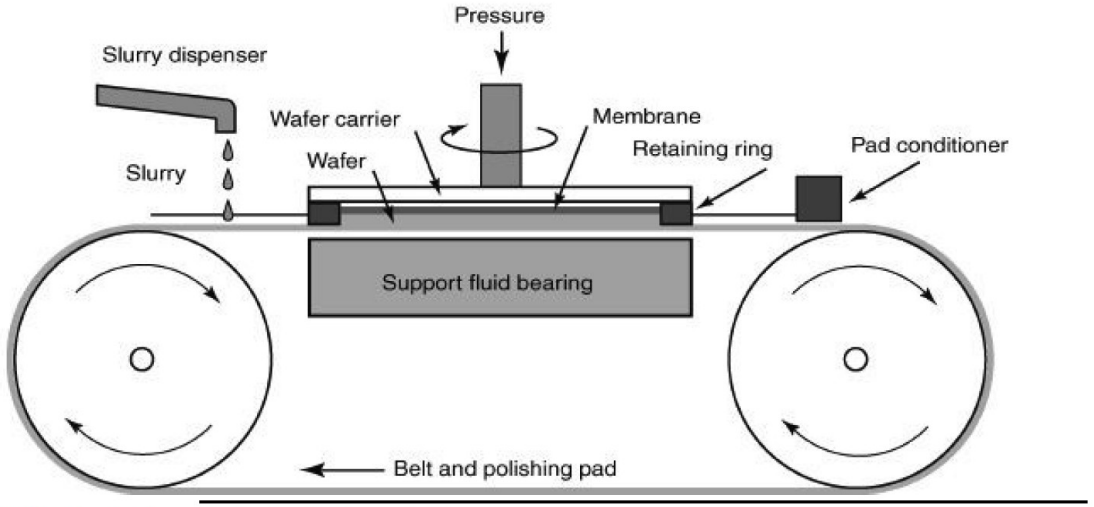
Chemical-mechanical polishing (CMP)





0.32 Chemical–mechanical polishing and planarization

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TOPIC 01

Slurry

Slurry chemistry is crucial in CMP due to its dual mechanical and chemical roles
Mechanical: abrasives (e.g., silica for oxide, alumina for metals) polish the surface
Abrasive sizes: 10-300 nm to polish without causing damage
Chemical: etchants like KOH or NH4OH used for silicon/silicon dioxide; metals like copper oxidized by slurry water
Basic materials reduce the oxidized film, removed by mechanical action
Additives: balance pH to control abrasive charges, reduce surface residues, improve flow (surfactants), and prevent metal redeposition (chelating agents)
Critical factors: slurry pH, wafer-pad interface flow, and etch selectivity for different materials

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TOPIC 01

Polishing Rates

Polishing rate is a key production measure in CMP
Influencing factors:
Pad material, slurry type, and abrasive size
Chemicals used for surface corrosion
Pad pressure, rotation rates, slurry flow rate, and viscosity
Temperature and humidity in the polishing chamber
Wafer diameter, pattern sizes, and surface materials
Balancing these factors is essential for optimizing polish speed while maintaining process control

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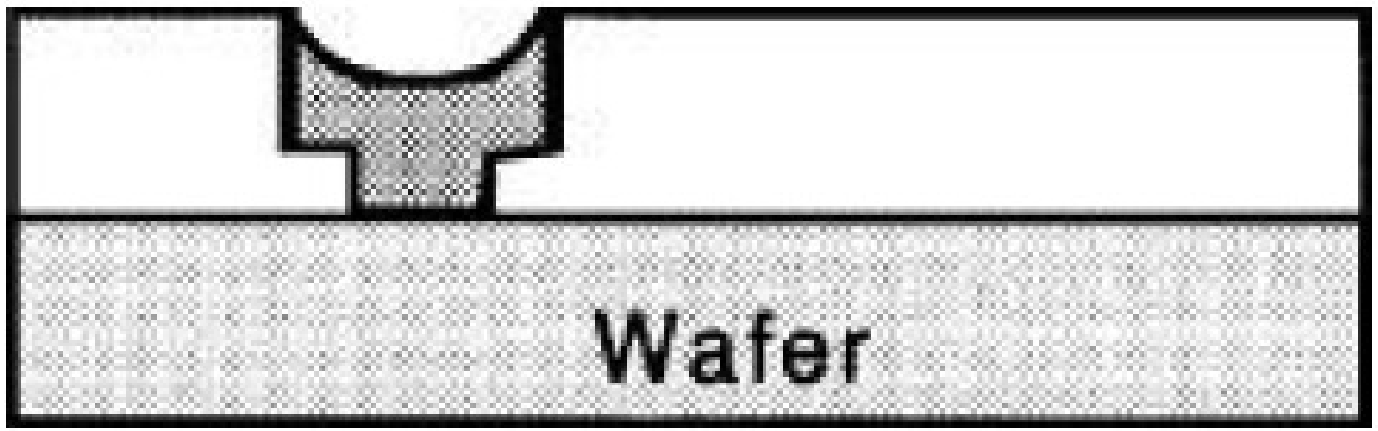
TOPIC 01

Planarity



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Evidence 1

Global planarity is the goal of CMP, but multimetal schemes, especially copper, pose challenges
Copper deposition in damascene trenches leads to lower density in the center, causing faster polishing and a "dish" shape
Copper density variations result in uneven polishing across the pattern

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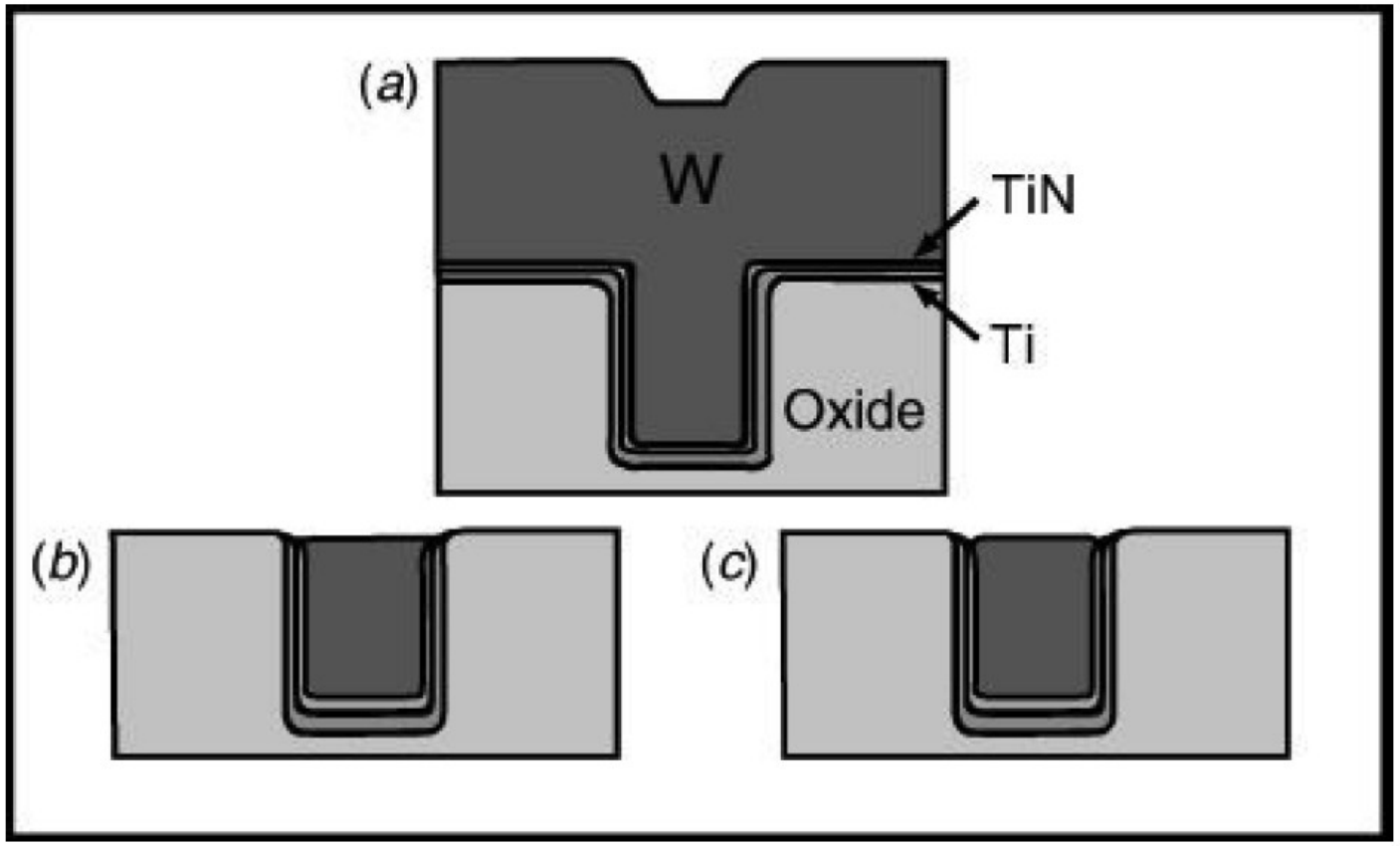
CMP Application in Tungsten plugs



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Evidence 1

Tungsten plugs also present issues, often recessed below surrounding oxide after initial CMP
An oxide buff is needed to fully planarize the surface after tungsten CMP

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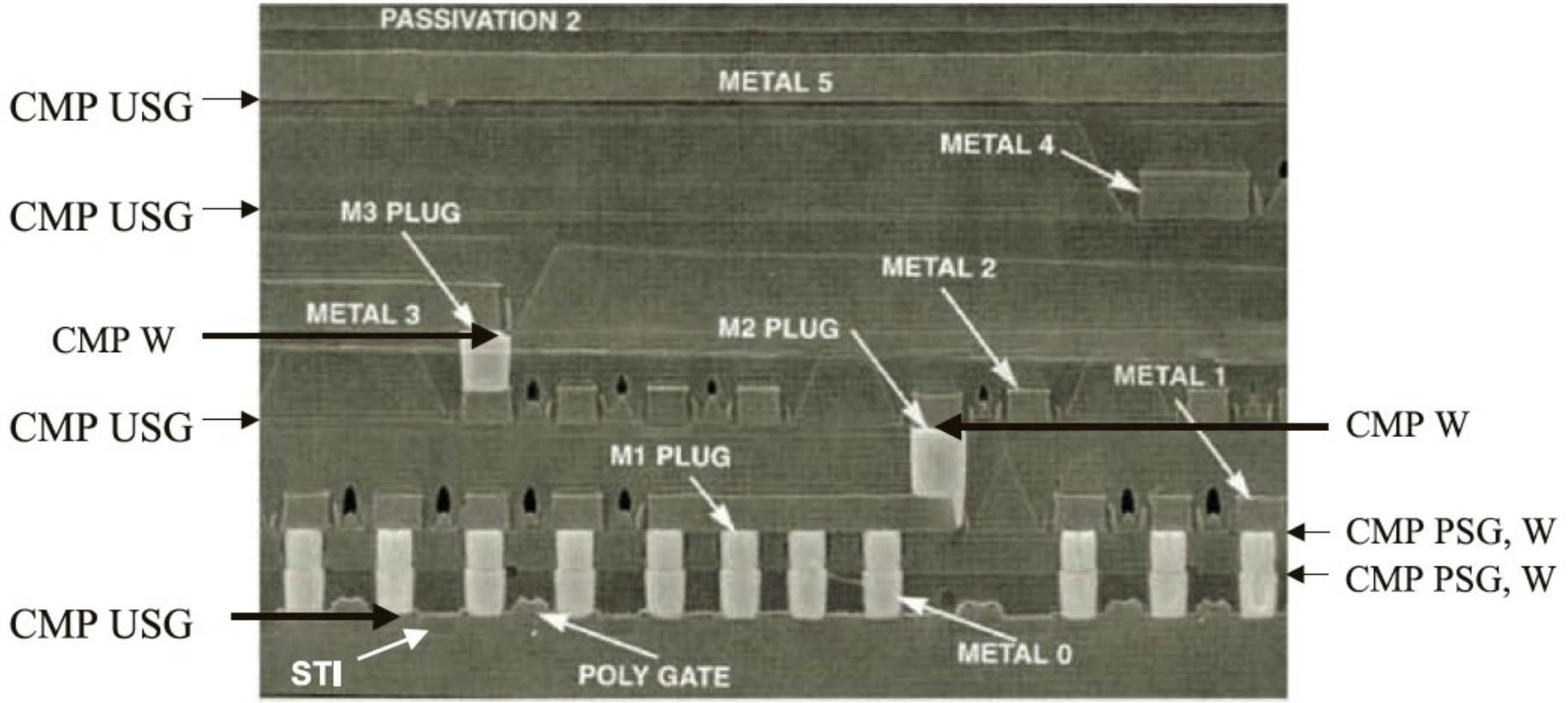
CMP Application in BEOL



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TOPIC 01

Post-CMP Clean

Clean wafer surfaces are crucial after CMP, with unique challenges from abrasive particles
Particles typically removed via mechanical brush cleaners or high-pressure water jets
Chemical cleaning uses standard FEOL methods
Slurry surfactants and pH adjustments create electrical repulsion, reducing particle contamination
Copper contamination is a key concern; it can degrade circuit performance if it enters the silicon
Copper residue levels should be reduced to 4×10^{13} atoms/cm^2

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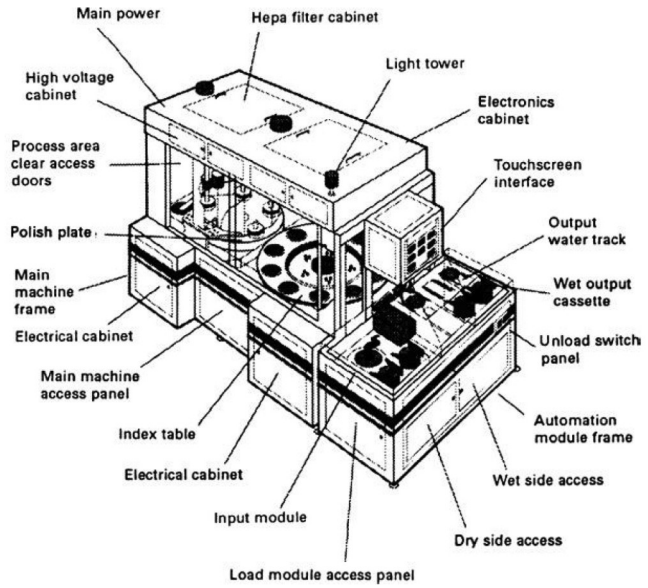
TOPIC 01

CMP Tools



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Evidence 1

Successful CMP requires a sophisticated integrated system beyond basic polishing units
Production tools include:
Automatic wafer-handling robots
On-board metrology and cleanliness detectors
End-point detection systems to signal layer removal or target depth
Post-CMP cleaning units may be integrated or connected via robots in a cluster design
Goal: achieve a "dry-in, dry-out" process for efficiency and cleanliness

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TOPIC 01

Etching

Photoresist pattern used as a mask for etching
Plasma etching is the standard process in IC fabrication
Etch gas (e.g., chlorofluorocarbons) injected into a low-pressure chamber
Plasma created by applying RF voltage between cathode and anode
Wafer placed on cathode, ions bombard wafer surface
Etching occurs anisotropically in selected regions
Silicon dioxide can also be etched similarly when covered by photoresist

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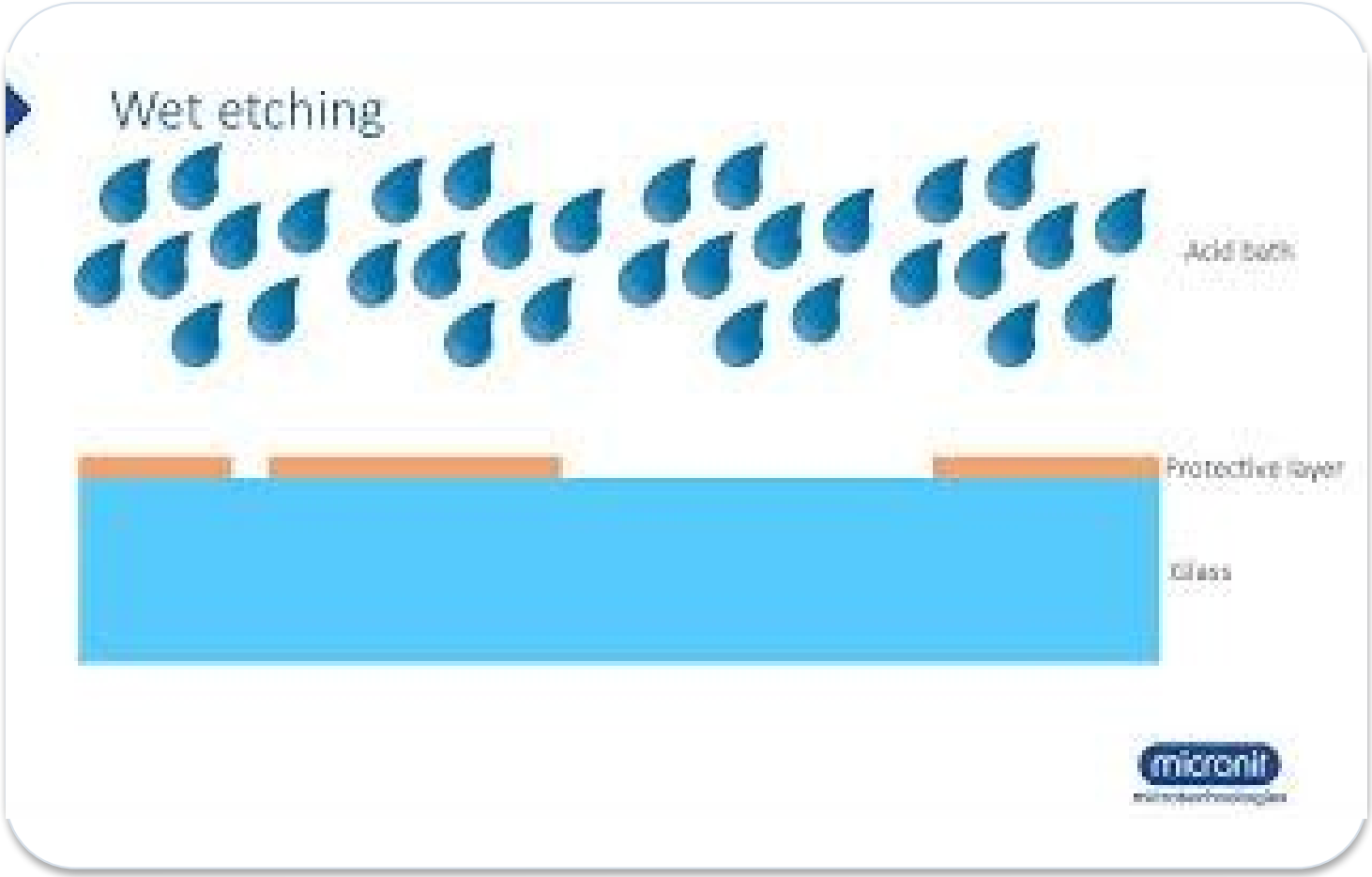
TOPIC 01

Wet Etching



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Evidence 1

Wet Etching: Immersion in chemical etchants, used for features larger than 3 µm.
Process Control: Heaters and agitation improve etch uniformity.
Etchant Selection: Based on selectivity (removing top layer without attacking underlying material).
Etch Time Variability: Influenced by temperature and resist adhesion.

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TOPIC 01

Incomplete Etch

1

Incomplete etch: portion of surface layer remains

Incomplete etch: portion of surface layer remains
Causes: short etch time, surface layer slowing etch, uneven surface
Wet etching: low temperature or weak solution
Dry etching: wrong gas mixture or improper system operation

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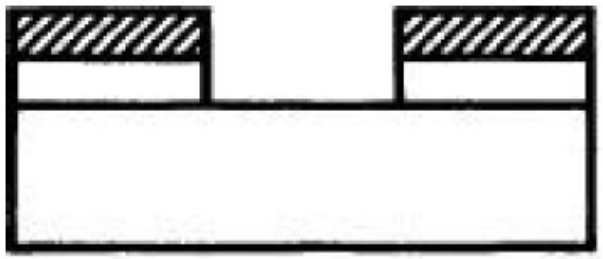
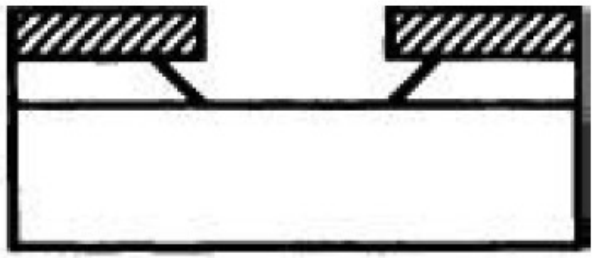
TOPIC 01

Overetch and Undercutting



Anisotropic etch



Isotropic etch

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Ideal etch: vertical sidewalls (anisotropic etching)

Ideal etch: vertical sidewalls (anisotropic etching)
Isotropic etching: material removed in all directions, causing sloped sides
Undercutting: surface layer eroded below resist edge

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Degrees of undercutting

Normal

Overetch

Overetch and resist lifting

1

Evidence 1

Severe undercutting: caused by excessive etch time, high temperature, or strong etch mixture
Weak photoresist adhesion also leads to undercutting
Dehydration, prime, soft-bake, and hard-bake steps prevent bond failure
Poor resist bond can cause severe or catastrophic undercutting

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TOPIC 01

Selectivity

1

Goal: preserve underlying surface during etching

Goal: preserve underlying surface during etching
Selectivity: ratio of etch rate of layer to underlying surface
High selectivity: minimal attack on underlying surface
Selectivity varies by material and method (e.g., oxide/silicon)
Critical for small contacts with high aspect ratios
Also applies to photoresist removal, especially in dry etching

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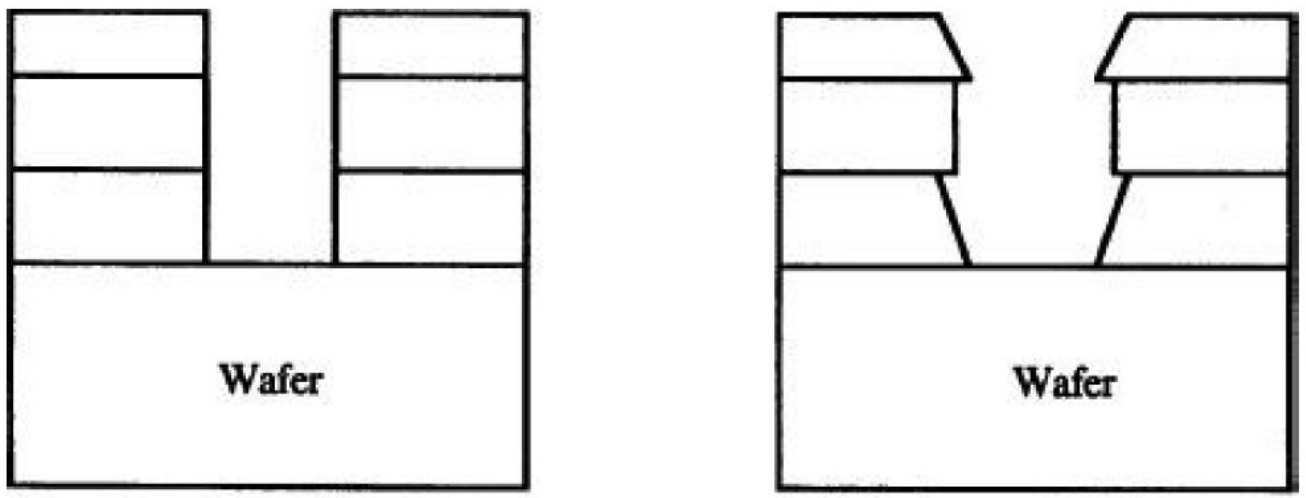
TOPIC 01

Etch Profile Control



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Evidence 1

Anisotropic etching produces vertical or near-vertical sidewalls
Challenge arises with multilayer stacks of different materials
Poor selectivity can lead to varying widths in the etched stack
Profile control in multilayer etching requires balancing etchant chemistry, power levels, system pressure, and design

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Summary of wet-etching process.

	Common Etchant	Etch Temp	Rate A/min	Method
SiO_2	HF & NH_4F (1:8)	Room	700	Dip & wetting agent predip
SiO_2	HF & NH_4F (1:8)	Room	700	Dip & wetting agent predip
SiO_2 (Vapox)	Acetic Acid & NH_4F(2:1)	Room	1000	Dip
Aluminum	H_3PO_4 : 16 HNO_3 : 1 Acetic : 1 H_2O : 2 Wetting agent	40–50°C	2000	a) Dip & agitation b) Spray
Si_3N_4 POLYSi	H_3PO_4 HNO_3 : 50 H_2O : 20 HF : 3	150–180°C Room	80 1000	Dip Dip

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TOPIC 01

Limitations of Wet Etch

1

Wet etching limits: pattern sizes ≥ 2 µm

Wet etching limits: pattern sizes ≥ 2 µm
Isotropic etching: sloped sidewalls
Requires rinse and dry steps
Hazardous/toxic chemicals
Contamination risk
Resist-wafer bond failure causes undercutting

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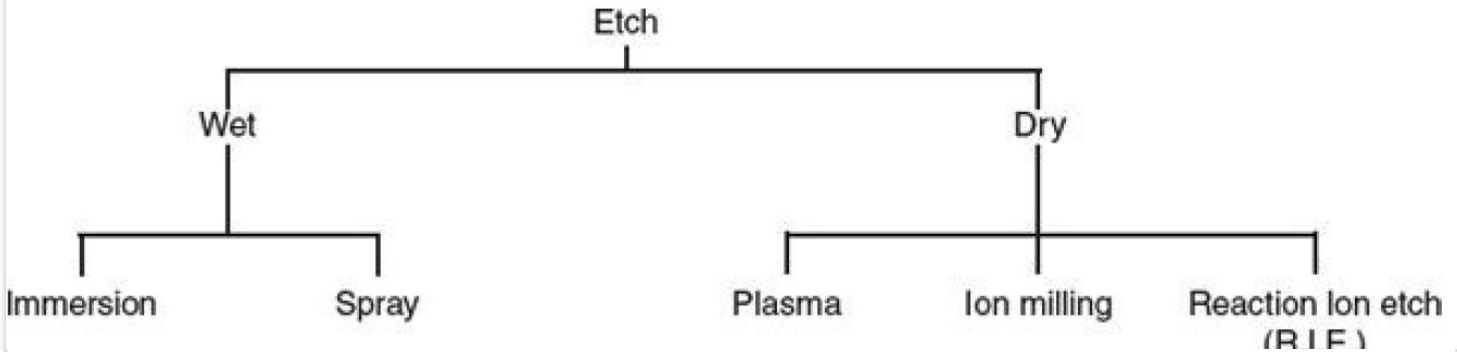
TOPIC 01

Dry Etching



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Dry etching: uses gases as the etch medium

Dry etching: uses gases as the etch medium
No wet chemicals or rinsing, wafers remain dry
Three techniques: plasma, ion beam milling, reactive ion etch (RIE)

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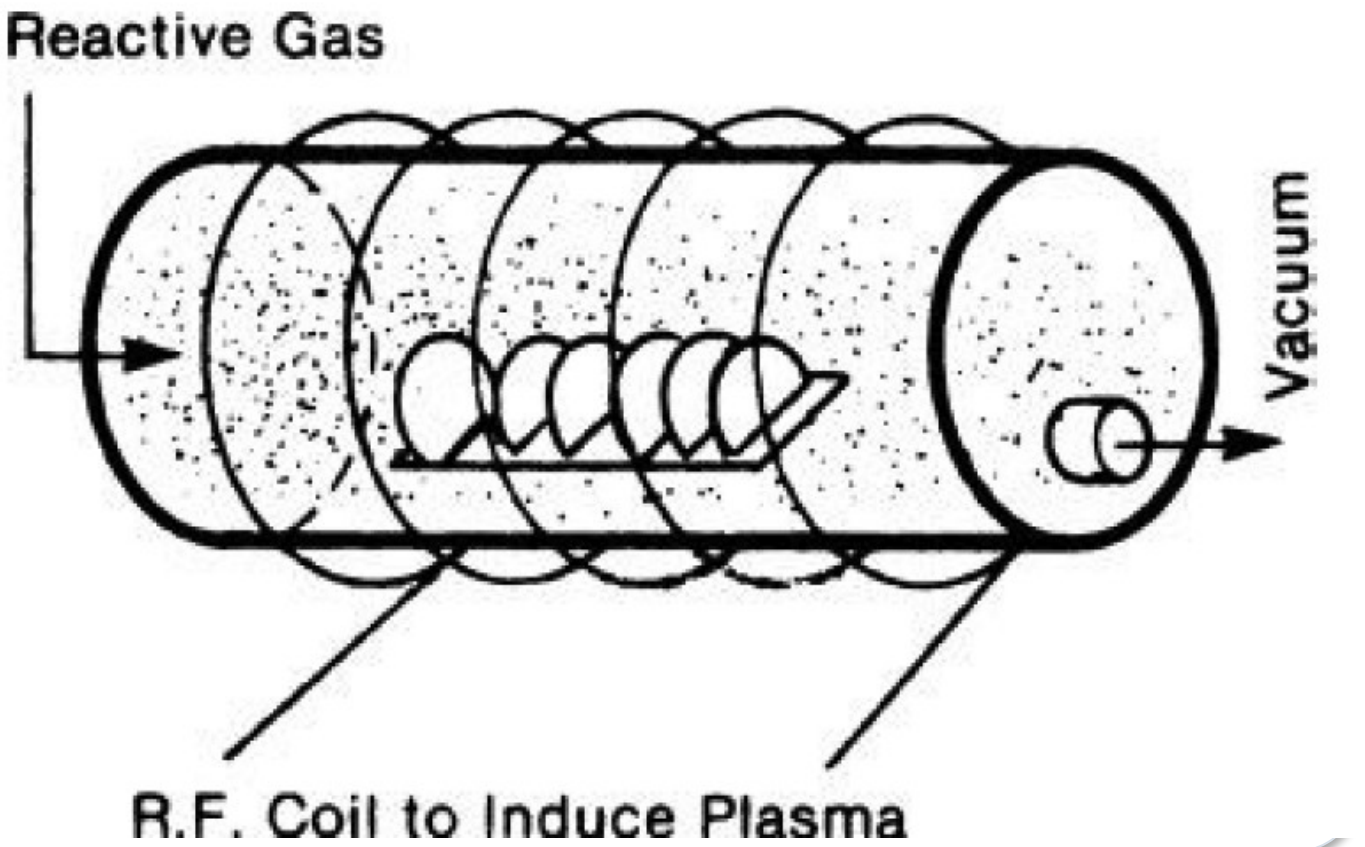
TOPIC 01

Plasma Etching



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Evidence 1

Plasma etching: chemical process using gases and plasma energy
Similar to wet etching but uses gases, not liquids
Requires: etchant gas (e.g., CF4 with oxygen) and energy source (RF field)
Process: vacuum chamber, gas supply, power source, end-point detector
Energized fluorine in plasma reacts with silicon dioxide, forming volatile byproducts
Byproducts removed by vacuum system

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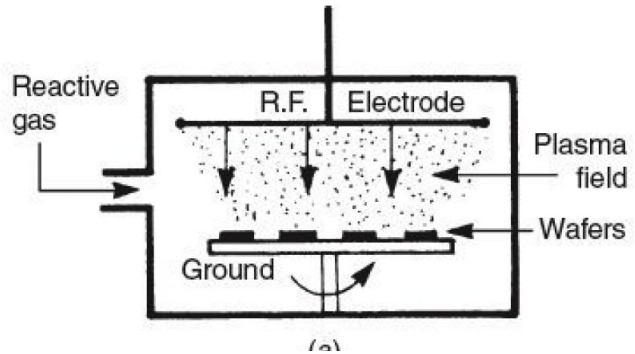
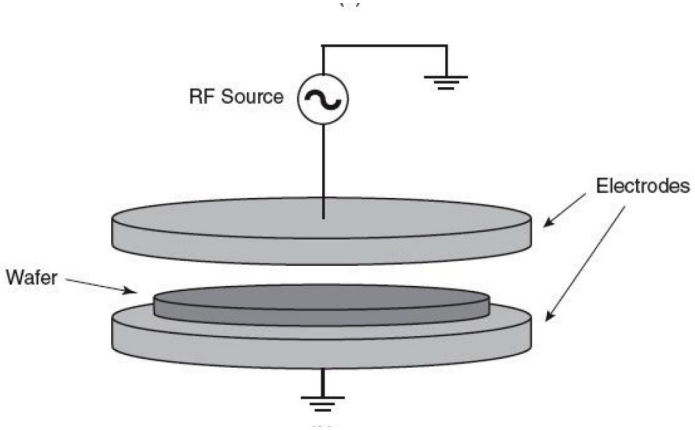
TOPIC 01

Plasma planar system





1

Plasma planar systems: used for more precise etching

Plasma planar systems: used for more precise etching
Wafers placed on grounded pallet under RF electrode
Etching occurs directly in the plasma field
More directional ions result in anisotropic etch (near-vertical sidewalls)
Wafer pallet rotation improves etching uniformity

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TOPIC 01

Plasma Etch Selectivity

Selectivity crucial in plasma etching, balanced with overetch
Overetch needed due to layer variations, uniformity issues, and microloading
Microloading: local etch rate varies with area size being etched
Overetch can reach 50-80% for metals, up to 200% for oxides/polysilicon
Critical factors: photoresist and underlying layer (e.g., silicon)
High resist-removal rates in dry etching, affecting small geometries
High-aspect-ratio patterns (up to 4:1) can slow etching at the bottom
Methods to control selectivity: gas formula, etch rate, gas dilution, end-point detectors (e.g., laser interferometer)

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Plasma etch chemicals.

Film	Etchant	Typical Gas Compounds
Al	Chlorine	BCl_3, CCl_4, Cl_2, $SiCl_4$
Mo	Fluorine	CF_4, SF_4, SF_8
Polymers	Oxygen DF_4, SF_4, SF_6	
Si	Chlorine, fluorine CF_4, SF_4, SF_6	BCl_3, CCl_4, Cl_2, $SiCl_4$
SiO_2	Chlorine, fluorine	CF_4, CHF_3, C_2F_6, C_3F_8
Ta	Fluorine	"
Ti	Chlorine, fluorine	"
W	Fluorine	"

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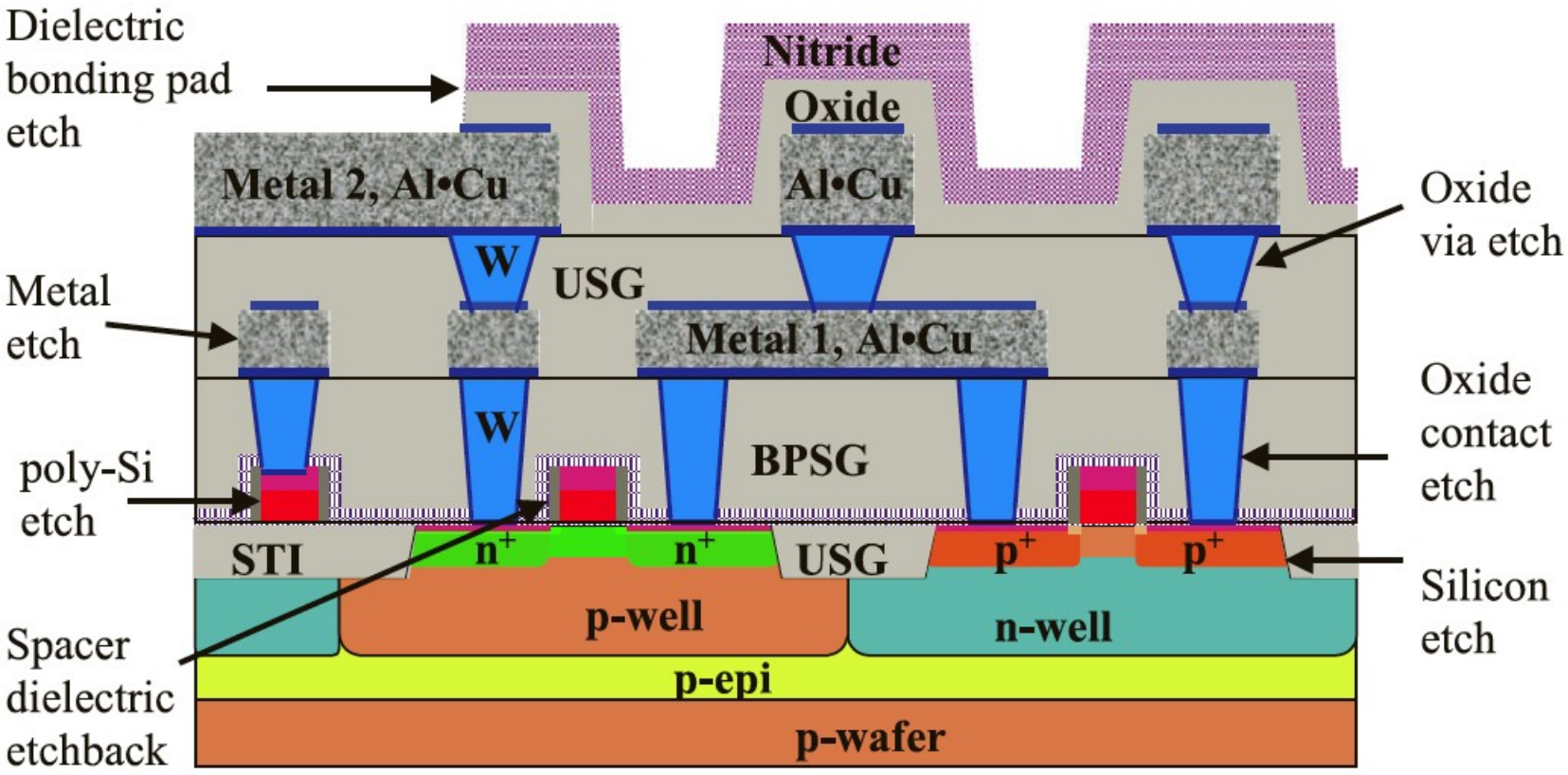
Plasma Etch in IC Fabrication



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TOPIC 01

Resist Stripping

After etching, resist layer is removed (stripped)
Wet chemical stripping preferred in FEOL to protect sensitive MOS gates
Plasma O2 stripping used more in BEOL, where sensitive parts are covered
Chemical choice for stripping depends on wafer surface, resist polarity, and resist condition
Resist removal needed after processes like wet etch, dry etch, and ion implantation
Challenges: high-temp hard bakes, plasma etch residues, sidewall polymers, and ion implantation crusting

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TOPIC 01

Sulfuric Acid and Oxidant Solutions

Sulfuric acid and hydrogen peroxide (SPM) are common wet strippers for nonmetallic surfaces (silicon dioxide, silicon nitride, polysilicon)
Effective for both negative and positive resists
Nitric acid sometimes added as an oxidant (typical 10:1 ratio with sulfuric acid)
Drawback: nitric acid turns the bath orange, masking carbon buildup
Resist removal through oxidation mechanism

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TOPIC 01

Dry Stripping

Dry-plasma resist stripping uses oxygen plasma to oxidize resist components into gases
Ashing: plasma process for removing organic residues
Plasma stripping: removes both organic and inorganic residues
Plasma generated by microwave, RF, or UV-ozone sources
Advantages: no wet chemicals, reduced handling
Disadvantages: ineffective at removing metallic ions, risk of radiation damage
Downstream strippers reduce radiation damage by generating plasma away from wafers
Wet stripping remains mainstream due to plasma limitations (metal ion removal, radiation concerns)

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TOPIC 01

Diffusion

Diffusion used for doping silicon in IC fabrication
Impurity atoms (boron, phosphorus) change conductivity type
Forms pn junctions in semiconductor devices
Wafers oxidized, windows opened with photolithography and etching
High-temperature furnace (~1100°C) used for dopant diffusion
Dopants diffuse into silicon due to a density gradient
Process stops as wafer cools, fixing dopants in place

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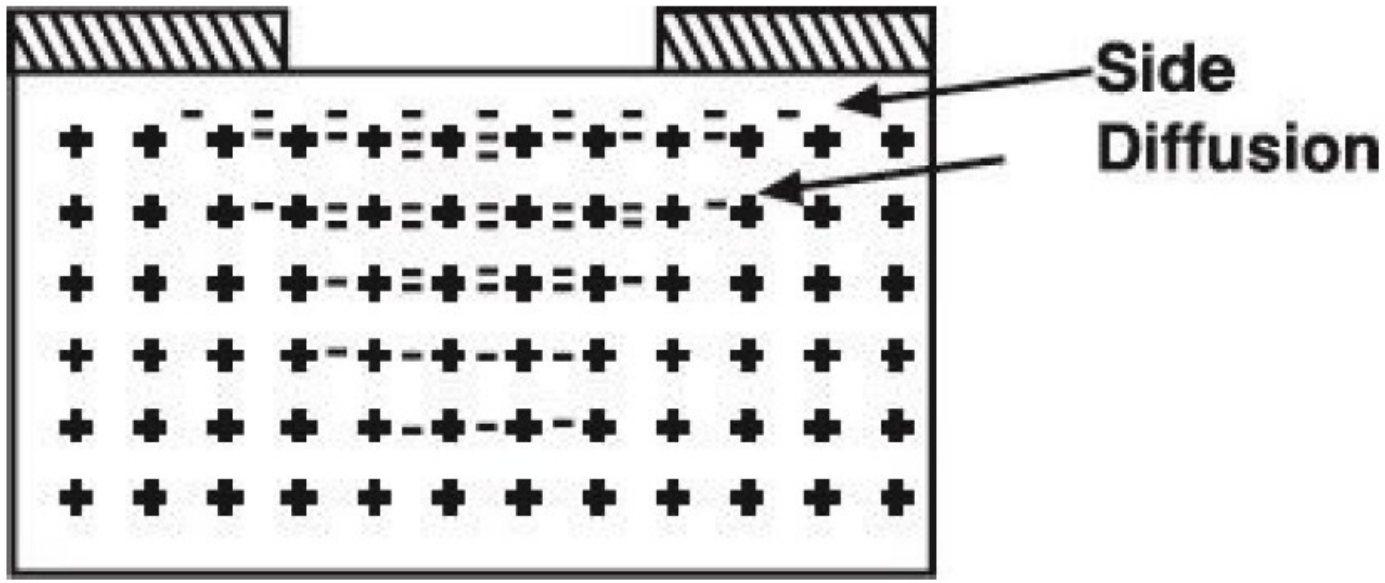
TOPIC 01

Lateral Diffusion



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Evidence 1

Dopant atoms in diffusion doping move in all directions, not just vertically
Lateral diffusion occurs under the oxide barrier, forming junctions
Lateral diffusion is about 85% of the vertical junction depth
Happens with both diffusion and ion implantation methods
Side diffusion impacts circuit density, discussed further in ion implantation

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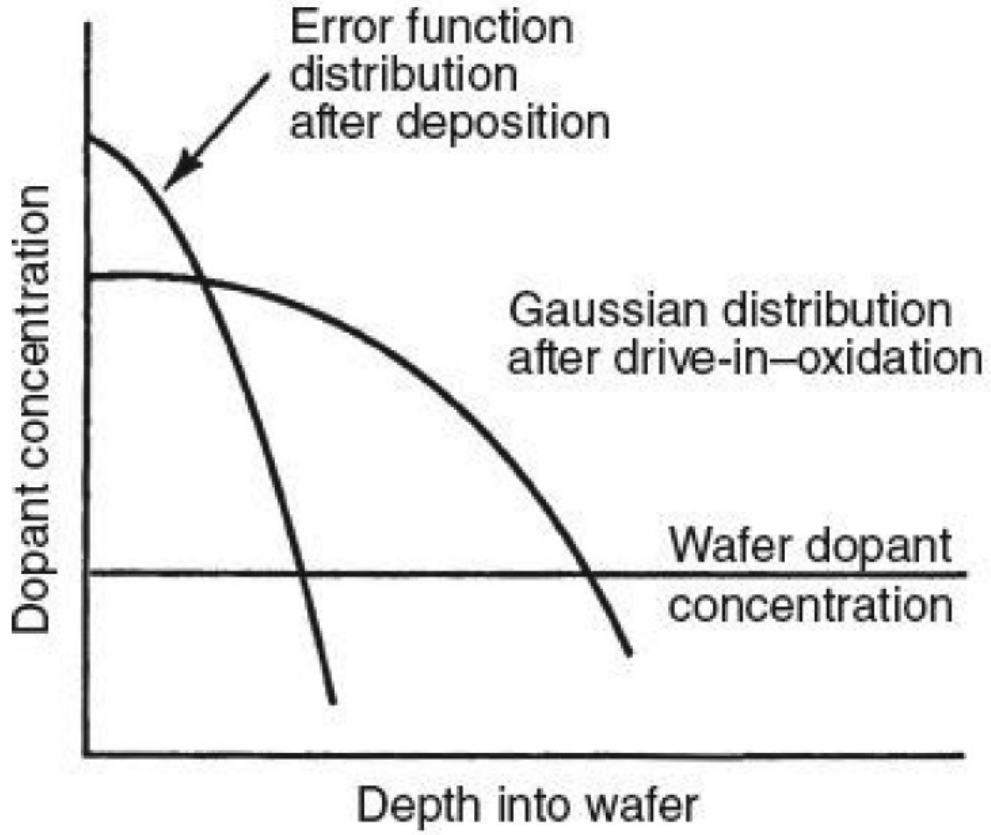
TOPIC 01

Drive-in Oxidation



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Evidence 1

Drive-in-oxidation step (also called drive-in, diffusion, reoxidation, or reox) serves two purposes:
Redistributes dopant deeper into the wafer
Grows new oxide on the exposed silicon surface
Dopant redistribution occurs through heat, without additional dopant source
Dopant atoms spread deeper and wider into the wafer while total dopant quantity remains constant
Surface concentration decreases, forming a Gaussian distribution

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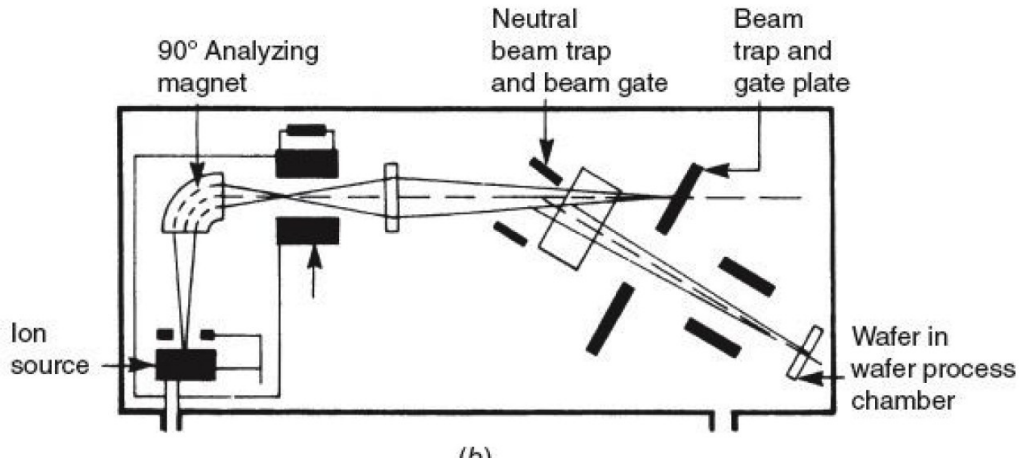
TOPIC 01

Ion Implantation

Diffusion is a chemical process. Ion implantation is a physical process,
Dopant ions accelerated and directed into the semiconductor surface
Ions lose energy through collisions, resting at various depths
Advantages: low-temperature process, precise doping layers
Photoresist or oxide layers block doping in selected regions
Disadvantage: silicon crystal damage from ion collisions
Damage repaired by thermal annealing at lower temperatures than diffusion



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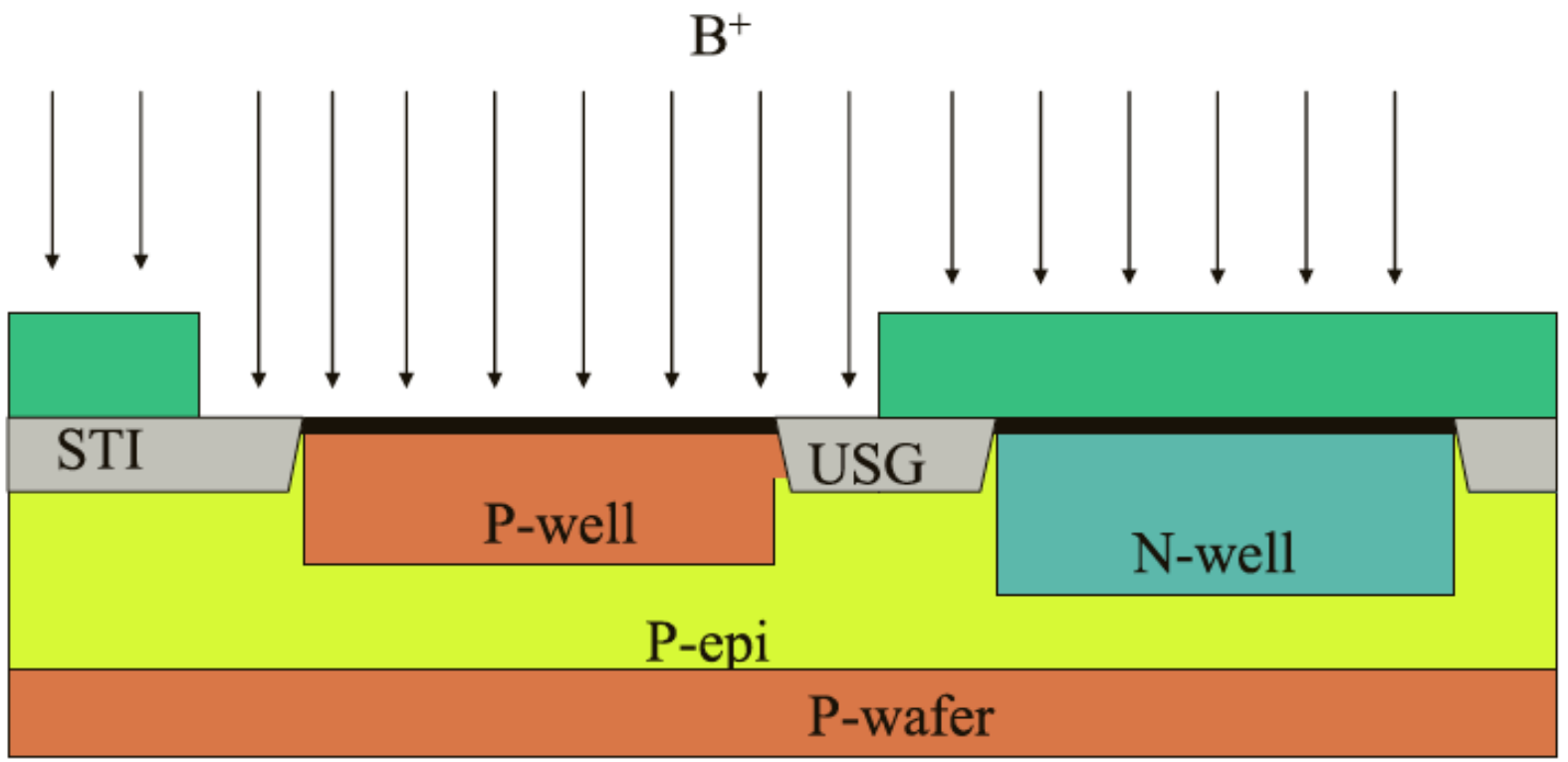
Well Ion Implantation



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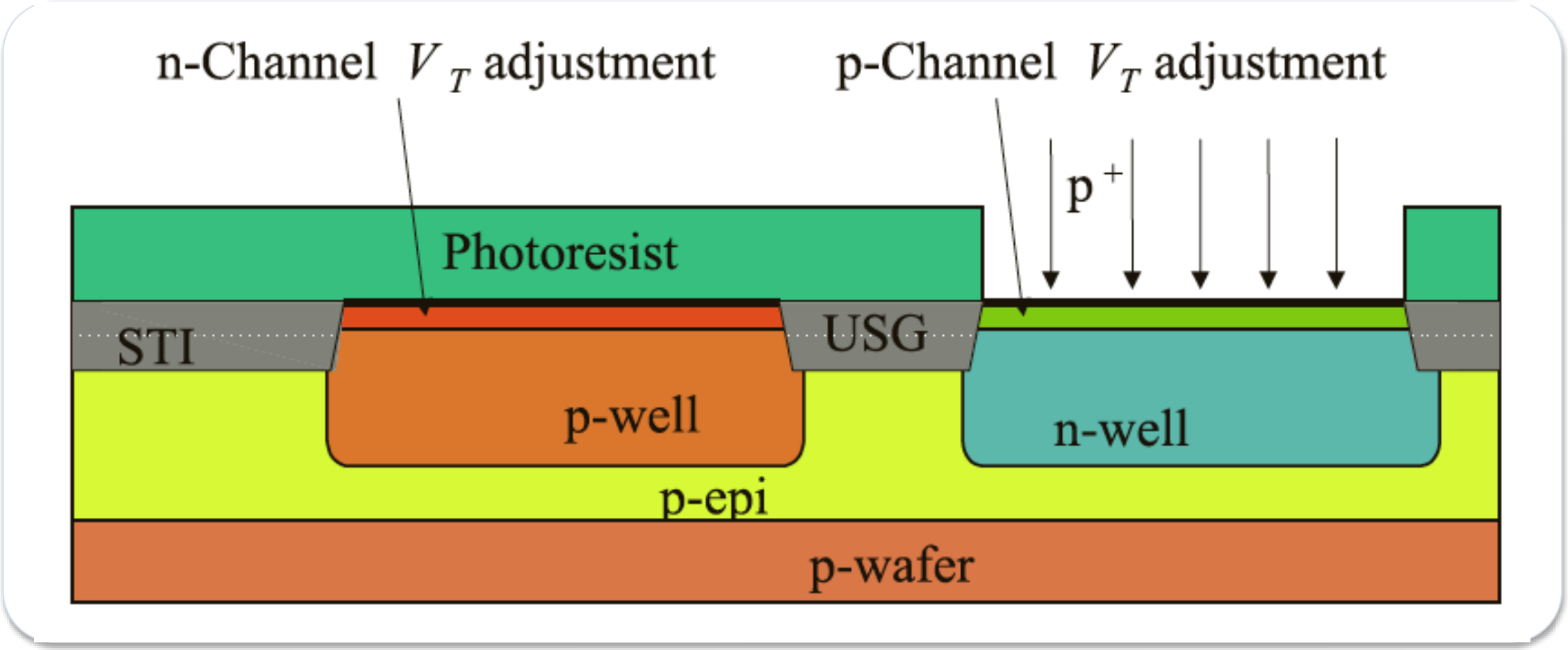
Vt Adjustment Ion Implantation



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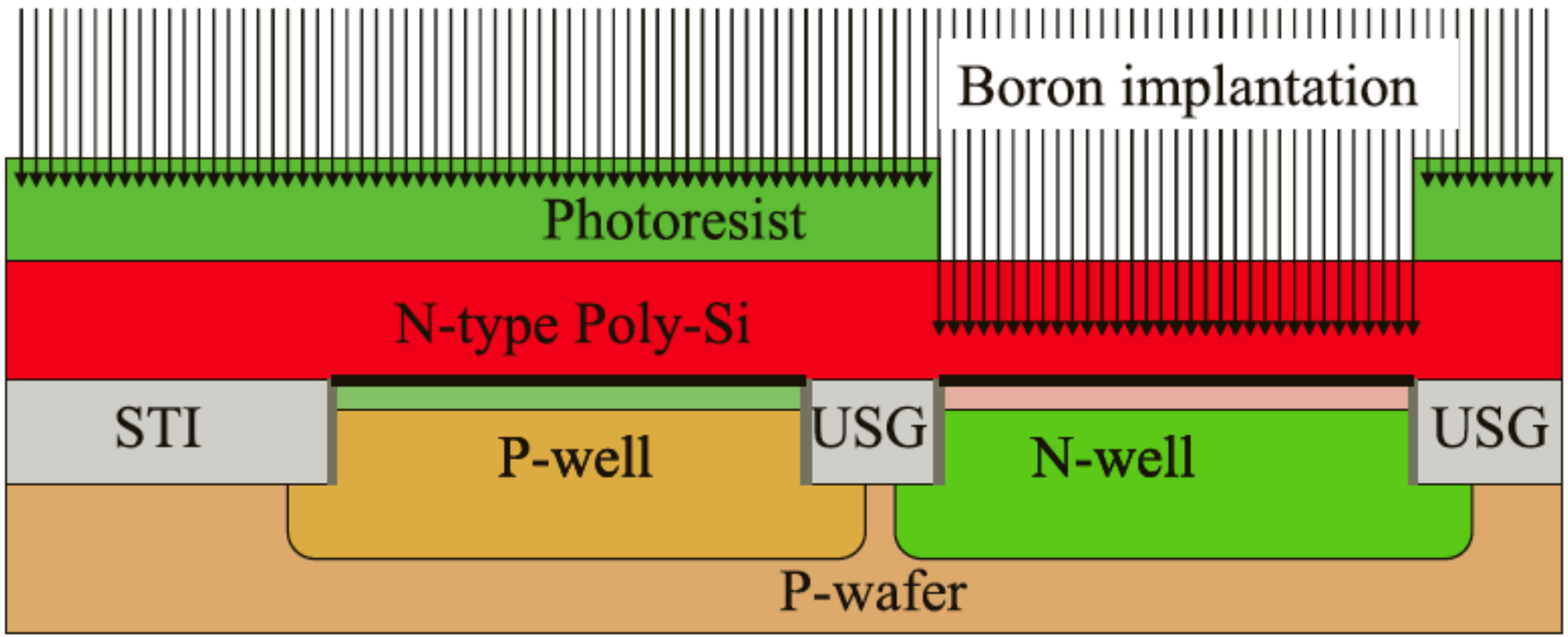
Poly Ion Implantation



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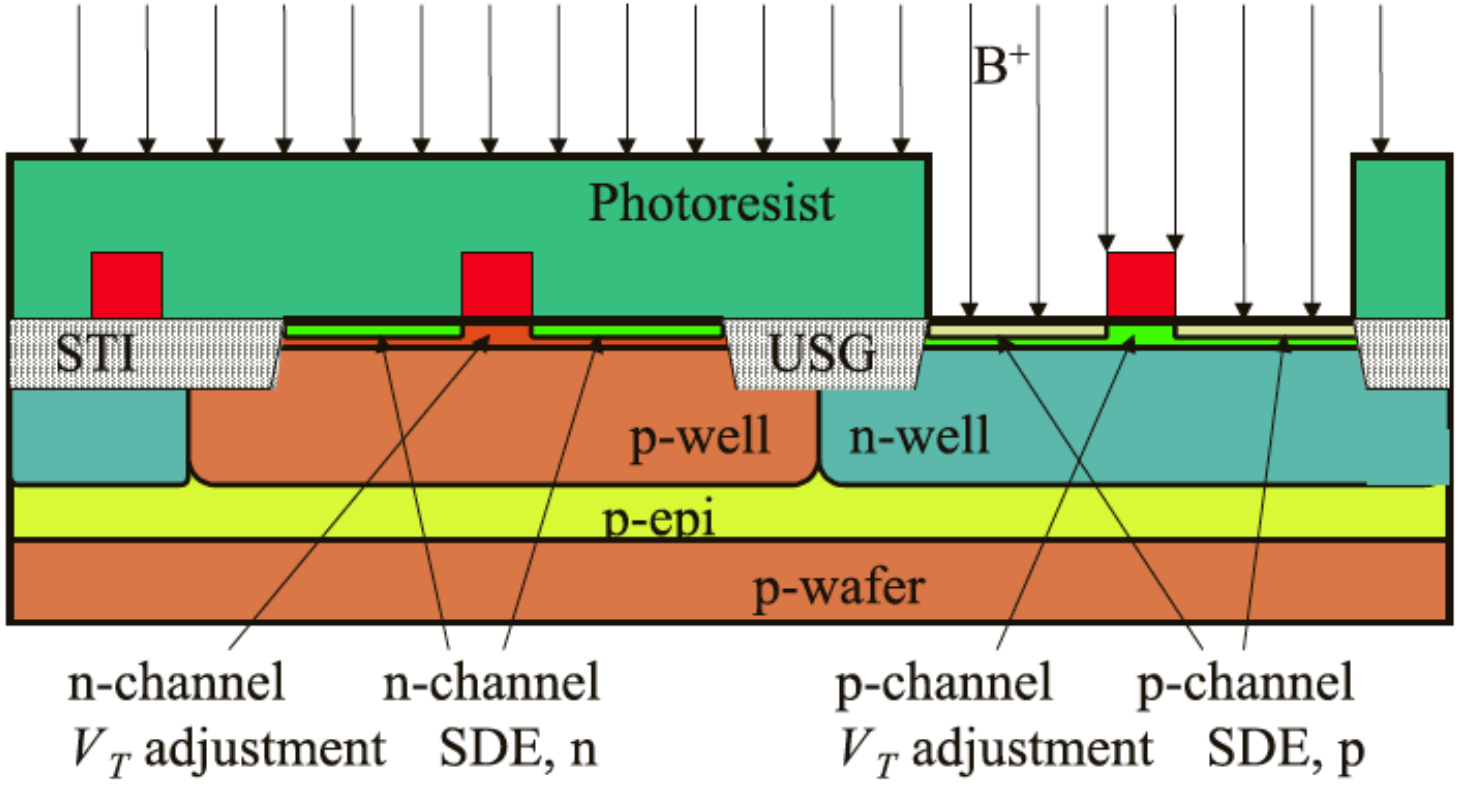
SDE Ion Implantation



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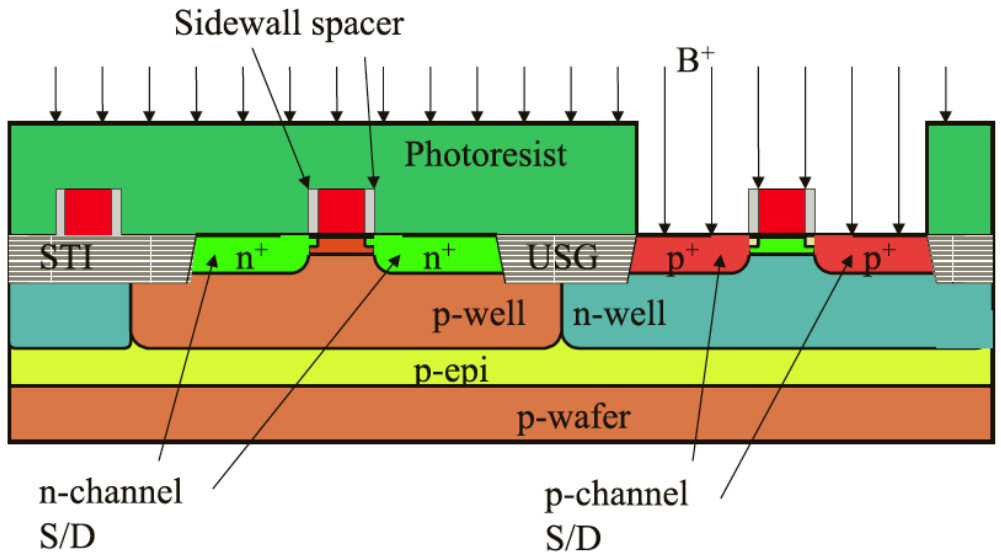
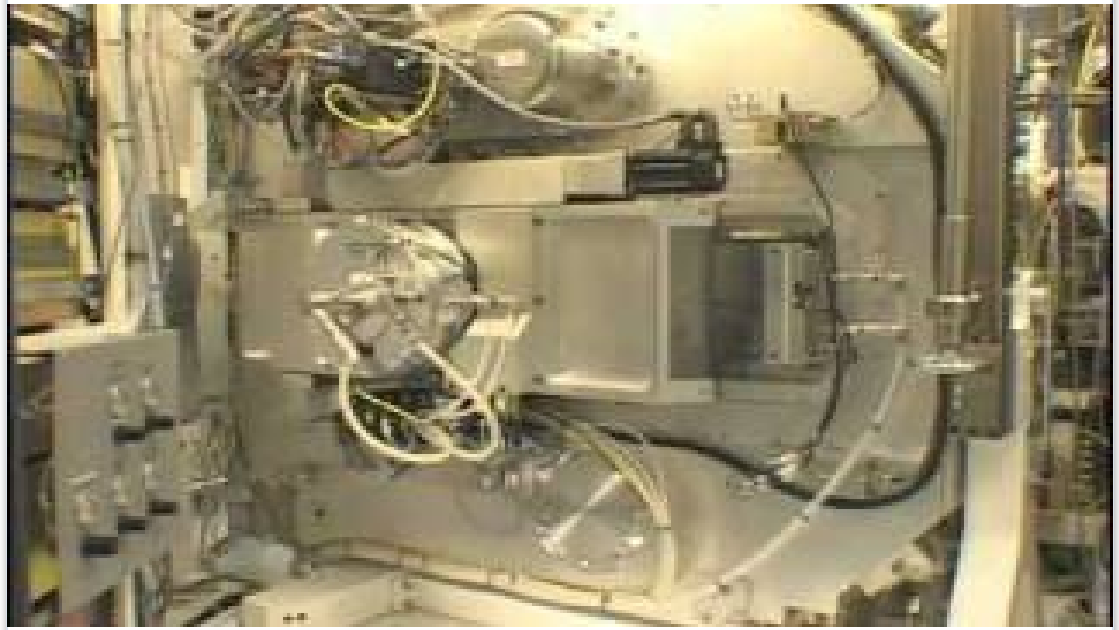
S/D Ion Implantation





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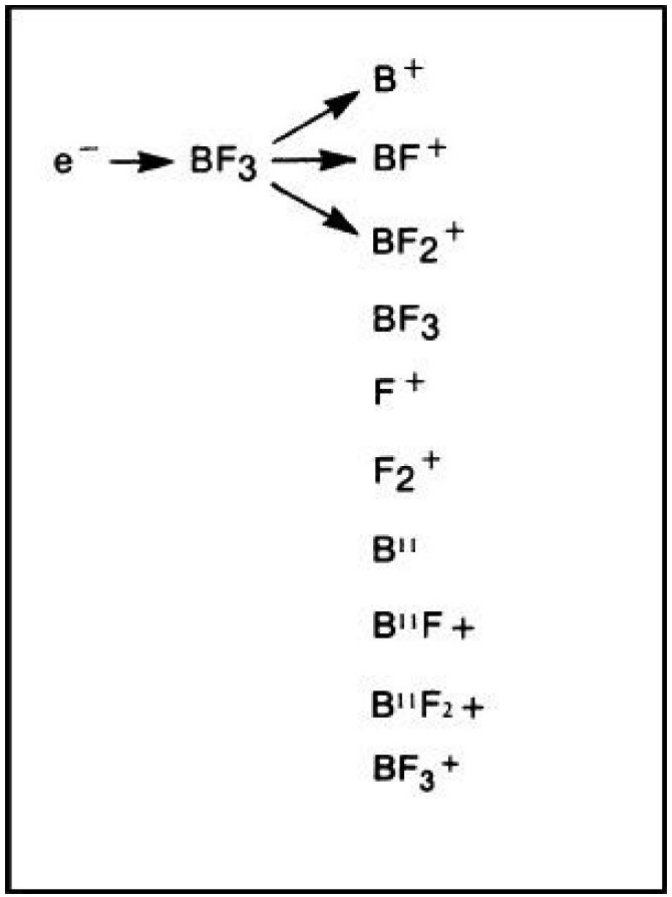
TOPIC 01

Ionization Chamber



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Evidence 1

Ions are atoms or molecules with a positive or negative charge
Ion implantation uses ionized atoms of the dopants
Ionization occurs in a vacuum chamber (~10^{-3} torr) fed by dopant source vapors
A heated filament generates electrons, which are attracted to a charged anode
As electrons travel to the anode, they collide with dopant source molecules, creating positively charged ions

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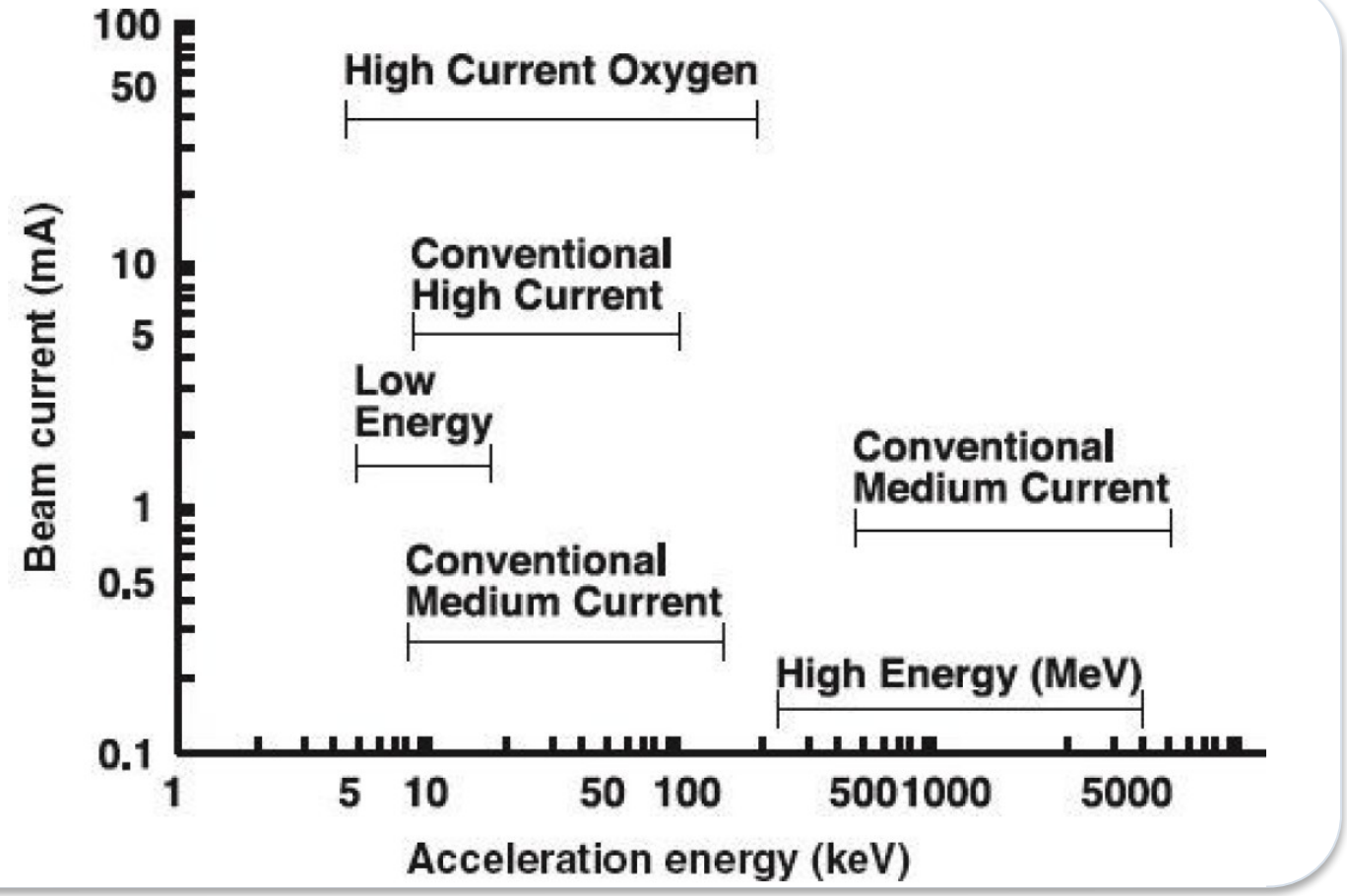
TOPIC 01

Wafer Charging



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Evidence 1

High-current implants can cause wafer charging, where excess positive charge builds on the wafer surface
Wafer charging attracts neutralizing electrons, potentially damaging dielectric layers, especially thin MOS gate dielectrics
Methods to reduce or neutralize charge:
Flood guns supplying electrons
Plasma bridge for low-energy electrons
Magnetic fields to control electron paths
High-energy implanters accelerate ions from 10 keV to 3.0 MeV with beam currents up to 1.0 mA
Oxygen-implant machines are used in SOI applications for high-current oxygen implantation

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TOPIC 01

Beam Focus

Upon exiting the acceleration tube, the ion beam separates due to charge repulsion, leading to uneven ion density and nonuniform wafer layers
Beam focusing is essential for successful implantation
Electrostatic or magnetic lenses are used to focus the ions into a small beam or parallel beams
Parallel beams are crucial for transistor-gate applications, as deviations can cause uneven dopant doses, impacting transistor performance

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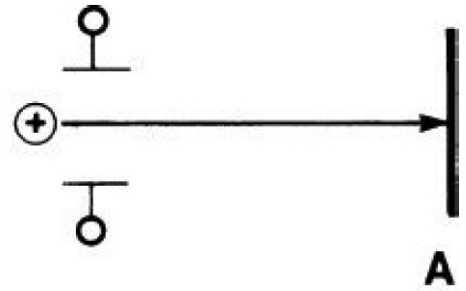
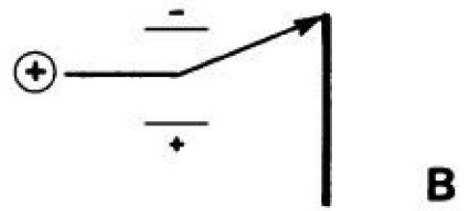
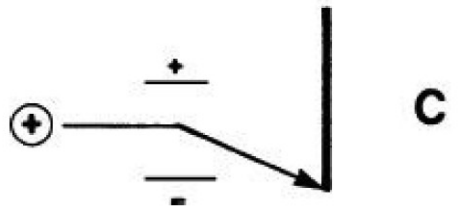
TOPIC 01

Beam Scanning







1

Evidence 1

Ion beams are smaller than the wafer (~1 cm), requiring scanning for uniform doping
Three methods for scanning: beam scanning, mechanical scanning, and shuttering (used alone or combined)
In beam scanning, the beam passes between electrostatic plates with controllable charges
By adjusting the positive and negative charges, the beam is swept across the wafer in a raster scan pattern to ensure uniform coverage

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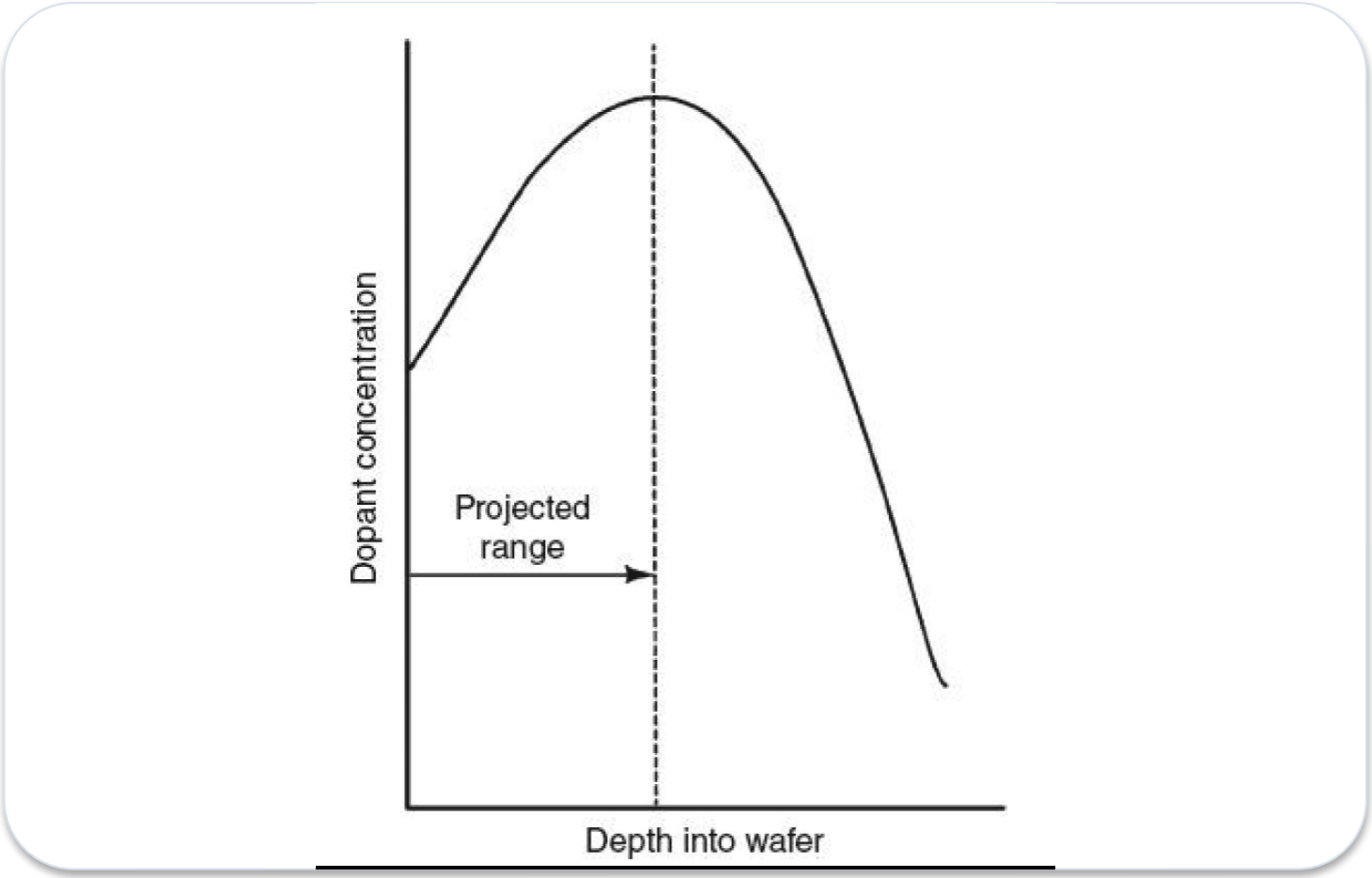
TOPIC 01

Dopant concentration profile after an ion implant.



1

Evidence 1

Ion distribution in ion implantation differs from diffusion-based doping
In ion implantation, dopant dose depends on beam current density and implant time
Ion location in the wafer is influenced by:
Ion energy and mass (heavier ions and higher energy penetrate deeper)
Wafer orientation (crystal planes with varying atom densities affect ion stopping)
Ions are slowed and stopped by:
Electronic interactions with wafer electrons
Physical collisions with wafer atom nuclei

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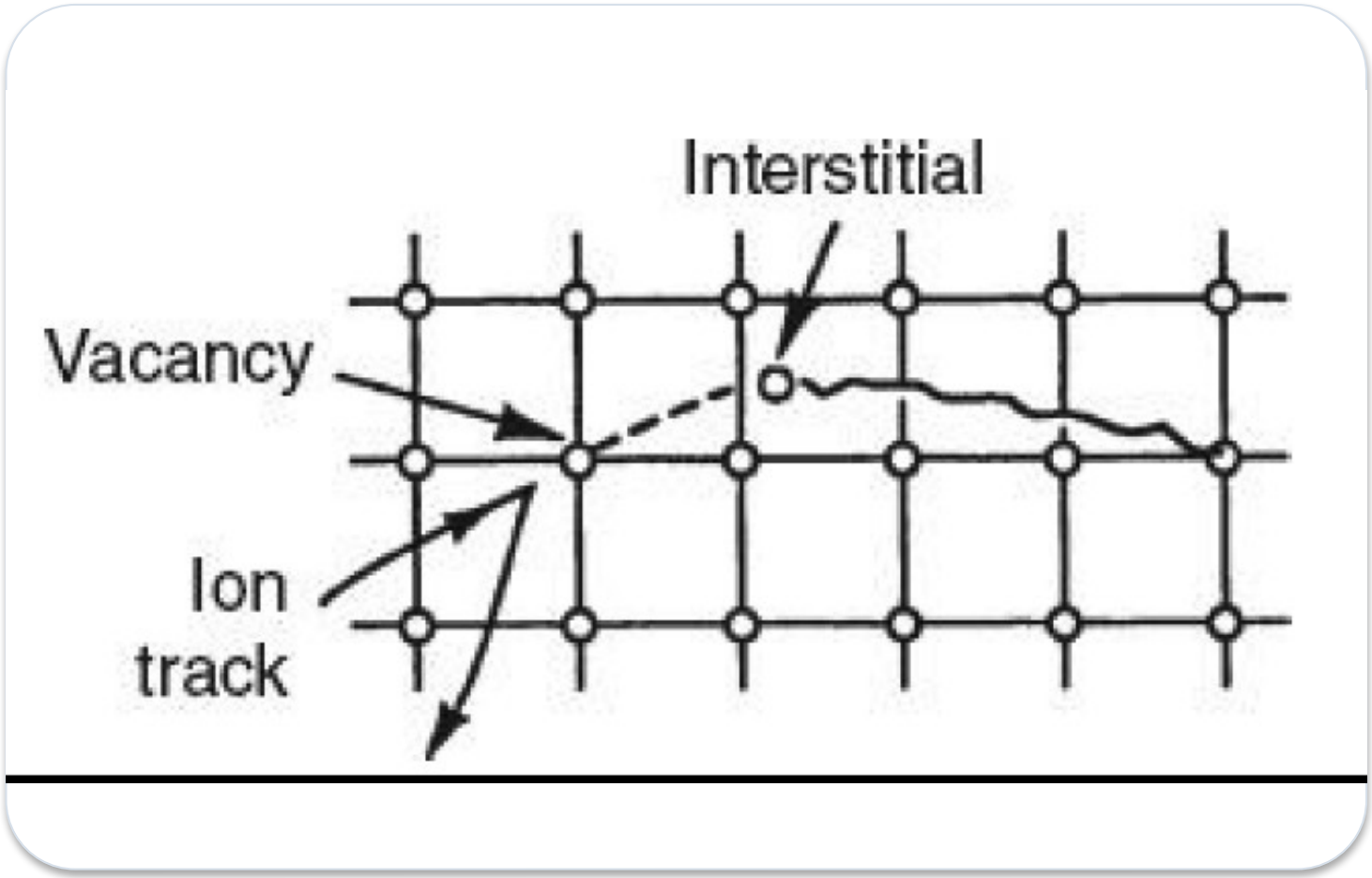
TOPIC 01

Crystal Damage



1

Evidence 1

Lattice damage: ions collide with host atoms, displacing them from their lattice sites
Damage clusters: displaced atoms further displace other substrate atoms, creating a cluster of displaced atoms
Vacancy-interstitial: the most common defect, where an incoming ion knocks a substrate atom from its lattice site, and the displaced atom settles in a non-lattice position

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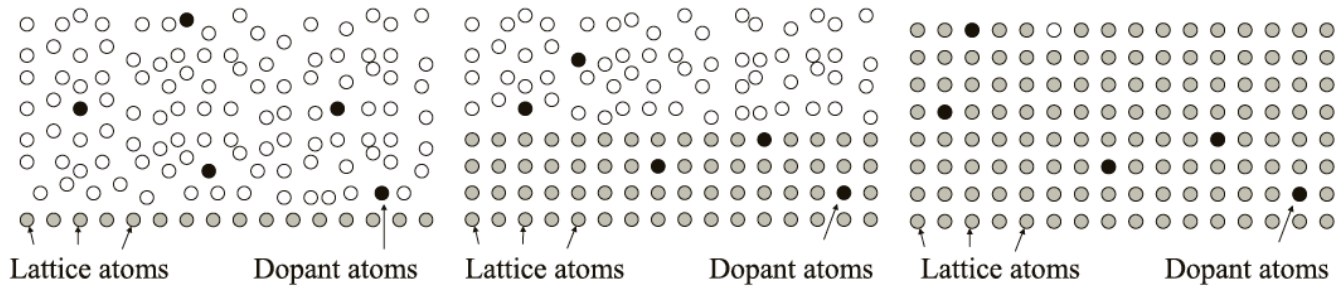
TOPIC 01

Annealing and Dopant Activation



1

Evidence 1

Crystal damage repair and electrical activation of dopants achieved through thermal annealing
Annealing temperature is kept below the dopant diffusion threshold to prevent lateral diffusion
Typical tube furnace annealing: 600–1000°C in a hydrogen atmosphere, lasting 15–30 minutes
RTP (Rapid Thermal Processing) is also used:

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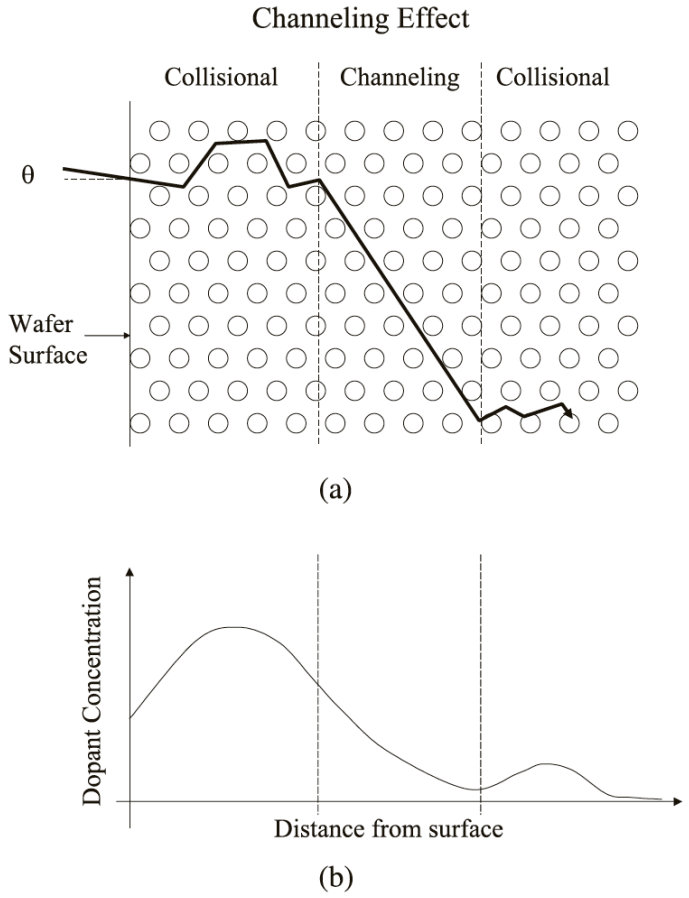
TOPIC 01

Channeling



1

Evidence 1

The crystalline structure of the wafer can cause channeling during ion implantation
Channeling occurs when the crystal's major axis aligns with the ion beam, allowing ions to penetrate much deeper than expected, up to 10 times the calculated depth
Ion concentration profiles in channeled areas show excessive dopant penetration

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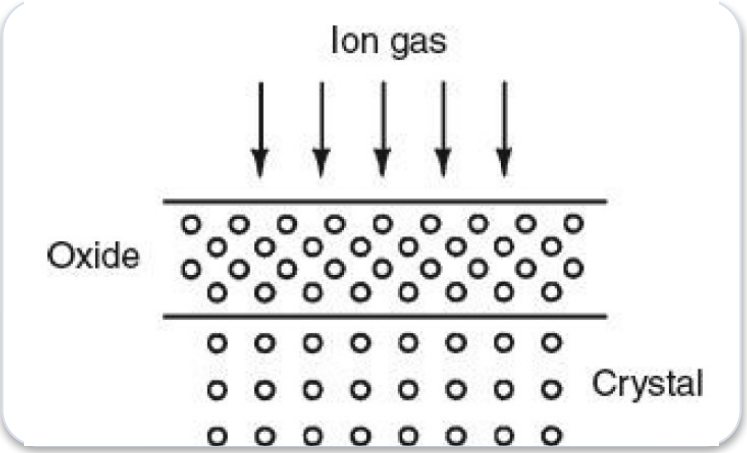
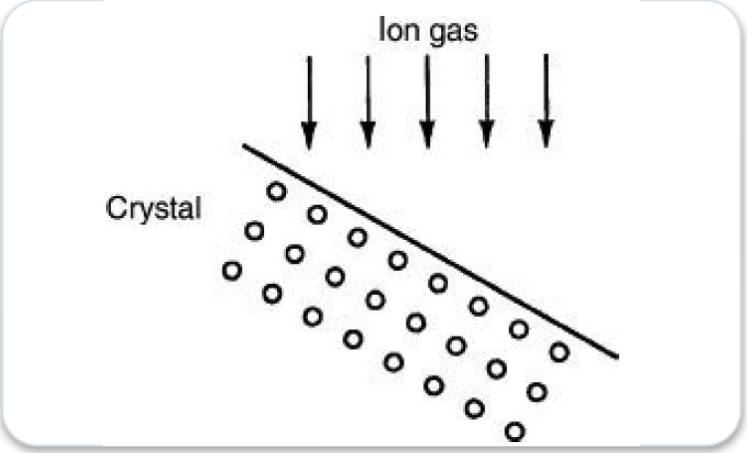
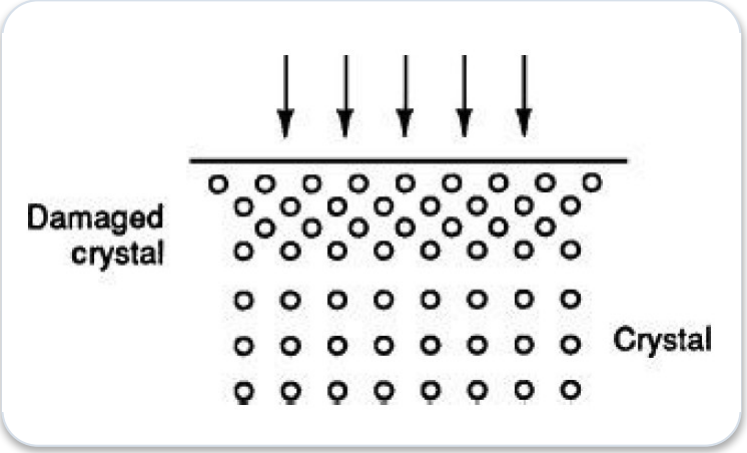
TOPIC 01

Techniques to minimize channeling







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Apply a blocking amorphous surface layer

Apply a blocking amorphous surface layer
Slightly misorient the wafer relative to the ion beam
Create a damage layer on the wafer surface

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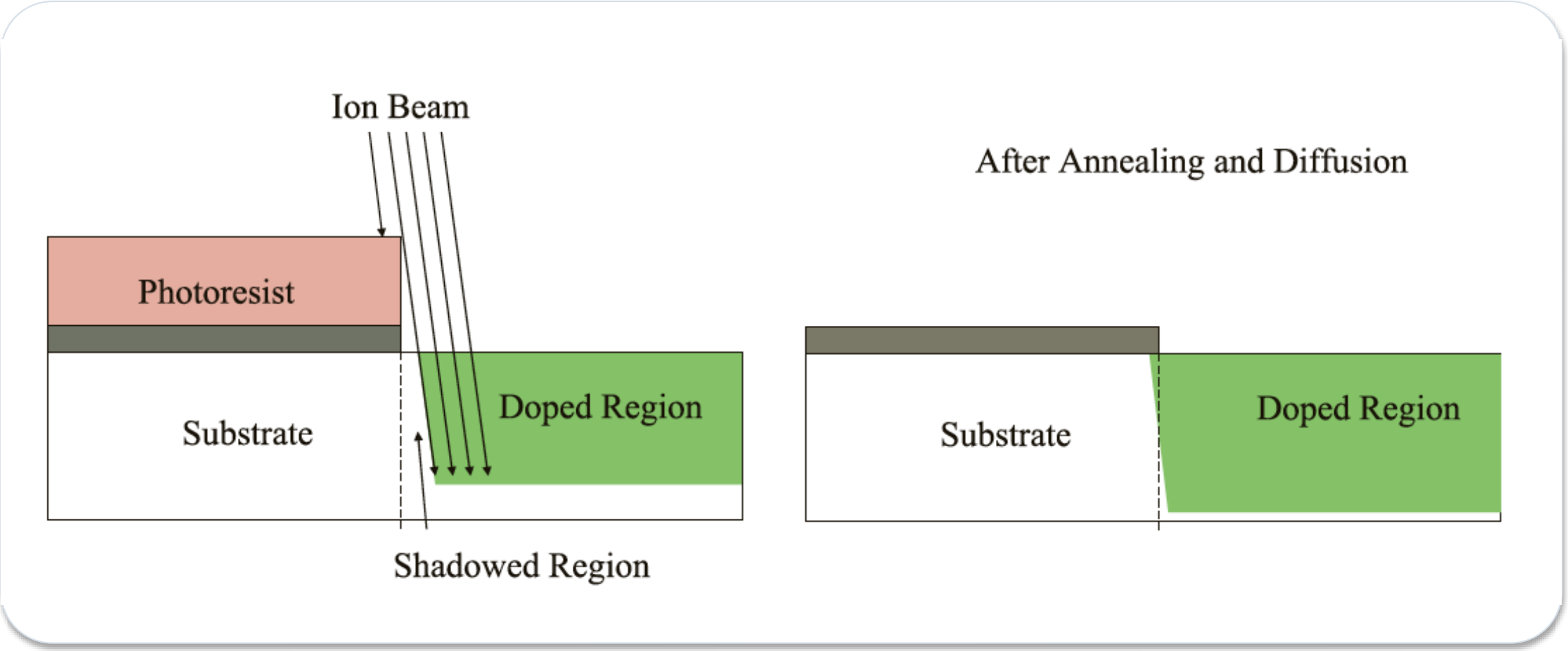
Shadowing Effect



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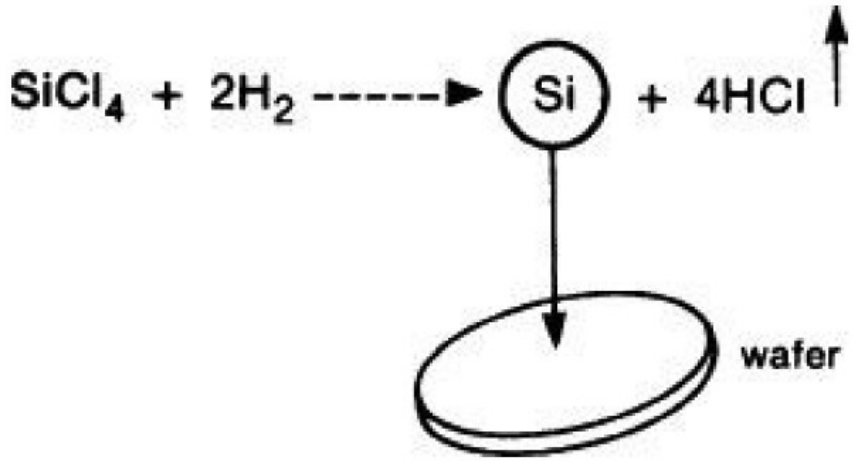
TOPIC 01

Chemical Vapor Deposition

Deposition: process of physically depositing material on the wafer surface
CVD (Chemical Vapor Deposition): primary technique for depositing films in semiconductor processing
CVD process:
Chemicals (C) containing desired atoms are mixed and reacted in a chamber to form a vapor (V)
Atoms/molecules from the vapor deposit (D) on the wafer surface, building up to form a film



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Examples of CVD reactions.

Pyrolysis $SiH_4 \rightarrow Si + 2H_2$

Reduction $SiCl_4 + 2H_2 \rightarrow SI + 4HCL$

Oxidation $SiH_4 + O_2 \rightarrow SiO_2 + 2H_2$

Nitridation $3SiH_2Cl_2 + 4NH_3 \rightarrow Si_3N_4 + pH + 6H_2$

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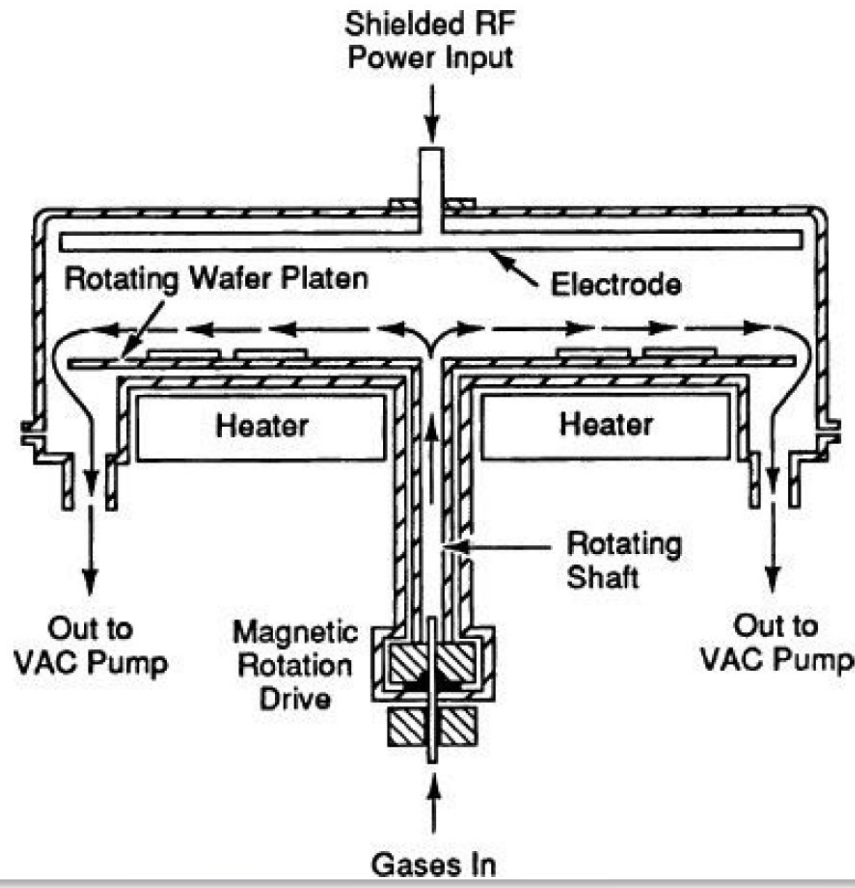
TOPIC 01

Plasma-Enhanced CVD (PECVD)



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Evidence 1

Plasma-Enhanced CVD (PECVD): Developed to replace silicon dioxide with silicon nitride, allowing for lower deposition temperatures.
Benefits: Enables deposition over aluminum layers without alloying.
PECVD Reactor: Physically similar to plasma etch systems, with a parallel-plate chamber and plasma source.
In Situ Cleaning: PECVD reactors can use plasma for etching and cleaning prior to deposition, reducing contamination.

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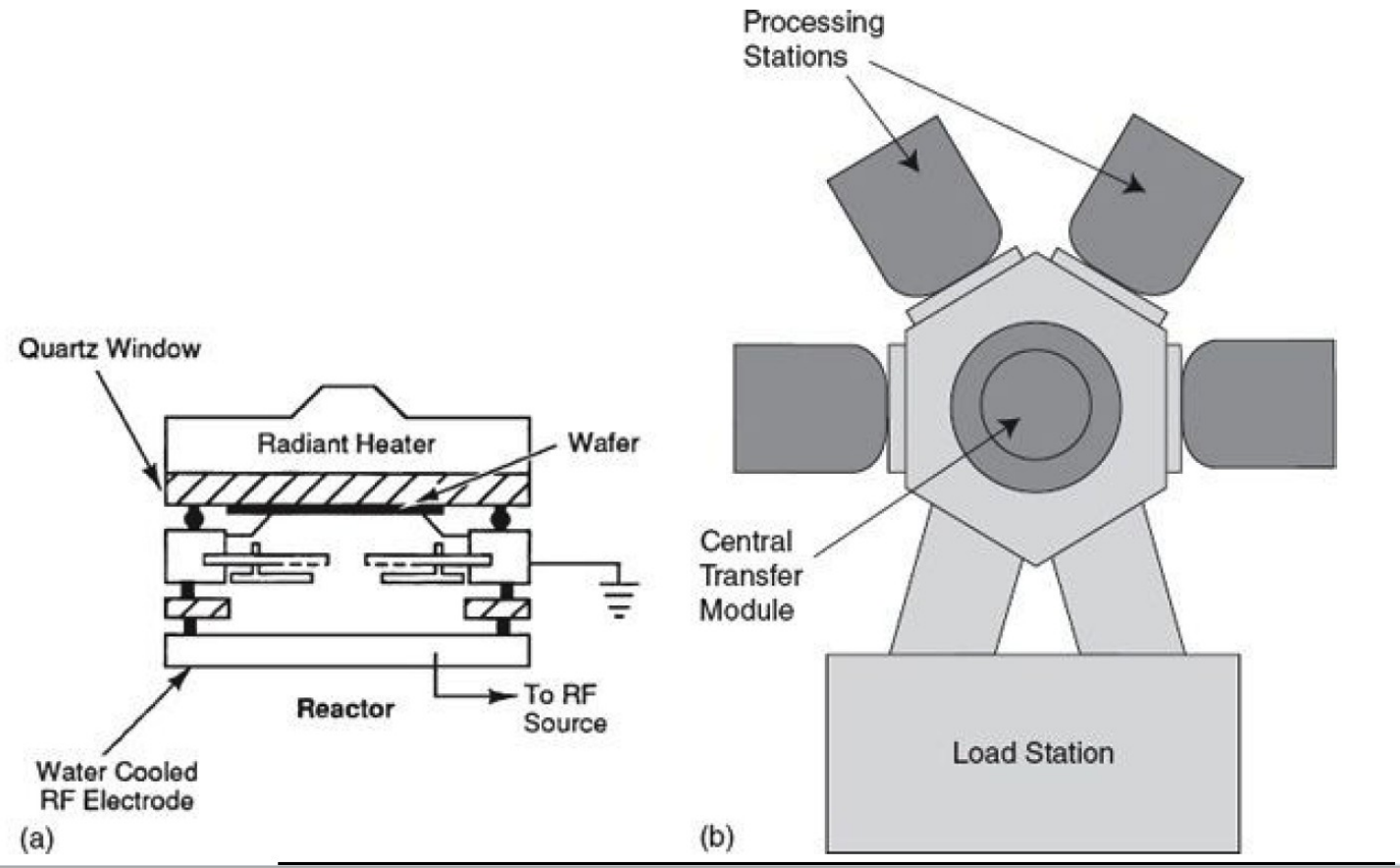
Multi-chamber PECVD Tool



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Epitaxial Process

Cycle	Temperature	Gas	Purpose
1	Room	N_2	Purge air from system
2	Room	H_2	Reduce any organic contaminants of wafers in system
3	(Heating)	N_2	Bring system to deposition temperature
4	Deposition Temperature	HCl	Etch wafer to prepare surface for epi deposition
5	Deposition Temperature	Source + Dopant + Carrier	Grow epitaxial film
6	(Heat Off)	N_2	Purge system of reactant gases

1

Evidence 1

Epitaxial Process: Involves rigorous wafer cleaning, deposition, and cooldown steps.
Process Steps: Gas-phase cleaning, deposition, and cooldown, all requiring precise control of temperature and gas flow.

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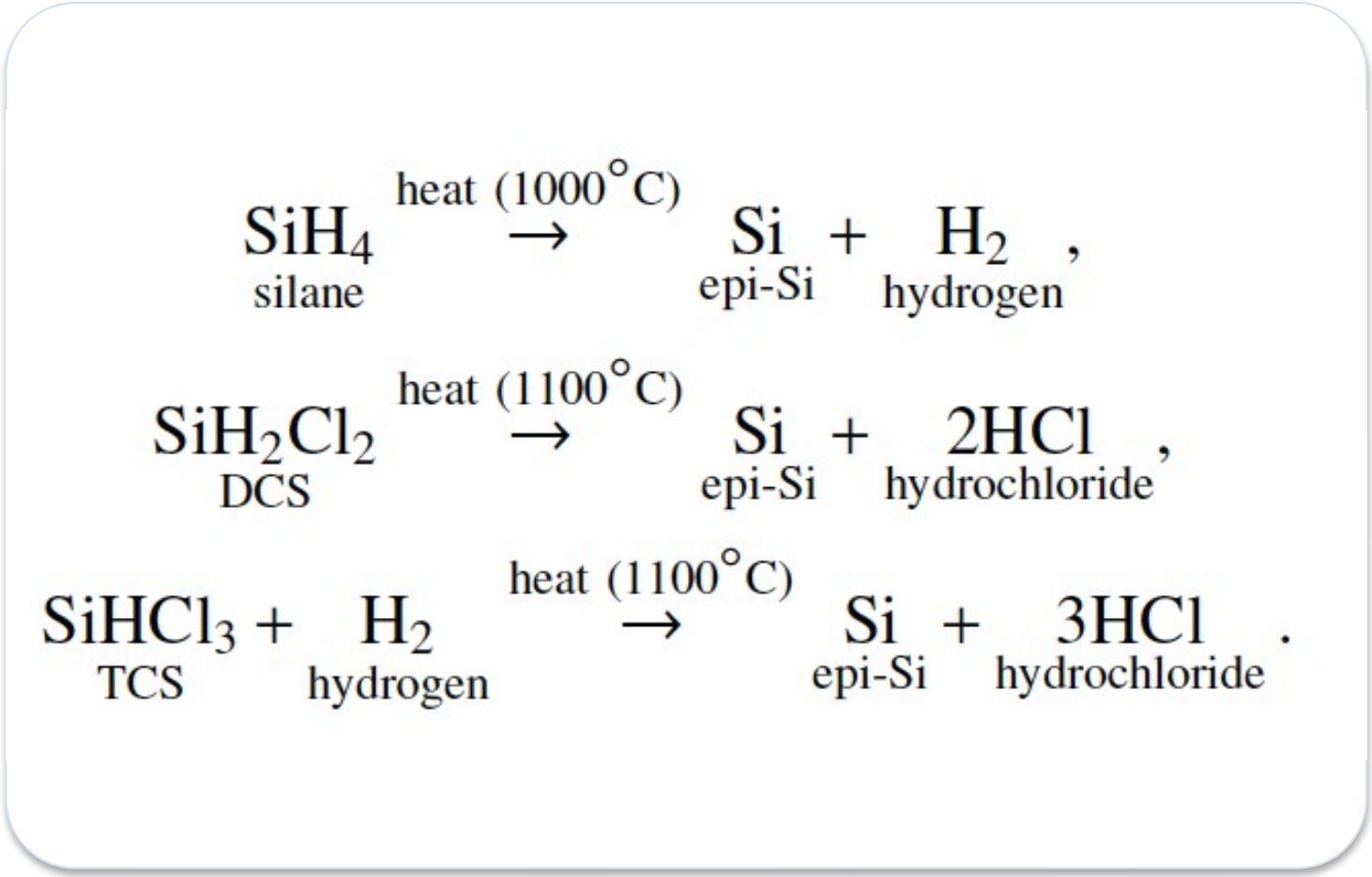
TOPIC 01

Epitaxial silicon deposition



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Evidence 1

Epitaxial silicon is a single-crystal silicon layer deposited on a single-crystal silicon wafer surface in a high-temperature process. An epitaxial silicon layer is needed for bipolar transistors, BiCMOS IC chips, and high-speed, advanced CMOS IC chips.

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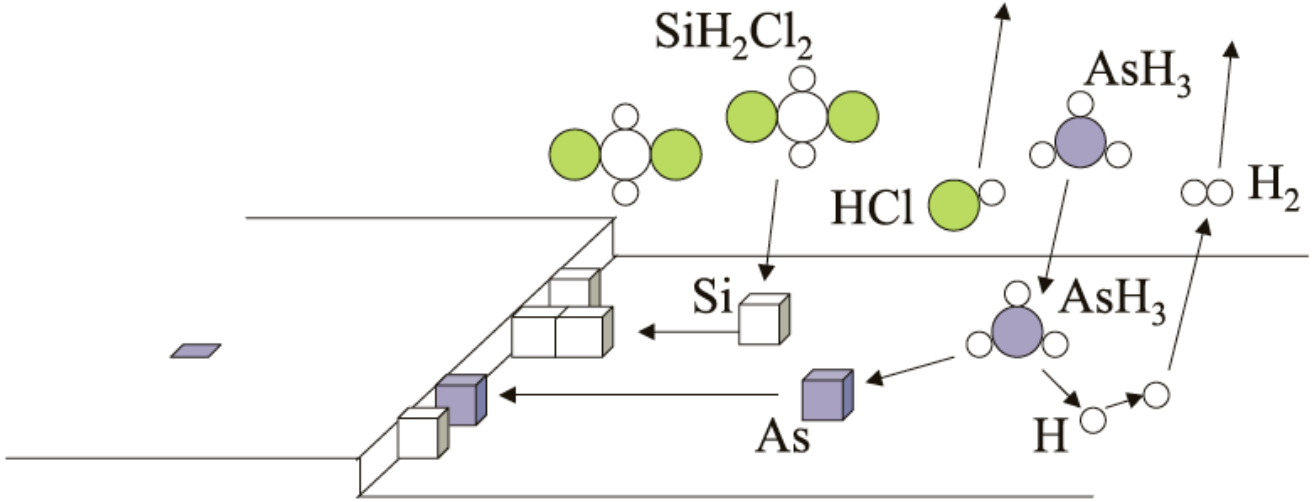
Epitaxial growth process



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Evidence 1

The process begins with precursors like DCS and AsH_3 entering the reactor. The steps include:
Precursor molecules diffuse to the wafer surface and adsorb.
They dissociate and react on the surface.
Adatoms (solid byproducts) migrate and bond with surface atoms, aligning with the substrate's crystal structure.
Volatile byproducts desorb from the hot surface and diffuse out of the reactor

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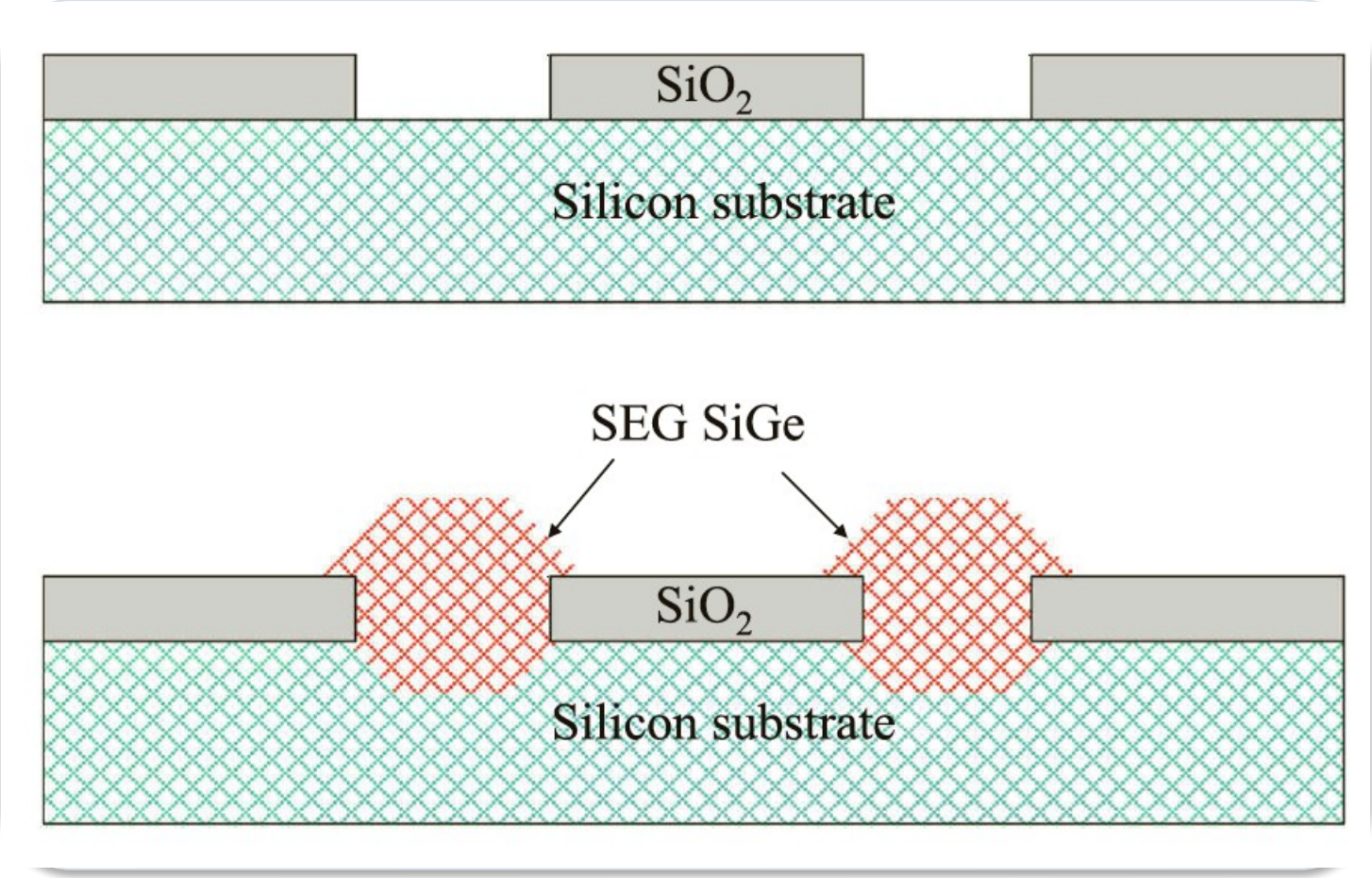
TOPIC 01

Selective Epitaxial Growth (SEG)



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Evidence 1

Selective Epitaxial Growth: Deposits epitaxial films through holes in dielectric layers, isolating devices. Advantages: Creates isolated devices, simplifying fabrication processes

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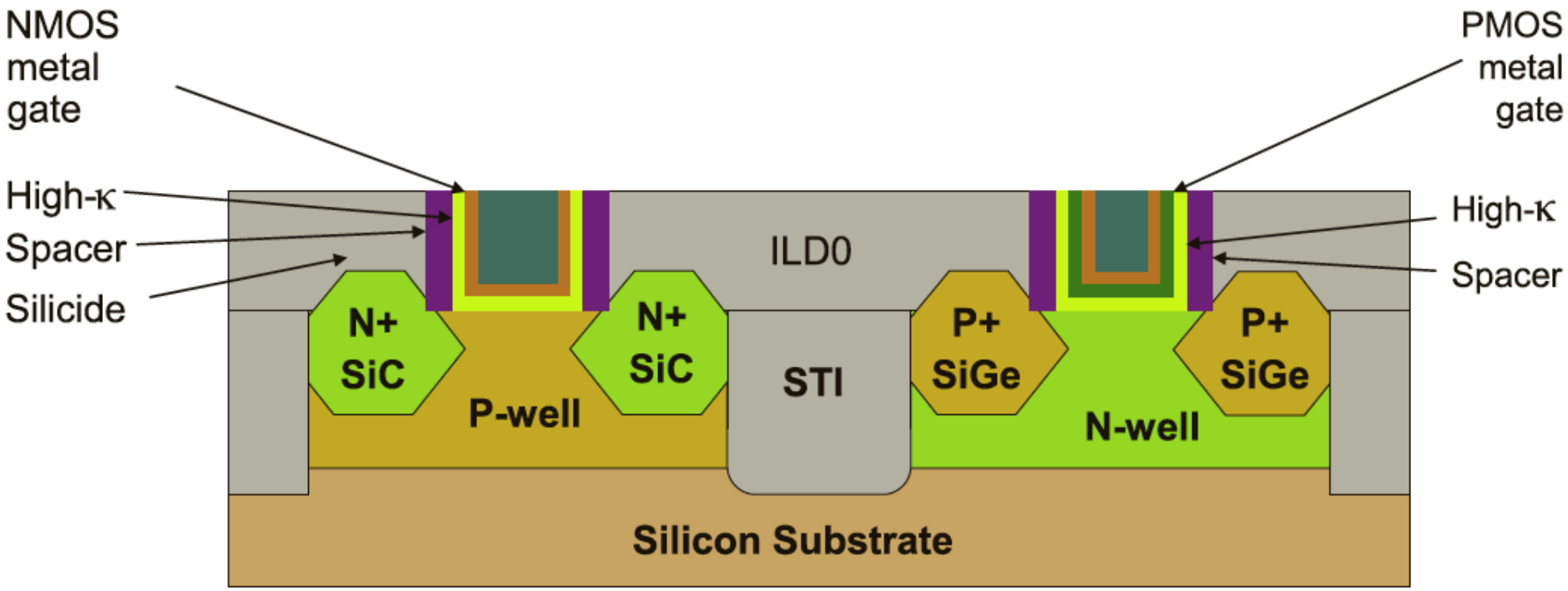
SEG Application



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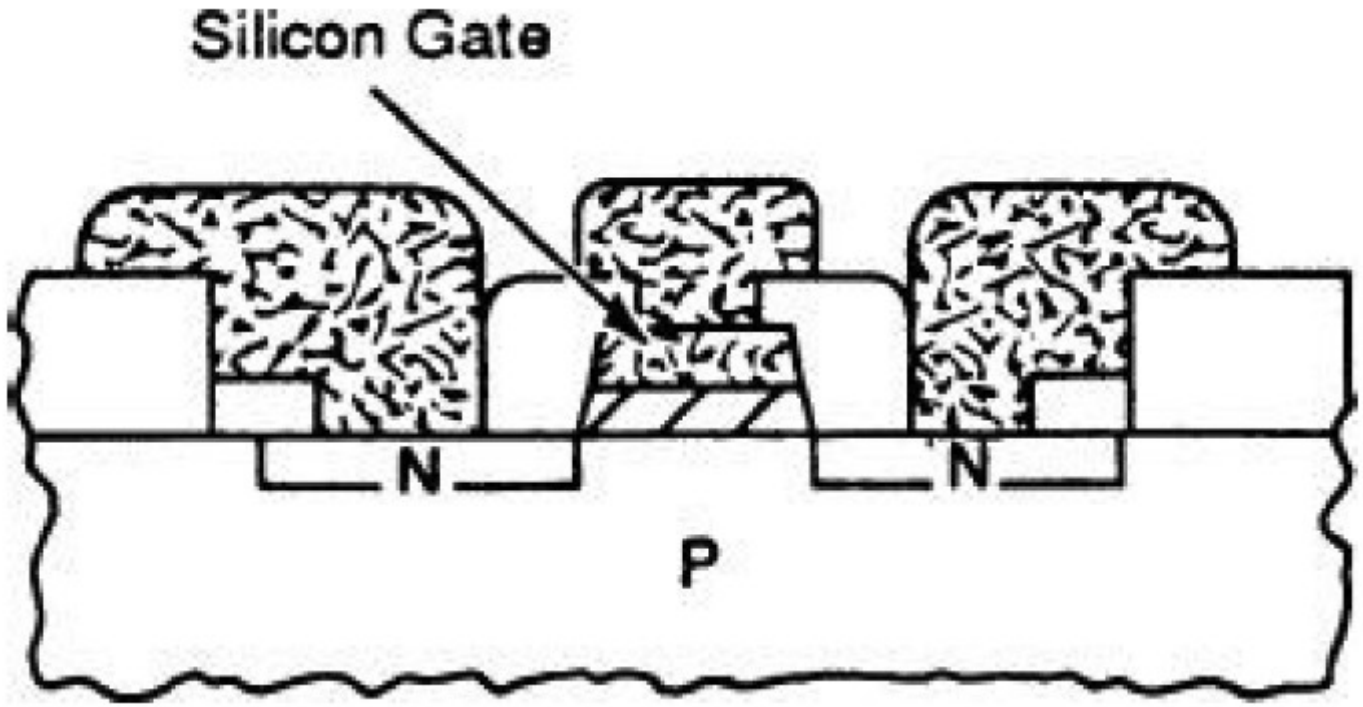
TOPIC 01

Polysilicon Deposition



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Evidence 1

Polysilicon Importance: Became crucial with the development of silicon-gate MOS devices.
Polysilicon Applications: MOS gates, load resistors, trench fills, multilayer poly, contact barrier layers, and bipolar emitters.

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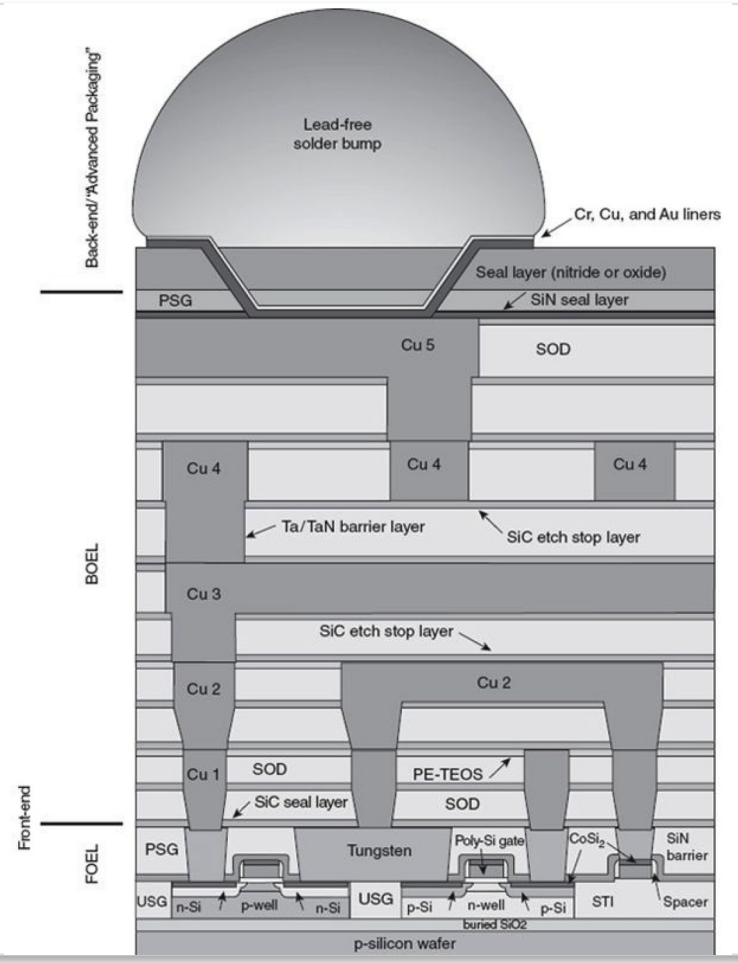
TOPIC 01

Multilevel Metal Schemes

Multilevel Metallization: Used to increase wiring density on chips by stacking multiple metal layers.
Key Components: Barrier layer, intermetallic dielectric layer (IDL), vias, and conducting plugs.
Challenges: Increased cost, lower yield, and planarization requirements.



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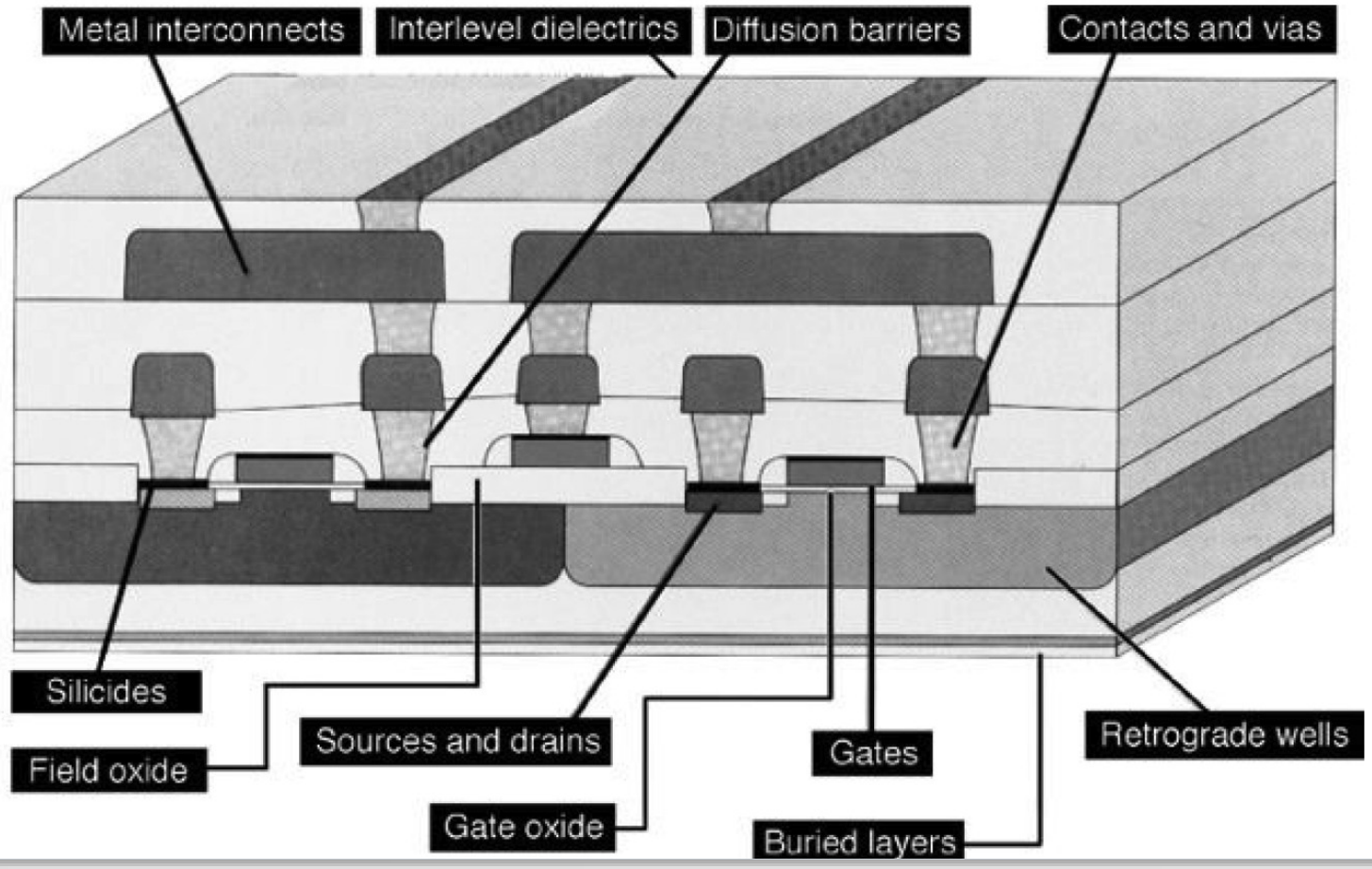
Two-metal structure.



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TOPIC 01

Barrier Metals

1

Key idea 1

Barrier Layers: Used to prevent alloying between silicon and aluminum or copper metallization.
Barrier Materials: TiW, TiN, Ta, and TaN are common barrier materials.
Applications: TiW used for aluminum metallization, TiN for both aluminum and copper

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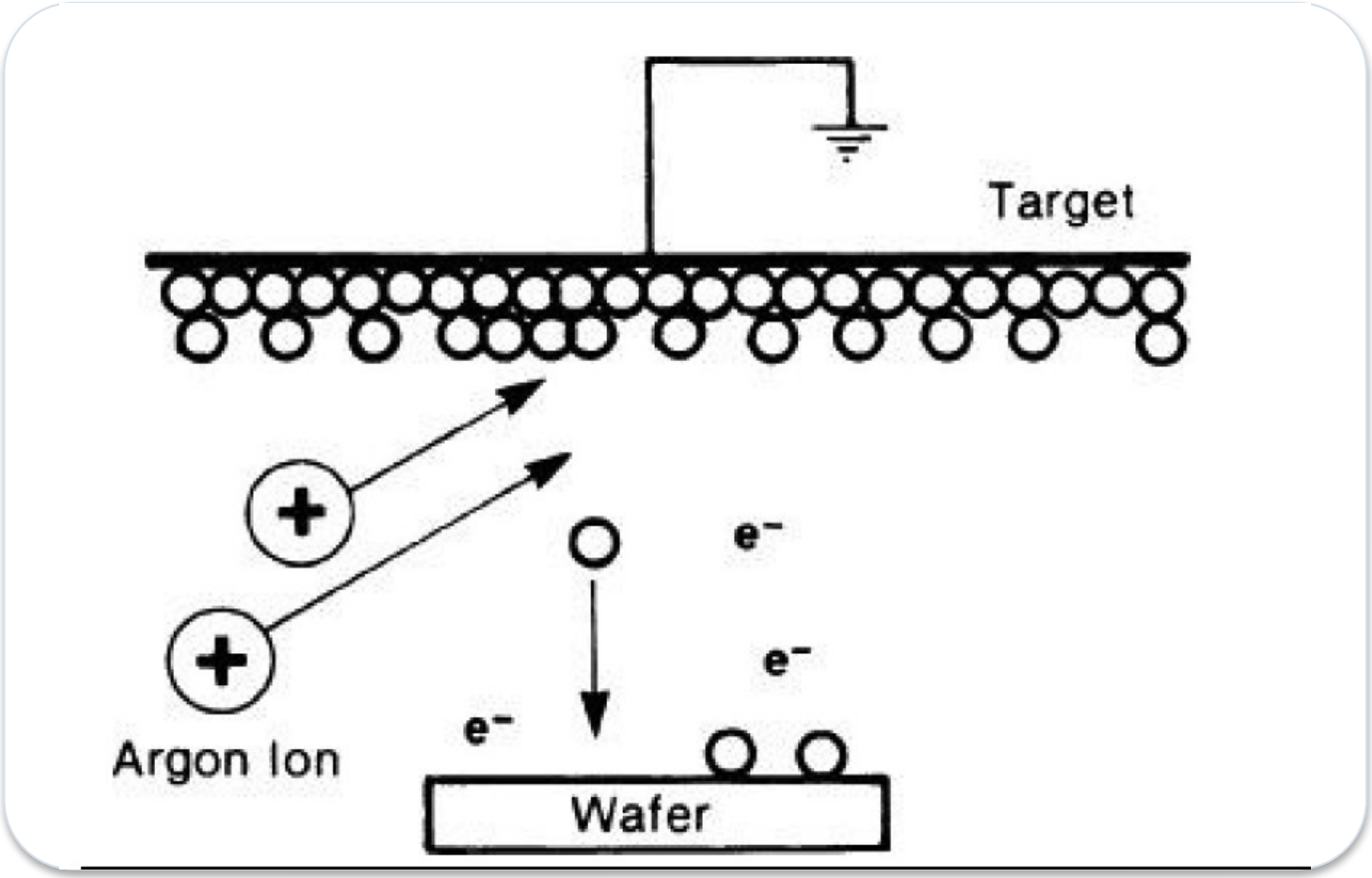
TOPIC 01

Sputtering Deposition



1

Evidence 1

Sputtering: A physical vapor deposition (PVD) process where ions bombard a target material, causing atoms to be ejected and deposited on a substrate.
Process Overview: Argon ions are accelerated towards a grounded target, knocking off atoms and depositing them on the wafer.

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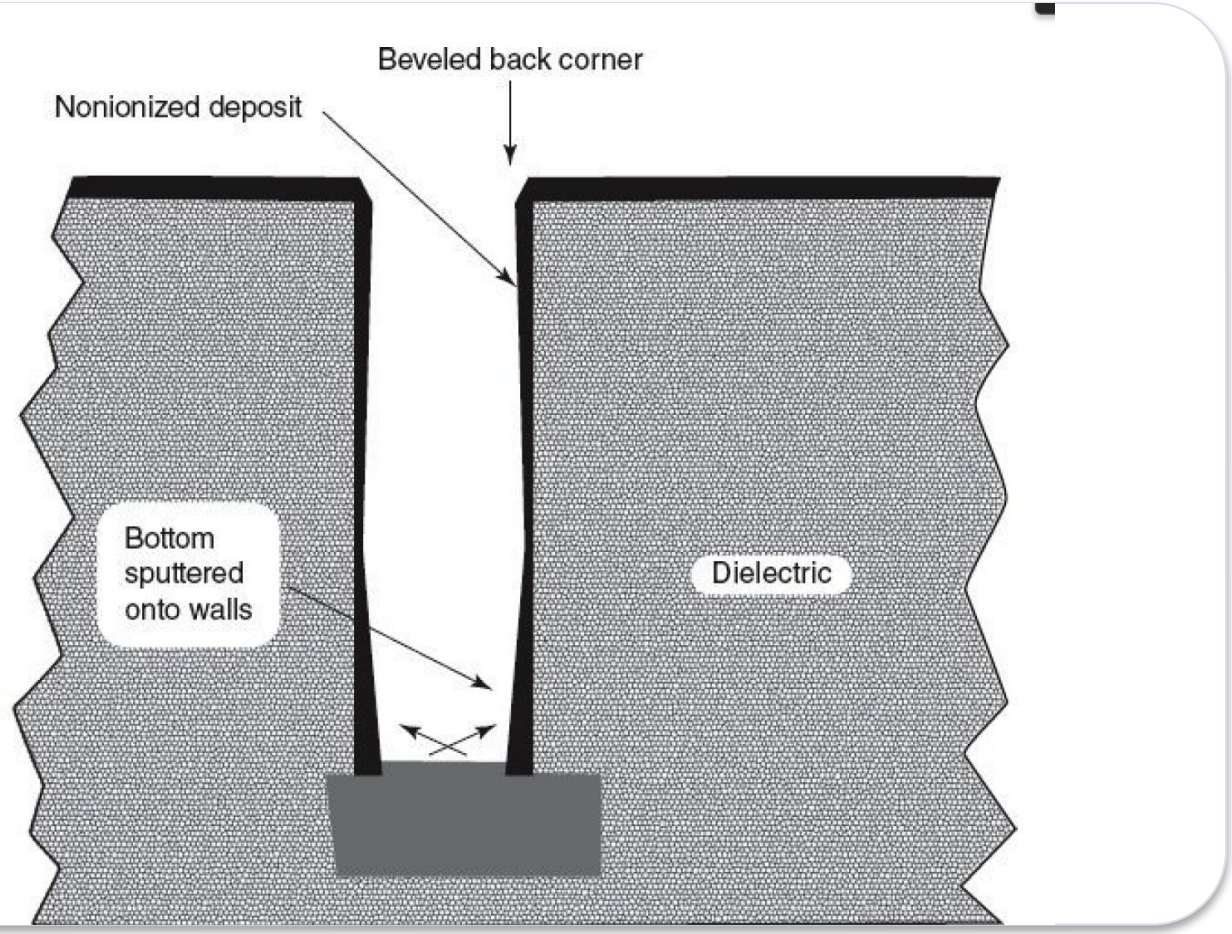
TOPIC 01

Resputtering



1

Evidence 1

Ionized Physical Vapor Deposition (I-PVD):
Enhances sputter deposition by introducing ions into the plasma, improving uniformity in deep holes.
Mechanism: Metal ions are attracted to the wafer, depositing directly into holes.
Resputtering: Ions can also sputter material at the bottom of the holes, depositing it on the sidewalls for better coverage.

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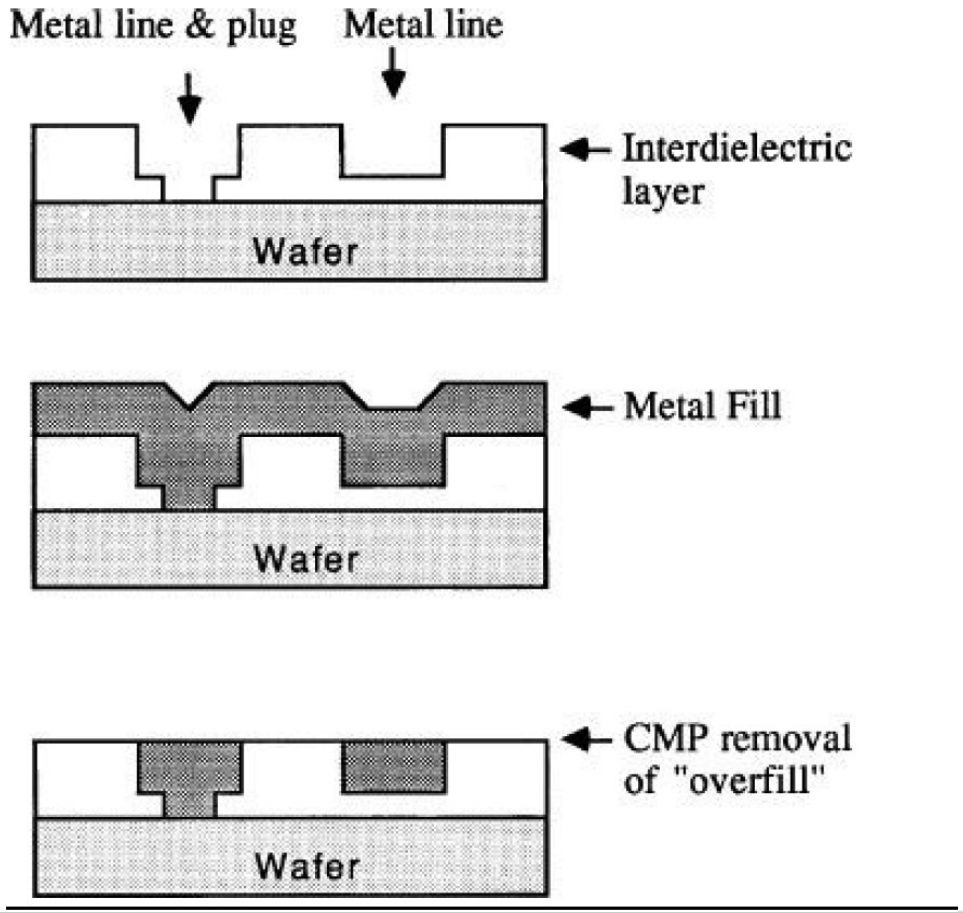
TOPIC 01

Dual-Damascene Copper Process



1

Evidence 1

Copper Metallization Challenges: Etching difficulty, high resistance to silicon, diffusion through silicon dioxide, and poor adhesion.
Damascene Process: Developed to overcome copper challenges, involving trench formation, copper plating, and chemical-mechanical polishing (CMP).
Advantages: Superior dimensional control and eliminates width variations from etching.

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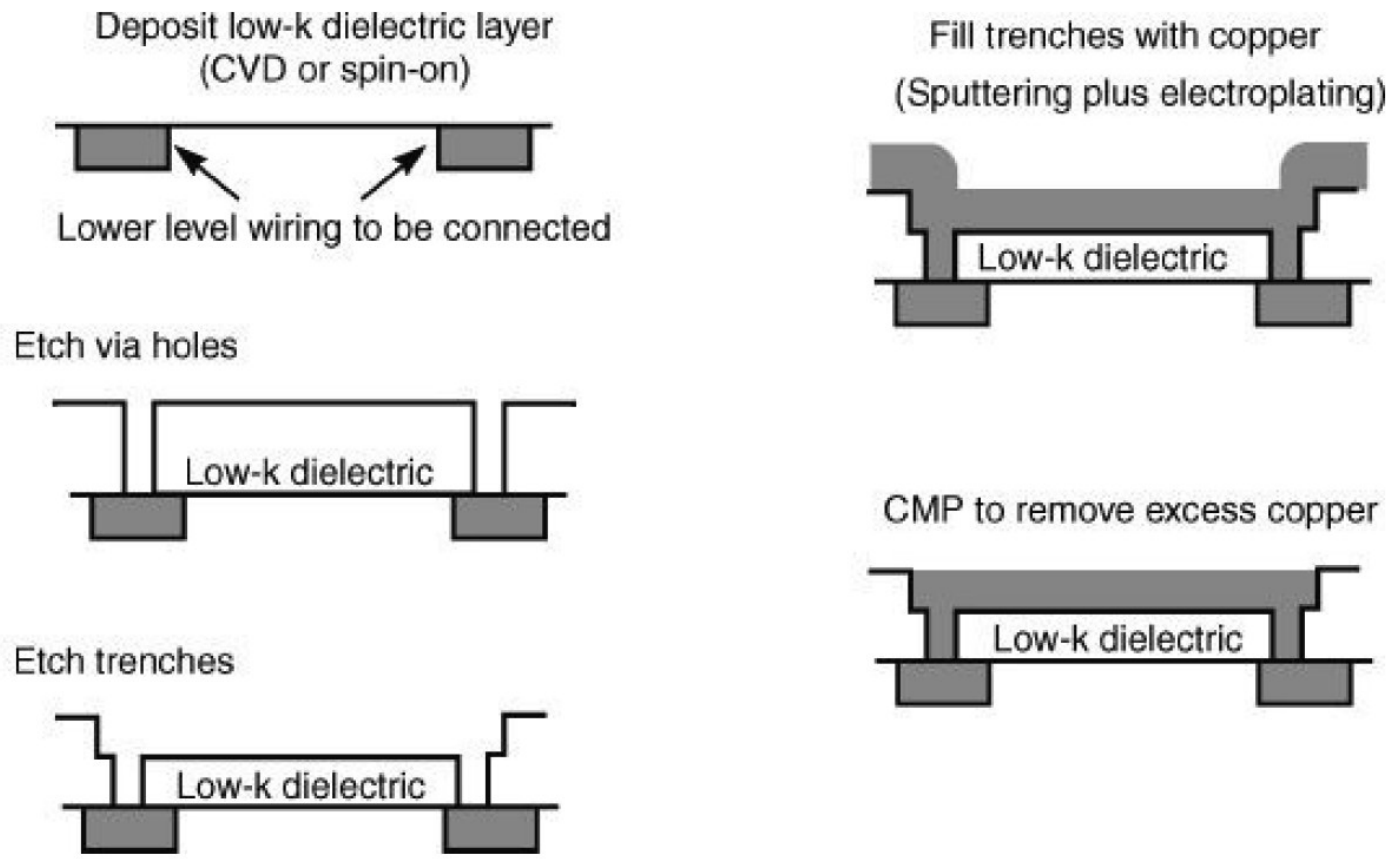
Dual-Damascene Copper Process



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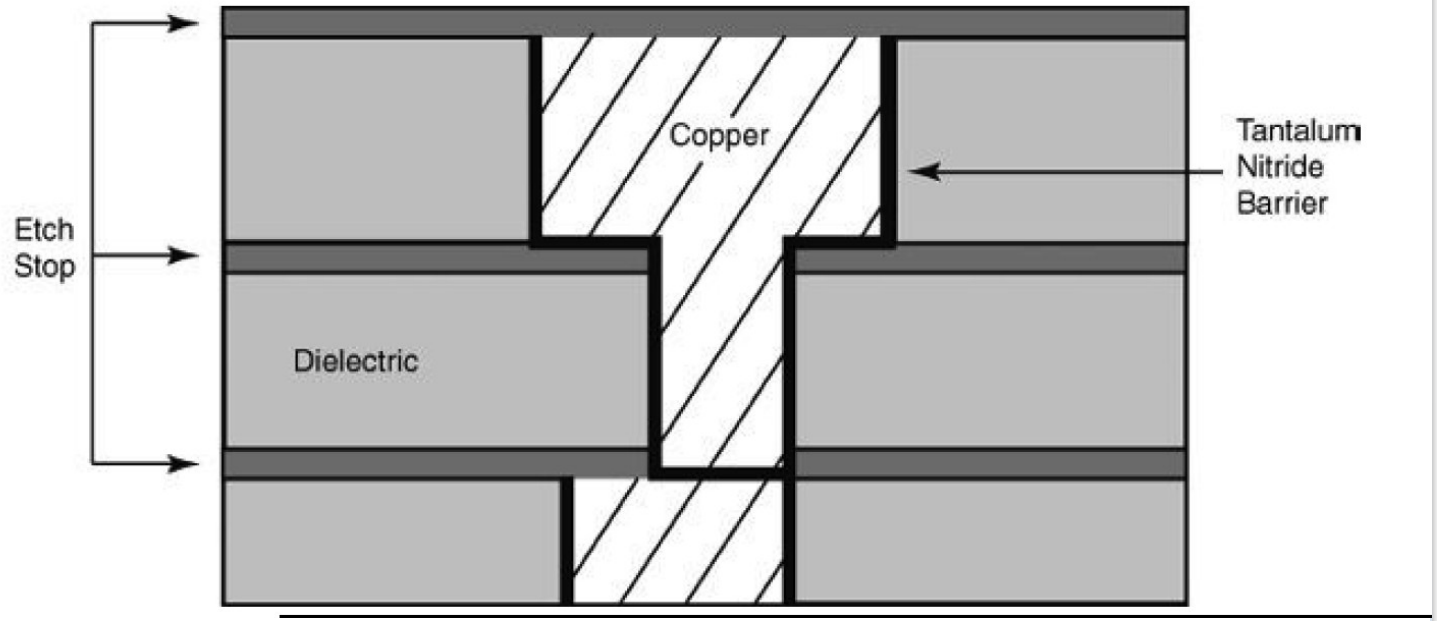
TOPIC 01

Barrier or Liner Deposition



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Evidence 1

Copper Diffusion Prevention: A "liner" layer of tantalum (Ta) is deposited in via holes to prevent copper diffusion into silicon.
Liner Deposition: Sputtering or CVD techniques are used, with the liner layer deposited on the bottom and sides of vias.

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TOPIC 01

Seed Deposition

1

Key idea 1

Copper Deposition: Electrochemical plating (ECP) is the preferred method for depositing copper.
Copper Seed Layer: A thin copper layer is deposited using PVD techniques to initiate ECP in via holes.
Uniformity Challenge: Achieving uniform seed layer deposition in high-aspect vias is crucial for successful ECP.

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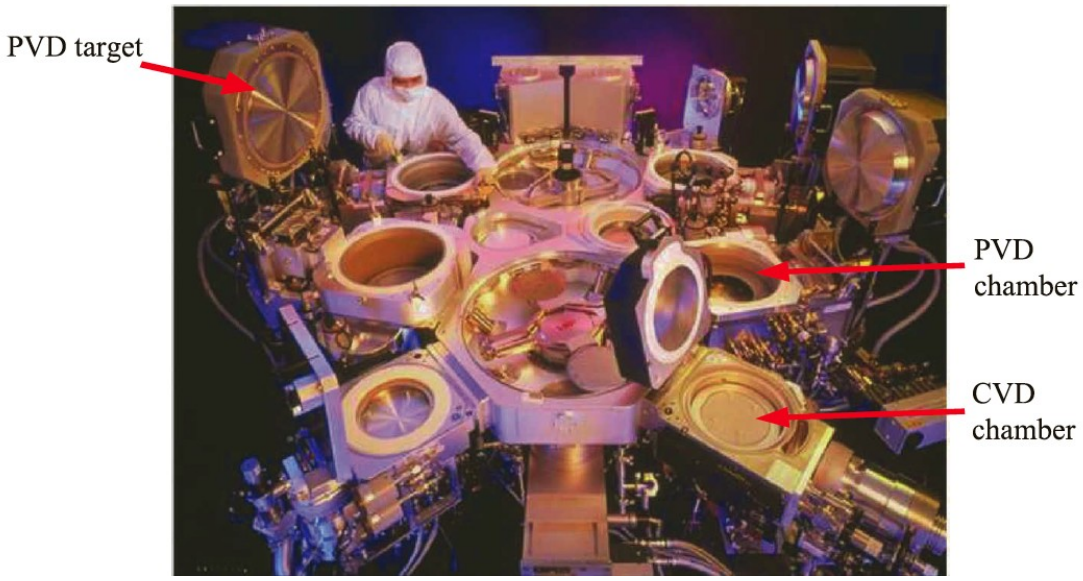
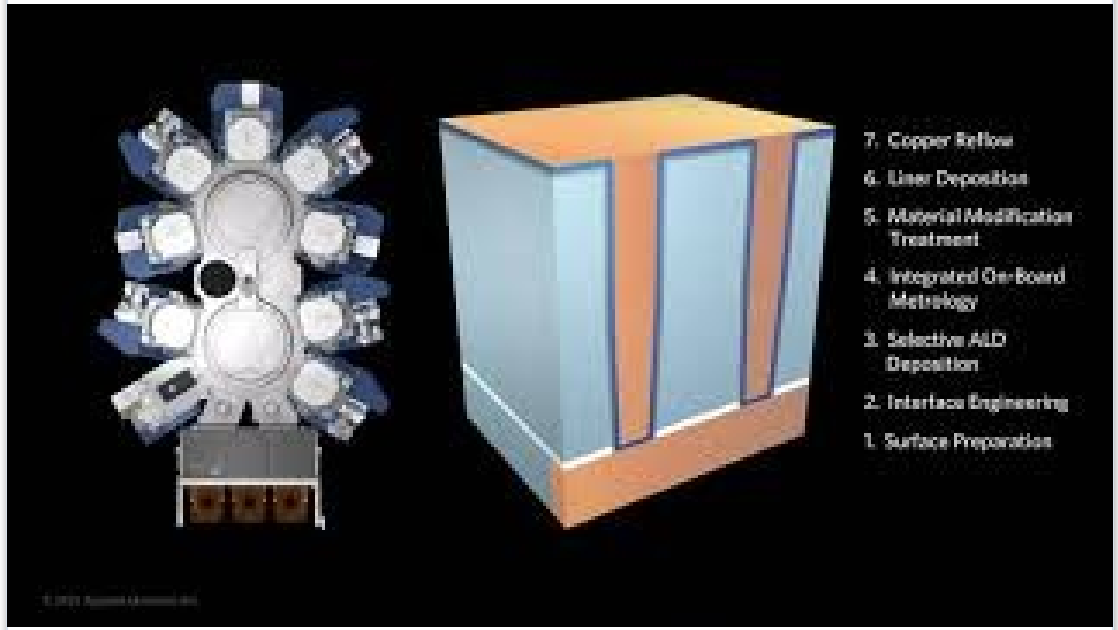
Multiple Chamber Tool





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TOPIC 01

Basic manufacturing process flows

Concepts · mechanisms · process evidence · integration decisions

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TOPIC 01

CMOS Process in the 1980s

In the 1980s, CMOS technology rapidly developed to meet the needs of digital devices like personal computers and LCDs:
Minimum feature size reduced from 3 μm to 0.8 μm, while wafer sizes grew from 100 mm to 150 mm.
LOCOS isolation with ion implantation was used to electrically isolate transistors.
PSG (Phosphosilicate Glass) served as the ILD0 and was reflowed at high temperatures (~1100°C).
Al-Si alloy was used for metal interconnections on tapered contacts, deposited by thermal or e-beam evaporation.
Horizontal furnaces were standard for processes like oxidation, LPCVD, and post-implantation annealing, with plasma etchers gradually replacing wet etching for smaller patterns.
Positive photoresist replaced negative photoresist to improve resolution, and projection alignment systems were used in photolithography

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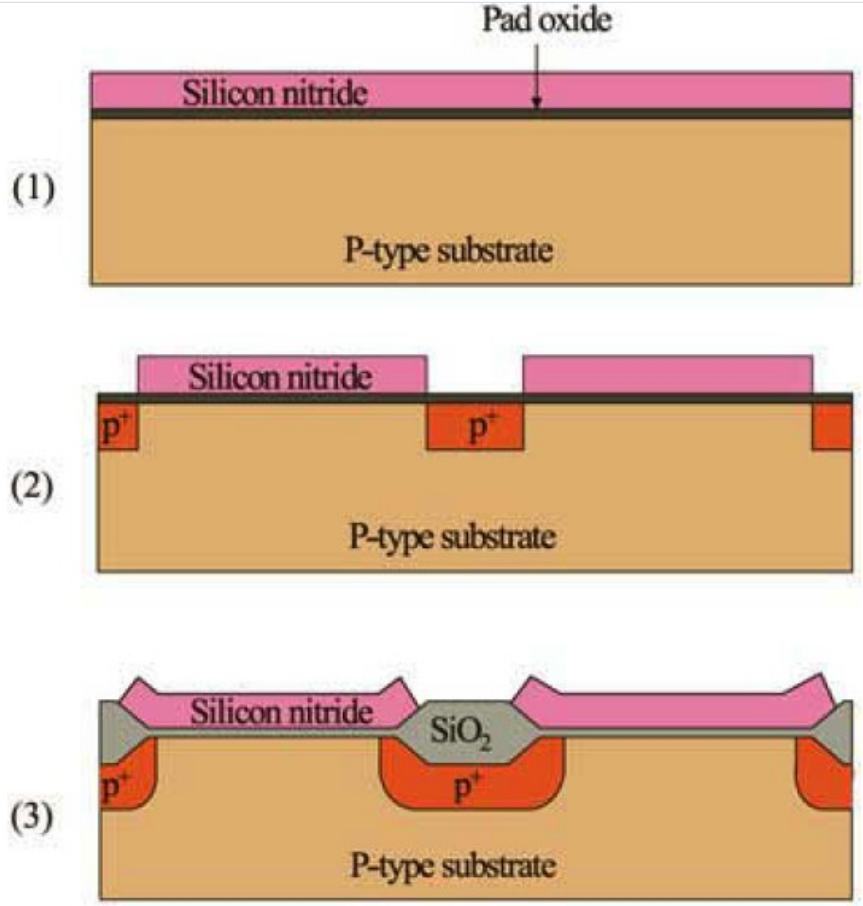
Isolation and Well Formation

- Bare wafer, p-type
- Clean
- Grow pad oxide
- LPCVD nitride (1)
- Mask 1, LOCOS
- Etch nitride and pad oxide
- Strip photoresist
- Clean
- Isolation implantation (boron) (2)



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Isolation and Well Formation

- Bare wafer, p-type
- Clean
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- Mask 1, LOCOS
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- Strip photoresist
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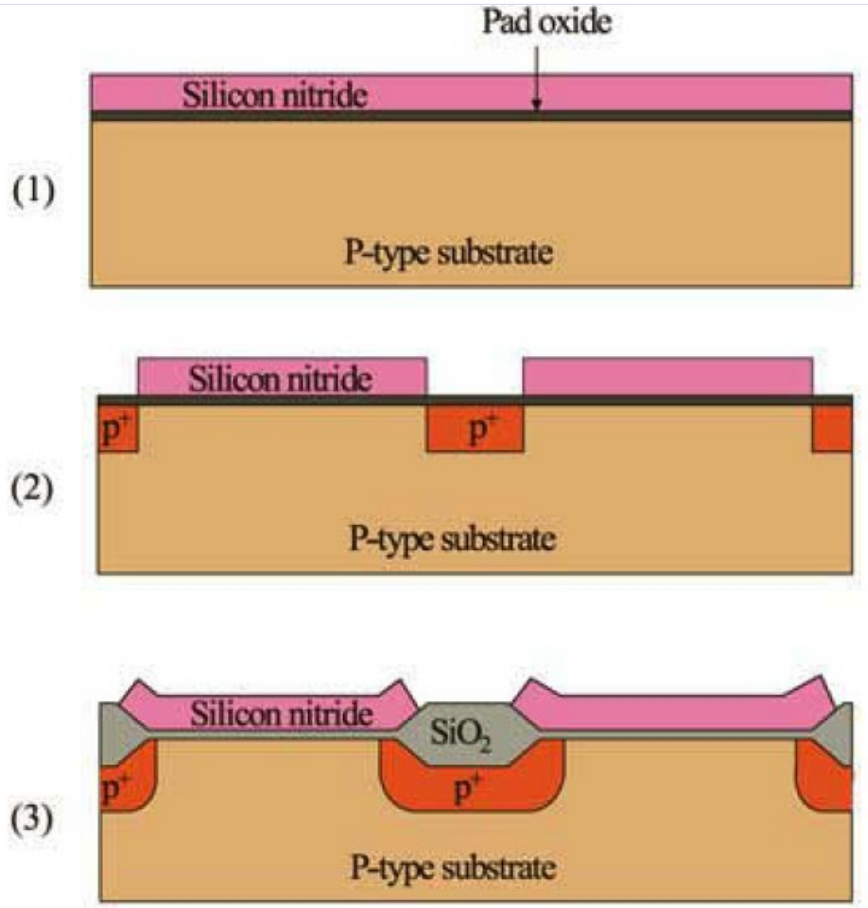

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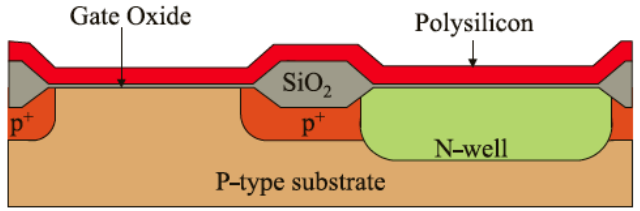
Transistor Formation

- Grow sacrificial oxide
- Strip sacrificial oxide
- Clean
- Grow gate oxide
- LPCVD polysilicon (7)
- Clean
- Mask 3, gate, and local interconnection
- Etch polysilicon (8)
- Strip photoresist
- Polysilicon annealing and oxidation
- Mask 4, n-channel S/D

(7)



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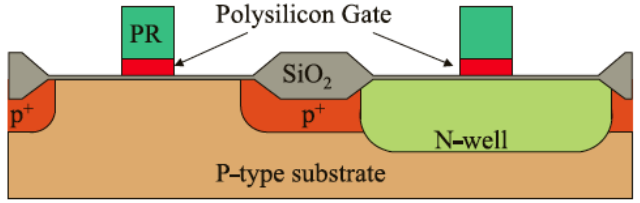

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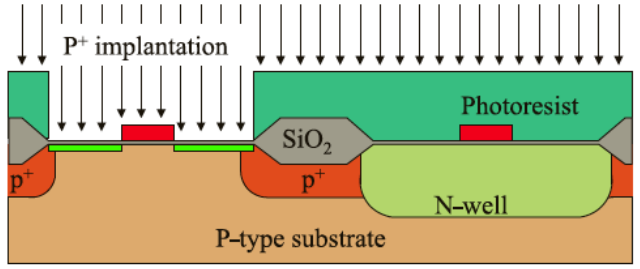

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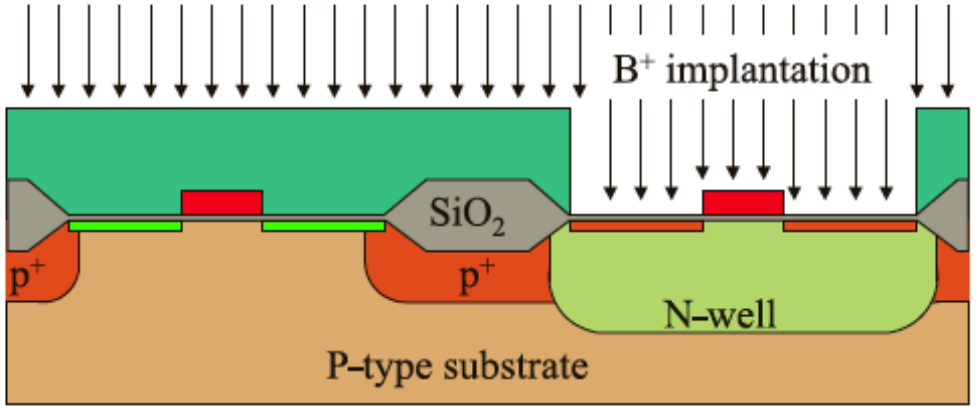
Transistor Formation

- n-channel implantation (phosphorus) (9)
- Strip photoresist
- Mask 5, p-channel S/D
- p-channel implantation (boron) (10)
- Strip photoresist
- Postimplantation annealing, furnace
- LPCVD nitride (11)

(10)



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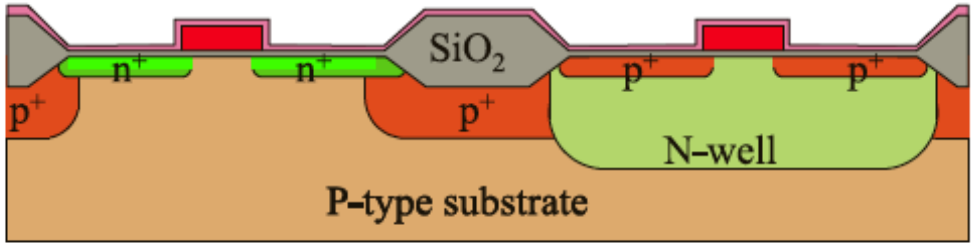

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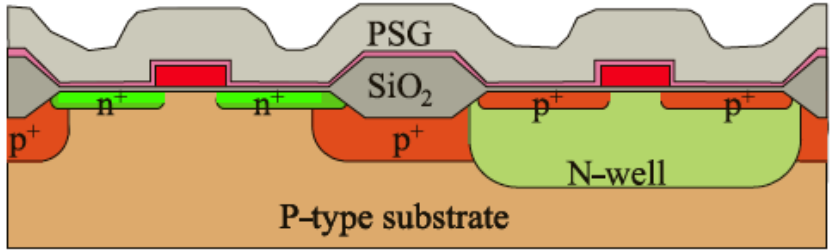
Interconnect Formation

- CVD PSG (12)
- PSG reflow, furnace (13)
- Clean
- Mask 6, contact holes
- Etch PSG
- Strip photoresist (14)
- Clean
- Deposit Al-Si alloy (15)

(12)



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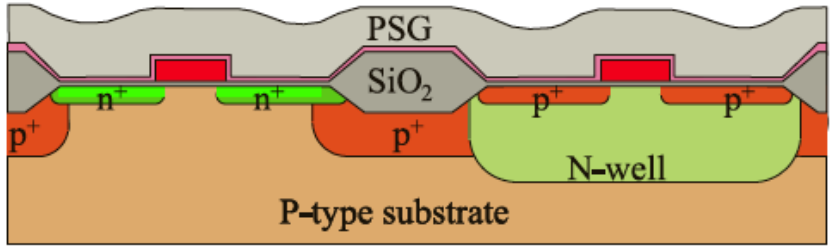

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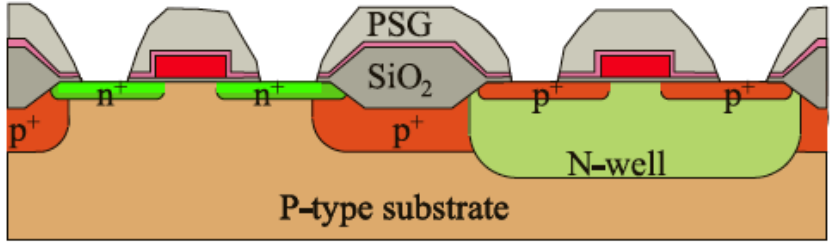

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Interconnect Formation

•Mask 7, metal interconnection

•Etch metal

•Strip photoresist

•Metal annealing (16)

•CVD oxide

•CVD nitride (17)

•Clean

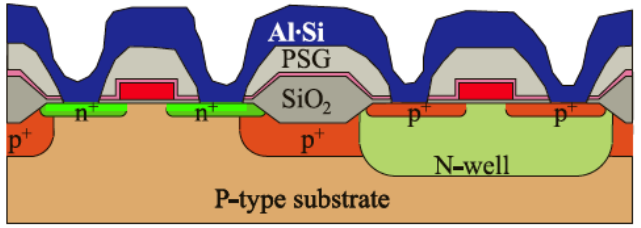
•Mask 9, bonding pad

•Etch nitride and oxide

•Strip photoresist

•Clean

(15)



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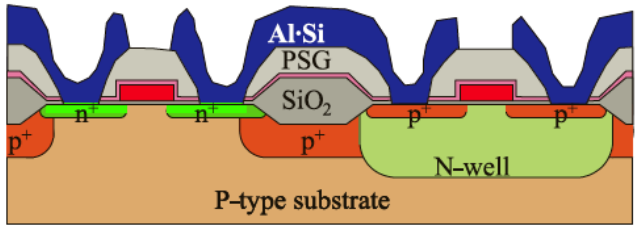

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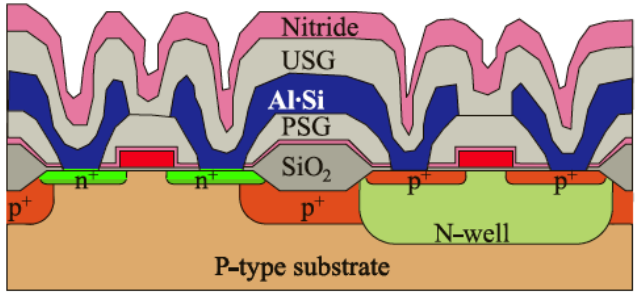

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TOPIC 01

CMOS Process in the 1990s

In the 1990s, CMOS IC processing advanced rapidly to meet the growing demands of digital electronics.
Key developments included:
Feature size shrank from 0.8 to 0.18 μm.
Wafer size increased from 150 mm to 300 mm.
Mid-1990s CMOS process flow featured a minimum feature size of 0.25 μm.
The process included steps like epitaxial-layer growth (performed by wafer manufacturers), STI, well and transistor formation, global interconnections, and passivation.
After wafer processing, the wafer is ready for testing, die sawing, sorting, packaging, and final testing.

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TOPIC 01

CMOS Process in the 1990s

In the 1990s, CMOS IC processing saw key advancements:
Photolithography shifted from UV to deep ultraviolet (DUV) at 248 nm using positive photoresists.
Steppers replaced older alignment systems, and integrated track-stepper systems became common for photoresist coating, baking, exposure, and development in one sequence.
Plasma etching became the standard for patterned etches, while wet etching was still used for blanket film stripping and quality control.
Vertical furnaces dominated for smaller footprints and better contamination control, and RTP gained popularity for post-implant annealing and silicide formation.

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TOPIC 01

CMOS Process in the 1990s

Sputtering replaced evaporation for metal deposition, with dc magnetron sputtering widely used in IC fabs.
Multilayer metal interconnections were developed to handle the increase in transistors.
CVD tungsten plugs were used for connecting layers, with titanium and titanium nitride as popular barrier and adhesion layers.
Titanium nitride also served as an anti-reflective coating (ARC) for better photolithography patterning of metal wires.

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TOPIC 01

Key Steps of CMOS Process

Wafer Preparation: Silicon wafer is prepared from an ingot.
Oxidation: Grow a silicon dioxide (SiO_2) layer on the wafer.
Photolithography: Apply photoresist, align mask, expose to UV light, and develop.
Etching: Remove material from exposed areas.
Doping (Ion Implantation): Introduce impurities to create n-type or p-type regions.
Gate Oxide Formation: Grow a thin oxide layer for the transistor gate.
Polysilicon Deposition: Deposit and pattern polysilicon for the gate structure.
Silicidation: Form silicide on source, drain, and gate for better conductivity.
Interlayer Dielectric Deposition: Add insulating layers between metal interconnects.
Metallization: Deposit and pattern metal for interconnections.
Passivation Layer: Apply a protective layer over the circuit.
Testing and Dicing: Test the wafer and dice it into individual chips.
Packaging: Bond the chip into a package and wire connections.
Final Testing: Perform final tests before shipment

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Shallow Trench Isolation

Shallow Trench Isolation

•Clean

•Grow pad oxide

•Deposit silicon nitride (1)

•Clean

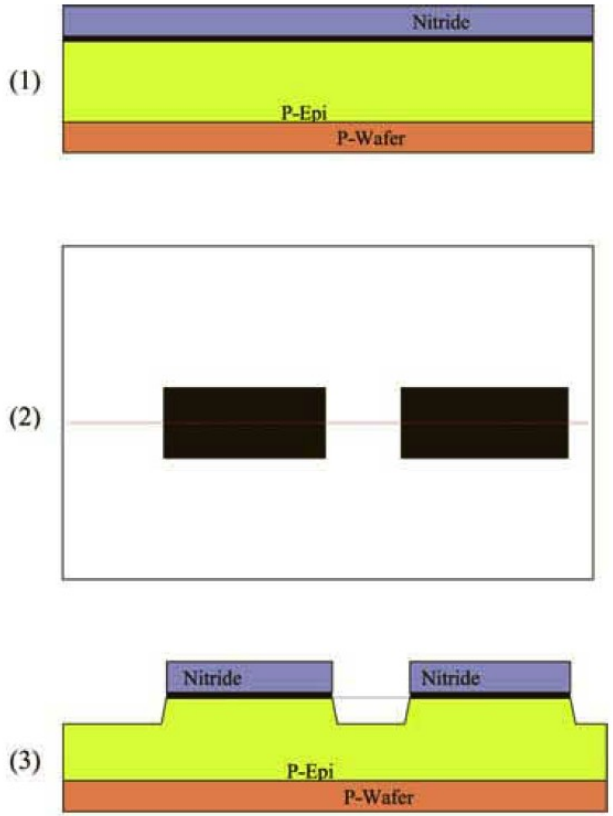
•Prep/spin-on photoresist/soft bake

•STI mask alignment and exposure (2)

•Postexposure bake/develop/hard bake

•Pattern inspection

•Etch nitride/pad oxide



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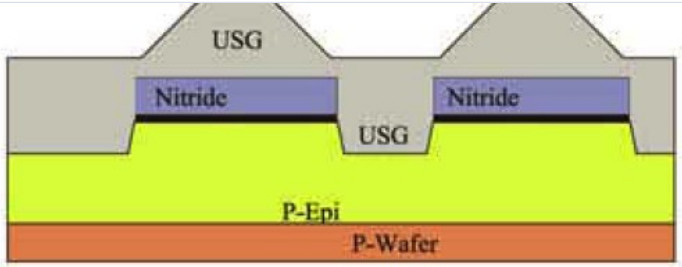
Shallow Trench Isolation

- Clean
- Etch silicon (3)
- Clean
- Grow barrier oxide
- HDP-CVD USG (4)
- CMP USG (5)
- Anneal USG
- Wet strip nitride and pad oxide (6)

(4)



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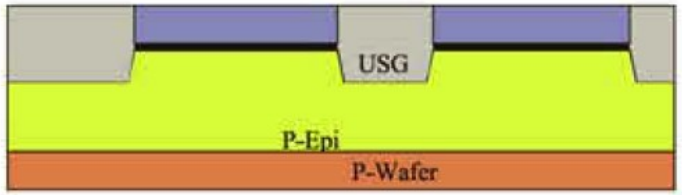

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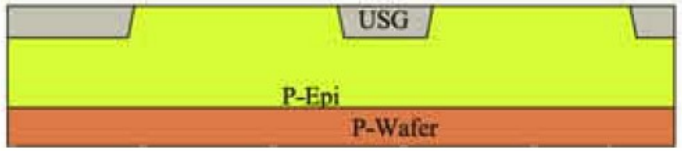

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Well Formation

Well Formation

•Clean

•Grow screen oxide

•Clean

•Prep/spin-on phtoresist/soft bake

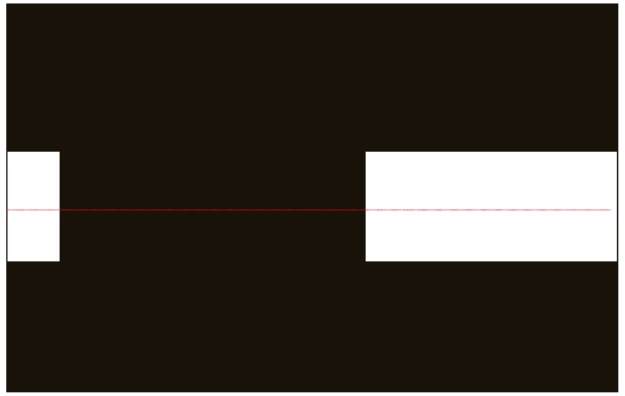
•n-well mask alignment and exposure (7)

•Postexposure bake/develop/hard bake

•Inspection

•n-well ion implantation, phosphorus (8)

(7)



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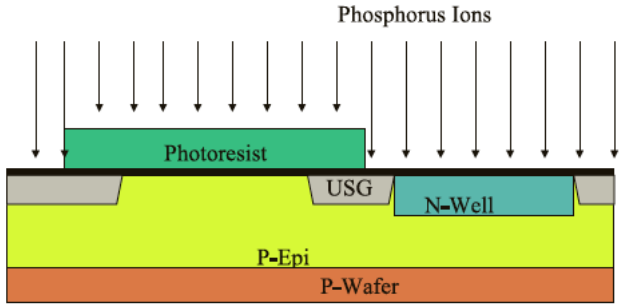

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Well Formation

•Strip photoresist

•Clean

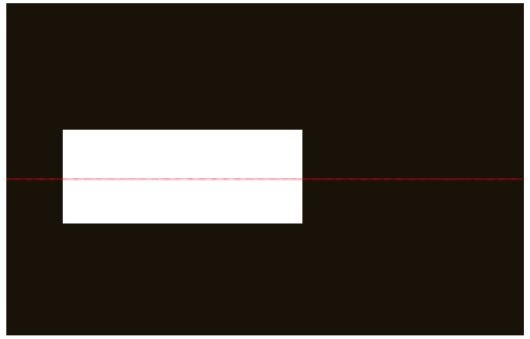
•Anneal and drive-in

•Clean

•Prep/spin-on photoresist/soft bake

•p-well mask alignment and exposure (9)

(9)

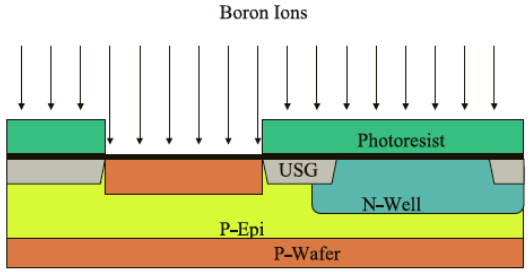


•Postexposure bake/develop/hard bake

•Inspection

•p-well ion implantation, boron (10)

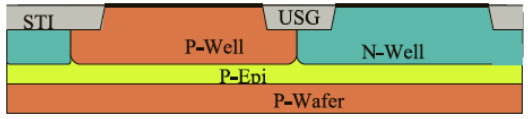
(10)



•Strip photoresist

•Anneal and drive-in (11)

(11)



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Well Formation

•Clean

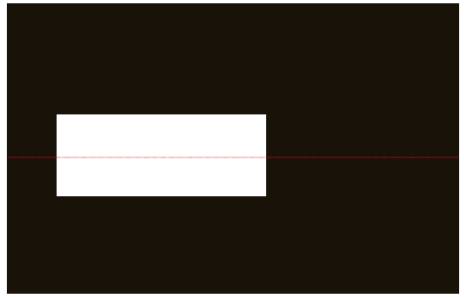
•Grow sacrificial oxide

•Clean

•Prep/spin-on photoresist/soft bake

•n-channel V_T mask alignment and exposure (12)

(12)

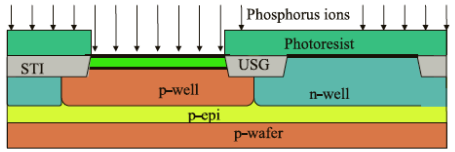


•Postexposure bake/develop/hard bake

•Pattern inspection

•n-channel V_T adjust implantation (13)

(13)

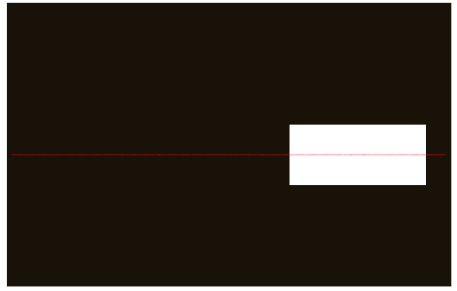


•Strip photoresist

•Clean

•Prep/spin-on photoresist/soft bake

(14)



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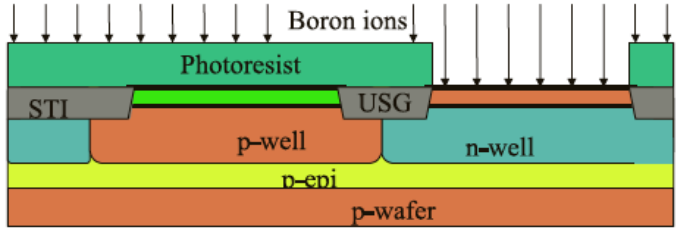
Well Formation

- p-channel V_T mask alignment and exposure (14)
- Postexposure bake/develop/hard bake
- Pattern inspection
- p-channel V_T adjust implantation (15)
- Strip photoresist/clean
- Rapid thermal annealing (16)
- Strip sacrificial oxide

(15)



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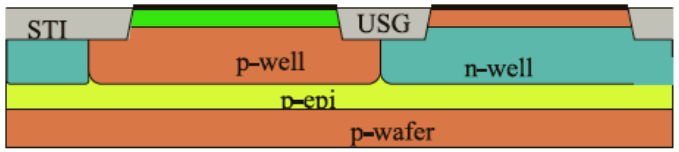

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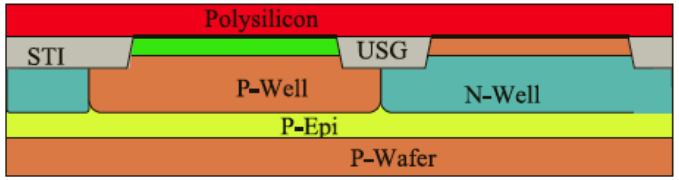

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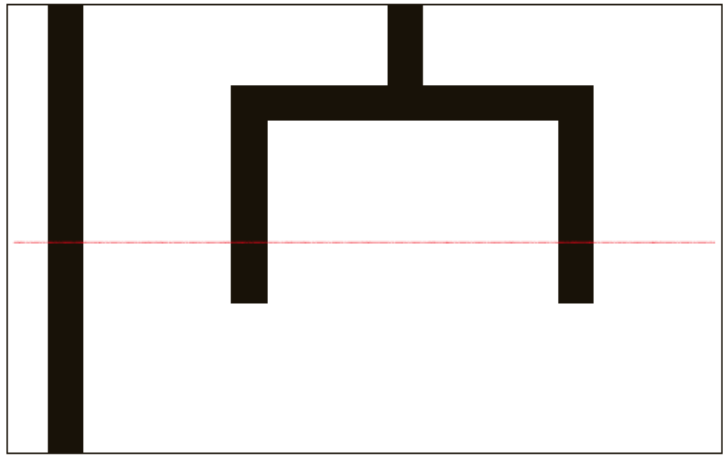
Transistor Formation

- Clean
- Strip native oxide
- Grow gate oxide
- Deposit polysilicon (17)
- Clean
- Prep/spin-on photoresist/soft bake
- Gate mask alignment and exposure (18)
- Postexposure bake/develop/hard bake

(18)



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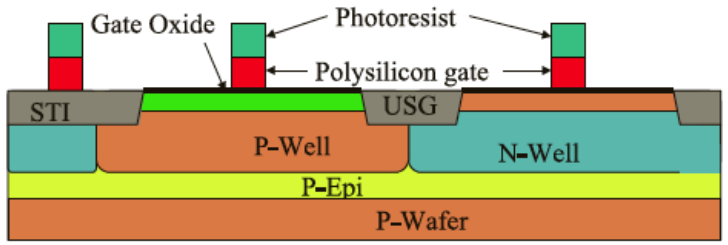

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Transistor Formation

•Inspection

•Etch polysilicon (19)

•Strip photoresist

•Clean

•Polysilicon anneal/oxidation

•Clean

•Prep/spin-on photoresist/soft bake

•n-channel LDD mask (20)

•Postexposure bake/develop/hard bake

•Inspection

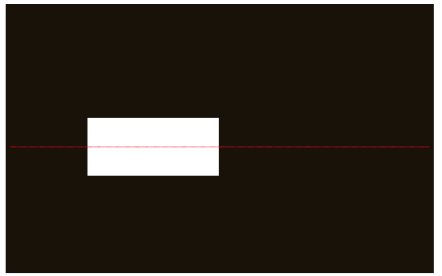
•n-channel LDD implantation (21)

•Strip photoresist

•Clean

•Prep/spin-on photoresist/soft bake

(20)



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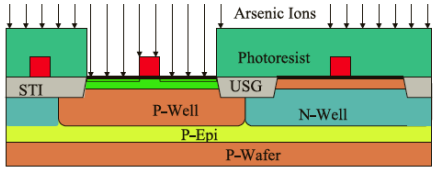

(22)

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Transistor Formation

•p-channel LDD mask (22)

•Postexposure bake/develop/hard bake

•Pattern inspection

•p-channel LDD implantation (23)

•Strip photoresist

•Clean

•LPCVD nitride

•CVD O_3-TEOS USG (24a)

•USG etchback, stop on nitride (24b)

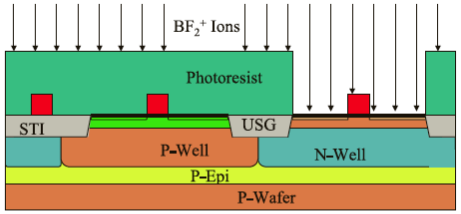
•Clean

•Prep/spin-on photoresist/soft bake

•n-channel S/D mask (25)

•Postexposure bake/develop/hard bake

(23)



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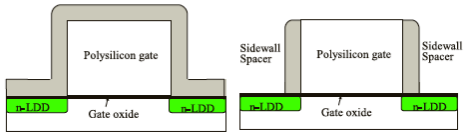

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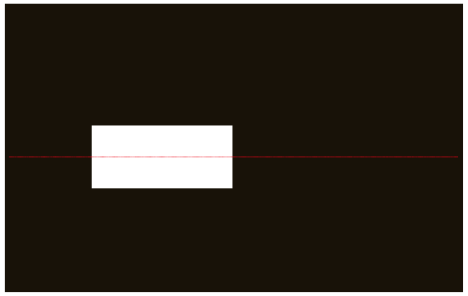

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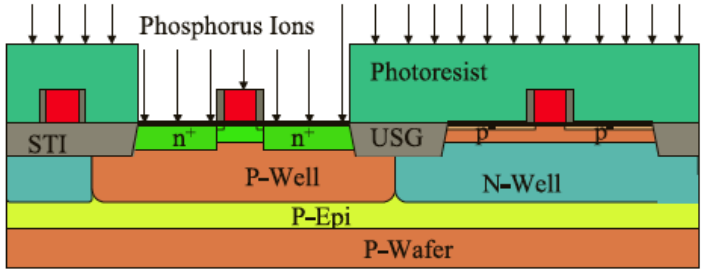
Transistor Formation

- Inspection
- n-channel S/D implantation (26)
- Strip photoresist
- Clean
- Rapid thermal annealing
- Clean
- Prep/spin-on photoresist/soft bake
- p-channel S/D mask (27)
- Postexposure bake/develop/hard bake

(26)



(27)

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Transistor Formation

•Inspection

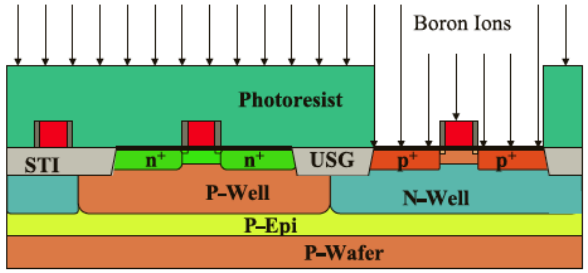
•p-channel S/D implantation (28)

•Strip photoresist

•Clean

•Rapid thermal annealing

(28)



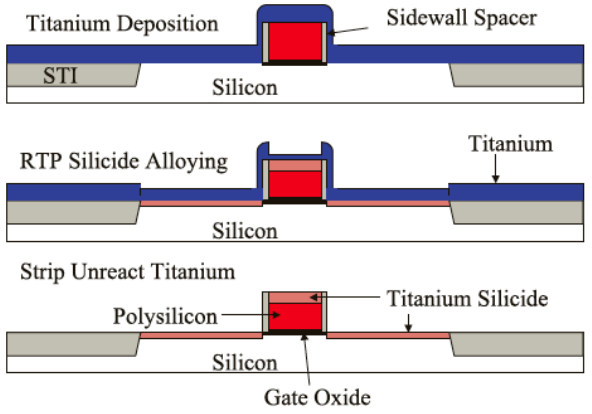
Global Interconnection

•Degas

•Argon sputtering preclean

•Sputtering deposit titanium (29a)

(29)



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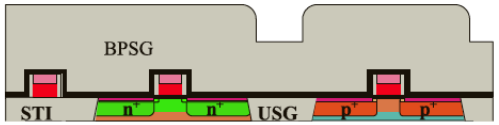
Backend Processes

- Rapid thermal anneal (29b)
- Wet strip titanium (29c)
- Second silicide annealing
- Deposit nitride
- O_3-TEOS based BPSG CVD (30)
- Reflow of BPSG (31)
- BPSG CMP
- Clean
- Prep/spin-on photoresist/soft bake
- Contact mask (32)
- Postexposure bake/develop/hard bake
- Inspection

(30)



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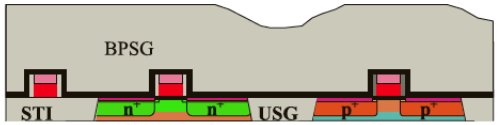

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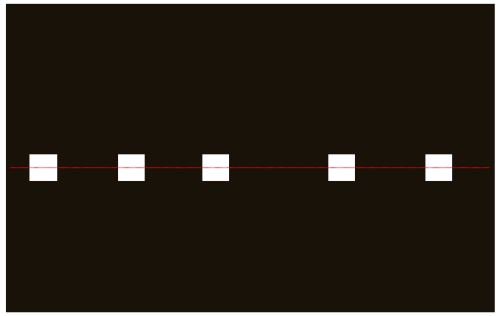

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Backend Processes

•Etch BPSG, Stop on silicide surface

•Strip photoresist (33)

•Clean

•Degas

•Argon sputtering pre-PDV clean

•Deposit Ti/TiN by both PVD and CVD

•CVD tungsten (34)

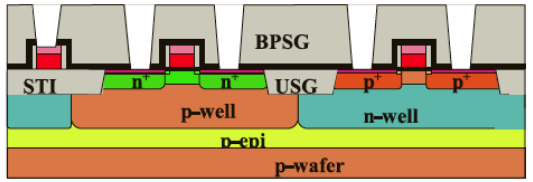
•Polish tungsten/TiN/Ti

•Clean

• Ar sputtering pre-PDV clean

•PVD Ti

(33)



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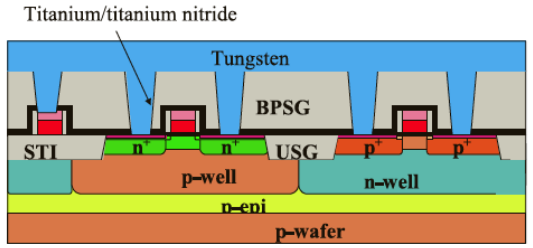

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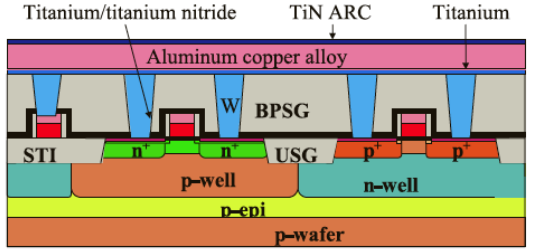

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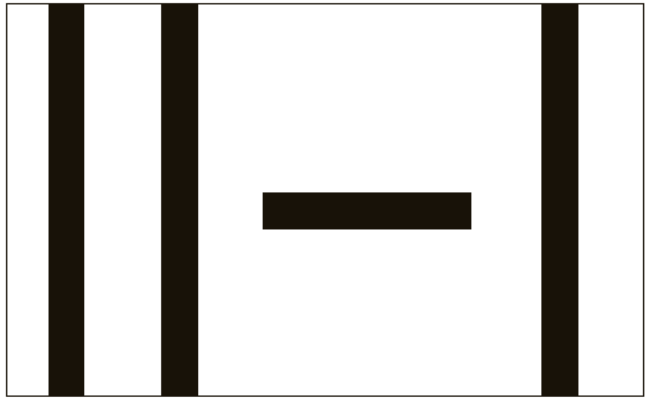
Backend Processes

- PVD of Al-Cu alloy
- PVD of TiN ARC layer (35)
- Clean
- Prep/spin-on photoresist/soft bake
- Metal-1 mask (36)
- Postexposure bake/develop/hard bake
- Inspection
- Etch metal
- Strip photoresist (37)

(36)



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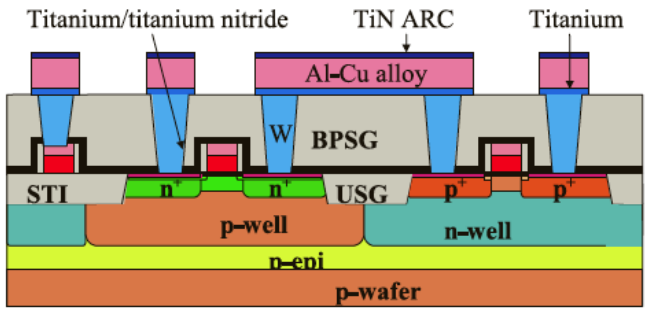

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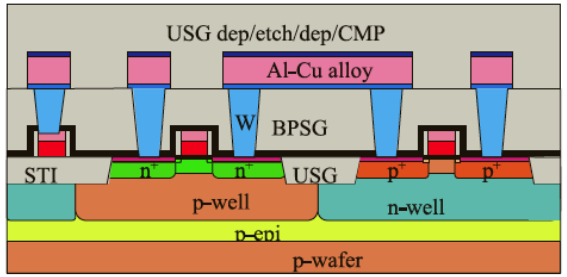
Backend Processes

- Metal anneal
- CVD USG
- Sputtering etchback USG
- CVD USG
- CMP USG (38)
- Clean
- Prep/spin-on photoresist/soft bake
- Via-1 mask (39)
- Postexposure bake/develop/hard bake
- Inspection
- Etch USG

(38)



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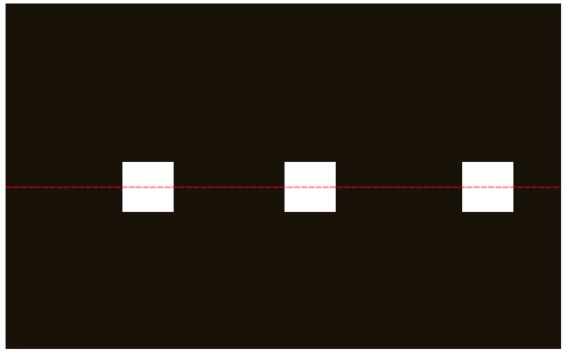

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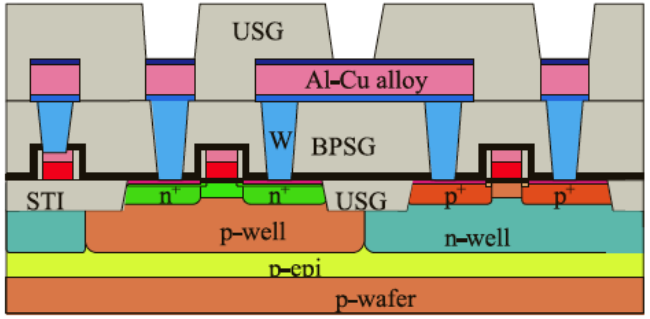
Backend Processes

- Strip photoresist (40)
- Degas
- Ar^+ sputtering preclean
- Deposit Ti/TiN
- Deposit tungsten
- Polish tungsten/TiN/Ti
- Ar^+ sputtering pre-clean
- Deposit Ti
- Deposit Al•Cu
- Deposit TiN (41)

(40)



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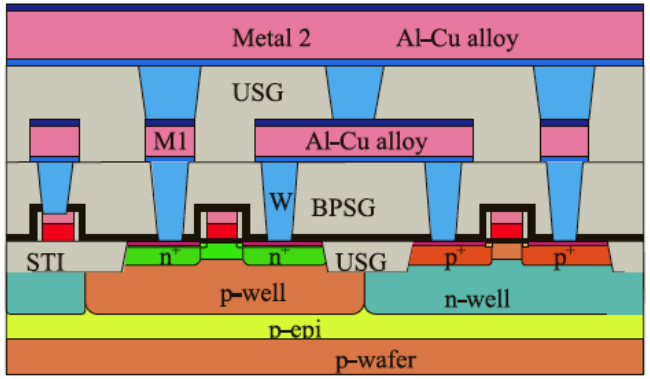

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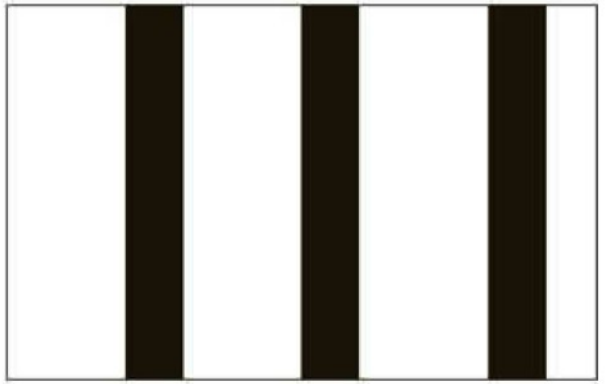
Backend Processes

- Clean
- Prep/spin-on photoresist/soft bake
- Metal-2 mask (42)
- Postexposure bake/develop/hard bake
- Inspection
- Etch metal 2
- Strip photoresist/clean (43)
- Metal annealing
- Deposit USG, PE-TEOS
- USG sputter etchback, Ar^+

(42)



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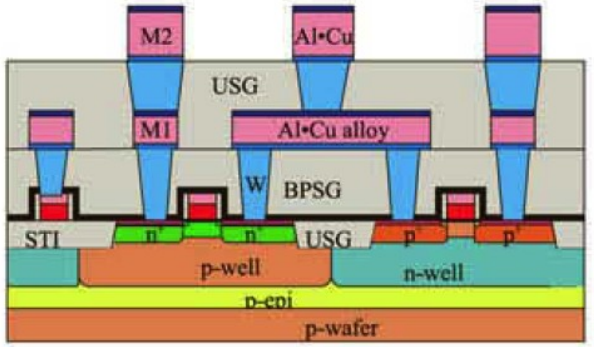

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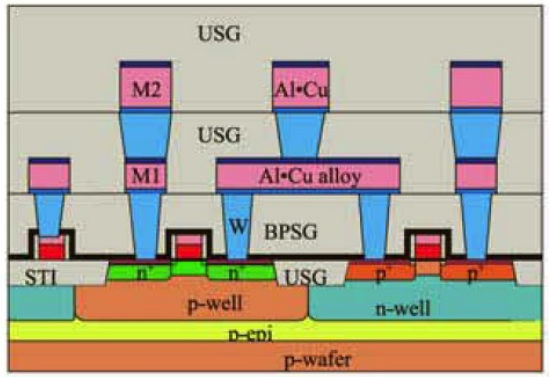
Backend Processes

- CVD USG, PE-TEOS
- CMP USG (44)
- Prep/spin-on photoresist/soft bake
- Via-2 mask (45)
- Postexposure bake/develop/hard bake
- Inspection
- Etch USG
- Strip photoresist (46)
- Clean
- Degas
- Ar^+ sputtering preclean

(44)



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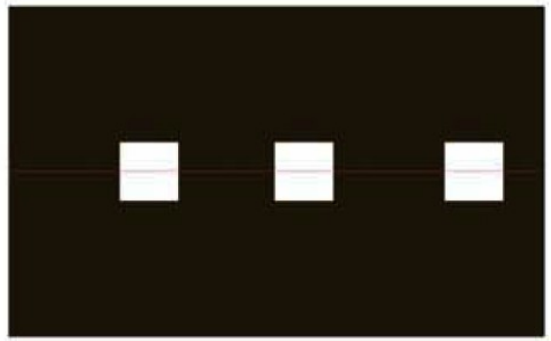

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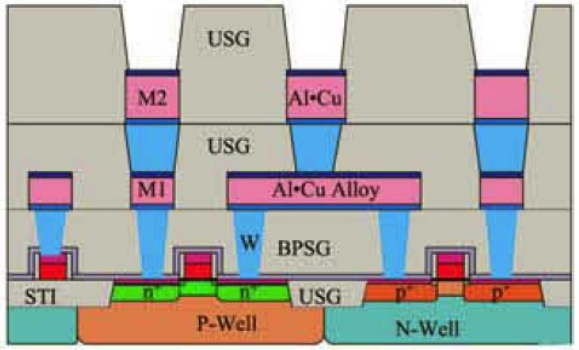
Backend Processes

- Deposit Ti/TiN
- Deposit tungsten
- Polish W/TiN/Ti
- Clean
- Degas
- Ar^+ sputtering preclean
- Deposit titanium
- Deposit Al-Cu
- Deposit titanium nitride (47)
- Clean

(46)



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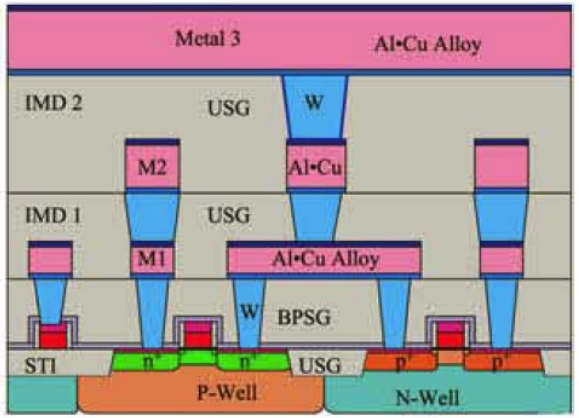

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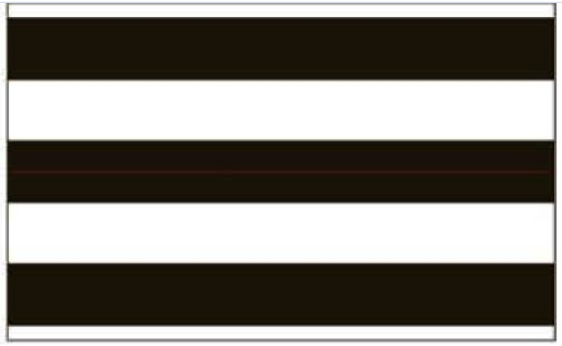
Backend Processes

- Prep/spin-coat photoresist/soft bake
- Metal-3 mask (48)
- Postexposure bake/develop/hard bake
- Inspection
- Etch metal 3
- Strip photoresist (49)
- Clean
- Metal annealing
- CVD USG, PE-TEOS
- Ar sputtering etchback
- CVD USG, PE-TEOS

(48)



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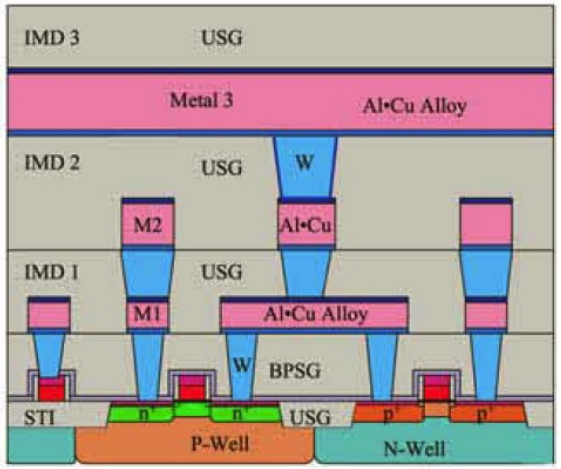

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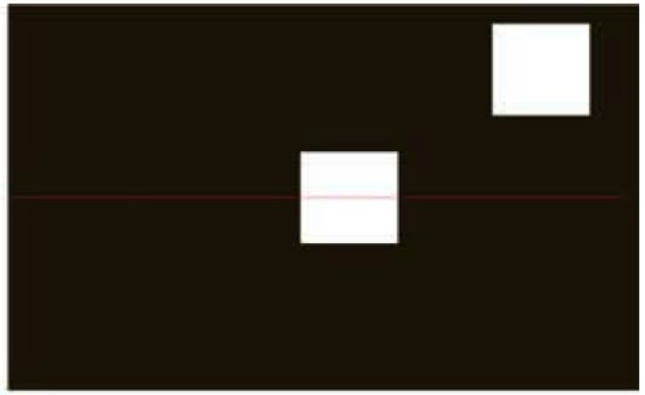
Backend Processes

- CMP USG
- Clean
- Prep/spin-coat photoresist/soft bake
- Via-3 mask (50)
- Postexposure bake/develop/hard bake
- Inspection
- Via-3 etch
- Strip photoresist (51)

(50)



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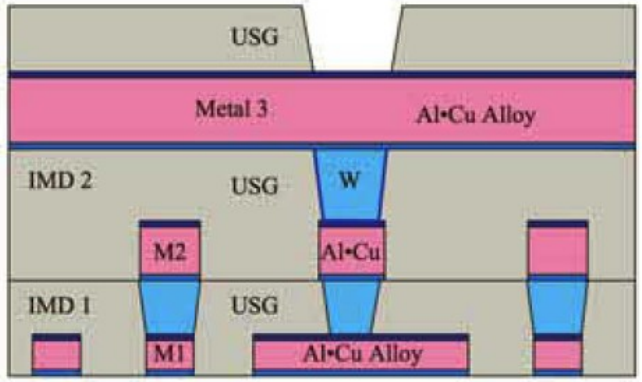

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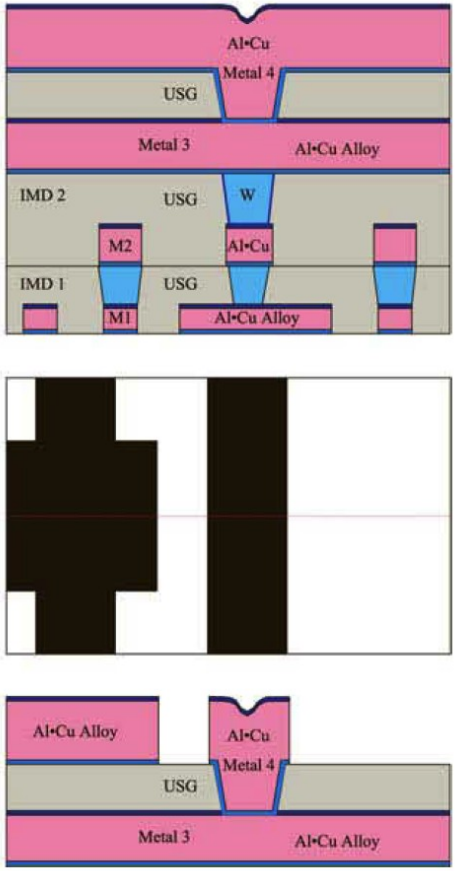
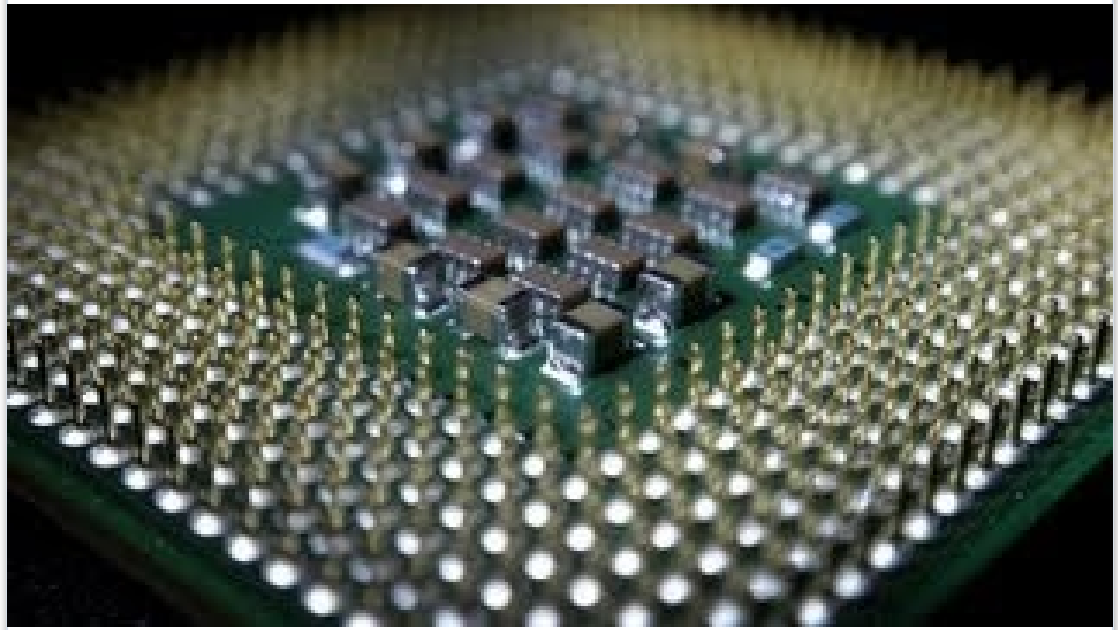
Backend Processes

- Clean
- Ar sputtering preclean (52)
- Deposit titanium
- Deposit Al-Cu
- Deposit titanium nitride (52)
- Clean/prep/spin-coat photoresist/soft bake
- Metal-3 mask (53)
- Postexposure bake/develop/hard bake/inspection (53)
- Etch metal 4
- Strip photoresist (54)
- Clean
- Metal annealing (54)





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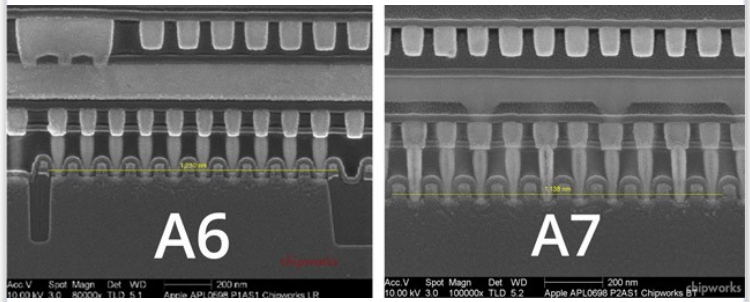
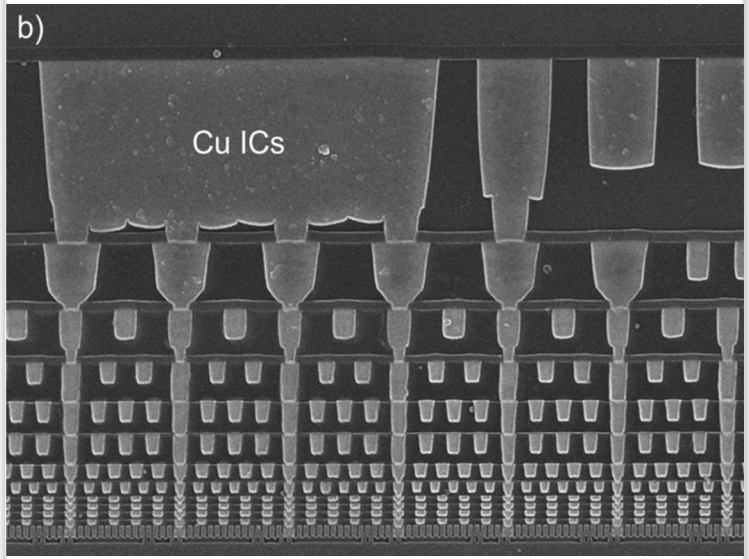
TOPIC 01

Cross Sections





1 FEOL

FEOL

2 BEOL

BEOL

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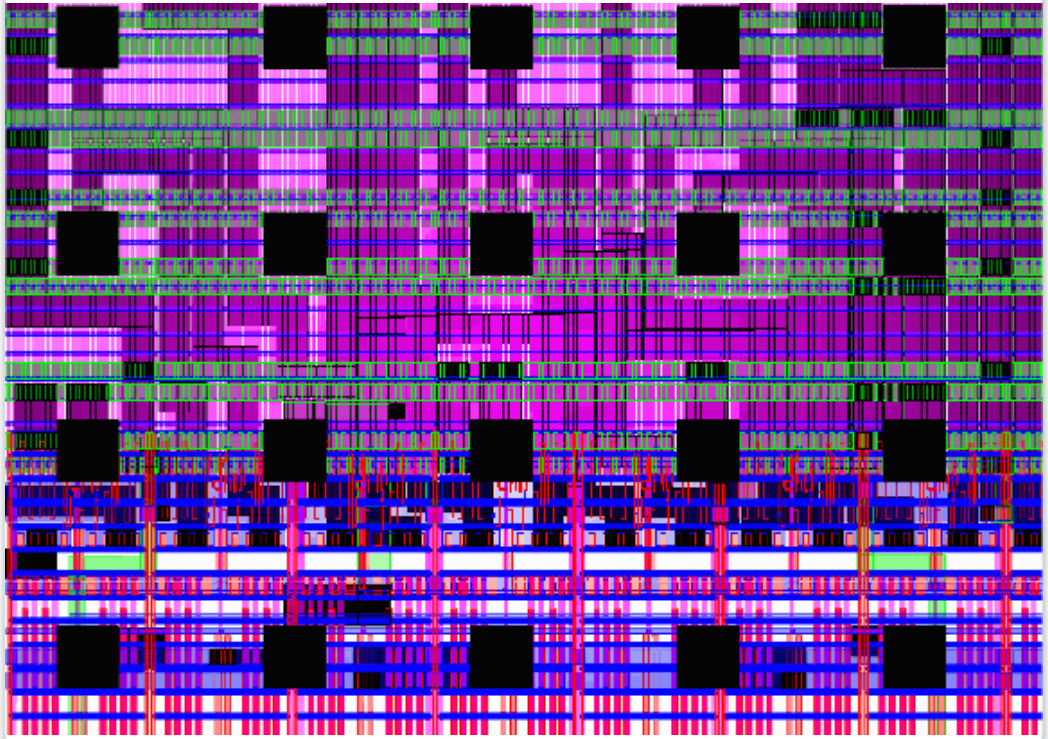
TOPIC 01

Layout and Design Rules

Chips are specified with set of masks
The masks are generated from layout
The layout obeys the design rule



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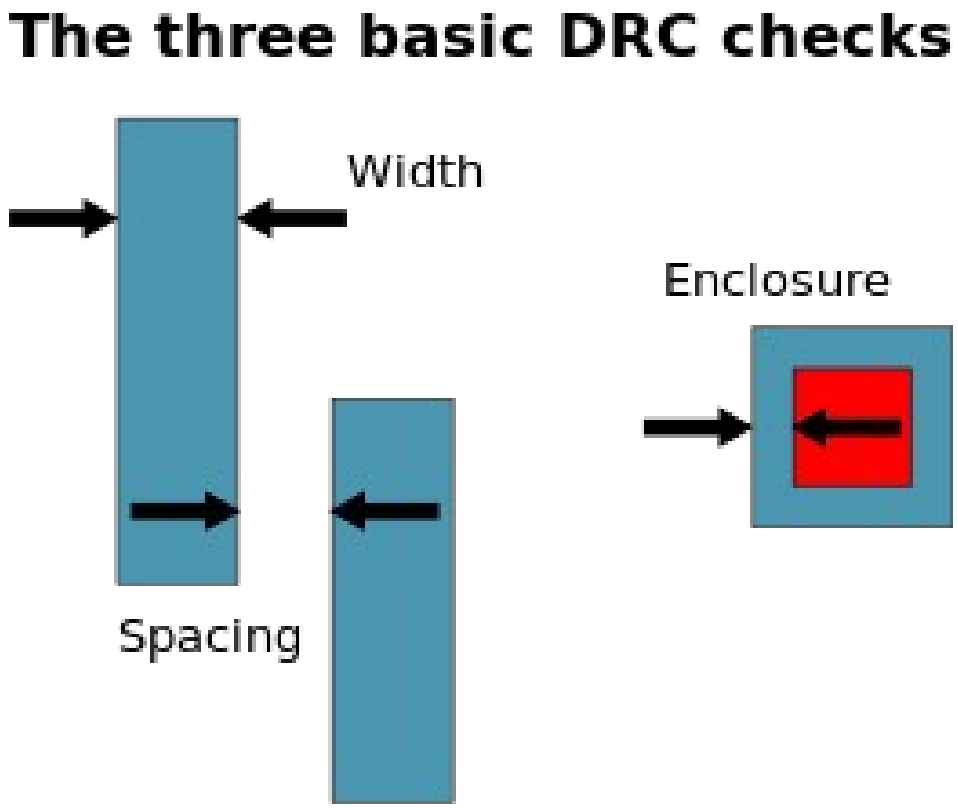
Design Rules Check



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Evidence 1

Design Rules are conservative rules for layout to ensure yield and reliability

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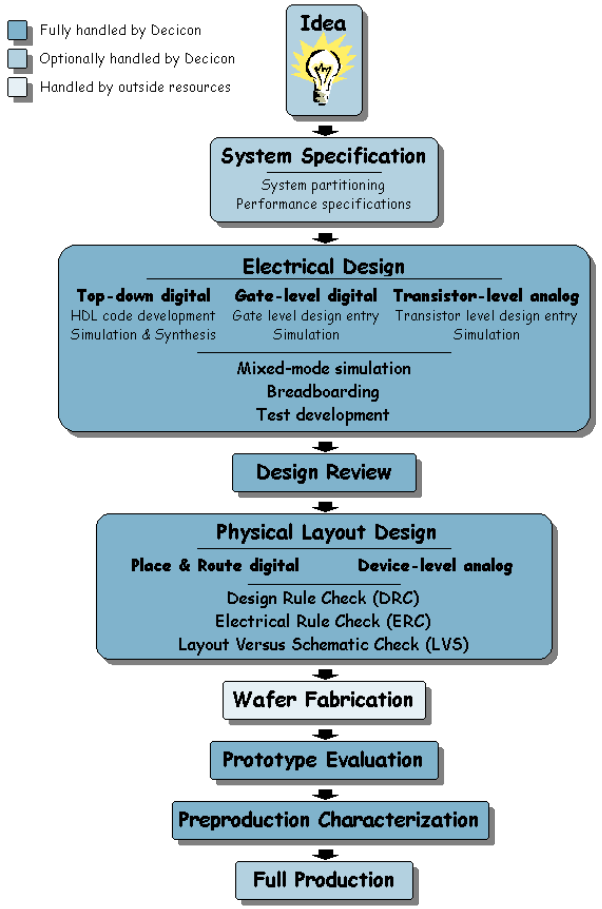
Design to Fabout Flow



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Singapore Semiconductor Landscape

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TOPIC 02

02

Process Integration Issues

Concepts · mechanisms · process evidence · integration decisions

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TOPIC 02

Topics

1 Process integration and yield issues

Process integration and yield issues
Device performance issues
Device reliability issues
Packaging
Follow Up Actions
What's Next

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TOPIC 02

Process integration and yield issues

Concepts · mechanisms · process evidence · integration decisions

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TOPIC 02

Isolations

1

Key idea 1

Local oxidation of silicon (LOCOS), and STI are the two isolation techniques that can be used in IC fabrication. p-type doped junctions also have been used to electronically isolate neighboring transistors.

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TOPIC 02

Local Oxidation of Silicon

Blanket field oxides used until 1970 had drawbacks:
High oxide step with sharp edges, difficult to cover during metal deposition.
Channel stop doping occurred before oxidation, complicating feature size reduction.
LOCOS (Local Oxidation of Silicon), introduced in the 1970s, improved the process:
Self-aligned field oxide with isolation doping after channel stop implantation.
Lower step height and sloped sidewalls, easing metal/polysilicon deposition.
LOCOS oxide thickness: 5,000–10,000 Å.

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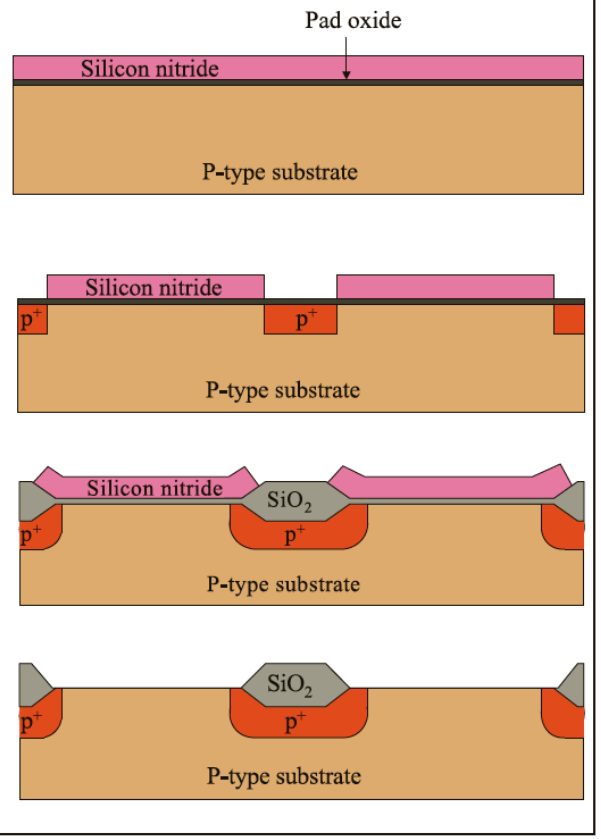
Local Oxidation of Silicon

- Bare wafer, p-type
- Clean
- Grow pad oxide
- LPCVD nitride
- Mask 1, LOCOS
- Etch nitride
- Strip photoresist
- Clean
- Isolation implantation, boron
- Wet oxidation, LOCOS formation
- Strip nitride and pad oxide
- Clean



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TOPIC 02

Local Oxidation of Silicon

LPCVD silicon nitride used as an oxidation mask for designated thick oxide growth, with pad oxide to relieve stress.
Plasma etch for nitride patterning and hot phosphoric acid for stripping after LOCOS formation.
A key disadvantage of LOCOS is the bird's beak:
Lateral oxide encroachment under the nitride due to isotropic diffusion of oxygen during thermal oxidation.
Bird's beak reduces transistor packing density, taking up about 0.5 μm on both sides.
Poly-buffered LOCOS (PBL) reduces bird's beak (by 0.1–0.2 μm) using a polysilicon buffer layer to consume lateral oxygen diffusion.

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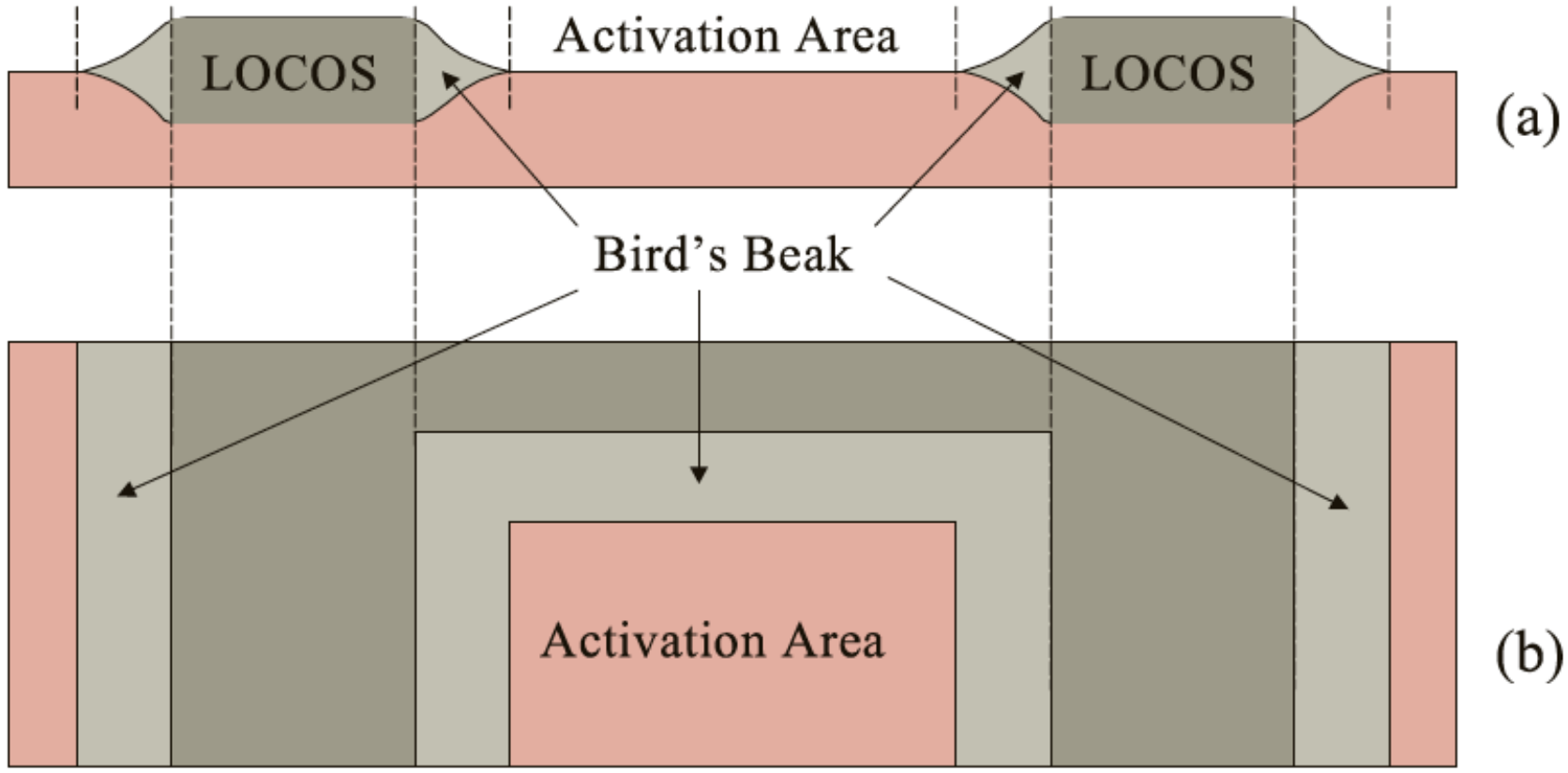
Local Oxidation of Silicon



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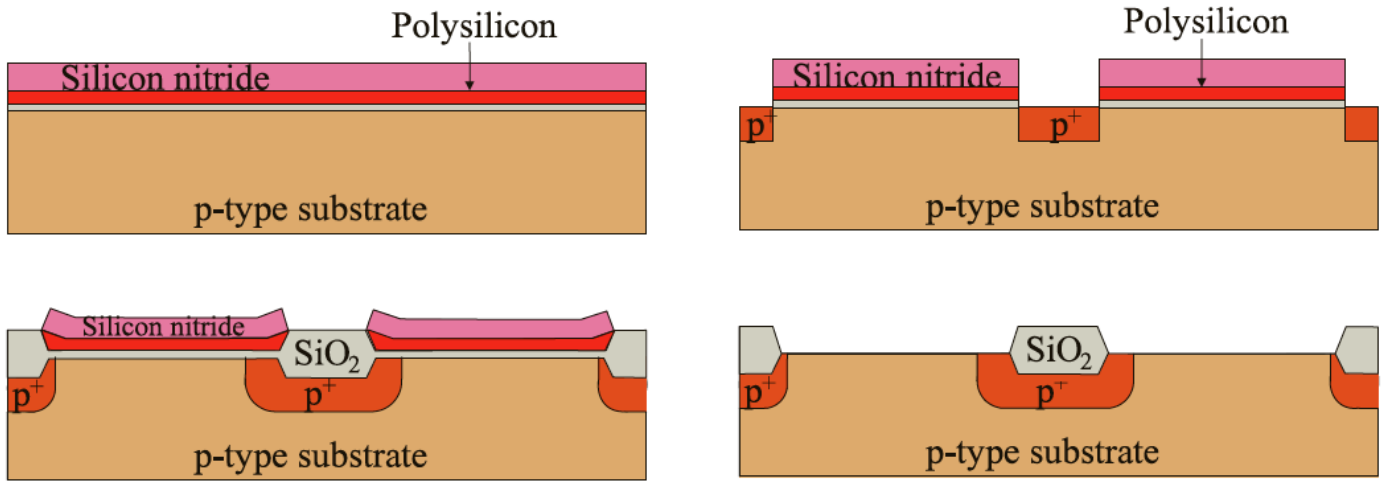
Local Oxidation of Silicon



1

Evidence 1

Poly-buffered LOCOS (PBL) reduces bird's beak (by 0.1–0.2 μm) using a polysilicon buffer layer to consume lateral oxygen diffusion.

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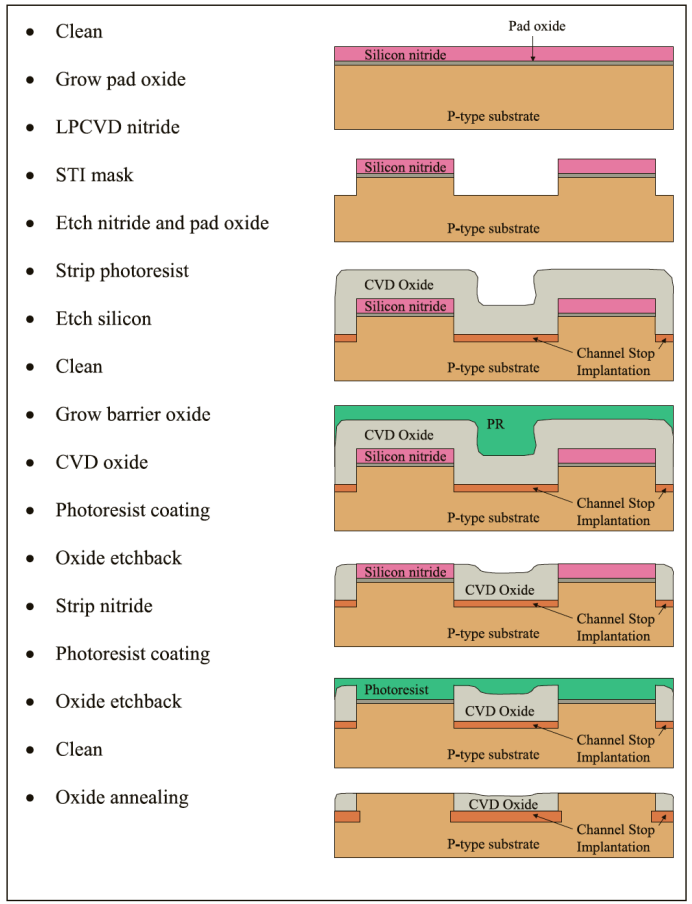
TOPIC 02

Shallow trench isolation



1

Evidence 1

Shallow Trench Isolation (STI) was developed to replace LOCOS, using CVD oxide trench fill instead of thermal oxidation, reducing oxide encroachment.
Early STI process:
LPCVD nitride used as a mask for silicon etching and as an oxide stop layer.
Trench is filled with CVD oxide, and channel stop implantation is applied.
Photoresist etchback planarizes the surface, stopping at the nitride layer.
Nitride is stripped with hot phosphoric acid, leaving oxide to isolate devices.
STI advantages:
Fewer oxide encroachment issues and better for smaller geometries (below 0.35 µm).
Replaced LOCOS when smaller feature sizes and highly planarized surfaces were required for photolithography resolution.

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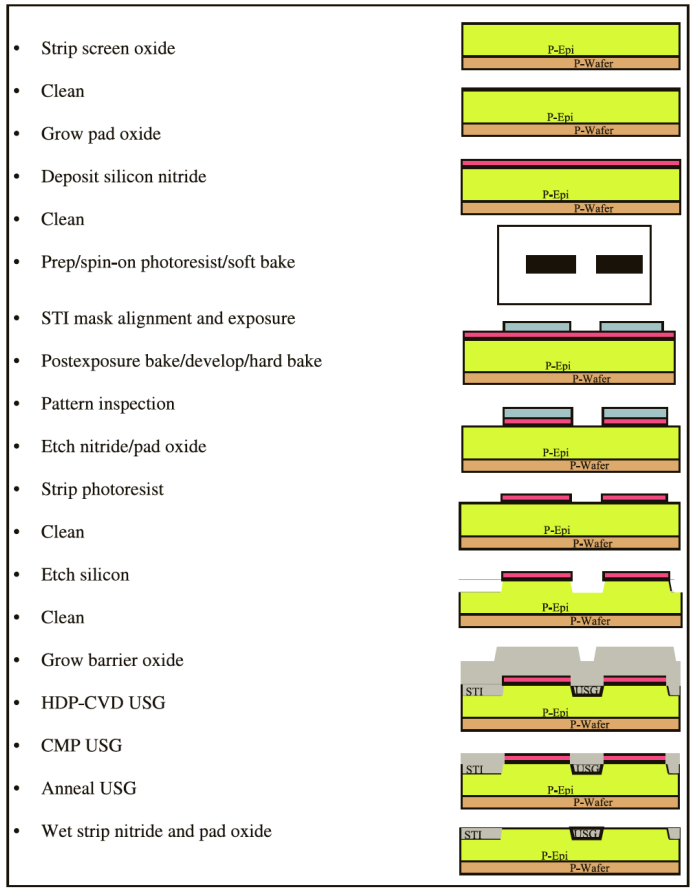
TOPIC 02

Shallow trench isolation



1

STI process steps:

STI process steps:
Oxidation, nitride deposition, etching, CVD oxide fill, CMP, annealing, and nitride stripping.
O3-TEOS chemistry used for trench fill, eliminating the need for thermal annealing due to dense oxide formation.
Challenges in STI: Single-crystal silicon etching, oxide CVD, and CMP become more difficult as feature sizes shrink.
Recent innovation: STI oxide is used to induce stress and strain in the silicon, improving device speed.

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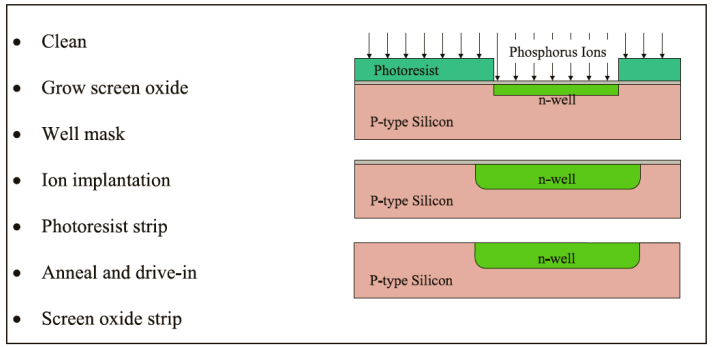
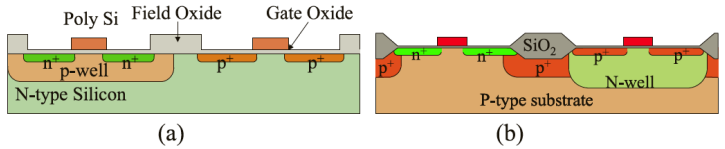
TOPIC 02

Single well





1

Evidence 1

Early CMOS IC processing used a single well (n- or p-well), depending on the wafer type.
Well formation involved high-energy, low-current ion implantation and thermal anneal/drive-in.
Photolithography steps for well formation: Photoresist coating, alignment, exposure, development, and pattern inspection.
p-well formation on an n-type wafer is similar to n-well formation on a p-type wafer.

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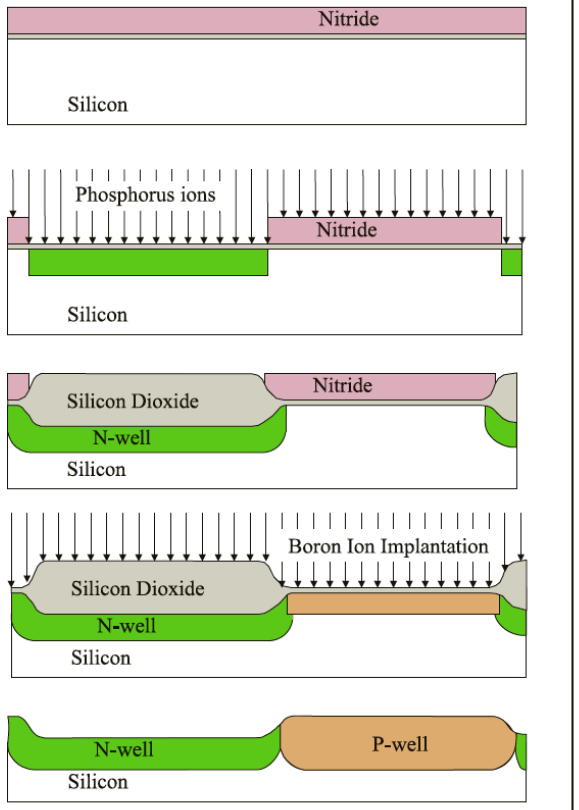
TOPIC 02

Twin wells

- Clean
- Grow pad oxide
- Deposit nitride
- n-well mask
- Etch nitride
- Strip photoresist
- n-well implantation
- Anneal/drive-in and oxidation
- Strip nitride
- p-well implantation
- Anneal and drive-in
- Strip oxide



1

Evidence 1

Self-aligned twin-well processing results in p- and n-wells at different levels, affecting photolithography resolution due to depth of field (DOF) issues.
Double-photomask twin wells are often used in advanced CMOS manufacturing to address this problem.
Both well implantation processes use high-energy, low-current implanters.
Furnaces are used for annealing and dopant drive-in processes in well formation.

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TOPIC 02

Self-aligned twin wells

Uses LPCVD silicon nitride to block oxygen and prevent ion penetration during well formation.
Reduces cost by eliminating a photolithography masking step.
Pad oxide is needed to relieve stress from the nitride layer.
Nitride is removed with hot phosphoric acid, selective to oxide
Advantages:
Eliminates a masking step, lowering production costs and improving chip yield.
Disadvantages:
Creates a non-flat surface, with n-well lower than the p-well due to oxide growth into the substrate.
In twin-well formation, n-well is implanted first because phosphorus diffuses slower than boron, preventing uncontrolled boron diffusion during annealing.

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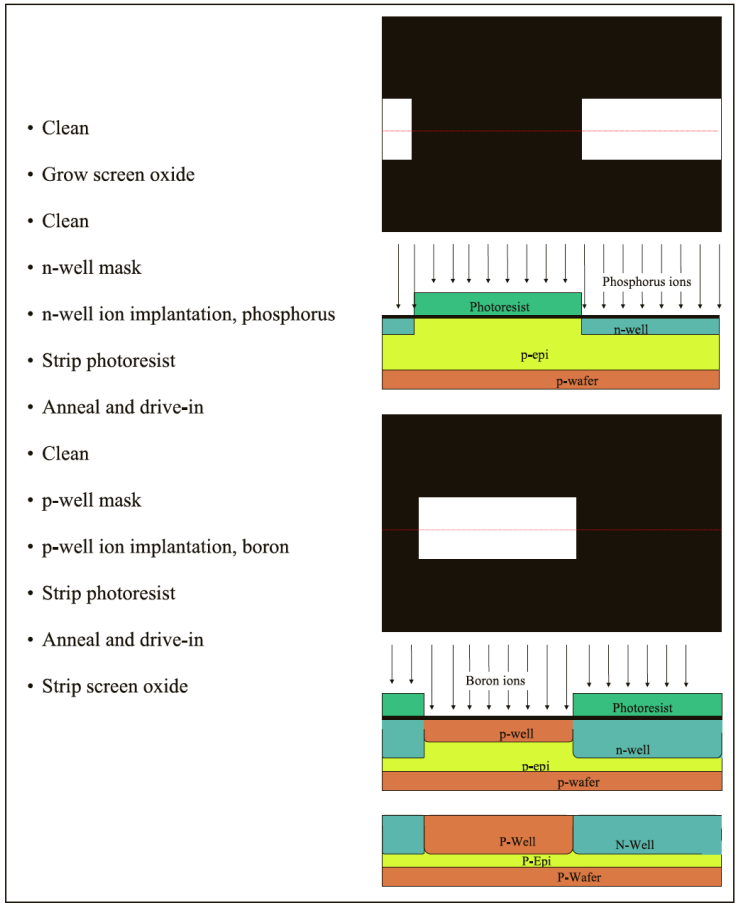
TOPIC 02

Self-aligned source/drain process



1

Evidence 1

Ion implantation introduced self-aligned source/drain (S/D) processes in MOSFET manufacturing.
Polysilicon replaced aluminum as the gate material due to aluminum's inability to withstand high post-implantation annealing temperatures.
Process steps:
Activation mask defines transistor areas on the field oxide.
Gate oxide growth and polysilicon deposition occur, followed by gate definition using a gate mask.
S/D ion implantation and thermal annealing complete the MOSFET formation.

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TOPIC 02

Lightly doped drain

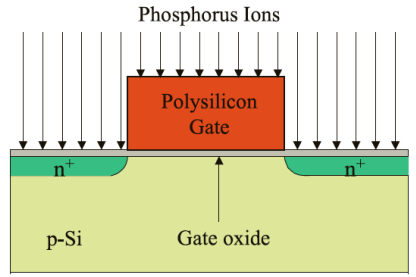


Figure 13.14 Formation of an nMOS with self-aligned S/D.

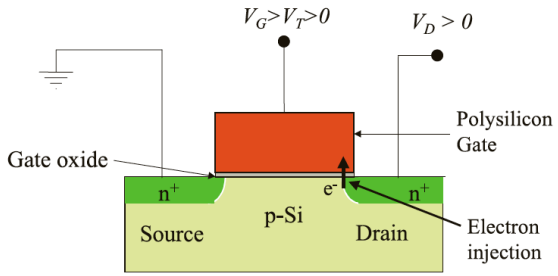
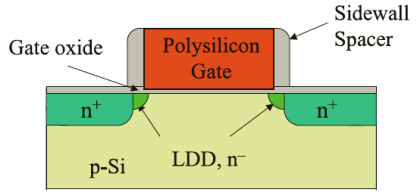


Figure 13.15 Hot electron effect of a MOS transistor.



1 Evidence 1

Hot electron effect occurs in gates <2 μm, causing gate leakage and reliability issues.
Solution: Lightly Doped Drain (LDD) or Source-Drain Extension (SDE).
Shallow, low-dopant junctions extend under the gate.
Sidewall spacers separate heavily doped S/D regions from the gate.
Reduces hot electron impact.

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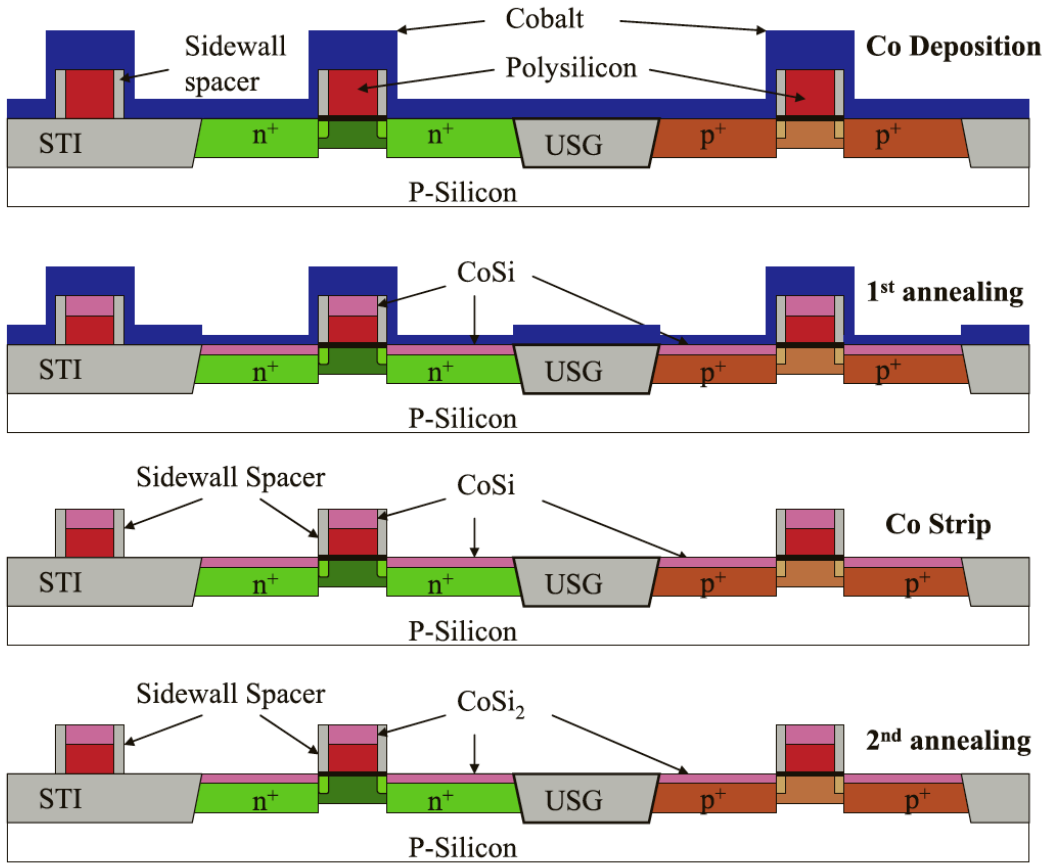
Self Aligned Cobalt Silicide Process



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TOPIC 02

Wafer Sort and Process Evidence

Wafer sort (also called die sort or electrical sort) tests chip performance after fabrication
Each chip is electrically tested for functionality and performance
The wafer is mounted on a vacuum chuck, with electrical probes contacting bonding pads
Probes are connected to power supplies that test circuits and record results
Testing is directed by a computer program
Wafer probers are automated, requiring minimal operator assistance after initial alignment

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TOPIC 02

Wafer Sort

Wafer sort (also called die sort or electrical sort) tests chip performance after fabrication
Each chip is electrically tested for functionality and performance
Wafer sort is a key step in yield calculation during chip production
The wafer is mounted on a vacuum chuck, with electrical probes contacting bonding pads
Testing is directed by a computer program
Wafer probers are automated, requiring minimal operator assistance after initial alignment

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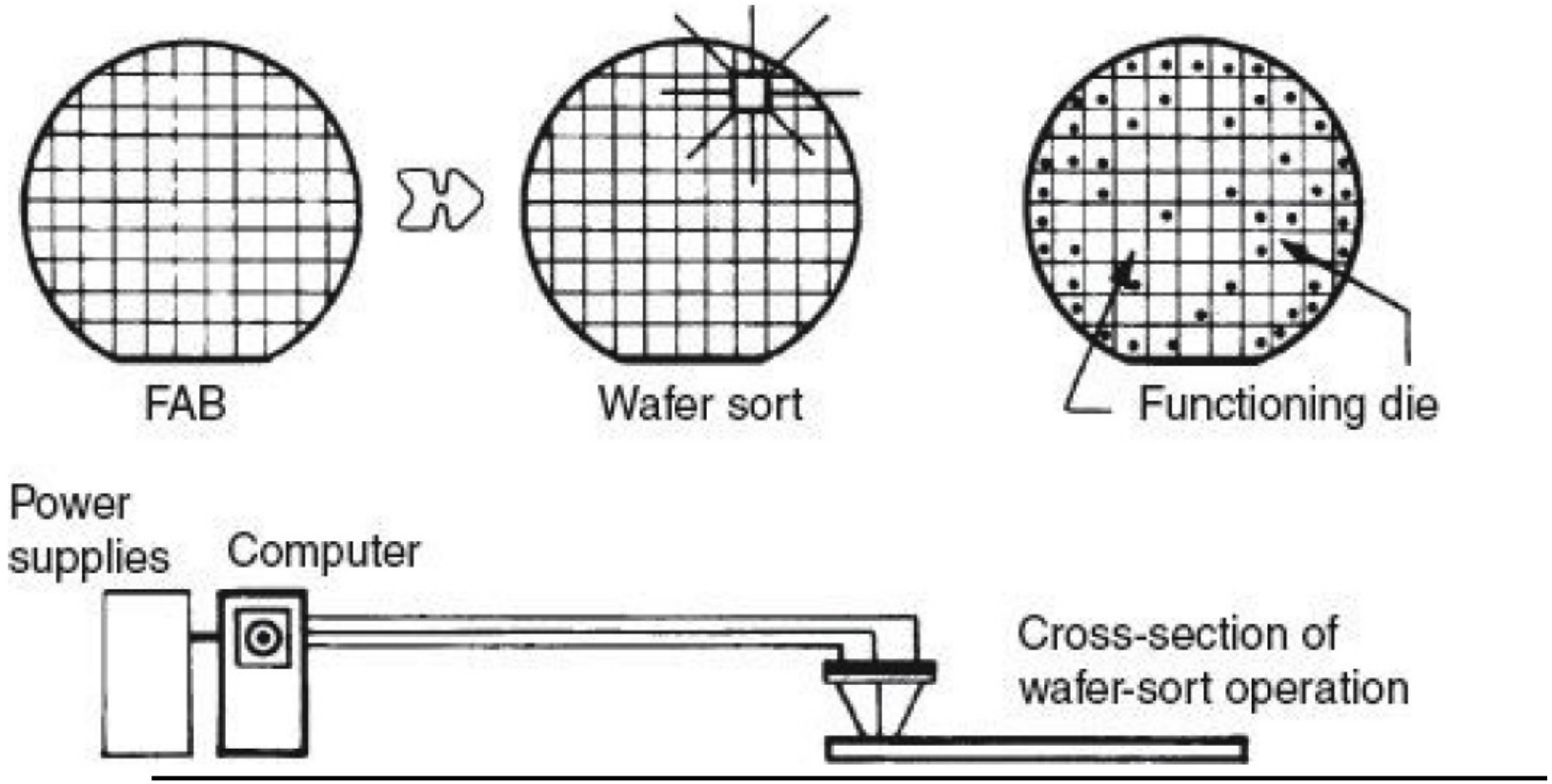
TOPIC 02

Wafer Sort



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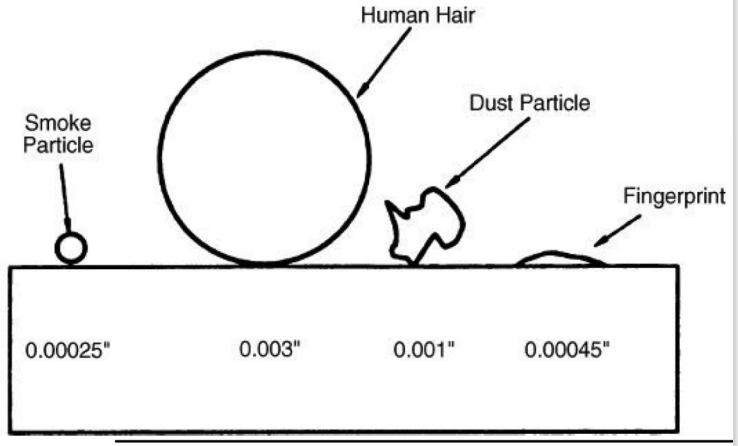
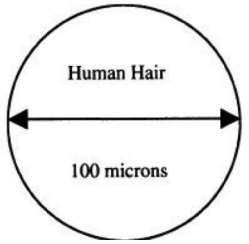
TOPIC 02

Particles





○ One centimeter = 10,000 microns

• One centimeter = 10,000,000 nanometers

1

Evidence 1

Dense integrated circuits are highly sensitive to contamination due to their small feature sizes and thin layers

Feature sizes are in the submicron to nanometer range (e.g., a human hair is ~100 µm)

Devices are vulnerable to airborne particulate contamination from workers, equipment, and chemicals

As feature sizes and films shrink, stricter control over particle size is required to prevent contamination

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TOPIC 02

Mobile ionic contaminants

Semiconductor devices require controlled resistivity, precise N-P junctions, and accurate doping
Small amounts of dopants achieve desired electrical properties
Mobile ionic contaminants (MICs), primarily metal ions, can disrupt device performance and reliability
MICs are electrically charged and highly mobile within semiconductor materials
These contaminants can cause device failure even after testing and shipping
Sodium is the most common mobile ionic contaminant in untreated chemicals and the most mobile in silicon
Controlling sodium is a critical objective in silicon processing
MIC contamination is particularly problematic in metal oxide silicon (MOS) devices
Specialty chemical suppliers have developed low-sodium-grade chemicals for MOS applications
Ultrapure water manufacturing also focuses on reducing MIC contamination

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TOPIC 02

Airborne Molecular Contaminants

Airborne molecular contaminants (AMCs) are fugitive molecules from process tools, chemical systems, or carried in by personnel
AMCs include gases, dopants, and chemicals like oxygen, moisture, organics, acids, and bases
They cause issues in delicate processes like photoresist exposure, shifting etch rates, and unwanted dopants affecting device parameters
AMCs can change the wetting characteristics of etchants, leading to incomplete etching
Detecting and controlling AMCs is crucial with increasing automation
FOUPs (wafer containers) are a source of AMC outgassing, as identified in the 2011 ITRS Yield Enhancement chapter

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Contamination-Caused Problems

1

Key idea 1

Contaminants affect the processing and devices in three specific performance areas include:
Device processing yield
Device performance
Device reliability

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Contamination-Caused Problems

1

The major contamination sources are:

The major contamination sources are:
Air
The production facility
Cleanroom personnel
Process water
Process chemicals
Process gases
Static charge
Process equipment

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Air cleanliness class standards

Class	Particles/ft^3				
	0.1 um	0.2 um	0.3 um	0.5 um	5 um
M-1	9.8	2.12	0.865	0.28	
1	35	7.5	3	1	
10	350	75	30	10	
100		750	300	100	
1000				1000	7
10,000				1000	70

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Typical class numbers for various environments

Environment	Class size	Maximum particle size (micron)
Projected-256 Mbit	0.01	<<0.1
Mini-environment	0.1	<0.1
ULSI Fab	1	0.1
VLSI Fab	10	0.3
VLF Station	100	0.5
Assembly Area	1000–10,000	0.5
House room	1000,000	
Outdoors	>500,000	

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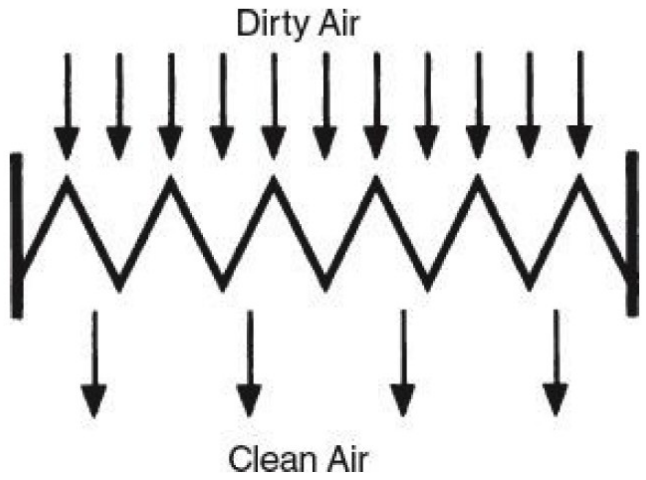
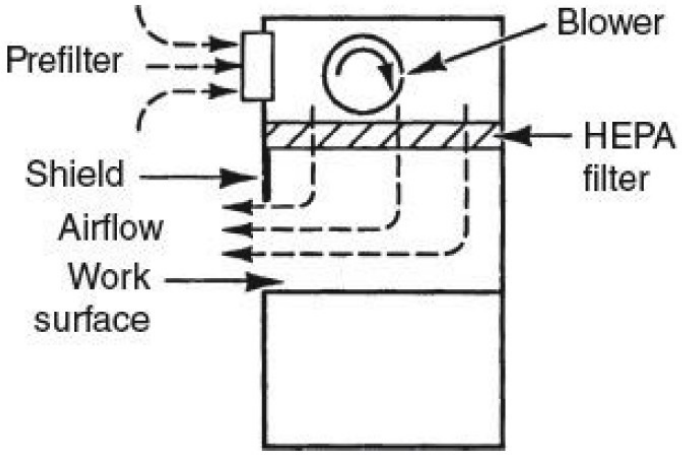
HEPA filter





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Evidence 1

Air is drawn through a prefilter by a fan and forced through a HEPA filter
The air exits the filter in a laminar flow pattern
A shield directs the laminar airflow over the wafers before exiting the hood
The setup is called a vertical laminar flow (VLF) station due to the laminar airflow

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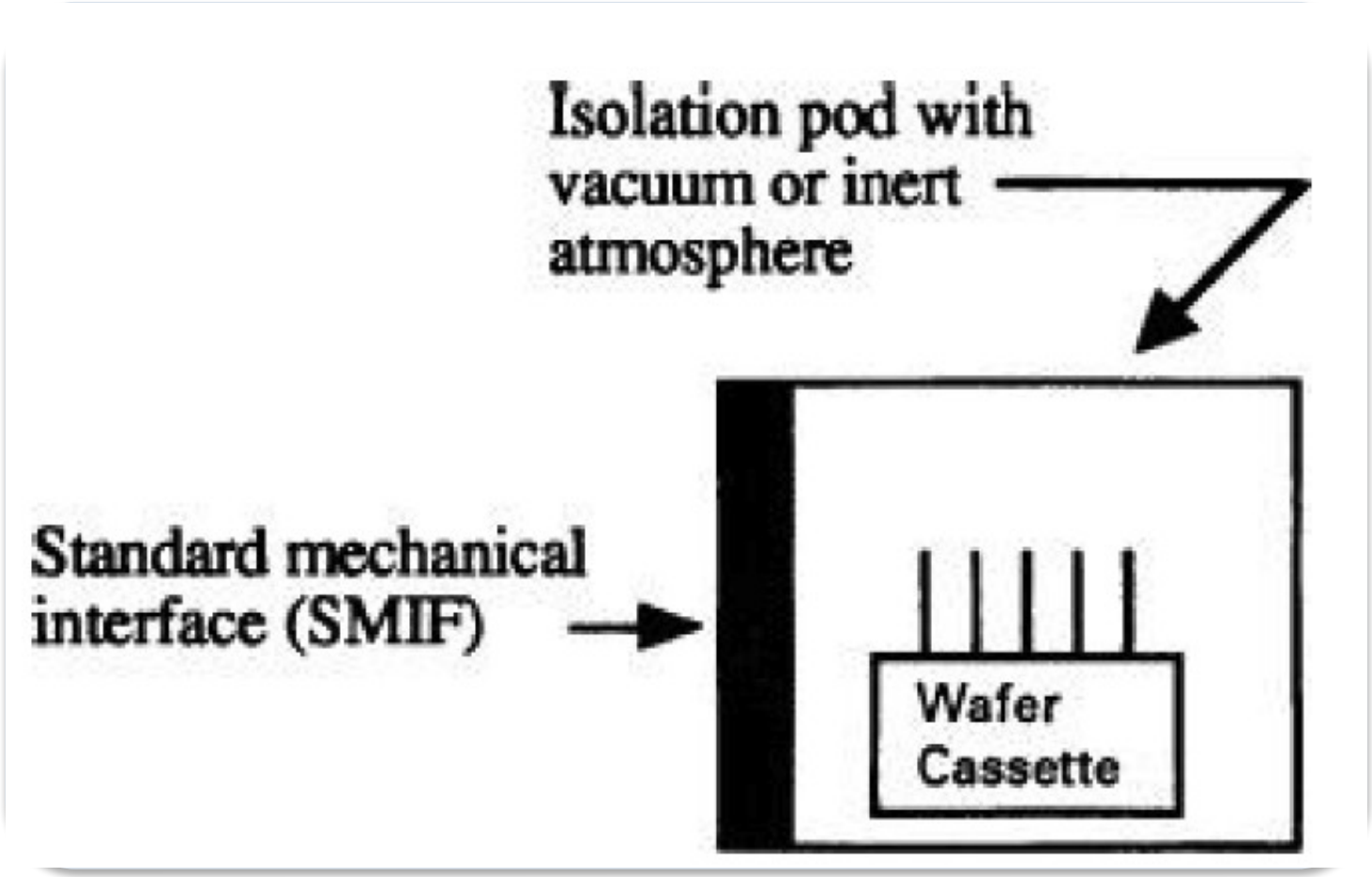
Micro-and Mini-Environments



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Evidence 1

Mid-1980s projections showed rising cleanroom costs with diminishing effectiveness
Focus shifted to isolating wafers in smaller environments for better control
Steppers and other process tools already employed clean microenvironments for wafer loading and unloading
This isolation approach became the new standard for improving contamination control
Mini-environments prevent wafer exposure to room air
SMIF (Standard Mechanical Interface) developed by HP in the 1980s isolates wafers in pressurized pods

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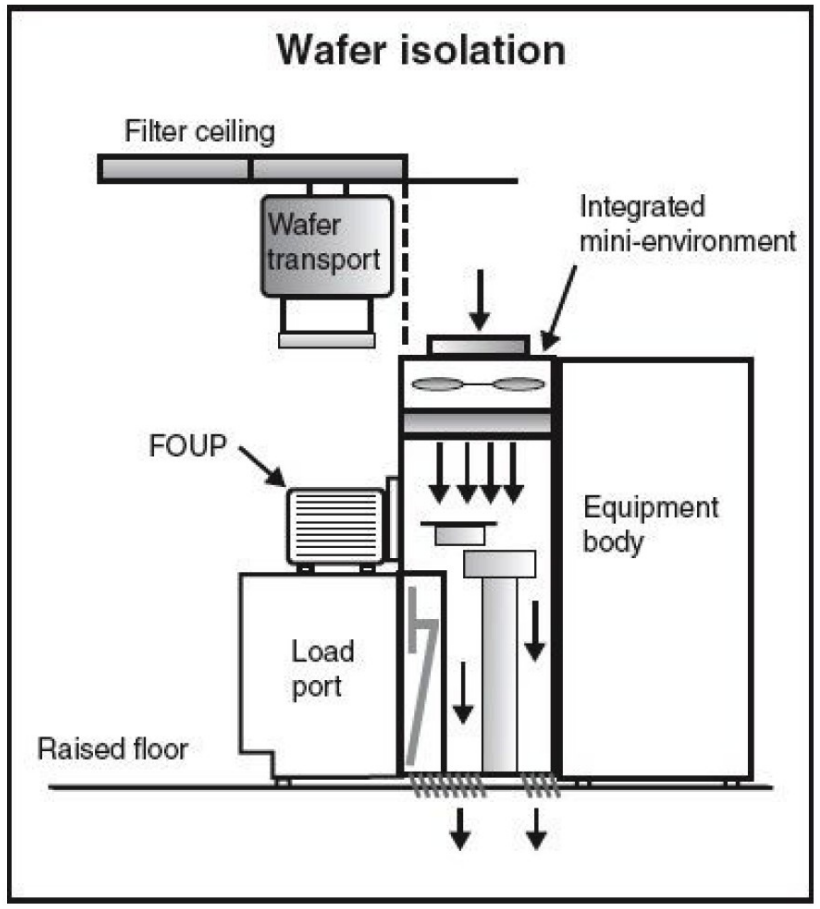
Micro-and Mini-Environments



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Evidence 1

WIT (Wafer Isolation Technology) includes wafer pods (e.g., FOUPs) for transport, isolated tool environments, and wafer loading mechanisms
FOUPs connect to tools but may be contaminated by outgassed chemicals
Mini-environments improve temperature, humidity, and cleanliness control, reducing contamination and yield loss
WIT lowers costs by reducing cleanroom standards and operator constraints
Larger wafer pods require robotic handling, increasing complexity
Layouts include storage (stockers) for wafers awaiting processing and zone off process areas to avoid cross-contamination

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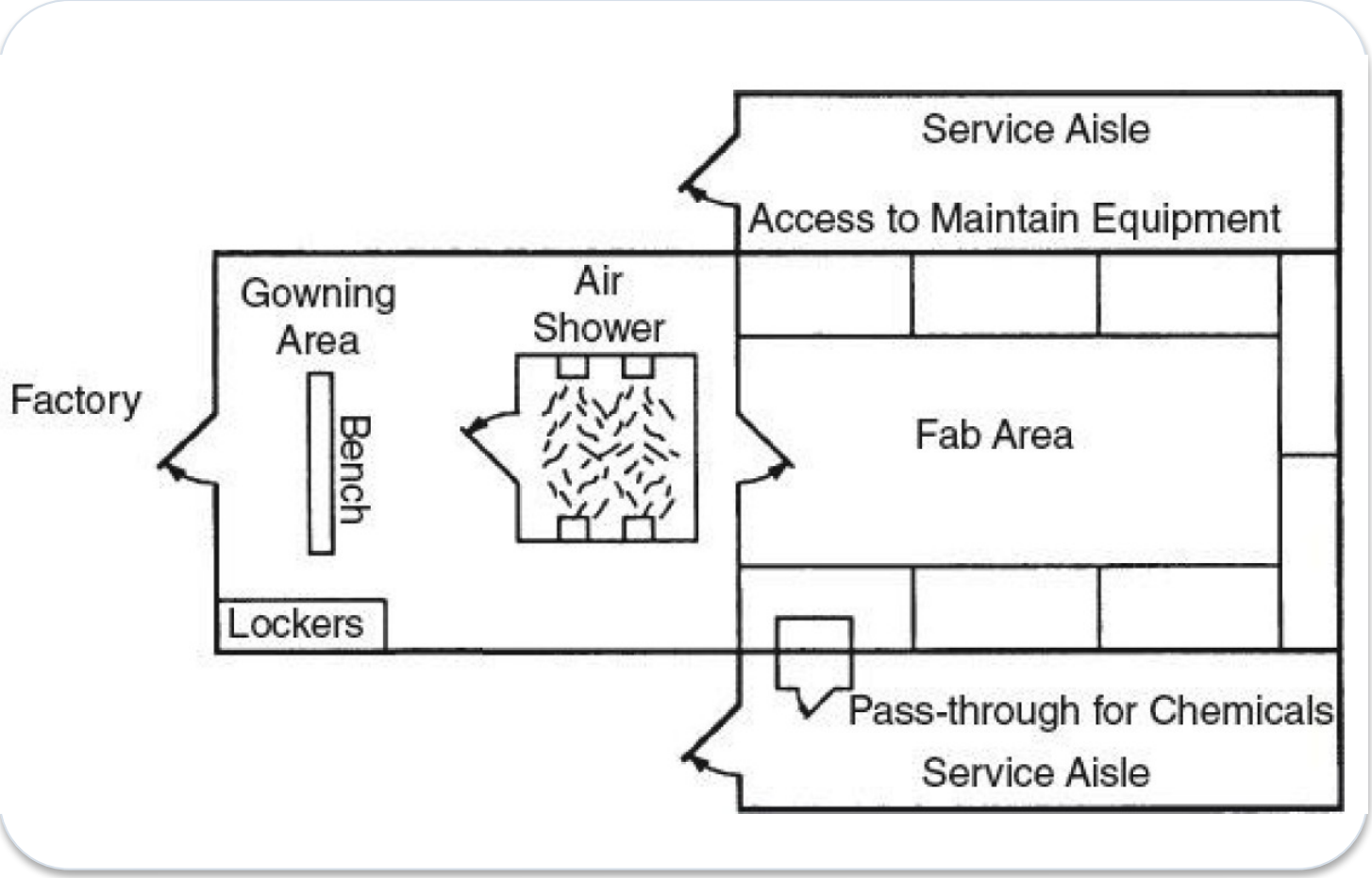
Cleanroom Elements



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Controlled airflow

Controlled airflow
Air filtration systems (e.g., HEPA filters)
Positive air pressure
Airlocks or vestibules
Cleanroom attire for personnel
Restricted access
Cleaning protocols
Equipment enclosures
Surface cleanliness maintenance

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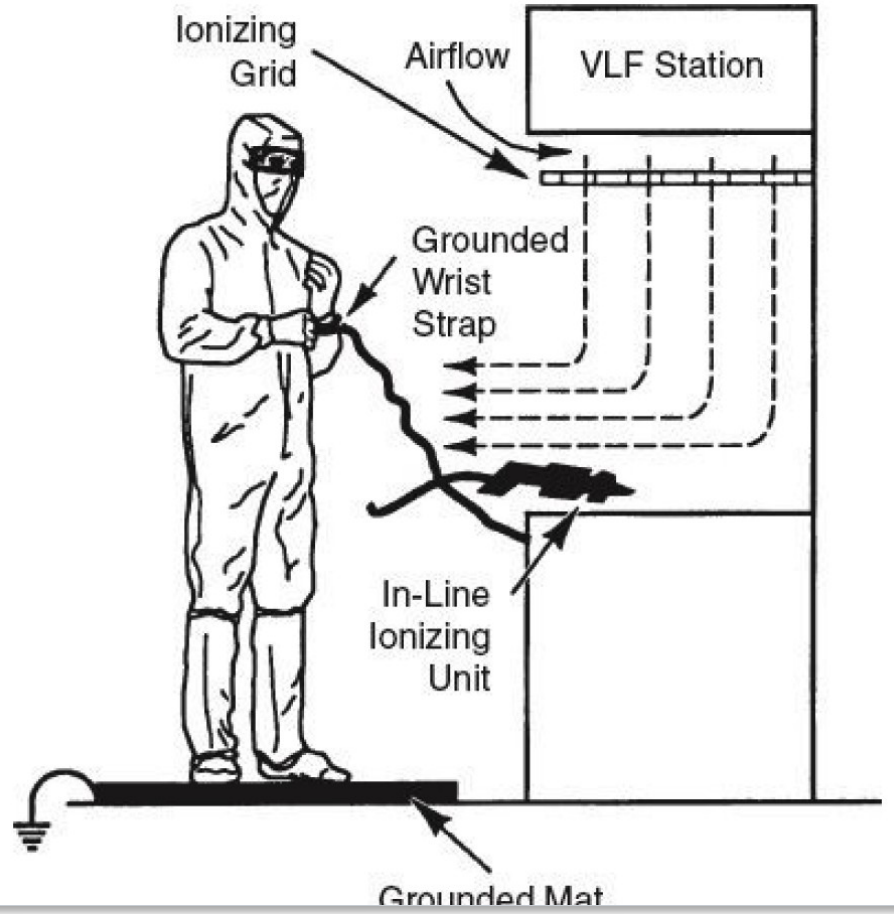
Static Control



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Evidence 1

Higher-density circuits with submicron features are highly vulnerable to static-attracted contamination
Static charges (up to 50,000V) build up on wafers, storage boxes, surfaces, and equipment, attracting particles from the air and garments
Statically held particles are difficult to remove with standard cleaning methods
Most static charge is caused by triboelectric charging, where materials in contact become charged when separated
One material becomes positively charged (losing electrons), the other negatively charged (gaining electrons)

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Process Water

Water quality parameter	Units	Typical municipal water supply	Typical ultrapure water product
Resistivity	M Ohms-cm	0.004	>18
pH	Units	i	6
TOC	ppb	3500	<10
Amonium	ppb	300	<1
Calcium	ppb	22000	<1
Magnesium	ppb	4000	<1
Potassium	ppb	4500	<1
Silica	ppb	4780	<10
Sodium	ppb	29000	<1
Chloride	ppb	15000	<1
Fluoride	ppb	740	<1
Sulfate	ppb	42000	<1

1

Evidence 1

Wafers undergo multiple etching and cleaning steps, each followed by a water rinse
Fabrication facilities use millions of gallons of water daily, requiring strict cleanliness standards
City water contains contaminants like dissolved minerals, particulates, bacteria, organics, dissolved oxygen, and silica
Reverse osmosis (RO) and ion-exchange systems remove contaminants
Deionized (DI) water has a resistivity of 18 MΩ-cm; ultrapure water (UPW) reaches 18.2 MΩ-cm at 25°C

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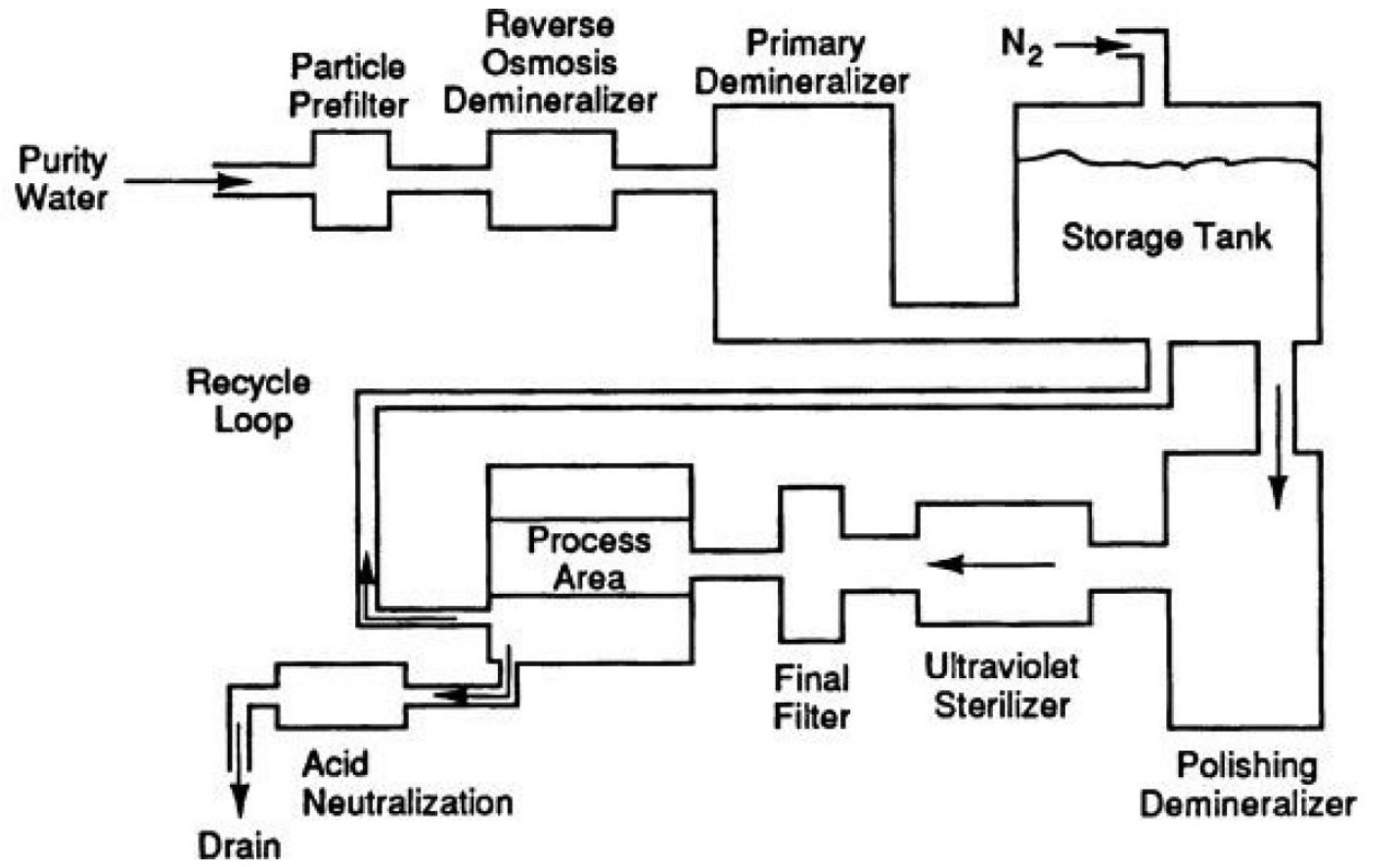
Deionized water system.



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RCA Clean

RCA cleans remain the standard for prefurnace cleaning, with chemical purity improvements over time
SC-1 and SC-2 steps can be reversed depending on the application
For oxide-free surfaces, an HF step is used before, between, or after RCA cleans
Adaptations include adding chelating agents (e.g., EDTA) to bind metallic ions and prevent redeposition
Diluted RCA solutions (e.g., SC-1: 1:1:50, SC-2: 1:1:60) are as effective as concentrated versions, reduce microroughness, and are cost-effective

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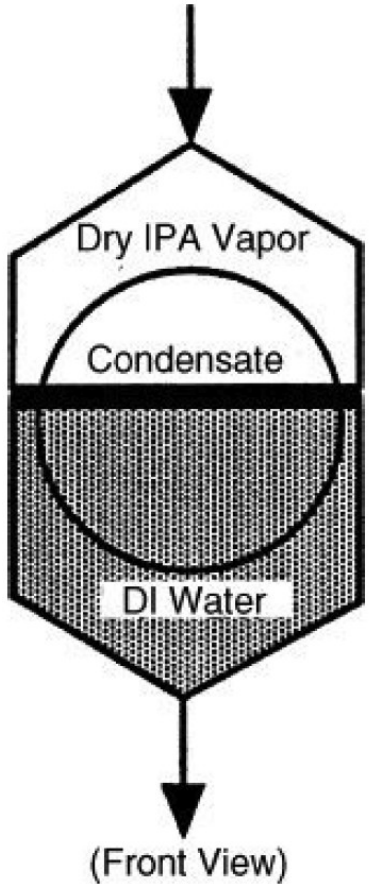
Isopropyl Alcohol (IPA) Vapor Drying



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Evidence 1

Alcohol drying uses IPA vapor to replace water on wafer surfaces
Chilled coils condense water vapor, leaving wafers dry
Direct displacement dryer moves wafers from a DI water bath directly into the IPA vapor zone for water displacement

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TOPIC 02

Yield and Process Variation

High yields are critical in semiconductor manufacturing, with typical yields ranging from 20% to 80% due to process complexity
Nonrepairable errors mean defective chips/wafers often can't be recovered
High capital costs and competition push for high-volume, high-yield processes
Yield is tracked at key points: after wafer fabrication, wafer sort, and final testing

Major yield measurement points

Manufacturing stage	Measured
Wafer fabrication- Yield =	$\frac{\text{\# Wafers out}}{\text{\# Wafers started}}$
Wafer sort- Yield =	$\frac{\text{\# Functioning (good) die}}{\text{\# Die on wafer(s)}}$
Packaging- Yield =	$\frac{\text{\# Packages die passing final test}}{\text{\# Good die started into packaging}}$

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Accumulative Wafer-Fabrication Yield

Step	Wafers in	Yield*	Wafers out	Accumulated yield
1. Field oxidation	1000	99.5	995	99.5
2. Source/drain mask	995	99.0	965	96.5
3. Source/drain doping	965	99.3	978	97.8
4. Gate region mask	978	99.0	968	96.8
5. Gate oxidation	968	99.5	964	96.4
6. Contact hole mask	964	94.0	906	90.6
7. Deposit metal layer	906	99.2	899	89.9
8. Metal layer mask	899	97.5	876	87.6
9. Alloy metal layer	876	100	876	87.6
10. Passivation layer deposition	876	99.5	872	87.2
11. Passivation layer mask	872	98.5	859	85.9

*Yield values are typical for the particular steps.

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Wafer-Fabrication Yield Limiters

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Key idea 1

Wafer-fabrication yield is influenced by five key factors:
Number of process steps
Wafer breakage and warping
Process variation
Process defects
Mask defects

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Wafer Breakage and Warping

Wafer breakage and warping are significant yield-limiting factors in fabrication:
Wafers are fragile, typically 300 mm in diameter and 800 µm thick, and require careful handling
Frequent handling and heat treatments increase the risk of breakage by inducing strain
Automatic handlers only process full wafers, so any breakage leads to rejection
Silicon wafers are easier to handle with automation, but gallium arsenide wafers are more prone to breakage, limiting yield
Warping from rapid heating/cooling distorts the surface, causing image projection errors during patterning

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Process Variation

During wafer fabrication, doping, layering, and patterning processes must meet strict physical and cleanliness standards, but variations still occur across wafers, batches, and days.
Process control aims to keep operations within specifications and maintain consistent process parameters (e.g., time, temperature, pressure)
Statistical process control techniques monitor these parameters
Inspections, tests, and equipment calibration help detect unwanted variations
Despite best efforts, some variation is inevitable, requiring process engineering and circuit design to accommodate these differences while ensuring functionality

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Process Defects

Process defects are isolated regions of contamination or irregularities on the wafer surface, often called spot defects or point defects.
Killer defects render circuits inoperable, while smaller defects may go undetected until wafer sort
Sources of defects include liquids, gases, air, personnel, machines, and water used in fabrication
Many defects occur during the patterning process, where particulates damage the fragile photoresist layer, causing pinholes in the wafer surface
Wafers are inspected frequently, and those exceeding contamination limits are rejected
The SIA ITRS sets a maximum defect density of 0.68 defects per cm^2 for 300-mm wafers

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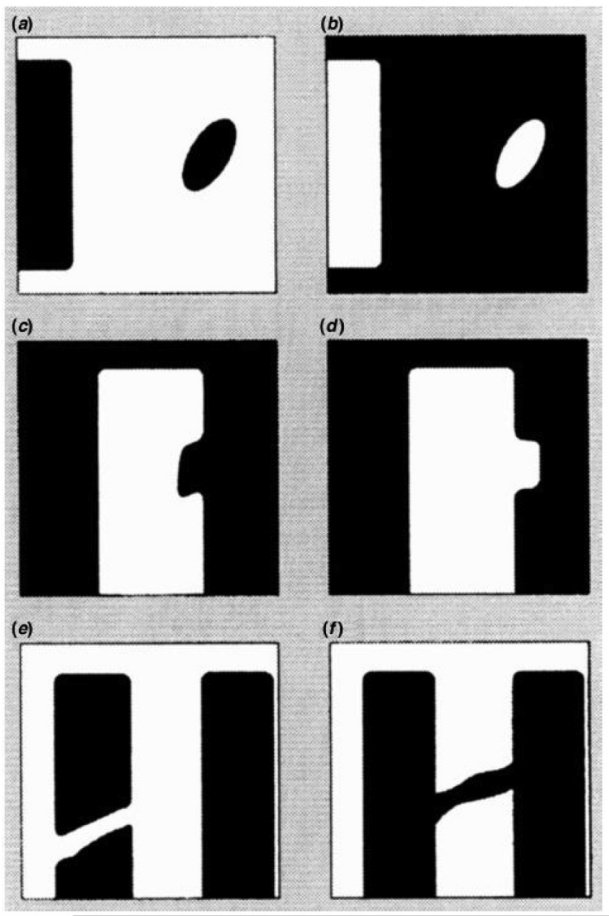
Mask Defects



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Evidence 1

Photomask or reticle defects can transfer directly to the wafer, causing issues during the patterning process. Three common defects include:
Contamination: Dirt or stains block light, printing as opaque areas on the wafer
Cracks: Cracks in the quartz plate block or scatter light, creating unwanted or distorted images
Pattern distortions: Defects like pinholes, chrome spots, pattern breaks, or bridges between patterns
These defects are more critical for circuits with smaller feature sizes, higher densities, and larger die sizes.

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Defect Detection

Black spots	Post-Cu CMP	Slurry/low-k Rxn
Etch residue	Post-trench etch	Post-etch clean
Low-k bumps	Post-low-k dep	Cu hillocks
Craters	Post-via etch	Etching of "bumps"
Cracking	Post-via etch	Stress of ILD
Resist poisoning	Via photo	Via patterning
Bond pad blistering	Low- dep	Adhesion loss

1

Evidence 1

Laser beams and e-beams are used for detecting smaller particles on wafer surfaces
Advantages:
Increased sensitivity to smaller particle sizes due to high-intensity reflected light
Helium and neon lasers are commonly used for detection
E-beams offer even smaller spot sizes, ideal for nano-sized patterns
E-beams are especially effective for copper and low-k dielectric metallization systems

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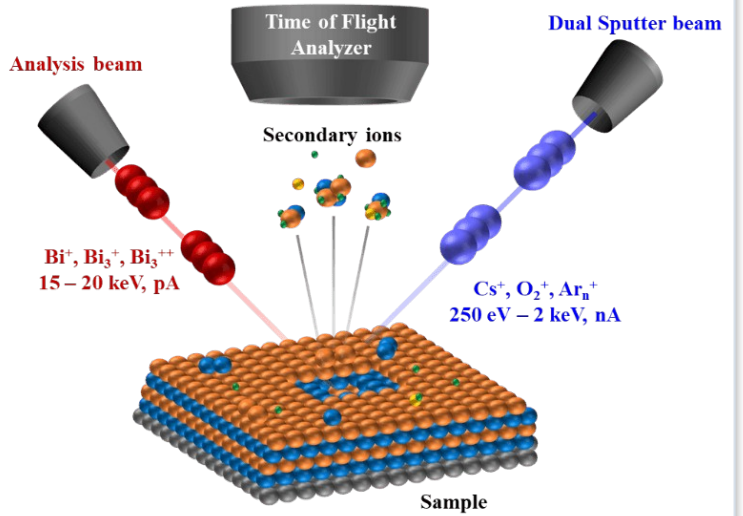
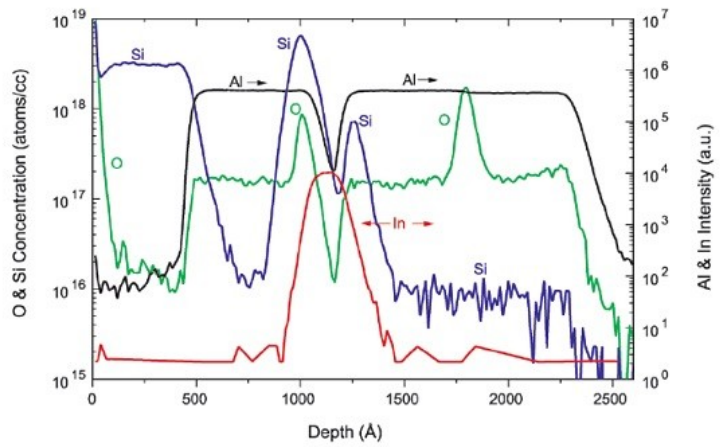
Secondary Ion Mass Spectrometry (SIMS)





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Evidence 1

Secondary Ion Mass Spectrometry (SIMS) bombards the wafer surface with high-energy ions (ion-sputtering)
Collisions produce secondary ions that identify the surface materials
SIMS detects dopants in the junction region
The junction is located where the dopant type disappears from the stream of secondary ions

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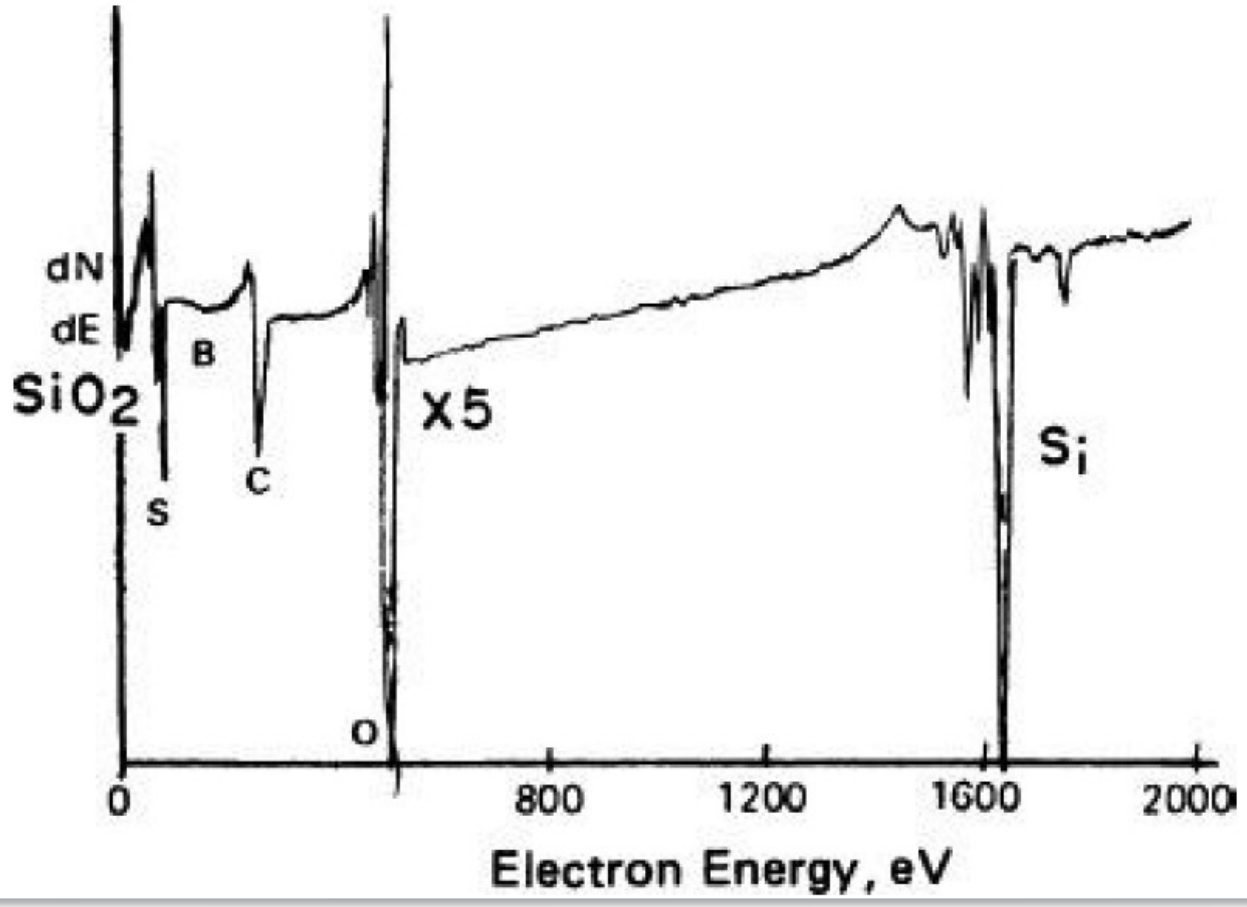
Auger Electron Spectroscopy



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Evidence 1

SEM releases a spectrum of secondary electrons when the electron beam hits the surface
Auger electrons are a subset released from the top nanometers, with energies specific to the emitting element
Different elements (e.g., sodium, chlorine) release distinct Auger electrons
Auger electron analysis allows for material identification, including surface contamination
The e-beam scans the wafer, and Auger electrons are analyzed for energy (wavelength)
Results are displayed on an x-y plot where energy peaks indicate the presence of specific elements

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Wafer-Sort Yield Factors

After fabrication, wafers undergo wafer sort testing to check each chip's electrical functionality and device specifications. Despite passing in-process checks, undetected defects and process variations can result in nonfunctional chips.
Factors influencing yield during wafer sort include:
Wafer diameter
Die size (area)
Number of processing steps
Circuit density
Defect density
Crystal defect density
Process cycle time

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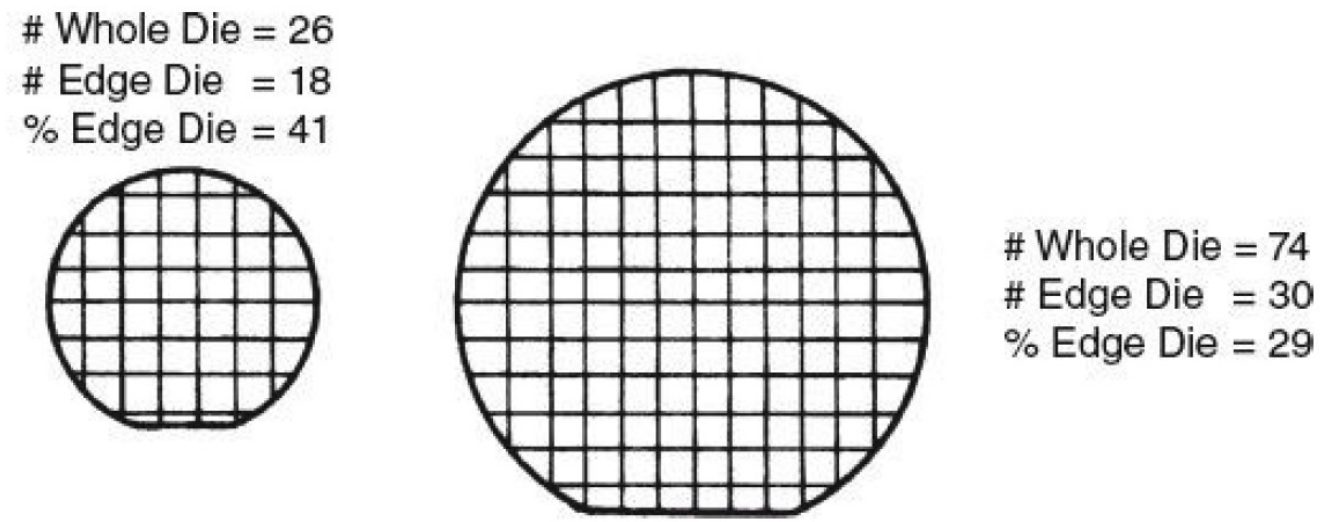
Wafer-Sort Yield Factors



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Evidence 1

The shift to larger-diameter wafers in the semiconductor industry has been driven by production efficiency and increasing die sizes.
Initial wafers were less than 1 inch in diameter
Standard wafer sizes evolved to 150 mm (6 in) in the 1980s, 200 mm in the 1990s, and 300 mm by 2012
Larger wafers increase production efficiency, as the incremental cost of processing larger wafers is outweighed by the significant increase in available chips per wafer

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Wafer Diameter and Die Size

Die size (mils)	Wafer diameter (mm)			
	150	200	300	450*
300 x 300	293	531	820	1845
400 x 400	113	165	550	1238
500 x 500	108	191	410	923

1

Evidence 1

Another reason for moving to larger-diameter wafers is the trend toward larger die sizes.
As die size increases without increasing wafer diameter, the percentage of whole die on the wafer decreases
To maintain good wafer-sort yields with larger die sizes, larger wafer diameters are needed
Larger wafers are more cost-effective, fitting more chips and improving production efficiency

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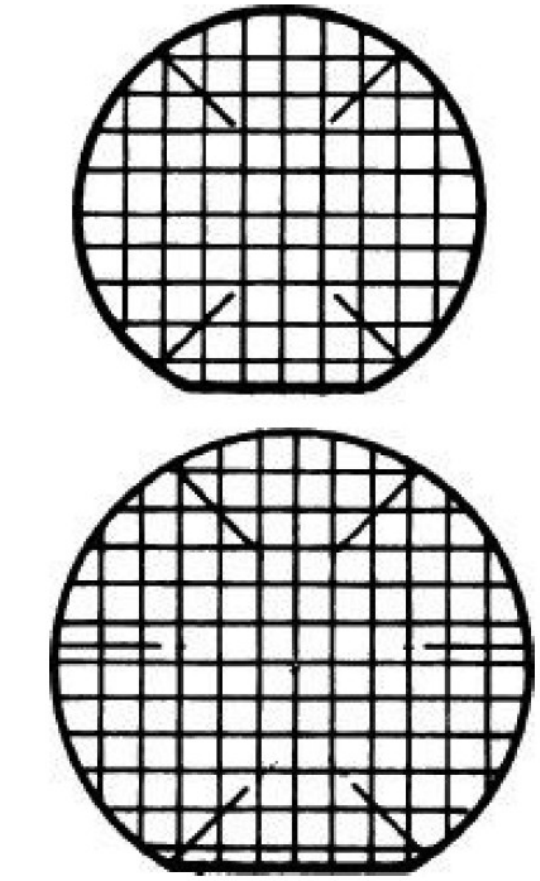
Crystal Defects



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Evidence 1

Crystal dislocations are point defects caused by discontinuities in the wafer's crystal structure, affecting wafer-sort yield alongside contamination and process defects.
Dislocations can form during fabrication, often nucleating from chips or abrasions on the wafer edge due to poor handling or equipment
These dislocations propagate toward the center during heat treatments (e.g., oxidations and diffusions), with their length depending on the wafer's thermal history
Wafers with more process steps or heat treatments develop more dislocation lines, impacting more chips
Larger-diameter wafers help mitigate this issue by leaving more unaffected die in the center

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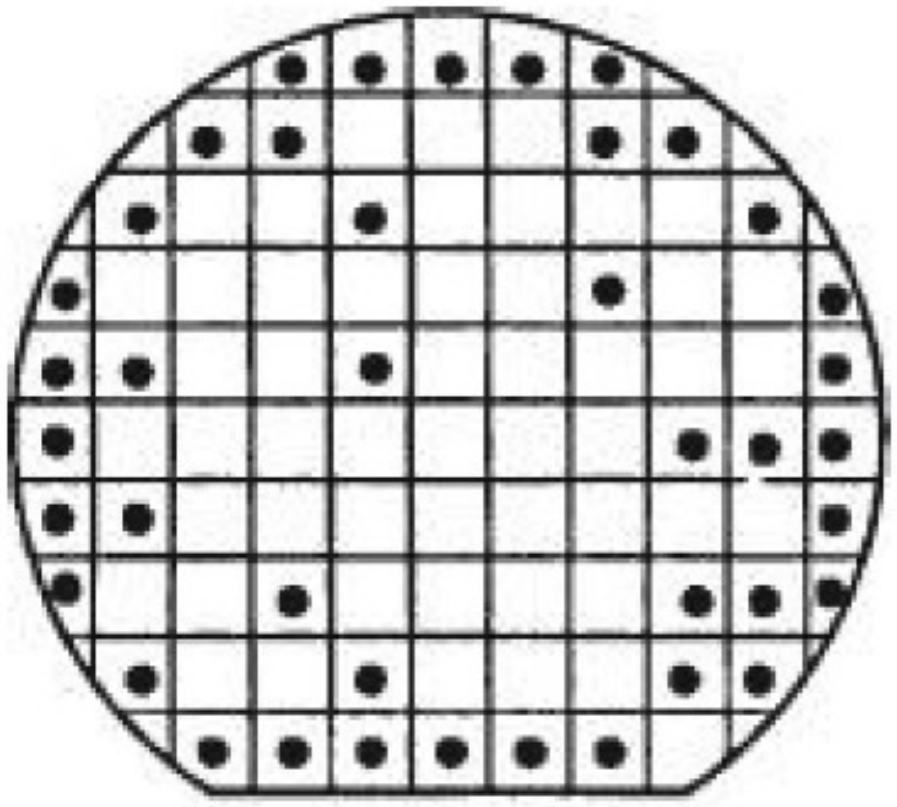
Process Variations



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Evidence 1

Process variations affect wafer-sort yield, with issues often concentrated around the wafer edges.
Key factors include:
Sampling limitations: Not all defects are detected during in-process inspections, leading to undetected problems that show up as failed devices at wafer sort
Temperature nonuniformity: In tube furnaces, edges heat and cool faster, causing variations at the wafer's outer edges
Contamination and handling: Physical contact and contamination are more frequent at wafer edges
Patterning uniformity: Full mask projection processes are less uniform at the edges, while steppers reduce image variations by exposing smaller areas

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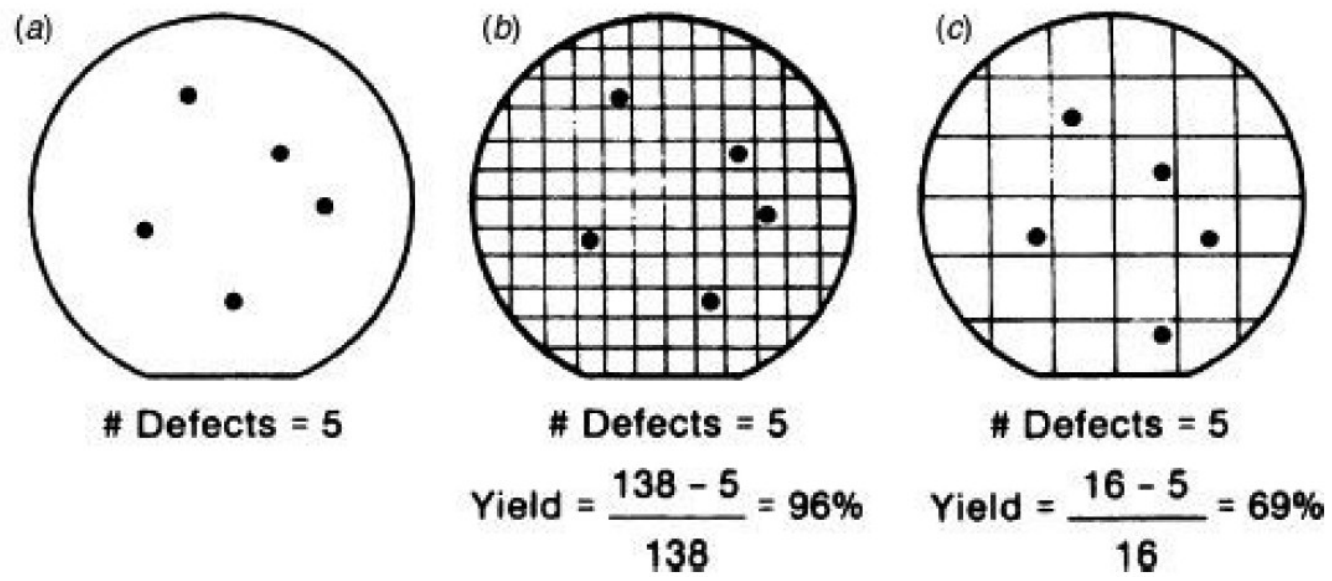
Die Area and Defect Density



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Evidence 1

Die size impacts wafer-sort yield in relation to defect density:
Background defect density: Defects on the wafer surface are present regardless of die size, device type, or process controls
Effect on yield: Larger die sizes result in lower yields at the same defect density, as more defects are likely to fall within each die area

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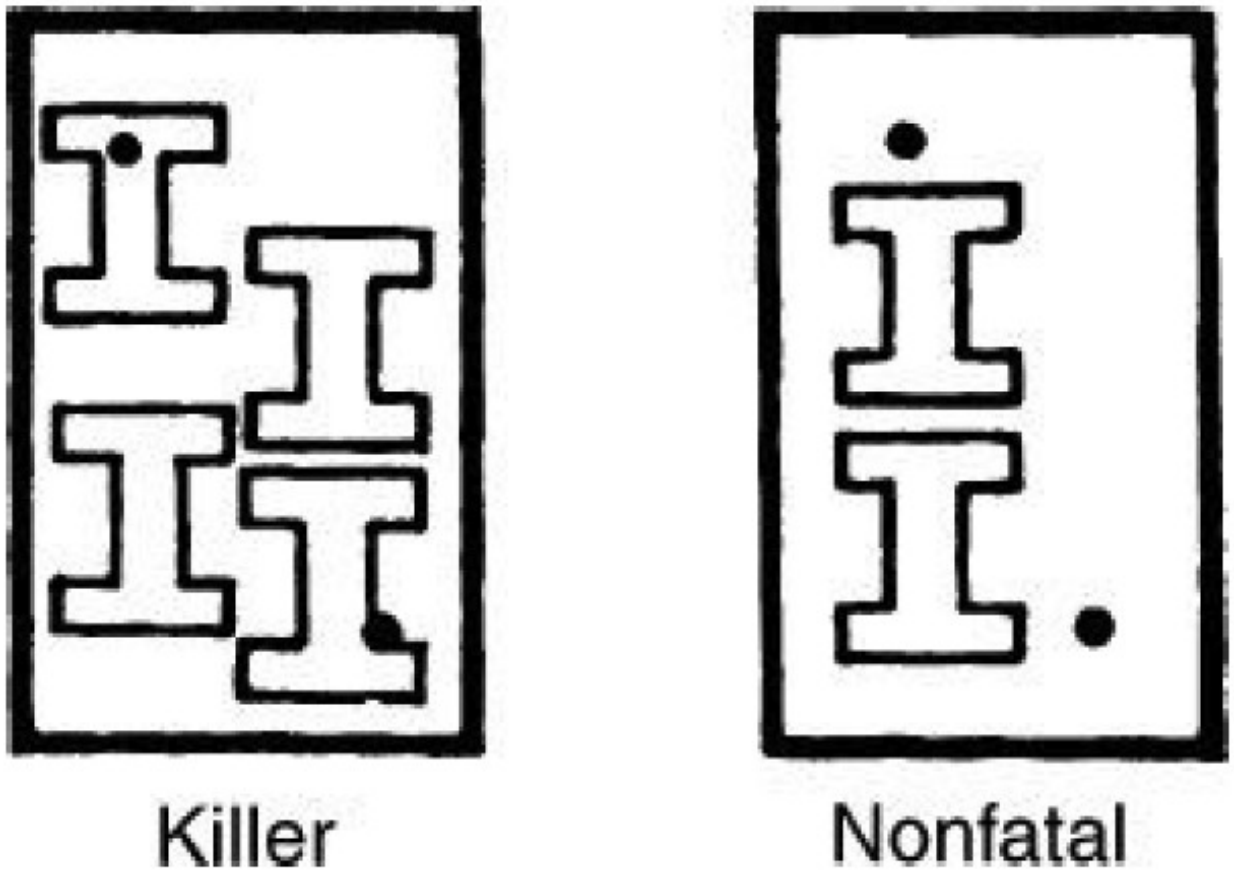
Circuit Density and Defect Density



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Evidence 1

Wafer surface defects cause die failures by disrupting critical circuit areas. While some defects may not impact nonsensitive parts, increasing circuit integration—due to smaller feature sizes and higher component density—raises the likelihood of defects occurring in active areas, thus reducing wafer-sort yield.

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Wafer-Sort Yield Formulas

Negative binomial

$$Y_r = \frac{1}{\left(1 + \frac{AD_0}{\alpha}\right)^{\alpha}}$$

Murphy

$$Y_r = \left(\frac{1 - e^{-AD_0}}{AD_0}\right)^2$$

Bose-Einstein

$$Y_r = \frac{1}{(1 + AD_0)^n}$$

Seeds

$$Y_r = e^{-\sqrt{AD_0}}$$

Poisson

$$P(k) = \frac{\lambda^k k e^{-\lambda}}{k!}$$

$$Y_r = P(0) = e^{-\lambda} = e^{-AD_0}$$

$$Y_r = \int_0^{\infty} F(D) e^{-AD} d$$

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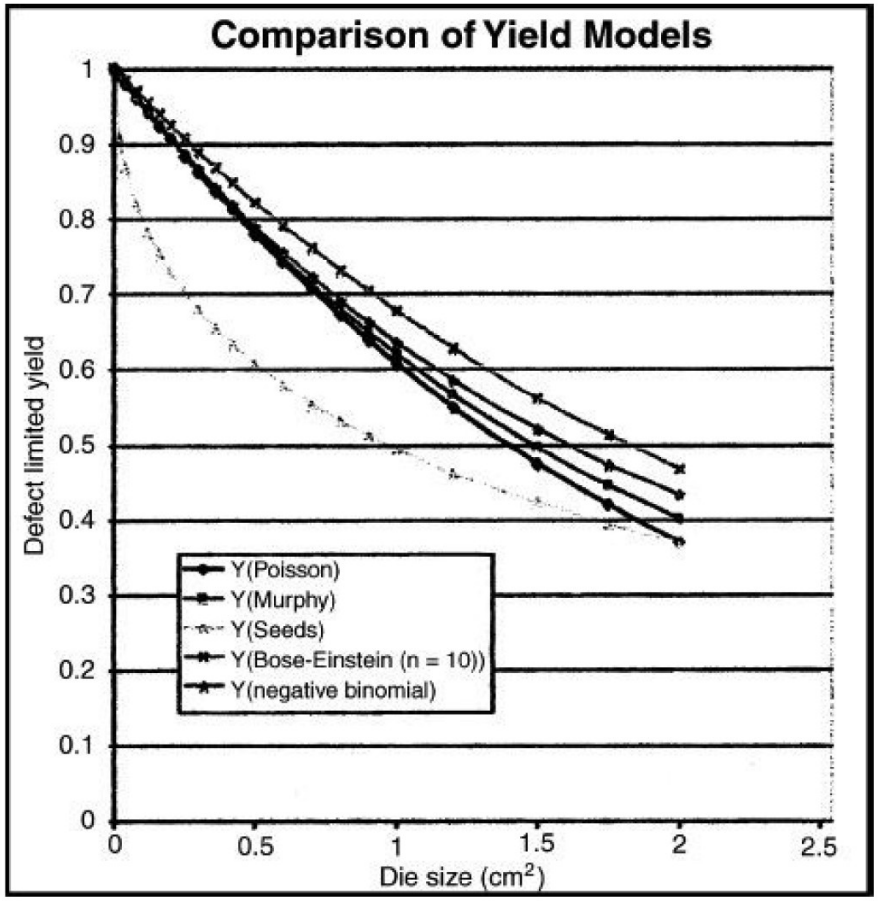
Comparison of Yield Models



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Evidence 1

The defect density in yield models differs from that found through optical inspection. Yield models focus on "killer defects", which include contaminants, surface, and crystal defects that destroy die.
Non-critical defects and cases where multiple defects fall in the same sensitive area are excluded from these models.

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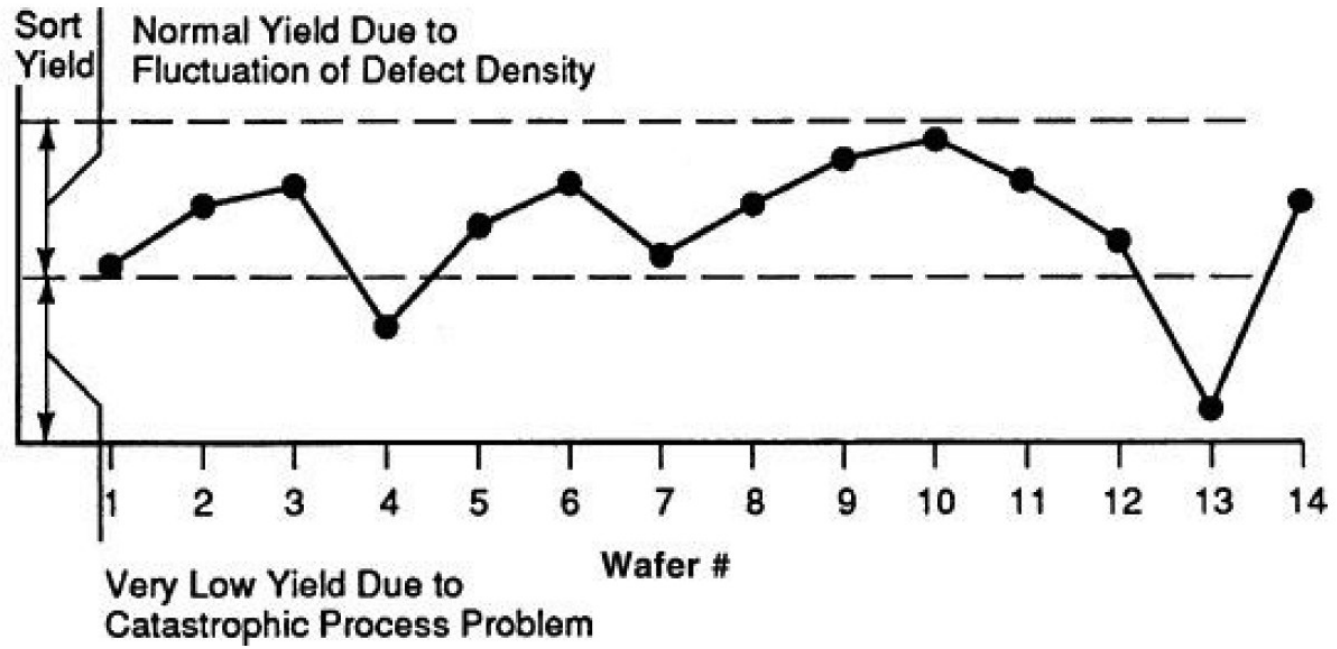
Yield Predictions



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Evidence 1

Yield predictions from formulas assume a controlled process. However, in practice, wafer-sort yield varies from wafer to wafer due to normal process variations during fabrication.

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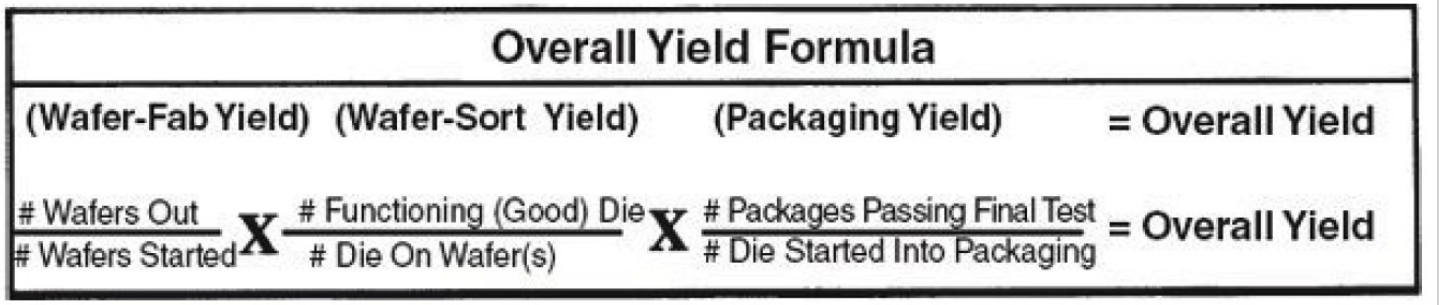
Overall Process Yields



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Evidence 1

The overall process yield is the product of three major yield points. This percentage represents the ratio of shipped die to the total number of whole die on the starting wafer, providing a comprehensive measure of process success

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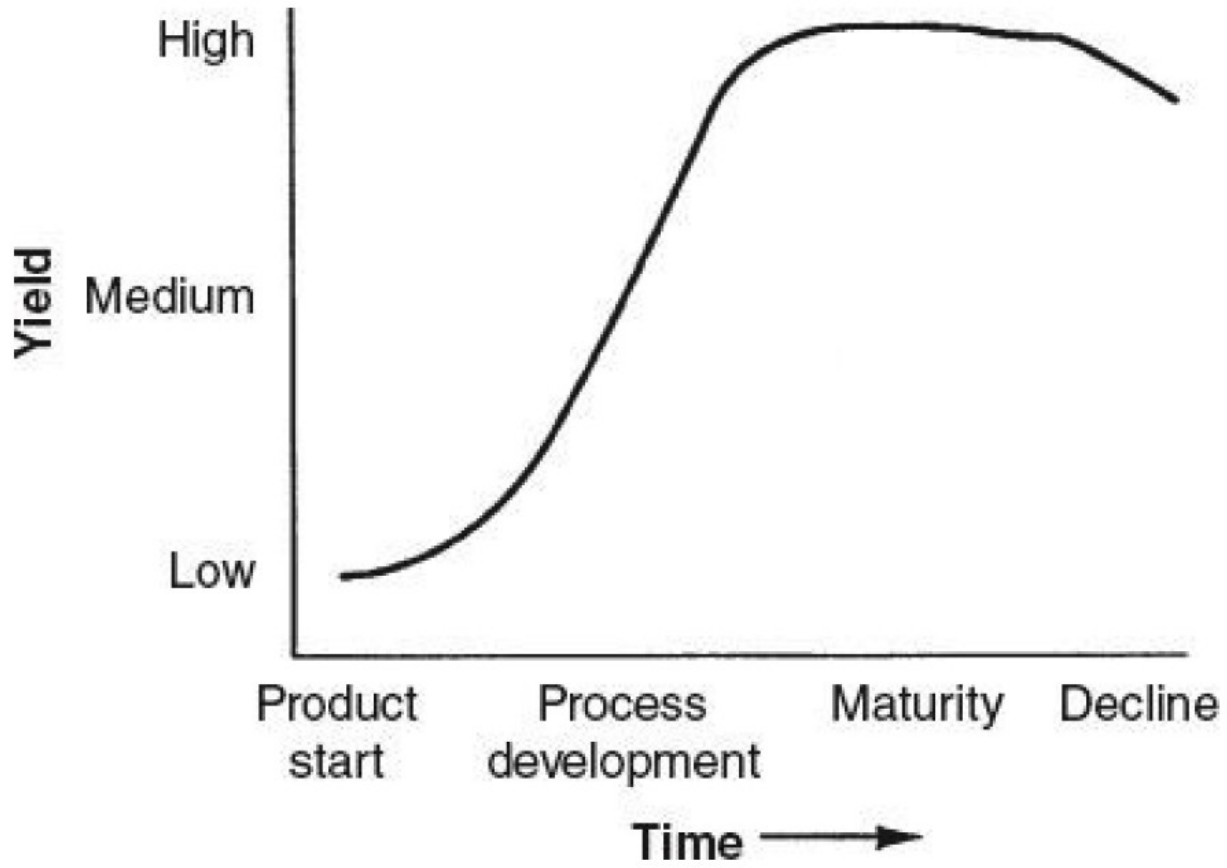
Overall Process Yields



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Evidence 1

Process yields typically follow an S-curve pattern over a product's manufacturing lifetime. Initially, yields rise slowly as early issues are resolved, followed by rapid improvement.
Eventually, yields level off due to process limits such as die size, integration level, circuit density, and defect density.
Overall yields can vary widely, from very low for new or poorly designed products to around 90% for simpler, mature products.
Yield performance is proprietary, as it directly impacts profit and production control.

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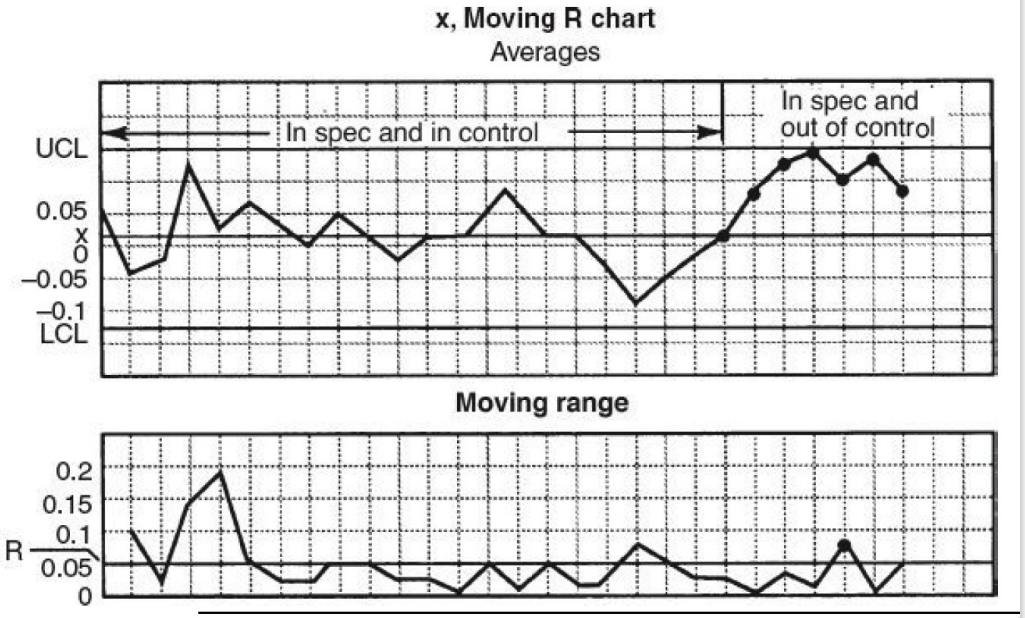
SPC (Statistical Process Control)

SPC (Statistical Process Control) ensures process control and yield improvement using statistical tools.
Monitoring and adjustments are needed to keep processes in control.
SPC uses averages and ranges to analyze data.



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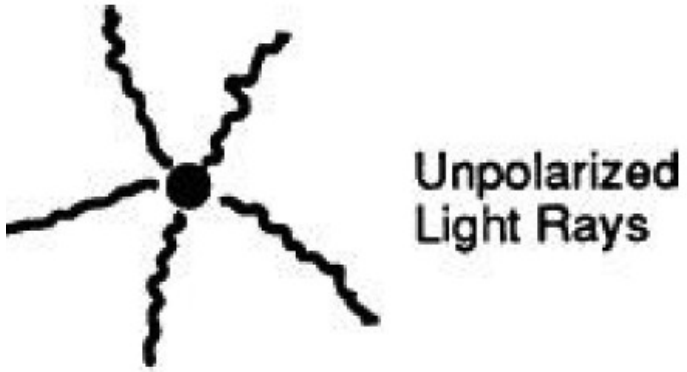
Ellipsometry



Unpolarized Light Rays

Polarized Light Rays

1

Evidence 1

Ellipsometry: A film-thickness measurement technique using polarized light and measuring beam rotation to determine thickness and refractive index.
Applications: Measuring thin oxides and multilayer thin-film stacks.
Advantages: High accuracy, especially for thin oxide

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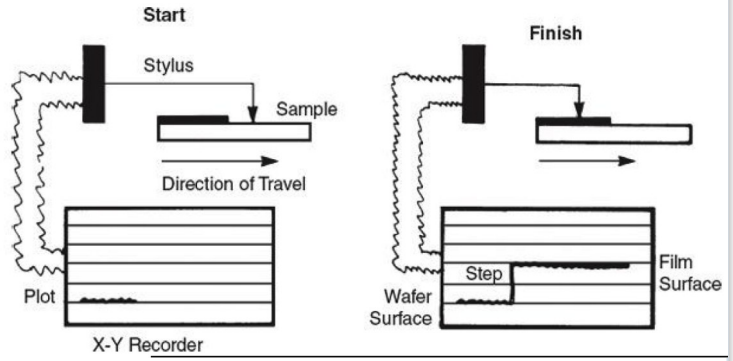
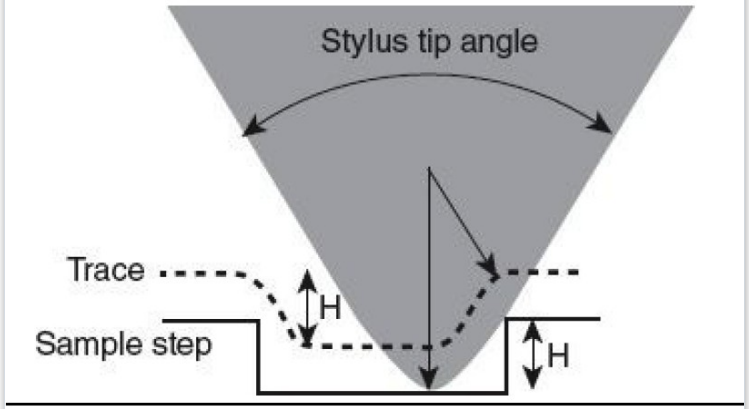
TOPIC 02

Stylus





1

Evidence 1

Mechanical Stylus Measurement: Used to measure film thickness of materials not suitable for optical techniques.
Method: A step is created in the film, and a stylus measures the height difference.
Data Acquisition: The stylus position is measured and recorded, providing film thickness information.

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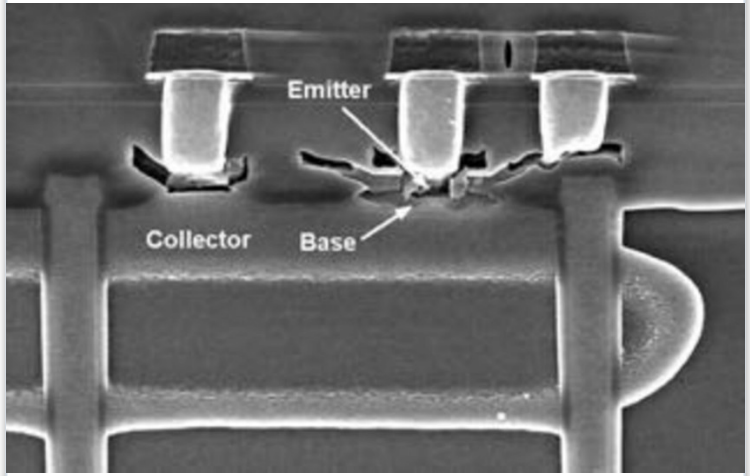
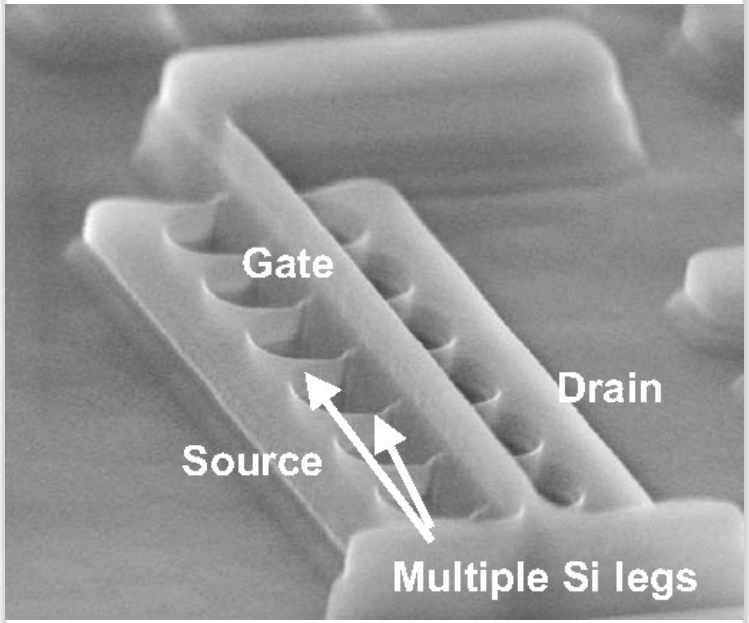
TOPIC 02

Scanning Electron Microscope (SEM)





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Evidence 1

SEM (Scanning Electron Microscope) is used to measure junction depths and film thickness
Wafer is cleaved, exposing the junction
Depth is measured from the SEM photograph using the scale factor
SEM and groove and stain methods offer a visual view of junctions and lateral diffusion not available with other techniques

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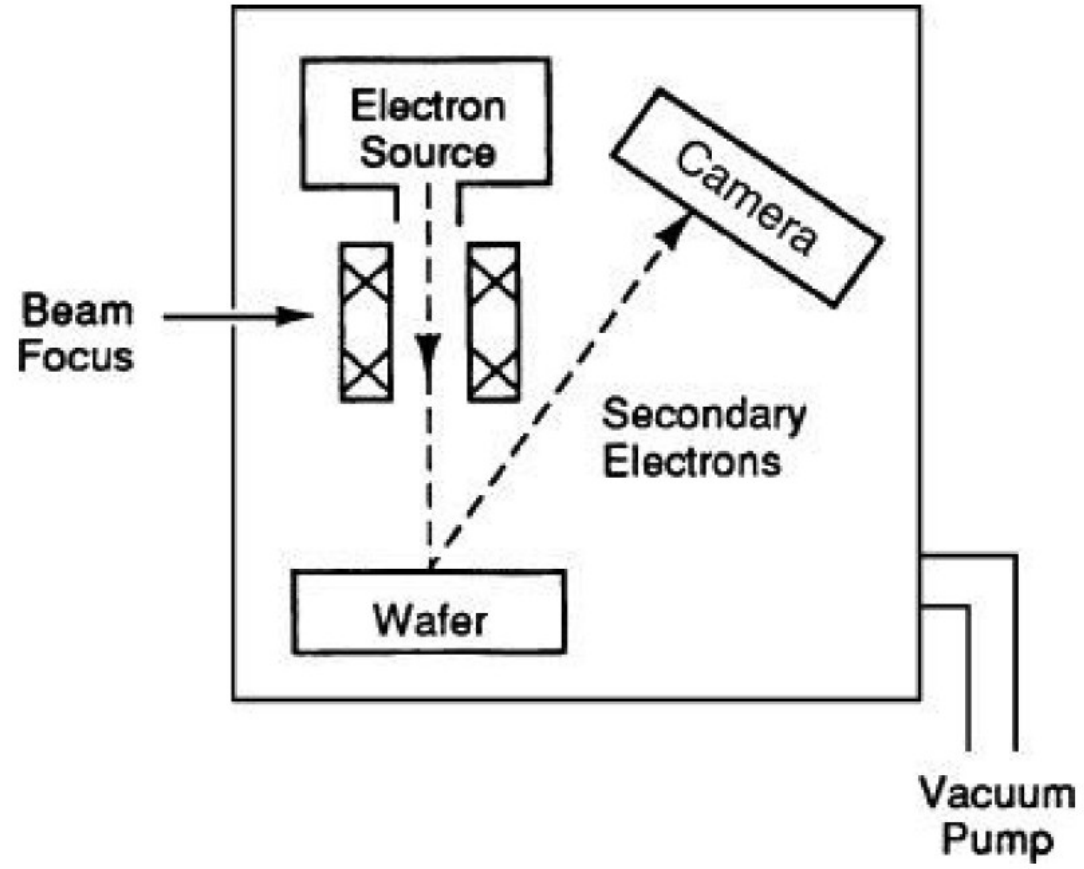
TOPIC 02

Scanning Electron Microscope (SEM)



1

Evidence 1

Uses an electron beam to create high-resolution images of surfaces in a vacuum.
Advantages: Submicrometer resolution, large depth of field, and high magnification.
Tilting Wafer Holder: Enables viewing of surfaces at different angles, enhancing three-dimensional perspective.]

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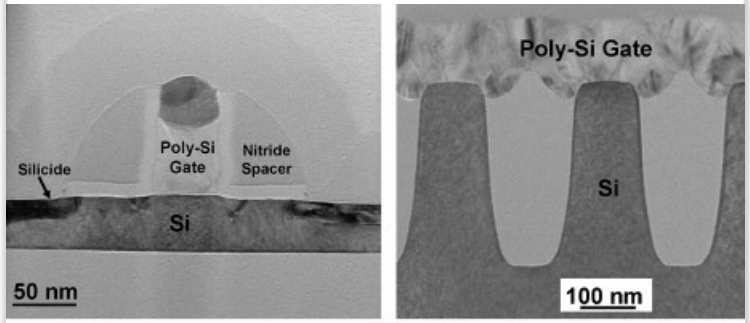
TOPIC 02

Transmission Electron Microscopy



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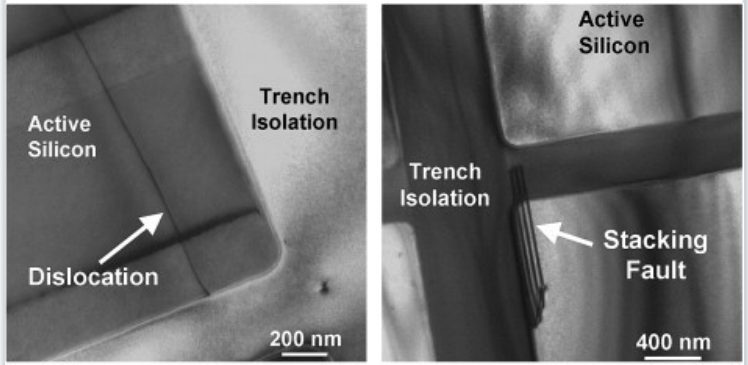

a b

1

Evidence 1

Transmission Electron Microscopy (TEM): Provides higher resolution than SEM by transmitting electrons through a thin sample, allowing for sub-angstrom resolution.
X-ray Spectroscopy (SEM-EDS): By analyzing X-rays emitted from the sample during SEM analysis, it provides information about the elemental composition of the surface.

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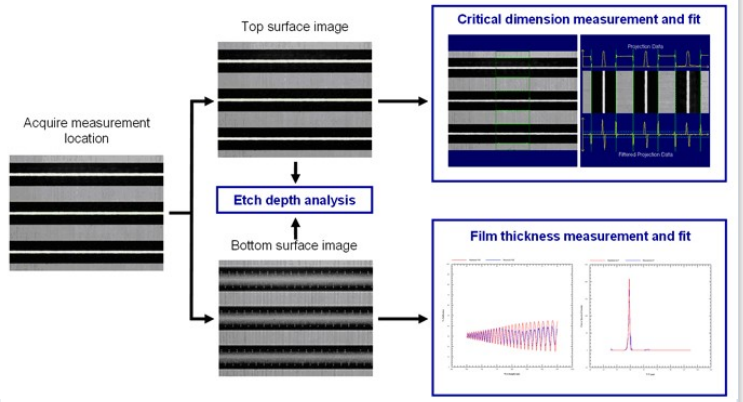
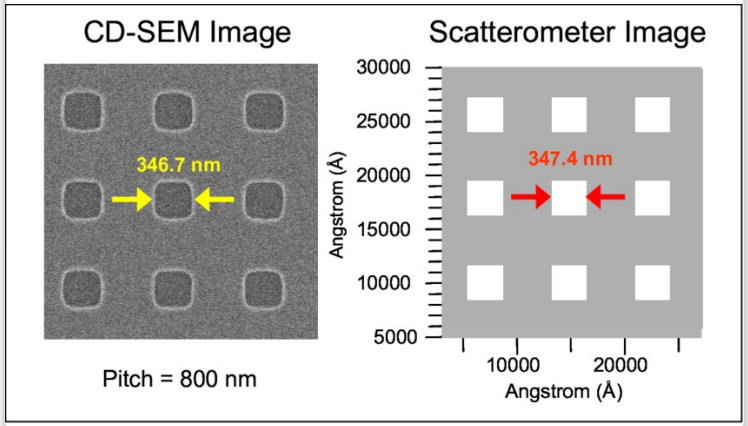
TOPIC 02

Critical Dimensions and Line-Width Measurements





1

Evidence 1

Critical dimensions are controlled by various processes:
Vertical dimensions: determined by doping and layering
Horizontal dimensions: defined by lithography
Critical dimensions measured during develop inspection and final inspection
Measurement techniques include:
Microscope image shearing (low-tech)
SEM (Scanning Electron Microscope)
Scatterometry
Atomic Force Microscope (AFM)
Device electrical measurements

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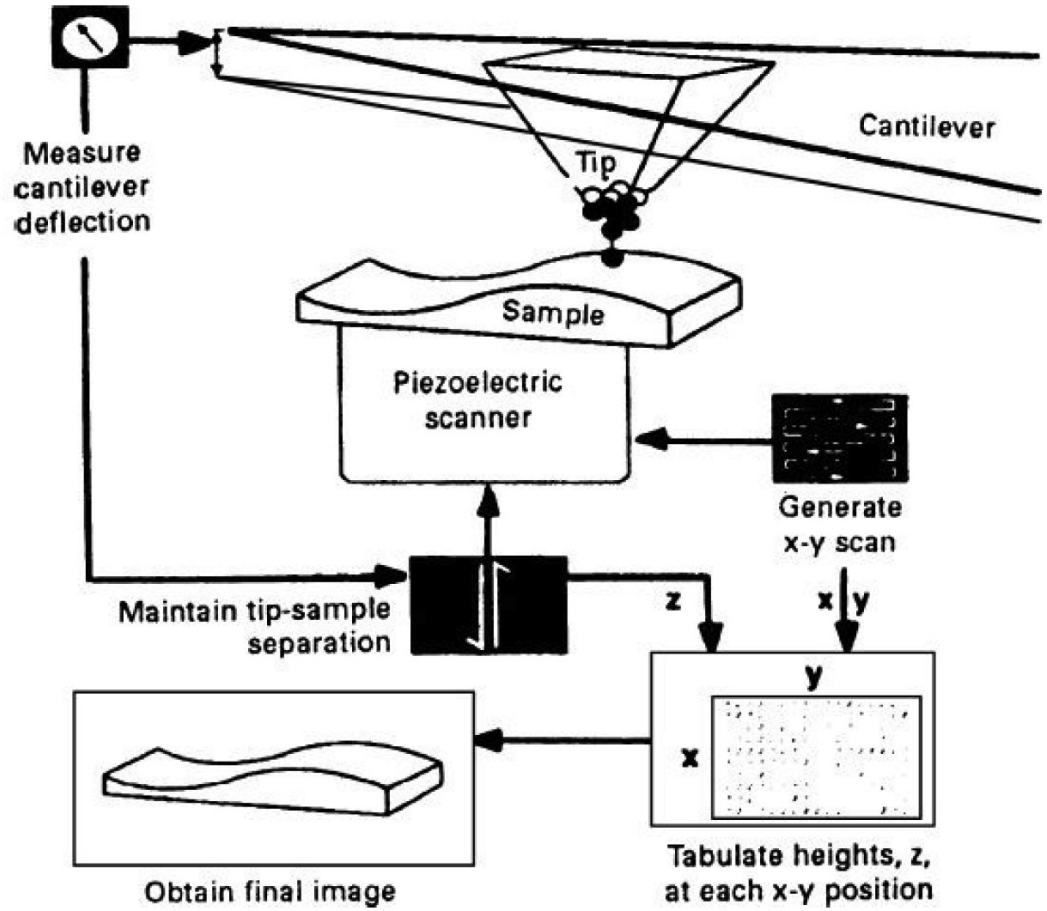
TOPIC 02

Atomic Force Microscopy (AFM)



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Evidence 1

Atomic Force Microscopy (AFM) is a surface profiling technique
AFM uses a delicately counterbalanced probe to scan the surface
The probe and surface separation is extremely small (~2 Å), allowing atomic forces to influence the probe
Sophisticated electronics track the probe's position as it moves across the surface
The result is a three-dimensional map of the surface

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TOPIC 02

Device performance issues

Concepts · mechanisms · process evidence · integration decisions

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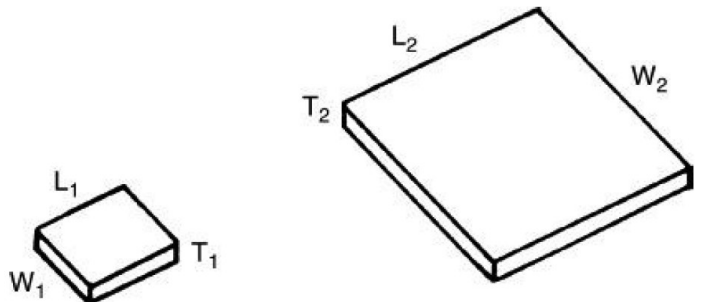
TOPIC 02

Four-Point Probe Measurement



$$T = \rho_s / R$$

1 **Evidence 1**

Used to measure resistivity of wafers, crystals, and thin layers of dopants.
Sheet Resistance (Rs): Electrical quantity measured on thin layers, expressed in ohms per square.
Ohms Per Square Concept: Sheet resistance is constant for any square of the same thin material, regardless of its size.

2 **For thin films, the formula is:**

For thin films, the formula is:
where T = layer thickness
ρs = resistivity
Rs = sheet resistance

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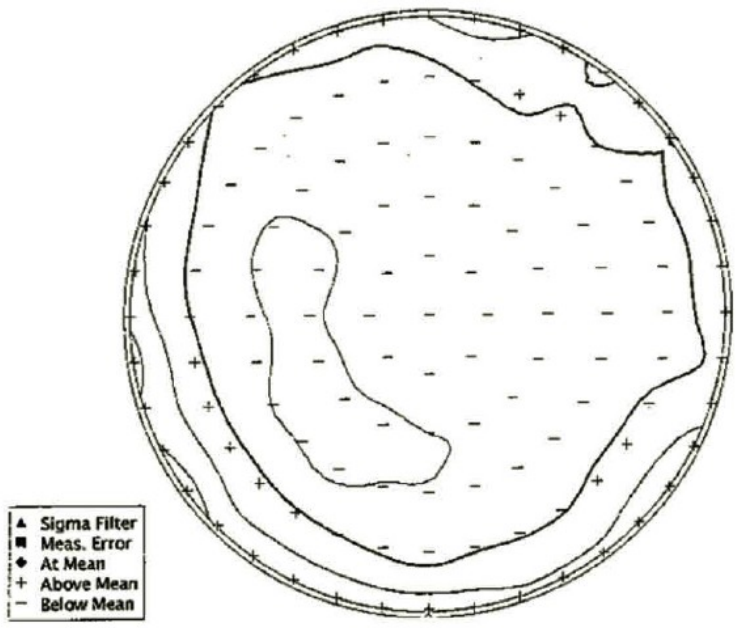
TOPIC 02

Sheet Resistance Wafer Map




```
Operator.............
Process..............
Wafer ID............. B0.5K-1
Slot................. 21
Lot ID...............

Mean................. 1.6466e+002
Std Dev.............. 1.5118e+000 / 0.918%
Minimum / Maximum... 1.63e+002/1.69e+002
Range................ 6.0529e+000
# Good / Sites...... 121 / 121

Sample Type......... Auto Range
Correct to Temp..... 23.0°C
Amperage............. 2.0784e+002 µA
Sigma Filter(U/L)... 3.0 / 3.0
Correlation.........
Units................ Ohms/Sq
Shift................
Equipment............
Process Date........ January 13, 2004
Process Time........ 20:33:17

Wafer Diam.......... 300.00 mm / 11.81 in
Test Diam........... 294.00 mm / 11.57 in
Test Pattern.........Contour Map, 121-Site
Probe................4.0 mil Tip - Type B

TCR Value........... 0.0000
Meas Temp(BF/AF/AVG) 23.63 / 23.75 / 23.69 °C
Voltage.............. 7.5033 mV
Target Mean......... 0.0000e+000
Warning Limit(U/L).. 0.0000e+000/0.0000e+000
Control Limit(U/L).. 0.0000e+000/0.0000e+000
Contour Interval.... 1.0000e+000%
```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TOPIC 02

Gate Oxide Electrical Measurement

Gate oxide integrity (GOI) refers to film uniformity and contamination
Measured using a destructive electrical test called oxide rupture voltage (BVox)
Test structure is the same as for capacitance-voltage (C/V) analysis
In the test, voltage is increased until the oxide breaks down, allowing current to flow from the gate electrode to the silicon
Maximum voltage before breakdown depends on oxide thickness, structural quality, and purity

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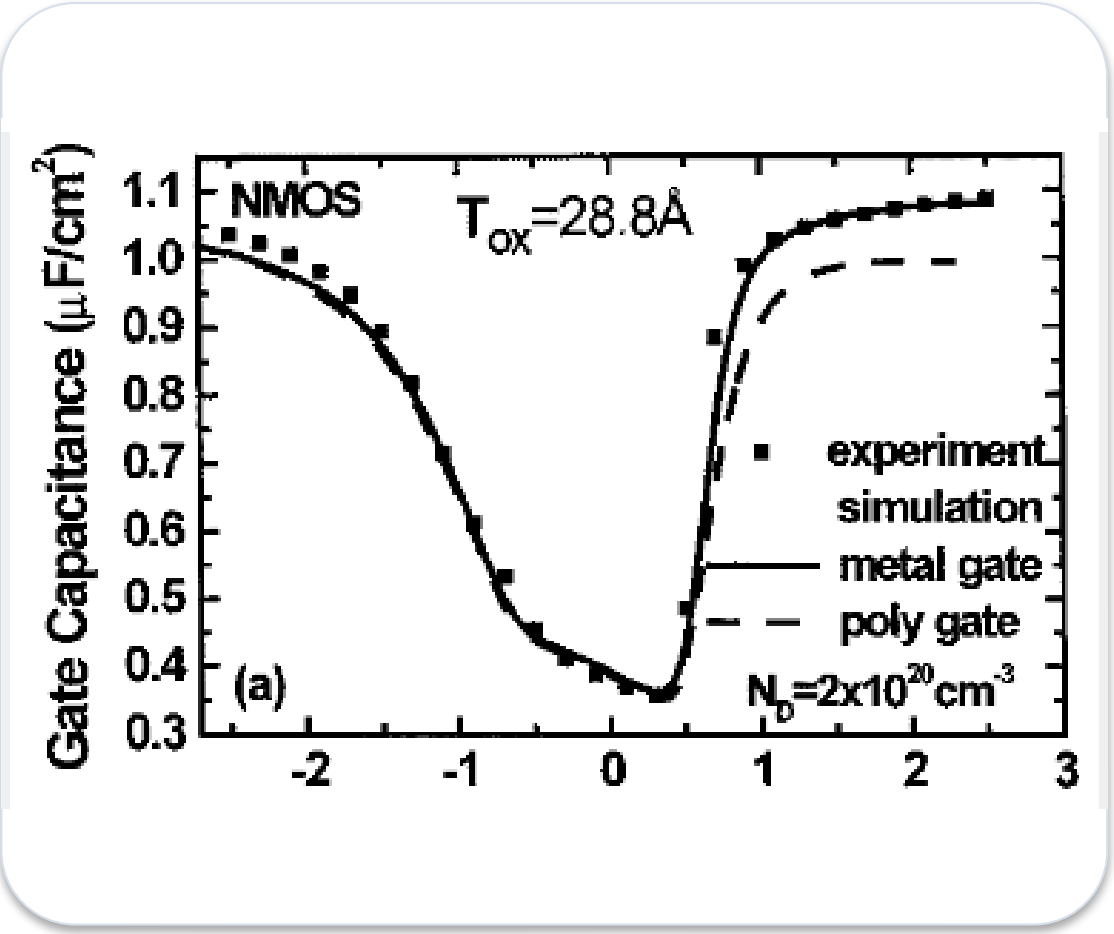
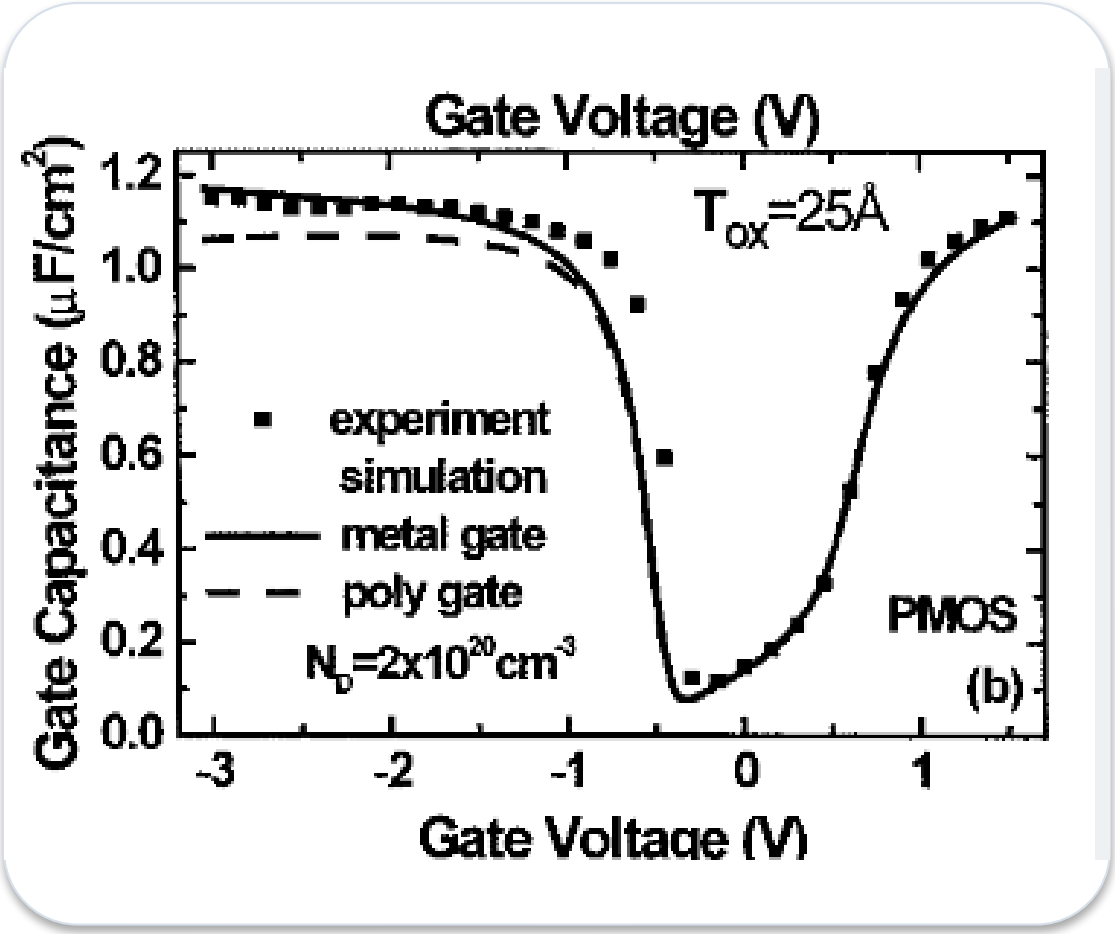
Gate Oxide Characterization





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TOPIC 02

Device electrical testing

Device electrical testing requires three main components:
Probe machine: positions needlelike probes on the devices
Switch box: applies correct voltage, current, and polarities to the device
Display method: shows the test results
For testing many devices, probe cards (like in wafer-sort) are used to position probes
Display options:
Curve tracer: an oscilloscope that displays voltage on the horizontal scale and current on the vertical scale, showing characteristic voltage/current traces
Digital voltmeter: for more specific measurements
High-production tests are performed digitally, with results stored, correlated, and displayed on a read-out

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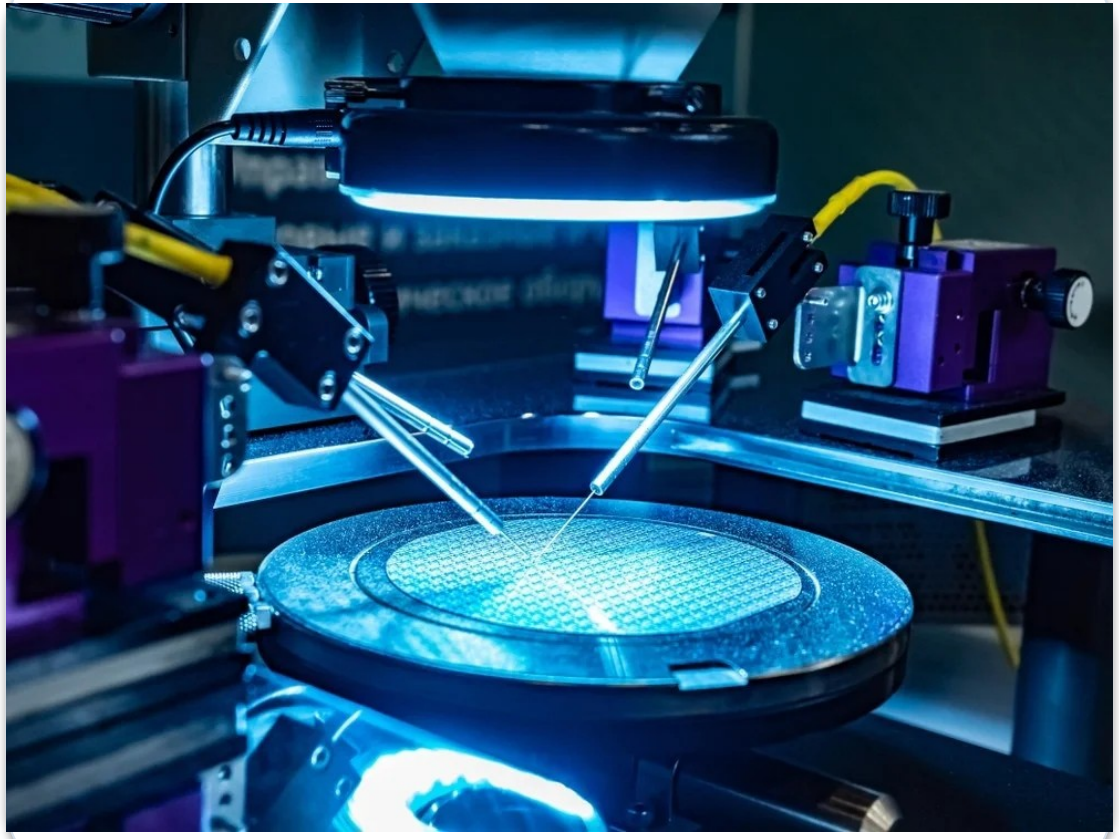
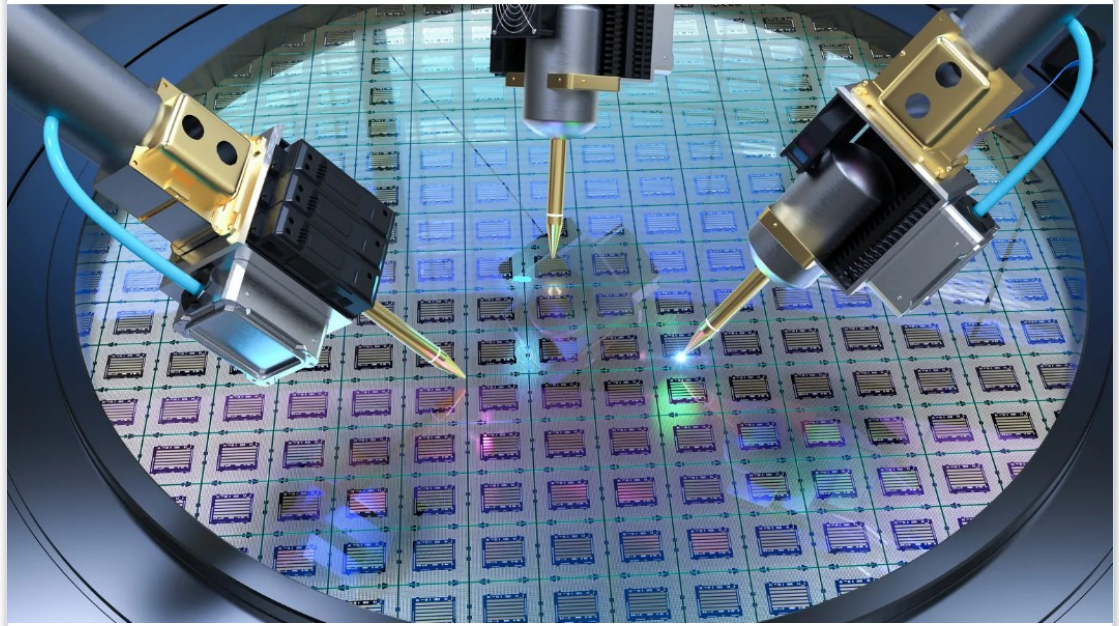
TOPIC 02

Device electrical testing





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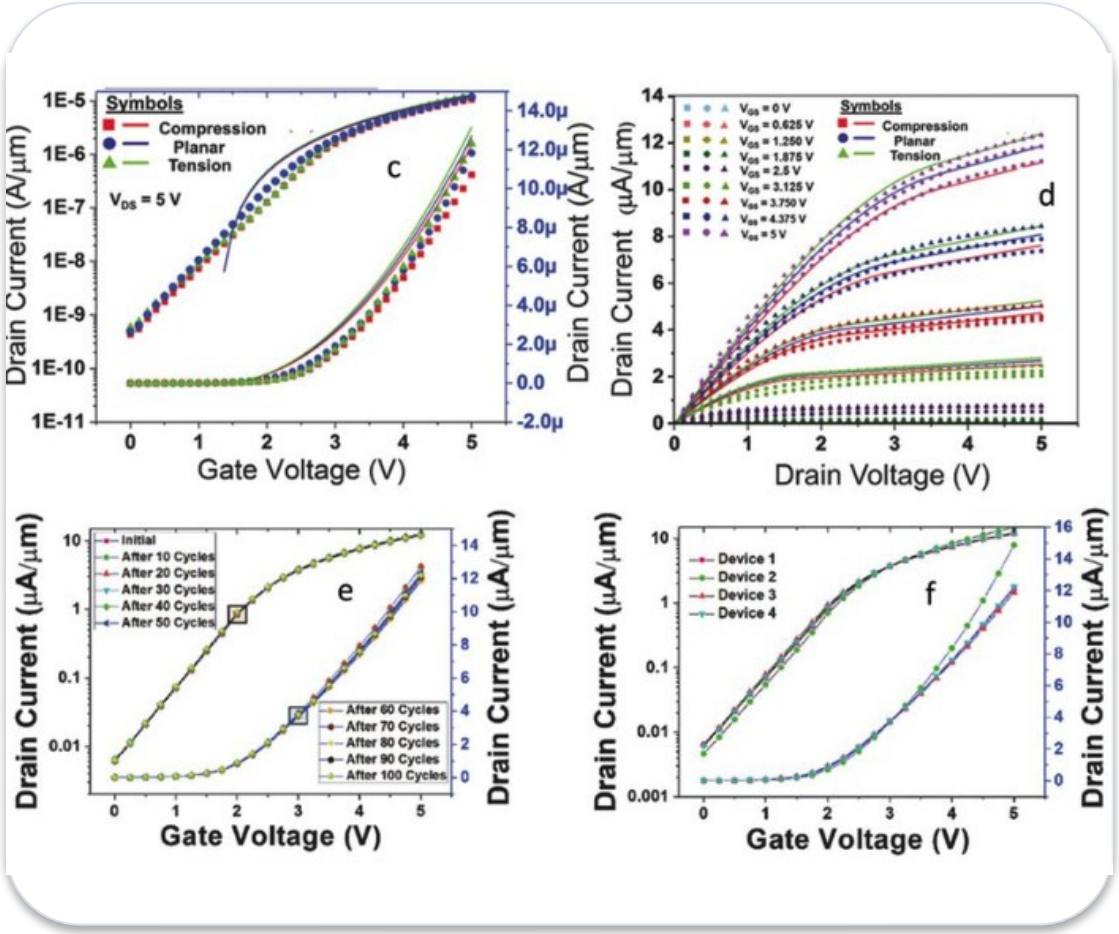
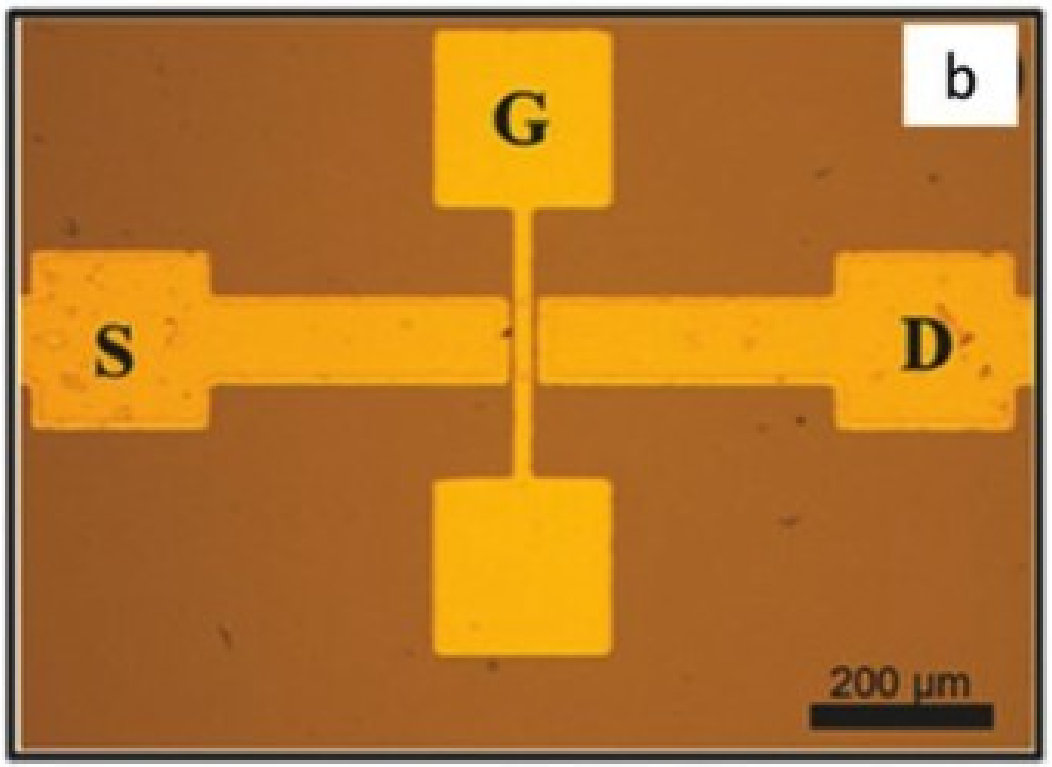
Device Performance Characterization





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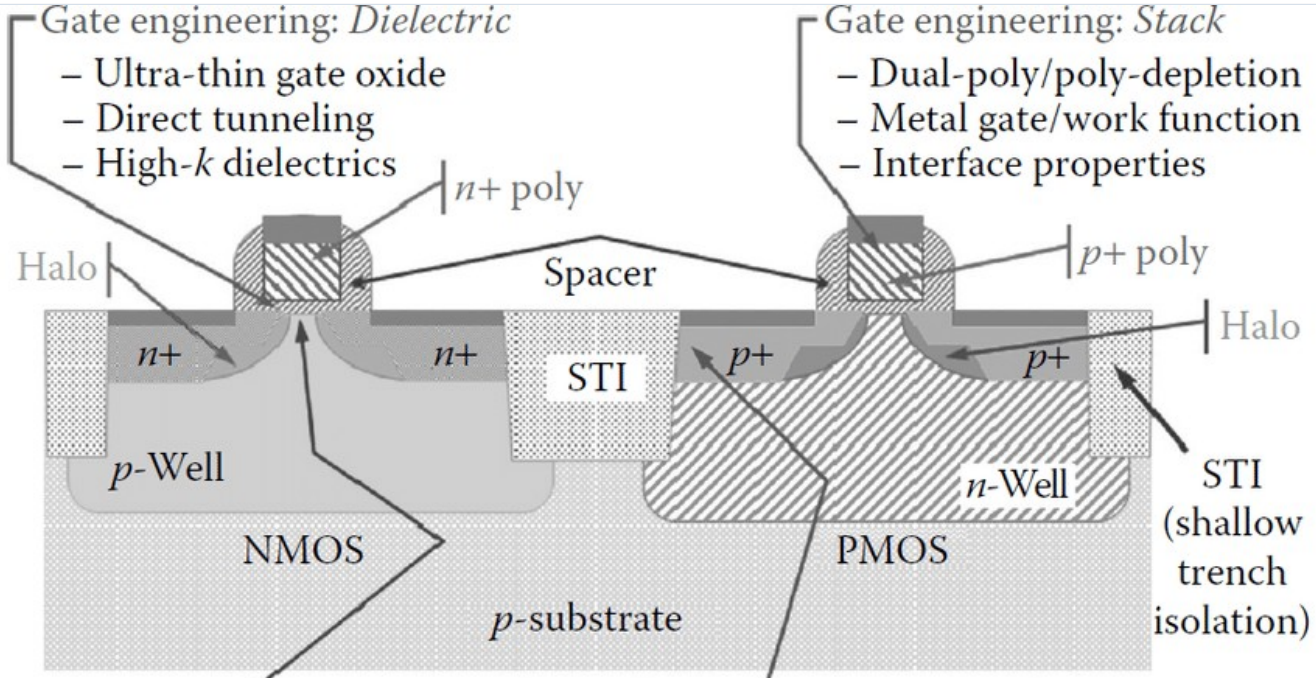
Device Engineering



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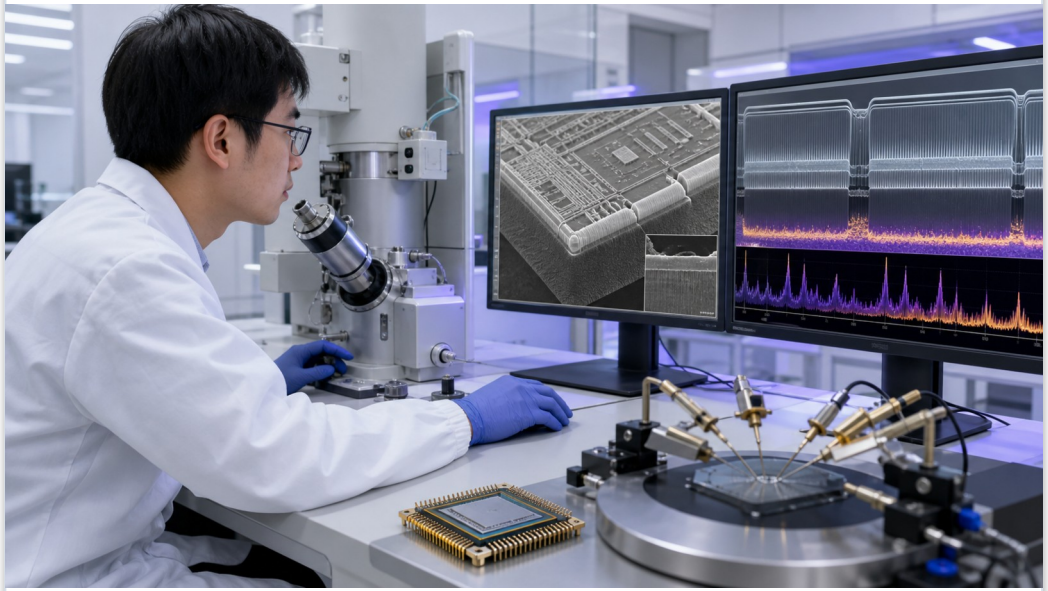
TOPIC 02

Device reliability issues

Concepts · mechanisms · process evidence · integration decisions



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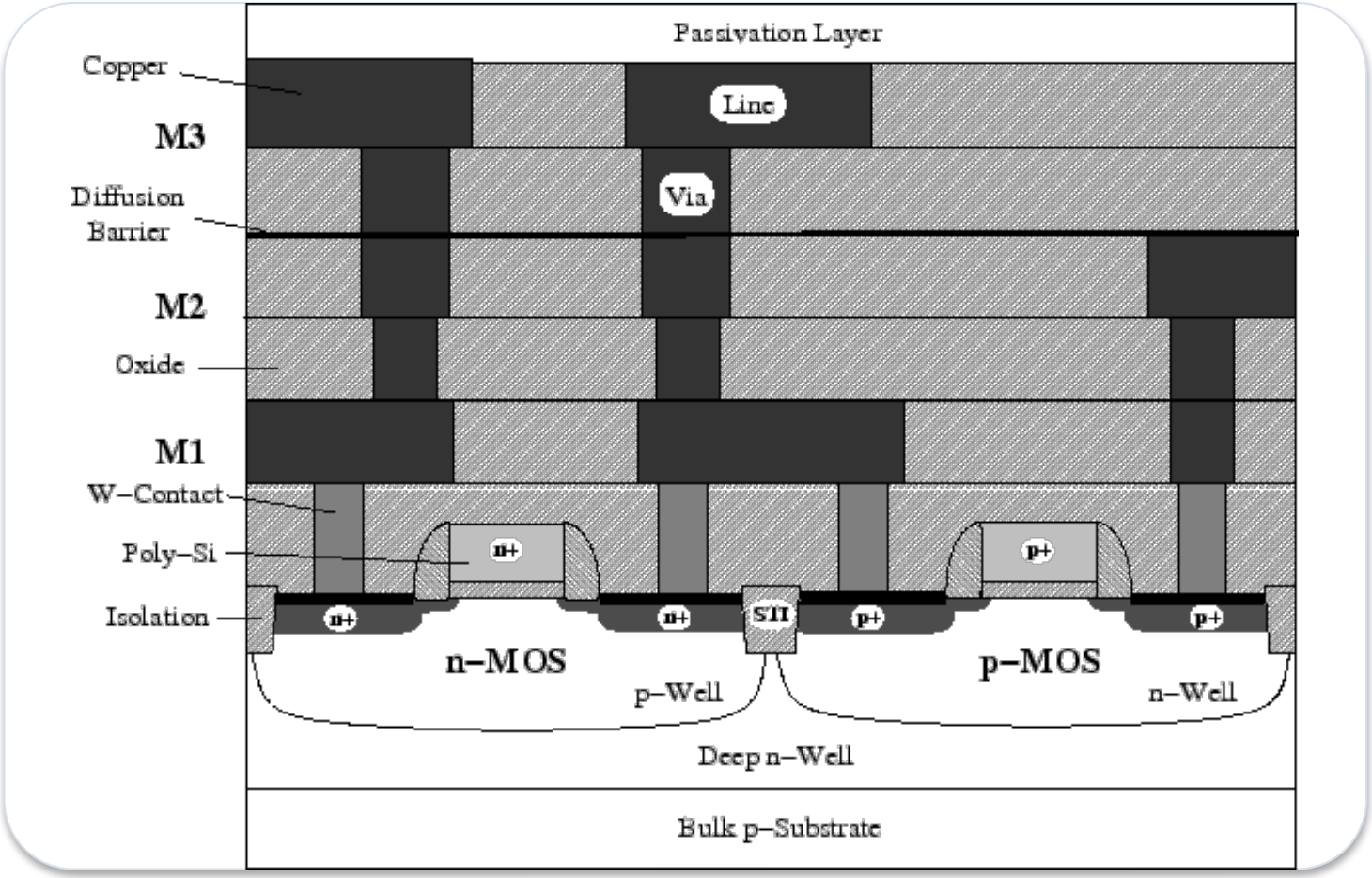
TOPIC 02

Device Reliability



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Hot Carrier Injection

Hot Carrier Injection
Time Dependent Dielectric Breakdown
Electromigration

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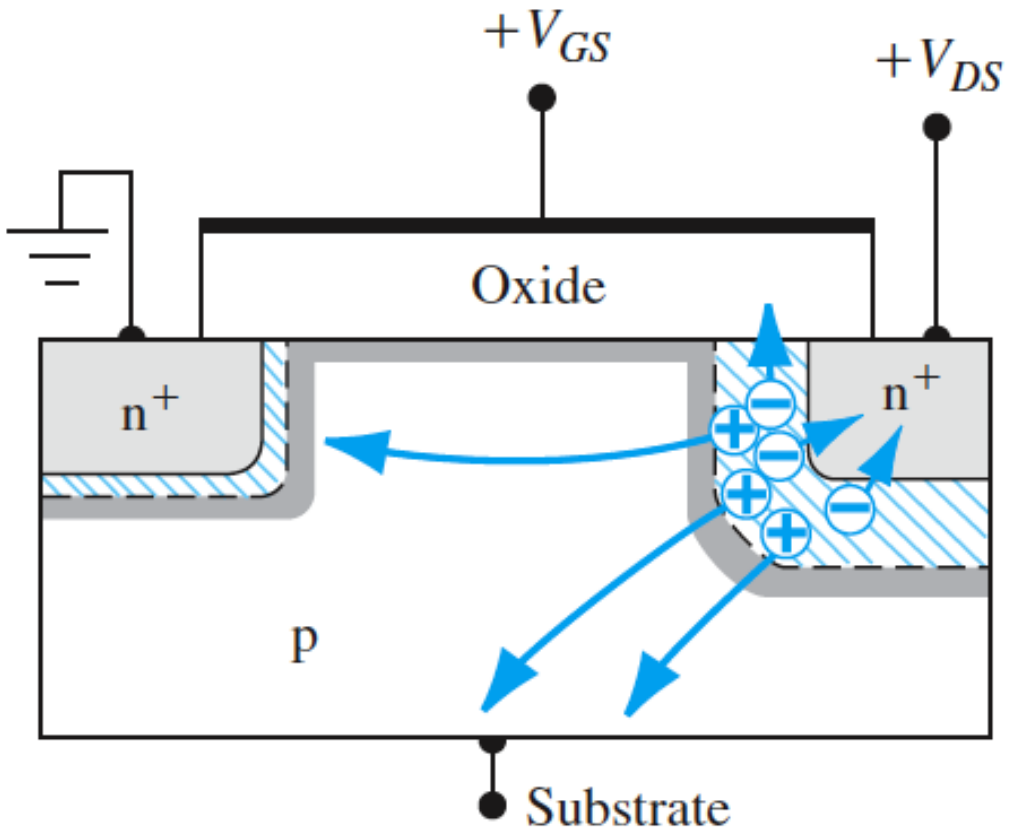
TOPIC 02

Hot Carrier Effects



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Evidence 1

Energetic electrons crossing the Si–SiO_2 interface can generate additional interface states
Interface states are likely caused by breaking silicon-hydrogen bonds, creating dangling silicon bonds
These interface states trap charge, leading to:
Threshold voltage shift
Increased surface scattering
Reduced carrier mobility
Hot-electron effects degrade the device over time, limiting its useful life

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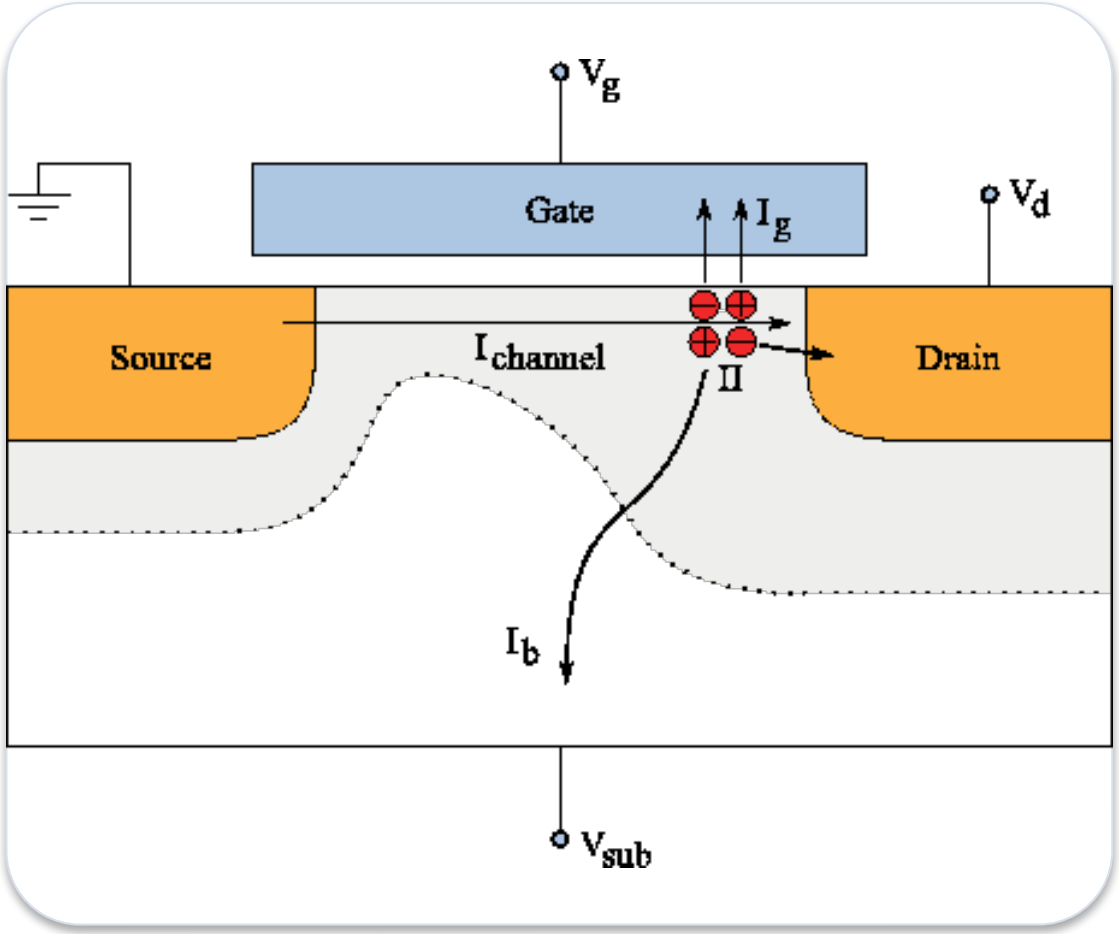
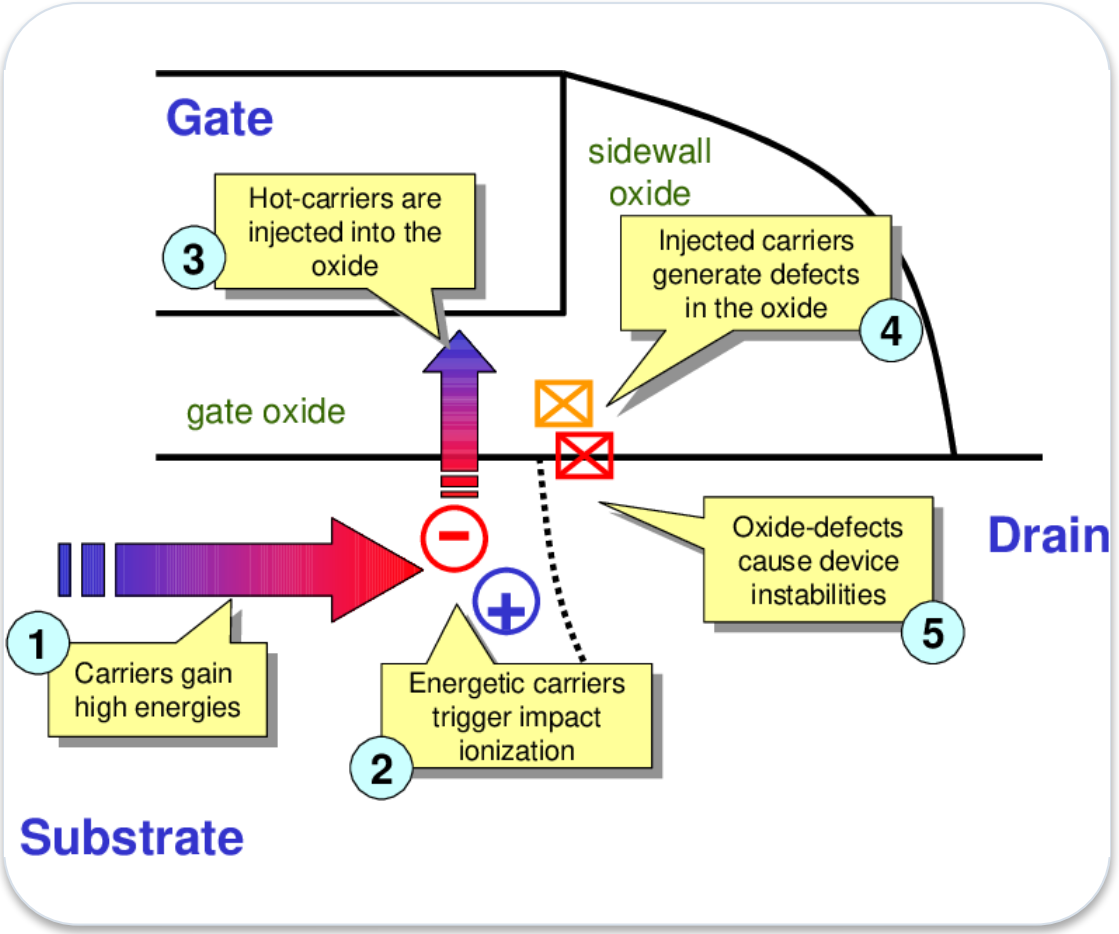
Hot Carrier Degradation





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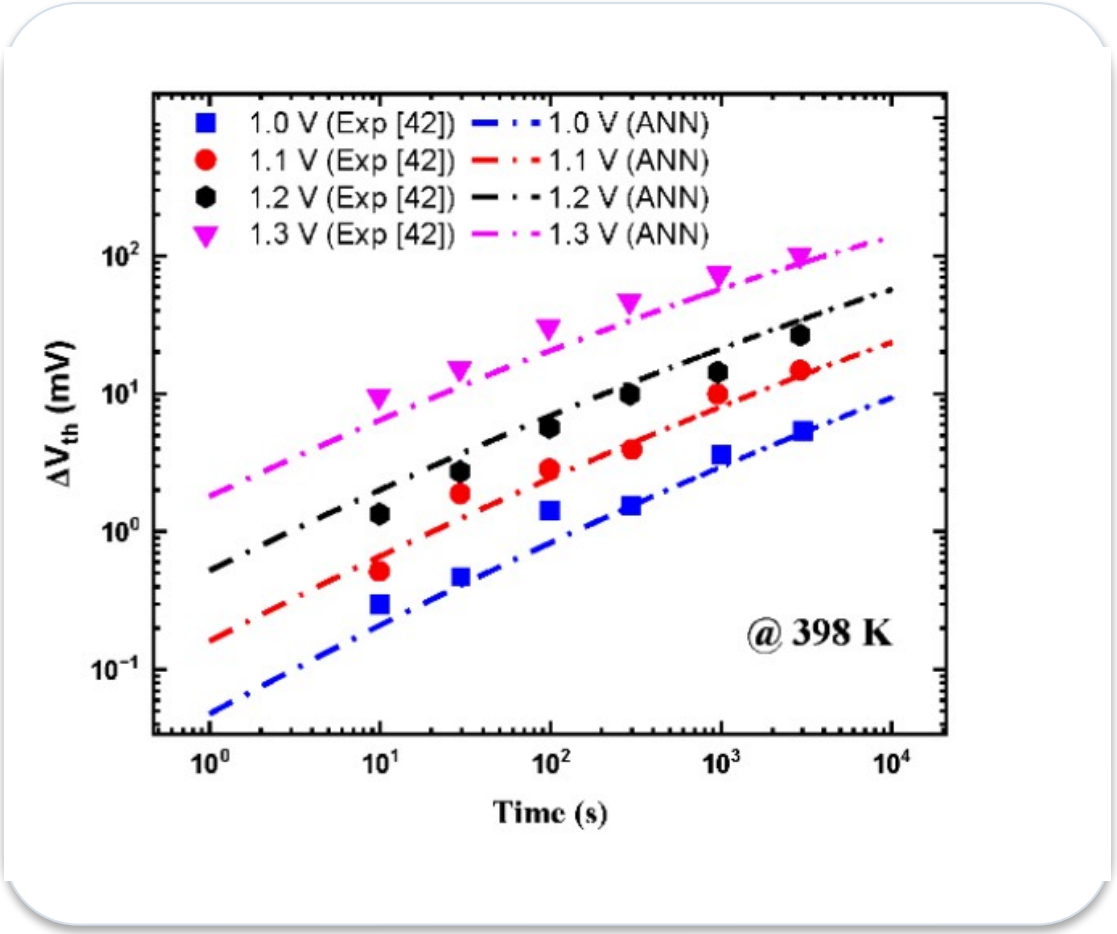
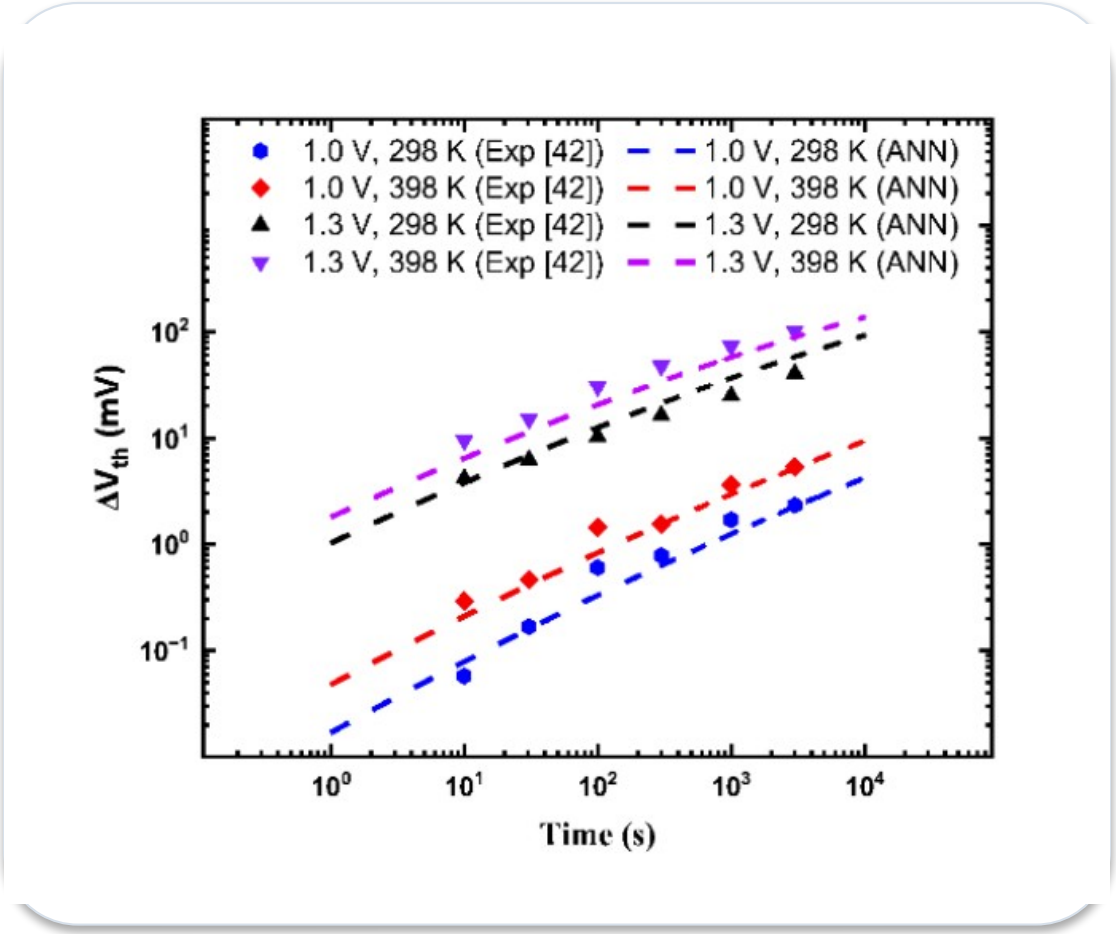
Hot Carrier Reliability Tests





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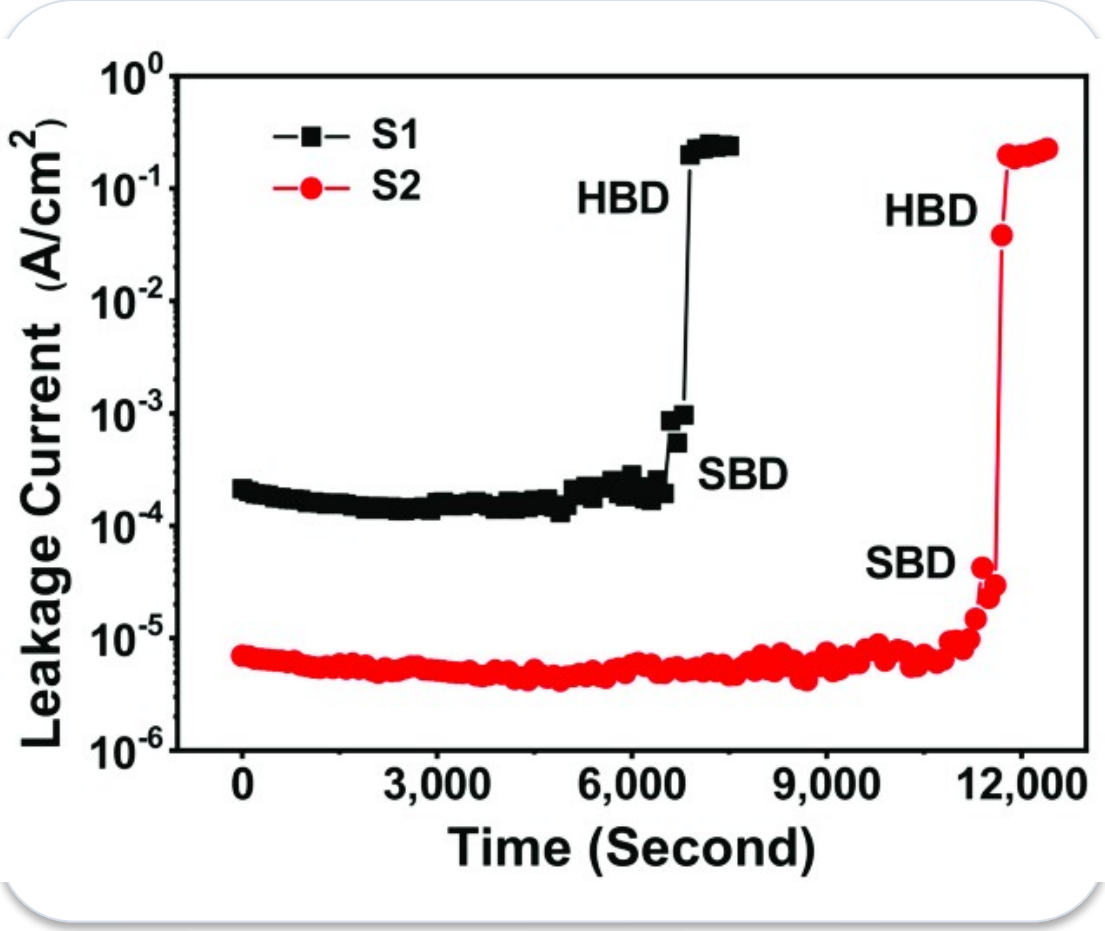
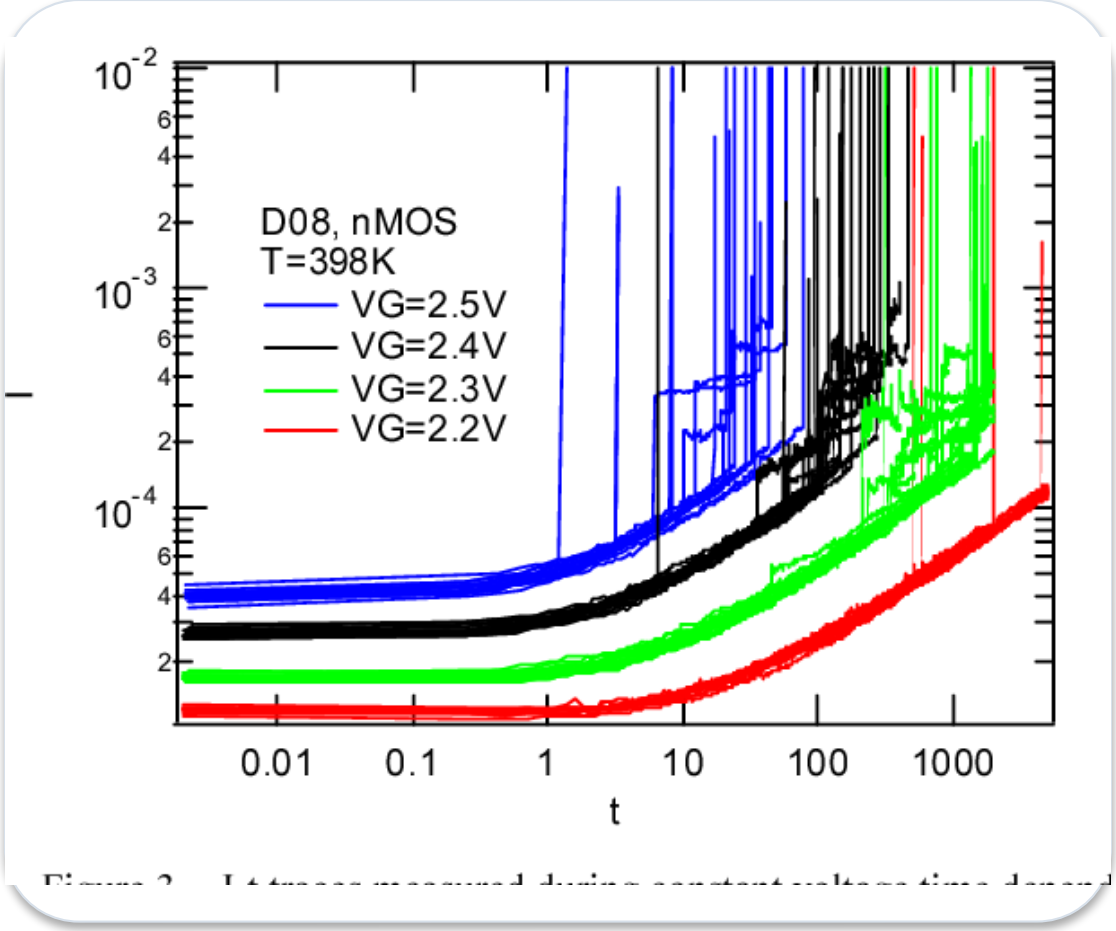
Time Dependent Dielectric Breakdown





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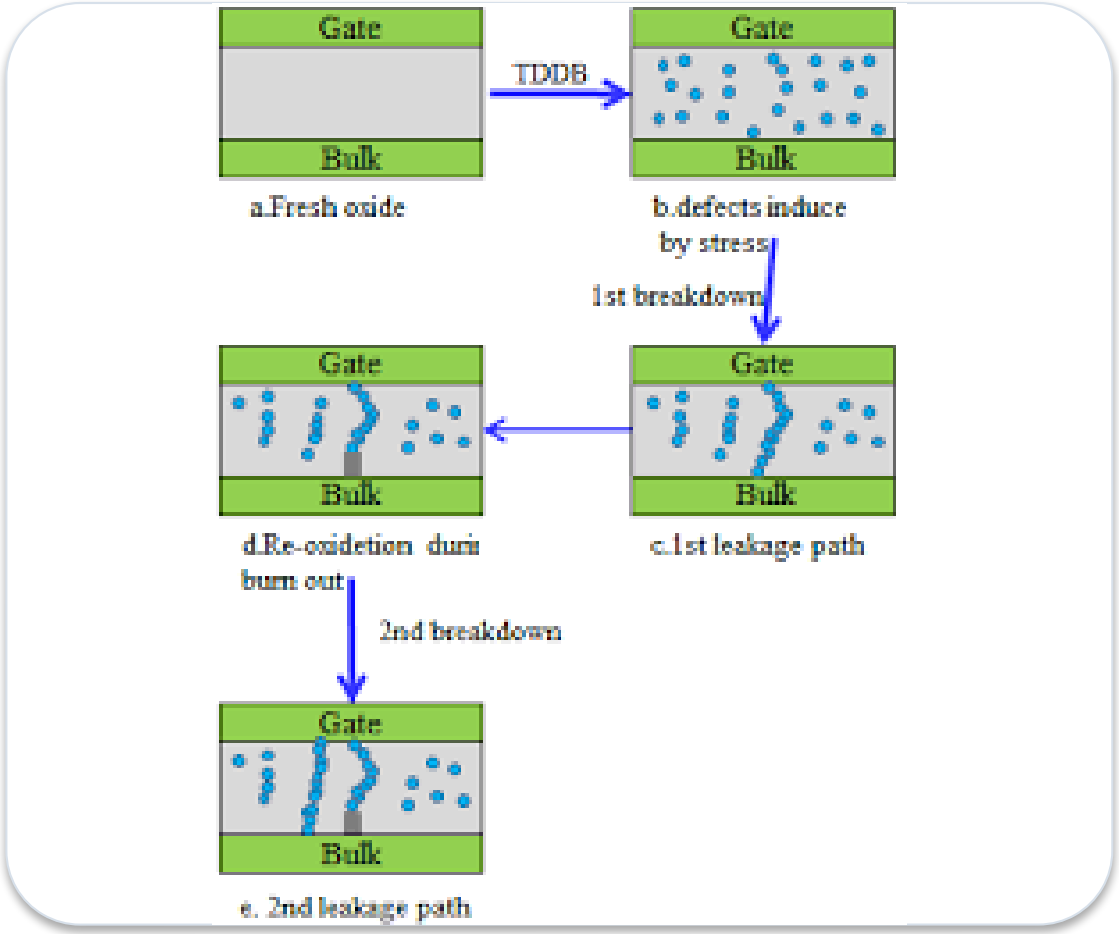
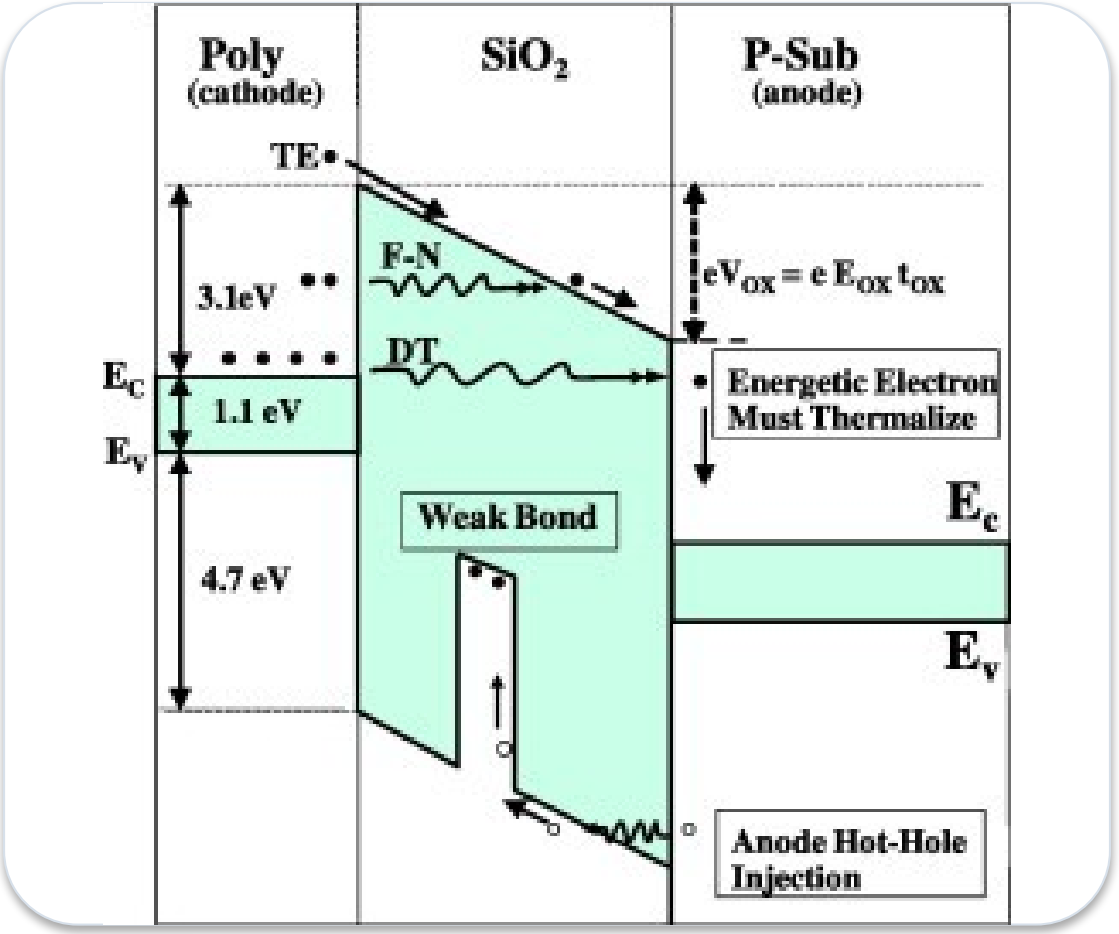
Time Dependent Dielectric Breakdown





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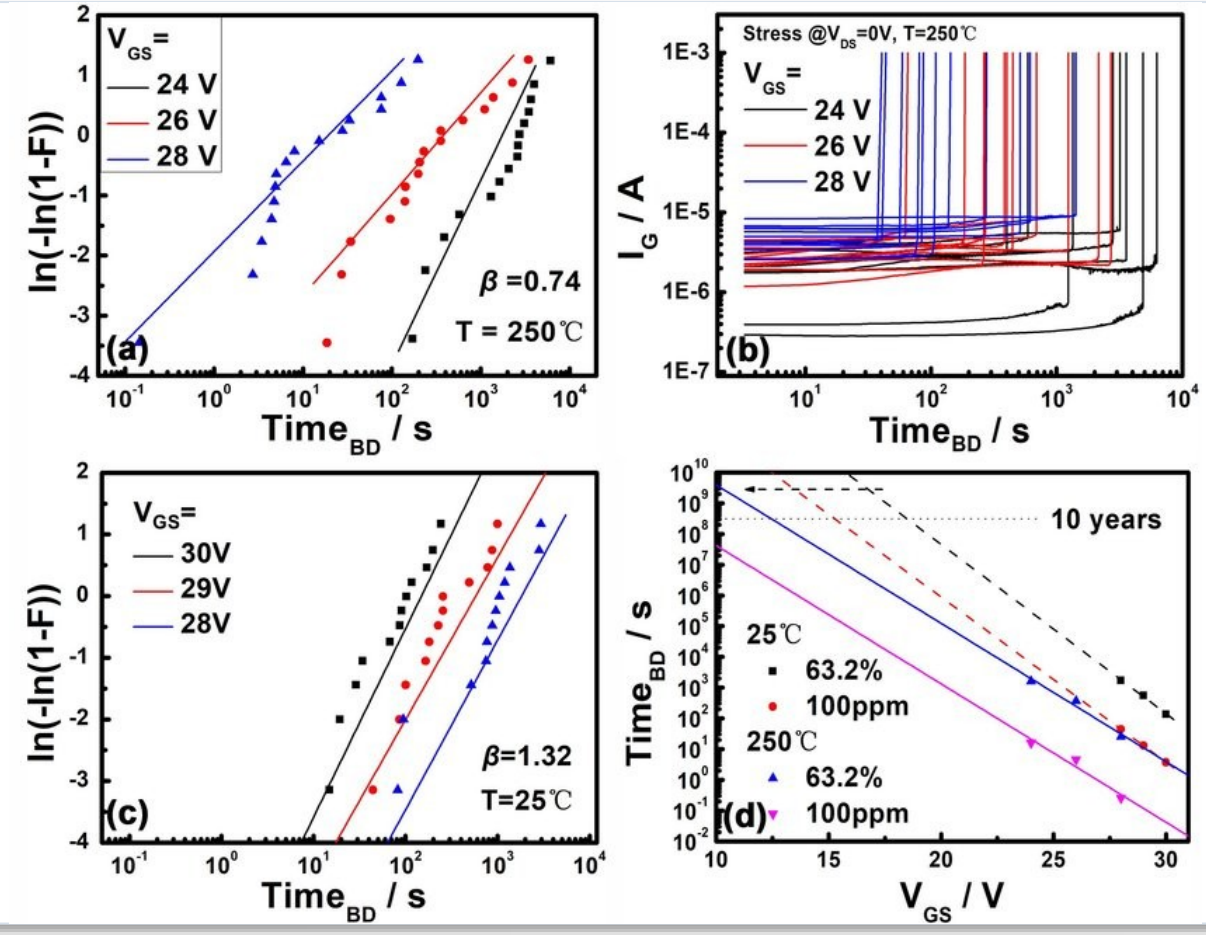
TDDB Reliability Test



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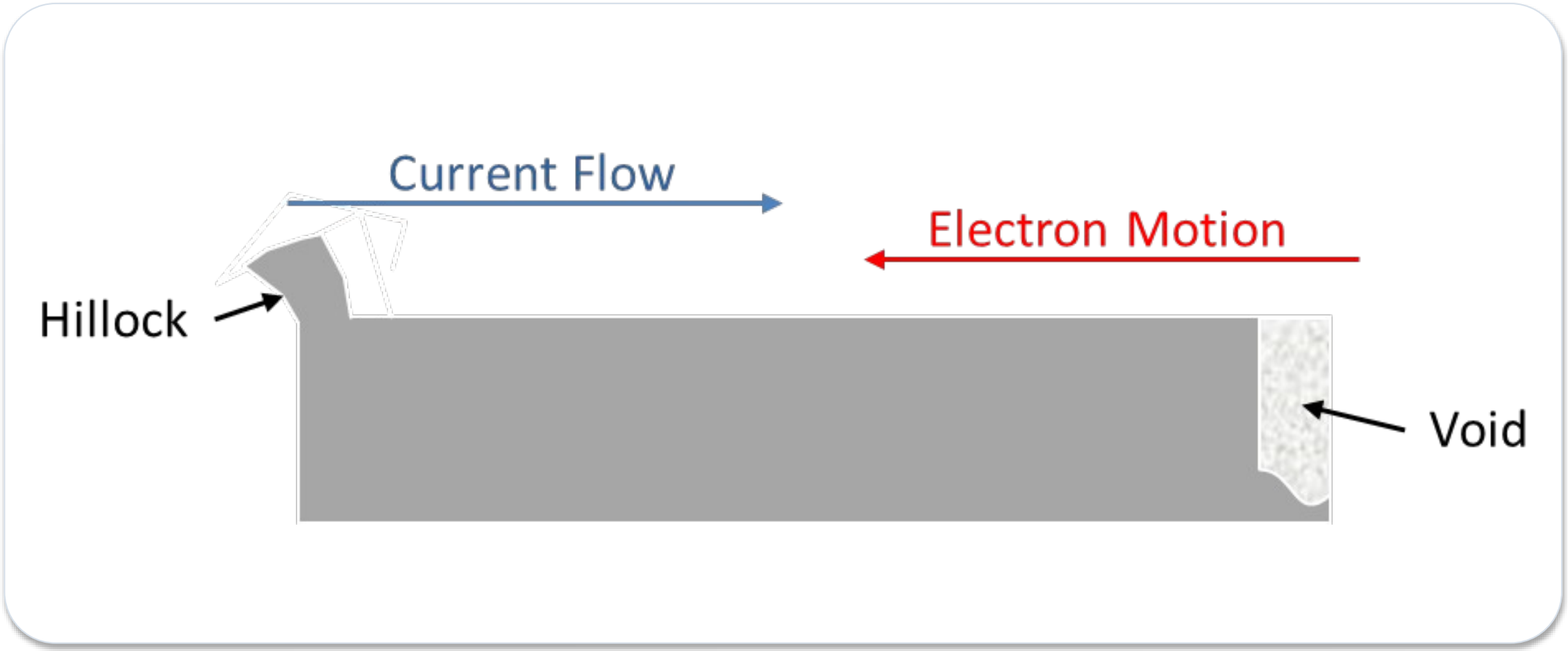
Electromigration



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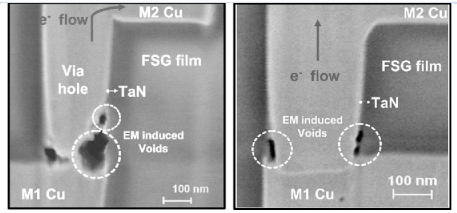
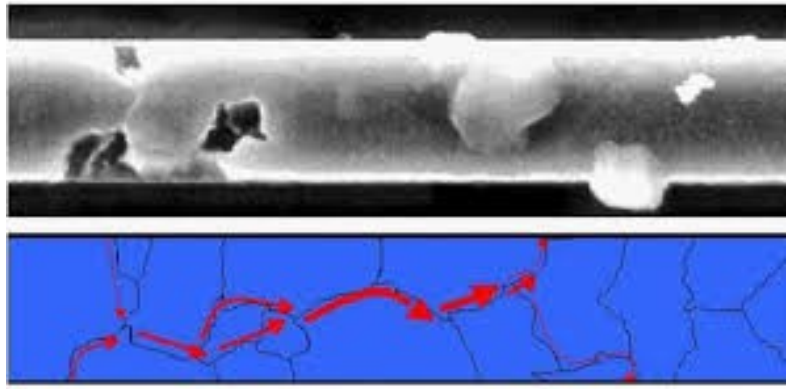
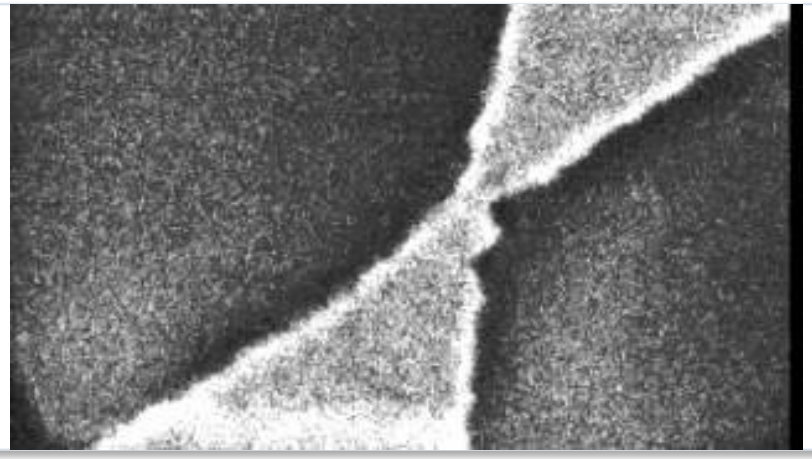
TOPIC 02

Electromigration Failures







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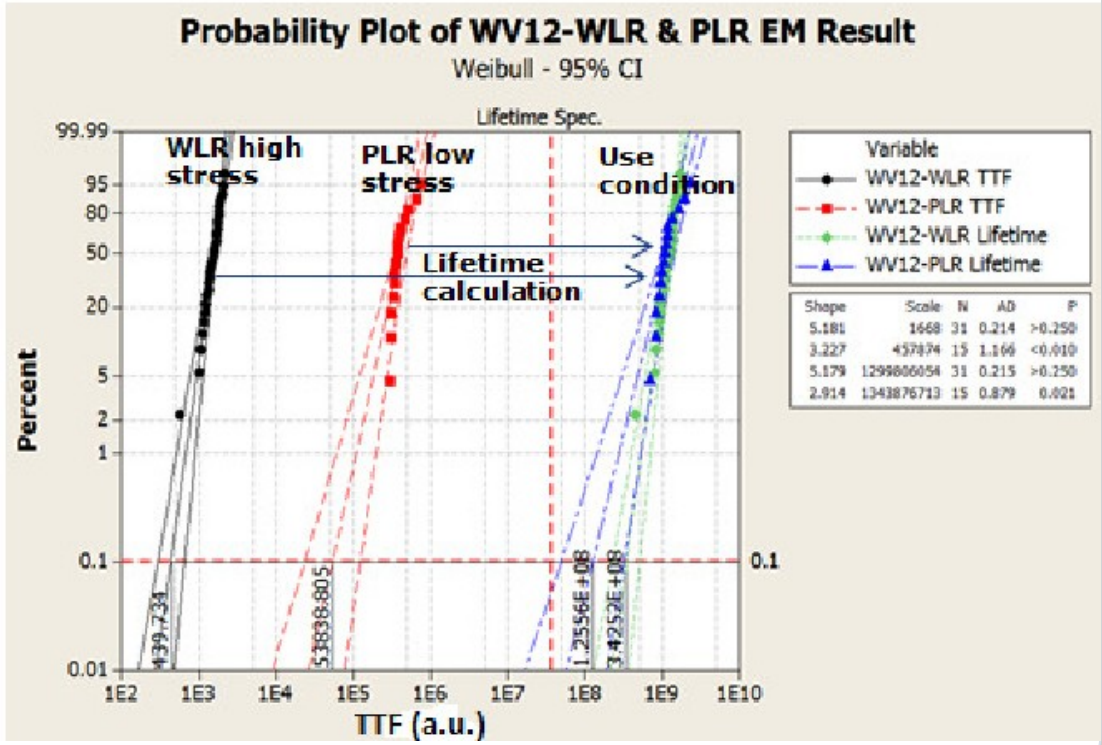
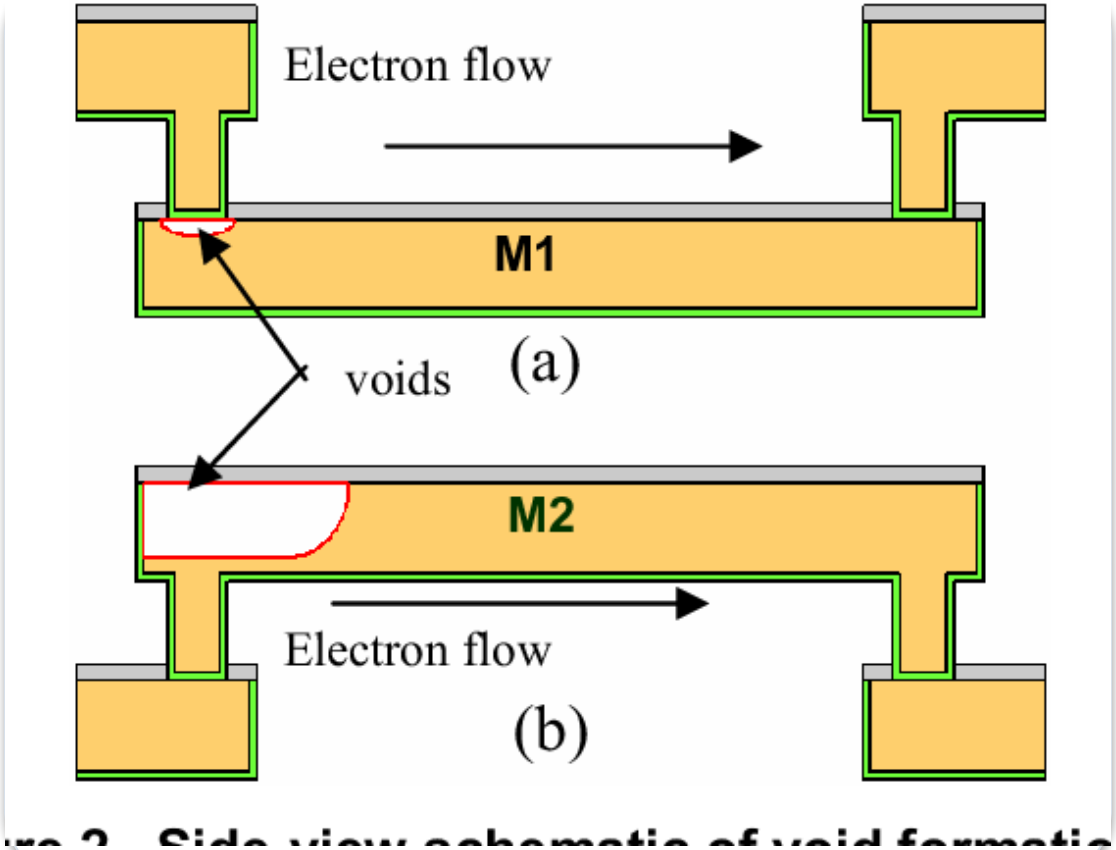
Electromigration Test





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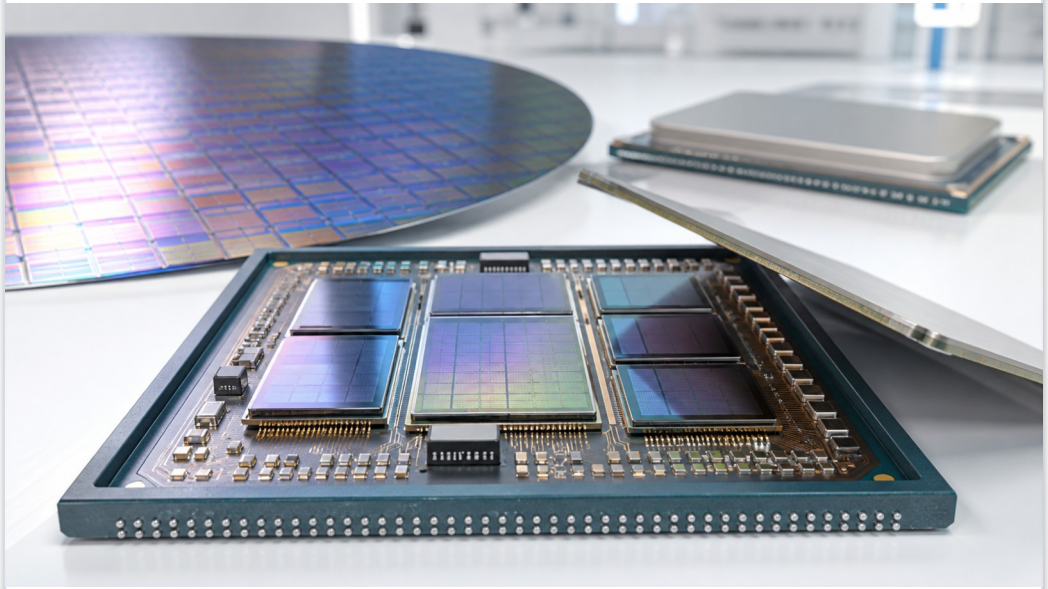
TOPIC 02

Assembly, Packaging and Final Test

Concepts · mechanisms · process evidence · integration decisions



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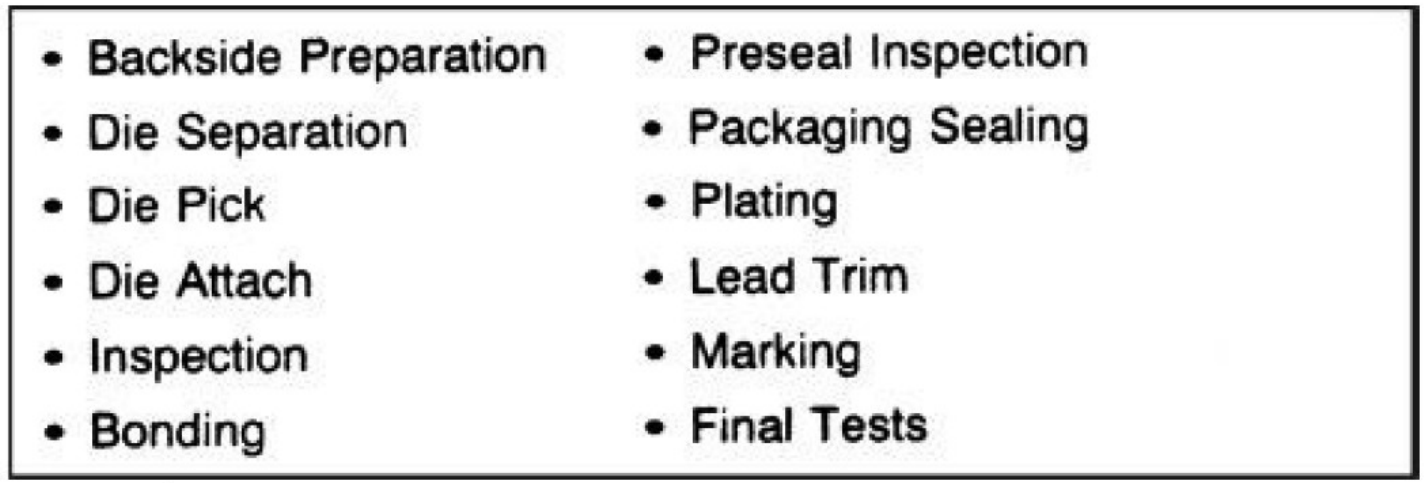
TOPIC 02

Basic Bonding Processes



1

Evidence 1

Packaging is a once-through process, unlike repeated wafer fabrication steps.
Three bonding methods for chip-to-package:
Wire bonding: most common for single chips.
TAB: connects leads to a circuit board via a lead frame.
Bump/flip-chip: uses solder balls, wafer is flipped and heat-processed.
3D packaging uses variations of these bonding methods.

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TOPIC 02

Prebonding Wafer Preparation

1

Key idea 1

After the final passivation layer and an alloy step in wafer fabrication, the circuits are complete. However, one or two additional processes may be performed on the wafer before transfer to packaging. These steps (wafer thinning and backside gold) are optional, depending on the wafer thickness and the particular circuit design.

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TOPIC 02

Wafer Thinning

Bump/ball bonding adds solder balls post-fabrication for wafer scale processes.
Thick wafers complicate die separation and packaging, solved by wafer thinning.
Thinning also removes unwanted electrical junctions and improves conduction.
Thinning (via grinding/CMP) reduces wafers to ~100 µm, crucial for 3D packaging.
Concerns: wafer scratching, breakage, and warping, which affect die separation and packaging.

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TOPIC 02

Backside Gold

1

Key idea 1

Another optional wafer process is adding a layer of backside gold. A layer of gold is required on wafers that are going to be attached to the package by eutectic techniques (see the “Die Attach” section).
The gold is usually applied in the fabrication area (after back grinding) by sputtering.

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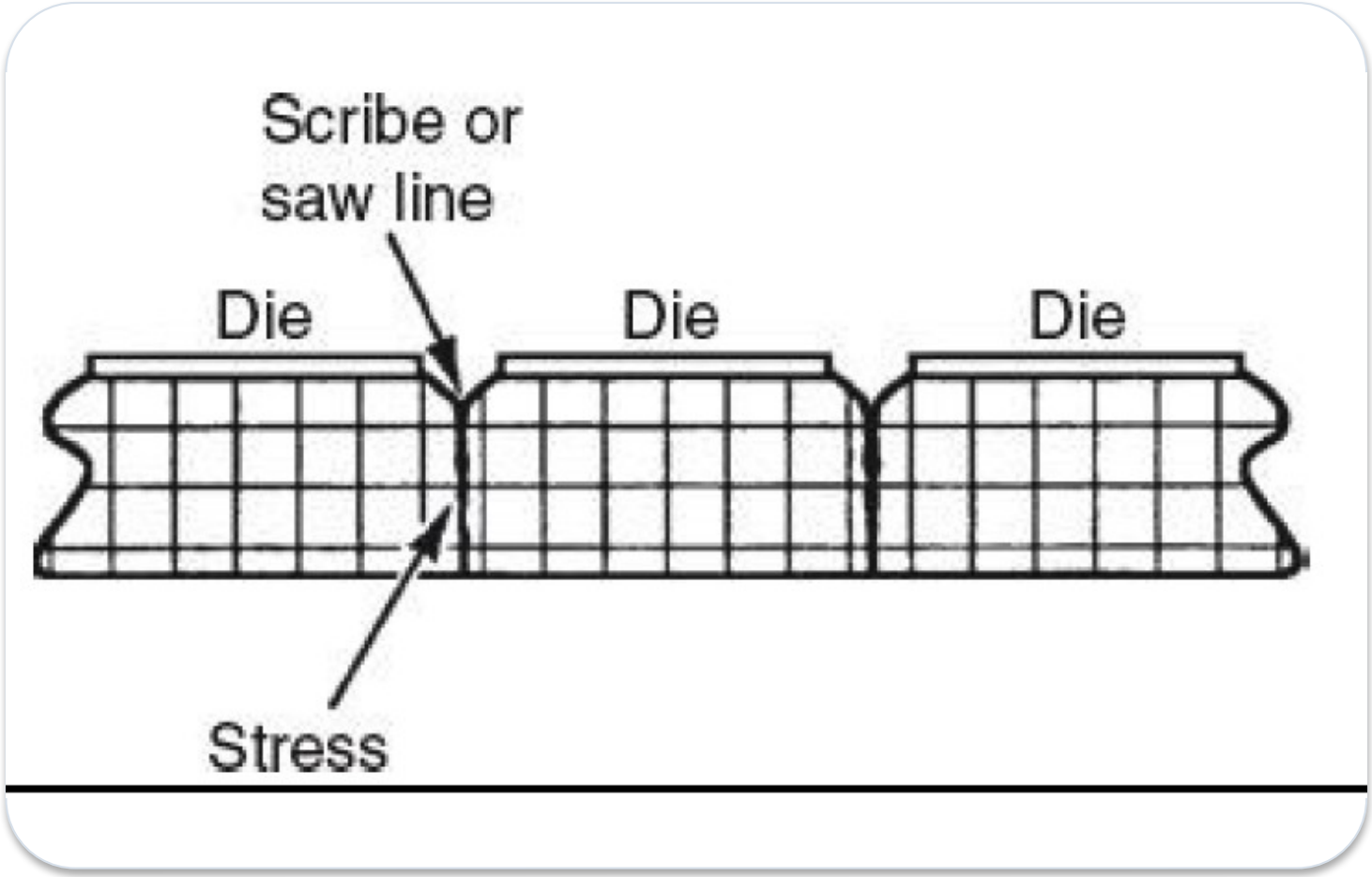
TOPIC 02

Die Separation



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Evidence 1

The traditional chip-packaging process starts with the separation of the wafer into individual dies. The two methods of die separation are scribing and sawing.
For bump or ball processes, the die separation occurs after the bump or ball system is established on the wafer.

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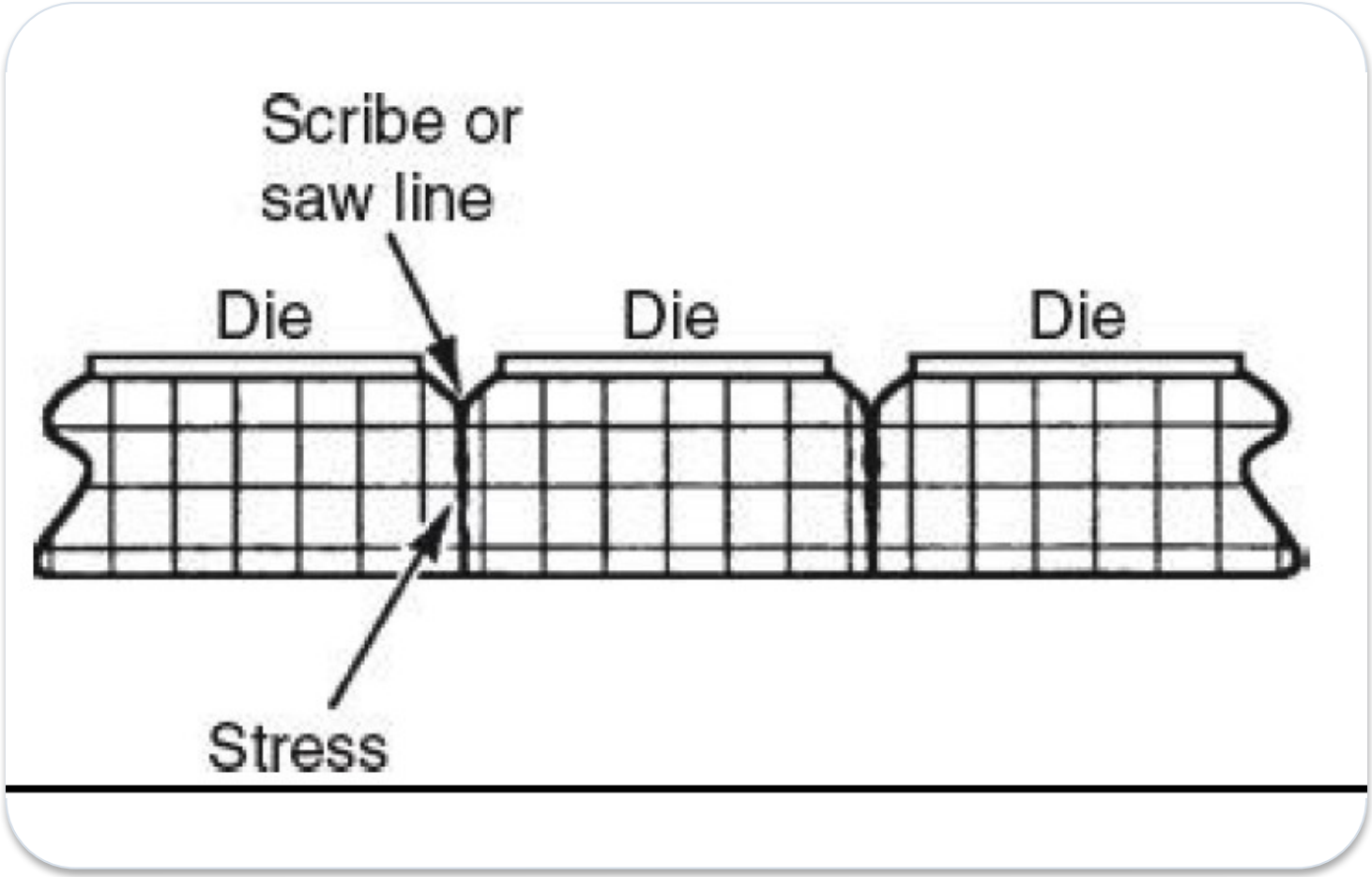
TOPIC 02

Sawing



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Evidence 1

The traditional chip-packaging process starts with the separation of the wafer into individual dies. The two methods of die separation are scribing and sawing.
For bump or ball processes, the die separation occurs after the bump or ball system is established on the wafer.

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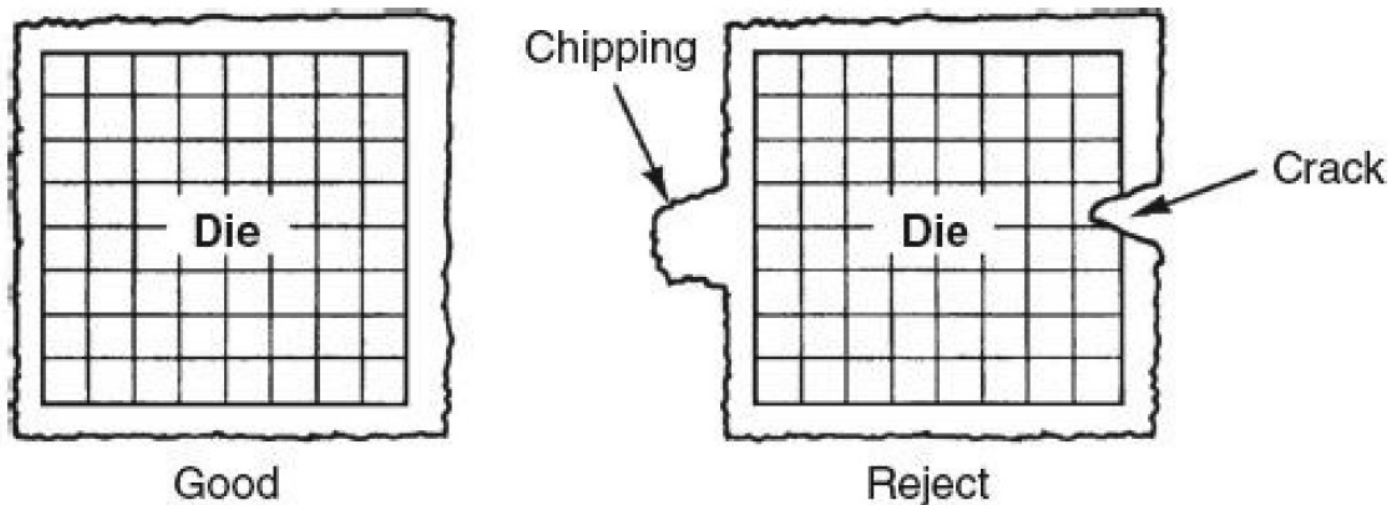
TOPIC 02

Sawing



1

Evidence 1

Sawing is the preferred die separation method for thicker wafers. Two techniques:
Partial cut followed by stress and roller separation.
Complete saw-through, often with wafers mounted on plastic film for stability.
Sawing provides cleaner edges and fewer cracks.

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TOPIC 02

Scribing

1

Key idea 1

Scribing, or diamond scribing, was the first production die-separation technique developed in the industry. It requires dragging a diamond-tipped scribe through the center of the scribe lines and separating the die by flexing the wafer.
Scribing becomes unreliable in wafers over 10 mils thick.

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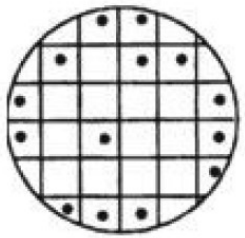
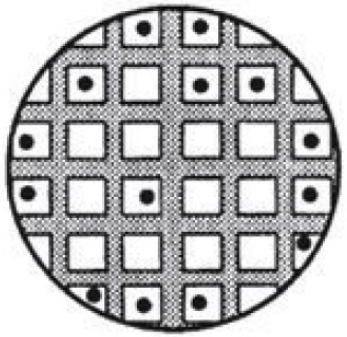
TOPIC 02

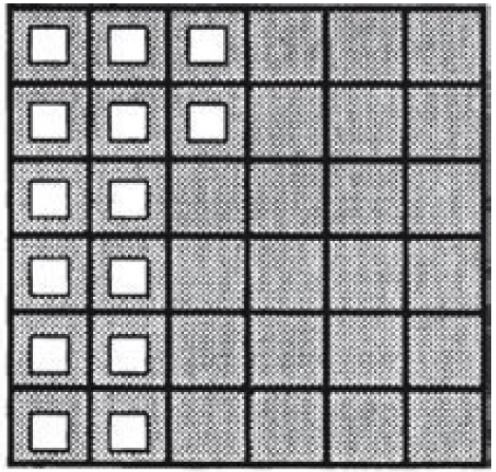
Die Pick and Place



After sort



After die separation



Plated die

1

Evidence 1

After sawing, the separated die are transferred to a station to select (pick) the functioning die.
In operation, a memory tape or disk that has recorded the locations of the good die (from wafer sort) and is loaded into an automatic “pick” tool
A vacuum wand picks up the good die and automatically places them in a sectioned plate for transfer to the next operation.

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TOPIC 02

Die Inspection

Before being committed to the rest of the process, the die are given an optical inspection. Of primary interest is the quality of the die edge, which should be free of chips and cracks. This inspection also sorts out surface irregularities, such as scratches and contamination. The identification of damaged die saves the expense and time of packaging a failed die.

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TOPIC 02

Die Attach

Before being committed to the rest of the process, the die are given an optical inspection.
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This inspection also sorts out surface irregularities, such as scratches and contamination.
The identification of damaged die saves the expense and time of packaging a failed die.

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TOPIC 02

Die Attach

1

Key idea 1

A crucial step in packaging, creating a strong bond between the chip and package.
Goals: Physical bond, electrical connection, and heat transfer.
Requirements: Durable, contaminant-free, and cost-effective.

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TOPIC 02

Eutectic Die Attach

1

Die Attach Methods: Eutectic and epoxy adhesives.

Die Attach Methods: Eutectic and epoxy adhesives.
Eutectic Method: Utilizes the eutectic alloy of gold and silicon, which melts at a lower temperature than either material alone.
Process: Gold is plated onto the die attach area and alloys with silicon when heated

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TOPIC 02

Epoxy Die Attach

Epoxy Die Attach: Uses thick liquid epoxy adhesives, offering flexibility in electrical and thermal conductivity.
Process: Epoxy is dispensed, the die is positioned, and the bond is cured in an oven.
Advantages: Cost-effective, easy to automate, and requires no stage heating.
Disadvantages: Potential decomposition at high temperatures and lower bonding strength compared to eutectics.
Successful Die Attach Indicators: Proper alignment, solid and uniform contact, and a void-free die attach area.

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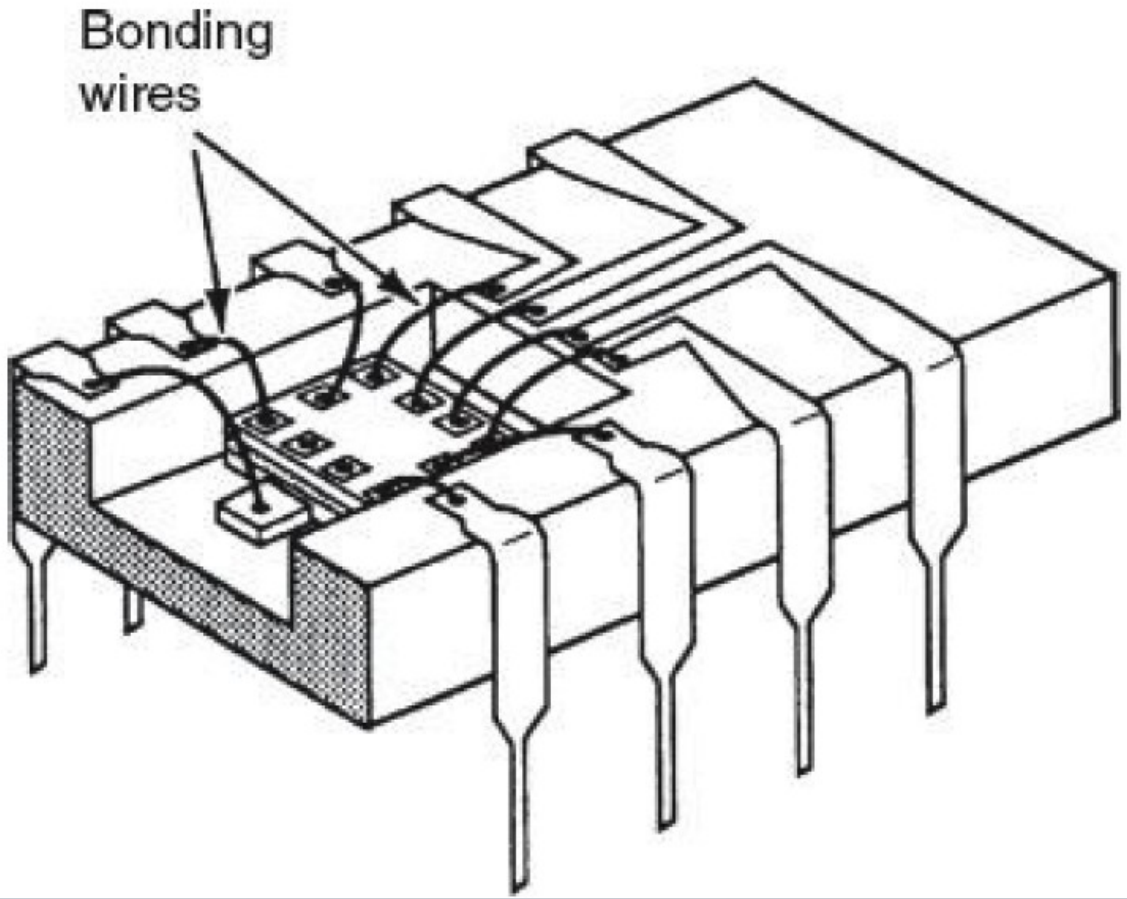
TOPIC 02

Wire Bonding



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Evidence 1

Wire Bonding: Critical process connecting chip bonding pads to package inner leads.
Process: A thin wire is bonded to the chip, spanned to the lead, and then bonded again.
Requirements: Precise wire placement, good electrical contact at both ends, and proper loop formation.
Pitch: Distance between adjacent wires.

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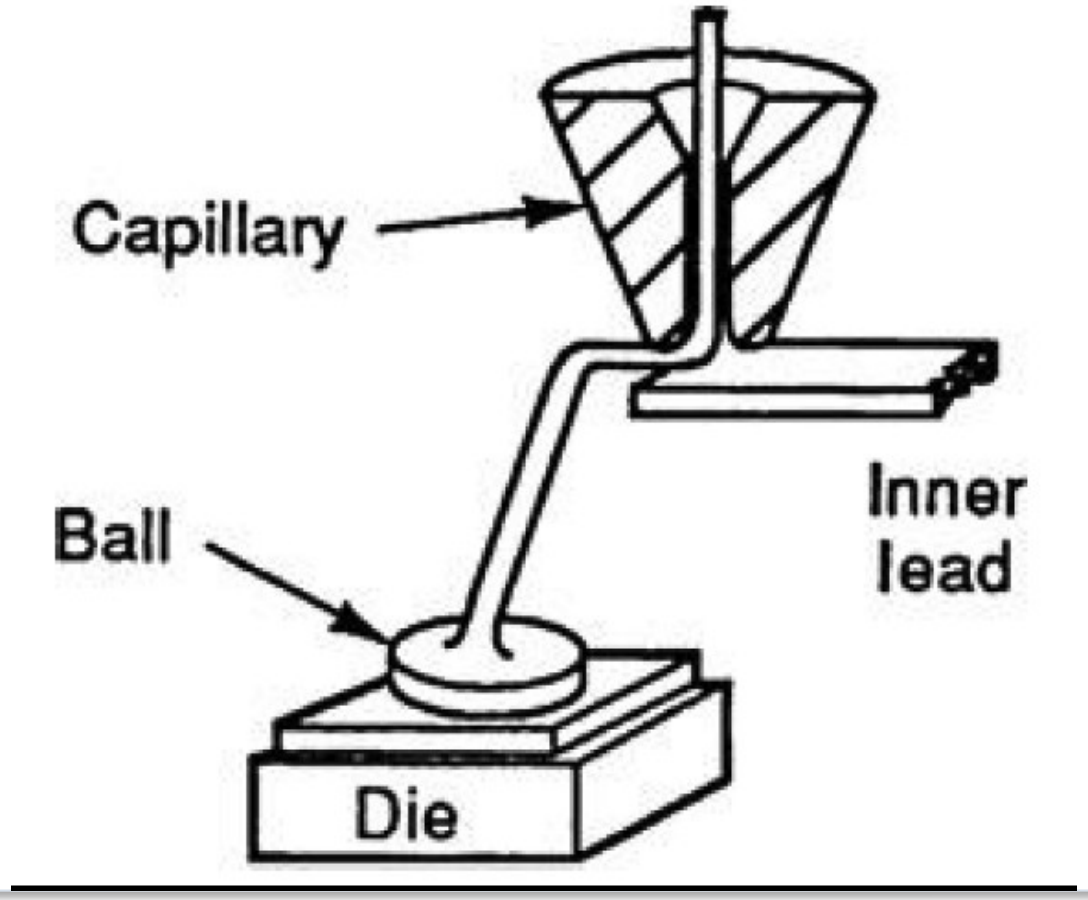
TOPIC 02

Gold Wire Bonding



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Evidence 1

Gold Wire Bonding: Preferred for its conductivity, heat transfer, and resistance to oxidation.
Methods: Thermocompression (TC) and thermosonic bonding.
Thermocompression Bonding: Uses heat and pressure to create a ball bond between the wire and bonding pad.
Process: A ball of melted gold wire is forced onto the pad, creating a strong bond.

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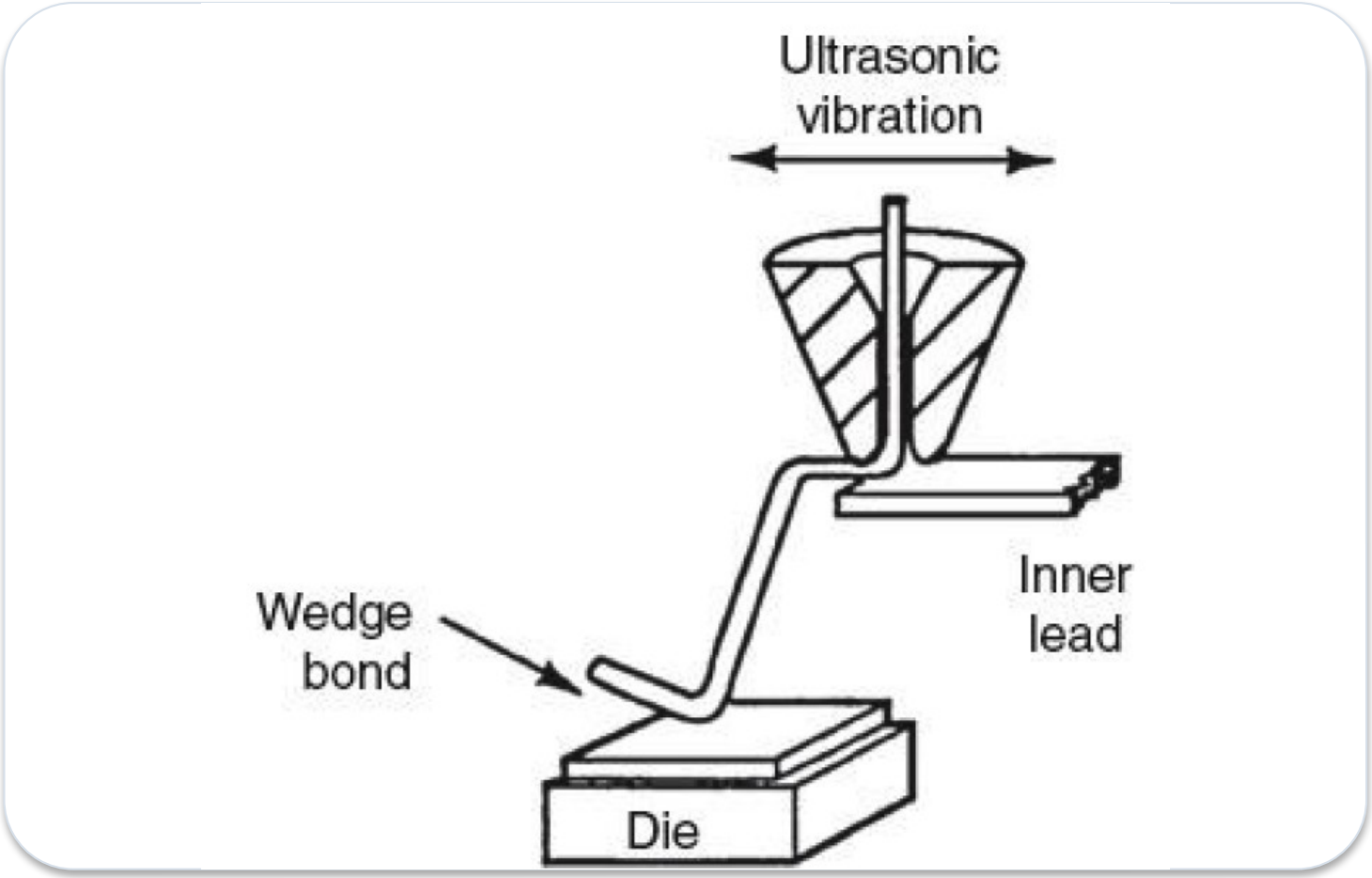
TOPIC 02

Aluminum Wire Bonding



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Evidence 1

Aluminum Wire Bonding: Cost-effective alternative to gold, offering advantages for compatibility with epoxy die attach.
Method: Ultrasonic or wedge bonding, where a wedge forces the wire onto the pad with ultrasonic energy.
Process: No ball is formed, the wire is bonded to the pad and lead using ultrasonic energy.

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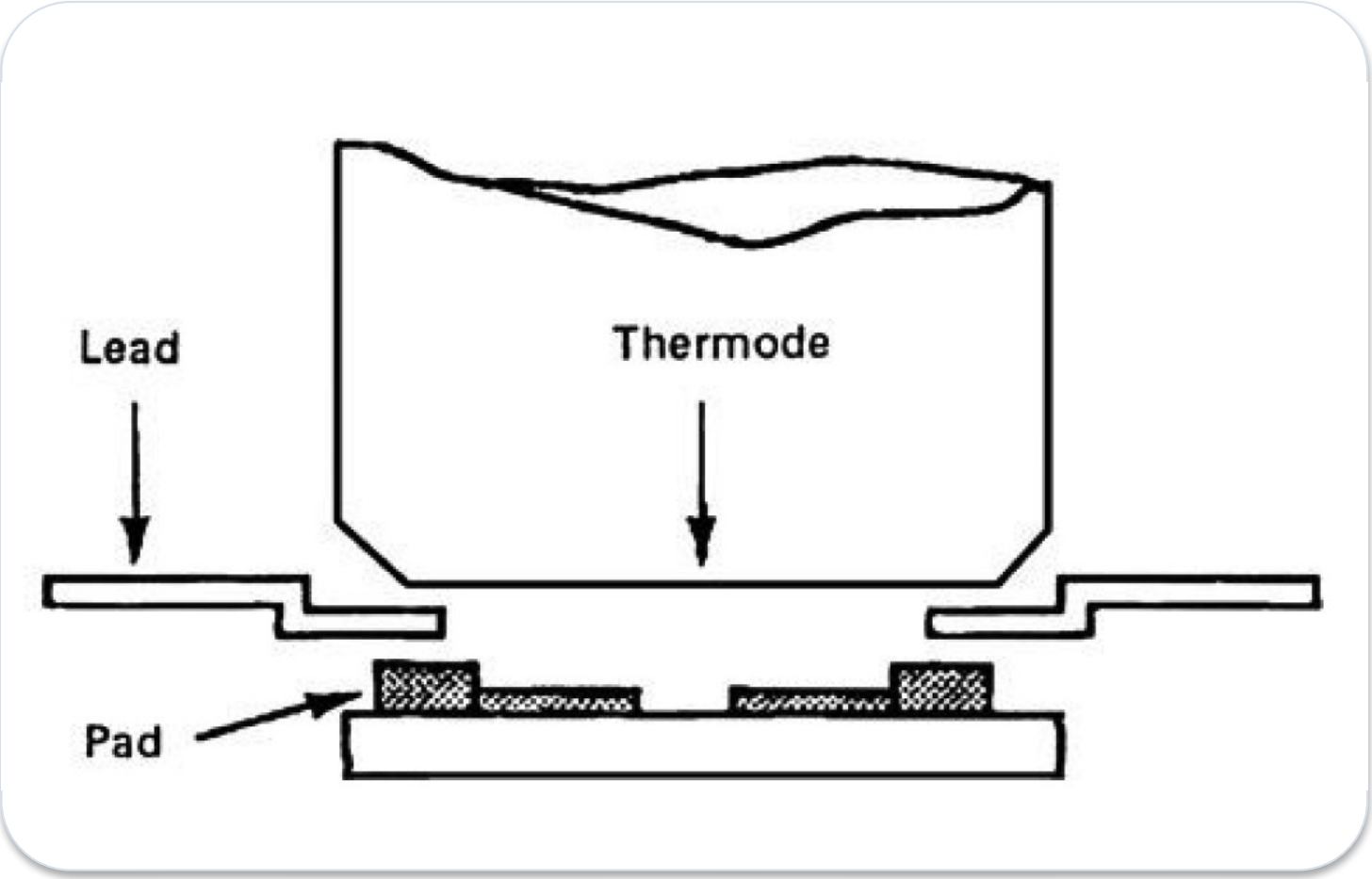
TOPIC 02

Tape Automated Bonding Process



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Evidence 1

Tape Automated Bonding (TAB): Used for connecting chips directly to PCBs, especially for thin devices.
Process: Leads are formed on a flexible tape, and the tape is positioned over the chip for bonding.
Applications: Thin devices, such as credit card-sized radios.

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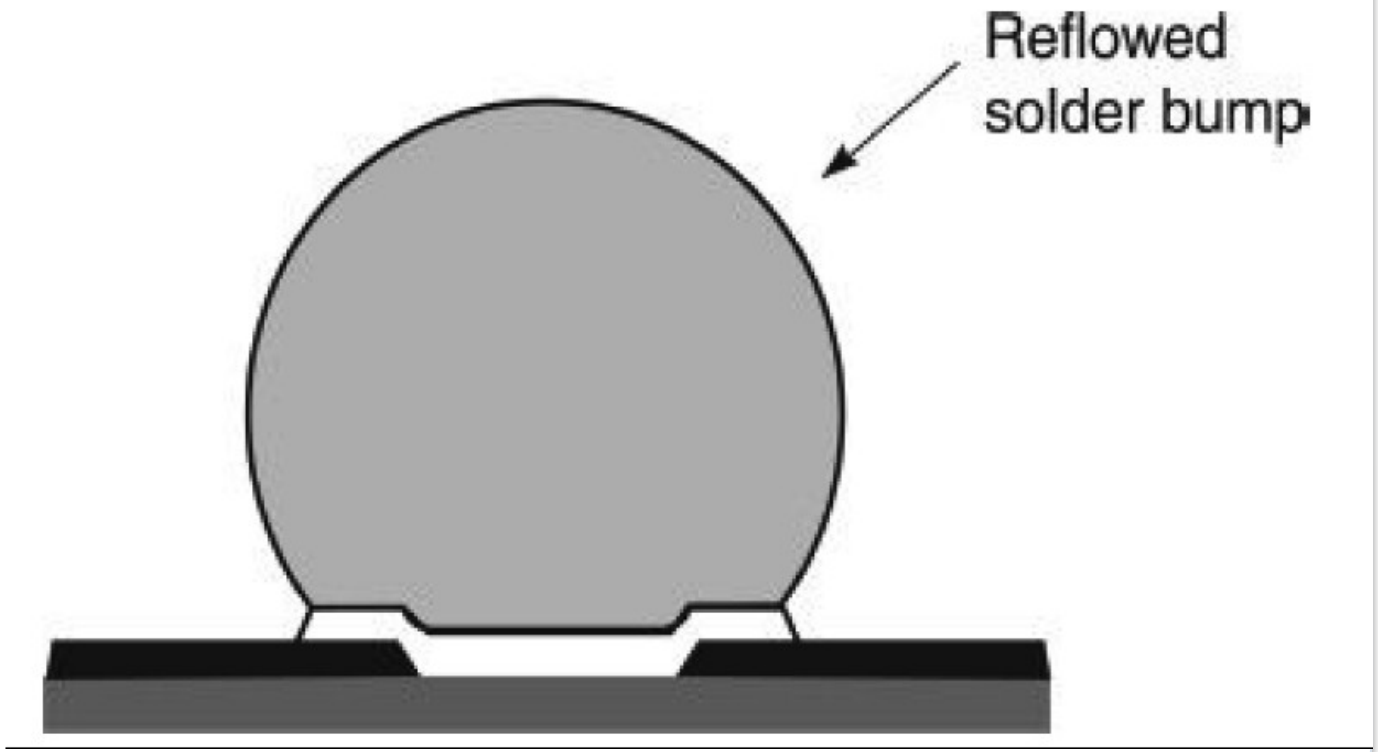
TOPIC 02

Bump or Ball FlipChip Bonding



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Evidence 1

Limitations of Wire Bonding: Electrical resistance, height limitations, potential for shorting, and limited space for connections.
Bump (Ball) Bonding: Addresses wire bonding limitations by replacing wires with metal bumps.
Advantages: Allows for bonding pads on both the edge and interior of the chip, increasing signal speed.
Process: Bumps are soldered to corresponding leads on the package or PCB after the chip is flipped over.

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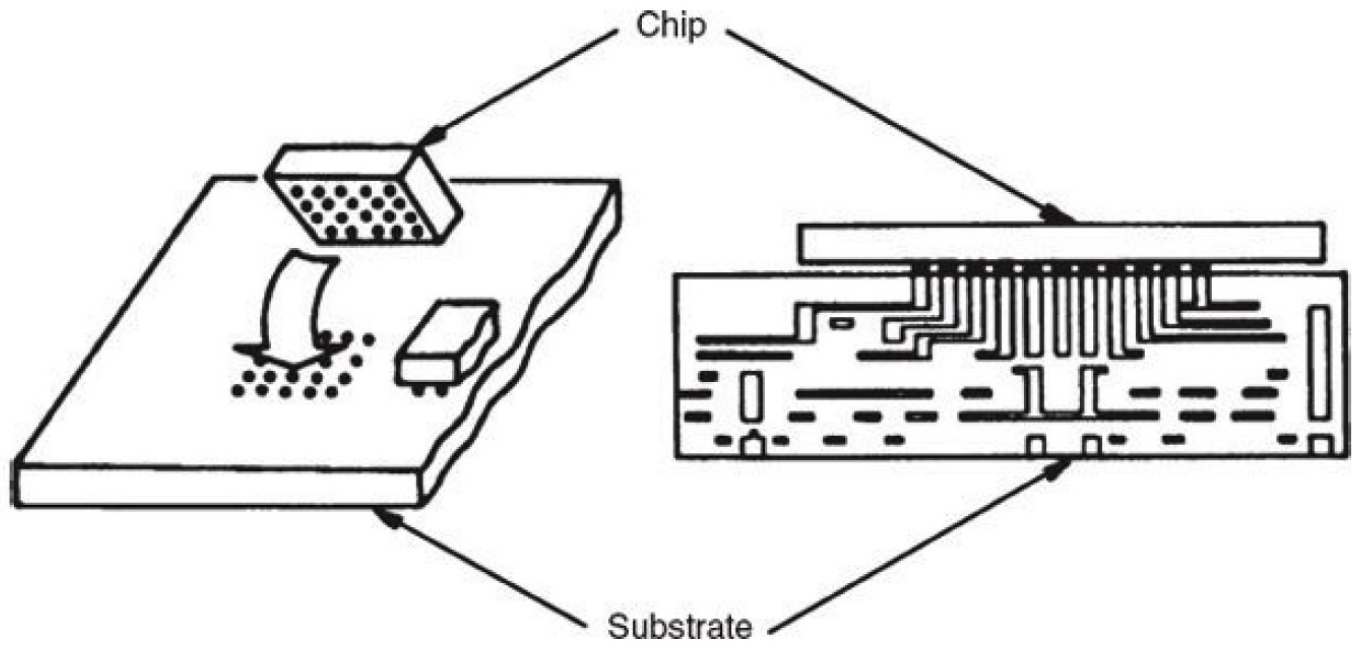
TOPIC 02

Bump or Ball FlipChip Bonding



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Evidence 1

Flip Chip Connection Technology: Involves depositing solder bumps on the chip during fabrication, flipping the chip, and bonding to package leads.
Advantages: Improved signal speed, higher density, and reduced height.
Underfill: Epoxy used to fill the gap between the chip and package, providing additional stress tolerance.
Wafer Preparation: Wafers are thinned and solder bumps are formed on the bonding pads

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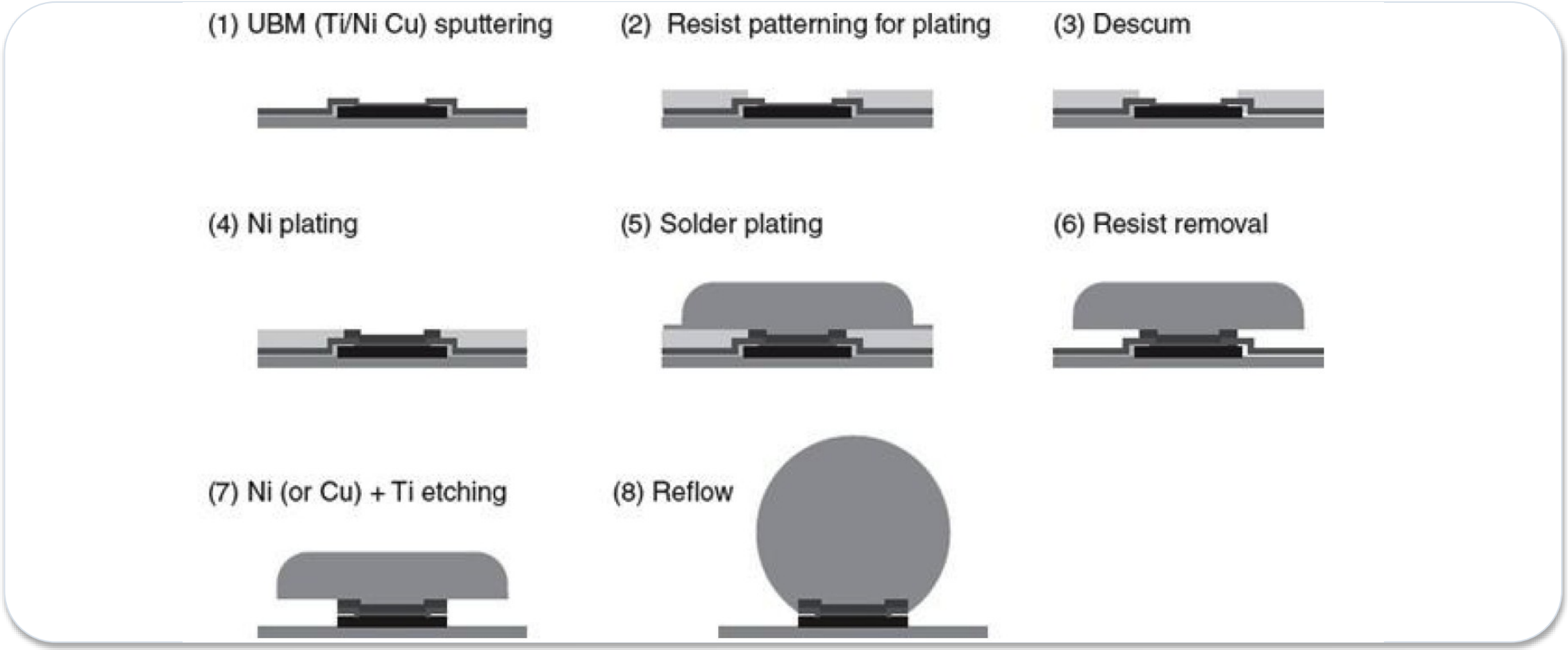
TOPIC 02

Bump or Ball Process



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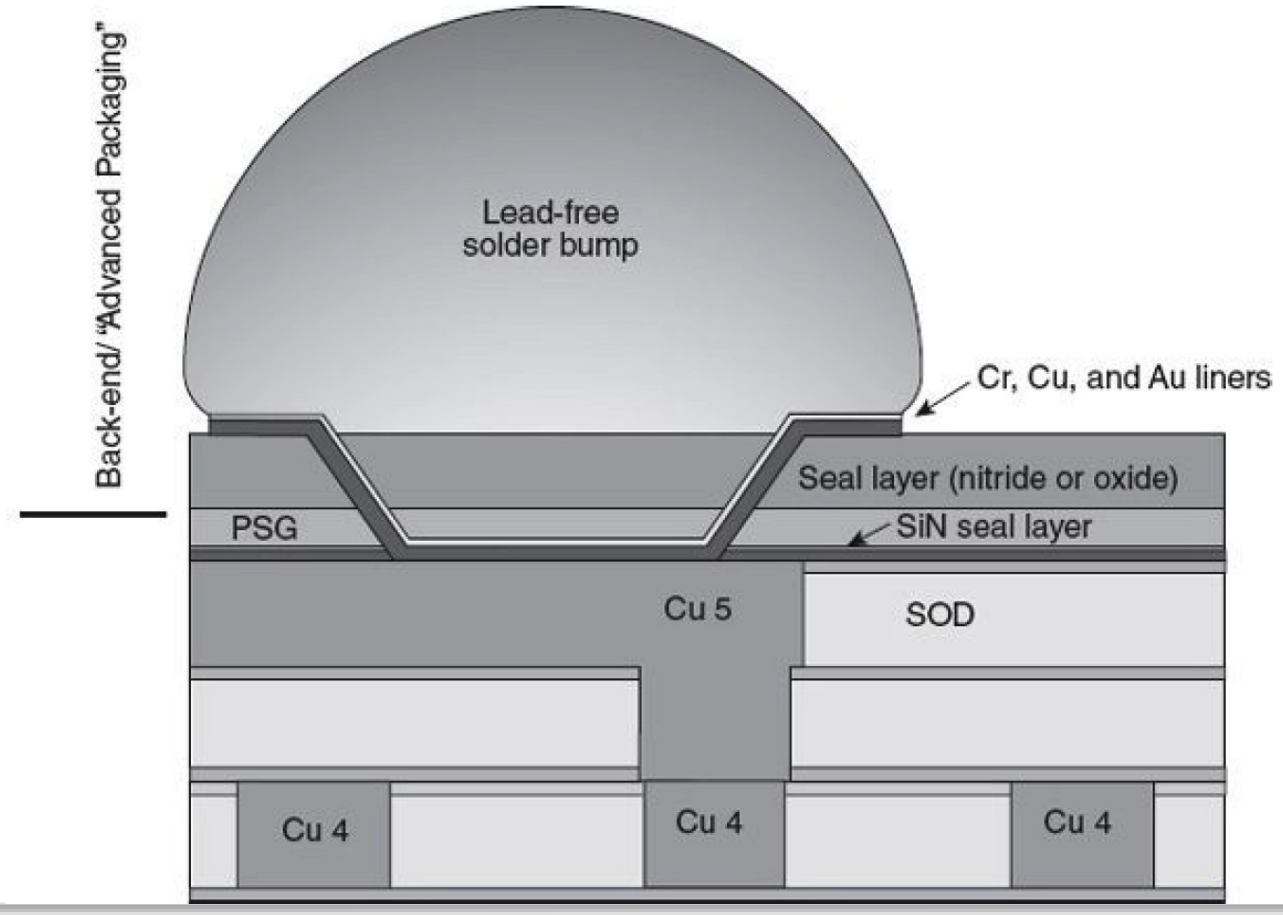
Copper Metallization (Damascene) Bump Bonding



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TOPIC 02

Reflow

1

Key idea 1

A heating of the wafer in the 350°C range in hydrogen enhances the electrical contact to the intermetal stack.
Also, surface tension pulls the solder deposit into a spherical shape, much like surface tension causes spherical soap bubbles.

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TOPIC 02

Die Separation and Die Pick and Place

1

Key idea 1

These two steps are the same as in the wire-bonding process.
Typically, the die are affixed to a flexible tape on a reel that allows automatic handling in the subsequent processes.

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TOPIC 02

Alignment of Die to Package

Packages used in flipchip systems are usually ceramic or organic.
Organic is the term used for packages based on the same materials used for circuit boards.
Sometimes they are referred to as “plastic” packages. The first step is flipping the die upside down and aligning it to the package such that the bumps on the die are positioned over the corresponding leads on the package.

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TOPIC 02

Postbonding and Preseal Inspection

Preseal/Precap Inspection: Performed after bonding to ensure package quality and identify potential reliability issues.
Inspection Categories: Commercial and high-reliability (Mil-Standard 883).
Inspection Criteria: Die attach quality, bond placement, bond shape, and chip surface condition.
High-Reliability Standards: More stringent requirements for chips used in space and military applications.

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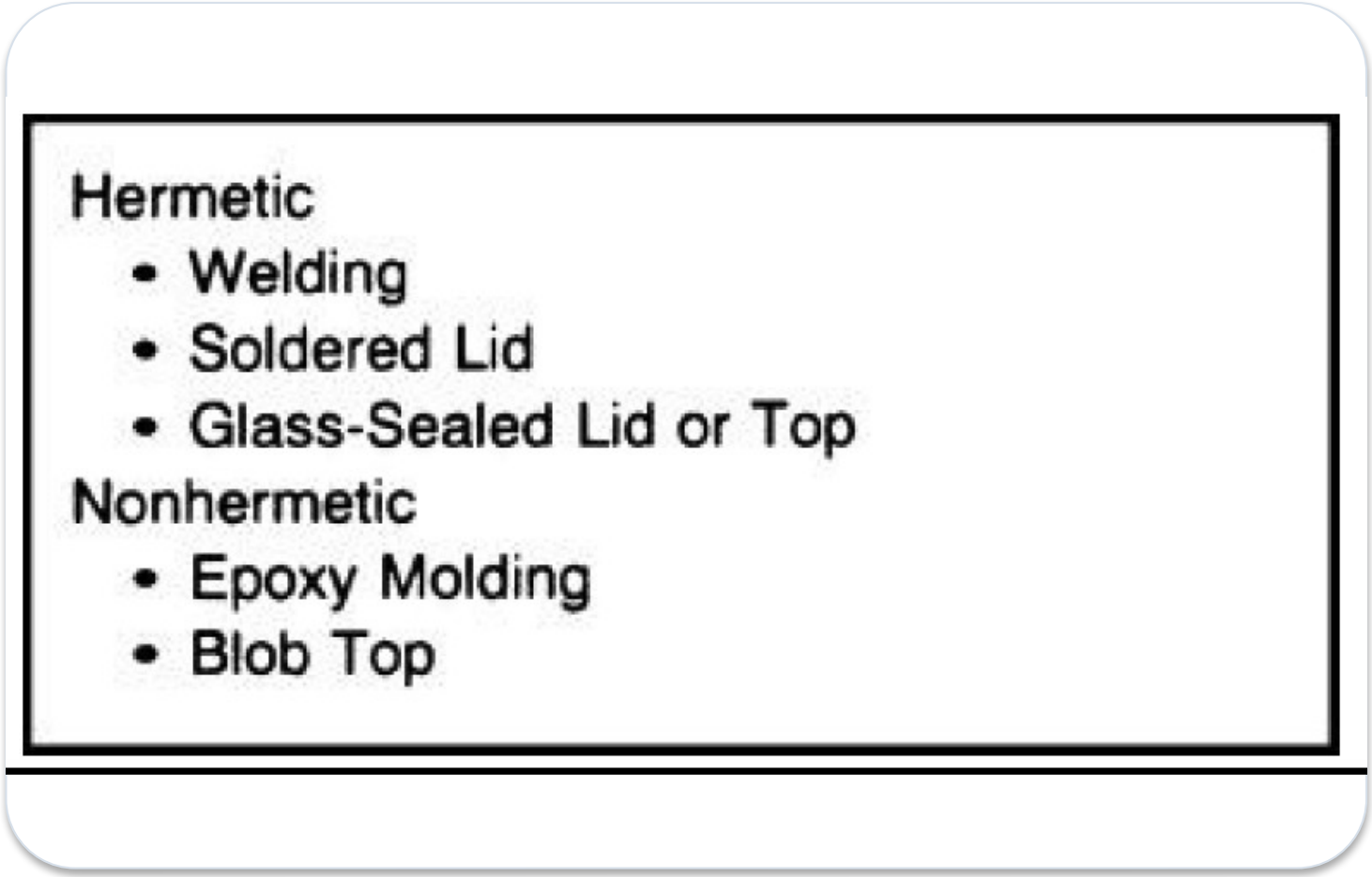
TOPIC 02

Sealing Techniques



1

Evidence 1

Packaging Sealing: Protects the chip from environmental factors, using various methods based on hermeticity and package type.
Methods: Welded seals, soldered seals, glass-sealed lids, CERDIP package construction, molded epoxy enclosures, and sealing layers.

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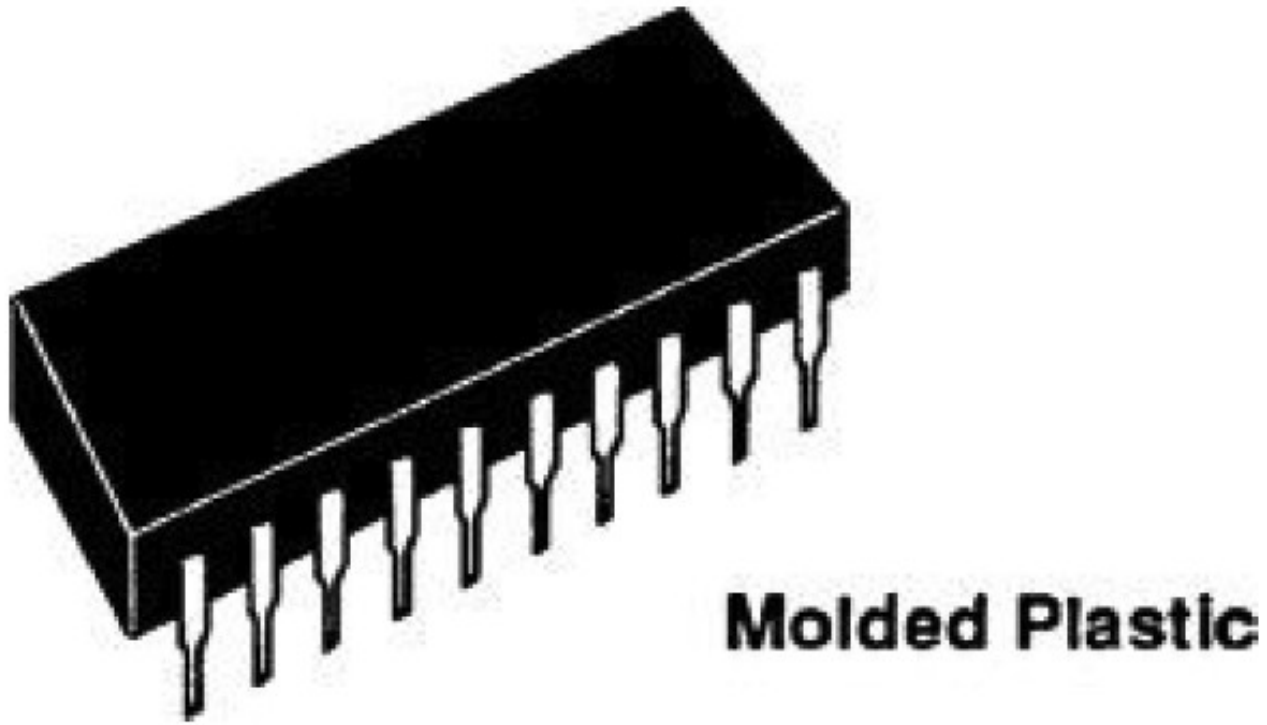
TOPIC 02

Molded Epoxy Enclosures



1

Evidence 1

Epoxy Molding: Used to create traditional plastic packages, providing non-hermetic protection.
Advantages: Lightweight, cost-effective, and efficient to manufacture.
Limitations: Non-hermetic seal, but research is ongoing to improve epoxy materials

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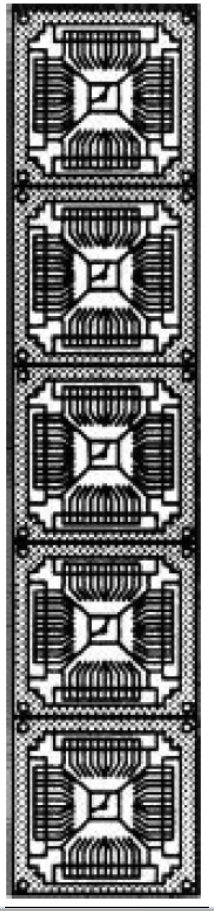
TOPIC 02

Lead Plating



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Evidence 1

Package Lead Finishes: Typically lead-tin solder, tin plate, or gold plate.
Functions: Improved solderability, protection from oxidation and corrosion during storage, and protection from corrosive agents during processing and use.

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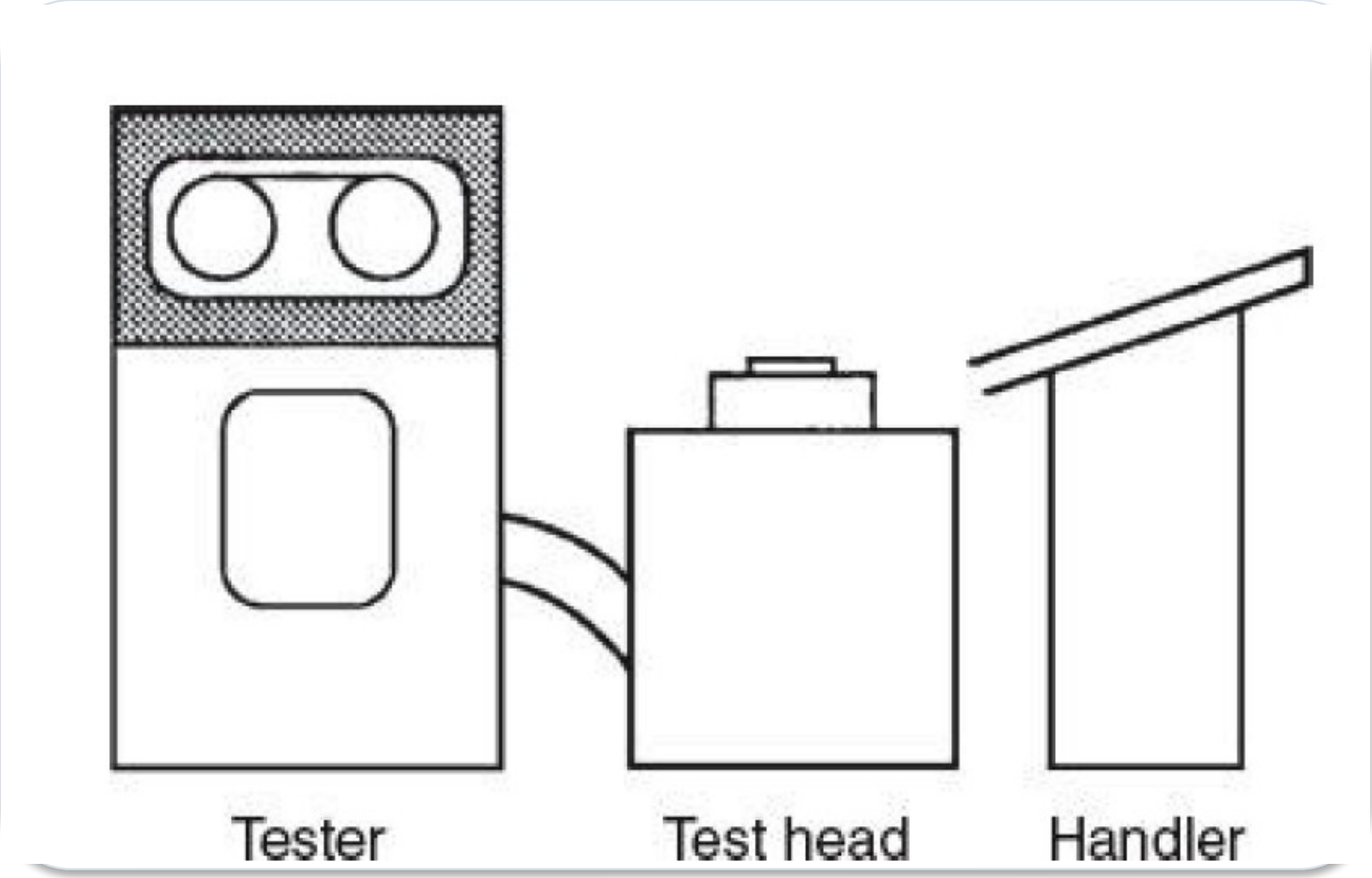
TOPIC 02

Final Testing



1

Evidence 1

At the conclusion of the packaging process, the completed package is put through a series of environmental, electrical, and reliability tests. These tests vary in type and specifications, depending on the customer and use of the packaged devices. The tests may be performed on all of the packages in a lot or on selected samples.

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TOPIC 02

Burn-In Tests

1

Key idea 1

Burn-In Test: Optional test used to identify potential failures in high-reliability devices.
Process: Packages are temperature-cycled under electrical bias.
Goal: Stress the electrical connections and drive contaminants into active circuitry, causing early failure.
Benefits: Increases reliability by identifying and removing devices prone to early failures.

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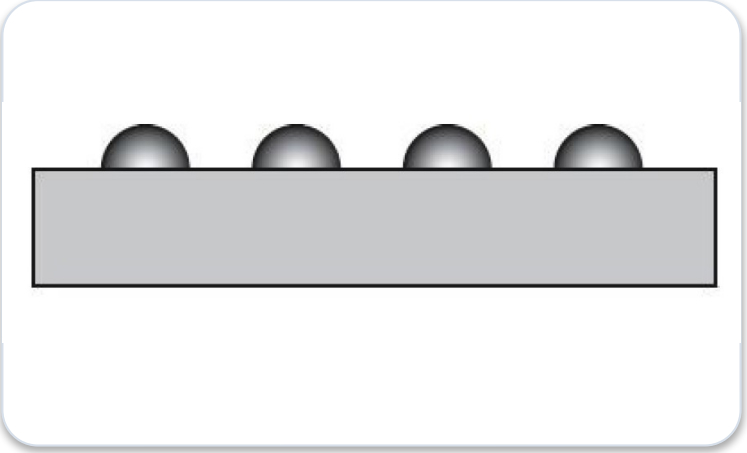
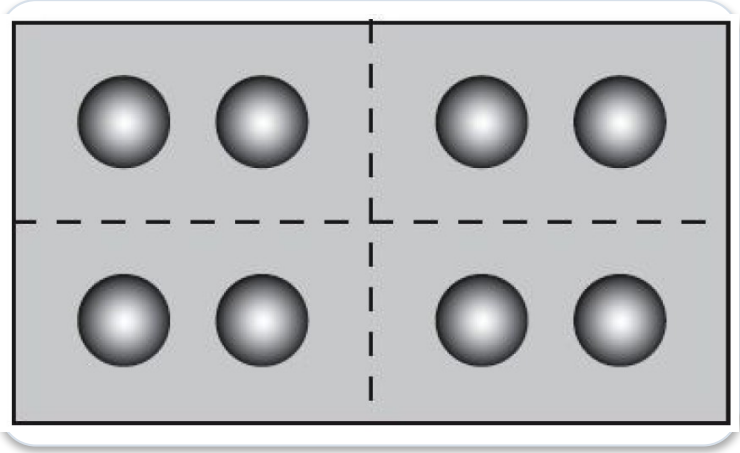
TOPIC 02

Chip-Scale Packages





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Evidence 1

Ideal Package: No package, as it adds electrical resistance, weight, and cost.
Chip-Scale Packages: Minimize package size, reducing cost and increasing density.
Challenges: Providing adequate mechanical and environmental protection and easy connection to PCBs.

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TOPIC 02

Follow up actions

Concepts · mechanisms · process evidence · integration decisions

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TOPIC 02

Process Integration Optimization

Process Monitoring and Control
Inline Testing: Continuously monitor key process parameters such as layer thickness, etching depth, and doping concentration during fabrication.
Statistical Process Control (SPC): Use SPC tools to track variations in the manufacturing process and ensure they remain within acceptable limits.
CMP (Chemical Mechanical Polishing) Optimization
Surface Planarity: Ensure that CMP leaves a smooth and planar surface for subsequent layers, particularly in multilayer interconnect systems.
Erosion and Dishing Control: Fine-tune CMP to minimize erosion and dishing effects that can lead to defective interconnect structures.
Stress and Strain Management
Stress Testing: Evaluate the stress and strain in thin films (e.g., silicon, oxides, metals) to prevent cracking, warping, or reliability issues over time.
Strain Engineering: Use techniques like strained silicon to enhance carrier mobility and improve device performance

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TOPIC 02

Process Integration Optimization

Material Compatibility:
Ensure compatibility between different layers (e.g., polysilicon, metal, dielectric) to avoid issues like delamination, stress, or oxidation problems.
Thermal Budget Management:
Optimize thermal cycles (e.g., during oxidation and annealing) to prevent damage to previously deposited layers or structures.
Interconnect Reliability:
Focus on preventing electromigration and stress migration in metal interconnects to improve long-term reliability.
Etch Stop Layer Optimization
Etch Stop Layer Integration: Ensure etch stop layers are properly implemented to avoid over-etching or damage to underlying structures.
Process Window Control: Keep the etching process within a narrow window to maintain uniformity and precision, especially with advanced nodes.

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TOPIC 02

Yield Improvement

Process Integration Debugging
Failure Analysis: For any chips that fail testing, perform failure analysis to identify the root cause of the defect (e.g., contamination, misalignment, or material defects).
Redesign or Adjustments: Based on failure analysis, adjust process steps or redesign aspects of the circuit to mitigate future issues.
Electrical Testing
Parametric Testing: Test individual components like transistors, resistors, and capacitors for electrical performance parameters (e.g., threshold voltage, leakage current, capacitance).
Yield Analysis: Analyze wafer yields and look for defects caused by process variations. Adjust processes to improve overall yields.

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TOPIC 02

Continuous Improvement

Defect Inspection and Reduction
Defect Inspection: Perform optical or scanning electron microscopy (SEM) inspections to detect physical defects such as particles, scratches, and pattern deviations.
Defect Classification: Classify defects based on their severity and location. This helps prioritize which defects require process improvements.
Continuous Improvement
Feedback Loop: Use the data collected from testing and inspection to feed back into the design and manufacturing process for continuous improvements.
Process Updates: As technologies evolve, update the CMOS process to integrate new materials, methods, or scaling approaches (e.g., FinFETs or gate-all-around structures).

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TOPIC 02

The Future

Concepts · mechanisms · process evidence · integration decisions

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TOPIC 02

Packaging

The future of Integrated Circuit (IC) packaging is driven by the need for higher performance, lower power consumption, smaller form factors, and increased functionality.
As the semiconductor industry continues to follow Moore's Law (though slowing down), the focus has shifted from traditional scaling of transistors to advanced packaging techniques to meet these needs.
Here are some key trends and technologies shaping the future of IC packaging:

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TOPIC 02

Heterogeneous Integration

Heterogeneous integration involves packaging multiple types of chips (e.g., logic, memory, analog, RF) and components into a single package, improving performance, power efficiency, and reducing size. This enables combining different technologies and materials that would be hard to integrate on a single monolithic chip.
System-in-Package (SiP): Multiple ICs are integrated into a single package, forming a miniaturized system with specific functionalities, such as combining sensors, logic, and memory.
3D Heterogeneous Integration: This involves stacking different types of chips (such as DRAM on top of logic) in a 3D arrangement, connected using through-silicon vias (TSVs) or hybrid bonding. This increases bandwidth while reducing latency and power consumption.

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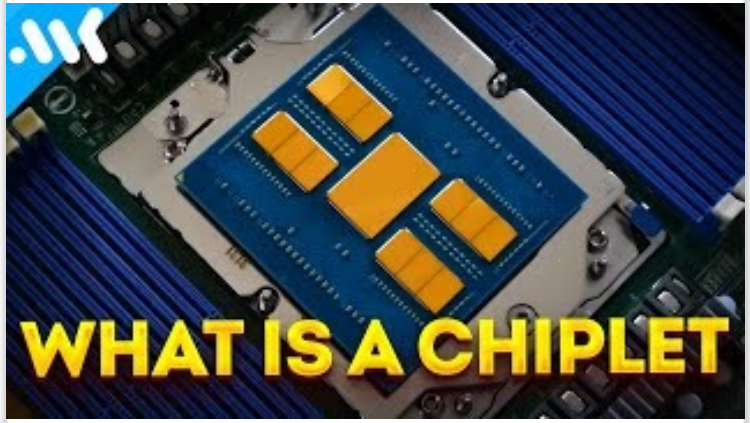
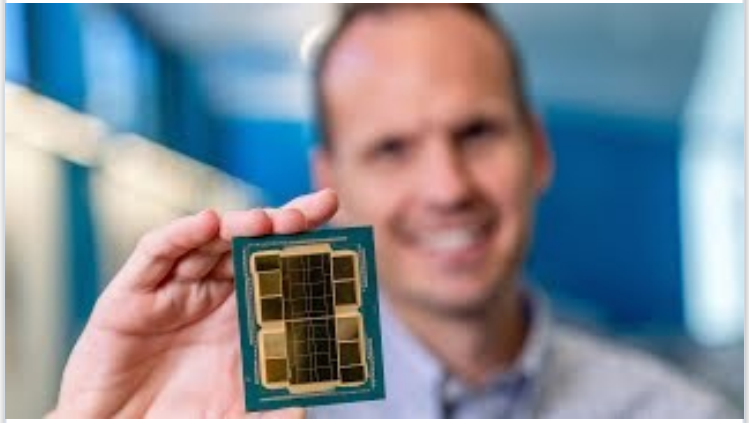
TOPIC 02

3D IC Packaging





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Evidence 1

In traditional ICs, components are placed side by side, leading to limitations in size and speed. 3D IC packaging, where dies are stacked on top of each other, allows for greater density and reduced interconnect lengths. It is highly beneficial for applications requiring high bandwidth and low latency, such as AI, machine learning, and high-performance computing.

Chiplet-based architectures: In this approach, multiple small functional blocks (chiplets) are connected through high-density interconnects within a single package. It allows for better yield, reduced cost, and flexibility by using different process nodes for different chiplets.

Die stacking and TSVs: Through-silicon vias enable vertical connections between stacked dies, offering significant performance improvements in data transfer rates between layers.

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TOPIC 02

Transistors

The future of transistors in IC (Integrated Circuit) manufacturing is driven by the demand for higher performance, lower power consumption, and smaller sizes, as well as new emerging applications such as artificial intelligence (AI), quantum computing, and 5G/6G communication.
As traditional CMOS (Complementary Metal-Oxide-Semiconductor) scaling slows down, the semiconductor industry is exploring new materials, architectures, and technologies to continue improving transistor performance. Here are some key trends and technologies expected to define the future of transistors in IC manufacturing:

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TOPIC 02

Gate-All-Around (GAA) transistors

Gate-All-Around (GAA) transistors are a key step forward from FinFETs (Fin Field-Effect Transistors), which are currently used in advanced nodes. In GAA transistors, the gate completely surrounds the channel, providing better control over the current and reducing leakage. This allows for smaller sizes and higher performance than FinFETs.
Nanowire and Nanosheet GAA FETs: GAA transistors will initially be implemented using nanowires or nanosheets. Samsung is already working on this technology with their 3nm GAA process, and other manufacturers like TSMC are expected to follow. Nanosheet transistors, in particular, allow designers to adjust the width of the nanosheets to balance performance and power efficiency.
Advantages of GAA FETs: Better electrostatic control of the channel, improved performance at smaller sizes, reduced leakage, and better power efficiency compared to FinFETs.

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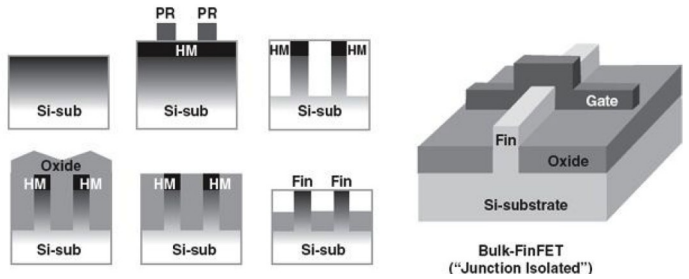
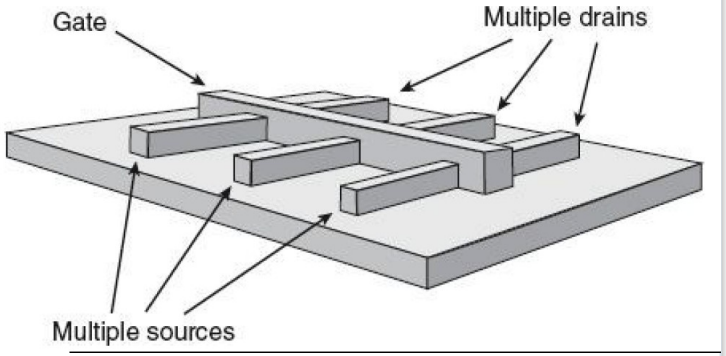
TOPIC 02

FinFET





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FinFET: vertical transistor with a 3D fin structure.

FinFET: vertical transistor with a 3D fin structure.
Raised fin covered with a high-k metal gate stack, increasing gate area by utilizing both sides and the top.
The "fin" is created by etching into the base material,
Multiple gates can also be added to the transistor for further innovation.

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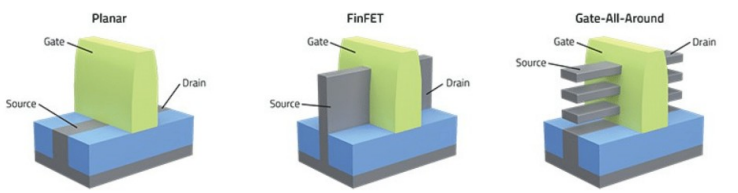
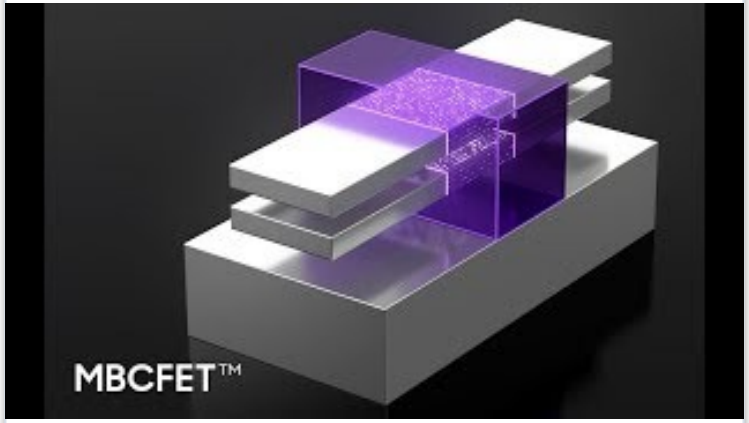
TOPIC 02

GAAFET





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Evidence 1

GAAFET, also known as a surrounding-gate transistor (SGT), is similar in concept to a FinFET except that the gate material surrounds the channel region on all sides. Depending on design, gate-all-around FETs can have two or four effective gates.

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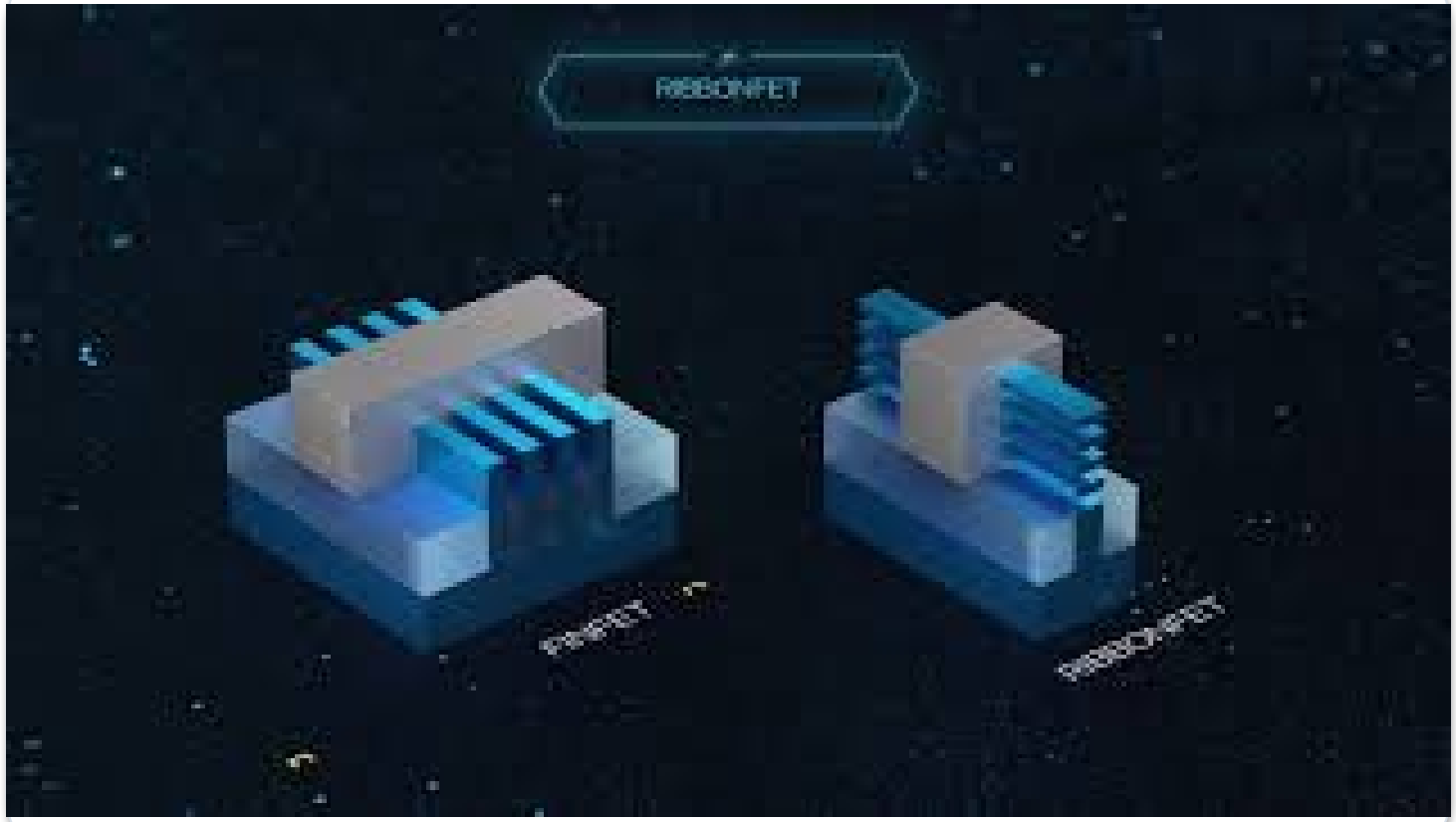
TOPIC 02

Vertical Transistors



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Evidence 1

Vertical transistors represent a potential future direction for transistor design. By stacking transistors vertically, it is possible to significantly increase transistor density without increasing the overall footprint. This can help extend Moore's Law beyond traditional scaling limits.
Vertical Nanowire FETs: These are transistors where the current flows vertically, reducing the chip’s area while maintaining or even improving performance. Vertical transistors can potentially overcome some of the limitations of traditional planar and even GAA designs.
Vertical Transport Field Effect Transistors (VTFETs): IBM and Samsung have introduced the concept of VTFETs, where transistors are built vertically rather than in a horizontal plane. This allows for greater density and potentially better power efficiency.

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TOPIC 02

2D Materials (beyond Silicon)

As the physical limits of silicon approach, new materials are being explored for transistors. Two-dimensional (2D) materials such as graphene, transition metal dichalcogenides (TMDs) (e.g., molybdenum disulfide), and black phosphorus offer promising electrical properties that could enable transistors to be scaled down further.
Graphene: A single layer of carbon atoms, graphene has excellent electrical properties, such as high electron mobility, making it a strong candidate for future transistor materials.
Molybdenum Disulfide (MoS2): A semiconductor from the TMD family, MoS2 is another promising 2D material because of its bandgap properties, which are more suitable for transistors than graphene.
Advantages of 2D materials: High electron mobility, potential for extreme miniaturization (down to atomic thickness), and better thermal management.

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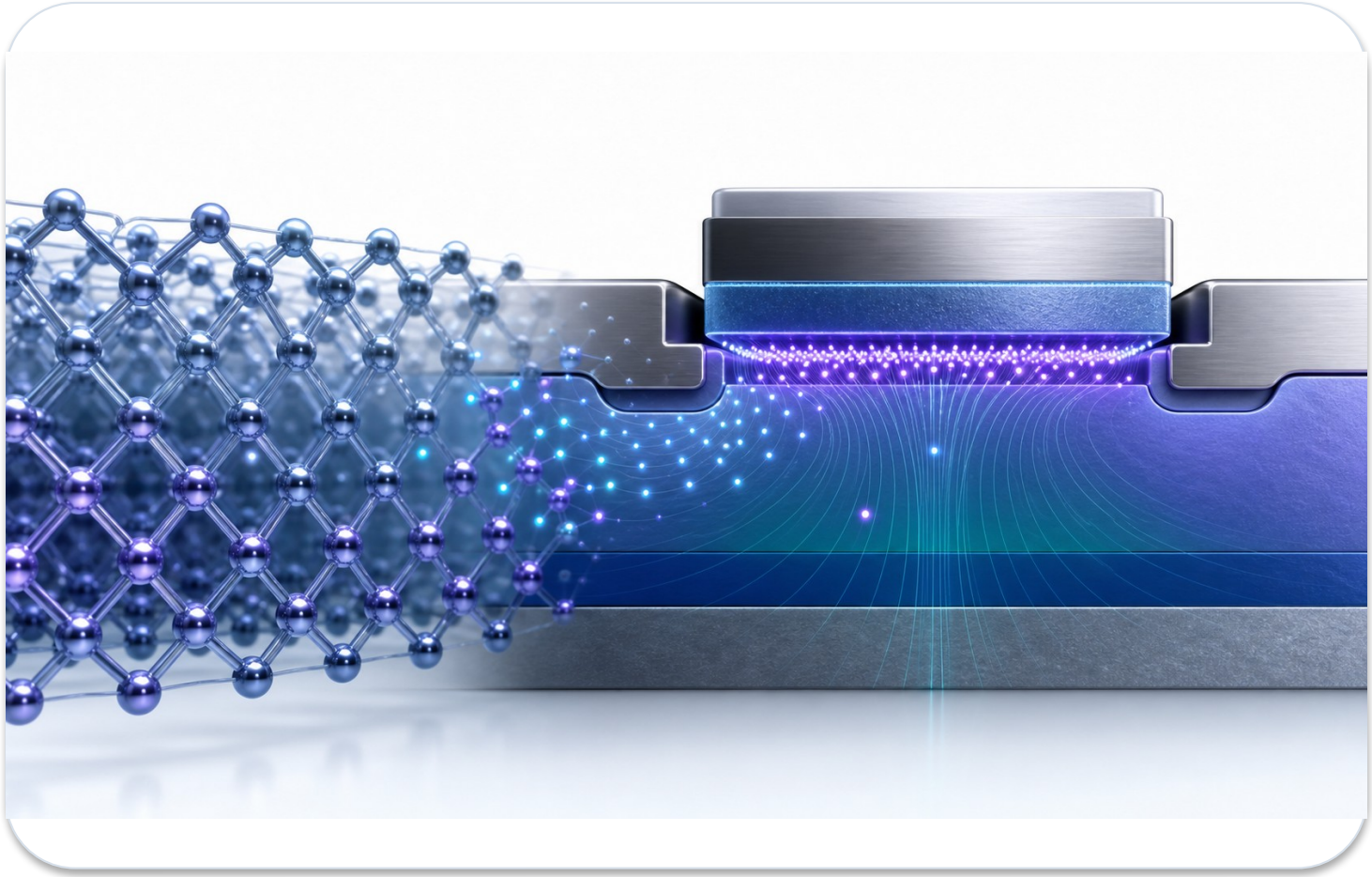
TOPIC 02

Quantum Computing Transistors



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Evidence 1

As quantum computing evolves, new types of transistors may be developed to support qubit-based architectures. These transistors would need to handle quantum states like superposition and entanglement, operating at cryogenic temperatures and with very low error rates.
Quantum Dots: Quantum dots can confine electrons or holes in a small space and could serve as the basis for transistors in quantum computing applications.
Superconducting Transistors: Used in quantum computing, these transistors operate at extremely low temperatures and use superconducting materials to enable qubit manipulation.

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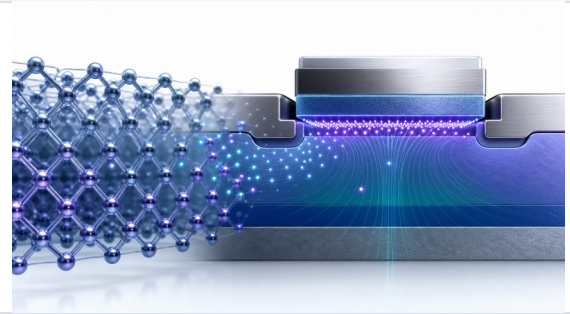
HANDS-ON ACTIVITY 1 · 35 MIN

Device Physics Evidence Map

A new process engineer must explain why doping, junction bias and oxide thickness matter before reviewing a CMOS traveller.



1. **Frame**
2. **Map**
3. **Analyse**
4. **Verify**
5. **Decide**

YOU'LL PRODUCE — **Completed device-physics evidence map with at least six mechanism-control-consequence links.**

Evidence: activities/01-device-physics-evidence-map · Detailed procedure: Learner Guide and activity folder

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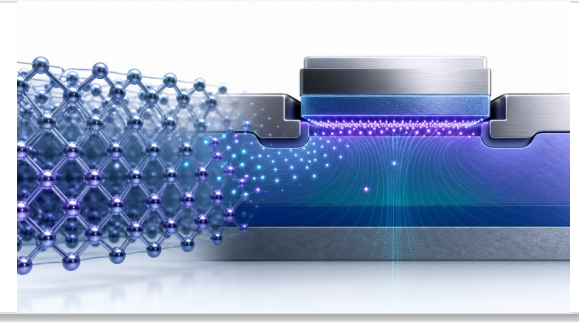
HANDS-ON ACTIVITY 2 · 40 MIN

Process Module Architecture

A fab is onboarding a mixed-experience team and needs a common architecture view that separates FEOL, MOL, BEOL and packaging loops.



1. **Frame**
2. **Map**
3. **Analyse**
4. **Verify**
5. **Decide**

YOU'LL PRODUCE — **Process architecture map with module boundaries, control gates and two feedback loops.**

Evidence: activities/02-process-module-architecture · Detailed procedure: Learner Guide and activity folder

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HANDS-ON ACTIVITY 3 · 50 MIN

CMOS Manufacturing Flow and Control Gates

A product transfer team must turn a high-level CMOS description into a controlled manufacturing flow for a pilot lot.

1. **Frame**
2. **Map**
3. **Analyse**
4. **Verify**
5. **Decide**

YOU'LL PRODUCE — **One-page CMOS process flow with control gates and three device-physics sequence rationales.**

Evidence: activities/03-cmos-manufacturing-flow · Detailed procedure: Learner Guide and activity folder

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HANDS-ON ACTIVITY 4 · 45 MIN

Lithography-Etch Interaction Case

Contact resistance rises after a lithography focus excursion and a longer-than-normal plasma over-etch.

1. **Frame**
2. **Map**
3. **Analyse**
4. **Verify**
5. **Decide**

YOU'LL PRODUCE — **Interaction matrix, causal chain and discriminating verification plan.**

Evidence: activities/04-lithography-etch-interaction · Detailed procedure: Learner Guide and activity folder

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HANDS-ON ACTIVITY 5 · 45 MIN

Doping and Thermal-Budget Interaction

Threshold voltage drifts low after a revised rapid thermal anneal recipe is introduced to improve activation.

1. **Frame**
2. **Map**
3. **Analyse**
4. **Verify**
5. **Decide**

YOU'LL PRODUCE **Thermal-budget comparison, evidence map and split-lot experiment design.**

Evidence: activities/05-doping-thermal-budget · Detailed procedure: Learner Guide and activity folder

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HANDS-ON ACTIVITY 6 · 55 MIN

Yield and Performance Metric Verification

A wafer-sort yield decline is concentrated near the edge while centre-die parametric results remain within specification.

1. **Frame**
2. **Map**
3. **Analyse**
4. **Verify**
5. **Decide**

YOU'LL PRODUCE — **Metric dashboard worksheet with calculations, spatial interpretation and verification conclusion.**

Evidence: activities/06-yield-performance-verification · Detailed procedure: Learner Guide and activity folder

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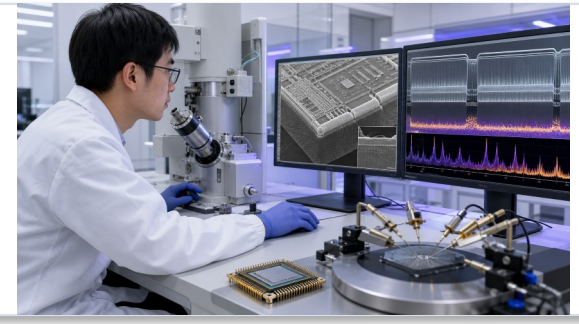
HANDS-ON ACTIVITY 7 · 45 MIN

Reliability Failure Analysis

Qualification finds early-life gate failures and resistance drift in upper-metal interconnects after accelerated stress.



1. **Frame**
2. **Map**
3. **Analyse**
4. **Verify**
5. **Decide**

YOU'LL PRODUCE **Failure-mechanism comparison and reliability verification statement.**

Evidence: activities/07-reliability-failure-analysis · Detailed procedure: Learner Guide and activity folder

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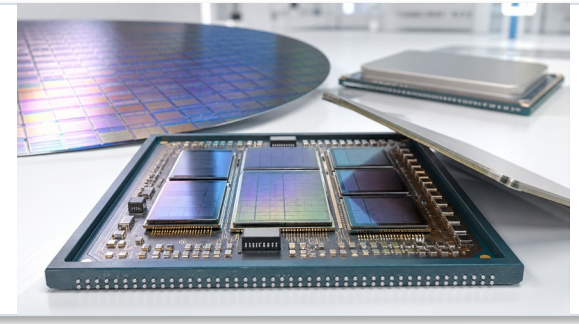
HANDS-ON ACTIVITY 8 · 60 MIN

Follow-Up Action and Integration Review

The integration review board must disposition a yield loss involving etch overrun, dielectric stress and weak SPC reaction discipline.



1. **Frame**
2. **Map**
3. **Analyse**
4. **Verify**
5. **Decide**

YOU'LL PRODUCE — **Integration review action register with containment, CAPA, owners and effectiveness checks.**

Evidence: activities/08-follow-up-action-review · Detailed procedure: Learner Guide and activity folder

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COURSE SUMMARY

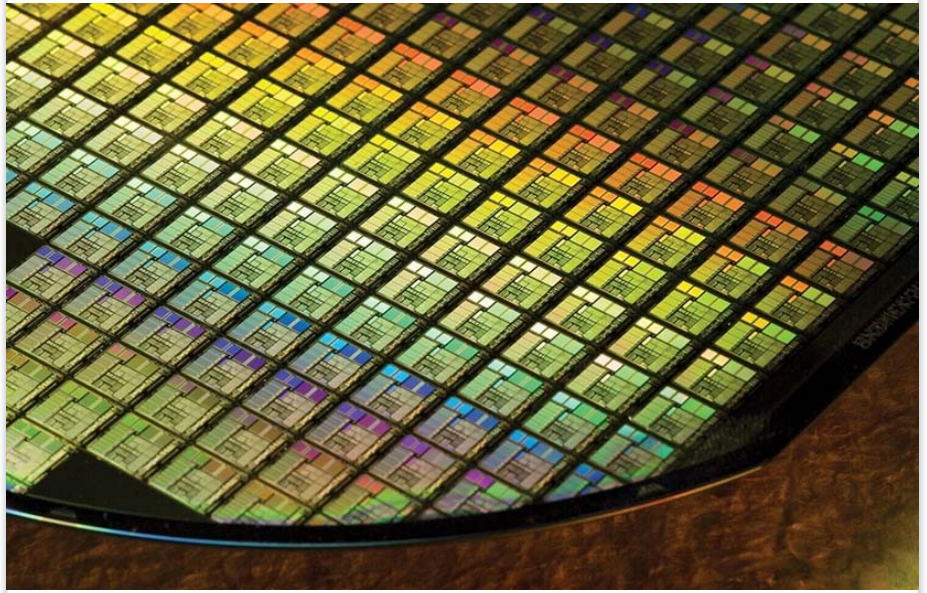
What You Can Now Do

1. **Explain devices**
   Relate carrier physics, junctions and transistor structures to electrical behaviour.

2. **Build the flow**
   Organise process modules into a coherent CMOS manufacturing architecture.

3. **Verify integration**
   Use yield, performance and reliability evidence to judge process outcomes.

4. **Determine action**
   Choose containment, corrective and preventive actions that address verified causes.



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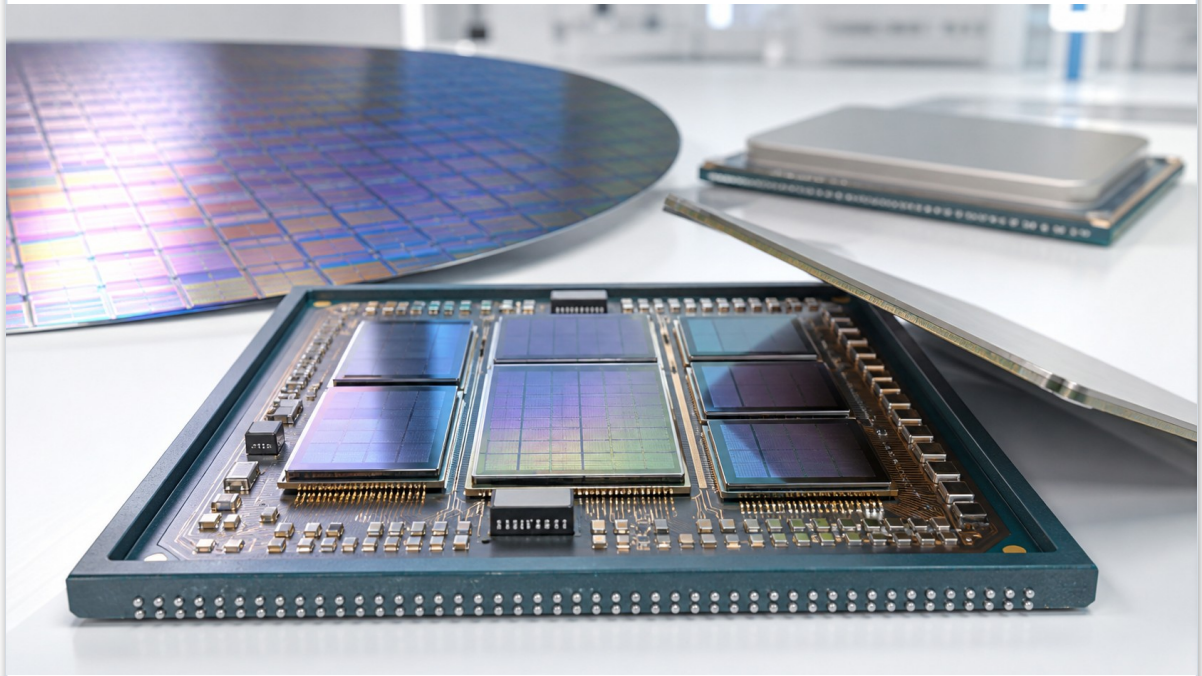
COURSE CLOSE

Courseware, Certification and Support



1 Courseware

Download the current learner materials and activities from the LMS.

2 Certification

Attend the full course to receive a Certificate of Attendance from Tertiary Infotech Academy.

3 Support

Email enquiry@tertiaryinfotech.com or call +65 6100 0613.

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Course Feedback

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Your trainer

Share what helped you learn and what we could improve.

Support

Email enquiry@tertiaryinfotech.com with any follow-up questions.

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Thank You

Questions and discussion

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